

ADDITIONAL

**UNDER THE EFFECT OF EARTHQUAKE
PRINCIPLES FOR DESIGNING BUILDINGS**

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SECTION 1 – GENERAL PROVISIONS

1.1. SCOPE

1.1.1 –The provisions of this Regulation apply to the design of newly constructed buildings under earthquake effects and the evaluation and reinforcement design of existing buildings.

1.1.2 –The provisions of this Regulation apply to the design of buildings made of cast-in-situ and prefabricated reinforced concrete, steel, light steel, masonry and timber materials under earthquake effects.

1.1.3 –Special rules for the design of high-rise buildings under earthquake influence**Chapter 13**It is defined in .

1.1.4 –Special rules for the design of insulated buildings under earthquake influences**Chapter 14**It is defined in .

1.1.5 –Special rules for the evaluation and strengthening of existing buildings under earthquake effects**Chapter 15**It is defined in .

1.1.6 –In buildings under the influence of earthquakes**1.1.4**Active and passive behavior control applications and related design rules other than insulation applications within the scope of this Regulation are outside the scope of this Regulation.

1.1.7 –Structures other than buildings and building-type structures such as bridges, dams, coastal and port structures, tunnels, pipelines, energy transmission lines, nuclear facilities, natural gas storage facilities, structures completely underground and other structures designed according to different calculation and safety principles than buildings are outside the scope of this Regulation.

1.1.8 –The evaluation and reinforcement of registered buildings and monuments with historical and cultural value under earthquake effects is outside the scope of this Regulation.

1.1.9 –Until special regulations are made for the design of buildings and building-type structures outside the scope of this Regulation under earthquake effects, the provisions given in the relevant Turkish Standards, as well as other internationally accepted equivalent standards, regulations, or technical regulations determined by their institutions, may be used, taking into account the principles set forth in this Regulation.

1.1.10 –The annexes to this Regulation are listed below:**(a)**And**(b)**It is defined in:

(a)*Information Supplement*The annexes, called the Annexes, have been prepared for the purpose of providing information for a better understanding and implementation of the relevant sections of the Regulation. These annexes are not mandatory parts of the Regulation.

(b)Other provisions included in the regulation*Additional's* are integral parts of the Regulation and their implementation is mandatory.

1.2. GENERAL PRINCIPLES

1.2.1 –According to this Regulation, the earthquake ground motion levels to be taken as basis in the design of new buildings and the evaluation of existing buildings under the effect of earthquakes.**Part 2**The earthquake effects corresponding to these earthquake ground motion levels are defined in the Council of Ministers decision dated 22/01/2018 and numbered 2018/11275. *Türkiye Earthquake Hazard Maps* is defined with.

1.2.2 –According to this Regulation, the design of new buildings and the evaluation of existing buildings under the effect of earthquakes will be based on: *Building Performance Targets with Earthquake Design Classes* (DTS) **Chapter 3** It is defined in .

1.2.3 –Buildings designed under the effect of earthquakes in accordance with this Regulation must comply with Turkish Standards in terms of material and workmanship conditions. *Ministry of Environment and Urbanization General Technical Specifications*, published in the Official Gazette dated 10.07.2013 and numbered 28703 *Construction Materials Regulation*, published in the Official Gazette dated 26.06.2009 and numbered 27270 *Regulation on Criteria to be Subject to Construction Materials* will be in accordance with the rules.

1.3. DESIGN SUPERVISION AND CONTROL ON SPECIAL TOPICS

1.3.1. Definition

1.3.1.1 –In the evaluation and design processes related to the implementation of this Regulation, civil engineers with theoretical and professional knowledge and experience in the relevant field will be assigned to work in all relevant design stages from the beginning to the completion of the project in matters requiring special expertise. *design supervision and control* service must be received.

1.3.1.2 –The educational conditions, professional qualifications and experience of those who will perform such services, their documentation and the procedures and principles regarding the execution of the service are determined by the Ministry of Environment and Urbanization.

1.3.2. Scope

In the implementation of this Regulation, the following **(a) to (g)** Topics mentioned in **1.3.1** defined in "*design supervision and control*" is subject to the service:

(a) Section 2 in the scope of **2.4.1** Site Specific Earthquake Hazard Analyses specified in

(b) Section 2 in the scope of **2.5** Description of Earthquake Ground Motions in Time History According to

(c) Section 2 and **Chapter 16** in the scope of **2.4.2** and **16.5** Site Specific Soil Behavior Analyses According to

(d) Section 5 in the scope of **5.6.6** According to Multimodal Push Methods and **5.7** Earthquake Calculation with Nonlinear Calculation Method in Time History According to

(e) Section 13 Earthquake Calculation and Design of High-Rise Building Structural Systems

(f) Section 14 Earthquake Calculation and Design of Insulated Building Support Systems According to

(g) Section 16 in the scope of **16.10** Nonlinear Structure-Pile-Soil Interaction Calculations in Time History According to

1.4. EXPERIMENT-BASED DESIGN

In cases where sufficient calculation models are not available, a large number of similar components are used, or in experimental studies to verify the assumptions used in the design, the principles given in Annex D of TS EN 1990 or equivalent international rules will be taken into account, and the certification procedures will be carried out in accordance with the principles given in Annex D of TS EN 1990.1.2.3It will be done according to.

1.5. REFERENCED STANDARDS, REGULATIONS AND/OR DOCUMENTS

In this Regulation, references are made to other standards and/or documents with or without a date. These references are indicated in appropriate places in the text and **Table 1.1**. In dated references, any amendments or revisions made later shall be applied on condition that the referenced Regulation is also amended or revised. The date of the referenced standard and/or document **Table 1.1** If not specified, the latest edition is used.

Table 1.1 – Referenced Standards, Regulations and/or Documents

EN, ISO, TS etc. No.	Ordinary	Cited Section
ÇYTHYE:2016	Regulation on Design, Calculation and Construction Principles of Steel Structures	Chapter 9
TS EN 1990:2002+A1:2005	Building Design Principles	Part 1
TS 500:2000 + T1:2001+T2:2002+T3:2014	Design and Construction Rules of Reinforced Concrete Structures	Chapter 7, Chapter 8, Chapter 15
TS 498	Structure The elements of In sizing Calculation Values of Loads To be taken	Chapter 7, Chapter 9, Chapter 10,
TS 708:2016	Steel – For Reinforced Concrete – Reinforcement Steel	Chapter 7
TS EN 13670	Construction of Concrete Structures	Chapter 7
TS EN 1992-1-1:2004 + AC:2009+A1:2015	Design of concrete structures - Part 1-1: General rules and regulations applicable to buildings	Chapter 7
TS EN 206	Concrete-Property, performance, manufacturing and suitability	Chapter 7
TSE CEN/TR 15177	Determination of Freeze-Thaw Resistance of Concrete - Internal Structural Damage	Chapter 8
ASTM A673	Standard Specification for Sampling Procedure for Impact Testing of Structural Steel	Chapter 9
TS EN 10025	Hot Rolled Structural Steels	Chapter 10
TS EN 10346	Hot Dip Continuously Coated Steel Flat Products - Technical Delivery Conditions	Chapter 10
TS EN ISO 1461	Hot Dip Galvanized Coatings on Iron and Steel Materials - Properties and Test Methods	Chapter 10
ASTM C1513	Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections	Chapter 10
TS EN 12369-1	Screed Based Boards - Characteristic Values for Design for Structural Purposes - Part 1: Osb, Particle Boards and Fibre Boards	Chapter 10

Table 1.1 (continued)

TS EN 12369-2	Wood Based Panels - Characteristic Values for Design for Structural Purposes - Part 2: Plywood	Chapter 10
TS EN 12369-3	Wood Based Boards - Characteristic Values for Design for Structural Purposes - Part 3: Solid Wood Boards	Chapter 10
TS EN 1993-1-3:2006 + AC:2009	Design of Steel Structures - Part 1-3: General Rules - for Cold-Formed Thin-Size Members and Sheet Metal Covering	Chapter 10
AISI S100	North American Specification for The Design of Cold-Formed Steel Structural Members	Chapter 10
TS EN ISO 898-1	Mechanical Properties of Fasteners - Manufactured from Carbon Steel and Alloy Steel - Part 1: Bolts, Screws and Studs	Chapter 10
TS EN 1996-1-1:2005 + A1:2012	Design of Masonry Structures - Part 1-1: General Rules for Buildings - Rules for Reinforced and Unreinforced Masonry	Chapter 11
TS EN 771-1	Masonry Units - Properties - Part 1: Clay Masonry Units	Chapter 11
TS EN 771-3	Masonry Units - Properties - Part 3: Concrete Masonry Units (Dense and Light Aggregate)	Chapter 11
TS EN 771-4	Masonry Units - Properties - Part 4: Aerated Concrete Masonry Units	Chapter 11
TS EN 771-6	Masonry units - Properties - Section 6: Natural Stone Masonry Units	Chapter 11
TS EN 771-5	Masonry units - Properties - Part 5: Engineered Stone Masonry Units	Chapter 11
TS EN 772-1	Masonry units - Test methods - Part 1: Determination of compressive strength	Chapter 11
TS EN 12602	Prefabricated Reinforced Aerated Concrete Building Elements	Chapter 11
TS EN 1015-11	Masonry mortar - Test methods - Part 11: Determination of compressive and flexural strength of hardened mortar	Chapter 11
TS EN 1052-1	Masonry – Test Methods – Part 1: Determination of Compressive Strength	Chapter 11
TS EN 1015-11	Masonry mortar - Test methods - Part 11: Determination of compressive and flexural strength of hardened mortar	Chapter 11
TS EN 1052-3	Masonry - Test Methods - Part 3: Determination of Initial Shear Strength	Chapter 11
TS EN 1052-4	Masonry-Test Methods-Part 4:Determination of Shear Strength in Joints with Waterproofing Layer	Chapter 11
TS EN 1995-1-1: 2004 + A1:2008+A2:2014	Design of timber structures - Part 1-1: General - General rules and regulations for buildings	Chapter 12
TS EN 12512	Timber Structures – Test Methods – Cyclic Loading Test for Joints Made with Mechanical Fasteners	Chapter 12
TS EN 1337-1	Bearings – Used in Structures – Part 1: General Design Rules	Chapter 14
TS EN 15129	Anti-Seismic Devices	Chapter 14

Table 1.1 (continued)

TS ISO/IEC 17025	General Conditions for the Competence of Testing and Calibration Laboratories	Chapter 14
TS EN 12504-1	Concrete - Testing of Concrete in Construction - Part 1: Core Samples - Coring, Inspection and Determination of Compressive Strength	Chapter 15

CHAPTER 2 – EARTHQUAKE GROUND MOTION

2.0. ICONS

DD-1	= Earthquake location with a 2% probability of being exceeded in 50 years (recurrence period 2475 years) movement level
DD-2	= Earthquake location with a 10% probability of being exceeded in 50 years (recurrence period 475 years) movement level
DD-3	= Earthquake location with a 50% probability of being exceeded in 50 years (recurrence period 72 years) movement level
DD-4	= Earthquake location with a 68% probability of being exceeded in 50 years (recurrence period 43 years) movement level
F_S	= Local ground action coefficient for short period region =
F_1	Local ground action coefficient for 1.0 second period =
g	Gravitational acceleration [$g=9.81 \text{ m/s}^2$]
$S_{ae}(T)$	= Horizontal elastic design spectral acceleration [g] =
$S_{aed}(T)$	Vertical elastic design spectral acceleration [g]
$S_{also}(T)$	= Horizontal elastic design spectral displacement [m] = Short period
S_{DS}	design spectral acceleration coefficient [dimensionless]
S_{D1}	= Design spectral acceleration coefficient for 1.0 second period [dimensionless] = Short
S_S	period map spectral acceleration coefficient [dimensionless]
S_1	= map spectral acceleration coefficient for 1.0 second period [dimensionless]
T	= Natural vibration period [s]
T_A	= Horizontal elastic design acceleration spectrum corner period [s]
T_{NAME}	= Vertical elastic design acceleration spectrum corner period [s] =
T_B	Horizontal elastic design acceleration spectrum corner period [s] =
T_{BD}	Vertical elastic design acceleration spectrum corner period [s]
T_L	= Transition to constant displacement region in horizontal elastic design spectrum period [s]
T_{LD}	= Transition to constant displacement region in vertical elastic design spectrum period [s]
T_p	= Dominant natural vibration period of the building [s]
$(V_s)_{30}$	= Average shear wave velocity in the upper 30 meters [m/s]

2.1. EARTHQUAKE HAZARD MAPS

2.1.1 –The data related to earthquake ground motions to be used as a basis in the design of buildings under earthquake effects are defined in this Section.

2.1.2 – 2.2 Earthquake data for four different earthquake ground motion levels defined in , put into effect by the Council of Ministers decision dated 22/01/2018 and numbered 2018/11275 *Türkiye Earthquake Hazard Maps* are defined by these maps. <https://tdth.afad.gov.tr/> can be accessed from the website at .

2.2. EARTHQUAKE GROUND MOTION LEVELS

Within the scope of this Regulation, four different earthquake ground motion levels are defined as follows.

2.2.1. Earthquake Ground Motion Level-1 (DD-1)

DD-1 *Earthquake Ground Motion*, the probability of exceeding the spectral magnitudes in 50 years is 2% and the corresponding recurrence period is 2475 years. *very rare* earthquake describes the ground motion. This earthquake ground motion, *largest earthquake ground motion considered* It is also called as.

2.2.2. Earthquake Ground Motion Level-2 (DD-2)

DD-2 *Earthquake Ground Motion*, the probability of exceeding the spectral magnitudes in 50 years is 10% and the corresponding recurrence period is 475 years. *sparse* earthquake describes the ground motion. This earthquake ground motion, *standard design earthquake ground motion* It is also called as.

2.2.3. Earthquake Ground Motion Level-3 (DD-3)

DD-3 *Earthquake Ground Motion*, the probability of exceeding the spectral magnitudes in 50 years is 50% and the corresponding recurrence period is 72 years. *chic* Earthquake describes ground motion.

2.2.4. Earthquake Ground Motion Level-4 (DD-4)

DD-4 *Earthquake Ground Motion*, the probability of exceeding the spectral magnitudes in 50 years is 68% (50% probability of exceeding in 30 years) and the corresponding recurrence period is 43 years. *very often* earthquake describes the ground motion. This earthquake ground motion, *service earthquake ground motion* It is also called as.

2.3. STANDARD EARTHQUAKE GROUND MOTION SPECTRUM

2.3.1. Definition

Earthquake ground motion spectra are calculated for a 5% damping ratio based on a specific earthquake ground motion level. **2.3.2, 2.3.3, 2.3.4** And **2.3.5** As explained in *map spectral acceleration coefficients* and *local ground effect coefficients* in standard format or depending on **2.4.1** According to *site-specific earthquake hazard analyses* are specifically identified with .

2.3.2. Map Spectral Acceleration Coefficients and Design Spectral Acceleration Coefficients

2.3.2.1 – Dimensionless *map spectral acceleration coefficients*, **2.2** For four different earthquake ground motion levels specified in *Türkiye Earthquake Hazard Maps* defined within the scope of:

(a) Short period *map spectral acceleration coefficient* S_s

(b) For 1.0 second period *map spectral acceleration coefficient* S_1

Corresponding to the geometric mean of earthquake effects in two horizontal directions perpendicular to each other *map spectral acceleration coefficients*, the reference ground condition for a given earthquake ground motion level [$(V_s)_{30}=760$ m/s] for a 5% damping ratio *map spectral accelerations* S_s divided by the acceleration due to gravity *dimensionless coefficients* has been defined as.

2.3.2.2 – 2.3.2.1 defined in *map spectral acceleration coefficients* S_s And S_1 , as follows *design spectral acceleration coefficients* S_{DS} And S_{D1} is converted to:

$$\begin{aligned} S_{DS} &= S_s F_s \\ S_{D1} &= S_1 F_1 \end{aligned} \quad (2.1)$$

Here F_s And F_1 **2.3.3** defined in *local ground effect coefficients* It shows the .

2.3.2.3 – Horizontal and vertical *elastic design spectrums*, **Equation (2.1)** obtained with *design spectral acceleration coefficients* Taking advantage of, respectively, **2.3.4** And **2.3.5** It is defined in .

2.3.3. Local Ground Effect Coefficients

2.3.3.1 – 16.4 defined in *local soil classes* Depending on *local ground effect coefficients* F_s And F_1 , respectively, **Table 2.1** And **Table 2.2** The spectral acceleration map in the tables is given in Linear interpolation can be done for intermediate values of the coefficients.

Table 2.1 – Local Ground Effect Coefficients for the short period region

Local Ground Class	For the short period region <i>Local Ground Effect Coefficient</i> F_s					
	$S_s \leq 0.25$	$S_s = 0.50$	$S_s = 0.75$	$S_s = 1.00$	$S_s = 1.25$	$S_s \geq 1.50$
Z A	0.8	0.8	0.8	0.8	0.8	0.8
ZB	0.9	0.9	0.9	0.9	0.9	0.9
ZC	1.3	1.3	1.2	1.2	1.2	1.2
ZD	1.6	1.4	1.2	1.1	1.0	1.0
Z E	2.4	1.7	1.3	1.1	0.9	0.8
ZF	Site specific ground behavior analysis will be performed.(See 16.5).					

Table 2.2 – Local Ground Effect Coefficients for 1.0 second period

Local Ground Class	For 1.0 second period <i>Local Ground Effect Coefficient</i> F_1					
	$S_1 \leq 0.10$	$S_1 = 0.20$	$S_1 = 0.30$	$S_1 = 0.40$	$S_1 = 0.50$	$S_1 \geq 0.60$
Z A	0.8	0.8	0.8	0.8	0.8	0.8
ZB	0.8	0.8	0.8	0.8	0.8	0.8
ZC	1.5	1.5	1.5	1.5	1.5	1.4
ZD	2.4	2.2	2.0	1.9	1.8	1.7
Z E	4.2	3.3	2.8	2.4	2.2	2.0
ZF	Site specific ground behavior analysis will be performed.(See 16.5).					

2.3.3.2 – Table 2.1 and/or Table 2.2 For soils falling into the ZF local soil class according to *ahaya specific ground behavior analysis* **16.5** It will be done according to.

2.3.4. Horizontal Elastic Design Spectrum

2.3.4.1 – For any earthquake ground motion level considered *horizontal elastic design acceleration spectrum* The coordinates of *horizontal elastic design spectral accelerations* $S_{ae}(T)$, natural Gravitational acceleration depending on the period of vibration $[g]$ in terms of **Equation (2.2)** It is defined in **(Figure 2.1)**:

$$\begin{aligned} S_{ae}(T) &= 0.4 + 0.6 \frac{T}{T_A} S_{DS} & (0 \leq T \leq T_A) \\ S_{ae}(T) &= S_{DS} & (T_A \leq T \leq T_B) \\ S_{ae}(T) &= \frac{S_{D1}}{T} & (T_B \leq T \leq T_L) \\ S_{ae}(T) &= \frac{S_{D1} T_L}{T^2} & (T_L \leq T) \end{aligned} \quad (2.2)$$

Here S_{DS} and S_{D1} defined in **2.3.2.2** design spectral acceleration coefficients's, T is the natural vibration period. Horizontal design spectrum corner periods T_A and T_B

Equation (2.3) with S_{DS} and S_{D1} is defined depending on:

$$T_A = 0.2 \frac{S_{D1}}{S_{DS}} \quad ; \quad T_B = \frac{S_{D1}}{S_{DS}} \quad (2.3)$$

Transition period to constant displacement region $T_L = 6$ s will be taken.

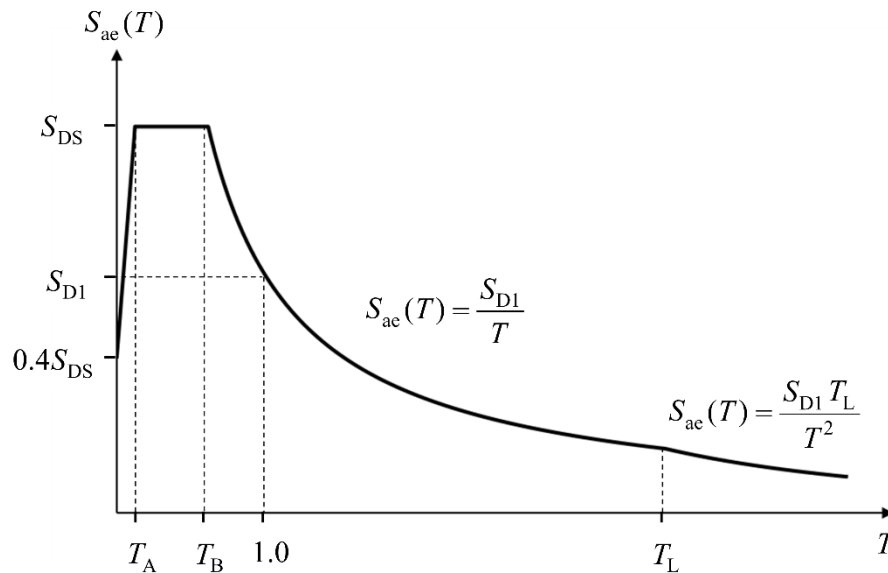


Figure 2.1

2.3.4.2 –For any earthquake ground motion level considered *horizontal elastic design displacement spectrum* The coordinates of *horizontal elastic design spectral displacements* $S_{\text{also}}(T)$, in meters [m] depending on the natural vibration period **Equation (2.4)** is defined by **(Figure 2.2)**:

$$S_{\text{also}}(T) = \frac{T_2}{4\pi^2} g S_{\text{ae}}(T) \quad (2.4)$$

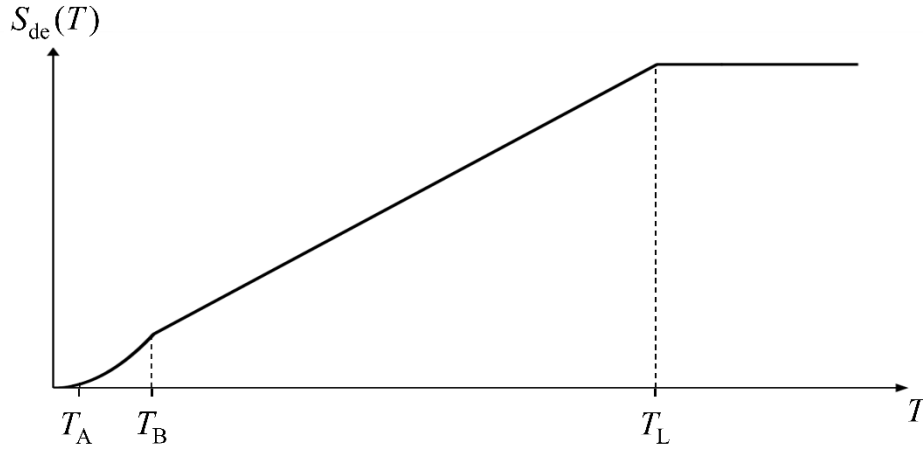


Figure 2.2

2.3.5 Vertical Elastic Design Spectrum

For any earthquake ground motion level considered *vertical elastic design acceleration spectrum* The coordinates of *vertical elastic design spectral accelerations* $S_{\text{aed}}(T)$, horizontal earthquake Gravitational acceleration depending on the short period design spectral acceleration coefficient defined for ground motion and the natural vibration period $[g]$ in terms of **Equation (2.5)** is defined by **(Figure 2.3)**:

$$\begin{aligned} S_{\text{aed}}(T) &= -0.32 + 0.48 \frac{T}{T_{\text{NAME}}} S_{\text{DS}} & (0 \leq T \leq T_{\text{NAME}}) \\ S_{\text{aed}}(T) &= 0.8 S_{\text{DS}} & (T_{\text{NAME}} \leq T \leq T_{\text{BD}}) \\ S_{\text{aed}}(T) &= 0.8 S_{\text{DS}} T \frac{T_{\text{BD}}}{T_{\text{LD}}} & (T_{\text{BD}} \leq T \leq T_{\text{LD}}) \end{aligned} \quad (2.5)$$

Equation (2.5) Located in T_{NAME} And T_{BD} vertical spectrum *corner periods* with T_{LD} period **Equation (2.6)** is given in .

$$T_{\text{NAME}} = \frac{T}{3} \quad ; \quad T_{\text{BD}} = \frac{T_B}{3} \quad ; \quad T_{\text{LD}} = \frac{T_L}{2} \quad (2.6)$$

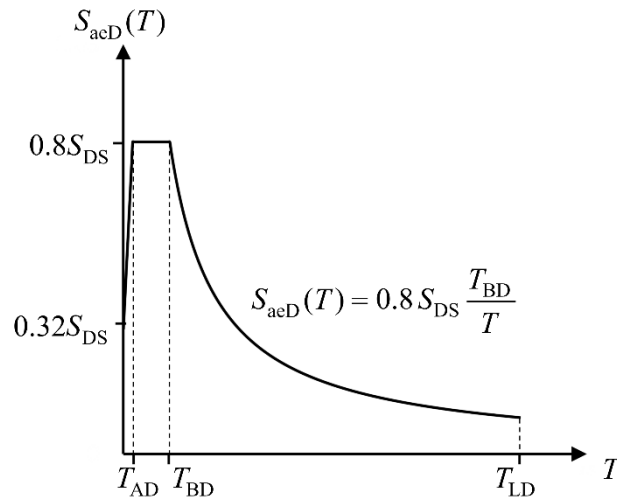


Figure 2.3

2.4. SITE SPECIFIC EARTHQUAKE GROUND MOTION SPECTRUM

2.4.1. Site-Specific Elastic Acceleration Spectrum

2.4.1.1 –In some special cases, *site-specific earthquake hazard analyses with site-specific earthquake ground motion spectra* Such spectra can be used in any case, depending on the preference of the project engineer.

2.4.1.2 –Site specific earthquake *ground motion* The ordinates of the spectrum are never **2.3.4** or **2.3.5** It shall not be smaller than 90% of the design spectrum ordinates defined in .

2.4.2. Site-Specific Soil Behavior Analyses

16.4 In the soils defined as ZF class in the local soil classification made according to **16.5** According to *site-specific ground behavior analyses* Such analyses may also be performed on soils of other classes, depending on the project engineer's preference.

2.5. DEFINITION OF EARTHQUAKE GROUND MOVEMENTS IN TIME DOMAIN

According to this regulation, the selection and use of earthquake records to define the necessary earthquake ground motions in one, two and three-dimensional earthquake calculations in the time domain of building load-bearing systems. *simple scaling* scaling with the method or *spectral correspondence* The rules that must be applied to ensure the transformation **2.5.1**, **2.5.2** And **2.5.3** It is defined in .

2.5.1. Selection of Earthquake Records

2.5.1.1 –The selection of earthquake records to be used in the earthquake calculation of the building load-bearing systems in the time history will be made by taking into account the earthquake magnitudes, fault distances, source mechanisms and local ground conditions compatible with the design-based earthquake ground motion level. If past earthquake records compatible with the design-based earthquake ground motion level are available in the region where the building is located, these records will be used first.

To determine the magnitude and fault distance information of earthquakes that contribute most to the site-specific earthquake hazard, *earthquake hazard separation* process can be used.

2.5.1.2 –In cases where a sufficient number or quality of earthquake records cannot be selected, time-history simulated ground motion records can be used. In the case of using such records, the seismic source, wave propagation and local ground properties of the site where the building is located will be taken into account. It will be shown that the model parameters to be used for simulation are compatible with the actual earthquake records recorded in earthquakes that occurred in the region in question.

2.5.1.3 –To be selected for one or two dimensional calculation *earthquake records* and will be selected for three-dimensional calculation *earthquake recording kits* The number of records or record sets to be selected from the same earthquake will not exceed three.

2.5.2. Scaling of Earthquake Records with Simple Scaling Method

2.5.2.1 – 16.5.2 to or 16.10 Apart from the ground response analyses to be carried out according to , the earthquake ground motions to be used in the calculation in the time history domain are obtained from the selected earthquake records. *simple scaling* can be obtained by the method below:

(a) For one or two-dimensional calculations, the average of the spectra of all selected records is $0.2 T_p$ and $1.5 T_p$ The amplitudes between the periods, **2.3.4** or **2.4.1** Design defined according to

The amplitudes of earthquake ground motions will be scaled according to the rule that the spectrum should not be smaller than the amplitudes of the earthquake in the same period range.

(b) For the three-dimensional calculation, the resultant horizontal spectrum will be obtained by taking the square root of the sum of the squares of the spectra of the two horizontal components of each selected earthquake record set. The average of the resultant spectra of all selected records will be $0.2 T_p$ and $1.5 T_p$ Periods

The amplitudes between, **2.3.4** or **2.4.1** The amplitudes of the earthquake ground motion components will be scaled according to the rule that the ratio of the amplitudes of the design spectrum defined according to the same period range should not be less than 1.3. This period range may vary for insulated buildings. **(See 14.14.4.2)** Scaling of both horizontal components will be done with the same scale factors.

2.5.2.2 –If the building characteristics require it with the available earthquake data and a site-specific earthquake ground motion spectrum is determined, the building is associated with a certain number of vibration periods. *conditioned average spectra* By making use of , earthquake records can be selectively scaled to provide less scatter in the analysis results.

2.5.3. Conversion of Earthquake Records to Achieve Spectral Matching

16.5.2 to or 16.10 Earthquake ground motions are calculated from selected earthquake records to be used in ground response analyses or other calculations in the time domain. **2.3.4** or **2.4.1** According to the design spectrum defined *spectral correspondence* The averages of the converted earthquake ground motion spectra will not be smaller than the design spectrum ordinates for all periods.

2.6. INSTALLATION OF ACCELERATION RECORDERS IN BUILDINGS

If deemed appropriate by the Disaster and Emergency Management Presidency, the Presidency or other public institutions or university organizations under the coordination of the Presidency will be allowed to place accelerometers in buildings, other structures or open areas belonging to public, private and legal entities for the purpose of measuring strong earthquake motion, and the owners or operators of the buildings or structures will be responsible for their protection.

CHAPTER 3 – EVALUATION OF BUILDINGS UNDER EARTHQUAKE EFFECT AND GENERAL PRINCIPLES FOR DESIGN

3.0. ICONS

BKS	= Building Usage Class =
BYS	Building Height Class
DD-1	= Earthquake location with a 2% probability of being exceeded in 50 years (recurrence period 2475 years) movement level
DD-2	= Earthquake location with a 10% probability of being exceeded in 50 years (recurrence period 475 years) movement level
DD-3	= Earthquake location with a 50% probability of being exceeded in 50 years (recurrence period 72 years) movement level
DD-4	= Earthquake location with a 68% probability of being exceeded in 50 years (recurrence period 43 years) movement level
DTS	= Earthquake Design Class
GÖ	= Collapse Prevention Performance Level
H_N	= Total Building Height [m]
h_i	= Floor height of the i'th floor of the building [m]
KH	= Controlled Damage Performance Level =
KK	Uninterrupted Use Performance Level =
I	Building Importance Factor
SH	= Limited Damage Performance Level
S_{DS}	= Short period design spectral acceleration coefficient [dimensionless]
$T_{p,all}$	= Dominant vibration in the direction of each of the building axes perpendicular to each other mode, calculated in the same direction for the entire building, including the basement floors <i>natural vibration period</i> [s]
$T_{p,top}$	= Dominant vibration in the direction of each of the building axes perpendicular to each other mode, the load-bearing system of the entire building is calculated in the same direction without taking into account all basement masses including the ground floor slab. <i>natural vibration period</i> [s]
$\sum A_{to}$	= Effective shear area in the earthquake direction considered at any floor [m ²]
$\sum A_g$	= At any floor, parallel to the earthquake direction taken into account The total of the cross-sectional areas of the load-bearing system elements working as a curtain [m ²]
$\sum A_k$	= Masonry filling parallel to the earthquake direction taken into account at any floor Total wall areas (excluding door and window openings) [m ²]
$\sum A_w$	= Sum of effective body areas of column cross-section at any floor [m ²]
$(\Delta_{xi})_{average}$	= Average reduced relative floor drift at the i'th floor of the building [m]

$(\Delta_{\text{fi}})_{\text{Max}}$ = Maximum reduced relative floor drift at the i'th floor of the building [m]

η_{bi} = Torsional Irregularity Coefficient defined at the i'th floor

η_i = Strength Irregularity Coefficient defined at the i'th floor

η_{that} = Stiffness Irregularity Coefficient defined at the i'th floor

3.1. BUILDING USE CLASSES AND BUILDING IMPORTANCE COEFFICIENTS

3.1.1. Building Usage Classes

3.2 defined in *Earthquake Design Classes* As a basis for determining the *Building Usage Classes* (BKS), depending on the purpose of use of the buildings **Table 3.1** It is defined in .

3.1.2. Building Importance Coefficients

Building Usage Classes Depending on *Building Importance Coefficients* **Table 3.1** It is defined in .

Table 3.1 – Building Usage Classes and Building Importance Factors

Building Use Class	Use of the Building Purpose	Building Importance Coefficient (<i>I</i>)
BKS = 1	Buildings that need to be used after an earthquake, buildings where people stay for long periods of time and in large numbers, buildings where valuables are stored and buildings containing hazardous materials a) Buildings that need to be used immediately after an earthquake (Hospitals, dispensaries, health centers, fire department buildings and facilities, PTT and other communication facilities, transportation stations and terminals, energy production and distribution facilities, provincial, district governorship and municipality administration buildings, first aid and disaster planning stations) b) Schools, other educational buildings and facilities, dormitories and dormitories, military barracks, prisons, etc. c) Museums d) Buildings where substances with toxic, explosive, flammable, etc. properties are located or stored	1.5
BKS = 2	Buildings where people stay densely and for short periods of time Shopping malls, sports facilities, cinemas, theaters, concert halls, places of worship, etc.	1.2
BKS = 3	Other buildings Other buildings that do not fall within the definitions given for BKS=1 and BKS=2 (Housing, workplaces, hotels, building type industrial structures, etc.)	1.0

3.2. EARTHQUAKE DESIGN CLASSES

3.1.1 defined in *Building Usage Classes* and DD-2 earthquake ground motion level for

2.3.2.2 defined in *Short Period Design Spectral Acceleration Coefficient* According to this Regulation, the design principles to be taken into account under earthquake effect are as follows: *Earthquake Design Classes* (DTS), **Table 3.2**

3.2 It will be determined according to.

Table 3.2 – Earthquake Design Classes (EDC)

DD-2 Short Period Design Spectral Acceleration Coefficient at Earthquake Ground Motion Level (S_{DS})	Building Usage Class	
	BKS = 1	BKS = 2, 3
$S_{DS} < 0.33$	DTS = 4a	DTS = 4
$0.33 \leq S_{DS} < 0.50$	DTS = 3a	DTS = 3
$0.50 \leq S_{DS} < 0.75$	DTS = 2a	DTS = 2
$0.75 \leq S_{DS}$	DTS = 1a	DTS = 1

3.3. BUILDING HEIGHT AND BUILDING HEIGHT CLASSES

3.3.1. Building Base and Building Height

3.3.1.1 – Below **(a)** And **(b)** which satisfies both conditions given in *buildings with basements* in *building base*, is defined at the floor slab level above the basement walls.

(a) Rigid basement walls surround the building from all sides or at least three sides,

(b) In the dominant vibration mode in the direction of each of the building axes perpendicular to each other, calculated for the entire building including the basement floors *natural vibration periods* calculated in the same direction without taking into account all basement masses including the ground floor slab in the same carrier system. *natural vibration period* The ratio of 'to' is less than 1.1 ($T_{p,all} \leq 1.1 T_{p,top}$).

3.3.1.2 – 3.3.1.1 Buildings with basements that do not meet any of the conditions given in *buildings without basements* in *building base* is defined at the upper foundation level.

3.3.1.3 – In terms of earthquake calculation *building height* H_n , **3.3.1.1** or **3.3.1.2** According to defined *building base* It is defined as the height measured from . In this definition, elevator machine rooms and similar small mass extensions located above the roof slab may not be taken into account.

3.3.2. Building Height Classes

3.3.2.1 – In designing buildings under earthquake effects, the heights of the buildings are eight *Building Height Class* (BYS). For buildings falling into these classes **3.3.1.3** Height ranges defined based on, **Table 3.2** in *Earthquake Design Classes* Depending on **Table 3.3** is given in.

3.3.2.2 – Table 3.3 Buildings specified as $BYS = 1$ **Chapter 13** According to *high rise buildings* will be classified as.

Table 3.3 – Building Height Classes and Defined According to Earthquake Design Classes
Building Height Ranges

Building Height Class	According to Building Height Classes and Earthquake Design Classes Defined Building Height Ranges [m]		
	DTS = 1, 1a, 2, 2a	DTS = 3, 3a	DTS = 4, 4a
BYS = 1	$H_N > 70$	$H_N > 91$	$H_N > 105$
BYS = 2	$56 < H_N \leq 70$	$70 < H_N \leq 91$	$91 < H_N \leq 105$
BYS = 3	$42 < H_N \leq 56$	$56 < H_N \leq 70$	$56 < H_N \leq 91$
BYS = 4	$28 < H_N \leq 42$	$42 < H_N \leq 56$	
BYS = 5	$17.5 < H_N \leq 28$	$28 < H_N \leq 42$	
BYS = 6	$10.5 < H_N \leq 17.5$	$17.5 < H_N \leq 28$	
BYS = 7	$7 < H_N \leq 10.5$	$10.5 < H_N \leq 17.5$	
BYS = 8	$H_N \leq 7$	$H_N \leq 10.5$	

3.4. BUILDING PERFORMANCE LEVELS

Building Performance Targets For building load-bearing systems under earthquake effect, based on the definition of *Building Performance Levels* 3.4.1, 3.4.2, 3.4.3, 3.4.4 It is defined in .

3.4.1. Uninterrupted Use (UC) Performance Level

This performance level corresponds to the situation where structural damage to the building load-bearing system elements does not occur or the damage remains negligible.

3.4.2. Limited Damage (LD) Performance Level

This performance level corresponds to the damage level where limited damage occurs in the building structural system elements, in other words, where nonlinear behavior remains limited.

3.4.3. Controlled Damage (CH) Performance Level

This level of performance corresponds to a level of damage that is not too severe and can mostly be repaired in the building load-bearing system elements to ensure life safety.

3.4.4. Migration Prevention (MI) Performance Level

This level of performance corresponds to the pre-collapse situation where there is severe damage to the building's structural system elements. Partial or complete collapse of the building is prevented.

3.5. BUILDING PERFORMANCE TARGETS AND DESIGN APPROACHES TO BE APPLIED UNDER EARTHQUAKE IMPACT

Under the influence of an earthquake *building performance targets*, 2.2 defined in *earthquake ground motion levels* and targeted under 3.4 Defined according to *building performance levels* It represents the.

3.5.1. Building Performance Targets

3.5.1.1 – 2.2 To be applied to buildings within the scope of this Regulation for the four earthquake ground motion levels defined in , *Earthquake Design Class* Defined for DTS = 1, 2, 3, 3a, 4, 4a *Normal Performance Targets* with *Earthquake Design Class* DTS = 1a, defined for 2a *Forward Performance Targets* **Table 3.4** and **Table 3.5** It is given in. Depending on the wishes of the building owner **Table 3.4** Further performance targets corresponding to earthquake ground motion levels in can be selected.

Table 3.4. For New or Existing Buildings According to Earthquake Design Classes Performance Targets and Evaluation/Design Approaches to be Applied

(a) New Cast-in-Situ Reinforced Concrete, Prefabricated Reinforced Concrete and Steel Buildings

(Except High-Rise Buildings – BYS ≥ 2)

Earthquake Location H. Level	DTS = 1.1a ⁽¹⁾ , 2, 2a ⁽¹⁾ , 3, 3a, 4, 4a		DTS = 1a ⁽²⁾ , 2a ⁽²⁾	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-3	--	--	SH	SGDT
DD-2	KH	DGT ⁽⁵⁾	KH	DGT ^(3,4)
DD-1	--	--	KH	SGDT

(b) New or Existing High-Rise Buildings (BYS = 1)

Earthquake Location H. Level	DTS = 1, 2, 3, 3a, 4, 4a		DTS = 1a, 2a	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-4	KK	DGT	--	--
DD-3	--	--	SH	SGDT
DD-2	KH	DGT ⁽³⁾	KH	DGT ^(3,4)
DD-1	GÖ	SGDT	KH	SGDT

(c) Existing Cast-in-Situ Reinforced Concrete, Prefabricated Reinforced Concrete and Steel Buildings

(Except High-Rise Buildings – BYS ≥ 2)

Earthquake Location H. Level	DTS = 1, 2, 3, 3a, 4, 4a		DTS = 1a, 2a	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-3	--	--	SH	SGDT
DD-2	KH	SGDT	--	--
DD-1	--	--	KH	SGDT

⁽¹⁾ BYS > 3 will be applied in existing buildings.

⁽²⁾ It will be applied in buildings with BYS = 2.3.

⁽³⁾ It will be done as a preliminary design.

⁽⁴⁾ F=1.5 will be taken and implemented.

⁽⁵⁾ See **3.5.2.2**.

3.5.1.2 – Table 3.4 Masonry, timber and light steel buildings not specified in , under the influence of DD-2 earthquake ground motion *Controlled Damage* (KH) will provide the performance target.

3.5.2. Evaluation/Design Approaches to be Applied

3.5.2.1 – Chapter 4 The accounting principles given in *Design by Strength* (With the DGT) approach
Chapter 5 The accounting principles given in *Evaluation and Design Based on Shapeshifting* (Scope of application of the (GDST) approach **Table 3.4** And **Table 3.5** is given in.

Table 3.5. Performance for Seismic Insulated Buildings According to Earthquake Design Classes Objectives and Evaluation/Design Approaches to be Applied

(a) New Earthquake Insulated Buildings – Superstructure

Earthquake Location H. Level	DTS = 1, 2, 3, 3a, 4, 4a		DTS = 1a, 2a	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-2	SH	DGT	KK	DGT
DD-1	--	--	--	--

(b) Existing Buildings to be Strengthened by Earthquake Isolation – Superstructure

Earthquake Location H. Level	DTS = 1, 2, 3, 3a, 4, 4a		DTS = 1a, 2a	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-2	KH	DGT	SH	DGT
DD-1	--	--	--	--

(c) New and Reinforced Earthquake Insulated Buildings – Insulation System and Infrastructure

Earthquake Location H. Level	DTS = 1, 2, 3, 3a, 4, 4a		DTS = 1a, 2a	
	Normal Performance The target	Evaluation/Design Approach	Advanced Performance The target	Evaluation/Design Approach
DD-2	--	--	--	--
DD-1	KK	SGDT ₍₁₎ –DGT ₍₂₎	KK	SGDT ₍₁₎ –DGT ₍₂₎

(1) It will be applied for the insulation system.

(2) It will be applied to infrastructure.

3.5.2.2 – Chapter 8 The preliminary design of fully post-tensioned prefabricated buildings described in is done with the DGT approach, and the final design is **8.4.3** According to , it will be done with the ŞGDT approach.

3.5.2.3 – Table 3.4 Design of masonry, timber and light steel buildings not specified in **Chapter 4** According to *Design by Strength* (It will be done with the DGT) approach.

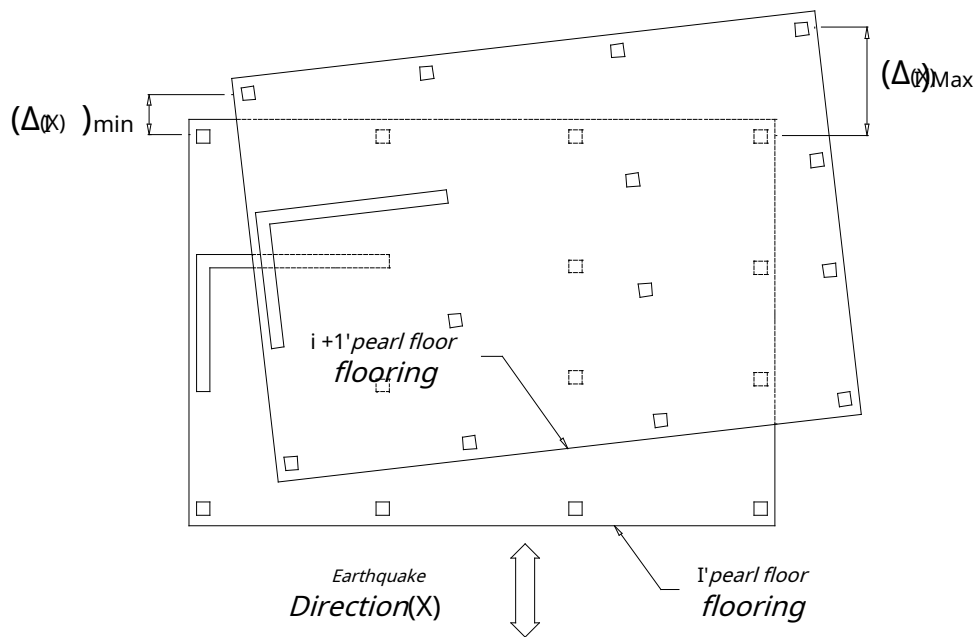
3.6. IRREGULAR BUILDINGS UNDER THE EFFECT OF EARTHQUAKES

3.6.1. Definition of Irregular Buildings

Design and construction of which should be avoided due to their negative behavior against earthquakes. *irregular buildings* Situations that create irregularities in plan and vertical direction regarding the definition of **Table 3.6**, the conditions related to these are **3.6.2** is given in .

Table 3.6 – Irregular Buildings

A – IRREGULAR CASES IN THE PLAN	Relating to Articles
A1 – Torsional Irregularity:For any of the two perpendicular earthquake directions, it is the ratio of the largest relative floor drift at any floor to the average relative drift in the same direction at that floor. <i>Torsional Irregularity Coefficient</i> η_{bi} If is greater than 1.2 (Figure 3.1). $[\eta_{bi} = (\Delta_{q0})_{Max} / (\Delta_{q0})_{average} > 1.2]$.Calculation of relative floor displacements,±%5 additional Taking into account the eccentricity effects,4.7' will be made according to.	3.6.2.1
A2 – Floor Discontinuities:On any floor slab(Figure 3.2); I – If the total of the space areas, including stairs and elevator shafts, exceeds 1/3 of the floor gross area, II – The presence of local floor gaps that make it difficult to safely transfer earthquake loads to vertical load-bearing system elements, III – The presence of sudden decreases in the in-plane rigidity and strength of the floor.	3.6.2.2
A3 – Presence of Protrusions in the Plan:The situation where both dimensions of the protruding parts in two perpendicular directions in the building floor plans are larger than 20% of the total plan dimensions of that floor of the building in the same directions.(Figure 3.3).	3.6.2.2
B – VERTICAL IRREGULARITY SITUATIONS	Relating to Articles
B1 – Strength Irregularity Between Adjacent Floors (Weak Floor): In reinforced concrete buildings, any floor in any of the two perpendicular earthquake directions <i>total effective cutting area's</i>, on the upper floor <i>total effective cutting area</i> Defined as the ratio of <i>Strength Irregularity Coefficient</i> η_i The situation when is less than 0.80. $[\eta_i = (\sum A)_{ei} / (\sum A)_{to\ i+1} < 0.80]$ Definition of effective shear area at any floor: $(\sum A)_{ei} = (\sum A) + (\sum A) + (0.15 \sum A)_{that}$ Notes:4.9.1.3(b) For walls defined in $A_k = 0$ will be taken.	3.6.2.3
B2 – Stiffness Irregularity Between Adjacent Floors (Soft Storey): For any of the two perpendicular earthquake directions, dividing the average relative floor drift ratio on any i'th floor, excluding basement floors, by the average relative floor drift ratio on the floor above or below defined <i>Stiffness Irregularity Coefficient</i> η_{that} The situation where is greater than 2.0. $[\eta_{that} = (\Delta_{q0}/h)_{I_{average}} / (\Delta_{q0}/h)_{i+1\ average} > 2.0$ or $\eta_{that} = (\Delta_{q0}/h)_{I_{average}} / (\Delta_{q0}/h)_{i-1\ average} > 2.0]$ Calculation of relative floor displacements,±%5 taking into account additional eccentricity effects,4.7' will be made according to.	3.6.2.1
B3 – Discontinuity of Vertical Elements of the Carrier System: The situation where the vertical elements of the carrier system (columns or curtains) are lifted at some floors and placed on the top or end of beams or gusseted columns, or the curtains on the upper floor are placed on the columns below.(Figure 3.4).	3.6.2.4



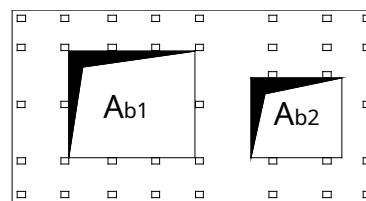
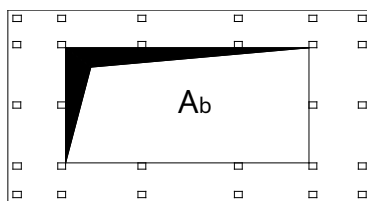
In case the slabs work as rigid diaphragms in their own planes

$$(\Delta x)_{\text{average}} = 1/2 [(\Delta x)_{\text{Max}} + (\Delta x)_{\text{min}}]$$

$$\text{Torsional irregularity coefficient: } \eta_{bi} = (\Delta x)_{\text{Max}} / (\Delta x)_{\text{average}}^{(X)}$$

$$\text{Torsional irregularity condition: } \eta_{bi} > 1.2$$

Figure 3.1



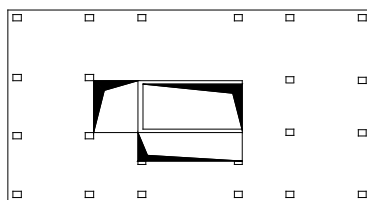
$$A_b = A_{b1} + A_{b2}$$

A2 type of disorder-I

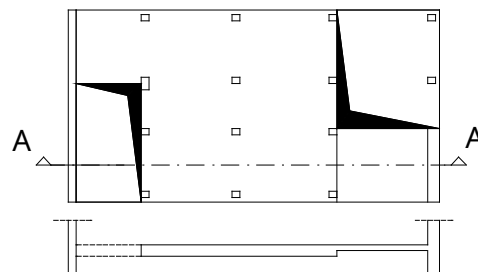
$$A_b / A > 1/3$$

Ab: Sum of void areas A:

Gross floor area



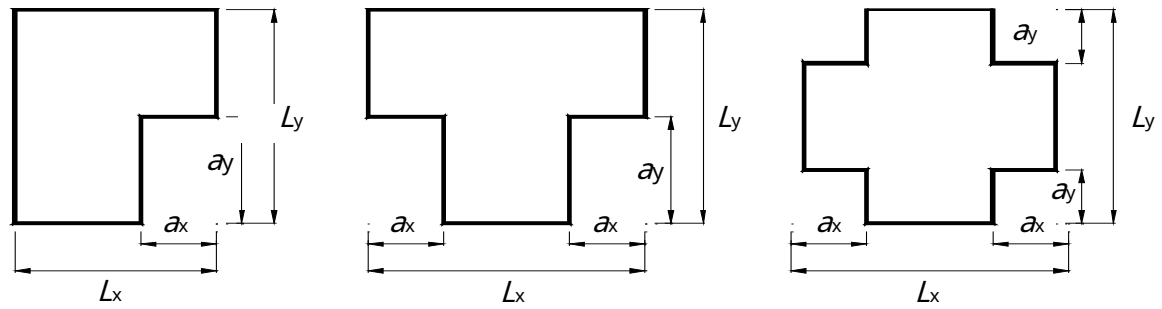
A2 type of disorder-II



Section AA

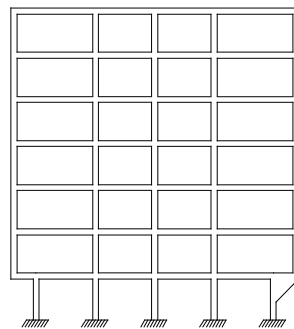
A2 type of disorder-II and III

Figure 3.2

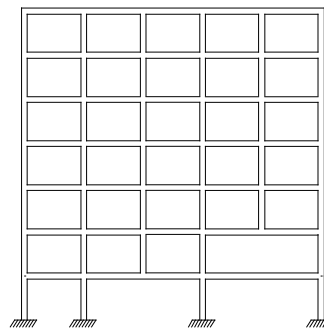


A3 type of disorder: $a_x > 0.2L_x$
and also $a_y > 0.2L_y$

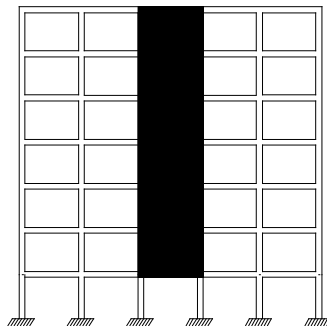
Figure 3.3



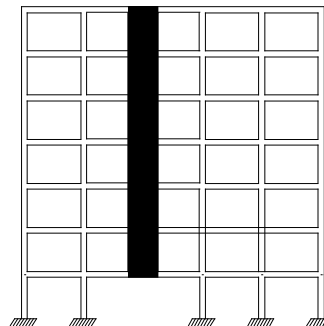
See.3.6.2.4 (a)



See.3.6.2.4 (b)



See.3.6.2.4 (c)



See.3.6.2.4 (d)

Figure 3.4

3.6.2. Conditions Regarding Irregular Buildings

Table 3.6 The conditions for irregularities defined in are set out below:

3.6.2.1 – A1AndB2 type of irregularities, **4.6** According to , irregularities are the factors that affect the selection of the earthquake calculation method.

3.6.2.2 – A2AndA3 In buildings with irregularities such as these, two-dimensional models were used to show that floor slabs can safely transfer earthquake forces between vertical structural system elements within their own planes. *plate(membrane)* or *shell* It will be modeled with finite elements (See.4.5.6.2).

3.6.2.3 – B1 In buildings with irregularity of this type, if the total of the infill wall areas on the i th floor is greater than those on the floor above, η_i Infill walls will not be taken into account in the calculation of η . Taking into account the smallest floor of $0.60 \leq (\eta_i)_{\min} < 0.80$

in the range of **Table 4.1** The load-bearing system behavior coefficient given in is $1.25(\eta_i)_{\min}$ with its value will be applied to the entire building in both earthquake directions by multiplying. However, η will never be < 0.60 . Otherwise, the strength and rigidity of the weak floor will be increased.

earthquake calculation will be repeated.

3.6.2.4–B3 Conditions for buildings with irregularities of this type are listed below: **(a)** to **(d)** It is stated in:

(a) Columns are never allowed to be placed on top of or at the ends of cantilever beams or gussets formed on the columns below on any floor of the building.

(b) In case the column is supported by a beam at both ends, **4.4.3** According to , vertical earthquake calculation is sufficient.

(c) Curtains on upper floors are never allowed to be placed on columns below.

(d) Shear walls are never allowed to be placed in the middle of the opening on beams within their own plane on any floor of the building.

INFORMATION ANNEX 3A – FOR PROPER DESIGN UNDER EARTHQUAKE IMPACT ORGANIZATION OF BUILDING CARRIER SYSTEMS

The following general rules should be taken into consideration when arranging building load-bearing systems. In this context: **3.6** defined in *irregular carrier systems in plan and vertical* should be avoided as much as possible.

3A.1. SIMPLICITY AND SIMPLICITY OF THE CARRIER SYSTEM

Due to uncertainties in earthquake ground motion, structural modeling and structural element behavior, as well as approximations in analysis and design methods, the basic rule of design under earthquake effect is that the load-bearing system should be as plain and simple as possible to ensure that the seismic behavior of the building is predictable.

3A.2. REGULAR AND SYMMETRIC ARRANGEMENT OF THE CARRIER SYSTEM

3A.2.1 –By arranging the carrier system regularly and symmetrically in plan, the inertia forces originating from the distributed masses in the floors can be transferred to the vertical carrier system elements in the most appropriate way. With the symmetric carrier system arrangement, eccentricities that may occur in terms of mass, rigidity and strength can be prevented and a predictable earthquake behavior can be achieved.

3A.2.2 –The carrier system should be arranged regularly in the vertical direction. In this context, sudden changes in floor rigidity and strength may occur. *soft floor* And *weak floor* regulations should be avoided whenever possible.

3A.3. PROVIDING OVER-DEPENDENCE FEATURE IN THE CARRIER SYSTEM

3A.3.1 –The hyper-instability behavior under static loads should also be valid under earthquake effects, and in this context, in case the strength of some load-bearing system elements decreases or even becomes disabled during an earthquake, a sufficient number of spare elements that can provide stable behavior in the system should be activated, in other words, the load-bearing system should be backed up.

3A.3.2 –It should not be overlooked that building blocks separated from each other by structural joints carry the risk of being damaged by collision in an earthquake. In this context;

(a) Apart from reasons such as preventing torsional irregularity, balanced arrangement of the carrier system elements, etc., separating the carrier system into independent blocks with structural joints should be avoided as much as possible, simply because of the plan lengths of the building or building blocks.

(b) The plan lengths of buildings or building blocks are for reinforced concrete elements, regardless of the earthquake calculation. **4.5.8** It can be determined according to the carrier system calculation to be made for uniform temperature change and shrinkage effects using the effective section stiffnesses defined in .

3A.4. SUFFICIENT STRENGTH AND RIGIDITY IN THE CARRIER SYSTEM

3A.4.1 –Considering that earthquake ground motion is effective in all directions in the plan, it is essential that the load-bearing system elements are preferably arranged in two main directions perpendicular to each other and have similar strength and rigidity.

3A.4.2 –In order to eliminate torsional irregularities that cause negative behavior in the carrier system and to prevent dangerous torsional vibrations, it is essential to provide sufficient torsional strength and rigidity. In this direction, a suitable solution is to arrange the carrier system elements with high rigidity and strength as much as possible around the building.

3A.5. SUFFICIENT DUCTILITY IN THE CARRIER SYSTEM

The design earthquake ground motion is defined in this Regulation to reduce the earthquake effects occurring on the building with the ductile behavior of the carrier system. *ductile design* And *capacity design* principles must be strictly adhered to.

3A.6. SUFFICIENT FLOOR RIGIDITY ON FLOORS AND TRANSITION FLOORS AND STRENGTH

3A.6.1 –In order to ensure that the inertial forces occurring in the floors during an earthquake are safely transferred to the vertical load-bearing system elements and at the same time that the earthquake effects are safely distributed among the vertical load-bearing system elements with different stiffnesses, it is essential that the floors have high in-plane stiffness and sufficient strength.

3A.6.2 –It must be shown through calculations that the in-plane forces are safely transferred from the slabs to the vertical carrier system elements. If necessary, additional connection reinforcement and transfer elements should be used in reinforced concrete slabs (See. **7.11**).

3A.6.3 –Large gaps in slabs should be avoided. In cases where gaps cannot be avoided, horizontal elements with sufficient rigidity and strength should be arranged at the edges of the gap to ensure that inertial forces are transferred to the vertical load-bearing system elements.

3A.6.4 –Especially located at the transition from floors with normal rigidity to very rigid basement floors and transferring most of the earthquake forces occurring on the upper floors to the surrounding walls in the basement floors. *transition tiles* It is essential to provide adequate in-plane rigidity and strength.

CHAPTER 4 – RESISTANCE OF BUILDINGS UNDER EARTHQUAKE EFFECT

CALCULATION PRINCIPLES FOR DESIGN

4.0. ICONS

A_t	=	Equivalent area used in the empirical natural vibration period calculation [m ²] Body
A_{wj}	=	cross-sectional area of j th bulkhead [m ²]
BYS	=	<i>Building Height Class</i>
C_h	=	Empirical coefficient used in the second order calculation
C_t	=	Coefficient used in the empirical natural vibration period calculation
C	=	The distance between the cross-sectional gravity centers of the connecting beam (hollow) curtain parts distance [m]
D	=	Excess Strength Coefficient
D_{lower}	=	Excess Strength Factor applied to the lower part of the building
$\bar{D}_{lower}(x)$	=	Equivalent Excess Strength Factor applied to the lower part of the building
$\bar{D}_{n,bottom}(x)$	=	Equivalent Strength Excess applied to the lower part of the building in the n th mode Coefficient
D_{top}	=	Excess Strength Factor applied to the upper part of the building
DTS	=	<i>Earthquake Design Class</i>
DD-1	=	Earthquake ground motion level with a probability of being exceeded in 50 years of 2% (recurrence period of 2475 years)
DD-2	=	Earthquake ground motion level with a probability of being exceeded in 50 years of 10% (recurrence period of 475 years)
DD-3	=	Earthquake location with a 50% probability of being exceeded in 50 years (recurrence period 72 years) movement level
D_{bi}	=	Additional eccentricity magnification factor on the i -th floor
$d_{\phi}^{(x)}$	=	(x) is the displacement caused by the fictitious load applied to the i th floor in the calculation of the dominant natural vibration period of the building in the earthquake direction [m]
TO_d	=	Directional coupling applied <i>design basis</i> Total earthquake effect
$TO_d^{(H)}$	=	Directional coupling applied <i>design basis</i> horizontal earthquake effect
$TO_d^{(x)}$	=	Under the influence of the earthquake in the (x) direction <i>design basis</i> earthquake effect
$TO_d^{(y)}$	=	Under the influence of the earthquake in the (y) direction <i>design basis</i> earthquake effect
$TO_d^{(z)}$	=	Under the influence of the earthquake in the (z) direction <i>design basis</i> earthquake effect
$F_{\phi}^{(x)}$	=	(x) Fictitious load applied to the i th floor in the calculation of the dominant natural vibration period of the building in the earthquake direction [kN]
$F_{iE}^{(x)}$	=	(x) Equivalent earthquake load acting on the i -th floor mass center in the earthquake direction [kN]

to	= Additional eccentricity of 5% [m]
$f_{jE}^{(Q)}$	= Equivalent seismic load acting on the jth finite element node [kN] Dead
G	= load effect
g	= Gravitational acceleration [m/s ²]
H	= Horizontal ground thrust effect
H_i	= Above the basement floors of the building <i>upper section</i> in the upper part of the i'th floor Height measured from base [m]
H_N	= Above the basement floors of the building <i>upper section</i> Total height of [m]
H_w	= Curtain height [m]
h_i	= Height of i'th floor [m]
I	= Building Importance Factor
$-w$	= Plan length of the curtain [m]
$-w_j$	= Plan length of jth curtain [m]
M_{Giant}	= Earthquake loads at the base of the reinforced concrete wall or braced frame occurring overturning moment [kNm]
$M_{i\theta}^{(X)}$	= On the i'th floor (X) the additional eccentricity effect is resisted in the direction perpendicular to the earthquake direction. <i>additional story torsional moment</i> [[kNm]
M_{he}	= Total overturning moment at the base due to earthquake loads for the entire building [kNm]
$M_{he}^{(X)}$	= (X) earthquake loads at the base for the entire building in the earthquake direction Total overturning moment [kNm]
M_1, M_2	= Bending moments obtained at the base of the curtain parts forming the tie beam curtain due to the earthquake effect [kNm]
m_i	= Total mass of the ith floor [t]
$m_{i\theta}$	= Mass moment of inertia of the ith floor [tm ²]
$m_j^{(S)}$	= The singular mass [t] acting on a typical finite element node j
m_t	= Above the basement floors of the building <i>upper section</i> Total mass of [t]
$m_{txn}^{(X)}$	= (X) is the vibration mode of the building in the x-axis direction for the earthquake direction. <i>base shear force modal effective mass</i> [[t]
$m_{txn,all}^{(X)}$	= Base shear force modal effective mass [t] calculated in the nth mode under the effect of earthquake in the (X) direction for the whole building (upper section + lower section)
$m_{txn,top}^{(X)}$	= Base shear force modal effective mass [t] calculated in the nth mode under the effect of earthquake in the (X) direction for the upper part of the building

$m_{\text{tyn}}^{(Y)}$	= (Y) is the vibration mode of the building in the y-axis direction for the earthquake direction. <i>base shear force modal effective mass</i> [[t]
N	= Above the basement floors of the building <i>upper section</i> Total number of floors in
M	= Shear forces occurring in the tie beams of the tie beam curtain under the effect of an earthquake The total of the equal tensile and compressive axial forces at the base of the curtain parts along the entire curtain height [kN]
n	= Live load participation coefficient
R	= <i>Carrier System Behavior Coefficient</i>
$R_a(T)$	= Predicted ductility capacity and period dependent <i>Earthquake Load Reduction Factor</i> Applied
R_{lower}	= to the lower part of the building <i>Carrier System Behavior Coefficient</i> Applied to the upper
R_{top}	= part of the building <i>Carrier System Behavior Coefficient</i> Applied to the lower part of the
$(R_a)_{\text{lower}}$	= building <i>Earthquake Load Reduction Factor</i>
$(R_a)_{n,\text{bottom}}$	= applied to the lower part of the building in the nth vibration mode <i>Earthquake Load Reduction Factor</i>
$\overline{(R_a)_{\text{lower}}}$	= Applied to the lower part of the building <i>Equivalent Earthquake Load Reduction Factor</i>
$\overline{(R_a)_{n,\text{bottom}}}$	= applied to the lower part of the building in the nth vibration mode <i>Equivalent Earthquake Load Reduction Factor</i>
$(R_a)_{\text{top}}$	= Applied to the upper part of the building <i>Earthquake Load Reduction Factor</i>
$(R_a)_{n,\text{top}}$	= applied to the upper part of the building in the nth vibration mode <i>Earthquake Load Reduction Factor</i>
Q	= Live load effect
S	= Snow load effect
$S_{ae}(T)$	= Horizontal elastic design spectral acceleration [g]
$S_{aR}(T)$	= Reduced design spectral acceleration [g]
S_{DS}	= Short period <i>design spectral acceleration coefficient</i> [[dimensionless]
T	= Natural vibration period [s]
T_B	= Horizontal elastic design acceleration spectrum corner period [s]
T_n	= Natural vibration period of nth mode [s]
T_{pA}	= Empirically calculated dominant natural vibration period [s]
$T_p^{(X)}$	= (X) the dominant natural vibration period of the building in the earthquake direction [s]
$U_I^{(X)}$	= (X) for any column or wall in the earthquake direction, at the i'th floor <i>reduced displacement</i> [[m]
$V_I^{(X)}$	= (X) in the earthquake direction on the i th floor <i>reduced ply shear force</i> [[kN]

$V_{x,all}^{(X)}$	= Calculated base shear force [kN] under the effect of earthquake in (X) direction for the whole building (upper section + lower section)
$V_{xn,all}^{(X)}$	= Calculated base shear force [kN] in the nth mode under the effect of earthquake in the (X) direction for the whole building (upper section + lower section)
$V_{x,top}^{(X)}$	= Calculated base shear force under the effect of earthquake in (X) direction for the upper part of the building [kN]
$V_{xn,top}^{(X)}$	= Base shear force calculated for the upper part of the building under the influence of the earthquake in the (X) direction in the nth mode [kN]
$V_{tE}^{(X)}$	= (X) earthquake direction affecting the entire building <i>Total Equivalent Earthquake burden</i> (base shear force) [kN]
$V_{tx}^{(X)}$	= With one of the modal calculation methods <i>direction</i> The biggest gain ever made in <i>total earthquake load</i> [[kN]
$W_j^{(S)}$	= The singular weight acting on a typical finite element node j [kN]
$W_{G,j}^{(S)}$	= Singular fixed weight [kN] acting on a typical finite element node j
$W_{Q,j}^{(S)}$	= Individual additional (moving) weight acting on a typical finite element node j [kN]
W_k	= Total weight acting on the kth floor [kN]
Y_M	= <i>Sufficient number of vibration modes</i>
α	= Empirical coefficient used for earthquake joint gaps
$\beta_{tE}^{(X)}$	= Equivalent base shear force magnification coefficient
$\beta_{II}^{(X)}$	= Second order magnification factor
$\Delta F_{WHAT}^{(X)}$	= (X) The building in the earthquake direction <i>What</i> affecting the th floor (top) <i>additional equivalent earthquake load</i> [[kN]
Δm_{θ}	= Floor mass moment of inertia increase [tm ²]
$\Delta_I^{(X)}$	= (X) is the displacement difference between two consecutive floors for any column or wall in the earthquake direction. <i>reduced relative storey drift</i> [[m]
$(\Delta_I^{(X)})_{average}$	= (X) is the average displacement difference between two consecutive floors in the earthquake direction. <i>reduced relative storey drift</i> [[m]
$\delta_I^{(X)}$	= (X) for the earthquake direction, for the columns or walls on the i-th floor of the building <i>effective relative storey drift</i> [[m]
$\delta_{i,max}^{(X)}$	= For the earthquake direction (X), the maximum value of the effective relative floor drifts on the ith floor of the building within the floor [m]
η_{bi}	= Torsional irregularity coefficient at the i-th floor

y_{TO}	= Used to determine the equivalent base shear force magnification coefficient empirical coefficient
λ	= Empirical coefficient used to limit relative story drifts
K	= Reinforced concrete and steel carriers in the definition of allowed relative storey drifts Coefficients used differently for systems
μ_k	= Ductility capacity anticipated for the carrier system
$V(X)$	= Equivalent seismic load reduction factor applied to the lower part of the building coefficient used in the calculation
$V_n(X)$	= Equivalent seismic load reduction applied in the nth mode for the lower part of the building The coefficient used in the calculation of the coefficient
$V_{lower}(X)$	= Reduced internal forces caused by the vibration of the lower part of the building itself The coefficient used to calculate
$V_{n,bottom}(X)$	= reduced internal vibration of the lower part of the building in the nth mode coefficient used to calculate forces
$V_{top}(X)$	= To calculate the internal forces transferred from the upper part of the building to the lower part coefficient used
$V_{n,top}(X)$	= internal forces transferred from the upper part of the building to the lower part in the nth mode The coefficient used to calculate
Ω	= In the curtain with tie beams <i>bond degree coefficient</i>
$\theta_{II,i}(X)$	= (X) defined for each ith floor in the earthquake direction <i>second degree indicator value</i>
$\theta_{II,max}(X)$	= (X) maximum defined in the earthquake direction <i>second order indicator value</i>

4.1. GENERAL

4.1.1. Purpose

The purpose of this Section is, **4.1.3** Buildings within the scope given in *Design by Strength* (DGT) approach will be used for design *linear calculation principles* is the explanation of.

4.1.2. Definition

One of the two main approaches for the design of building load-bearing systems under earthquake effects. *Design by Strength* (In the DGT approach:

(a) Reduced seismic loads corresponding to the defined structural system ductility capacity are determined for a specific anticipated performance target.

(b) The load-bearing system under reduced earthquake loads *linear earthquake calculation* The internal forces of the element found from this calculation are reduced, if necessary. *excess strength* combined with internal forces from other loads, also taking into account *strength demands* is obtained.

(c) Element *strength demands*, the element internal force capacities defined for the anticipated performance target (*endurance capacities*) is compared with .

(d) Obtained from earthquake calculation *relative storey drifts* compared with the allowable limits.

(to) The design is completed by showing that the strength demands are below the strength capacities and also that the relative story drifts are below the allowed limits. Otherwise, the element sections are changed and the calculation is repeated to reach the result.

4.1.3. Scope

Explained in this section *Design by Strength* (DGT) accounting principles, **Chapter 13** Explained in *High-Rise Buildings's* design, **Chapter 14** Explained in *Earthquake Insulated Buildings's* design and **Chapter 15** Explained in *Assessment and Reinforcement of Existing Buildings* outside the scope of this Regulation **Table 4.1** In the design of all buildings specified in *basic accounting principles* will be applied as. From the DGT calculation principles, *High-Rise Buildings's* (**Chapter 13**) And *Earthquake Insulated Buildings's* (**Chapter 14**) will be partially utilized in its design.

4.1.4. Performance Targets

4.1.4.1 – Table 4.1 In all buildings covered by this Section, **Table 3.4(a)** in *Normal Performance Target* defined as *Controlled Damage* (In order to achieve the performance target (KH), earthquake calculations will be made using the DGT calculation principles given in this Section under the influence of DD-2 earthquake ground motion.

4.1.4.2 – Table 3.4(a) According to *Earthquake Design Class* DTS=1a, DTS=2a and also *Building Height Class* In buildings with BYS=2, BYS=3;

(a) Under the influence of DD-2 earthquake ground motion $\bar{I}=1.5$ is taken into account in the design made with the DGT calculation principles in this Section. *preliminary design* will be taken into consideration.

(b) The pre-designed building carrier system is now under the influence of the DD-1 earthquake ground motion. **Table 3.4(a)** in *Advanced Performance Target* defined as *Controlled Damage* (KH) performance target and also under the DD-3 earthquake *Limited Damage* (SH) to achieve performance target **Chapter 5** According to *Evaluation and Design Based on Shapeshifting* (($\bar{I}$ GDT) approach and if necessary, the design will be repeated to ensure the targeted performance.

4.2. EARTHQUAKE LOAD COEFFICIENTS AND CAPACITY DESIGN PRINCIPLES

4.2.1. Earthquake Load Reduction Coefficient

4.2.1.1 – Design by Strength within the framework of, *predicted ductility capacity – strength demand* relationship and the resulting *earthquake load coefficients* Definition of **ANNEX 4A** is given in .

4.2.1.2 – ANNEX 4A According to the definition made in , the linear elastic earthquake loads will be reduced as a basis *Earthquake Load Reduction Coefficient* $R_a(T)$ is defined as follows:

$$R_a(T) = \frac{R}{I} \quad T > T_B \quad (4.1a)$$

$$R_a(T) = D + \frac{-R}{-I} - D \frac{T}{-T_B} \quad T \leq T_B \quad (4.1b)$$

Here R and D Table 4.1 defined in *Carrier System Behavior Coefficient* with *Excess Strength Coefficient*'s, T Table 3.1 defined in *Building Importance Coefficient*'s, T_n natural of the system the vibration period and T_B , Equation (2.3) defined by *spectrum corner period*. It shows the .

4.2.2. Capacity Design Principles

Design by Strength In addition to the rules given in this Section, in the design of building load-bearing systems within the framework of *capacity design principles* is taken into consideration. *Capacity design* The design approach envisages that the nonlinear ductile behavior in the load-bearing system should be limited to certain clearly defined elements (or sections) and that all other elements should have sufficient resistance capacity in accordance with this behavior. *Capacity Design Principles* The rules regarding the implementation of are given in the relevant sections of this Regulation.

4.3. APPLICATION LIMITS OF CARRIER SYSTEMS, CARRIER SYSTEM BEHAVIOR COEFFICIENTS AND EXCESS RESISTANCE COEFFICIENTS

4.3.1. Application Limits of Carrier Systems

Design by Strength Building load-bearing systems to which the calculation principles given in this section can be applied and for these systems Table 3.3 According to the definition allowed in *Building Height Classes* (BYS), 4.3.3 Depending on the ductility levels defined in , Table 4.1 is given in .

4.3.1.1 – High-Rise Buildings (Carrier systems and calculation principles for $BYS=1$) Chapter 13 It is defined in .

4.3.1.2 – Table 4.1' also A21, A22 And C21, C22 In the carrier systems symbolized by , only buildings with $DTS = 4$ are allowed *Building Height Class* $BYS \geq$ Can be increased to 2.

4.3.2. R and D Coefficients According to the Type of Carrier System

4.3.2.1 – Cast-in-place and prefabricated reinforced concrete, steel, light steel, masonry, wooden building load-bearing systems and 4.3.3 For various ductility levels defined in ANNEX 4 defined in *Carrier System Behavior Coefficient R* And *Excess Strength Coefficient D* Table 4.1 is given in .

4.3.2.2 – In composite column systems, given for steel carrier systems R And D coefficients will be used.

4.3.2.3 – 3.3.1 According to the definition given in , in buildings with basements surrounded by rigid external walls, in the lower section where the basement floors are located (R/I) = 2.5 and $D=1.5$ will be taken.

4.3.2.4 – In reinforced concrete shear walls and/or steel braced frame buildings with $DTS = 1, 1a, 2, 2a$, in any direction below (a) And (b) If one of the two conditions defined in is not met, accordingly *Carrier System Behavior Coefficient R* instead of $(4/5)R$ will be taken into consideration. *Excess Strength Coefficient D* There will be no changes in .

4.5.4.5 The tie beam curtain system that meets the condition given in will be considered as a single curtain.

(a) Base overturning moment of a single wall or steel braced frame in the carrier system M_{GIANT} , the total earthquake loads for the entire building in that direction base overturning moment M_{he} It will not be more than 1/3 of the.

(b) Base overturning moment of the curtain/curtains or steel braced frame/frames located on each side axis of the building M_{GIANT} or M_{GIANT} The sum of the 's is Total base overturning caused by earthquake loads for the entire building in the direction moment M_{he} It shall not be less than 1/6 of the.

4.3.3. High Ductility Level, Limited and Mixed Structural Systems

4.3.3.1 – Reinforced concrete and steel carrier systems, **Table 4.1** In terms of ductility levels, as given in *High ductility level carrier systems, Structural systems with limited ductility level And ductility level mixed load-bearing systems* It is divided into three classes.

4.3.3.2 – *High ductility level And limited ductility level* Definitions and conditions to be complied with regarding cast-in-place and prefabricated reinforced concrete, steel, light steel and timber load-bearing systems, respectively, **Chapter 7, Chapter 8, Chapter 9, Chapter 10 And Chapter 12** is given in .

4.3.3.3 – *Ductility level mixed carrier systems, limited ductility level frame carrier systems high ductility level* These are systems created by using reinforced concrete walls or steel braced frames.

4.3.4. Conditions Regarding Ductility Levels of Structural Systems

4.3.4.1 – Regarding limited ductility level and mixed systems;

(a) In buildings classified as DTS=1a, DTS=2a, DTS=3a and DTS=4a *Structural systems with limited ductility level* Other limitations associated with such carrier systems

4.3.4.3 It is stated in .

(b) $BYS \leq$ In buildings classified as 6 and DTS=1a and DTS=2a *ductility level mixed load-bearing systems* cannot be used.

4.3.4.2 – Carrier systems in perpendicular directions *ductility levels* It is necessary for the are the same. However, they are different in perpendicular directions. R coefficients and their corresponding D coefficients can be used. **Table 4.1** The maximum allowable limit according to *Building Height Class*, will be determined as the unfavorable of the given ones according to two directions.

4.3.4.3 – All earthquake effects are moment transferring *limited ductility level* load-bearing systems supported by reinforced concrete frames (**Table 4.1** also **A31, B31, C31** carrier systems) will only be used in buildings with DTS=3 and DTS=4. Carrier systems consisting of reinforced concrete frames with single-direction gear floors with or without filling (asmolen) also do not contain curtain walls, *Structural systems with limited ductility level* and will only be used in buildings with DTS=3 and DTS=4. Such carrier systems, *high ductility level* reinforced concrete tie beam (hollow) and/or hollow walls or *high ductility level* arranged with steel eccentric and/or central braced frames *ductility level mixed* can be done as systems (**Table 4.1** also **A2, B2, C2** carrier systems).

4.3.4.4 – In load-bearing systems containing only beamless floors, all earthquake effects are felt in reinforced concrete buildings. *high ductility level* curtains with tie beams (with holes) and/or without holes

or *limited ductility level* will be covered by gapless curtains (**Table 4.1**'also **A12, A13** And **A32** (In steel buildings, *high ductility level* centrally and/or eccentrically braced or buckling-restricted braced frames or *limited ductility level* Central braced frames will be used (**Table 4.1**'also **C12, C13** And **C32** (bearing systems). Calculations for such systems will be made in two stages. In the first stage calculation, frame columns will be taken as hinged from the bottom and top. In the second stage calculation, the connections of these elements will be modeled as monolithic. Internal forces in curtains, columns, diagonals and slabs will be calculated as the unfavorable one of the two stages. Relative floor drifts will be obtained from the second stage calculation.

4.3.4.5 – High ductility level Cast-in-place or prefabricated reinforced concrete shear walls with tie beams (with holes) or without holes and central, eccentric or buckling-restricted steel braced frames for moment transferring *high ductility level* In buildings where the frames are used together, the total overturning moments at the base of the walls or braced frames due to seismic loads shall not be less than 40% and more than 75% of the total overturning moment at the base due to seismic loads for the entire building:

$$0.40 M_{he} < \sum M_{GIANT} < 0.75 M_{he} \quad (4.2)$$

If the upper bound condition in this relation is not met, **Table 4.1** All earthquake effects in *high ductility level* defined for cases where the walls are covered by curtains or braced frames. *R* And *D* The maximum allowed *BYS* with coefficients will be taken into account. In case the lower limit condition is not met, **Table 4.1** Given in *R* And *D* No changes will be made to the coefficients, but one more than the maximum permitted *BYS* will be taken into account.

4.3.4.6 – Reinforced concrete and steel ductility level mixed in carrier systems, *high ductility level* The total of the overturning moments at the base of the reinforced concrete shear walls with tie beams (with holes) or without holes and the central, eccentric or buckling-restrained steel braced frames due to seismic loads shall not be less than 75% of the total overturning moment at the base due to seismic loads for the entire building:

$$\sum M_{GIANT} \geq 0.75 M_{he} \quad (4.3)$$

If this condition is not met, **Table 4.1** All earthquake effects in *limited ductility level* defined for situations where it is met by frames *R* And *D* The highest permitted *BYS* with coefficients will be taken into account.

4.3.4.7 – Limited ductility level void-free reinforced concrete walls or *limited ductility level* moment resisting frames with central steel braces *limited ductility level* In buildings where it is used with reinforced concrete or steel frames **Equation (4.3)** The condition given in will be met. Otherwise **4.3.4.6** The rule given in will be applied.

4.3.4.8 – 4.3.2.4, 4.3.4.5 And 4.3.4.6 Base overturning moments of the curtains to be used in M_{GIANT} , for gapless curtains **4.5.3.7(d)** or **4.5.3.8(c)** According to, tie beam (for hollow) curtains **4.5.4.3** It will be calculated according to. Total overturning moment resulting from earthquake loads for the entire building M_{he} whereas **4.7, 4.8.2** or **4.8.3** According to obtained will be done.

Table 4.1. Structural System Behavior Coefficient for Building Structural Systems, Strength Excess Factor and Permitted Building Height Classes

Building Support System	Carrier System Behaviour Coefficient R	Strength The excess Coefficient D	Allowed Building Height Classes BYS
A. CAST-IN-SITU REINFORCED CONCRETE BUILDING CARRIER SYSTEMS			
A1. High Ductility Level Carrier Systems			
A11. All earthquake effects are moment transferring <i>high ductility level</i> buildings with reinforced concrete frames	8	3	BYS ≥ 3
A12. All earthquake effects <i>high ductility level</i> Buildings with reinforced concrete shear walls with tie beams (hollows)	7	2.5	BYS ≥ 2
A13. All earthquake effects <i>high ductility level</i> Buildings with reinforced concrete walls without gaps	6	2.5	BYS ≥ 2
A14. Moment transferring effects of earthquake <i>high ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings where reinforced concrete shear walls with tie beams (hollow) are supported together (See 4.3.4.5)	8	2.5	BYS ≥ 2
A15. Moment transferring effects of earthquake <i>high ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings where they are supported by reinforced concrete shear walls without gaps (See Fig.4.3.4.5)	7	2.5	BYS ≥ 2
A16. All earthquake effects are caused by articulated connections at roof level and not exceeding 12 m in height. <i>high ductility level</i> Single-storey buildings supported by reinforced concrete columns	3	2	-
A2. Ductility Level Mixed Structural Systems (See 4.3.4.1, 4.3.4.6)			
A21. Moment transferring effects of earthquake <i>limited ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings where reinforced concrete shear walls with tie beams (hollow) are supported together (See 4.3.1.2)	6	2.5	BYS ≥ 4
A22. Moment transferring effects of earthquake <i>limited ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings where they are supported by reinforced concrete shear walls without gaps (See Fig.4.3.1.2)	5	2.5	BYS ≥ 4
A23. Moment transferring effects of earthquake <i>Single direction gear slab with or without filling (hollow) of limited ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings supported by reinforced concrete shear walls with tie beams (hollows)	6	2.5	BYS ≥ 6
A24. Moment transferring effects of earthquake <i>Single direction gear slab with or without filling (hollow) of limited ductility level</i> with reinforced concrete frames <i>high ductility level</i> Buildings that are faced with reinforced concrete shear walls without gaps	5	2.5	BYS ≥ 6
A3. Structural Systems with Limited Ductility Level (See 4.3.4.1, 4.3.4.3, 4.3.4.7)			
A31. All earthquake effects are moment transferring <i>limited ductility level</i> buildings with reinforced concrete frames	4	2.5	BYS ≥ 7
A32. All earthquake effects <i>limited ductility level</i> Buildings with reinforced concrete walls without gaps	4	2	BYS ≥ 6
A33. Moment transferring effects of earthquake <i>limited ductility level</i> with reinforced concrete frames <i>limited ductility level</i> Buildings that are faced with reinforced concrete shear walls without gaps	4	2	BYS ≥ 6

Table 4.1 (continued)

Building Support System	Carrier System Behaviour Coefficient R	Strength The excess Coefficient D	Allowed Building Height Classes BYS
B. PREFABRICATED REINFORCED CONCRETE BUILDING CARRIER SYSTEMS			
B1. High Ductility Level Carrier Systems			
B11. All earthquake effects are connected to moment transferring elements. <i>high ductility level</i> buildings covered with prefabricated frames			
MAB1, MAB2 type moment transferring connections:	7	2.5	BYS ≥ 4
MAB3, MAB4 type moment transferring connections:	5	2.5	BYS ≥ 6
B12. Connections of earthquake effects transferring moment <i>high ductility level</i> with pregenerated frameworks, <i>high ductility level</i> Buildings supported by reinforced concrete shear walls with cast-in-place tie beams (hollow) (See Fig. 4.3.4.5)			
MAB1, MAB2 type moment transferring connections:	7	2.5	BYS ≥ 2
MAB3, MAB4 type moment transferring connections:	5	2.5	BYS ≥ 6
B13. Connections of earthquake effects transferring moment <i>high ductility level</i> with pregenerated frameworks, <i>high ductility level</i> Buildings where the walls are supported by cast-in-situ, void-free reinforced concrete shear walls (See Fig. 4.3.4.5)			
MAB1, MAB2 type moment transferring connections:	6	2.5	BYS ≥ 2
MAB3, MAB4 type moment transferring connections:	5	2.5	BYS ≥ 6
B14. The connections of the vertical loads are made with hinged prefabricated and two-way frames, and all earthquake effects are <i>high ductility level</i> Buildings where cast-in-place void-free and/or tie beam (void) reinforced concrete shear walls are used	4	2	BYS ≥ 7
B15. All earthquake effects are caused by articulated connections at roof level and not exceeding 12 m in height. <i>high ductility level</i> single-storey buildings supported by columns	3	2	-
B2. Ductility Level Mixed Structural Systems (See 4.3.4.1, 4.3.4.6)			
B21. Connections of earthquake effects transferring moment <i>limited ductility level</i> with pregenerated frameworks, <i>high ductility level</i> Buildings supported by reinforced concrete shear walls with (hollow) or solid cast-in-place tie beams			
MAB1, MAB2 type moment transferring connections:	5	2.5	BYS ≥ 5
MAB3, MAB4 type moment transferring connections:	4	2.5	BYS ≥ 6
B3. Structural Systems with Limited Ductility Level (See 4.3.4.1, 4.3.4.7)			
B31. All earthquake effects are connected to moment transferring elements. <i>limited ductility level</i> buildings covered with prefabricated frames	3	2	BYS = 8
B32. Connections of earthquake effects transferring moment <i>limited ductility level</i> prefabricated frames, cast in place <i>limited ductility level</i> Buildings that are faced with reinforced concrete shear walls without gaps	3	2	BYS ≥ 7
B33. All earthquake loads are resisted by prefabricated reinforced concrete vertical double-walled panels. <i>limited ductility level</i> buildings	4	2	BYS ≥ 6
B34. All earthquake loads are resisted by prefabricated reinforced concrete vertical single-wall panels. <i>limited ductility level</i> buildings	3	2	BYS ≥ 7

Table 4.1 (continued)

Building Support System	Carrier System Behaviour Coefficient R	Strength The excess Coefficient D	Allowed Building Height Classes BYS
C. STEEL BUILDING CARRIER SYSTEMS			
C1. High Ductility Level Carrier Systems			
C11. All earthquake effects are moment transferring <i>high ductility level</i> buildings with steel frames	8	3	BYS ≥ 3
C12. All earthquake effects <i>high ductility level</i> Buildings supported by eccentric or buckling-restrained centrally braced steel frames	8	2.5	BYS ≥ 2
C13. All earthquake effects <i>high ductility level</i> buildings supported by centrally braced steel frames	5	2	BYS ≥ 4
C14. Moment transferring effects of earthquake <i>high ductility level</i> with steel frames <i>high ductility level</i> eccentric or buckling-restrained centrally braced steel frames or <i>high ductility level</i> Buildings where reinforced concrete shear walls with tie beams (hollow) are supported together (See 4.3.4.5)	8	3	BYS ≥ 2
C15. Moment transferring effects of earthquake <i>high ductility level</i> with steel frames <i>high ductility level</i> centrally braced steel frames or <i>high ductility level</i> Buildings where they are supported by reinforced concrete shear walls without gaps (See Fig. 4.3.4.5)	6	2.5	BYS ≥ 2
C16. All earthquake effects are caused by articulated connections at roof level and not exceeding 12 m in height. <i>high ductility level</i> Single-storey buildings supported by steel columns	4	2	-
C2. Ductility Level Mixed Structural Systems (See 4.3.4.1, 4.3.4.6)			
C21. Moment transferring effects of earthquake <i>limited ductility level</i> with steel frames <i>high ductility level</i> eccentric or buckling-restrained centrally braced steel frames or <i>high ductility level</i> Buildings where reinforced concrete shear walls with tie beams (hollow) are supported together (See 4.3.1.2)	6	2.5	BYS ≥ 4
C22. Moment transferring effects of earthquake <i>limited ductility level</i> with steel frames <i>high ductility level</i> centrally braced steel frames or <i>high ductility level</i> Buildings where they are supported by reinforced concrete shear walls without gaps (See Fig. 4.3.1.2)	5	2	BYS ≥ 4
C3. Structural Systems with Limited Ductility Level (See 4.3.4.1, 4.3.4.7)			
C31. All earthquake effects are moment transferring <i>limited ductility level</i> buildings with steel frames	4	2.5	BYS ≥ 7
C32. All earthquake effects <i>limited ductility level</i> Buildings with centrally braced steel frames	3	2	BYS = 8
C33. Moment transferring effects of earthquake <i>limited ductility level</i> with steel frames <i>limited ductility level</i> Buildings supported by centrally braced steel frames	4	2	BYS ≥ 7

Table 4.1 (continued)

Building Support System	Carrier System Behaviour Coefficient R	Strength The excess Coefficient D	Allowed Building Height Classes BYS
D. LIGHT STEEL BUILDING CARRIER SYSTEMS			
D1. High Ductility Level Carrier Systems			
Light steel buildings with high ductility level where all earthquake effects are resisted by screwed, bolted sheet metal, OSB or plywood wall panels.	4	2	BYS = 8
D2. Structural Systems with Limited Ductility Level (See 4.3.4.1)			
Light steel buildings with limited ductility level where all earthquake effects are resisted by cladding or braced panels containing gypsum boards.	3	2	BYS = 8
E. MASONRY BUILDING CARRIER SYSTEMS			
E1. High Ductility Level Carrier Systems			
E11. Reinforced masonry buildings	4	2	BYS ≥ 7
E12. Reinforced aerated concrete panel buildings	4	2	BYS ≥ 7
E2. Structural Systems with Limited Ductility Level (See 4.3.4.1)			
E21. Besieged masonry buildings	3	2	BYS = 8
E22. Unreinforced masonry buildings	2.5	1.5	BYS = 8
F. WOODEN BUILDING CARRIER SYSTEMS			
F1. High Ductility Level Carrier Systems			
Wooden buildings with high ductility level where all earthquake effects are resisted by nailed or screwed OSB or plywood wall panels.	4	2	BYS ≥ 7
F2. Structural Systems with Limited Ductility Level (See 4.3.4.1)			
Wooden buildings with limited ductility level where all earthquake effects are resisted by glued wall panels connected with nails, screws and bolts or by wooden braces.	3	2	BYS = 8

4.3.4.9 –Except for the basement perimeter curtains, $H_w / -w \leq$ In gapless curtains with 2.0 **Table 4.1** Given in R Internal forces calculated according to the coefficients, $[3 / (1 + H_w / -w)]$ It will be enlarged by multiplying it by the coefficient. However, this coefficient will not be greater than 2.

4.3.4.10 –Rigid reinforced concrete curtains used around the basement floors of buildings, **Table 4.1** It shall not be considered as part of the curtain or curtain-frame systems included in (See **4.3.5.1**).

4.3.5. Application of Excess Strength Factors

4.3.5.1 –*Excess Strength Coefficient D* , **ANNEX 4A** As defined in , it is the coefficient expressing the excess of yield strength over design strength.

4.3.5.2 –High or limited level of load-bearing system elements *Ductile* against his behavior (*bending moment, tensile force and so on*) in the calculation of reduced internal forces *Excess Strength Coefficient* will not be used ($D=1$).

4.3.5.3 –Carrier system elements *non-ductile* against his behavior (*shear force in reinforced concrete elements, forces acting on joints in steel elements and similar*) for reduced internal forces *Excess Strength Coefficient* will be used as a multiplier ($D > 1$). However, *High level of ductility* in carrier systems *D* Internal forces magnified by coefficients, *capacity design principle* In the sections defined (allowed) as required by *flowing* situation *Compatible with* shall not be taken greater than the internal forces.

4.3.5.4 –In building carrier systems *Excess Strength Coefficient* Detailed rules regarding the implementation of are given in the relevant sections of the Regulation.

4.3.5.5 –Reduced internal forces acting in-plane on building floors, for the relevant carrier system **Table 4.1** defined in *Excess Strength Coefficient* will be applied.

4.3.5.6 –Excess Strength Factors in structural system elements in basement floors surrounded by rigid external walls **4.10.1** will be taken into consideration according to.

4.3.5.7 –Rules regarding the consideration of excess strength in the forces transferred from the building load-bearing systems to the foundations **4.10.3** It is defined in .

4.3.6. Different Upper and Lower Sections of Buildings *R* And *D* Using Coefficients

Different from each other in upper and lower parts *R* And *D* In buildings where coefficients are used **4.3.6.1** or **4.3.6.2** The calculation will be made according to the rules given in **3.3.1** These rules can also be applied to buildings with basements surrounded by external rigid walls according to the definition given in . Alternatively **4.7.5** or **4.8.5** Calculations can also be made according to the rules explained in .

4.3.6.1 – 4.7 Explained in *Equivalent Earthquake Load Method* with the entire carrier system (*upper section + subsection*) in the calculation made taking into account;

(a) Upper section which is against the ductile behavior of the structural system elements in *reduced internal forces*, for the upper section **Table (4.1)** Selected from R_{top} And D_{top} coefficients and taken into account

(X) the dominant natural vibration period in the earthquake direction $T_{(x)}$ Depending on **Equation (4.1)** from calculated *Earthquake Load Reduction Factor* $(R_a)_{top}$ will be obtained using .

(b) Upper section The carrier system elements in *non-ductile* reduced internal forces opposing the behavior, **(a)** Internal forces obtained in D_{top} obtained by multiplying the coefficient by will be done.

(c) Subsection which is against the ductile behavior of the structural system elements in *reduced internal forces* for *equivalent earthquake load reduction factor* $(R_a)_{lower}$ **Equation (4.4)** will be determined by:

$$(\bar{R})_{a \text{ sub}} = \frac{(R_a)_{top}}{V_{(X)}} \quad (4.4)$$

In this equation $v_{(x)}$ The coefficient is given below:

$$V_{(X)} = V_{(X) \text{ top}} + V_{(X) \text{ lower}} \quad (4.5a)$$

$$V_{(X) \text{ top}} = \frac{V_{(X)}}{v_{(X) \text{ all}}} ; \quad V_{(X) \text{ lower}} = (1 - v_{(X) \text{ top}}) \frac{(R_a)_{top}}{(R_a)_{lower}} \quad (4.5b)$$

Equation (4.5a) The first term in $v_{(x)pp}$, *upper section* from *subsection* Reduced forces transferred to, second term $v_{(x)lower}$ where as *subsection* Calculate the reduced forces caused by the vibration of corresponds to the coefficients used for $(R_a)_{lower}$ *subsection* For the carrier system in **Table (4.1)** Selected from R_{lower} And D_{lower} coefficients and $T_{(x)p}$ Depending on **Equation (4.1)** Calculated from *Earthquake Load Reduction Factor* $v_{(x)top}$ indicates, $v_{(x)top}$, under unmitigated earthquake loads *upper section* The base shear force of *entire carrier system's (upper section+subsection)* is defined as the ratio of the base shear force.

(d) Subsection The carrier system elements in *non-ductile* reduced internal forces opposing behavior **(c)** The internal forces obtained in are defined below. *equivalent strength excess coefficient* $\bar{D}_{(x)lower}$ will be obtained by multiplying by:

$$D_{(x)lower} = \frac{0.6v_{(x)top}D_{top} + v_{(x)lower}D_{lower}}{V_{(x)}} \quad (4.6)$$

4.3.6.2 – 4.8 Explained in *Modal Calculation Methods* with the entire carrier system (*upper section+subsection*) in the calculation made by taking into account,

(a) Equation (4.4), Equation (4.5) And Equation (4.6) All operations in , for each n th vibration mode and the corresponding natural vibration period T_n will be implemented taking into account the n th mode

Equation (4.5b) Instead of the base shear force ratio in , the shear forces opposing these forces in the same mode are *modal effective masses* The ratio of can also be used (See. **4B.1.4**).

(b) $R_{lower} < R_{top}$ for all vibration modes in buildings with rigid basements **Equation (4.5b)** from $v_{(x)n,top}$ If the calculation of is avoided, v always gives less favorable results. $v_{(x)n,top} = 0$ It can be assumed that, in this case, *subsection* For the n th mode, the following simplifications can be made:

$$(\bar{R})_{n, \text{moment, bottom}} \equiv (R_a)_{n, \text{bottom}} ; \quad D_{(x)n, \text{bottom}} \equiv D_{lower} \quad (4.7)$$

4.3.2.3 for basements according to $D_{lower} = 1.5$ will be taken.

4.4. DEFINITION OF EARTHQUAKE EFFECTS AND COMBINATION WITH OTHER EFFECTS

4.4.1. Reduced Design Acceleration Spectrum Under Horizontal Earthquake Effect

4.7 And 4.8.2 In the calculation methods given in *reduced earthquake loads in the horizontal direction* will be used to determine *reduced design acceleration spectrum's* specific T natural vibration The coordinate for the period *reduced design spectral acceleration* $S_{ae}(T)$, **Equation (4.8)** is defined by:

$$S_{ae}(T) = a_e \frac{S(T)}{R_a(T)} \quad (4.8)$$

Here $S_{ae}(T)$, **2.2** For the DD-2 earthquake ground motion defined in **Equation (2.2)** determined by *horizontal elastic design spectral acceleration* $n_i, R_a(T)$ where as **Equation (4.1)** defined by *Earthquake Load Reduction Factor* It shows the .

4.4.2. Combining Earthquake Effects in Horizontally Perpendicular Directions

4.4.2.1 –Seismic calculation of the carrier system under the effect of horizontal earthquake **4.7** or **4.8.2** If it is done with one of the methods given in , the earthquake effects resulting from earthquakes defined in the horizontally perpendicular (X) and (Y) directions **Equation (4.9)** will be combined as defined in:

$$\begin{aligned} TQ_{(H)} &= \pm TQ_{(X)} \pm 0.3 TQ_{(Y)} \\ TQ_{(H)} &= \pm 0.3 TQ_{(X)} \pm TQ_{(Y)} \end{aligned} \quad (4.9)$$

Here $TQ_{(X)}$ And $TQ_{(Y)}$ earthquakes in perpendicular (X) and (Y) directions in any section under the influence **4.10** Earthquake effects defined and calculated separately according to , $TQ_{(H)}$ whereas *directional coupling* applied *Horizontal earthquake effect as a basis for design* It symbolizes the.

4.4.2.2 –Seismic calculation of the carrier system under the effect of horizontal earthquake **4.8.3** With the method given in *in the time domain* In case of construction, earthquake components in horizontal (X) and (Y) directions perpendicular to each other **2.5** Since they are defined simultaneously according to , combined horizontal earthquake effect $TQ_{(H)}$, is obtained directly as a result of this calculation.

4.4.3. Vertical Earthquake Effect

4.4.3.1 –Vertical earthquake calculations for buildings classified as DTS=1, DTS=1a, DTS=2 and DTS=2a and containing the following elements are carried out: *local vertical vibration modes* Based on *only for these elements* **2.3.5** defined in *vertical elastic acceleration spectrum* According to **4.8.2'** also will be done using the method given. *Vertical earthquake effect E_{zd}* In this way, all carriers in the account for systems $R/I=1$ and $D=1$ will be taken.

- (a) Buildings containing beams with a horizontal projection of 20 m or more in spans,
- (b) Buildings containing consoles with a horizontal projection of 5 m or more of their openings,
- (c) Buildings containing columns resting on beams,
- (d) Buildings with columns inclined to the vertical.

4.4.3.2 – 4.4.3.1 In the carrier system parts other than the elements specified in and **4.4.3.1** in buildings outside the definition *vertical earthquake effect E_{zd}* , without any special calculation, **Equation (4.10)** will be calculated approximately with .

$$TQ_{zd} \approx (2/3) S_{DS} G \quad (4.10)$$

Here G the dead load effect, S_{DS} whereas **2.3.2** defined in *short period design spectral acceleration coefficient* it shows the .

4.4.4. Combining Earthquake Effects with Other Effects

4.4.4.1 –Load combinations including earthquake effects to be taken as basis in the design of the load-bearing system elements **Equation (4.11)** And **Equation (4.12)** is defined by:

$$G + Q + 0.2 S + TQ_{(H)} \pm 0.3 TQ_{(Z)d} \quad (4.11)$$

$$.9 G + H + TQ_{(H)} - 0.3 TQ_{(Z)d} \quad (4.12)$$

Here Q live load effect, S snow load effect, H whereas **Chapter 16** Horizontal ground defined in symbolizes the thrust. Horizontal earthquake effect $TO(H)$ and **4.4.2** According to , vertical earthquake effect $TO(Z)$ and whereas **4.4.3** It will be determined according to.

4.4.4.2 –In steel and light steel buildings;

(a) Design with Load and Resistance Coefficients In case of application of the YDKT approach **Equation (4.11)** instead of 1.2 G to be taken, **Equation (4.12)** will be used as is.

(b) Design with Safety Factors In case of application of the GKT approach, **Chapter 16** Apart from its basic design, according to **Chapter 9** And **Chapter 10** The load combinations given in will be used.

4.5. RULES FOR MODELLING THE SUPPORT SYSTEM FOR LINEAR CALCULATION

Design by Strength to be done within the scope of *linear* Rules for modeling load-bearing systems to be used in calculations are given in this section.

4.5.1. General Modeling Rules

4.5.1.1 –Building load-bearing systems are always *three dimensional* will be modeled as .

4.5.1.2 –*Two horizontal lines perpendicular to each other* The earthquake effect in the direction will always be taken into account. The vertical earthquake effect is also **4.4.3** will be taken into account according to.

4.5.1.3 –The damping ratio shall be taken as 5% unless otherwise stated.

4.5.1.4 –The modeling rules given here can also be applied to non-earthquake loading cases.

4.5.2. Modeling of Beams and Columns

4.5.2.1 –Beams and columns, *frame (bar) finite elements* They will be modeled as . All 6 degrees of freedom will be taken into account at the nodes where the columns and beams meet. In case the slabs are modeled as rigid diaphragms, the degrees of freedom corresponding to the rigid motion will be removed.

4.5.2.2 –Reinforced concrete columns and beam *effective section stiffnesses* **4.5.8** It will be determined according to.

4.5.3. Modelling of Reinforced Concrete Void-Free Shear Walls

4.5.3.1 –Reinforced concrete void-free walls are generally *console* They are vertical carrier system elements that work as

4.5.3.2 –Rectangular reinforced concrete shear walls are carrier system elements defined as having a ratio of length to thickness of at least 6 (six).

4.5.3.3 –In reinforced concrete walls with cross-sectional shape I, T, L, U or C, at least one wall arm in each direction **4.5.3.2** Otherwise, the carrier system element will not be considered as a curtain in that direction. However, in I, T, L, U or C section curtains, the curtain arm (or arms) **4.5.4.5A** provider of *tie beam curtain* In case of curtain piece (or pieces), **4.5.3.2** The condition given in may not be applied.

4.5.3.4 – Chapter 7 defined for reinforced concrete design of shear wall section *curtain end zones* is modeled as a column and the body area between them is modeled as a very rigid beam. *sliding frame* models will not be used for curtains.

4.5.3.5 –Modelling techniques in which the curtain arms are modelled and calculated separately for curtains with a cross-sectional shape of T, L, U or C will not be used for curtains.

4.5.3.6 –Reinforced concrete walls, **4.5.3.7** And **4.5.3.8** It will be modeled by one of the methods given in .

4.5.3.7 –Reinforced concrete shear walls with a rectangular, I, T, L, U or C cross-sectional shape contain degrees of freedom for both in-plane and out-of-plane displacements. *shell finite elements* They will be modeled with .

(a) All 6 degrees of freedom will be taken into account at the nodes where the shell finite elements meet.

(b) The finite element dimensions shall be selected to ensure that the internal force distribution is calculated with sufficient accuracy.

(c) Regarding in-plane and out-of-plane behavior *effective section stiffnesses* **4.5.8** It will be determined according to.

(d) In shear walls with a rectangular, I, T, L, U or C cross-sectional shape, the resultants of the finite element nodal point forces are taken as basis in the reinforced concrete section calculation at the cross-section gravity center. *equivalent bar section effects* (bending/torsional moments, shear forces, axial force) will be obtained. The bending moment obtained in this way at the base of the wall, **4.3.4.5, 4.3.4.6** And **4.3.4.7** curtain in *base overturning moment* M_{GIANT} will be used as.

4.5.3.8 –Curtains with a rectangular, I, T, L, U or C cross-sectional shape are those with a axis passing through the cross-sectional centre of gravity, provided that the ratio of the largest curtain arm length in the plan to the total curtain height does not exceed 1/2. *equivalent bar* They can be modeled as finite elements. In this case;

(a) At the nodal points where the shear wall parts meet the beam and/or floor finite elements in the plan at floor levels *dependant* degrees of freedom, *three-dimensional rigid body motion* to be defined at the section center of gravity in a way that satisfies the condition *main node* 6 in *independent* degrees of freedom *kinematics* will be connected as .

(b) Bending and shearing of shear walls modeled as equivalent bar *effective section stiffnesses* **4.5.8** It will be determined according to.

(c) In the calculation of reinforced concrete section, the bar section effects (bending/torsion moments, shear forces, axial force) are obtained directly at the section gravity center. The bending moment obtained at the base of the curtain wall, **4.3.4.5, 4.3.4.6** And **4.3.4.7** curtain in *base tipping moment* M_{GIANT} will be used as.

4.5.4. Modeling of Reinforced Concrete Tie Beam (Hollow) Shear Walls

4.5.4.1 –Reinforced concrete *tie beam (hollow) curtains*, two gapless curtain parts are connected together with short and very high shear strength tie beams. *a single curtain* These types of curtains act as cantilever-like vertical carrier system elements. Thanks to their strong connecting beams, these types of curtains exhibit a behavior between the frames and the gapless curtains that work as cantilevers (**See 4.5.4.5**).

4.5.4.2 –The cross-sectional shapes of the shear wall parts forming the tie beam shear wall can be rectangular or U or C shaped as in building cores (**Figure 4.1**).

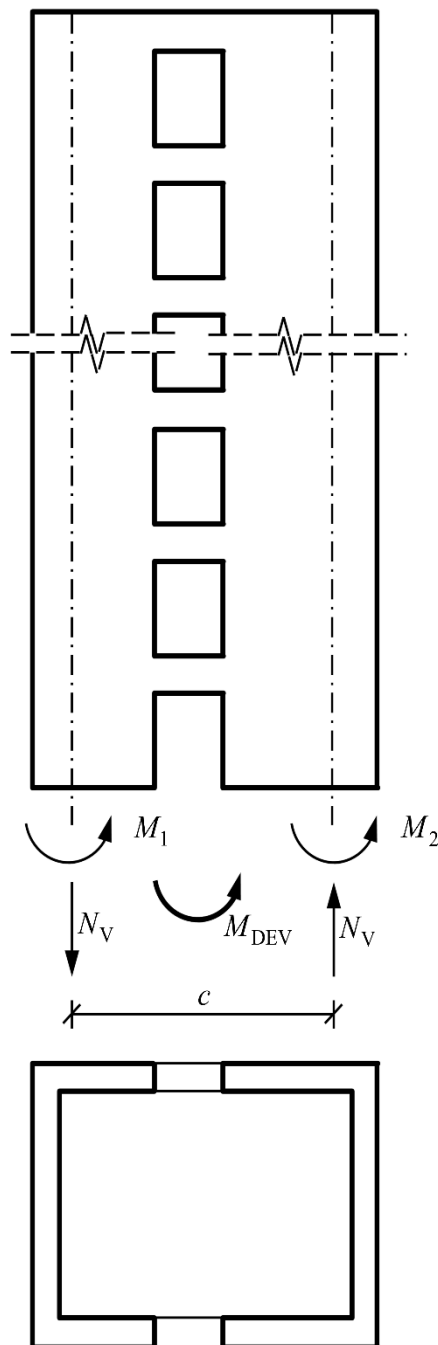


Figure 4.1

4.5.4.3 –Base overturning moment of the tie beam (hollow) shear wall **Equation (4.13)** is calculated with:

$$M_{GIANT} = M_1 + M_2 + c N_w \quad (4.13)$$

Here M_{GIANT} Total overturning moment at the base of the tie beam (hollow) shear wall, M_1 And M_2

Bending moments obtained at the base due to earthquake effect in the curtain parts forming the tie beam curtain, N_w where a shear forces occurring in the connecting beams during earthquake

The total of the total height of the curtain wall opposes the equal tensile and compressive axial forces formed at the base of the curtain parts. c , shows the distance between the cross-sectional gravity centers of the curtain parts (**Figure 4.1**). **Equation (4.13)** Obtained from size **4.3.4.5** And **4.3.4.6** curtain in *base overturning moment* M_{GIANT} will be used as.

4.5.4.4 –The basis for defining the tie beam (hollow) curtains *bond degree coefficient* Ω , **Equation (4.14)** is given in.

$$\Omega = \frac{c N_w}{M_{GIANT}} = \frac{c N_w}{M_1 + M_2 + c N_w} \quad (4.14)$$

4.5.4.5 –Tie beam (hollow) curtain, **Equation (4.14)** Given in *bond degree coefficient's* **Equation (4.15)** It is defined as the carrier system element that satisfies the condition defined in:

$$\Omega \geq \frac{1}{3} \quad (4.15)$$

Equation (4.15) If the condition defined in is not met, each of the curtain parts *gapless curtain* In addition to this condition, in order to prevent excessive axial forces from occurring in the curtain parts, $\Omega \leq 2$ Efforts should be made to ensure that all three conditions are met.

4.5.4.6 –The curtain parts that form the reinforced concrete tie beam curtain, **4.5.3.7** or **4.5.3.8** It will be modeled according to.

4.5.4.7 –Tie beams can be modeled as bar elements. Tie beam *effective section stiffnesses* **4.5.8** It will be determined according to.

4.5.5. Modeling of Basement Shear Walls

4.5.5.1 –In buildings with basements, basement walls arranged all around the building, transfer all or a large part of the inertial forces formed in the upper floors to the transition floors (See Fig. **4.5.7**) and transfer them to the foundation, and at the same time, they are the carrier system elements that resist ground thrusts during an earthquake.

4.5.5.2 –Basement curtains **4.5.3.7** According to , the shell will be modeled with finite elements.

4.5.6. Modeling of Floors

4.5.6.1 –Floor coverings;

(a) Transfers the inertial forces created by the masses on the floors due to the effect of earthquake accelerations to the vertical carrier system elements, together with the beams, if any, thanks to their high in-plane rigidity,

(b) At the same time, and usually more importantly, they are horizontal carrier system elements that ensure that the earthquake loads acting on the building are distributed among the vertical carrier system elements according to their rigidity. The floors, which may also contain gaps of various sizes, have their own planes.

It is essential that they are modeled appropriately so that the load transfer within them can be determined accurately.

4.5.6.2 – 3.6.2.2 According to **A2** and **A3** type of irregularities and/or pavements *rigid diaphragm* In buildings where it is not planned to operate as a floor and in reinforced concrete beamless floor systems, the floors will be modeled with two-dimensional finite elements.

4.5.6.3 – 3.6.2.2 According to **A2** and **A3** It is expected that no irregularities of this type are present and that no significant in-plane deformations will occur. *Regular in plan* reinforced concrete floors in buildings *rigid diaphragm* can be modeled as. The rigid diaphragm model, **4.5.10** According to *additional eccentricity effect* it will also be used in the calculation to be taken into account.

4.5.6.4 – Rigid diaphragm As a result of the calculation made according to the model, the force transferred from the floor to any vertical load-bearing system element (column or wall) in any direction will be calculated as the difference of the shear forces obtained for that element in the relevant direction at the floors below and above the floor.

4.5.6.5 – 4.5.6.2 or **4.5.6.4** According to , it will be shown by calculation that the in-plane forces obtained from the earthquake calculation are safely transferred from the slabs to the vertical carrier system elements. If necessary, additional connection reinforcement and transfer elements will be used in reinforced concrete slabs.

4.5.7. Modeling of Floors in Transition Floors

4.5.7.1 – 3A.6.4 As stated in , it is located at the transition from normal floors to very rigid basement floors and has to transfer all or a large part of the inertia forces formed in the upper floors suddenly to the surrounding walls in the basement floors. *transition floor slabs* It is essential to provide sufficient in-plane stiffness and strength in . This condition also applies to other transition floors where sudden stiffness changes occur for other reasons.

4.5.7.2 – 3.6.2.2 According to **A2** and **A3** Regardless of the presence of irregularities, the floors of the transition floors are laid with sufficient floor thicknesses, **4.5.6.2** It will be modeled according to.

4.5.7.3 – As a result of the earthquake calculation, the forces transferred to the rigid perimeter walls in the basement floors will be calculated and it will be shown that the transition slabs have sufficient strength for this transfer. If necessary, additional connection reinforcement will be arranged for the transfer elements in the slabs and the load transfer to the walls.

4.5.8. Effective Section Stiffness of Reinforced Concrete Bearing System Elements

4.5.8.1 – Design by Strength in modeling the section properties of reinforced concrete load-bearing system elements within the scope of **Table 4.2** Given in *effective section stiffness factors* will be used.

4.5.8.2 – Table 4.2 Both multipliers given in will be taken into account in the calculation model.

4.5.8.3 – Effective section stiffness factors will only be applied in calculations under loads that are included in earthquake effective load combinations and that enter these combinations.

4.5.9. Modeling of Masses

4.5.9.1 –In case the carrier system elements are modeled as rod, plate (membrane) or shell finite elements *singular nodal masses*, are assigned as the resultants of distributed masses in the spans of connected finite elements. The individual masses at the finite element nodes are defined to correspond to only two horizontal or additional vertical translational degrees of freedom.

4.5.9.2 –Singular mass acting on a typical finite element node j $m_j^{(S)}$ Equation (4.16) will be calculated.

$$m_j^{(S)} = w_{(S)j} + n w_{Q,j}^{(S)} \quad ; \quad m_j^{(S)} = \frac{w_j^{(S)}}{g} \quad (4.16)$$

Here $w_{(S)j}$ And $w_{Q,j}^{(S)}$ Resultant fixed load and moving load acting on finite element node j indicates the load. Equation (4.16) Located in *moving load mass participation coefficient*, n , Table 4.3 For fixed equipment weights in industrial buildings $n=1$ will be taken, but crane lifting loads will not be taken into account in calculating the floor weights. 30% of the snow loads will be taken into account in calculating the roof floor weight. Regarding non-structural elements and equipment 6.1.3 The condition given in will be taken into consideration.

Table 4.2. Effective Section Stiffness Factors of Reinforced Concrete Bearing System Elements

Reinforced Concrete Carrier System Element	Effective Section Stiffness The multiplier	
	<i>Axial</i>	<i>Slip</i>
<i>Curtain – Floor (In-Plane)</i>		
Curtain	0.50	0.50
Basement curtain	0.80	0.50
Floor	0.25	0.25
<i>Curtain – Floor (Out of Plane)</i>	<i>Bending</i>	<i>Cutting</i>
Curtain	0.25	1.00
Basement curtain	0.50	1.00
Floor	0.25	1.00
<i>Rod element</i>	<i>Bending</i>	<i>Cutting</i>
Tie beam	0.15	1.00
Frame Beam	0.35	1.00
Frame Column	0.70	1.00
Curtain (equivalent rod)	0.50	0.50

Table 4.3. Moving Load Mass Participation Coefficient

Purpose of Use of the Building	n
Warehouse, warehouse, etc.	0.80
School, dormitory, sports facility, cinema, theater, concert hall, place of worship, restaurant, store, etc.	0.60
Residence, workplace, hotel, hospital, parking lot, etc.	0.30

4.5.9.3 –Floor coverings **4.5.6.4** According to within their own planes *rigid diaphragm* If they are modeled as *floor masses*, at the floor mass center *main node* In-plane three *independent rigid motion degrees of freedom* Independent degrees of freedom are usually chosen as two horizontal translation degrees of freedom and a rotational degree of freedom about the vertical axis passing through the main node. In the calculation of floor masses, **Equation (4.16)** The masses corresponding to the vertical degrees of freedom in the floor will be taken into account. **4.5.9.2** It will be defined as in .

4.5.10. Modeling the Effect of Additional Eccentricity

4.5.10.1 –To take into account possible uncertainties in the effects of earthquake ground motion on the building and the rigidity and mass distribution of the load-bearing system. *additional eccentricity effect* has been defined.

4.5.10.2 –Floor coverings **4.5.6.4** If they are modeled as rigid diaphragms in their own planes,

(a) **4.5.9.3** According to the floor mass center (*main node*) defined *floor mass* Earthquake calculations will be made in the direction of each earthquake based on the

(b) Floor mass center (*main node*) horizontal earthquake loads acting on the building will be shifted by +5% and –5% of the floor size in the direction perpendicular to the earthquake direction taken into account, and an earthquake calculation will also be made for these cases.

(c) Earthquake calculation **4.7** In case of doing it according to , instead of shifting the earthquake load for ease of modelling, it is shifted to the floor mass centre (*main node*) affecting equivalent earthquake load F_{xi} together with **Equation (4.17)** given with *additional story torsional moment* it is appropriate to take into consideration.

$$M_{ix} = F_{xi} \cdot \Delta x \quad (4.17)$$

Here Δx indicates an additional eccentricity of 5%.

(d) Earthquake calculation **4.8** According to , in case of using modal methods, instead of shifting the earthquake load for ease of modelling, it is shifted to the floor mass centre (*main node*) defined *floor mass* together with, *floor mass moment of inertia* m_{i0} to **Equation (4.18)** given by Δm_{i0} It is appropriate to add the increment.

$$\Delta m_{i0} = m_{i0} \cdot \Delta x \quad (4.18)$$

4.5.10.3 –Earthquake calculation **4.7** or **4.8.2** According to , in case of a one-way earthquake, additional eccentricity is taken into account for each direction. **4.8.3** According to , in case the construction is carried out under the effect of a two-direction earthquake acting simultaneously, the eccentricities for each direction will be applied separately.

4.5.10.4 –Floor coverings **4.5.6.2** Two-dimensional, including the degrees of freedom for displacements within their own planes according to *plate (membrane)* in case of modeling with finite elements,

(a) With this model, earthquake calculations will be made without the effect of eccentricity, and internal forces and displacements will be obtained in the slabs and in the carrier system elements outside the slabs. The sizes obtained for the slabs will be taken into account in the slab design.

(b) In order to take into account the eccentricity effect, the in-plane finite element degrees of freedom *rigid diaphragm* will be assumed and 4.5.10.2 The floor mass centers will be shifted as defined in . A second earthquake calculation based on rigid diaphragm modeling will be performed to determine the effect of the additional eccentricity on the structural system elements other than the slabs and beams.

(c) Internal forces and displacements as a basis for design for load-bearing system elements other than slabs and beams (a) And (b) The envelope of the ones obtained in will be determined as (unfavorable ones).

4.6. SELECTION OF LINEAR CALCULATION METHOD

4.6.1. Linear Calculation Methods

Design by Strength linear calculation methods to be used within the scope, details 4.7 Explained in *Equivalent Earthquake Load Method* with details 4.8 Explained in *Modal Calculation Methods* is.

4.6.2. Selection of Calculation Method

4.6.2.1 –Details 4.8 Explained in *Modal Calculation Methods* Any of (*Mod Merge Method* or *Mod Collection Method*) can be used in the earthquake calculation of all buildings within the scope of this Section.

4.6.2.2 –Details 4.7 Explained in *Equivalent Earthquake Load Method* Buildings where can be applied Table 4.4 is given in.

Table 4.4. Buildings to which the Equivalent Seismic Load Method can be applied

Building Type	Permitted Building Height Class	
	DTS = 1, 1a, 2, 2a	DTS = 3, 3a, 4, 4a
Torsional irregularity coefficient at each storey $\eta_b \leq$ It meets the 2.0 condition and also B2 buildings without type irregularity	BYS $\geq$ 4	BYS $\geq$ 5
All other buildings	BYS $\geq$ 5	BYS $\geq$ 6

4.7. LINEAR EARTHQUAKE CALCULATION WITH EQUIVALENT EARTHQUAKE LOAD METHOD

Equivalent Earthquake Load Method, will be applied separately for earthquakes affecting the building in the (X) and (Y) earthquake directions perpendicular to each other. The following relations are given for the (X) earthquake direction. For the definitions of building base and building height in buildings with and without basements 3.3.1 will be taken as basis.

4.7.1. Determination of Total Equivalent Earthquake Load

4.7.1.1 –In the earthquake direction (X) considered, the effect on the entire building *total equivalent earthquake load* (base shear force), $V_{(x)}$, Equation (4.19) will be determined by.

$$V_{(x)} = m S_{t \text{ ar}} (T_{(x)}) \geq 0.04 m_t W O R K_{D s} g \quad (4.19)$$

Here $S_{aR}(T_g)$ in the earthquake direction considered (X) **4.7.3** The building calculated according to the dominant natural vibration period T_g Taking into account **Equation (4.8)** Calculated from *Reduced Design Spectral Acceleration* It shows the S_{ds} whereas *short period* for **2.3.2.2** defined in *design spectral acceleration coefficient* is.

4.7.1.2 – Equation (4.19) in m_t the building **Equation (4.20)** corresponds to the total mass calculated by:

$$m_t = \sum_{i=1}^N m_i \quad (4.20)$$

Here m_i is the total mass of the i'th floor slab.

4.7.2. Determination of Equivalent Earthquake Loads Acting on Floors

4.7.2.1 – Equation (4.19) The total equivalent earthquake load calculated by is the sum of the equivalent earthquake loads acting on the building floors. **Equation (4.21)** is expressed by:

$$V_{tE}^{(X)} = \Delta F_{tE}^{(X)} + \sum_{i=1}^N F_{iE}^{(X)} \quad (4.21)$$

4.7.2.2 – The building N affecting the t h floor (top) *additional equivalent earthquake load* $\Delta F_{tE}^{(X)}$ The value of **Equation (4.22)** will be determined by.

$$\Delta F_{tE}^{(X)} = 0.0075 N V_{tE}^{(X)} \quad (4.22)$$

4.7.2.3 – Total equivalent earthquake load $\Delta F_{tE}^{(X)}$ the rest of the part, N including the t h floor including building floors **Equation (4.23)** will be distributed with (**Figure 4.2a**).

$$F_{iE}^{(X)} = (V_{tE}^{(X)} - \Delta F_{tE}^{(X)}) \frac{m_i H_i}{\sum_{j=1}^N m_j H_j} \quad (4.23)$$

4.7.2.4 – Floor coverings **4.5.6.4** According to *rigid diaphragm* If modeled as **Equation (4.23)** calculated with $F_{iE}^{(X)}$ Equivalent earthquake load, i'th floor *main node* na will be affected in line with the earthquake taken into account.

4.7.2.5 – Floor coverings **4.5.6.2** According to *plate (membrane)* is the equivalent earthquake load acting on the j th node at the i 'th floor in the case of modeling with finite elements. **Equation (4.24)** will be calculated with:

$$f_{jE}^{(Q)} = \frac{F_{iE}^{(X)}}{m_i} m_j^{(S)} \quad (4.24)$$

Here $m_j^{(S)}$, j th node **Equation (4.16)** is the singular mass defined by .

4.7.2.6 – Total overturning moment at the base of the building due to earthquake loads **Equation (4.25)** is calculated with:

$$M_{\text{te}} = \sum_{i=1}^N F_{iE}^{(X)} H_i \quad (4.25)$$

4.7.3. Determination of the Dominant Natural Vibration Period of the Building

4.7.3.1 – Equivalent Earthquake Load Method In all buildings where 's is applied **Equation (4.19)** The dominant natural vibration period of the building in the earthquake direction (X) located in expressive $T_{p(X)}$, unless a more precise calculation is made, **Equation (4.26)** will be calculated with .

$$T_{p(X)} = 2\pi \frac{\sum_{i=1}^N m_i d_{\text{phi}}^{(X)2}}{\sum_{i=1}^N F_{(X)} d_{\text{phi}}^{(X)}} \quad (4.26)$$

Here, the fictitious load acting on the ith floor is $F_{\text{phi}}^{(X)}$, **Equation (4.23)** in $(V_{\text{te}}^{(X)} - \Delta F_{\text{WHAT}}^{(X)})$ in its place any value (for example 100) will be obtained by placing.

4.7.3.2 – The building **Equation (4.26)** The dominant natural vibration period calculated by $T_{p(X)}$'s earthquake the largest value to be taken into account in the calculation, **4.7.3.4** Given in T_{pA} shall not be more than 1.4 times the period.

4.7.3.3 – DTS = 1, 1a, 2, 2a and DTS ≥ The dominant natural vibration period in buildings with 6 and all buildings with DTS = 3, 3a, 4, 4a is, **4.7.3.1** directly, without calculation from **4.7.3.4** Given in empirical T_{pA} can be taken as the period ($T_{p(X)} \cong T_{pA}$).

4.7.3.4 – Empirical dominant natural vibration period **Equation (4.27)** will be calculated with:

$$T_{pA} = C_t H_N^{3/4} \quad (4.27)$$

(a) In buildings whose carrier system consists only of reinforced concrete frames $C_t = 0.1$, in buildings consisting of steel frames or braced steel frames $C_t = 0.08$ in all other buildings $C_t = 0.07$ will be taken.

(b) In buildings where all earthquake effects are covered by reinforced concrete walls. C_t coefficient **Equation (4.28a)** will be calculated with:

$$C_t = \frac{0.1}{\sqrt{A_t}} \leq 0.07 \quad (4.28a)$$

In this relation A_t equivalent area **Equation (4.28b)** is given in:

$$A_t = \sum_j A_{wj} - 0.2 + \frac{w_j}{H_N} \leq \sum_j A_{wj} \quad (4.28b)$$

4.7.4. Torsion Calculation in Equivalent Earthquake Load Method

On any i'th floor of the building **Table 3.6** defined in **A1** In case of irregularity of type 1.2 $\eta_{bi} \leq 2.0$, **4.5.10.2** According to this floor applied $\pm 5\%$ additional eccentricity for both earthquake directions **Equation (4.29)** Given in D_{bi} will be enlarged by multiplying by the coefficient.

$$D_{bi} = \frac{-\eta_{bi}^2}{-1.2} \quad (4.29)$$

4.7.5. Calculation of Buildings with Basements Using the Equivalent Earthquake Load Method

3.3.1 According to the definition given in , in buildings with basements surrounded by rigid external walls, the upper part of the building and the lower part with the basement are together. *a single common carrier system* In the earthquake calculation of such buildings, one of the two methods given below can be used:

(a) **4.3.6.1** The calculation method explained in ,

(b) **4.7.5.1, 4.7.5.2** And **4.7.5.3** Explained in *two loading states* calculation method (**Figure 4.2**).

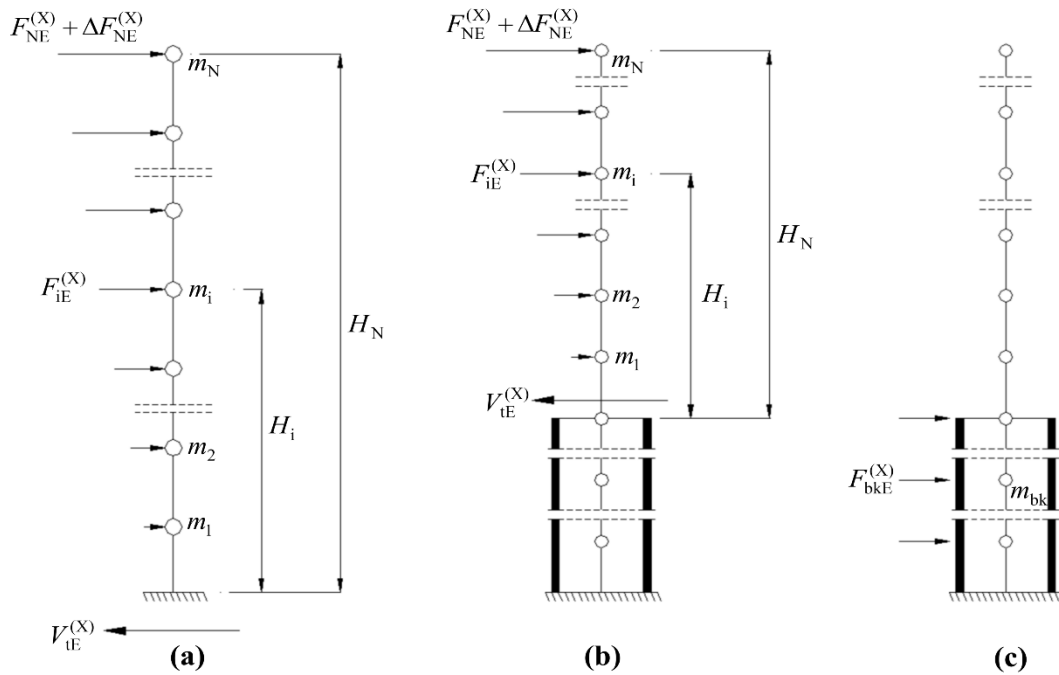


Figure 4.2

4.7.5.1 –In buildings with basements, in terms of horizontal rigidity *upper section* with relatively very rigid *subsection* (basement floors) have very different characteristics in terms of dynamic behavior and strength. Approximate earthquake calculations that can be applied to such buildings with modal calculation methods *two-load-case calculation approach* In , the upper part of the building and the lower part with the basement are modeled together as a single carrier system, but *upper section* with *subsection* Since the vibrations of are in very different modes, the earthquake calculation is done separately for two loading conditions:

4.7.5.2 –In case of first installation *common single carrier system* in the model **4.7.2.3** or **4.7.2.5** Equivalent earthquake loads calculated according to only *upper section* is affected by (**Figure 4.2b**). The top of the account for the section **Table (4.1)** Selected from R_{top} And D_{top} coefficients and earthquake direction $T_{(x)}$ p according to the dominant vibration period **Equation (4.1)** Calculated from *earthquake load reduction coefficient* $(R_a)_{top}$ will be used. As a result of the calculation for the first loading case, both *top section* de, and also *subsection* also *reduced internal forces* is obtained.

4.7.5.3 –In the case of the second load, again *common single carrier system* in the model only *subsection* The masses of the basement floors in , **Equation (4.8)** also $T=0$ is obtained by putting the reduced spectral acceleration S_{BR} Approximate equivalent earthquake loads acting on these floors by multiplying by (0) is calculated (**Figure 4.2c**). In the account *subsection* (for basement) **Equation (4.1)** Calculated from *earthquake load reduction coefficient* $(R_a)_{lower}=D_{lower}= 1.5$ will be used. For the second loading case, as a result of the calculation, *subsection* in *reduced internal forces* is obtained.

4.7.5.4 –In buildings with basements *internal forces underlying the design* **4.10.1** It is defined in .

4.8. LINEAR EARTHQUAKE CALCULATION WITH MODAL CALCULATION METHODS

4.8.1. Modal Calculation Methods

4.8.1.1 –Based on the modal behavior of the carrier system under the effect of an earthquake *Modal Calculation Methods*, **4.8.2** Based on the earthquake spectrum given in *Mod Merge Method* And **4.8.3** Based on the time domain given in *Mod Collection Method*. Detailed explanations for these methods **ANNEX 4B** For the definitions of building base and building height in buildings with and without basements. **3.3.1** will be taken as basis.

4.8.1.2 –In modal calculation methods, the following must be taken into account: *sufficient number of vibration modes*, Y_M ,

(a) **ANNEX 4B** According to (X) and (Y) earthquake directions calculated for each mode *base shear force modal effective masses* It will be determined according to the rule that the sum of is not less than 95% of the total mass of the building.

$$\sum_{n=1}^{Y_M} m_{(X)n} \geq 0.95 m_t \quad ; \quad \sum_{n=1}^{Y_M} m_{(Y)n} \geq 0.95 m_t \quad (4.30)$$

However, all modes whose contribution is greater than 3% will be taken into account.

(b) The largest of the Y_M s calculated for both directions will be taken into account in the three-dimensional calculation.

4.8.2. Earthquake Calculation with Mode Combination Method

4.8.2.1 –*Mod Merge Method* In the direction of a given earthquake *earthquake design spectrum* The maximum values of the response magnitudes in each vibration mode considered are calculated by the modal calculation method. The maximum values of the response magnitudes calculated for sufficient vibration modes, but not simultaneously, are calculated by using *largest modal behavior magnitudes* then statistically combined *largest behavioral magnitudes* Approximate values of are obtained. Details of the method **ANNEX 4B** is given in .

4.8.3. Earthquake Calculation with Mode Summation Method in Time History

4.8.3.1 – Mod Collection Method In case it is considered that the earthquake acts simultaneously in two horizontal directions perpendicular to each other, the modal response magnitudes of each vibration mode are *modal calculation method in time history* calculated for sufficient vibration mode. *Simultaneous modal behavior magnitudes* then directly collected in the time domain *Change in behavioral magnitudes over time* and to be taken as a basis in design *The largest value* is obtained. Details of the method **ANNEX 4B** is given in .

4.8.3.2 – Mod Collection Method In:

(a) Since the mode contributions are directly collected in the time domain, there is no need to apply statistical mode combination rules.

(b) Since horizontal ground motion components perpendicular to each other can be taken into account simultaneously,

4.4.2 Approximately as defined in *directional coupling* there is no need to apply the rules.

4.8.4. Magnification of Reduced Internal Forces and Displacements According to Equivalent Base Shear Force

4.8.4.1 – For any (X) earthquake direction $V_{(x)} < V_{tx}$ $\frac{V_{(x)}}{V_{tx}}$ In case of occurrence, **4.8.2** or

4.8.3 All reduced internal force and displacement magnitudes obtained by the modal calculation method applied according to **Equation (4.31)** given with *equivalent base shear force magnification coefficient* β_{te} will be enlarged by multiplying by .

$$\beta_{te}^{(x)} = \frac{y_{to} V_{te}^{(x)}}{V_{tx}^{(x)}} \geq 1 \quad (4.31)$$

Here $V_{(x)} / V_{tx}$ According to the Equivalent Earthquake Load Method **Equation (4.21)** calculated with *total equivalent earthquake load* (base shear force), $V_{(x)} / V_{tx}$ whereas **4.8.2** or **4.8.3** According to *x direction* Obtained in *Total earthquake load* shows. y_{to} The multiplier will be taken as follows:

(a) **Table 3.6** defined in **A1, B2** or **B3** If at least one of the irregularities of type γ is present in the building $y_{to} = 0.90$ will be taken.

(b) **Table 3.6** If none of the irregularities defined in are present in the building $y_{to} = 0.80$ will be taken.

4.8.4.2 – 3.3.1 In buildings with basements surrounded by rigid external walls according to the definition given in *equivalent base shear force magnification factor*, only above the basement floors of the building *upper section* will be calculated for.

4.8.5. Calculation of Buildings with Basements Using Modal Calculation Methods

3.3.1 According to the definition given in , in buildings with basements surrounded by rigid shear walls from the outside, the upper part of the building and the lower part with the basement will be modeled together as a common carrier system. In the seismic calculation of such buildings, one of the two methods given below can be used:

(a) **4.3.6.2** The calculation method explained in ,

(b) **4.8.5.1, 4.8.5.2** And **4.8.5.3** Explained in *two loading states* calculation method.

4.8.5.1 – 4.7.5.1 Similar to the method explained in , the approximate method that can be applied for linear earthquake calculations of buildings with basements with modal calculation methods *two-load-case calculation approach* In , the upper part of the building and the lower part with the basement are modeled together as a single carrier system, but *upper section* with *subsection* Earthquake calculation due to vibrations in very distant modes *two loading conditions* are made separately as follows:

4.8.5.2 – In case of first loading of the account, *common single carrier system* in the model only *upper section* Modal calculation is made by considering the masses of . In this case, the sufficient number of vibration modes is only *upper section* Calculated based on the total mass of *effective mass participation rates* It will be determined according to. For the upper section in the account **Table (4.1)** Selected from R_{top} And D_{top} According to the coefficients of each m for 'th vibration mode **Equation (4.1)** Calculated from *earthquake load reduction coefficient* (R_a)_{m, upper} will be used. In the first loading case, both *upper section* de, both also *subsection* also *reduced internal forces* is obtained.

4.8.5.3 – In case of second loading of the account, *common single carrier system* in the model only *subsection* The analysis is performed by considering the masses of . In this case, the sufficient number of vibration modes is only *subsection* Calculated based on the total mass of *effective mass participation rates* According to will be determined. In the account *subsection* (for (basement) $R_{lower/I}=2.5$ and $D_{lower}=1.5$ for each n for 'th vibration mode **Equation (4.1)** Calculated from *earthquake load reduction coefficient* (R_a)_{n, bottom} will be used.

4.8.5.4 – In buildings with basements *internal forces underlying the design* **4.10.1** It is defined in .

4.9. LIMITATION OF RELATIVE STOREY DIVERSATIONS, SECOND ORDER EFFECTS AND EARTHQUAKE JOINTS

4.9.1. Calculation and Limitation of Effective Relative Story Drifts

4.9.1.1 – (X) for any column or wall in the earthquake direction, two consecutive floors expressing the displacement difference between *reduced relative storey drift*, $\Delta_i^{(X)}$, **Equation (4.32)** will be obtained with .

$$\Delta_i^{(X)} = u_i^{(X)} - u_{i-1}^{(X)} \quad (4.32)$$

Equation (4.32) also $u_i^{(X)}$ And $u_{i-1}^{(X)}$, the i th and $(i-1)$ th building for the typical (X) earthquake direction at the ends of any column or curtain on the floors *reduced earthquake loads* shows the horizontal displacements calculated according to . However, in this calculation **4.7.3.2** The condition given in and also **Equation (4.19)** The minimum equivalent earthquake load condition defined in will not be taken into account.

4.9.1.2 – For typical (X) earthquake direction, for columns or walls on the i -th floor of the building *effective relative storey drift*, $\delta_i^{(X)}$, **Equation (4.33)** will be obtained with .

$$\delta_i^{(X)} = \frac{R}{I} \Delta_i^{(X)} \quad (4.33)$$

4.9.1.3 –For each earthquake direction, on the columns or walls of any i'th floor of the building, **Equation (4.33)** calculated with $\delta_{(X)}$ The maximum effective relative storey drift within the storey large value $\delta_{(X)\max}$, below **(a)** or **(b)** will meet the conditions given in .

(a) In case of infill walls and facade elements made of brittle material, with or without voids, completely adjacent to the frame elements, without any flexible joints or connections between them:

$$\lambda_{i,\max} \leq \frac{\delta_{(X)}}{h_i} \leq 0.008 \kappa \quad (4.34a)$$

(b) In case of infill walls made of brittle material and frame elements having flexible joints between them, facade elements being connected to external frames with flexible connections or infill wall elements being independent from the frame:

$$\lambda_{i,\max} \leq \frac{\delta_{(X)}}{h_i} \leq 0.016 \kappa \quad (4.34b)$$

However, in this case, the in-plane horizontal drift capacities of jointed infill wall elements, flexible infill wall elements and flexible connected facade elements **Equation (4.34b)** The limit value given in is met **To 1.4** It will be documented based on testing according to. An example of a flexible joint application for infill walls **ANNEX 4C** is given in .

4.9.1.4 – Equation (4.34) The coefficient λ in is the dominant vibration period of the building in the earthquake direction considered. **2.2DD-3** earthquake ground motion defined in **2.3.4.1** Calculated according to *elastic design spectral acceleration's*, **DD-2** earthquake ground motion *elastic design spectral acceleration* 'what is the ratio. **Equation (4.34)** Located in **K** The coefficient will be taken as $\kappa = 1$ for reinforced concrete buildings and $\kappa = 0.5$ for steel buildings.

4.9.1.5 –In single-storey buildings where all earthquake loads are carried by steel frames whose connections can transfer reversible moments, **Equation (4.34)** The limits defined by can be increased by a maximum of 50%.

4.9.1.6 – Equation (4.34) If the condition given in is not met on any floor of the building, the earthquake calculation will be repeated by increasing the rigidity of the carrier system.

4.9.2. Second Order Effects

4.9.2.1 –In the earthquake direction considered (X), on each i-th floor **Equation (4.35)** with *Second Rank Indicator Value* $\theta_{(X) II,i}$ will be calculated.

$$\theta_{(X) II,i} = \frac{(\Delta_{(X)})_{\text{average}} \sum_{k=1}^N w_k}{V_{(X)} h_i} \quad (4.35)$$

In this relation $(\Delta_{(X)})_{\text{average}}$, columns and walls on the i'th floor (X) in the earthquake direction as the average value of the calculated reduced relative storey drifts within the storey **4.9.1** It will be found according to .

4.9.2.2 –Calculated θ for all floors (X) II, The maximum value of $\theta_{(X)II,max}$ of Equation (4.36) in If the given condition is met, second order effects need not be taken into account in the calculation of the internal forces used in the design.

$$\theta_{(X)II,max} \leq 0.12 \frac{D}{C_h R} \quad (4.36)$$

In this situation/oca/Second order effects can be taken into account in the element design according to the applicable reinforced concrete and steel codes. Equation (4.36) in R And D, for the building carrier system

Table 4.1 Given in carrier system behavior coefficient And excess strength coefficient's, C_h whereas represents a coefficient defined depending on the nonlinear hysteretic behavior of the load-bearing system. In reinforced concrete buildings $C_h=0.5$ in buildings with steel and composite columns $C_h=1$ will be taken.

4.9.2.3 –Calculated θ for all floors (X) II, The maximum value of $\theta_{(X)II,max}$ of Equation (4.36) in If the given condition is not met, all internal forces for the earthquake direction (X) are given below. Equation (4.37) defined by second order magnification factor $\beta_{(X)II}$ with II will be increased by multiplication.

$$\beta_{(X)II} = 0.88 + \frac{C_h R}{D} \theta_{(X)II,max} \geq 1 \quad (4.37)$$

Another option that can be applied in this case is to renew the earthquake calculation by increasing the rigidity and/or strength of the carrier system appropriately.

4.9.2.4 –The above operations will also be performed for the earthquake direction (Y) perpendicular to (X). **3.3.1** In buildings with basements surrounded by rigid external walls according to the definition given in second order effects, above the basement floors of the building upper section will be taken into consideration for.

4.9.3. Earthquake Joints

Excluding the effects of foundation displacements and rotations due to different ground settlements and temperature changes, the conditions regarding the joint spaces to be left between building blocks or existing old buildings and newly constructed buildings only for earthquake effects are stated below:

4.9.3.1 – 4.9.3.2 Unless a less favorable result is obtained according to , the joint spaces shall not be less than the value obtained by multiplying the square root of the sum of the squares of the displacements obtained in the neighboring blocks or buildings for each floor by the coefficient α defined below. The floor displacements to be considered shall be at the nodes where the columns or walls are connected.

calculated reduced $u_{(X)}$ will be the average of the displacements within the floor. The current old

If it is not possible to make calculations for the building, the displacements of the old building will not be smaller than the values calculated for the new building on the same floors.

(a) If the floor slabs of neighboring buildings or building blocks are at the same level on all floors, $\alpha = 0.25 (R/I)$ will be taken.

(b) If the floor slabs of neighboring buildings or blocks of buildings are at different levels, even on some floors, $\alpha = 0.5$ for the entire building (R/I) will be taken.

4.9.3.2 –The minimum joint gap to be left will be at least 30 mm up to a height of 6 m, and at least 10 mm will be added to this value for every 3 m height after 6 m.

4.9.3.3 –The joints between the building blocks will be arranged in a way that will allow the blocks to operate independently of each other in all directions during an earthquake.

4.9.3.4 –In case two separate building blocks or two parts of a building that are different in terms of seismic behavior are connected to each other by a bridge or similar element, the displacement capacity of the movable support on one of the blocks connected by the said element in both earthquake directions and directions shall be at least $1.5(R/I)$ will be solid.

4.10. INTERNAL FORCES BASED ON DESIGN AND FORCES TRANSFERRED TO FOUNDATIONS

In Design by Strength **Equation (4.9)** in $TQ_{(H)}$ Against the horizontal earthquake effect shown by about to come *internal forces (strength demands) with forces transferred to the foundations* will be determined as follows.

4.10.1. Internal Forces Based on Design in Buildings with Basements

3.3.1 According to the definition given in , in buildings with basements surrounded by rigid external walls, the basements above the basement floors *upper section* and basement floors are located in *subsection* also *internal forces underlying the design* will be determined as follows:

4.10.1.1 –*Upper section* which is against the ductile behavior of the structural system elements in *internal forces underlying the design*;

(a) 4.3.6 In case of calculation using the method explained in , **4.3.6.1(a)** or **4.3.6.2** are the internal forces defined in .

(b) 4.7.5 or **4.8.5** In case of calculation with the methods explained in , in the first loading case *upper section* are the internal forces obtained in .

4.10.1.2 –*Upper section* Against the non-ductile behavior of the structural system elements in *internal forces underlying the design*;

(a) 4.3.6 In case of calculation using the method explained in , **4.3.6.1(b)** or **4.3.6.2** are the internal forces defined in .

(b) 4.7.5 or **4.8.5** In case of calculation with the methods explained in , in the first loading case *upper section* Internal forces obtained in D_{top} will be obtained by multiplying by .

4.10.1.3 –*Upper section* Corresponding to the ductile or non-ductile behavior of the structural system elements in *internal forces underlying the design*, in case of necessity **4.9.2** defined in *second degree magnification factor* ($\beta_{(x)} \geq 1$) and *Modal Calculation Methods* In case of use of , also **4.8.4** defined in *equivalent base shear force magnification factor* ($\beta_{(x)} \geq 1$) multiplied by will be enlarged.

4.10.1.4 –*Subsection* which is against the ductile behavior of the structural system elements in *internal forces underlying the design*;

(a) 4.3.6 In case of calculation using the method explained in **4.3.6.1(c)** or **4.3.6.2** are the internal forces defined in .

(b) 4.7.5 or 4.8.5 In case of calculations using the methods explained in , the internal forces obtained from the second loading case are 4.7.5.2 or 4.8.5.2 In the first loading case in *subsection* It is the sum of the internal forces obtained in .

4.10.1.5 – *Subsection* Against the non-ductile behavior of the structural system elements in *internal forces underlying the design*;

(a) 4.3.6 In case of calculation using the method explained in 4.3.6.1(d) or 4.3.6.2 are the internal forces defined in .

(b) 4.7.5 or 4.8.5 In case of calculations using the methods explained in , the internal forces obtained from the second loading case D_{lower} multiplied by 4.7.5.2 or 4.8.5.2 The first in in loading states *subsection* The internal forces obtained in $0.6D_{top}$ It will be obtained by adding the multiplication with .

4.10.2. Internal Forces Based on Design in Buildings Without Basements

3.3.1 According to the definition given in *buildings without basements*'s internal forces underlying the design, 4.10.1.1' without taking the basement into consideration only *upper section* are the internal forces defined for .

4.10.3. Forces Transferred to Foundations

Chapter 16 Within the scope of the design of the foundations with the bearing capacity approach, the forces to be transferred from the building to the foundation will be determined as follows.

4.10.3.1 – 3.3.1 According to the definition given in , in buildings without basements or in buildings with basements, where the critical wall height starts from the upper foundation level,

(a) The bending (overturning) moment transferred from the wall to the foundation is the bending moment at the wall base section. *upper section* Belonging to D_{op} will be obtained by multiplying the coefficient by . However, this bending moment, *high ductility level* The shear force transferred from the reinforced concrete wall to the foundation shall not be taken greater than the yield moment at the wall base section. 7.6.6.3 is the force defined according to .

(b) In such buildings, other internal force components of the shear walls and internal forces to be transferred to the foundation from other elements other than the shear walls, 4.10.1.1 According to the ductile design, the inner forces $0.6D_{top}$ It will be obtained by multiplying and magnifying by .

4.10.3.2 – 3.3.1 According to the definition given in , in buildings with basements, in cases where the critical shear wall height starts above the foundation top level, the internal forces to be transferred from all elements to the foundation, including the bending moments and shear forces transferred from the shear walls,

4.10.1.5 It will be calculated according to .

4.10.4. Structure – Pile – Soil Interaction for Pile Foundations

Designed according to strength for pile foundations *structure – pile – soil interaction* details of their accounts 16.10 and ANNEX 16C is given in .

INFORMATION ANNEX 4A – YIELD STRENGTH, DESIGN STRENGTH AND EARTHQUAKE LOAD COEFFICIENTS

4A.0. ICONS

D	= Excess Strength Coefficient
$f_d(\mu, T)$	= The ductility capacity of the carrier system depending on the period and the anticipated should be <i>design strength</i>
$f_{to}(T)$	= Calculated for the carrier system <i>linear (elastic) strength demand</i>
$f_y(\mu_k, T)$	= Predicted ductility capacity and period dependent <i>yield strength</i>
I	= Building Importance Coefficient
R	= <i>Carrier System Behavior Coefficient</i>
$R_a(T)$	= Predicted ductility capacity and period dependent <i>Earthquake Load Reduction Factor</i>
$R_y(\mu_k, T)$	= Predicted ductility capacity and period dependent <i>Yield Strength Reduction Coefficient</i>
T	= Natural vibration period [s]
T_B	= Horizontal elastic design acceleration spectrum corner period [s]
μ_k	= Ductility capacity anticipated for the carrier system

4A.1. YIELD STRENGTH AND YIELD STRENGTH REDUCTION COEFFICIENT

Design by Strength In the framework, for a modal single degree of freedom system *predicted ductility capacity – strength demand* relationship and the definition of *earthquake load coefficients* are given below (**Figure 4A.1**).

4A.1.1. Yield Strength

Design by Strength In his approach, *predicted ductility capacity* μ_k Depending on the carrier system, *yield strength* $f_y(\mu_k, T)$, **Equation (4A.1)** is defined by:

$$f_y(\mu_k, T) = \frac{f_{to}(T)}{R_y(\mu_k, T)} \quad (4A.1)$$

Here $f_{to}(T)$ calculated for the carrier system *linear (elastic) strength demand* n_i , T the natural vibration period of the system, $R_y(\mu_k, T)$ whereas **4A.1.2** defined in *Yield Strength Reduction Coefficient* R_y it symbolizes the.

4A.1.2. Yield Strength Reduction Coefficient

Yield Strength Reduction Coefficient $R_y(\mu_k, T)$, *equal displacement rule* in accordance with *not very rigid* for carrier systems *predicted ductility capacity* μ_k is taken equal to:

$$R_y(\mu_k, T) = \mu_k \quad T > T_B \quad (4A.2a)$$

High rigidity for carrier systems in this Regulation **Equation (4A.2b)** Based on:

$$R_y(\mu_k, T) = 1 + (\mu_k - 1) \frac{T}{T_B} \quad T \leq T_B \quad (4A.2b)$$

Here T_B , **Part 2'** also **Equation (2.3)** defined by *spectrum corner period* It shows the .

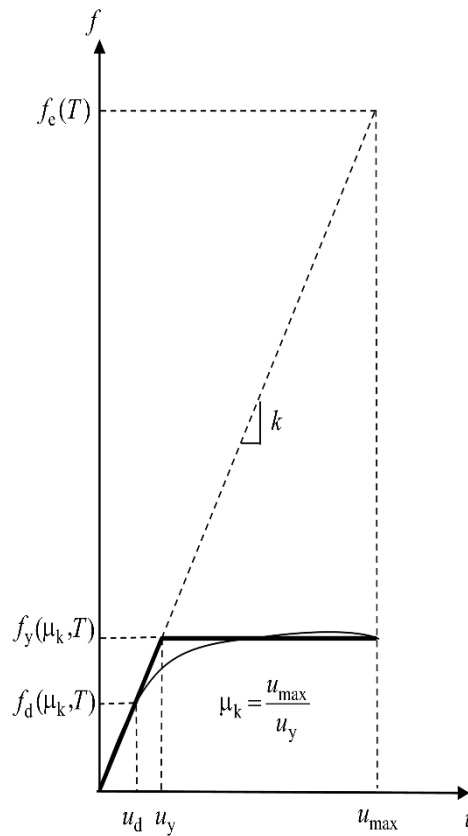


Figure 4A.1

4A.2. DESIGN STRENGTH AND EXCESS STRENGTH COEFFICIENT

4A.2.1. Design Strength

Design by Strength in, predicted ductility capacity' depending on, Section design with bearing capacity approach For the carrier system to have design strength

$f_d(\mu_k, \tau)$ **Equation (4A.3)** is defined by:

$$f_d(\mu_k, T) = \frac{f_y(\mu_k, T)}{D} \quad (4A.3)$$

4A.2.2. Excess Strength Coefficient

Equation (4A.3) *in D, Excess Strength Coefficient.* This coefficient expresses the excess of yield strength over design strength.

4A.3. CARRIER SYSTEM BEHAVIOR COEFFICIENT AND EARTHQUAKE LOAD REDUCTION COEFFICIENT

4A.3.1. Carrier System Behavior Coefficient

Carrier System Behavior Coefficient R , predicted ductility capacity μ_k to, **Equation (4A.3)** in defined Excess Strength Coefficient D and **Table 3.1** defined in *Building Importance Factor* Depending on **Equation (4A.4)** is defined by:

$$\frac{R}{I} = \mu_k D \quad (4A.4)$$

4A.3.2. Earthquake Load Reduction Coefficient

Design by Strength in carrying capacity approach for section design, specific parameters selected for each type of carrier system still opposing a ductility capacity Earthquake Load Reduction Coefficient $R_a(T)$ **Equation (4A.5a)** is defined by;

$$R_a(T) = \frac{f_d(T)}{f_d(\mu_k, T)} \quad (4A.5a)$$

or **Equation (4A.1)** And **Equation (4A.3)** By taking advantage of **Equation (4A.5b)** It can also be expressed as in:

$$R_a(T) = DR_y(\mu_k, T) \quad (4A.5b)$$

In conclusion **Equation (4A.2)**, **Equation (4A.4)** And **Equation (4A.5)** By taking advantage of *Earthquake Load Reduction Coefficient* $R_a(T)$ to be used in practice **Equation (4A.6)** is expressed by:

$$R_a(T) = \frac{R}{I} \quad T > T_B \quad (4A.6a)$$

$$R_a(T) = D + \frac{-R}{-I} - D \frac{T}{-T_B} \quad T \leq T_B \quad (4A.6b)$$

ANNEX 4B – MODAL CALCULATION METHODS

4B.0. ICONS

- $a_{nR}^{(X,Y)}(t)$ = nth earthquake ground motion components under the joint effect of (X) and (Y) at the same time
time-dependent reduced linearity of the vibration mode *modal pseudo-acceleration* [m/s²]
- $d_n^{((X,Y))}(t)$ = Time-dependent linear regression of the nth vibration mode under the joint effect of the (X) and (Y) earthquake ground motion components at the same time *modal displacement* [m]
- $d_n^{-(X,Y)}(t)$ = Time-dependent linear regression of the nth vibration mode under the joint effect of the (X) and (Y) earthquake ground motion components at the same time *modal speed* [m/s]
- $d_n^{-(X,Y)}(t)$ = Time dependent vibration of nth vibration mode under the joint effect of (X) and (Y) earthquake ground motion components at the same time *linear modal acceleration* [m/s²]
- $f_{ixn,max}^{(X)}$ = For the earthquake direction (X), the nth natural vibration mode of the building in the x-axis direction acting on the i'th floor *maximum modal earthquake load* [kN]
- $f_{ixn}^{(X,Y)}$ = At the same time, the nth earthquake ground motion components (X) and (Y) are under the joint effect. acting on the i'th floor in the x direction in vibration mode *modal earthquake load* Variation of with respect to time [kN]
- H_i = Above the basement floors of the building *upper section* in the upper part of the i'th floor
Height measured from base [m]
- $M_{oxn,max}^{(X)}$ = (X) for the earthquake direction, the building has the nth vibration mode in the x-axis direction
Belonging *maximum modal base overturning moment* [kNm]
- $M_{oxn}^{(X,Y)}(t)$ = nth earthquake ground motion components under the joint effect of (X) and (Y) at the same time in vibration mode, *modal base overturning moment* Variation of with respect to time [kNm]
- m_i = Total mass of the ith floor [t]
- $m_{i\theta}$ = Mass moment of inertia of the ith floor [tm²]
- $m_{ixn}^{(X)}$ = (X) is the nth natural vibration mode of the building in the x-axis direction for the earthquake direction. *i'th floor modal effective mass* [t]
- $m_{iyn}^{(X)}$ = (X) is the nth natural vibration mode of the building in the y-axis direction for the earthquake direction. *i'th floor modal effective mass* [t]
- $m_{i\theta n}^{(X)}$ = nth natural vibration around the z axis of the building for the (X) earthquake direction
mode *i-th floor modal effective mass moment of inertia* [tm²]
- $m_j^{(S)}$ = The singular mass [t] acting on a typical finite element node j
- $m_{txn}^{(X)}$ = (X) is the vibration mode of the building in the x-axis direction for the earthquake direction. *base shear force modal effective mass* [t]

$m_{\text{ty}}^{(Y)}$	= (Y) For the earthquake direction, the building's y-axis direction <i>base shear force modal effective mass</i> [[t]
$r_{\text{Max}}^{(X)}$	= (X) Any response magnitude (displacement, (relative story drift, internal force component) corresponding to the combined typical/ <i>largest modal behavior magnitude</i>
$r_n^{(X)}$	= No behavior for earthquake direction in nth natural vibration mode (X) Typical magnitude (displacement, relative story drift, internal force component) corresponding to <i>unit modal behavior magnitude</i>
$r_{n,\text{max}}^{(X)}$	= No behavior for earthquake direction in nth natural vibration mode (X) Typical magnitude (displacement, relative story drift, internal force component) corresponding to <i>largest modal behavior magnitude</i>
$r^{(X,Y)}(t)$	= Under the joint effect of earthquake ground motion components (X) and (Y) at the same time corresponding to any behavioral magnitude (displacement, relative story drift, internal force component) <i>typical behavioral magnitude</i> Change of with respect to time
$r_n^{(X,Y)}(t)$	= At the same time, the nth earthquake ground motion components (X) and (Y) are under the joint effect. typical response magnitude (displacement, relative story drift, internal force component) in vibration mode <i>modal behavior magnitude</i> Change of with respect to time
$S_{aR}(T_n)$	= Reduced design spectral acceleration of the nth vibration mode [g]
T_n	= Natural vibration period of nth mode [s]
$u_g^{(X)}(t)$	= (X) earthquake The ground acceleration component defined in the direction of time change [m/s ²]
$u_g^{(Y)}(t)$	= (Y) the ground acceleration component defined in the earthquake direction with respect to time change [m/s ²]
$V_{\text{txn,max}}^{(X)}$	= (X) is the vibration mode of the building in the x-axis direction for the earthquake direction. <i>maximum modal base shear force</i> [[kN]
$V_{\text{txn}}^{(X,Y)}(t)$	= At the same time, under the joint effect of (X) and (Y) earthquake ground motion components in the nth vibration mode, <i>modal base shear force</i> Change of with respect to time
YM	= <i>Sufficient number of vibration modes</i>
β_m	= The ratio of the mth and nth natural vibration periods
$\Phi_{i(X)n}$	= The nth natural vibration mode shape amplitude in the earthquake direction at the i-th floor (X)
Φ_{ixn}	= The nth natural vibration mode shape amplitude in the x-axis direction on the i-th floor The
Φ_{Iyn}	= nth natural vibration mode shape amplitude in the y-axis direction on the i-th floor
$\Phi_{i\theta n}$	= The nth natural vibration mode shape amplitude as rotation around the z-axis on the i-th floor
$\Gamma_n^{(X)}$	= For the earthquake direction (X), the nth vibration mode <i>modal contribution factor</i>

- ξ_n = Modal damping ratio of the nth vibration mode
- ω_n = Natural vibration angular frequency of the nth vibration mode [rad/s]
- ρ_m = In the Perfect Quadratic Combination Rule, the mth and nth natural vibration modes are cross correlation coefficient

4B.1. MODAL CALCULATION PARAMETERS

4B.1.1 – 4B.1.4, 4B.1.5 and 4B.1.6 defined in *modal calculation parameters*, is calculated based on the earthquake direction taken into consideration and the information obtained from the free vibration calculation of the carrier system, independent of the earthquake data, and is given below: **4B.2** and **4B.3** Both described in *modal calculation method* are also the sizes used in .

4B.1.2 – Modal calculation parameters, is defined below only for the horizontal earthquake direction (X). The same parameters can be defined similarly for the earthquake direction (Y) perpendicular to (X).

4B.1.3 – In the definition of modal calculation parameters, the degrees of freedom of the carrier system are:

(a) In the case of modeling floor slabs as rigid diaphragms, the displacements defined in the x and y horizontal directions at the mass center of any ith floor slab and the rotation around the vertical axis passing through the floor mass center were taken into account and this freedom was calculated as follows: floor mass corresponding to degrees m_i with times mass moment of inertia $m_i \theta$ has been defined.

(b) Floor slabs are not taken as rigid diaphragms and **4.5.6.2** Two-dimensional, including the degrees of freedom for displacements within their own planes according to *plate (membrane)* in case of modeling with finite elements, m_i finite element instead of floor masses at the nodes $m_i(s)$ masses will be taken into account.

4B.1.4 – Modal Contribution Factor and Base Shear Force Modal Effective Mass. For the given earthquake direction (X), the nth vibration mode *modal contribution factor* $\Gamma_n^{(X)}$ with the x-axis of the building in line with *base shear force modal effective mass* $m_n^{(X)}$, **Equation (4B.1)** is defined by:

$$\Gamma_n^{(X)} = \frac{\sum_{i=1}^N m_i \Phi_{i(X)n}}{\sum_{i=1}^N (m_i \Phi_{ixn}^2 + m_i \Phi_{iyn}^2 + m_i \theta \Phi_{i\theta n}^2)} ; \quad m_n^{(X)} = \Gamma_n^{(X)} \sum_{i=1}^N m_i \Phi_{ixn} \quad (4B.1)$$

4B.1.5 – Floor Modal Effective Masses. For a typical nth vibration mode for the given earthquake direction (X), above **4B.1.3** of the degrees of freedom defined in *floor modal effective masses* **Equation (4B.2)** is defined by:

$$m_{ixn}^{(X)} = m_i \Phi_{ixn} \Gamma_n^{(X)} ; \quad m_{iyn}^{(X)} = m_i \Phi_{iyn} \Gamma_n^{(X)} ; \quad m_{i\theta n}^{(X)} = m_i \theta \Phi_{i\theta n} \Gamma_n^{(X)} \quad (4B.2)$$

4B.1.6 – Unit Modal Behavior Magnitude. For a given earthquake direction (X), any response magnitude (displacement, relative storey drift, internal force component) corresponding to the typical *unit modal behavior magnitude* $r_n^{(X)}$, **Equation (4B.2)** with defined *floor modal effective masses* It is obtained by a static calculation in which is applied as a load in its own direction.

4B.2. EARTHQUAKE CALCULATION WITH MODE COMBINATION METHOD

4B.2.1 –Mod Merge Method, is explained below for the (X) earthquake direction. A similar calculation will be made for the (Y) earthquake direction perpendicular to (X).

4B.2.2 –Maximum response magnitudes obtained separately for horizontal (X) and (Y) earthquake directions **4.4.2** According to *directional coupling* will be applied.

4B.2.3 –For a given earthquake direction (X), in a typical nth vibration mode, corresponding to any response magnitude (displacement, relative story drift, internal force component) *maximum modal behavior magnitude* $r_{n,max}^{(X)}$, **Equation (4B.3)** is calculated with:

$$r_{n,max}^{(X)} = r_n^{-(X)} S_{aR}(T_n) \quad (4B.3)$$

Here $r_n^{-(X)}$ is the typical nth natural vibration period T_n for **Equation (4.8)** Obtained from *reduced design spectral acceleration* It shows the .

4B.2.4 –Internal force components, displacement and relative story drift, etc. *to be applied separately for each*, for each vibration mode **4B.2.3** The largest non-simultaneous modal contributions calculated according to will be statistically combined as described below:

(a) The most general mode combining rule is *Perfect Square Combination (FQA) Rule* **Equation (4B.4)** is given in.

$$r_{Max}^{(X)} = \sqrt{\sum_{m=1}^{YM} \sum_{n=1}^{YM} r_{m,max}^{(X)} \rho_{mn} r_{n,max}^{(X)}} \quad (4B.4)$$

Here $r_{m,max}^{(X)}$ And $r_{n,max}^{(X)}$, for typical m-th and n-th vibration modes **4B.2.3** calculated with *largest modal behavior magnitudes* ρ_{mn} belongs to these modes *cross correlation coefficient* It shows the .

(b) Above **Equation (4B.4)** Located in *cross correlation coefficient* **Equation (4B.5a)** is given in:

$$\rho_{mn} = \frac{8 \sqrt{\xi_m \xi_n} \beta_m^{3/2} \beta_n^{3/2}}{(1 - \beta_m^2)^2 + 4 \xi_m^2 \beta_m^2 (1 + \beta_m^2) + 4 (\xi_m^2 + \xi_n^2) \beta_m^2 \beta_n^2} ; \quad \beta_m = \frac{T_n}{T_m} \quad (4B.5a)$$

Here β_m , is the ratio of the mth and nth natural vibration periods considered, ξ_m and ξ_n belong to the same modes and can be taken differently from each other. *modal damping ratios* It shows the .

(c) If the modal damping ratios are assumed to be the same in all modes, *cross correlation coefficient* **Equation (4B.5b)** can be simplified as given by:

$$\rho_{mn} = \frac{8 \xi_m^2 (1 + \beta_m^2) \beta_m^{3/2}}{(1 - \beta_m^2)^2 + 4 \xi_m^2 \beta_m^2 (1 + \beta_m^2)} \quad (\xi_m = \xi_n = \xi) \quad (4B.5b)$$

(d) β for all modes considered $\beta < 0.8$ is met,

Equation (4B.4) Instead of the combination rule given in **Equation (4B.6)** Given in *Square Root Sum of Squares (SRSS) Rule* available.

$$r_{\text{Max}}^{(X)} = \sqrt{\sum_{n=1}^{YM} (r_{n,\text{max}}^{(X)})^2} \quad (4B.6)$$

This combination rule, **Equation (4B.4)** in $\rho_m = 0$ ($m \neq n$) and $\rho_m = 1$ ($m = n$) corresponds to the special case.

4B.2.5 –For a typical n th vibration mode for a given earthquake direction (X), the load-bearing system moves along the x-axis. *maximum modal base shear force* $V_{(X)\text{txn,max}}$ and against it *maximum base overturning moment* $M_{(X)\text{oxn,max}}$ **Equation (4B.7)** is calculated with:

$$V_{(X)\text{txn,max}} = \sum_{i=1}^N f_{i\text{xn,max}}^{(X)} = m_{\text{txn}} S_{\text{aR}}^{(X)}(T_n) \quad ; \quad M_{(X)\text{oxn,max}} = \sum_{i=1}^N f_{i\text{xn,max}}^{(X)} H_i \quad (4B.7)$$

Combining the mode contributions of these quantities **4B.2.4** It will be done according to.

4B.3. EARTHQUAKE WITH MODE COLLECTION METHOD IN TIME DOMAIN ACCOUNT

4B.3.1 –(For a typical n th vibration mode for an earthquake simultaneously acting in (X) and (Y) directions, no response magnitude (displacement, relative storey drift, internal force components) corresponding to typical *modal behavior magnitude* Change of with respect to time $r_{(X,Y)}^{(n)}(t)$, **Equation (4B.8)** is calculated with:

$$r_n^{(X,Y)}(t) = \bar{r}_n^{(X)} a_{nR}^{(X,Y)}(t) \quad (4B.8)$$

Here $\bar{r}_n^{(X)}$, for the earthquake direction (X) selected as the calculation reference direction **4B.1.6** in typical defined *unit modal behavior magnitude* $a_{nR}^{(X,Y)}(t)$ is for the n th vibration mode below **Equation (4B.9)** reduced time-dependent defined in *modal pseudo-acceleration* It opposes.

$$a_{nR}^{(X,Y)}(t) = \omega_n^2 \frac{d_{(X,Y)}^{(n)}(t)}{a T_n} \quad ; \quad \omega_n = \frac{2\pi}{T_n} \quad (4B.9)$$

Equation (4B.9) at ω_n , is the natural angular frequency of the typical n th vibration mode, $d_{(X,Y)}^{(n)}(t)$ whereas **4B.3.2** calculated in *modal displacement* it shows the .

4B.3.2 – Eq.(4B.9) Located in *linear modal displacement*, $d_{(X,Y)}^{(n)}(t)$, at the same time (X) and (Y) belongs to the n th mode under the combined effect of earthquake ground motion components. *modal single degree of freedom system* of **Equation (4B.10)** It is obtained from the linear solution of the equation of motion in the time domain:

$$d_{n(X,Y)}^{(n)}(t) + 2\xi_n \omega_n d_{n(X,Y)}^{(n)}(t) + \omega_n^2 d_{n(X,Y)}^{(n)}(t) = -\ddot{u}_g^{(X)}(t) - \frac{\Gamma^{(Y)}}{\Gamma^{(X)}} \ddot{u}_g^{(Y)}(t) \quad (4B.10)$$

Here $\ddot{u}_g^{(X)}(t)$ And $\ddot{u}_g^{(Y)}(t)$ in the perpendicular (X) and (Y) earthquake directions **4B.3.5** According to defined ground accelerations $\ddot{u}_g^{(X,Y)}(t)$ And $d_{(X,Y)}^{(n)}(t)$ is the time dependent variable of the n th vibration mode. *linear modal velocity and acceleration*. The time increment to be taken as basis in the calculation T_n It will not be greater than /10.

4B.3.3 –(Changes in response magnitudes such as internal force components, displacement and relative story drift under the effect of the earthquake defined in the (X) and (Y) directions, with respect to time, $r_{(X,Y)}(t)$, for each vibration mode **4B.3.1** The simultaneous modal contributions calculated according to , $r_{(X,Y)}(t)$, will be obtained by direct addition:

$$r_{(X,Y)}(t) = \sum_{n=1}^M r_{(X,Y)}^n(t) \quad (4B.11)$$

4B.3.4 –(In a typical nth vibration mode under the effect of an earthquake defined in the (X) and (Y) directions, the load-bearing system moves in the x-axis direction. *modal base shear force* and against this incoming *base overturning moment* Changes of over time $V_{(X,Y)}(t)$ And $M_{(X,Y)}(t)$

Equation (4B.12) is calculated with:

$$V_{(X,Y)}(t) = \sum_{i=1}^N f_{ixn}^{(X,Y)}(t) = m_{txn}^{(X)} a_{nR}^{(X,Y)}(t) ; \quad M_{(X,Y)}(t) = \sum_{i=1}^N f_{ixn}^{(X,Y)}(t) H_i \quad (4B.12)$$

The time domain addition of mode contributions of these quantities is also **4B.3.3** It will be done according to.

4B.3.5 –Mode Addition Method in time history At least 11 earthquake ground motion sets will be used in linear calculations to be made with. Acceleration records in two horizontal directions perpendicular to each other will be simultaneously affected in the direction of the main axes (X) and (Y) of the carrier system. Then, the axes of the acceleration records will be 90°. The calculation will be repeated by rotating. Selection and scaling of earthquake records to be used in calculations **2.5** It will be done according to.

4B.3.6 –Behavior magnitudes are calculated from each of at least $2 \times 11 = 22$ calculations. **4B.3.3** It will be found as the average of the largest absolute values of the results to be obtained according to .

INFORMATION ANNEX 4C – FLEXIBLE CONNECTION FOR INFILL WALLS DETAIL EXAMPLE

4C.1 –In order to prevent damage to the infill wall under frequently repeated DD-3 and DD-4 earthquake ground movements, flexible joints are created between the brittle infill walls and the adjacent columns/walls. These joints should be filled with a flexible material that does not prevent the wall from changing shape.

4C.2 –An example detailing for flexible joints that can be applied for this purpose **Figure 4C.1**. The flexible joint is provided by a C-profile connected to the inner faces of the column/wall and the lower face of the upper beam/slab with an anchor along the height of the column/wall. This profile also prevents the out-of-plane movement of the wall during an earthquake. In the application of the detail, precautions regarding fire, heat, sound and water insulation should also be taken.

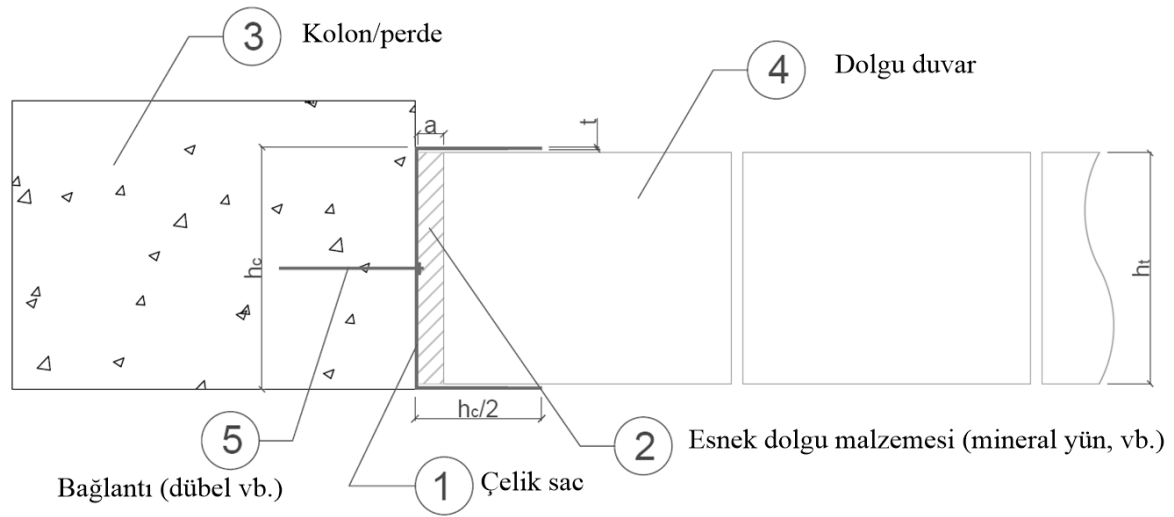


Figure 4C.1

CHAPTER 5 – DEFORMATION OF BUILDINGS UNDER THE EFFECT OF EARTHQUAKES ACCOUNTING PRINCIPLES FOR EVALUATION AND DESIGN

5.0. ICONS

$A_{s,h}$	= Transverse reinforcement area (rectangular section) [mm ²] =
A_{os}	Area of transverse reinforcement (circular section) [mm ²]
a_l	= Longitudinal reinforcement supported by a stirrup arm or crossbar horizontally distance between axes [mm]
b_{he}	= Section dimension between the axes of the stirrups encasing the core concrete [mm] = Core
b_k	dimension (distance between the axes of the outermost transverse reinforcement) [mm]
D	= Distance between the spiral/winding reinforcement axes [mm] =
DTS	<i>Earthquake Design Class</i>
DD-1	= Earthquake location with a 2% probability of being exceeded in 50 years (recurrence period 2475 years) movement level
DD-2	= Earthquake location with a 10% probability of being exceeded in 50 years (recurrence period 475 years) movement level
DD-3	= Earthquake location with a 50% probability of being exceeded in 50 years (recurrence period 72 years) movement level
DD-4	= Earthquake location with a 68% probability of being exceeded in 50 years (recurrence period 43 years) movement level
d_b	= Longitudinal reinforcement diameter (average in tension) [m]
TO_d	= Directional combination applied <i>design basis</i> Total earthquake effect
$TO_d^{(H)}$	= Directional combination applied <i>design basis</i> horizontal earthquake effect
$TO_d^{(Z)}$	= Vertical earthquake effect
$(EI)_{to}$	= Column, beam, beam or wall modeled according to stacked plastic behavior <i>effective section stiffness</i>
f_{ce}	= Average (expected) compressive strength of concrete [MPa]
f_{ck}	= Characteristic compressive strength of concrete [MPa]
f_{to}	= Calculated for the carrier system <i>linear (elastic) strength demand</i>
f_y	= Carrier system <i>yield strength</i>
f_{eat}	= Average (expected) yield strength of steel [MPa]
f_{yk}	= Characteristic yield strength of steel [MPa]
f_{YWE}	= Average (expected) yield strength of transverse reinforcement [MPa]
G	= Dead load effect
h	= Section height [m]
h_{he}	= Section dimension between the axes of the stirrups surrounding the core concrete [mm]

I	= Building Importance Factor
L_p	= Plastic joint length [m] =
L_s	Cutting clearance [m]
M_y	= Effective yield moment [kNm]
m_i	= Total mass of the i th floor [t]
$m_{i\theta}$	= mass moment of inertia of the i 'th floor [tm ²]
N	= Above the basement floors of the building <i>upper section</i> Total number of floors in =
n	Live load participation coefficient
R_y	= Yield Strength Reduction Coefficient
Q	= Live load effect
Q_{to}	= Effective live load effect =
S	Snow load effect
S	= Winding reinforcement spacing [m]
S_{DS}	= Design spectral acceleration coefficient defined for the short period region =
α_{if}	Winding reinforcement efficiency coefficient
$\epsilon_c^{(GO)}$	= <i>Prevention of Immigration</i> allowed confined concrete unit for performance level shortening limit
$\epsilon_c^{(KH)}$	= <i>Controlled Damage</i> Allowable confined concrete unit shrinkage for performance level limit
$\epsilon_c^{(SH)}$	= <i>Limited Damage</i> Allowable confined concrete unit shrinkage for performance level limit
$\epsilon_s^{(GO)}$	= <i>Prevention of Immigration</i> permissible reinforcing steel unit for performance level deformation limit
$\epsilon_s^{(KH)}$	= <i>Controlled Damage</i> permissible reinforcing steel unit for performance level deformation limit
$\epsilon_s^{(SH)}$	= <i>Limited Damage</i> permissible reinforcing steel unit for performance level deformation limit
ϵ_{This}	= Unit elongation of reinforcement corresponding to maximum strength
φ_y	= Yield curvature [m ⁻¹]
φ_u	= Pre-collapse curvature [m ⁻¹]
ω_{we}	= Effective winding reinforcement to mechanical reinforcement ratio
$\rho_{s,h}$	= Volumetric ratio of transverse reinforcement in the direction considered = The smaller of
$\rho_{sh,min}$	the volumetric ratio of transverse reinforcement in the two horizontal directions
θ_y	= For flow condition <i>displaced axis rotation</i> [[rad]

$\theta_p^{(KH)}$	= <i>Controlled Damage</i> allowable plastic rotation limit for performance level [rad]
$\theta_p^{(GO)}$	= <i>Prevention of Immigration</i> permissible plastic rotation for performance level limit [rad]
$\theta_p^{(SH)}$	= <i>Limited Damage</i> allowable plastic rotation limit for performance level [rad]

5.1. EVALUATION AND DESIGN APPROACH BASED ON SHAPE CHANGE

5.1.1. Definition

One of the two main approaches for the design of building load-bearing systems under earthquake effects. *Evaluation and Design Based on Shapeshifting* (In the SGGT approach:

- (a) Internal force-strain relations compatible with nonlinear modeling approaches are determined for existing or previously pre-designed load-bearing system elements.
- (b) Foreseen performance target(s) Under the earthquake ground motion(s) selected in accordance with the calculation of the load-bearing system, static or dynamic incremental methods in the time history are used, *nonlinear ductile* behavioral requests for change of form with *crispy* behavioral strength demands is obtained.
- (c) The obtained demands for shapeshifting and internal strength, defined as being in line with the anticipated performance target(s) deformation and strength capacities compared with.
- (d) For existing buildings, it can be demonstrated that the deformation and strength demands are below or exceed the corresponding deformation and strength capacities. *evaluation by shapeshifting* is completed.
- (to) If the deformation and strength demands for new or retrofitted existing buildings are below the corresponding deformation and strength capacities. *design by shapeshifting* Otherwise, the element sections are changed and the calculation is repeated and re-evaluated, and so on. *design by shapeshifting* is concluded.

5.1.2. Scope

5.1.2.1 – Explained in this section *Evaluation and Design Based on Shapeshifting*–SGGT calculation principles, **Chapter 13** Explained in *High-Rise Buildings's* design, **Chapter 14** Explained in *Earthquake Insulated Buildings's* design and **Chapter 15** Explained in *Assessment and Reinforcement of Existing Buildings* will be applied for.

5.1.2.2 – 5.1.3.2 In the buildings specified in, **Chapter 4** According to *Design by Strength*–After the preliminary design is made by applying the DGT calculation principles, also explained in this section *Evaluation and Design Based on Shapeshifting*–Performance evaluation will be made using SGGT calculation principles.

5.1.2.3 – Chapter 8 After the preliminary design of fully post-tensioned prefabricated buildings defined in was made with the DGT approach, **8.4.3** According to , performance evaluation will be made with the SGGT approach.

5.1.3. Performance Targets

5.1.3.1 – Evaluation and Design Based on Shapeshifting–Within the scope of ŞGDT **Chapter 13** Explained in *High-Rise Buildings* In the design of and **Chapter 15** Explained in *Assessment and Reinforcement of Existing Buildings* Performance targets to be taken as basis in **Table 3.4** In, **Chapter 14** Explained in *Earthquake Insulated Buildings* The performance targets to be taken as basis in the design of are: **Table 3.5** It is defined in .

5.1.3.2 – Table 3.4(a) According to *Earthquake Design Class* $DTS = 1a$, $DTS = 2a$ and *Building Height Class* In buildings with $BYS = 2$, $BYS = 3$;

(a) Under the influence of DD-2 earthquake ground motion $I=$ Taking 1.5 **Chapter 4A** design made with DGT accounting principles in *preliminary design* will be taken into consideration.

(b) The pre-designed building carrier system is now under the influence of the DD-1 earthquake ground motion. **Table 3.4(a)** in *Advanced Performance Target* defined as *Controlled Damage* (KH) performance target and also under DD-3 earthquake ground motion *Limited Damage* (SH) in accordance with this Section to achieve the performance target *Evaluation and Design Based on Shapeshifting* (ŞGDT) approach and the design will be improved if necessary.

5.2. DEFINITION OF EARTHQUAKE EFFECTS AND COMBINATION WITH OTHER EFFECTS

5.2.1. Earthquake Effect

5.2.1.1 – Evaluation and Design Based on Shapeshifting within the scope of the details **5.6** Explained in *Pushing Methods* The earthquake effect to be taken into account, **2.3.4** or **2.4.1** Given in *horizontal elastic design acceleration spectrum* or **2.5** Selected and scaled according to *earthquake ground movements* will be taken into consideration.

5.2.1.2 – Evaluation and Design Based on Shapeshifting within the scope of the details **5.7** Explained in *Nonlinear Calculation Method in Time History* Earthquake effect in, **2.5** Selected and scaled according to *earthquake ground movements* will be taken into consideration.

5.2.2. Combining Earthquake Effects with Other Effects

5.2.2.1 –The combination of the earthquake effect and the vertical load effect is taken as a basis for the evaluation of the load-bearing system elements. **Equation (5.1)** It is defined in:

$$G + Q + 0.2S + TQ_{(H)} + 0.3 TQ_{(Z)} \quad (5.1)$$

Here G the dead load effect, S snow load effect, $TQ_{(Z)}$ whereas **4.4.3** Vertical earthquake determined according to shows its effect. *Effective live load* effect, **Table 4.3** defined by *Moving Load Mass Participation Coefficient* n by using $Q_{to} = nQ$ will be calculated as Horizontal earthquake effect $TQ_{(H)}$ d

5.2.2.3 It is defined in .

5.2.2.2 –In this Section *nonlinear calculation methods* Before the earthquake calculation to be made with, **Equation (5.1)** also $TQ_{(H)}$ Static vertical loads outside the carrier system *incremental* as applied *nonlinear static calculation* will be made. The internal forces and deformations obtained from this calculation will be taken into account as initial values in the horizontal earthquake calculation. In newly constructed and strengthened buildings, nonlinear

deformations are not allowed. However, in the evaluation of existing buildings, non-linear deformations (if any) will also be taken into account as starting values.

5.2.2.3 – Equation (5.1) in $T_{O(H)}$ below (a) and (b) It is defined as follows:

(a) Nonlinear horizontal earthquake calculation 5.6 Given in *Pushing Method* to be done with in case of $T_{O(H)}$ Separately calculated effects in (X) and (Y) earthquake directions 4.4.2.1 to It opposes the horizontal earthquake effect obtained by combining according to.

(b) Nonlinear horizontal earthquake calculation 5.7 According to *in the time domain* In case of construction, earthquake components in horizontal (X) and (Y) directions perpendicular to each other 2.5 According to Since they are defined simultaneously, the combined horizontal earthquake effect $T_{O(H)}$ d, this account is obtained directly as a result.

5.3. NONLINEAR BEHAVIOR MODELS

5.3.1. Stacked Plastic Behavior Model

5.3.1.1 – *Frame (bar) finite elements* columns, beams and 4.5.3.8 As a nonlinear behavior model in reinforced concrete shear walls that meet the geometric condition given in *Stacked Plastic Behavior (Plastic Hinge) Model* available.

5.3.1.2 – *Stacked Plastic Behavior Model* It is assumed that plastic deformations occur in a uniformly distributed manner throughout finite-length regions where the internal forces reach their plastic capacity. *Plastic joint length* called *plastic deformation zone* The length of (L_p), section size in the working direction (h) will be taken equal to half of ($L_p - 0.5h$).

5.3.1.3 – The length of the plastic deformation zones of elements that undergo plastic deformation only under axial force shall be taken equal to the free length of the relevant element.

5.3.1.4 – Representing the bulk plastic deformation *plastic joint's*, theoretically 5.3.1.2 As stated in *plastic deformation zone* It should be placed in the middle of . However, in practical applications for beams and columns 5.4.2.3, for curtains 5.4.3.1 The approximate idealizations stated in are permissible.

5.3.1.5 – Reinforced concrete plastic joint sections *effective yield moments* The conditions for defining are given below (a), (b), (c) is given in:

(a) Material strengths 5.4.1.5 will be taken according to.

(b) In the calculation of effective yield moment, the compressive strain of concrete can be taken as 0.0035 and the strain of reinforcement steel as 0.01.

(c) Axial forces resulting from vertical loads will be taken into account in the calculation of effective yield moment.

5.3.1.6 – In bilinear internal force-plastic deformation relations of reinforced concrete and steel sections, the hardening effect (increase in plastic moment due to increase in plastic rotation) can be abandoned.

5.3.1.7 – 5.6 And 5.7 According to the time domain *nonlinear* Elasto-plastic standard for steel structural systems as a cyclic behavior model in earthquake calculation

cycle model, for reinforced concrete load-bearing systems *the previous maximum-oriented model*, models derived from it can be used to allow for online stiffness reduction.

5.3.2. Diffuse Plastic Behavior Models

5.3.2.1 –As a nonlinear behavior model for the columns, beams and reinforced concrete walls that form the load-bearing system of the building *Diffuse plastic behavior model* can also be used.

5.3.2.2 –Models of distributed plastic behavior include finite-length end regions (*plastic deformation zones*) or can be used to consider nonlinear deformations continuously (distributed) over the entire length of the element.

5.3.2.3 –In the section, the concrete or structural steel is modeled with sufficiently small cells, the steel reinforcement bars are modeled as single cells, and the nonlinear axial stress-strain relations can be cyclically considered in each cell. *section cell (fiber)* The model can be used in practice especially for nonlinear modeling of reinforced concrete walls. With this model, the cross-section of reinforced concrete walls with complex cross-sections (T, L, U or C shapes) can be divided into various regions in plan, and a two-dimensional structure can be created in each wall arm by considering finite length parts along the wall in vertical direction. *finite element mesh* can be created.

5.3.2.4 –Used for modeling the nonlinear behavior of reinforced concrete shear walls under bending and axial force effects. *section cell (fiber)* parallel to the model on the screen *shear deformations* also, **5.3.2.3** mentioned in *finite element mesh* connected in parallel to *linear* One *sliding model* can be approximately calculated with . The shear modulus to be used in this model is **Table 4.2** can be obtained from.

5.3.2.5 –Material strengths **5.4.1.5** will be taken according to.

5.3.2.6 – 5.7 According to the time domain *nonlinear* in earthquake calculation *concrete and steel fibers* Generally accepted models will be used as cyclical behavior models.

5.4. RULES FOR MODELLING THE SUPPORT SYSTEM FOR NONLINEAR CALCULATION

5.4.1. General Modeling Rules

5.4.1.1 –Building load-bearing systems are always *three dimensional* will be modeled as .

5.4.1.2 – *Two horizontal lines perpendicular to each other* The earthquake effect in the direction will always be taken into account.

5.4.1.3 –The linear damping ratio shall be taken as 5% unless otherwise stated.

5.4.1.4 –The deformed carrier system caused by axial forces *second order effects* will be taken into consideration.

5.4.1.5 –Below are the basic models for evaluation and design based on deformation: **(a)** And **(b)**
The material strengths given in will be taken as basis:

(a) Evaluation of existing buildings according to deformation of concrete and reinforcement steel
Chapter 15 defined in *Current strengths* will be taken as basis.

(b) Evaluation and design of new buildings according to deformation of concrete, reinforcement steel and structural steel. Table 5.1 defined in *expected (average) strengths* basis will be taken. In the table f_{ce} and f_{ck} average and characteristic compressive strengths of concrete, f_{eat} and f_{yk} shows the average and characteristic yield strengths of steel.

5.4.1.6 –In new reinforced concrete load-bearing systems with high ductility level to be evaluated in accordance with this Section, due to the preliminary design made according to capacity design principles and other ductile design rules, the effects of online strength reduction, shear damage, column-beam joint area damage, insufficient reinforcement overlap length and insufficient wrapping reinforcement may not be taken into account in the load-bearing system and cyclic behavior models.

5.4.1.7 –Modeling and calculation rules for evaluation and retrofit design of existing buildings Chapter 15 is given in.

Table 5.1. Expected (Average) Material Strengths

Concrete	$f_{ce} = 1.3 f_{ck}$
Reinforcement steel	$f_{eat} = 1.2 f_{yk}$
Structural steel (S235)	$f_{eat} = 1.5 f_{yk}$
Structural steel (S275)	$f_{eat} = 1.3 f_{yk}$
Structural steel (S355)	$f_{eat} = 1.1 f_{yk}$
Structural steel (S460)	$f_{eat} = 1.1 f_{yk}$

5.4.2. Modeling of Beams and Columns

5.4.2.1 –Beams and columns, *frame (bar) finite elements* Nonlinear behavior is modeled as finite element ends defined *plastic deformation zones* in 5.3.1 According to *clumped plastic behavior* with the model or 5.3.2 According to *diffuse plastic behavior* can be modeled with models. In practice, it is accumulated *plastic joint*. The use of the model can generally be considered sufficient.

5.4.2.2 –At the nodes where columns and beams meet, all six degrees of freedom will be considered. In the case of modeling slabs as rigid diaphragms, those degrees of freedom corresponding to rigid motion will be removed.

5.4.2.3 –In columns and beams *plastic joints*, can be placed just outside the column-beam joint area, in other words, at the ends of the clear openings of columns or beams. However, it should be taken into account that plastic hinges may also form in beam openings due to the effect of vertical loads.

5.4.2.4 –Plastic on the ends *joints* Reinforced concrete columns and beams will be modeled as linear elements along the length between . These elements *effective section stiffnesses* 5.4.5 It will be determined according to.

5.4.3. Modeling of Reinforced Concrete Shear Walls

5.4.3.1 –Earthquake calculation 5.6 In case of non-linear pushing methods, 4.5.3.8 As a nonlinear behavior model in reinforced concrete solid shear walls or in each of the shear wall parts forming the connecting beam (hollow) shear wall that provides the geometric condition given in *Stacked Plastic Behavior (Plastic Hinge) Model* available.

(a) In this situation *plastic joints*, can be placed at the lower end of the shear wall section on each floor. In case there are rigid perimeter shear walls in the basement floors of the buildings, the plastic sections of the shear walls continuing from these shear walls to the upper floors should be defined starting at least from the first basement floor floor.

(b) Plastic *joints* The curtains will be modeled as linear elements along the length between . These elements *effective section stiffnesses* 5.4.5 It will be determined according to.

(c) At the nodal points where the curtain parts forming the complex section (T, L, U or C shaped) are connected in plan with the beam and/or floor end elements at the floor levels. *dependant* degrees of freedom, *three-dimensional rigid body motion* to be defined at the section center of gravity in a way that satisfies the condition *main node* Six in *independent* degrees of freedom *kinematics* will be connected as .

5.4.3.2 –Earthquake calculation 5.7 According to , in case of nonlinear method in time history and **Chapter 13** In high-rise buildings within the scope of the plastic joint model will not be used for the gapless and connecting beam (gapped) curtain parts. For these elements **5.3.2** defined in *section cell (fiber)* model must be used.

5.4.3.3 –Nonlinear models of tie beams of tie beam (void) shear walls 5.4.2 According to *rod element* or with special modeling methods.

5.4.4. Modeling of Basement Walls and Floors

Except in special cases, nonlinear modeling of basement walls and building floors around the building is not required. In normal cases, for these elements **Table 4.2** Given in *effective section stiffnesses* Equivalent linear modeling will be done with **4.5.5, 4.5.6** And **4.5.7** All modeling rules given in will be followed.

5.4.5. Effective Section Stiffness of Reinforced Concrete Bearing System Elements

5.4.5.1 –In-plane and out-of-plane behavior of linearly modeled walls and slabs *effective section stiffness factors* 4.5.8 in **Table 4.2** will be taken from.

5.4.5.2 –Modelling of columns, beams, tie beams and shear walls according to stacked plastic behaviour *effective section stiffnesses* **Equation (5.2)** It will be determined according to.

$$(EI)_{\bar{\theta}_0} = \frac{M_y}{\theta_y} \frac{L_s}{3} \quad (5.2)$$

Here M_y and θ_y plastic joints at the ends of the rod element *effective yield moments* with *flow rotations* Shows the averages of . L_s is the cutting aperture (in the section moment/shear force ratio); it can be taken as approximately half the span for columns and beams, and half the distance from the base of each floor to the top of the shear wall for shear walls. **Equation (5.2)** The plastic hinge yield rotation θ located in **Equation (5.3)** will be calculated with:

$$\theta_y = \frac{\varphi}{3} \frac{L_s + 0.0015\eta}{1 + 1.5 \frac{h}{L_s}} + \frac{\varphi_y d_b t_b^2}{8 \sqrt{f_{ce}}} \quad (5.3)$$

Here φ_y in the plastic joint section *effective yield curvature* In beams and columns $\eta = 1$, and $\eta = 0.5$ will be taken for curtains. h is the section height. *Reinforcement for yield condition*

stripping in the last term expressing the rotation δ the average diameter of the reinforcement steel clamped at the support (node or foundation), f_{ce} And f_{eat} shows the average (expected) compressive strength of the concrete and the average yield strength of the reinforcement.

5.4.6. Modeling of Masses

Modeling of masses 4.5.9 It will be done according to.

5.4.7. Modeling the Effect of Additional Eccentricity

On any i 'th floor of the building Table 3.6 defined in A1 type irregularity and torsional irregularity coefficient η_{bi} In case of > 1.5 , 4.5.10.2 The additional defined in

The eccentricity effect will also be taken into account in this Section. On each floor *additional floor mass moment of inertia* 4.5.10.2(b) and will be calculated according to 4.5.10.3 will be taken into consideration.

5.5. SELECTION OF NON-LINEAR CALCULATION METHOD

5.5.1. Non-Linear Calculation Methods

Evaluation and Design Based on Shapeshifting Non-linear calculation methods to be used within the scope, details 5.6 Explained in *Pushing Methods* with 5.7 Explained in *Nonlinear Calculation Method in Time History*'s.

5.5.2. Selection of Calculation Method

5.5.2.1 –Details 5.6.2 to 5.6.5 Explained in *Single Mode Propulsion Methods*, Table 3.3 According to *Building Height Class* $BYS \geq 5$ and 5.6.2.2 It can be used for buildings that meet the conditions given in 5.6.6 defined in *Multimodal Push Methods* is $BYS \geq$ Can be used for all buildings with 2.

5.5.2.2 –Details 5.7 Explained in *Nonlinear Calculation Method in Time History*, can be used in earthquake calculations of all buildings. This method Chapter 13 According to *high rise buildings* (Table 3.3) Its use is mandatory for buildings with $BYS = 1$.

5.6. EARTHQUAKE CALCULATION WITH NONLINEAR PUSHING METHODS

5.6.1. Pushing Methods

5.6.1.1 –Within the scope of this Regulation, 5.6.2 to 5.6.5 Explained in *Single Mode Propulsion Methods* And 5.6.6 defined in *Multimodal Push Methods* can be used in nonlinear earthquake calculations.

5.6.1.2 –As in all nonlinear methods, in the initial step of the calculation (step 0), 5.2.2.2 Under non-earthquake loadings as stated in *nonlinear incremental static calculation* is done. The internal forces and deformations obtained from this calculation will be taken into account as initial values in the earthquake calculation. Non-linear deformations caused by vertical loads are not allowed in newly constructed buildings.

5.6.1.3 –The ductile behavior obtained as a result of the calculations made with the Push Methods is contrary to the *basis for evaluation* plastic deformations (e.g. plastic rotations) and non-ductile

Internal forces opposing the (brittle) behavior are compared with the permissible limit values for the selected performance level. *evaluation by shapeshifting* is completed.

5.6.2. Single Mode Propulsion Methods

5.6.2.1 – Single Mode Propulsion Methods, Chapter 4 The linear described in *Mod Merge Method* Nonlinearity of the single-mode implementation of *incrementals* is the equivalent.

5.6.2.2 – Single Mode Propulsion Methods In order for it to be implemented, the following (a) And (b) Both of the conditions given in must be met:

(a) Based on linear elastic behavior without considering additional eccentricity at any floor **Chapter 3, Table 3.5** Calculated according to *torsional irregularity coefficient's* $\eta_{bi} < 1.4$ condition must be met.

(b) The first (dominant) vibration mode calculated based on linear elastic behavior in the earthquake direction considered *base shear force effective mass* The ratio of to the total building mass (excluding the masses of basement floors surrounded by rigid shear walls) must be at least 0.70.

5.6.2.3 – Single Mode Propulsion Methods, in proportion to the dominant vibration mode shape in the earthquake direction considered. *earthquake displacement demand* Under the influence of seismic load increments applied monotonically step by step up to the limit, the loads occurring in the carrier system *displacement, plastic deformation (plastic rotation, elongation, etc.) and internal force increases* with these *cumulative (cumulative)* values are calculated. In the last step, the cumulative values corresponding to the earthquake demand are calculated, *dimensions based on deformation and evaluation* is obtained as .

5.6.2.4 – In This Section Single Mode Propulsion Methods, is explained for the case where rigid diaphragms are assumed for the floors at each floor and the degrees of freedom are defined as the horizontal displacement components in two perpendicular directions at the floor mass center and the rotation around the vertical axis. In case the degrees of freedom corresponding to the in-plane deformations in the floor slabs are taken into account, single-mode pushover methods can be adapted within the framework of the same principles.

5.6.3. Fixed Single Mode Thrust Method

Fixed Single Mode Thrust Method In the earthquake direction considered, the impact on the floors at each thrust step *earthquake load increments*, which is determined in the first step after the non-earthquake loading and is not changed during the pushover calculation. *fixed mode shape* They are defined proportionally to . As a result of the push calculation, the coordinates are *top displacement – base shear force* the one which... *thrust curve* Then, the coordinate transformation applied to this curve is used to obtain the coordinates *modal displacement – modal pseudo-acceleration* the one which... *modal capacity diagram* is obtained. In the final stage of the calculation, this diagram is used as a basis for calculating the modal displacement demand under the defined earthquake effect and the internal force and plastic deformation demands occurring in the carrier system accordingly. Details of the method **ANNEX 5B** is given in .

5.6.4. Variable Single Mode Thrust Method

Variable Single Mode Thrust Method In the earthquake direction considered, the impact on the floors *earthquake load increments* compatible with these *floor displacement increments*, non-earthquake

At each pushing step after the loading, the previously formed plastic joints are renewed by taking into account *free vibration calculation* Obtained from *variable mode shape* are defined proportionally to . In this method **5.6.3** As stated in *thrust curve* Without the need for drawing *modal capacity diagram* is obtained directly. The last stage of the calculation is **5.6.3** Details of the method are as stated in **ANNEX 5B** is given in .

5.6.5. Obtaining the Modal Displacement Demand of the Earthquake in Single Mode Pushover Methods

5.6.5.1 –The earthquake *modal displacement request*'s achievement, under the given earthquake effect *modal capacity diagram* represented by *maximum displacement of a modal single degree of freedom system* It goes against the account of.

5.6.5.2 –Modal displacement demand of the earthquake;

(a) In a modal single degree of freedom system *Nonlinear Spectral Displacement* can be obtained as .

(b) The modal single degree of freedom system can be obtained from the time history calculation under earthquake effect.

Both methods **ANNEX 5B** It is explained in .

5.6.6. Multimodal Push Methods

In the nonlinear earthquake calculation, below **5.6.6.1** And **5.6.6.2** Provided that the conditions defined in are met, the international practice literature includes *Multimodal Push Methods* can also be used.

5.6.6.1 –*Multi-Mode Pushing Method* According to the given design spectrum of *in a special case* using initial (elastic) stiffnesses *linear calculation* All internal forces and displacements obtained as a result of their application are based on the same design spectrum. **4.8.2** It will be shown in the calculation report that it is exactly the same as the sizes obtained according to .

5.6.6.2 –Used *Multi-Mode Pushing Method* In case the independent fixed modal load vectors defined for different vibration modes of the building are applied to the structure incrementally and separately, the resulting modal internal forces will not be statistically combined, **4B.2.4** According to , the combined modal element end displacements and yield rotations will be calculated in accordance with .

5.7. EARTHQUAKE CALCULATION WITH NONLINEAR CALCULATION METHOD IN TIME DOMAIN

5.7.1. Definition

Nonlinear calculation in time history; the differential equation set expressing the equations of motion of the carrier system under the influence of earthquake ground motion with time increments *step by step direct integration* During this process, the change of the system stiffness matrix with time due to nonlinear behavior is taken into account.

5.7.2. Selection and Scaling of Earthquake Records

5.7.2.1 –In the time domain At least eleven earthquake ground motion sets will be used in nonlinear calculations. Acceleration records in two horizontal directions perpendicular to each other will be simultaneously affected in the direction of the main axes (X) and (Y) of the carrier system. Then, the axes of the acceleration records will be rotated 90° and will be returned and the calculation will be repeated.

5.7.2.2 –Selection and scaling of earthquake records to be used in calculations It will be done according to.

5.7.3. Matters to be Considered in the Calculation

5.7.3.1 –Nonlinear modeling of the carrier system It will be done according to the principles given in .

5.7.3.2 – Apart from the energy consumption related to nonlinear behavior in the structural system elements, the dynamic energy loss in linear structural and non-structural elements will be represented by the viscous damping matrix in the equations of motion. The viscous damping matrix, *Rayleigh Proportional Damping Matrix* or *Modal Damping Matrix* will be created as .

5.7.3.3 – As in all nonlinear methods, in the initial step of the calculation, under non-earthquake loadings as stated in *nonlinear incremental static calculation* The internal forces and non-linear deformations (not allowed in newly constructed buildings) obtained from this calculation will be taken into account as initial values in the earthquake calculation.

5.7.3.4 – The time increment to be used in the calculation will be chosen to allow adequate representation of all vibration modes contributing to the dynamic behavior.

5.7.4. Deformation and Internal Force Demands as Basis for Evaluation

The deformation demands that form the basis of evaluation in elements with ductile behavior and the internal force demands that form the basis of evaluation in elements without ductile behavior are the analyses performed (at least $2 \times 11 = 22$ analyses) will be calculated as the average of the largest absolute values of the results obtained from each.

5.8. EVALUATION OF SHAPES AND INTERNAL FORCES

5.8.1. Allowable Deformation and Internal Force Limits for New Reinforced Concrete Building Elements

5.8.1.1 –Prevention of Immigration (GO) Performance Level To be used in the performance evaluation for **Chapter 13** Given in *High-Rise Buildings* given in this Section for new reinforced concrete building elements, including *diffuse plastic behavior* to the model

Concrete and reinforcement steel calculated according to *total unit deformation* $\epsilon_{(GO)}$ and $\epsilon_{s(GO)}$ Below are the permitted limits for (a) And (b) It is defined in:

(a) *Prevention of Immigration* Concrete unit shortening for performance level: In

rectangular section columns, beams and shear walls:

$$\epsilon_{(GO)} = 0.0035 + 0.04 \sqrt{\omega_{we}} \leq 0.018 \quad (5.4a)$$

In circular sections:

$$\varepsilon_c^{(GO)} = 0.0035 + 0.07 \sqrt{\omega_{we}} \leq 0.018 \quad (5.4b)$$

The first term in these relations corresponds to the unit shortening of the unconfined concrete (shell concrete). ω_{we} effective winding reinforcement to mechanical reinforcement ratio. It shows:

$$\omega_{we} = \alpha_{inf} \rho_{sh,min} \frac{f_{yWE}}{f_{ce}} \quad (5.4c)$$

Equation (5.4c) α_{inf} winding reinforcement efficiency coefficient, $\rho_{sh,min}$ the smaller of the volumetric transverse reinforcement ratios in two horizontal directions in a rectangular section, f_{yWE} average of transverse reinforcement shows the (expected) yield strength:

$$\alpha_{inf} = 1 - \frac{\sum a_{sh}}{6b_s h_{sh}} \leq 1 - \frac{s_{sh}}{2b_{sh}} \leq 1 - \frac{s_{sh}}{2h_{sh}} \quad ; \quad \rho_{s,h} = \frac{A_{s,h}}{b_s s} \quad (5.4d)$$

Equation (5.4d) $A_{s,h}$ and $\rho_{s,h}$ the area and volumetric ratio of transverse reinforcement in the direction considered, b_s core size in the vertical direction (distance between the outermost transverse reinforcement axes), s transverse reinforcement spacing, b_{sh} and h_{sh} confined concrete dimensions measured from the axes of the confining reinforcement, a_{sh} axes of longitudinal reinforcement supported by a stirrup arm or crossbar shows the distance between. *Effectiveness coefficient of circular winding reinforcement* **Equation (5.4e)** is given in:

$$\alpha_{inf} = 1 - \frac{s_{sh}}{2D_{sh}} \quad ; \quad \rho_{s,h} = \frac{2A_{os}}{D s} \quad (5.4e)$$

Here A_{os} area of spiral/winding reinforcement, s the spacing of transverse reinforcement or the pitch of the spiral, D_{sh} whereas is the distance between the axes of spiral/wound reinforcement. For circular stirrups, $n = 2$, for spiral reinforcement, $n = 1$.

(b) Prevention of Immigration Reinforcing steel strain for performance level:

$$\varepsilon_s^{(GO)} = 0.4 \varepsilon_{This} \quad (5.5)$$

Here ε_{This} It shows the unit elongation corresponding to the tensile strength (See **ANNEX 5A**).

5.8.1.2 – Migration Prevention (MI) Performance Level To be used in the performance evaluation for **Chapter 13** Given in *High-Rise Buildings* given in this Section for new reinforced concrete building elements, including *clumped plastic behavior* calculated according to the model *plastic rotations* The permissible limit for the axial force acting on the section and **ANNEX 5A** As a result of the curvature analysis to be made by taking into account the concrete and reinforcement steel models given in **Equation (5.6)** will be calculated with .

$$\theta_p^{(GO)} = \frac{2}{3} (\varphi_u - \varphi_y) L_p \leq 0.5 \frac{L_p}{L_s} + 4.5 \varphi_u d_b \quad (5.6)$$

Here φ_u , **5.8.1.1** With the concrete and reinforcement steel unit deformations given in **ANNEX 5A** in obtained from the analysis made by using the given concrete and reinforcement steel models and taking into account the axial force acting on the section. *Total curvature before collapse* It shows the . **Equation (5.6)** The last term in , *for the post-yield (before-failure) condition, the yield elongation penetration-related reinforcement pullout rotation* is against.

5.8.1.3 – Controlled Damage (KH) Performance Level For use in the performance evaluation of new reinforced concrete building elements, the allowable values for concrete and reinforcement steel calculated using the calculation methods given in this Section *total unit Transformations* $\epsilon_{(KH)}$ and $\epsilon_{(KH)}$ Plastic rotation with $\theta_{(KH)}$ borders, *Prevention of Immigration* for performance level **5.8.1.1** and **5.8.1.2** Depending on the values defined in **Equation (5.7)** It is defined in:

$$\epsilon_c^{(KH)} = 0.75 \epsilon_c^{(GO)} ; \quad \epsilon_{(KH)} = 0.75 \epsilon_{(GO)} \quad (5.7a)$$

$$\theta_{(KH)} = 0.75 \theta_{(GO)} \quad (5.7b)$$

5.8.1.4 – Limited Damage (SH) Performance Level The calculation methods given in this Section will be used in the performance evaluation of new reinforced concrete building elements.

Allowable concrete and reinforcement steel calculated with *total unit deformations* $\epsilon_{(SH)}$ and $\epsilon_{(SH)}$ s

Equation (5.8a) It is defined in:

$$\epsilon_{(SH)} = 0.0025 ; \quad \epsilon_{(SH)} = 0.0075 \quad (5.8a)$$

5.4.5.2 In the calculation made using the effective section stiffnesses defined in *SH performance level* Plastic joint formation will not be allowed in the carrier system for:

$$\theta_p^{(SH)} = 0 \quad (5.8b)$$

5.8.1.5 – The internal force demands calculated for the Prevention of Collapse performance condition with the calculation methods given in this Section for new reinforced concrete building elements, for the relevant elements. **Chapter 7** It will be shown that the internal force capacities are smaller than the internal force capacities defined in . However, in the calculation of the internal force capacities, the characteristic material strengths are used instead of **Table 5.1** Given in *average (expected) material strengths* Relevant rules for reinforced concrete high-rise buildings **Chapter 13** is given in.

5.8.1.6 – Allowable unit deformation and internal force limits in the evaluation of existing building elements **Chapter 15** It is defined in .

5.8.2. Allowable Deformation and Internal Force Limits for New Steel Building Elements

5.8.2.1 – Chapter 13 Given in *High-Rise Buildings* The permissible limits for deformations calculated using the calculation methods given in this Section in new steel building elements, including **ANNEX 5C** is given in .

5.8.2.2 – Internal force demands calculated with the calculation methods given in this Section for new steel building elements, for the relevant elements **Chapter 9** It will be shown that the internal force capacities are smaller than the internal force capacities defined in . However, in the calculation of the internal force capacities, the characteristic

instead of material strengths **Table 5.1** Given in *average (expected) material strengths* will be taken as basis.

5.8.3. Allowable Deformation and Internal Force Limits for Reinforced Concrete and Steel Piles

5.8.3.1 – 16.10 Nonlinear analysis performed under the influence of DD-1 earthquake ground motion with Method I or Method II according to *structure – pile – soil interaction* The limits allowed for plastic rotations obtained in reinforced concrete or steel piles as a result of the calculation are given below. **(a),(b),(c)** It is defined in:

(a) Plastic rotation limits in reinforced concrete piles **5.8.1.2** And **5.8.1.3** 35% of the limits given in will be taken.

(b) Plastic rotation limits in the reinforced concrete plug made for the connection of the steel pile to the foundation **5.8.1.2** And **5.8.1.3** 50% of the limits given in will be taken.

(c) Plastic rotation limits in steel piles **ANNEX 5C** 50% of the limits given in will be taken.

5.8.3.2 – 16.10 Nonlinear analysis performed under the influence of DD-1 earthquake ground motion with Method I or Method II according to *structure – pile – soil interaction* internal force demands obtained as a result of the calculation, **Chapter 7** And **Chapter 9** It will be shown that the internal force capacities defined for columns are smaller than the internal force capacities. However, instead of the characteristic material strengths, the internal force capacities are calculated. **Table 5.1** Given in *average (expected) material strengths* will be taken as basis.

5.9. FINALIZATION OF DESIGN ACCORDING TO SHAPE CHANGE

5.1.3.2 As stated in, **Chapter 4** Given in *Design by Strength* The dimensions and reinforcement of the vertical load-bearing elements (wall and column) and foundations determined in the preliminary design with the approach will not be reduced as a result of the non-linear calculation made according to this Section. If a reduction is desired, both designs will be repeated by changing the sections. For other load-bearing elements (beam, tie beam, etc.), if deemed necessary, reductions can be made. However, in this case, the non-linear calculation made according to this Section will be repeated.

ANNEX 5A – STRESS – DEFORMATION FOR CONCRETE AND REINFORCEMENT STEEL RELATIONS

5A.0. SYMBOLS

A_s	= Longitudinal reinforcement area
a_l	= Distance between the axes of vertical reinforcement around the section
b_{he}	= The cross-sectional dimension between the axes of the stirrups that surround the core concrete.
TO_c	= Elasticity modulus of concrete
TO_s	= Modulus of elasticity of reinforcement steel
f_c	= Concrete compressive stress in confined concrete
f_{cc}	= Confined concrete strength
f_{co}	= Compressive strength of unconfined concrete
f_{to}	= Effective wrapping pressure
f_s	= Stress in reinforcement steel
f_{sy}	= Yield strength of reinforcement steel
f_{tis}	= Tensile strength of reinforcement steel
f_{wow}	= Yield strength of transverse reinforcement
h_{he}	= The cross-sectional dimension between the axes of the stirrups that surround the core concrete.
k_{to}	= Confinement Efficiency Coefficient =
S	Transverse reinforcement spacing
ρ_s	= volumetric ratio of total transverse reinforcement (in rectangular sections $\rho_s = \rho_x + \rho_y$)
ρ_x, ρ_y	= Transverse reinforcement volume ratio in relevant directions
ϵ_c	= Concrete pressure unit deformation
ϵ_{cu}	= Maximum compressive strain in confined concrete
ϵ_{sy}	= Yield unit deformation of reinforcement steel
ϵ_s	= Unit deformation of the reinforcement steel at the beginning of hardening
ϵ_{tis}	= Unit deformation of reinforcement steel at break

5A.1. CONDONED AND UNCONDONED CONCRETE MODELS

Nonlinear Methods In the evaluation according to deformation, the following stress-strain relations are defined for confined and unconfined concrete to be used in cases where no other model is selected. **(Figure 5A.1).**

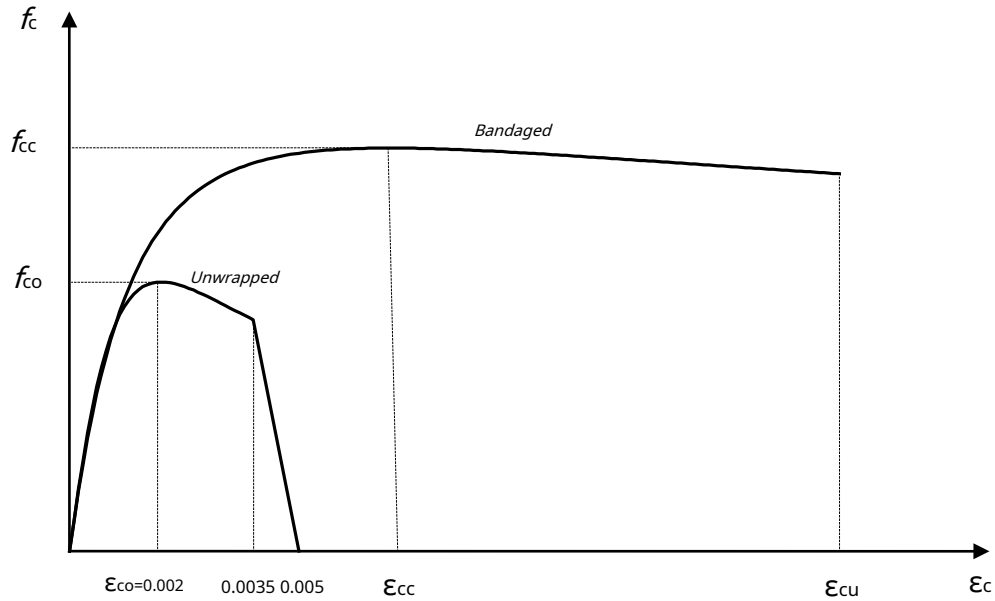


Figure 5A.1

(a) Concrete compressive stress in confined concrete f_c , pressure strain ϵ_c The function of aspect Equation (5A.1)' is given by the relation:

$$f_c = \frac{f_{cc} r}{r - 1 + \chi_r} \quad (5A.1)$$

The confined concrete strength in this relation Equation (5A.2) is given in .

f_{cc} Unconfined concrete strength with f_{co} Relationship between

$$f_{cc} = \lambda_c f_{co} \quad ; \quad \lambda_c = 2.254 \sqrt{1 + 7.94 \frac{f_{co}}{f_{cc}}} - 2 \frac{f_{co} - 1}{f_{cc}} \quad (5A.2)$$

Here f_{co} effective wrapping pressure, for two perpendicular directions in rectangular sections Equation (5A.3) It can be taken as the average of the values given in :

$$f_{ex} = k_{to} \rho_x f_{wyw} \quad ; \quad f_{article} = k_{to} \rho_y f_{wyw} \quad (5A.3)$$

In these relations f_{wyw} yield strength of transverse reinforcement, ρ_x and ρ_y transverse in the relevant directions volumetric ratios of reinforcement, k_{to} whereas Equation (5A.4) defined in winding efficiency coefficient rate It shows the .

$$k_{to} = \frac{1}{6} \left(1 - \frac{\sum a_i^2}{b_{ne} h_{ne}} \right) - \frac{s}{2 b_{ne}} \left(1 - \frac{s}{2 h_{ne}} \right) - \frac{A_s}{b_{ne} h_{ne}} \quad (5A.4)$$

Here a the distance between the axes of the longitudinal reinforcement around the cross-section, b_{ne} And h_{ne} stomach the cross-sectional dimensions between the axes of the stirrups that surround the concrete, s the distance between the axes of the stirrups in the longitudinal direction, A_s indicates the longitudinal reinforcement area. Equation (5A.1) Normalized concrete strain in x with r Relationships related to the variable Equation (5A.5) And Equation (5A.6) is given in .

$$\chi_c = \frac{\varepsilon}{\varepsilon_{cc}} \quad ; \quad \varepsilon_{cc} = \varepsilon_{co}[1 + 5(\lambda_c - 1)] \quad ; \quad \varepsilon_{co} \approx 0.002 \quad (5A.5)$$

$$r = \frac{TO_c}{TO_c - TO_{sec}} \quad ; \quad TO_c \approx 5000 f_{cp} \sqrt{\quad} [MPa] \quad ; \quad TO_{sec} = \frac{f_{cc}}{\varepsilon_{cc}} \quad (5A.6)$$

5A.2. REINFORCEMENT STEEL MODEL

Nonlinear Methods for reinforcement steel to be used in the evaluation according to deformation **Equation (5A.7)** The stress-strain relations in are defined **(Figure 5A.2)**:

$$\begin{aligned} f_s &= TO_s \varepsilon_s & (\varepsilon_s \leq \varepsilon_{sy}) \\ f_s &= f_{sy} & (\varepsilon_{sy} < \varepsilon_s \leq \varepsilon_{s.h.}) \\ f_s &= f_{this} - (f_{this} - f_{sy}) \frac{(\varepsilon_{this} - \varepsilon_s)^2}{(\varepsilon_{this} - \varepsilon_{s.h.})^2} & (\varepsilon_{s.h.} < \varepsilon_s \leq \varepsilon_{this}) \end{aligned} \quad (5A.7)$$

Modulus of elasticity of reinforcement steel $TO_s = 2 \times 10^5$ MPa. Other information about reinforcement steels **Table 5A.1** is given in .

Table 5A.1. Information on Reinforcement Steels

Quality	f_y (MPa)	ε_{sy}	$\varepsilon_{s.h.}$	ε_{this}	f_{this} / f_{sy}
S220	220	0.0011	0.011	0.12	1.20
S420	420	0.0021	0.008	0.08	1.15 - 1.35
B420C	420	0.0021	0.008	0.08	1.15 - 1.35
B500C	500	0.0025	0.008	0.08	1.15 - 1.35

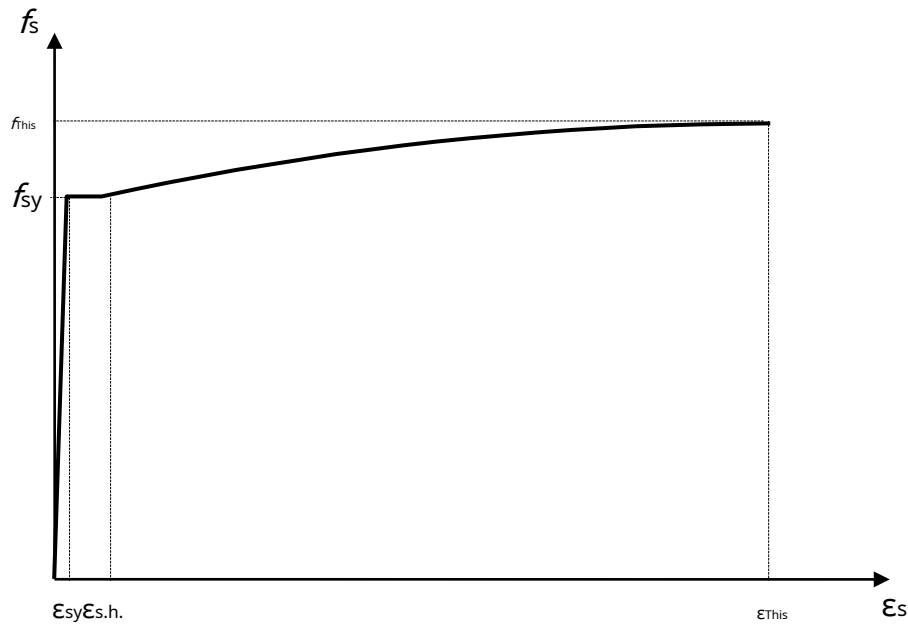


Figure 5A.2

ANNEX 5B – SINGLE MODE THRUST CALCULATION METHODS

5B.0. ICONS

$a_1^{(X,k)}$	= (X) for the earthquake direction in the kth impulse step of the first mode <i>modal single degree of freedom system</i> of modal pseudo-acceleration [m/s ²] =
a_{y1}	For the first mode <i>yield pseudo-acceleration</i> [m/s ²]
C_R	= <i>Spectral displacement ratio</i>
$d_1^{(X,k)}$	= (X) for the earthquake direction in the kth impulse step of the first mode <i>modal single degree of freedom system</i> of modal displacement [m]
$d_{1,max}^{(X)}$	= (X) is the maximum value of the modal single degree of freedom system for the earthquake direction. displacement [m]
$m_{ix1}^{(X,1)}$	= (X) for the earthquake direction in the x-axis direction in the first thrust step determined and never changed during the push calculation <i>fixed mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass [t]
$m_{tx1}^{(X,1)}$	= (X) for the earthquake direction in the x-axis direction in the first thrust step determined and never changed during the push calculation <i>fixed mode shape</i> Calculated according to <i>base shear force modal effective mass</i> [t]
$m_{iy1}^{(X,1)}$	= (X) for the earthquake direction in the y-axis direction in the first thrust step determined and never changed during the push calculation <i>fixed mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass [t]
$m_{i01}^{(X,1)}$	= (X) is determined in the first thrust step around the z axis for the earthquake direction and never changed during the push calculation <i>fixed mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass moment of inertia [tm ²]
$m_{ix1}^{(X,k)}$	= (X) for the earthquake direction at every kth thrust step in the x-axis direction renewed with free vibration calculation <i>variable mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass [t]
$m_{iy1}^{(X,k)}$	= (X) for the earthquake direction at every kth thrust step in the x-axis direction renewed with free vibration calculation <i>variable mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass [t]
$m_{i01}^{(X,k)}$	= (X) for the earthquake direction at every kth thrust step in the x-axis direction renewed with free vibration calculation <i>variable mode shape</i> Calculated according to <i>i</i> 'th floor modal effective mass moment of inertia [tm ²]
$S_{ae}(T_1)$	= First natural vibration period T_1 against <i>linear elastic spectral acceleration</i> [[g]
$S_{also}(T_1)$	= First natural vibration period T_1 against <i>linear elastic spectral displacement</i> [m]
$S_{said}(T_1)$	= First natural vibration period T_1 against <i>nonlinear spectral displacement</i> [m]
T_B	= Horizontal elastic design acceleration spectrum corner period [s]
T_1	= Natural vibration period of the first mode [s]

$U_{ix1}^{(X,k)}$	= (X) for the earthquake direction at the i'th floor in the k'th thrust step x axis Calculated displacement in direction [m]
$U_{Nx1}^{(X,k)}$	= (X) for the earthquake direction at the kth thrust step at the Nth floor (top of the building) Calculated displacement in x-axis direction [m]
$U_g^{(X)}(t)$	= (X) the ground acceleration component defined in the earthquake direction with respect to time change [m/s ²]
$V_{tx1}^{(X,k)}$	= (X) for the earthquake direction in the x-axis direction in the kth thrust step calculated <i>base shear force</i> [[kN]
$\Delta a_1^{(X,k)}$	= (X) for the earthquake direction in the kth impulse step of the first mode <i>modal single degree of freedom system</i> of <i>modal pseudo-acceleration increment</i> [m/s ²]
$\Delta d_1^{(X,k)}$	= (X) for the earthquake direction in the kth impulse step of the first mode <i>modal single degree of freedom system</i> of <i>modal displacement</i> [[m]
$\Delta f_{ix1}^{(X,k)}$	= (X) for the earthquake direction at the i'th floor in the k'th thrust step x axis acting in the direction of <i>earthquake load increase</i> [[kN]
$\Delta f_{iy1}^{(X,k)}$	= (X) y-axis at the i'th floor in the k'th thrust step for the earthquake direction acting in the direction of <i>earthquake load increase</i> [[kN]
$\Delta f_{i\theta1}^{(X,k)}$	= (X) earthquake direction for the kth thrust step at the i'th floor around the z axis <i>effective earthquake moment increase</i> [[kNm]
$\Delta u_{ix1}^{(X,k)}$	= (X) for the earthquake direction at the i'th floor in the k'th thrust step x axis Calculated displacement increment in direction [m]
$\Delta u_{iy1}^{(X,k)}$	= (X) y-axis at the i'th floor in the k'th thrust step for the earthquake direction Calculated displacement increment in direction [m]
$\Delta u_{i\theta1}^{(X,k)}$	= (X) earthquake direction for the kth thrust step at the i'th floor around the z axis calculated rotation increment [m]
$\Phi_{ix1}^{(1)}$	= determined in the first push step at the i-th floor and never during the push calculation unchanged <i>fixed mode shape</i> The amplitude of in the x direction
$\Phi_{ix1}^{(f)}$	= calculated along the x-axis in the i'th floor in the k'th push step assumed to be equal to the displacement approx. <i>mode shape amplitude</i>
$\Phi_{iy1}^{(1)}$	= determined in the first push step at the i-th floor and never during the push calculation unchanged <i>fixed mode shape</i> The amplitude of in the y-direction
$\Phi_{i\theta1}^{(1)}$	= determined in the first push step at the i-th floor and never during the push calculation unchanged <i>fixed mode shape</i> Rotation amplitude of about the z-axis
$\Phi_{ix1}^{(f)}$	= updated with free vibration calculation at every kth push step on the i'th floor <i>variable mode shape</i> The amplitude of in the x direction
$\Phi_{iy1}^{(f)}$	= updated with free vibration calculation at every kth push step on the i'th floor <i>variable mode shape</i> The amplitude of in the y-direction

$\Phi_{i\theta 1}^{(f)}$	= updated with free vibration calculation at every kth push step on the i'th floor <i>variable mode shape</i> Rotation amplitude of about the z-axis
$\Gamma_1^{(X,1)}$	= (X) earthquake direction determined in the first thrust step and thrust calculation never changed throughout <i>fixed mode shape</i> Calculated according to <i>modal contribution factor</i>
$\Gamma_1^{(X,k)}$	= (X) with free vibration calculation for each kth impulse step for the earthquake direction renewed <i>variable mode shape</i> Calculated according to <i>modal contribution factor</i>
$F_1^{(X,k)}$	= (X) earthquake direction and for the first vibration mode at the kth impulse step calculated <i>approximate modal contribution factor</i> =
ξ_1	Modal damping ratio of the first vibration mode
$\mu(R_y, T_1)$	= <i>Yield Strength Reduction Coefficient</i> and according to the first natural vibration period calculated <i>demand for ductility</i>
$\omega_1^{(f)}$	= First one found from the free vibration calculation renewed at every kth thrust step mode <i>natural angular frequency</i> [rad/s]

5B.1. MODAL CAPACITY WITH FIXED SINGLE MODE PUSHING METHOD OBTAINING THE DIAGRAM

5B.1.1 –Fixed single mode pushing methodIn the considered earthquake direction (X), the impact on the floors in the kth thrust step*earthquake load increments*, which is determined in the first step after the non-earthquake loading and is not changed during the pushover calculation.*fixed mode shape*Calculated according to*floor modal effective masses*are expressed in terms of:

$$\Delta f_{ix1}^{(X,k)} = m_{ix1}^{(X,1)} \Delta a_{ix1}^{(X,k)} \quad ; \quad \Delta f_{iy1}^{(X,k)} = m_{iy1}^{(X,1)} \Delta a_{iy1}^{(X,k)} \quad ; \quad \Delta f_{i\theta 1}^{(X,k)} = m_{i\theta 1}^{(X,1)} \Delta a_{i\theta 1}^{(X,k)} \quad (5B.1)$$

Here $m_{ix1}^{(X,1)}$, $m_{iy1}^{(X,1)}$ And $m_{i\theta 1}^{(X,1)}$ **ANNEX 4B in Equation (4B.2)** The first floor effective masses are given by are the first mode equivalents (n = 1) calculated in step (k = 1):

$$m_{ix1}^{(X,1)} = m \Phi_{ix1}^2 \Gamma_1^{(X,1)} \quad ; \quad m_{iy1}^{(X,1)} = m \Phi_{iy1}^2 \Gamma_1^{(X,1)} \quad ; \quad m_{i\theta 1}^{(X,1)} = m \Phi_{i\theta 1}^2 \Gamma_1^{(X,1)} \quad (5B.2)$$

In these relations $\Gamma_1^{(X,1)}$, for the earthquake direction (X) and the first vibration mode considered in the first push step **Equation (4B.1)** Calculated from *modal contribution factor*'s.

5B.1.2 –In the kth push step defined between two consecutive joint formations, the unknown quantity belongs to the first mode.*modal single degree of freedom system* of **Equation (5B.1)** Located in *modal pseudo-acceleration increment* $\Delta a_{ix1}^{(X,k)}$ This size is a new plastic that forms at the end of each step. joint **5.3.1** defined in *flow condition* It is calculated from . The obtained *modal pseudo-acceleration increment*, is added to the pseudo-acceleration value found at the end of the previous step and is given in the kth step. cumulative modal pseudo-acceleration at the end $a_{ix1}^{(X,k)}$ is obtained. In the traditional push calculation $a_{ix1}^{(X,k)}$, (X) in the direction of the earthquake *base shear force* $V_{ix1}^{(X,k)}$ written for **Equation (5B.3)** is obtained from:

$$a_{ix1}^{(X,k)} = \frac{V_{ix1}^{(X,k)}}{m_{ix1}^{(X,1)}} \quad (5B.3)$$

In this relation *base shear force modal effective mass* $m_{(x,1)tx1}$, in the x-axis direction **Equation (5B.2)** also *mode shape in the first thrust step* and defined according to *all push account constant throughout* taken as *floor modal effective masses* $m_{(x,i)ix1}$ It is the sum of 's in all floors.

5B.1.3 –Belonging to the first mode *modal single degree of freedom system* of *modal displacement* $d_{(x)k}$ can be calculated from the horizontal displacement in the x direction at any i'th floor from the pushover calculation. In the traditional pushover calculation, the top displacement at the N'th floor is used for this purpose:

$$d_{(x,k)} = \frac{U_{(x,k)}^{N \times 1}}{\Phi_{(1)}^{N \times 1} \Gamma_1^{(x,1)}} \quad (5B.4)$$

5B.1.4 –In the traditional push calculation, first *base shear force – top displacement* as a relationship *thrust curves* is drawn (**Figure 5B.1a**) and then the coordinates of this curve **Equation (5B.3)** And **Equation (5B.4)** Transformed according to *modal single degree of freedom system* Belonging to *modal pseudo acceleration – modal displacement* as a relationship *modal capacity diagram* is obtained (**Figure 5B.1b**).

5B.1.5 –In the traditional pushover calculation, the modal displacement is changed at each step **Equation (5B.4)** belongs to the linear system in the initial step with *fixed mode shape* In order to partially eliminate the approximation made due to the fact that it is obtained depending on , the displacement increment obtained at any kth push step is approximately equal to that of *variable mode shape amplitude* can be taken as. In this context **Equation (5B.5)** The relation in is written for a typical ith floor with degrees of freedom in the x-direction:

$$\Phi_{(f)}^{ix1} \approx U_{(x,k)}^{ix1} - U_{(x,k-1)}^{ix1} \quad (5B.5)$$

In this situation *modal displacement* **Equation (5B.4)** Regardless of the top displacement as in , **Equation (5B.6)** can be obtained as follows:

$$d_{(x,k)}^{(x)} = d_{(x,k-1)}^{(x)} + \frac{1}{\Gamma_1^{(x,k)}} \quad (5B.6)$$

Here $\Gamma_{(x,k)}^{(x)}$, **Equation (5B.5)** By using the approximate mode shape defined in **Equation (4B.1)** calculated at each push step with *approximate modal contribution factor* is.

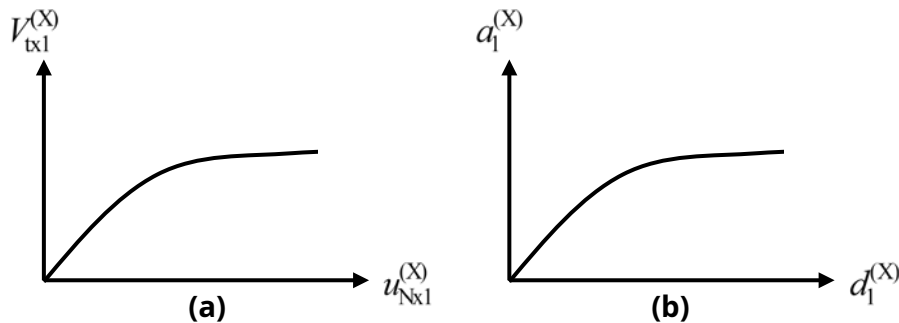


Figure 5B.1

5B.1.6 – Caused by vertical loads on the deformed carrier system *second order effects* In buildings where may be important, *Fixed Mode Pushing Method* Since is based on the base shear force, these effects cannot be taken into account in a coherent manner, **5B.2** Given in *Variable Mode Pushing Method* should be used.

5B.2. MODAL CAPACITY WITH VARIABLE SINGLE MODE PUSHING METHOD OBTAINING THE DIAGRAM

5B.2.1 – *Variable single mode push method* In the earthquake direction considered (X), the impact on the floor *earthquake load increments*, obtained from the free vibration calculation renewed at each kth thrust step after non-earthquake loading *variable mode shape* Calculated according to *floor modal effective masses* is expressed in terms of:

$$\Delta f_{ix1}^{(X,k)} = m_{ix1}^{(X,k)} \Delta a_{ix1}^{(X,k)} ; \Delta f_{iy1}^{(X,k)} = m_{iy1}^{(X,k)} \Delta a_{iy1}^{(X,k)} ; \Delta f_{i\theta1}^{(X,k)} = m_{i\theta1}^{(X,k)} \Delta a_{i\theta1}^{(X,k)} \quad (5B.7)$$

Here $m_{ix1}^{(X,k)}$, $m_{iy1}^{(X,k)}$ and $m_{i\theta1}^{(X,k)}$ free vibration renewed at each kth impulse step

depending on the mode shape **ANNEX 4B'** also **Equation (4B.2)** are the first mode equivalents of the effective masses calculated with (n=1):

$$m_{ix1}^{(X,k)} = m_{ix1} \Phi_{ix1}^2(\gamma_{ix1}^{(X,k)}) ; m_{iy1}^{(X,k)} = m_{iy1} \Phi_{iy1}^2(\gamma_{iy1}^{(X,k)}) ; m_{i\theta1}^{(X,k)} = m_{i\theta1} \Phi_{i\theta1}^2(\gamma_{i\theta1}^{(X,k)}) \quad (5B.8)$$

In these relations $\gamma_{ix1}^{(X,k)}$, for each kth thrust step for the earthquake direction (X) considered depending on the renewed first vibration mode **Equation (4B.1)** Calculated from *modal contribution factor* is.

5B.2.2 – At each kth push step, above **Equation (5B.7)** given floor in *earthquake load increments* The storey displacement increases under the influence of **Equation (5B.9)** is obtained with:

$$\Delta u_{ix1}^{(X,k)} = \Phi_{ix1}^{(f)} \Gamma_{ix1}^{(X,k)} \Delta d_{ix1}^{(X,k)} ; \Delta u_{iy1}^{(X,k)} = \Phi_{iy1}^{(f)} \Gamma_{iy1}^{(X,k)} \Delta d_{iy1}^{(X,k)} ; \Delta u_{i\theta1}^{(X,k)} = \Phi_{i\theta1}^{(f)} \Gamma_{i\theta1}^{(X,k)} \Delta d_{i\theta1}^{(X,k)} \quad (5B.9)$$

5B.2.3 – Eq.(5B.7) belongs to the kth push step in *modal pseudo-acceleration increment* $\Delta a_{ix1}^{(X,k)}$ with **Equation (5B.9)** Located in *modal displacement increment* $\Delta d_{ix1}^{(X,k)}$ between *step-by-step linear* relationship **Equation (5B.10)** is given in:

$$\Delta a_{ix1}^{(X,k)} = (\omega_{ix1}^{(f)})^2 \Delta d_{ix1}^{(X,k)} \quad (5B.10)$$

Here $\omega_{ix1}^{(f)}$ from the free vibration calculation of the carrier system, which is renewed at every kth thrust step first mode found *natural angular frequency* is.

5B.2.4 – As the unknown quantity in the kth push step defined between two consecutive joint formations, *modal single degree of freedom system* of **Equation (5B.7)** The first fashion in Belonging *modal pseudo-acceleration increment* $\Delta a_{ix1}^{(X,k)}$ can be taken (*force-based calculation*). However, instead, it enables direct calculation of internal forces, **Equation (5B.9)** Located in *modal displacement increment* $\Delta d_{ix1}^{(X,k)}$ It is preferable to take as the unknown quantity (*displacement based calculation*). In both cases, the unknown is taken as *modal increment*, the new plastic joint formed at the end of step **5.3.1** defined in *flow condition* It will be calculated from .

After calculating the modal increment taken as unknown, other *modal increment* in **Equation (5B.10)** is obtained from.

5B.2.5 –The modal pseudo-acceleration and modal displacement increments obtained in the k th pushing step are added to the values obtained at the end of the previous step to obtain the cumulative values of these quantities. **Equation (5B.11)** is obtained as follows:

$$\begin{aligned} a_1^{(X,k+1)} &= a_1^{(X,k)} + \Delta a_1^{(X,k)} \\ d_1^{(X,k+1)} &= d_1^{(X,k)} + \Delta d_1^{(X,k)} \end{aligned} \quad (5B.11)$$

Like this *variable load or displacement distribution thrust method*, without the need to draw the thrust curve *modal capacity diagram* is obtained directly. In this diagram, in the k th push step between two consecutive joint formations *step-by-step linear behavior* Representing slope of the line segment, **Equation (5B.10)** According to $(\omega_1^{(k)})^2$ is equal to **(Figure 5B.2a)**.

5B.2.6 –*Second order effects* Representing *geometric stiffness matrix* Free vibration of if taken into account in the calculation, **Equation (5B.10)** in $(\omega_1^{(k)})^2$, in other words modal The slope of the capacity diagram is in the forward thrust steps where plastic deformations in the carrier system increase. *negative* can be obtained as values **(Figure 5B.2b)**.

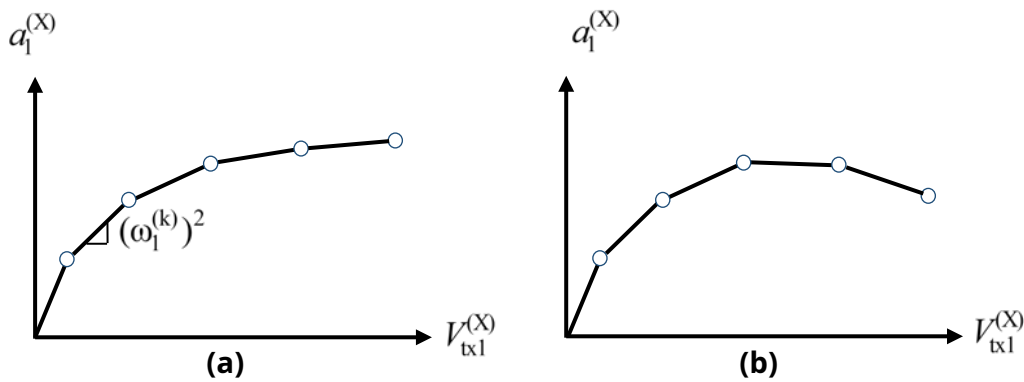


Figure 5B.2

5B.3. LINEAR DEMAND OF MODAL DISPLACEMENT OF EARTHQUAKE OBTAINING AS NON-SPECTRAL DISPLACEMENT

The earthquake *modal displacement request's* achievement, under the given earthquake effect *modal capacity diagram* represented by *maximum displacement of a modal single degree of freedom system* It goes against the account of.

5B.3.1 –The maximum displacement in a modal single degree of freedom system is, *nonlinear spectral displacement* is defined as:

$$d_{1,max}^{(X)} = S_{said}(T_1) \quad (5B.12)$$

Here $d_{1,max}^{(X)}$ *maximum displacement of a modal single degree of freedom system* ni, $S_{said}(T_1)$ whereas first natural vibration period of the carrier system T_1 and opposing **Equation (5B.13)** defined by *nonlinear spectral displacement* it shows the .

$$S_{said}(T_1) = C_R S_{also}(T_1) \quad (5B.13)$$

Here $S_{also}(T_1)$, **Equation (2.5)** defined by *elastic design spectral displacement* $\mu(R_y, T_1)$, C_R whereas **Equation (5B.14)** defined in *spectral displacement ratio* It shows the .

5B.3.2 – Eq.(5B.13) Located in *spectral displacement ratio* C_R , **Equation (5B.14)** It is defined in:

$$C_R = \frac{\mu(R_y, T_1)}{R_y} \quad (5B.14)$$

Here *yield strength reduction coefficient* showing the R_y , *design according to strength* for the approach **ANNEX 4A** Unlike the definition given in , it is not a quantity defined depending on the predicted ductility capacity, but a quantity obtained directly from the pushover calculation. *yield strength* It expresses a magnitude that depends on:

$$R_y = \frac{f_y}{f_{y1}} = \frac{S_y(T_1)}{a_{y1}} \quad (5B.15)$$

In this relation f_{y0} And $S_{ae}(T_1)$ *elastic strength demand* and those who oppose him *elastic spectral acceleration* i , f_{y1} And a_{y1} whereas *yield strength* and those who oppose him *yield pseudo-acceleration* It represents (**Figure 5B.4**).

5B.3.3 – Equation (5B.14) Located in $\mu(R_y, T_1)$, *yield strength* and natural vibration period expressed as depending on *demand for ductility* is. To calculate this size **ANNEX 4A** in **Equation (4A.2)** The following relations are obtained by writing the given relations in reverse order:

(a) *Ductility demand of earthquake* $\mu(R_y, T_1)$, *equal displacement rule* in accordance with *not very rigid* for carrier systems *Yield Strength Reduction Coefficient* R_y is taken equal to:

$$\mu(R_y, T_1) = R_y \quad T_1 > T_B \quad (5B.16a)$$

(b) *High rigidity* for carrier systems **Equation (4A.2b)** from **Equation (5B.16b)** The relation in is obtained:

$$\mu(R_y, T_1) = 1 + (R_y - 1) \frac{T_B}{T_1} \quad T_1 \leq T_B \quad (5B.16b)$$

5B.3.4 – Eq.(5B.14) defined in *spectral displacement ratio* C_R , **Equation (5B.16)** By taking advantage of **Equation (5B.17)** It is expressed as follows:

$$C_R = 1 \quad T_1 > T_B \quad (5B.17a)$$

$$C_R = \frac{1 + (R_y - 1) \frac{T_B}{T_1}}{R_y} \geq 1 \quad T_1 \leq T_B \quad (5B.17b)$$

5B.3.5 – Figure 5B.3 And **Figure 5B.4** and coordinates of the first (dominant) vibration mode in *modal displacement-modal pseudo-acceleration* (d_1, a_1) the one which *modal capacity diagram* with coordinates *spectral displacement-spectral acceleration* (S_{also}, S_{ae}) the one which *linear earthquake spectrum* are drawn together.

(a) Figure 5B.3 The situation shown in , **Equation (5B.13)** together with **Equation (5B.17a)** In this case, no action is taken on the modal capacity diagram. without doing, only the natural vibration period in the first thrust step $T_1 > T_B$ or $(\omega_1^{(1)})^2 \leq \omega_B^2$ It is sufficient to show that the condition is met.

(b) On the other hand **Figure 5B.4** The situation shown in , **Equation (5B.13)** together with **Equation (5B.17b)** It is against the implementation of . In this case *spectral displacement ratio* C_R , consecutive For this purpose, the modal capacity diagram will be calculated with the following approach: **Figure 5B.4a** As shown in , before $C_R = 1$ and the two lines are taken as *elasto-plastic* is converted into a diagram. Conversion The equality of the areas under the diagrams is taken as basis in the process. The approximate yield pseudo-acceleration a_{ey1} by using **Equation (5B.15)** from R_y and accordingly **Equation (5B.17b)** from C_R And **Equation (5B.13)**'s $S_{aid}(T_1)$ is calculated. Accordingly *elasto-plastic* The diagram is recreated (**Figure 5B.4b**) and found again a_{y1} The same processes are repeated based on the steps where the results are close enough. The sequential approximation is terminated.

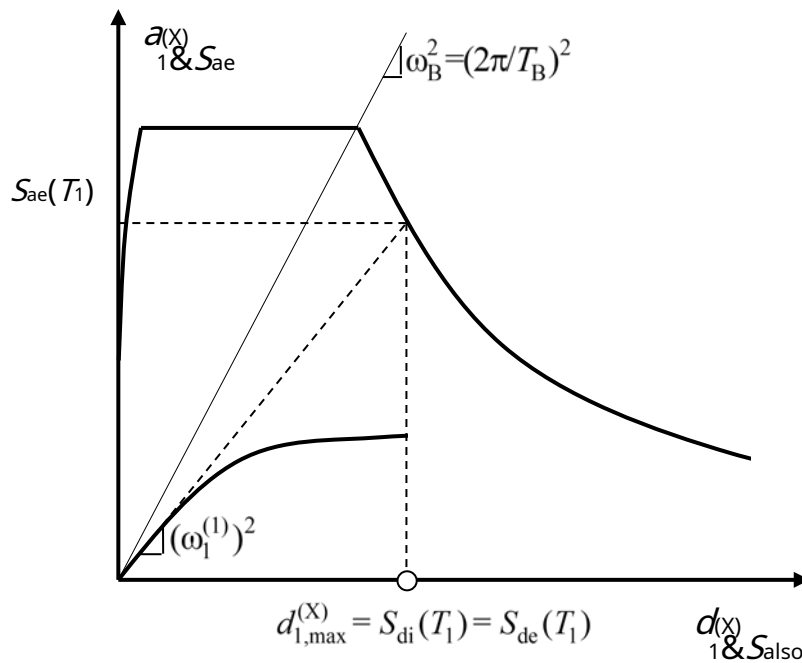


Figure 5B.3

5B.3.6 –The earthquake modal displacement request's **Equation (5B.13)** And **Equation (5B.17)** By taking advantage of **Equation (5B.12)** According to the account below **(a)** And **(b)** It does not apply in the situations defined in .

(a) In cases where the nearest fault is less than 15 km away from the building, **2.5** Selected and scaled according to *near-field earthquake records* by using **5B.4** According to *time domain accounting* will be done.

(b) *Second order effects* If the slopes of the modal capacity diagram after yielding are negative due to **2.5** Selected and scaled according to *earthquake records* by using **5B.4** According to *time domain accounting* will be done.

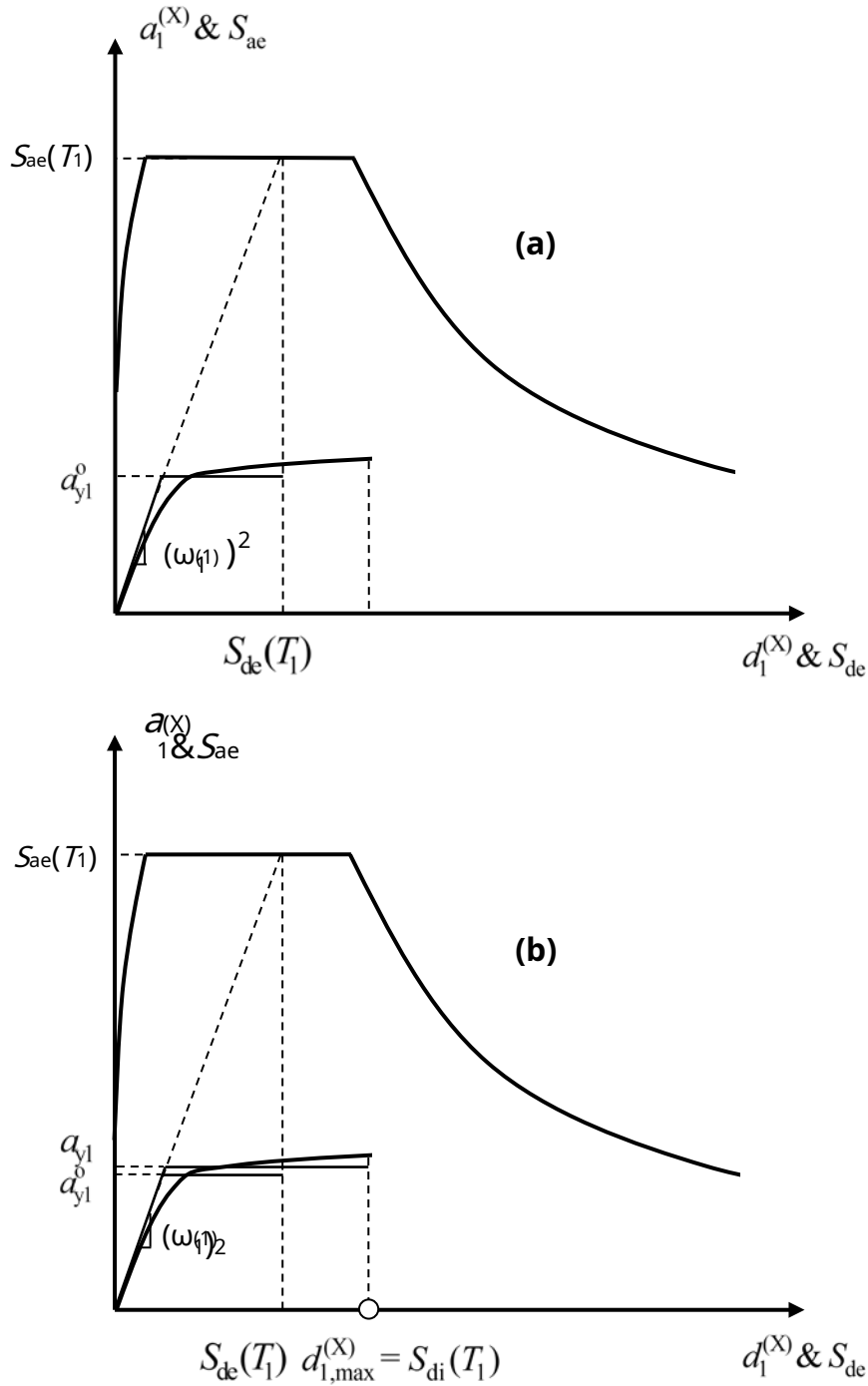


Figure 5B.4

5B.4. TIME DEFINITION OF MODAL DISPLACEMENT DEMAND OF EARTHQUAKE OBTAINING FROM NONLINEAR CALCULATION IN THE FIELD

5B.4.1 –The earthquake *modal displacement demand*, as the maximum displacement of the modal single degree of freedom system in the time history *nonlinear calculus* can also be obtained from **5B.1** or **5B.2** The modal capacity diagram obtained in **5.3.1.7** By adapting a suitable hysteretic behavior model to be selected according to **Equation (5B.18)** The equation of motion given in increments is solved.

$$\Delta d_{T(x)}(t) + 2\xi \omega \Delta d_{T(x)}(t) + \Delta a_{T(x)}(t) = -\Delta \ddot{u}_g(t) \quad (5B.18)$$

Displacements of the carrier system and the resulting deformations and internal forces, **Equation (5B.18)**

Taking the maximum value of the modal displacement obtained from the solution of the equation of motion in **Equation (5B.4)** or **Equation (5B.9)** with **Equation (5B.11)** can be obtained from.

5B.4.2 –At least eleven earthquake ground motions will be used in the nonlinear time history calculation in the earthquake direction considered. Selection of earthquake records to be used **2.5** Defined in. For conversion of records **2.5.3** The modal displacement will be calculated as the average of the largest absolute values of the results obtained from each of the eleven analyses performed.

ANNEX 5C – ALLOWED DEFORMATION FOR STEEL MEMBERS BORDERS

5C.0. ICONS

A_{bec}	= Buckling restrained steel cross-sectional area [m ²] =
A_c	Cross section area [m ²]
d	= Beam cross-section height [m] =
d_c	Column cross-section height [m]
TO	= Structural steel modulus of elasticity, $E = 2 \times 10^8$ [kN/m ²]
F_{eat}	] = Expected yield stress of structural steel [kN/m ²]
I_b	= Beam section moment of inertia [m ⁴] =
I_k	Column section moment of inertia [m ⁴]
K_{he}	= Rotational rigidity of the joint area with end plate and head plate [kNm/rad] =
-b	Beam length [m]
-bec	= Buckling restrained steel cross length [m] =
-k	Column length [m]
-c	= Diagonal length [m]
M_{pe}	= Plastic moment capacity [kNm]
P	= Axial load on the steel section at maximum plastic rotation [kN] = Expected
P_c	axial compressive strength of the steel element [kN]
P_{bec}	= Axial tensile and compressive strength of the buckling restrained steel brace [kN]
P_{eat}	= Expected axial yield force of the steel member [kN]
t_p	= Shear zone thickness [m]
θ	= Displaced axis rotation [rad] =
θ_y	Yield rotation [rad]
V_{eat}	= Expected shear force capacity [kN] =
W_p	Plastic strength moment [m ³]
EI_{bn}	= Beam stiffness for systems with end plate and head plate connections [kNm ²] = Yield
Δ_c	displacement corresponding to the axial compressive strength of the steel element [m] = Yield
Δ_T	displacement corresponding to the axial tensile strength of the steel element [m] = Yield
Δ_y	displacement of the steel element with restrained buckling [m]

5C.1. DEFORMATION LIMITS OF STEEL BEAMS AND COLUMNS

In steel beams and columns (bending), the deformation limits will be calculated in terms of yield rotation. In the calculation of yield rotations, the following **5C.1.1** for steel beams in, **5C.1.2** In , the connections given for steel columns will be used.

5C.1.1. Calculation of Yield Rotation for Steel Beams

For all section types:
$$\theta_y = \frac{W.F.-}{6EI_b} \quad (5C.1)$$

5C.1.2. Calculation of Yield Rotation for Steel Columns

For all section types:
$$\theta_y = \frac{W.F.-}{6EI_k} - \frac{P-}{P_{eat-}} \quad (5C.2)$$

Table 5C.1 for steel beams in, **Table 5C.2** in , the allowable deformation limits for steel columns are given as plastic rotation for different performance levels. Ductility level in steel beams and columns **Table 9.3** Considering the cross-sectional conditions defined in will be determined by taking into account the PIP_c Non-linear deformations will not be allowed in columns > 0.50 .

Table 5C.1. Plastic Rotation Limits in Steel Beams

Beam (Bending)	Deformation Limits		
	SH	KH	GÖ
High Ductility Level (See Table 9.3)	$1\theta_y$	$6\theta_y$	$9\theta_y$
Limited Ductility Level (See Table 9.3)	$0.25\theta_y$	$3\theta_y$	$4\theta_y$

Table 5C.2. Plastic Rotation Limits in Steel Columns

Column (Bending)	Plastic Rotation Limits [rad]		
	SH	KH	GÖ
$PIP_c < 0.20$ (See Table 9.3)			
High Ductility Level	$1\theta_y$	$6\theta_y$	$9\theta_y$
Limited Ductility Level	$0.25\theta_y$	$3\theta_y$	$4\theta_y$
$0.20 \leq PIP_c \leq 0.50$ (See Table 9.3)			
High Ductility Level	$1.5 (1 - 1.66 PIP_c) \theta_y$	$9 (1 - 1.66 PIP_c) \theta_y$	$13.5 (1 - 1.66 PIP_c) \theta_y$
Limited Ductility Level	$0.25\theta_y$	$0.7\theta_y$	$1\theta_y$
Slip Zone	$1\theta_y$	$9\theta_y$	$12\theta_y$

5C.2. COLUMN-BEAM JOINTS OF MOMENT-BEARING FRAMES SHAPE CHANGE BOUNDARIES OF REGIONS

For column-beam joint areas in moment-resisting steel frame systems **Table 5C.3** The plastic rotation limits given in shall be taken into account. Full strength bolted end plate connections and bolted head plate connections shall be modeled using rotation springs representing the joint stiffness. Rotation spring stiffness **5C.2.1** According to , the beam stiffness is **5C.2.2** will be calculated according to:

**Table 5C.3. Plasticity of Column-Beam Joint Area in Moment Resisting Steel Frames
Rotation Limits**

Column-Beam Connection Type	Plastic Rotation Limits [rad]		
	SH	KH	GÖ
Full Strength Bolted Face Plate Joint			
Facing plate leakage	0.01	0.025	0.04
Leaking of bolts	0.008	0.012	0.018
Bolted Header Plate Joint			
Yielding of the net section of the header plate or bolts in shear	0.008	0.02	0.03
Fracture of the header plate weld or plate	0.003	0.008	0.012
Full penetration butt weld joint	0.026 - 0.00063d	0.0323 - 0.00045d	0.043 - 0.00060d
Welded weakened beam cross-section joint	0.025 - 0.00015d	0.0525 - 0.00023d	0.070 - 0.00030d

5C.2.1. Rotational Stiffness of End Plate and Head Plate Connections

Rotational rigidity:
$$K_{he} = \frac{M_{pe}}{0.005} \quad (5C.3)$$

5C.2.2. Beam Stiffness of Face Plate and Head Plate Connected Systems

New beam stiffness:
$$EI_{bn} = \frac{1}{\frac{3}{-b K_{he}} + \frac{1}{EI_b}} \quad (5C.4)$$

5C.3. SHAPE CHANGE OF CENTRAL BRACED STEEL FRAMES BORDERS

In centrally braced steel frame systems, the deformation limits for steel braces are expressed in terms of axial plastic deformations. **Table 5C.4** Flow deformation of diagonal members under axial compression force is given in **5C.3.1**. In , the yield deformation under axial tensile force is **5C.3.2**. In buckling restrained steel braces, axial element stiffness will be obtained by serially connecting the core element that is allowed to yield and the transfer elements that are not allowed to yield. In the stiffness calculation, the gusset plates to which the buckling restrained braces are connected and the column-beam region can be assumed to be fully rigid. Yield deformation in buckling restrained steel braces, in the light of the information given above, **5C.3.3** is given in.

5C.3.1. Yield Deformation Under Axial Compression Force

Flow deformation:
$$\Delta_c = \frac{P_c}{EA_c} \quad (5C.5)$$

5C.3.2. Yield Deformation Under Axial Tensile Force

Flow deformation:
$$\Delta_T = \frac{P_{eat\ c}}{EA_c} \quad (5C.6)$$

5C.3.3. Yield Deformation of Buckling Restrained Steel Brace

Flow deformation:
$$\Delta_y = \frac{P_{bec}}{EA_{bec}} \quad (5C.7)$$

Table 5C.4 Strain Limits of Centrally Braced Steel Frames

Element Type	Axial Plastic Strain Limits		
	SH	KH	GÖ
Diagonal Members in Compression (Except Eccentric Diagonals)			
<i>Delicate Sections</i> $\frac{KL}{r} \geq 4.2 \sqrt{\frac{TO}{F_y}}$			
W, I, 2L, 2C sections: in-plane buckling	$0.5\Delta_c$	$7\Delta_c$	$10\Delta_c$
2L, 2C sections: out-of-plane buckling	$0.5\Delta_c$	$6\Delta_c$	$9\Delta_c$
HSS, pipe, box sections	$0.5\Delta_c$	$6\Delta_c$	$9\Delta_c$
L section	$0.5\Delta_c$	$9\Delta_c$	$12\Delta_c$
<i>Full Sections</i> $\frac{KL}{r} \leq 2.1 \sqrt{\frac{TO}{F_y}}$			
HD, I, 2L, 2C sections: in-plane buckling	$0.5\Delta_c$	$6\Delta_c$	$8\Delta_c$
2L, 2C sections: out-of-plane buckling	$0.5\Delta_c$	$5\Delta_c$	$7\Delta_c$
HSS, pipe, box sections	$0.5\Delta_c$	$5\Delta_c$	$7\Delta_c$
Diagonal Members in Tension (Except Eccentric Diagonals)			
W section	$0.5\Delta_T$	$10\Delta_T$	$13\Delta_T$
2L section	$0.5\Delta_T$	$9\Delta_T$	$12\Delta_T$
HSS section	$0.5\Delta_T$	$8\Delta_T$	$11\Delta_T$
Pipe section	$0.5\Delta_T$	$7\Delta_T$	$9\Delta_T$
L section	$0.5\Delta_T$	$8\Delta_T$	$10\Delta_T$
Beams and columns under tension	$0.5\Delta_T$	$6\Delta_T$	$7\Delta_T$
Buckling-resistant steel braces	$1\Delta_y$	$10\Delta_y$	$13.3\Delta_y$

5C.4. ECCENTRICALLY BRACED STEEL FRAMES AND STEEL TIE BEAMS LIMITS OF SYSTEMS' SHAPING

In eccentrically braced steel frame systems and reinforced concrete shear walls with steel tie beams, the deformation limits for tie beams are in plastic rotation type. **Table 5C.5** In reinforced concrete shear walls with steel tie beams, the stiffness of the steel tie beams will be taken as 60% of the total tie beam stiffness.

**Table 5C.5. In Eccentric Braced Steel Frames and Steel Tie Beam Systems
Plastic Rotation Limits**

Element Type	Plastic Rotation Limits [rad]		
	SH	KH	GÖ
Steel Tie Beam			
$t \leq 1.6 \frac{M_{pe}}{V_{eat}}$	0.005	0.12	0.15
$t \geq 2.6 \frac{M_{pe}}{V_{eat}}$	Table 5C.1 It is the same as the steel beams in .		
$1.6 \frac{M_{pe}}{V_{eat}} < t < 2.6 \frac{M_{pe}}{V_{eat}}$	Linear interpolation will be done.		

CHAPTER 6 – NON-STRUCTURAL BUILDING UNDER EFFECT OF EARTHQUAKE DESIGN PRINCIPLES OF ELEMENTS

6.0. ICONS

A_{ie}	= Under DD-2 earthquake ground motion, the element or equipment on the i-th floor will move to the floor. the maximum total acceleration acting on the region to which it is attached
B_{to}	= Magnification factor applied to the element or equipment
F_{ie}	= Acting horizontally to the centre of gravity of the element or equipment and Equivalent seismic load that can be distributed on the equipment in proportion to its mass
g	= Gravitational acceleration [9.81
h_i	m/s ²] = height of i'th floor [m]
h_x, h_y	= Upper and lower connection points of non-structural elements and equipment at relevant floor level height from base [m]
I	= Building Importance Coefficient
k_{to}	= Effective stiffness coefficient of the element or equipment and/or its
m_{to}	connection = Mass of the element or equipment in working order [t]
R	= Carrier system behavior coefficient
R_{to}	= Response coefficient defined for the element or equipment =
S_{DS}	Short period design spectral acceleration coefficient [dimensionless]
T_{to}	= Natural vibration period of the element or equipment [s]
T_n	= nth natural vibration period of the building [s]
T_p	= Dominant natural vibration period of the building in the earthquake direction considered [s]
U_i	= Reduced earthquake in line with the earthquake taken into account on the ith floor of the building Calculated horizontal displacement under loads
U_{of}	= Reduced earthquake in line with the earthquake taken into account on the ith floor of the building Horizontal displacement of the nth mode calculated under loads
$\delta_{to}^{(x)}$	= Typical (X) earthquake direction for non-structural elements and equipment effective relative storey drift
$\delta_{i,max}^{(x)}$	= (X) effective relative floor drifts at the ith floor of the building for the earthquake direction Maximum value within the floor [m]

6.1. GENERAL RULES

6.1.1 –This applies to all kinds of protrusions (such as balconies, parapets, chimneys, consoles) that are connected to the load-bearing system but work independently, facade and partition panels, architectural elements, mechanical and electrical equipment and their connections to the structure, which may harm people or the structural system of the building or prevent the use of the building in case of damage during an earthquake.

It is mandatory to make an earthquake calculation according to the rules given in the section. However, the furniture in the building and the equipment temporarily located in the building and not connected to the building and *Earthquake Design Class* seismic calculations are not required for non-structural elements in buildings with DTS=4.

6.1.2 –Non-structural elements and equipment must be permanently attached to the structure and the connection elements must be capable of withstanding the equivalent seismic loads and displacements given in this Section. Additional capacity due to friction will not be taken into account in the calculation of the connection elements (e.g. welds, bolts, dowels, rivets, etc.) connecting the equipment to the structure under the effect of an earthquake. The connection elements must have the strength to ensure uninterrupted load transfer from the equipment to the structure.

6.1.3 –If the weight of a non-structural element or equipment is greater than 10% of the total weight of the floor it is located on, the element or equipment will be considered as part of the building load-bearing system. In this case, the mass of the element or equipment and the rigidity properties of its connection to the building will be taken into account in the seismic calculation of the building load-bearing system.

6.2. EQUIVALENT EARTHQUAKE LOADS

6.2.1 –Equivalent seismic load acting horizontally to the centre of gravity of the element or equipment and acting on the element or equipment F_{ie} is defined by the following relation:

$$F_{ie} = m_{ie} \frac{A_{ie} E U}{R_{to}} \quad (6.1)$$

Here m_{ie} the mass of the element or equipment in working order, A_{ie} DD-2 earthquake ground motion

The maximum total acceleration acting on the area where the element or equipment on the i-th floor is

connected to the floor, B_{to} the magnification factor applied to the element or equipment, R_{to} element or

the behavior coefficient defined for the hardware, I whereas **Table 3.1** Shows the building importance

coefficient defined in B_{to} And R_{to} coefficients for non-structural architectural elements **Table**

6.1, for mechanical and electrical equipment **Table 6.2** is given in .

6.2.2 –The maximum total acceleration acting on the element or equipment shall be defined as the maximum value obtained from the following calculations:

(a) For building carrier system **4.7** As a result of the linear earthquake calculation to be made according to , at the location of any i-th floor element or equipment, the calculated in the relevant direction

absolute acceleration value A_{ie} **Equation (6.2)** will be calculated with .

$$A_{ie} (R/I) = \frac{2\pi}{T_p} U_i \quad (6.2)$$

Here T_p is the dominant natural vibration period of the building in the earthquake direction considered.

R **Table 4.1** is the carrier system behavior coefficient defined in U_i the earthquake taken into account

is the horizontal displacement calculated according to the reduced earthquake loads on the i'th floor of the building.

(b) For building carrier system **4.8** In the calculation made according to A_{ie} , **4.8.1.2** According to for each YM vibration mode, at the i'th floor of the building, at the location of the element or equipment, the relevant

absolute modal floor acceleration calculated in the direction A_{ien} . It will be taken as the square root of the sum of the squares of 's. A_{ien} **Equation (6.3)** will be calculated with .

$$A_{ien} = (R/I) \cdot \frac{2\pi}{T_n} \cdot u_i \quad (6.3)$$

Here T_n is the natural vibration period of the building, u_i is the earthquake taken into account shows the horizontal modal displacement calculated according to reduced earthquake loads at the i 'th floor of the building.

Table 6.1. Magnification and Behavior Coefficients for Architectural Elements

Architectural Element	B_{to}	R_{to}
Non-structural masonry interior walls and partitions	1.0	1.5
Other non-structural interior walls and partitions	1.0	2.5
Cantilever elements without lateral support or with lateral support below the centre of gravity (parapets, cantilever interior walls, chimneys etc.)	2.5	2.5
Cantilever elements with lateral support above the centre of gravity (parapets, cantilever external walls, chimneys etc.).	1.0	2.5
External walls and connections	1.0	2.5
Facade cladding panels	1.0	1.5
Attics independent of the building system	2.5	3.5
Suspended ceilings	1.0	2.5
Storage cabinets and laboratory equipment	1.0	2.5
Access floors	1.0	1.5
Billboards	2.5	2.5
Other rigid architectural elements	1.0	2.5
Other flexible architectural elements	2.5	2.5

(c) For building carrier system 4.8.3 or 5.8 As a result of the earthquake calculation in the time history to be made according to , the average of the absolute maximum accelerations calculated for eleven ground motion sets in the relevant direction at the location of the element or equipment on the i 'th floor of the building A_{ie} will be defined as.

(d) In special cases where it is deemed necessary to take into account the rigidity and mass of the element or equipment and/or its connections, the element or equipment is located in its location. **6.2.2 (c)** The acceleration function obtained in the time domain as defined in The floor spectrum is derived using the element or equipment T_{to} Spectral acceleration corresponding to the natural period of vibration A_{ie} can be calculated as . T_{to} is the natural vibration period **Equation (6.4)** will be calculated with .

$$T_{to} = 2\pi \sqrt{\frac{m_{to}}{k_{to}}} \quad (6.4)$$

Here k_{to} is the effective stiffness coefficient of the element or hardware and/or its connection. In this case **Equation (6.1)** Magnification Factor defined in $B_{to} = 1$ will be taken.

6.2.3 – Equation (6.1) Equivalent earthquake load calculated with, **Equation (6.5)** It will not be taken smaller than the value given in .

$$F_e \geq 0.3 m_o \text{WORK}_{0.5g} \quad (6.5)$$

Table 6.2. Magnification and Behavior Coefficients for Mechanical and Electrical Equipment

Mechanical or Electrical Equipment	B_{to}	R_{to}
Mechanical systems made of sheet metal such as ventilation, heating and cooling systems.	2.5	6.0
Mechanical systems made of flexible materials such as water heaters, water coolers, heat exchange systems.	1.0	2.5
Elements such as engines, turbines, pumps, compressors, etc.	1.0	2.5
Elevators and escalator components	1.0	2.5
Generators, transformers, and similar electrical equipment	1.0	2.5
Control panels, instrument cabinets, junction and switchgear boxes, and similar equipment made of thin sheet metal	2.5	6.0
Communication equipment, computers, devices and control systems	1.0	2.5
Chimneys, towers, cooling and electrical systems built into the roof supported horizontally below the center of gravity	2.5	3.0
Chimneys, towers, cooling and electrical systems built into the roof supported horizontally above the center of gravity	1.0	2.5
Lighting systems	1.0	1.5
Other mechanical and electrical systems	1.0	1.5
Vibration isolated equipment	2.5	2.5
Internally insulated equipment.	2.5	2.0
Suspended equipment supported by or internally insulated vibration isolating suspension systems.	2.5	2.5
Pipe and tube systems made of materials with low deformation capacity (e.g. cast iron, glass, rigid plastic)	2.5	3.0
Channel distribution systems made of materials with high deformation capacity and connected to each other by welding or brazing.	2.5	9.0
Channel distribution systems made of materials with high deformation capacity and connected to each other by materials other than welding or brazing.	2.5	6.0
Channel distribution systems made of materials with low deformation capacity (e.g. cast iron, glass, rigid plastic etc.)	2.5	3.0
Electrical transmission pipes, water pipes, rigidly connected cable trays	1.0	2.5
Suspended cable trays	2.5	6.0

6.2.4 –Equivalent earthquake load is the dead load of the element or equipment, the service loads carried by the element and (separately in two horizontal earthquake directions perpendicular to each other) $\pm 0.3 m_o \text{WORK}_{0.5g}$

It will be applied with a vertical equivalent earthquake load of magnitude.

6.2.5 –For elements or equipment attached to the building carrier system with suspension type connections (such as chains, cables), instead of the equivalent earthquake loads specified above, a load equal to 1.4 times the weight of the element or equipment will be applied in both horizontal and vertical directions.

6.3. LIMITATION OF DISPLACEMENTS

6.3.1 –In cases where non-structural elements and equipment are connected to two separate points of the same structure that may undergo different displacements or to points in two separate load-bearing systems, the effects resulting from the relative displacements occurring between the connection points during the earthquake should also be taken into account. Relative displacements, **Chapter 4** or **Chapter 5** It will be obtained by using the results of the calculation method to be applied according to .

6.3.2 –Effective relative story drift for non-structural elements and equipment for typical (X) earthquake direction δ_{x0} **Equation (6.6)** will provide.

$$\delta_{to}^{(x)} \left(\leq h_x - h_y \right) \leq \frac{\delta_{i,max}^{(x)}}{h_i} \quad (6.6)$$

Here h_x And h_y upper and lower connection points of non-structural members and equipment, respectively the height from the relevant floor base, $\delta_{(x)i,max}/h_i$, depending on the method used **4.9.1.3** permission in shows the maximum given relative storey drift ratio.

CHAPTER 7 – CAST-IN-SITU REINFORCED CONCRETE BUILDING UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR THE DESIGN OF CARRIER SYSTEMS

7.0. ICONS

In the dimensional expressions in this section where the following symbols are used, the forces are: *Newton*[[N], lengths *millimeter*[mm] and stresses *Mega Pascal*[MPa] = [N/mm²] unit.

A_c	= Gross cross-sectional area of the column or wall end area
A_{ch}	= Each piece of curtain, floor or floor of a gapless curtain, a curtain with a tie beam gross cross-sectional area of each slab piece in a hollow slab
A_{ck}	= Core concrete area within the measurement taken from outside to outside of the winding reinforcement
$\sum A_{to}$	= Effective shear area in the earthquake direction considered at any floor
$\sum A_g$	= At any floor, parallel to the earthquake direction taken into account The sum of the cross-sectional areas of the load-bearing system elements working as a curtain = the cross-
A_{os}	sectional area of the spiral/winding reinforcement
$\sum A_p$	= Sum of plan areas of all floors of the building
A_{s1}	= To accommodate the negative moment of the beam on one side of the column-beam joint Total area of tensile reinforcement placed on top for
A_{s2}	= Column-beam node point A_{s1} On the other side, the positive moment of the beam with respect to Total area of tensile reinforcement placed underneath to meet
A_h	= Transfer equipment
A_{sb}	= Connection fittings
A_{sd}	= Total area of each cross reinforcement bundle in the tie beam
$A_{s,h}$	= along the height corresponding to the transverse reinforcement spacing, at the column or shear wall end considering the cross-sectional area values of all stirrup arms and crossties in the region b_k The sum of the projections perpendicular to
A_w	= Column cross-section effective body area (column projections perpendicular to the earthquake) (excluding area)
$\sum A_w$	= Effective body areas of column cross-section at any floor A_w The sum of the 's
A_{wp}	= Vertical punching reinforcements (stirrups) located on the punching perimeter total cross-sectional area of the arms or shear keys
a	= Between the stirrup arms and/or crossties in the column or wall end area horizontal distance
b_j	= In the earthquake direction considered, the beam stuck in the joint area If it is the same width as the column or extends from both sides of the column, the column width; otherwise, twice the smaller of the distances from the vertical center axis of the beam to the column edges (cannot exceed the sum of the beam width and the height of the joint)

b_k	= For each of the horizontal directions perpendicular to each other, column or wall end area cross-sectional dimension of the core (distance between the outermost cross-bar axes)
b_w	= Beam body width, bulkhead body thickness
D	= Circular column core diameter (distance between spiral/winding reinforcement axes) = Excess Strength Factor
d	= Useful height of the beam =
TO	Earthquake effect
f_{CD}	= Design compressive strength of concrete
f_{ck}	= Characteristic cylinder compressive strength of concrete
f_{ctd}	= Design tensile strength of concrete
f_{yd}	= Design yield strength of longitudinal reinforcement
f_{yk}	= Characteristic yield strength of longitudinal reinforcement
f_{ywd}	= Design yield strength of transverse reinforcement
f_{ywk}	= Characteristic yield strength of transverse reinforcement
H_{cr}	= Curtain critical height
H_w	= Total curtain wall measured from the top of the foundation or ground floor slab height
G	= Constant load effect
h	= Cross-sectional dimension of the column in the earthquake direction considered
h_k	= Beam height
-b	= Interlocking length given for tensile reinforcement in TS 500
-n	= The free height of the column between the beams, the height of the beam between the column or wall the free space between the faces
-w	= Plan length of the curtain or the tie beam curtain part
M_a	= At the lower end of the free height of the column, the basis for calculating the column shear force moment taken
$(M_d)_t$	= Vertical loads and earthquake multiplied by load coefficients at the base section of the curtain Calculated moment under the joint effect of the loads
M_{pa}	= At the lower end of the free height of the column f_{ck} , f_{yk} and increase in strength of steel moment capacity calculated taking into account
M_{pi}	= Left end of beam i on the column face f_{ck} , f_{yk} and considering the increase in strength of steel Positive or negative moment capacity calculated by taking
M_{pj}	= Right end of beam j on the column face f_{ck} , f_{yk} and considering the increase in strength of steel negative or positive moment capacity calculated by taking
$\sum M_p$	= The beams joining the node oppose the rotation of the node in the same direction. sum of moment capacities

M_{pu}	= At the upper end of the free height of the column f_{ck} , f_{yk} and increase in strength of steel moment capacity calculated taking into account
$(M_p)_t$	= At the base section of the curtain f_{ck} , f_{yk} and considering the increase in strength of steel calculated moment capacity
$M_{r.a.}$	= At the lower end of the free height of the column or wall f_{cd} And f_{yd} According to calculated bearing capacity moment
M_{ri}	= Left end of beam/On the column or curtain face f_{cd} And f_{yd} Calculated according to positive or negative bearing capacity moment
M_{rj}	= Right end of beam/On the column or curtain face f_{cd} And f_{yd} Calculated according to negative or positive bearing capacity moment
$(M_r)_t$	= At the base section of the curtain f_{cd} And f_{yd} Bearing capacity moment calculated according to
M_{ru}	= At the upper end of the free height of the column or wall f_{cd} And f_{yd} According to calculated bearing capacity moment
$M_{\bar{u}}$	= At the upper end of the column's free height, in the calculation of column shear force based moment
N_d	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Calculated axial force under
N_{dm}	= Under the combined effect of vertical loads and earthquake loads (live loads in TS 498) The maximum of the calculated axial pressure forces (taking into account the live load reduction coefficients defined for
Q	= Live load effect
S	= Transverse reinforcement spacing, spiral/wound reinforcement
U_p	pitch = Punching perimeter in the slab
V_c	= Contribution of concrete to shear strength
V_d	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Calculated shear force under
V_{dy}	= Vertical loads not multiplied by load coefficients at any section of the beam simple beam shear force generated
V_{to}	= The basis for calculating transverse reinforcement in columns, beams, joints and walls. shear force
V_{first}	= Building/All columns on the 'th floor' are in line with the earthquake taken into account. Chapter 4 The sum of the cutting forces calculated according to
V_{work}	= Building/On the 'th floor', Eq.7.3 Both the bottom and top nodes of in the columns provided at the points, in the earthquake direction taken into account Chapter 4 The sum of the cutting forces calculated according to
V_{arm}	= Above and below the node Chapter 4 Column shear calculated according to the smaller of the forces
V_r	= Shear force resistance of column, beam or wall section

V_t	= Chapter 4 Total earthquake load acting on the building (base shear force) according to
V_{good}	= Simple beam support shear force under vertical loads in seismic condition
α_i	= Any I calculated on the 'th floor' $V_{work} / V_{First rate}$
β_v	= Dynamic magnification factor of shear force in the wall =
φ	Reinforcement diameter
γ	= Angle of the cross reinforcement bundle used in the connection beam with the
ρ	horizontal = Tensile reinforcement ratio at the top or bottom of the beam support
μ	= Coefficient of cutting friction
ρ_s	= Volumetric ratio of spiral reinforcement in column [$\rho_s = 4A_{os}(\rho_s)$]
$\rho_{s,h}$	= Volumetric ratio of horizontal web reinforcement in the wall [$(\rho_{s,h})_{min} = 0.0025$]

7.1. SCOPE

7.1.1 –Dimensioning and equipping of the load-bearing system elements of reinforced concrete buildings under the effect of earthquakes shall be carried out primarily in accordance with the rules specified in this section, together with the relevant standards and regulations in force on this subject. Rules regarding reinforced concrete building foundations **Chapter 16** is given in .

7.1.2 –Horizontal load-bearing systems of reinforced concrete buildings within the scope of this section may consist of only frames, only walls, or a combination of frames and walls.

7.1.3 –Reinforced concrete buildings with a concrete class higher than C80 are outside the scope of this regulation. In addition, steel-reinforced concrete composite columns in which steel profiles are used as reinforcement in the carrier system elements are outside the scope of this section. **Chapter 9** is subject to the rules given in .

7.2. GENERAL RULES

7.2.1. Classification of Reinforced Concrete Carrier Systems

Horizontal load-bearing systems of reinforced concrete buildings in terms of their earthquake-resistant behavior, **7.2.1.1, 7.2.1.2, 7.2.1.3** It is divided into three classes defined in .

7.2.1.1 –Below **(a), (b) And (c)** Reinforced concrete carrier systems specified in, *Systems with High Ductility Level* defined as:

(a) 7.3, 7.4 And 7.5 Frame type carrier systems formed by columns and beams sized and equipped according to the rules specified in ,

(b) 7.6 Carrier systems consisting of void-free or void-filled (with connecting beams) curtain walls, equipped according to the dimensions,

(c) Above **(a) And (b)** Shear wall-framed load-bearing systems created by combining two types of systems in .

7.2.1.2 –Below **(a), (b) And (c)** Reinforced concrete carrier systems specified in, *Systems with Limited Ductility Level* defined as:

(a) 7.7,7.8And7.9Frame type carrier systems formed by columns and beams sized and equipped according to the rules specified in ,

(b) 7.10Carrier systems consisting of gapless curtains sized and equipped according to ,

(c)Above**(a)**And**(b)**Shear wall-framed load-bearing systems created by combining two types of systems in .

7.2.1.3 –Ductility Level Mixed Structural SystemsThese are systems created by using frame carrier systems with limited ductility level together with reinforced concrete shear walls with high ductility level.

7.2.1.4 –Conditions for high ductility, limited and mixed load-bearing systems4.3.4is given in.

7.2.2. Relevant Standards

Cast-in-place reinforced concrete load-bearing systems within the scope of this section, together with the rules specified in this Section,**Part 2,3,4And5**The calculation rules in , the loads foreseen in TS 498, the rules in TS 500, TS 708 and TS EN 13670 will be used in the project. In special cases where the rules given in the relevant standards are different, the rules in this section will be taken as basis.

7.2.3. Section Stiffnesses to be Used in the Carrier System Calculation

In the carrier system account**4.5.8**'The effective section stiffnesses of the load-bearing system elements given in will be used.

7.2.4. Method to be used in Section Calculations

Within the scope of this section, in earthquake resistant dimensioning of reinforced concrete elements and reinforcement calculations, it is mandatory to use the assumptions given in TS 500, the stress distribution in concrete and the elasticity modulus. In cases where concrete classes higher than C50 are used, the stress distribution in concrete and the elasticity modulus will be used in the calculations of the bending moment of the sections and the bearing capacity under axial load as given in TS EN 1992-1.

7.2.5. Material

7.2.5.1 –Concrete with a strength lower than C25 cannot be used in all reinforced concrete buildings to be constructed within the scope of this Regulation.

7.2.5.2 –In all reinforced concrete buildings to be constructed within the scope of this Regulation, it is mandatory to use concrete that has been quality controlled, maintained and placed with a vibrator according to the definition in TS 500. However, self-compacting concrete can also be used.

7.2.5.3 –In reinforced concrete elements that will withstand the effects of earthquakes;

(a)Concrete classes C25 to C80 will be used from the concretes given in TS EN 206. If the concrete compressive strength needs to be determined at ages other than 28 days for special purposes, TS EN 206 will be taken as basis.

(b)B420C and B500C ribbed reinforcement steels given in TS 708 will be used. In addition to the conditions given in TS 708, the "tensile strength/yield strength" ratio should be less than 1.35.

be ($R_m/R_{to} < 1.35$) and S420 concrete steel can also be used provided that the equivalent carbon content does not exceed 0.55%.

7.2.6. Interlocking Length of Rebars

Unless otherwise stated in this section, the required interlocking lengths for reinforced concrete reinforcement shall be calculated according to the rules given in TS 500.

7.2.7. Welded, Sleeved and Chemically Anchored Joints and Connections

7.2.7.1 –It is mandatory that the lap welded joints of longitudinal reinforcements are made by welders certified by the relevant organizations. Butt weld joints are not allowed. The carbon equivalent of the reinforcement steel to be welded shall not exceed the 0.50% limit value given in TS 500.

7.2.7.2 –Welded and sleeved longitudinal reinforcement splices have adequate performance under monotonic and cyclic loads. **Chapter 8 ANNEX 8A** The rules given in , shall be documented based on international standards and regulations. The characteristic breaking strength of the joint, found by testing, shall not be lower than the characteristic breaking strength given in TS 500, taking into account the net cross-sectional area of the added reinforcement bars in the case of a sleeved joint.

7.2.7.3 –Welding of transverse reinforcement to longitudinal reinforcement is not allowed.

7.2.7.4 –It is not allowed to weld steel window and door frames, dowels, connection plates, installation elements, machinery and equipment to longitudinal and transverse reinforced concrete reinforcement.

7.2.7.5 –Technical regulations such as internationally accepted standards and regulations will be used in the design and application of sprout planting and reinforcement attachments.

7.2.8. Special Earthquake Stirrups and Crossties

Stirrups used in columns, column-beam joint areas, shear wall end areas and beam wrapping areas of all reinforced concrete systems with high or limited ductility levels. *special earthquake stirrup*, and the kippers *special earthquake kipper* Conditions required for special earthquake stirrups and crossties **Figure 7.1** is given in .

7.2.8.1 –Special earthquake stirrups must be placed at both ends. *135 degree curved* There will be hooks. In special earthquake crossties, a 90 degree curved hook can be made at one end. In this case, crossties with hook bends of 135 degrees and 90 degrees on one face of the column or wall will be arranged in a staggered manner both horizontally and vertically. The inner bend diameter of the 135 degree curved hooks will be at least 5ϕ , where ϕ represents the transverse reinforcement diameter. The straight length of the hooks will not be taken less than 6ϕ and 80 mm in ribbed bars from the last tangent point in the bend (**Figure 7.1**).

7.2.8.2 –Special earthquake ties will grip the longitudinal reinforcement from the outside and their hooks will close around the same longitudinal reinforcement. The diameter and spacing of the special earthquake ties will be the same as the diameter and spacing of the stirrups. The ties will definitely surround the longitudinal reinforcement and the external stirrup at both ends. The ties and ties will be tightly connected to the longitudinal reinforcement so that they will not slip while the concrete is being poured.

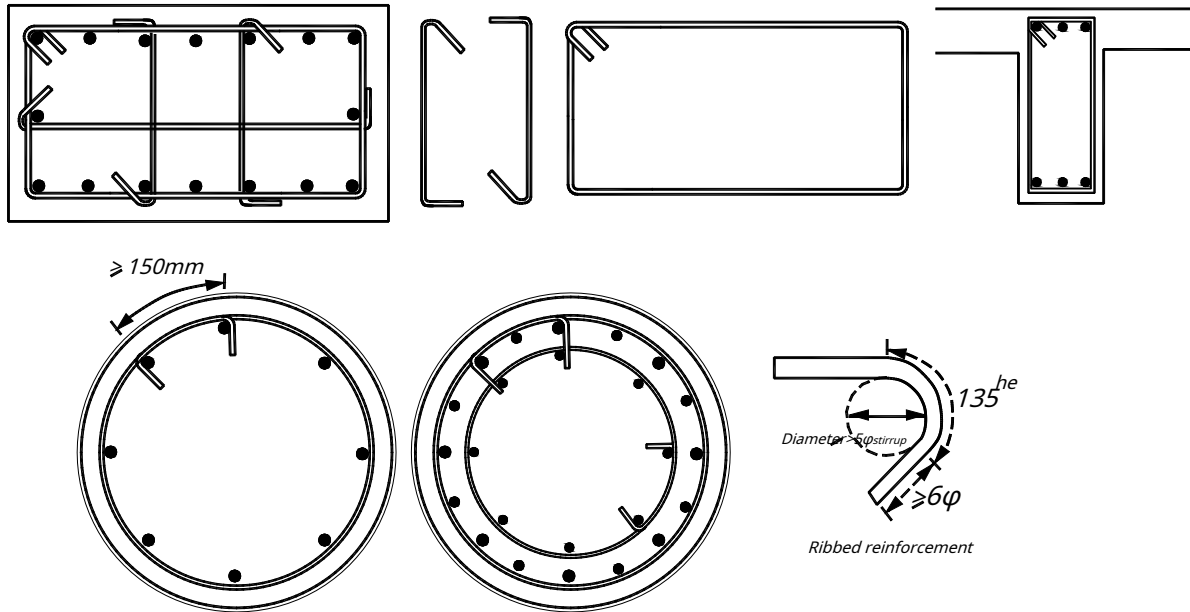


Figure 7.1

7.3. COLUMNS WITH HIGH DUCTILITY LEVEL

7.3.1. Cross-Section Conditions

7.3.1.1 –The minimum cross-section dimension of rectangular columns shall not be less than 300 mm and the diameter of circular columns shall not be less than 350 mm.

7.3.1.2 –Gross cross-sectional area of the column, N_{dm} Taking into account the live load reduction coefficients defined for live loads in TS 498, G And Q vertical loads and T the combined effect of the earthquake effect $G+Q+T$ being the maximum of the axial compressive forces calculated under, $A_c \geq N_{dm}/(0.40 f_{ck})$ condition will be met.

7.3.2. Longitudinal Reinforcement Conditions

7.3.2.1 –Longitudinal reinforcement area in columns shall not be less than 1% and more than 4% of the gross section. Reinforcement area shall not be less than $\phi 14$ in columns and less than 6 in circular columns. will not be used.

7.3.2.2 –In sections with overlapped joints, the total longitudinal reinforcement ratio will not exceed 6%.

7.3.3. Arrangement of Longitudinal Reinforcement

7.3.3.1 –Lap splices of column longitudinal reinforcement shall be made in the middle third of the clear height of the column. Length of the lap splice -bNot to be less than. Overlapping

The spacing of transverse reinforcement to be placed along the splice shall not be greater than 1/3 of the smallest dimension of the column and 150 mm.

7.3.3.2 –In case of change of column section between floors, the slope of longitudinal reinforcement with respect to vertical in column-beam joint area shall not be greater than 1/6. In case of greater change of section or in the columns of the top floor; the longitudinal reinforcement of the column below shall not be

The interlocking length inside the beam on the opposite side is 1.5-It will not be taken shorter than ' and 40φ. On the opposite side, the beam

If not available, the clamping will be provided by making a downward bend on the opposite face of the column if necessary. The length of the 90 degree horizontal hook or the downward bending vertical hook will be at least 12φ (**Figure 7.2**).

7.3.3.3 –The longitudinal distance between sleeved or welded joints made in adjacent longitudinal reinforcement shall not be less than 600 mm.

7.3.4. Transverse Reinforcement Conditions

7.3.7.6 Conditions regarding the minimum transverse reinforcement to be used in columns, unless a more unfavourable situation occurs, *colon wrapping regions* for **7.3.4.1** and in *colon mid region* for **7.3.4.2** is given in (**Figure 7.3**). Throughout the entire column, **7.2.8** defined in *special earthquake stirrups* And *special earthquake ties* will be used.

7.3.4.1 –Special at the lower and upper ends of each column *hugging areas* The length of each confinement zone shall not be less than 1/6 of the column free height, 1.5 times the largest section dimension of the column and 500 mm, measured upwards from the top floor level or downwards from the bottom face of the beam with the largest height connected to the column. In cantilever columns, the confinement zone shall be created at the lower end of the column and its length shall not be less than twice the largest dimension of the column. The conditions for transverse reinforcement to be used in confinement zones are given below. **(a)** to **(d)** These reinforcements shall be continued inside the foundation for a height not less than the minimum dimension of the column. However, in columns supported on cup foundations, the transverse reinforcement in the confinement zone shall be continued along the height of the cup.

(a) Transverse reinforcement with diameter smaller than φ8 shall not be used in confinement zones., The spacing of stirrups and crossties in the longitudinal direction shall not be greater than 1/3 of the smallest section dimension, 150 mm, greater than six times the diameter of the longitudinal reinforcement and less than 50 mm. The horizontal distance between the stirrup arms and/or crossties shall not be a, shall not be taken larger than 25 times the stirrup diameter. The pitch of continuous circular spirals shall not be larger than 1/5 of the hub diameter and 80 mm. In circular columns, all wrapping reinforcement shall be provided by circular transverse reinforcement placed around the perimeter.

(b) In stirrup columns $N_d > 0.20 A_c f_{ck}$ (pressure) in the wrapping areas minimum total transverse reinforcement area, **Equation (7.1)** It will be calculated to provide the unfavorable condition given in . In this calculation, the column core size b_k , both directions will be taken into account separately for (**Figure 7.3**):

$$\begin{aligned} A_{s,h} &\geq 0.30 s b_k \left[\left(\frac{A_c}{A_c} \right) - 1 \right] \left(\frac{f_{ck}}{f_{yk}} \right) \\ A_{s,h} &\geq 0.075 s b_k \left(\frac{f_{ck}}{f_{yk}} \right) \end{aligned} \quad (7.1)$$

(c) In circular reinforced columns $N_d > 0.20 A_c f_{ck}$ (pressure) wrapping in case of Minimum volumetric ratio of transverse reinforcement in zones, **Equation (7.2)** It will be calculated to provide the least favorable of the conditions in .

$$\begin{aligned} \rho_s &\geq 0.45 \left[\left(\frac{A_c}{A_c} \right) - 1 \right] \left(\frac{f_{ck}}{f_{yk}} \right) \\ \rho_s &\geq 0.12 \left(\frac{f_{ck}}{f_{yk}} \right) \end{aligned} \quad (7.2)$$

(d) $N_d \leq 0.20 A_c f_{ck}$ in the column wrapping regions in case of **Equation (7.1)** And **Equation (7.2)** At least 2/3 of the transverse reinforcement given shall be used as minimum transverse reinforcement.

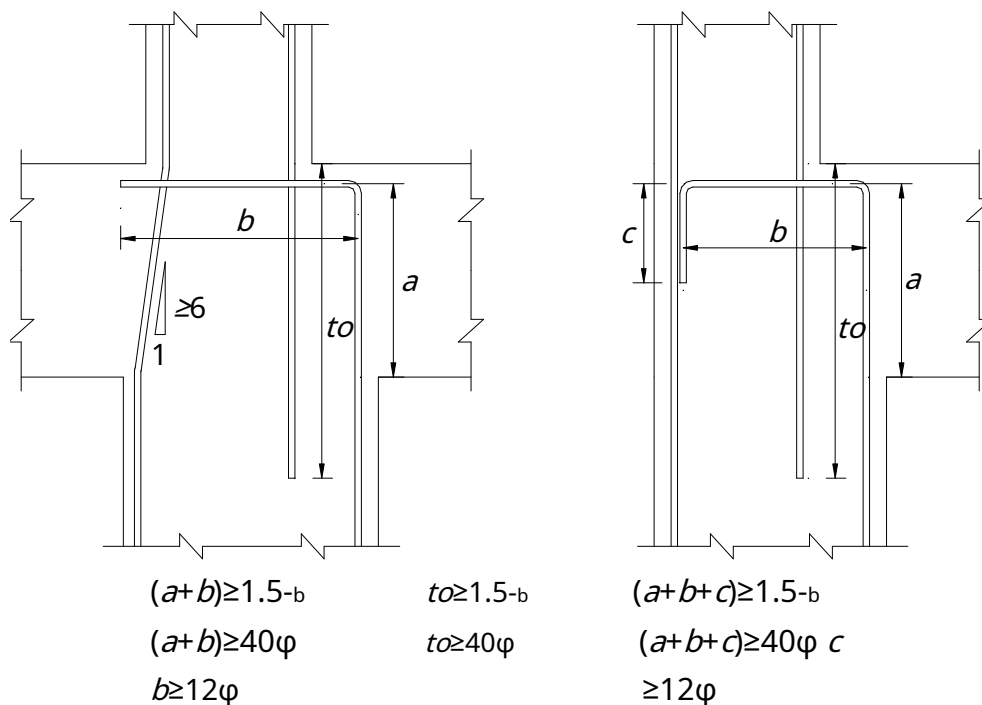


Figure 7.2

7.3.4.2 - Column middle region, is the region between the wrapping zones defined at the lower and upper ends of the column (**Figure 7.3**). Transverse reinforcement with diameter less than $\phi 8$ in the middle area of the column. The stirrup, crossbar or spiral spacing along the column shall not be taken as greater than half of the smallest cross-section dimension and 200 mm. The horizontal distance between the stirrup arms and/or crossbars shall be a , It shall not be more than 25 times the stirrup diameter.

7.3.5. Condition that Columns are Stronger than Beams

7.3.5.1 -In load-bearing systems consisting of only frames or a combination of shear walls and frames, the total of the bearing capacity moments of the columns connected to each column-beam node shall be at least 20% greater than the total of the bearing capacity moments of the beams connected to that node in the sections on the column face. **Figure 7.4**):

$$(M_{r.a.} + M_{r.u}) \geq 1.2 (M_{r.i} + M_{r.j}) \quad (7.3)$$

7.3.5.2 - Equation (7.3), will be applied separately for each earthquake direction and in a way that will give unfavorable results for both directions of the earthquake (**Figure 7.4**). Column bearing capacity moments. In the calculation, the one that minimizes these moments in accordance with the direction of the earthquake N_d axial forces shall be taken into account.

7.3.5.3 - Equation (7.3) Special cases regarding the application of are listed below (a), (b) And (c) It is stated in:

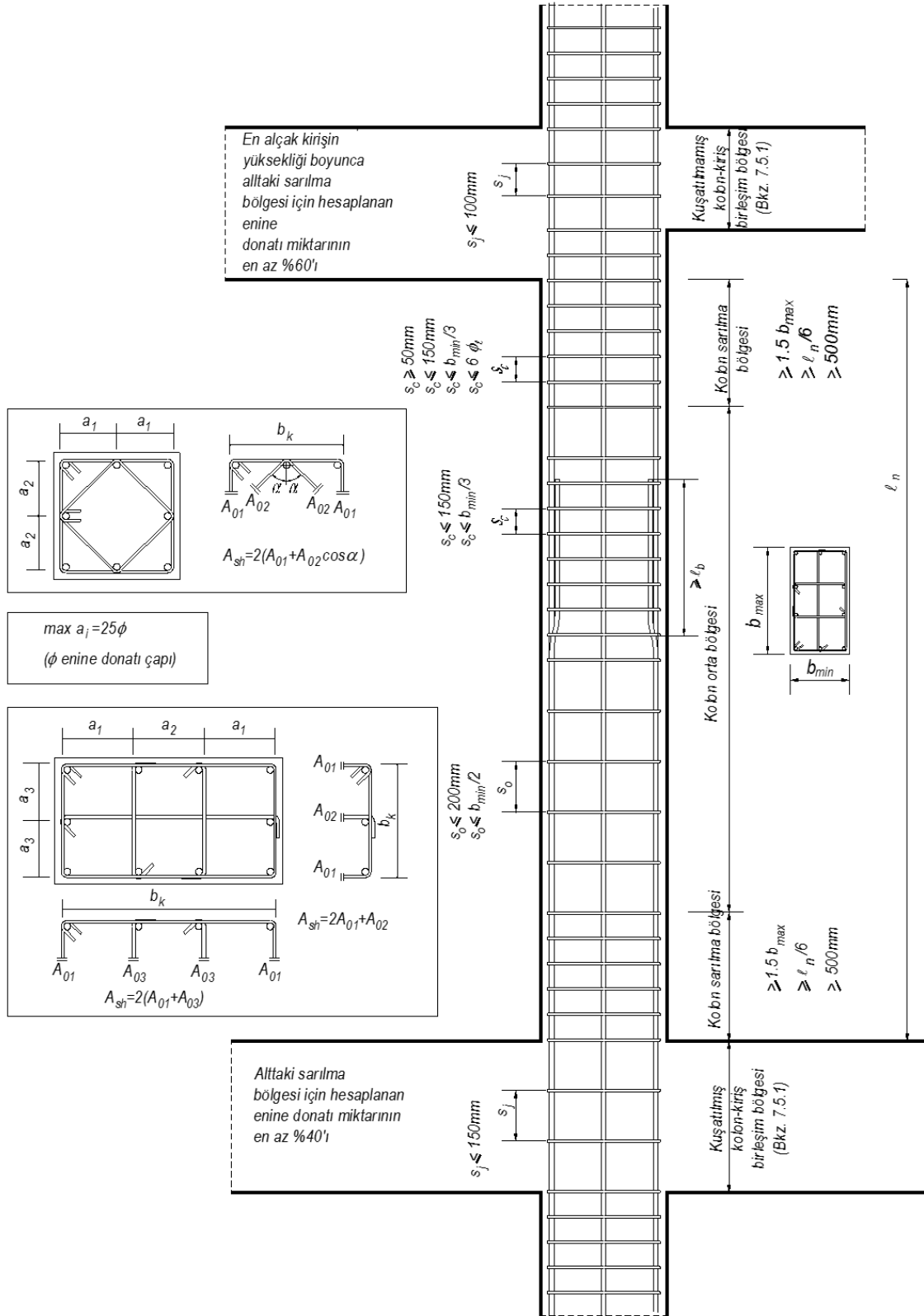


Figure 7.3

(a) In both columns that join the node point $N_d \leq 0.10 A_c f_{ck}$ in case of occurrence, **Equation (7.3)** It is not mandatory to provide .

(b) In single-storey buildings and in multi-storey buildings, at the joints where the columns do not continue to the upper floor **Equation (7.3)** It will not be checked whether or not the is provided.

(c) If the wall to which the beams are attached acts like a column in the weak direction, **Equation (7.3)** It will not be checked whether or not the is provided.

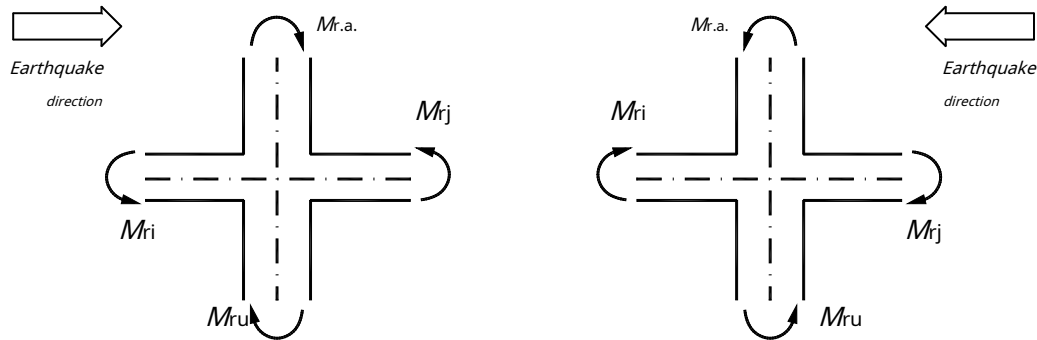


Figure 7.4

7.3.6. The Condition of Columns Being Stronger Than Beams is Not Provided in Some Columns

7.3.6.1 – In carrier systems consisting of only frames or a combination of shear walls and frames, any movement of the building in the earthquake direction taken into account. On the 'th floor, **Equation (7.4)** Provided that , at some nodes below and/or above the relevant floor **Equation (7.3)** It is permissible that 's not provided.

$$\alpha_1 = V_{work} / V_{First} \geq 0.70 \quad (7.4)$$

$N_d \leq 0.10 A_c f_{ck}$ at the ends of the columns that meet the condition, **Equation (7.3)** Even if not provided, these columns V_{work} can be taken into account in the calculation of .

7.3.6.2 – Equation (7.4) If is provided, $0.70 \leq \alpha_1 \leq 1.0$, **Equation (7.3)**'s both bending moments and shear forces acting on the columns provided at both the lower and upper nodes ($1/\alpha_1$) will be increased by multiplying the ratio. **Equation (7.3)** Columns that do not satisfy , It will be reinforced under the vertical load and earthquake effects occurring in the sections.

7.3.6.3 – On any floor **Equation (7.4)** If this cannot be provided, all frames in carrier systems consisting of only frames or a combination of curtains and frames **Table 4.1** According to *ductility level limited frame* will be taken into consideration. **7.2.1.3** As mentioned above, it is also possible to use frames with limited ductility level together with walls with high ductility level as mixed ductility level systems.

7.3.7. Shear Safety of Columns

7.3.7.1 – Shear force to be used as basis for transverse reinforcement calculation in columns V_{to} , **Equation (7.5)** will be calculated with .

$$V_{to} = (M_a + M_u) / l_n \quad (7.5)$$

Equation (7.5)in M_a and M_u To calculate , at the lower and/or upper ends of the column
Equation (7.3) In case of providing **7.3.7.2**, if not provided, **7.3.7.3** will be applied (**Figure 7.5**). With vertical loads D calculated from the earthquake augmented by the total cutting force, **Equation (7.5)** calculated with V_o If it is less than , V_o
This cutting force will be used instead.

Kat No.	M_u 'nün hesaplanması		M_a 'nün hesaplanması	
	Kolon üst ucunda Denk. 7.3'ün sağlaması durumu	Kolon üst ucunda Denk. 7.3'ün sağlamaması durumu	Kolon alt ucunda Denk. 7.3'ün sağlamaması durumu	Kolon alt ucunda Denk. 7.3'ün sağlaması durumu
i+1				
i				
i-1				
	$\Sigma M_p = M_{pi} + M_{pj}$ $M_u = \frac{M_{hu(i)}}{M_{hu(i)} + M_{ha(i+1)}} \Sigma M_p$			
	$M_{hu(i)}$ i'inci kat kolonu üst ucunda Bölüm 3 ve Bölüm 4'e göre bulunan moment $M_{ha(i)}$ i'inci kat kolonu alt ucunda Bölüm 3 ve Bölüm 4'e göre bulunan moment			

Figure 7.5

7.3.7.2 - Equation (7.3) The sum of the moment capacities at the ends of the beams connected to the node where is provided ΣM_p moment will be calculated:

$$\Sigma M_p = M_{pi} + M_{pj} \quad (7.6)$$

In cases where more precise calculations are not made, $M_{pi} \approx 1.4 M_{ri}$ and $M_{pj} \approx 1.4 M_{rj}$ receivable. ΣM_p moment, at the ends of the columns that meet at the node **Chapter 4** According to , the moments obtained will be distributed to the columns in proportion and as a result of the distribution, the lower or upper parts of the relevant column will be
The moment obtained at the tip, **Equation (7.5)** in M_a or M_u will be taken into consideration.
For both directions of the earthquake **Equation (7.6)** will be applied separately and the largest obtained in each direction ΣM_p value will be taken into account in distribution.

Equation (7.3) Despite the fact that 's provided **Equation (7.5)** in M_a or M_u 's account, to stay on the safe side, **7.3.7.3** It can also be done according to .

7.3.7.3 – Equation (7.3) The moments at the ends of the columns connected to the node where is not provided will be calculated as the moment capacities of the columns and **Equation (7.5)** in M_a and/or M_u . Moment capacities are used in cases where more precise calculations are not made. $M_{pa} \approx 1.4 M_{r.a.}$ And $M_{pu} \approx 1.4 M_{r.u.}$ can be taken as M_{pa} and M_{pu} . In calculating the moments, the one that makes the moments the largest in accordance with the direction of the earthquake axial forces shall be taken into account.

7.3.7.4 – At the lower end of the columns connected to the foundation M_a moment too, **7.3.7.3** According to will be calculated as moment capacities.

7.3.7.5 – Equation (7.5) The cutting force calculated by V_o , shear force calculated under the combined effect of vertical loads and earthquake loads multiplied by load coefficients V_d smaller than will not be taken and also **Equation (7.7)** will meet the conditions given. **Equation (7.7)** If the second condition in is not met, the earthquake calculation will be repeated by enlarging the section dimensions as necessary.

$$\begin{aligned} V_o &\leq V_r \\ V_o &\leq 0.85 A_w \sqrt{f_{ck}} \end{aligned} \quad (7.7)$$

7.3.7.6 – Column transverse reinforcement V_o Contribution of concrete to shear strength in the calculation according to shear force, V_c , will be determined according to TS 500. However, **7.3.4.1** The column defined in In the calculation of transverse reinforcement in confinement zones, the shear force consisting of only earthquake loads should be greater than half of the total shear force in the seismic situation and at the same time in time $N_d \leq 0.05 A_c f_{ck}$ If the condition is met, the contribution of concrete to the shear strength is $V_c = 0$ will be taken.

7.3.8. Conditions Regarding Short Columns

Short columns may be formed due to the carrier system or due to the gaps left between the columns in the infill walls (**Figure 7.6** In cases where the formation of short columns cannot be prevented, the shear force to be used as the basis for the calculation of transverse reinforcement **Equation (7.5)** will be calculated with .

Equation (7.5) Bending moments at the upper and lower ends of the short column $M_a \approx 1.4 M_{r.a.}$

And $M_u \approx 1.4 M_{r.u.}$ will be calculated as, l_n is taken as the free length of the short column and calculated cutting force **Equation (7.7)** will satisfy the conditions given in . Throughout the short column, **7.3.4.1** Minimum transverse reinforcement and placement conditions defined for the confinement zones of columns shall be applied. In case the infill walls are completely adjacent to the columns, the transverse reinforcement shall be continued throughout the entire story height in columns that become short columns (**Figure 7.6**).

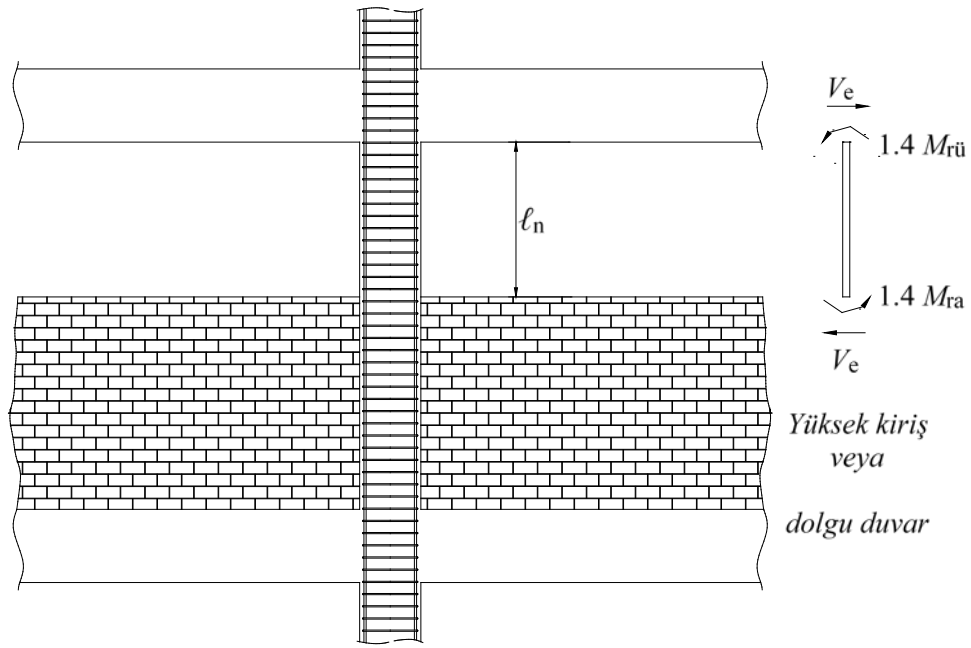


Figure 7.6

7.4. BEAMS WITH HIGH DUCTILITY LEVEL

7.4.1. Cross-Section Conditions

7.4.1.1 -Conditions regarding the cross-sectional dimensions of beams forming a frame with columns or connected to the walls in their own planes are given below.(a)to(d)is given in:

- (a)The beam web width shall be at least 250 mm. The beam web width shall not exceed the sum of the beam height and the perpendicular width of the column or wall to which the beam is connected.
- (b)The beam height shall not be less than 3 times the floor thickness and 300 mm. Elements that do not meet this condition shall be modeled together with the floor elements in the analysis, shall be reinforced as beams, but shall not be accepted as frame beams. In addition, the beam height shall not be more than 3.5 times the beam web width.
- (c)In cases where the beam height is more than 1/4 of the clear span, longitudinal web reinforcement shall be placed on both sides of the beam web along the beam height. The total longitudinal web reinforcement area shall not be less than 30% of the largest of the sum of the upper and lower longitudinal reinforcement areas at the right or left support sections. The web reinforcement diameter shall not be less than 12 mm and its spacing shall not be more than 300 mm. Horizontal web crossties shall be placed at intervals not exceeding 600 mm along the beam height and 400 mm along the beam axis. Similar to the interlocking of longitudinal reinforcements, the interlocking of web reinforcement shall also be carried out.7.4.3.1'of
- (b)And(c)paragraphs shall apply.
- (d)Regarding the beam width and height above(a),(b)And(c)The limitations specified in are not mandatory for reinforced concrete beams connected to columns by hinges, tie beams of tie beam (void) walls and secondary beams fixed to frame beams other than at the beam-column nodes.

7.4.1.2 –In the carrier system elements to be dimensioned and equipped as beams, the design axial pressure force $N_d \leq 0.10 A_c f_{ck}$ It is mandatory to meet the condition. Otherwise, this elements **7.3** It will be sized and equipped as a column according to .

7.4.2. Longitudinal Reinforcement Conditions

7.4.2.1 –For minimum ratio of tensile reinforcement at beam supports **Equation (7.8)** The condition given with will be complied with.

$$\rho \geq 0.8 f_{ctd} / f_{yd} \quad (7.8)$$

7.4.2.2 –The diameter of longitudinal reinforcement shall not be less than 12 mm. At least two reinforcement bars shall be continuously located at the top and bottom of the beam throughout the beam span.

7.4.2.3 –Earthquake Design Class; In load-bearing systems with DTS = 1, 1a and DTS = 2, 2a, beam The lower reinforcement at the support cannot be less than 50% of the upper reinforcement at the same support. However, in other cases this ratio can be reduced to 30%.

7.4.2.4 –The tensile reinforcement ratio in the openings and supports shall not exceed the maximum value given in TS 500 and 2%.

7.4.3. Arrangement of Longitudinal Reinforcement

7.4.3.1 –Conditions for placing and clamping longitudinal reinforcement are given below: **(a), (b)** And **(c)** is given in **(Figure 7.7)**:

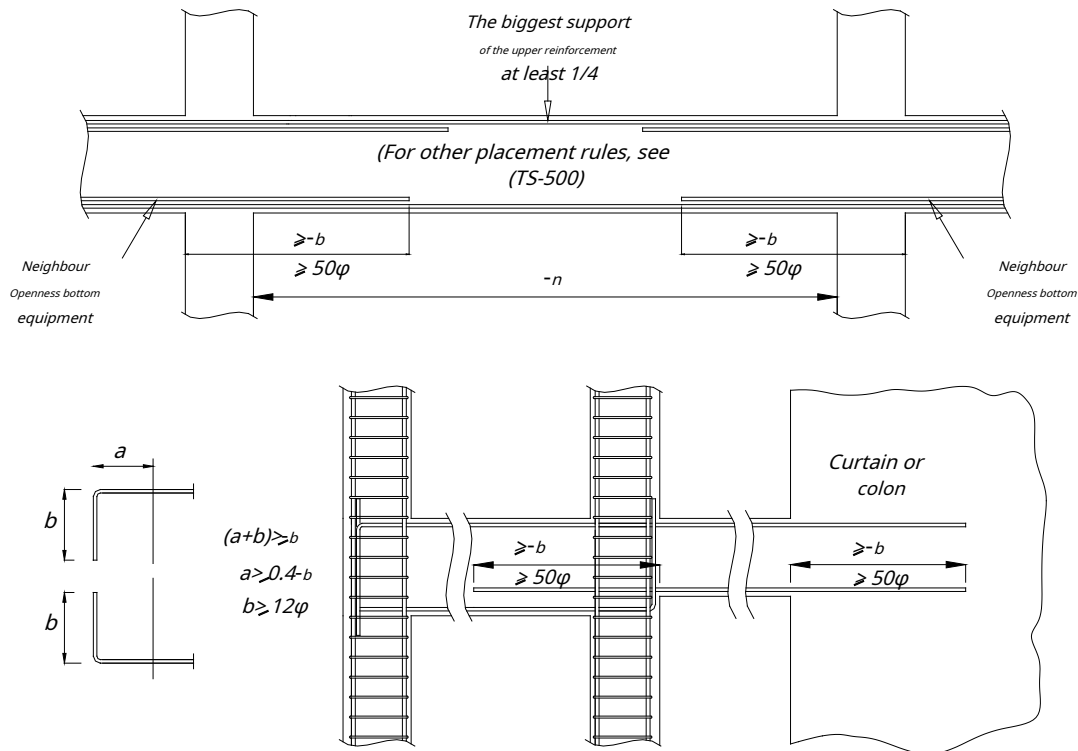


Figure 7.7

(a) At least 1/4 of the larger of the support top reinforcements at both ends of the beam shall be continued continuously along the entire beam. The remaining part of the support top reinforcement shall be arranged in accordance with TS 500 so that no uncompensated moment is left along the beam.

(b) In cases where the beams connected to the column do not continue on the other side of the column, the lower and upper reinforcement in the beams will be extended to the opposite surface of the column's core wrapped with stirrups and bent 90 degrees from the inside of the stirrups. In this case, the total length of the horizontal part of the longitudinal reinforcement remaining inside the column and the 90-degree bent vertical part will be the length specified in TS 500.

straight clamping length -b The horizontal part of the 90 degree hook shall not be less than 0.4-b, and the vertical part shall not be less than 12φ. In curtains and a flat clamping length of the measurement

-b In columns with a diameter greater than 50φ, the clamping of longitudinal reinforcement can be achieved straight without making a 90-degree hook.

(c) In case the beams are connected to the columns from both sides, the beam bottom reinforcements shall be at least 50φ from the column face adjacent to the opening, at least as per the interlocking angle given in TS 500.

height -b will be extended. Due to reasons such as the height difference in the beams, this possibility docking in the absence of the above (b) According to paragraph 1, it will be done as defined for cases where the beam does not continue on the other side of the column.

7.4.3.2 – Conditions for adding longitudinal reinforcement are given below: (a) And (b) is given in:

(a) **7.4.4.** Lap splices will not be made in critical areas where the reinforcement is likely to reach yielding, such as beam confinement areas, column-beam joint areas and lower reinforcement areas in the middle of the span as defined in . In places where lap splices will be made outside these areas, **7.2.8** defined in *special earthquake stirrups* The spacing of these stirrups will not exceed 1/4 of the beam height and 100 mm. There is no need to use special earthquake stirrups in the joints of the upper assembly reinforcement outside the confinement areas in the opening.

(b) Sleeved joints or lap welded joints shall be applied by skipping only one reinforcement in a section and the longitudinal distance between the centres of two adjacent joints shall not be less than 600 mm.

7.4.4. Transverse Reinforcement Conditions

In beam supports, the area twice the beam height from the column face, *Hug Zone* is defined as. Throughout this region **7.2.8** defined in *special earthquake stirrups* Transverse reinforcement with diameters smaller than φ8 shall be used in confinement zones.

will not be used and the distance of the first stirrup to the column face will be maximum 50 mm. **7.4.5.3** Unless a less favourable value is obtained according to , the stirrup spacing shall not exceed 1/4 of the beam effective height, eight times the diameter of the smallest longitudinal reinforcement and 150 mm (**Figure 7.8**). Outside the confinement zone, the transverse reinforcement conditions given in TS 500 will be complied with. The spacing of the stirrup arms in the direction perpendicular to the beam axis will not exceed 350 mm.

7.4.5. Shear Safety of Beams

7.4.5.1 – The shear force to be taken as basis for the calculation of transverse reinforcement in beams, V_{to} , the earthquake from the left separately for cases where it acts to the right or from right to left and will give unfavorable results, **Equation (7.9)** will be found with (**Figure 7.9**).

$$V_{to} = V_{dy} \pm (M_{pi} + M_{pj}) / -n \quad (7.9)$$

Moment capacities at beam ends, $M_{pi} \approx 1.4 M_{ri}$ and $M_{pj} \approx 1.4 M_{rj}$ can be taken as. Vertical calculated from the earthquake with loads D and the sum of the increased cutting force, **Equation (7.9)** calculated with V_o . If it is less than V_o , instead of this shear force will be used.

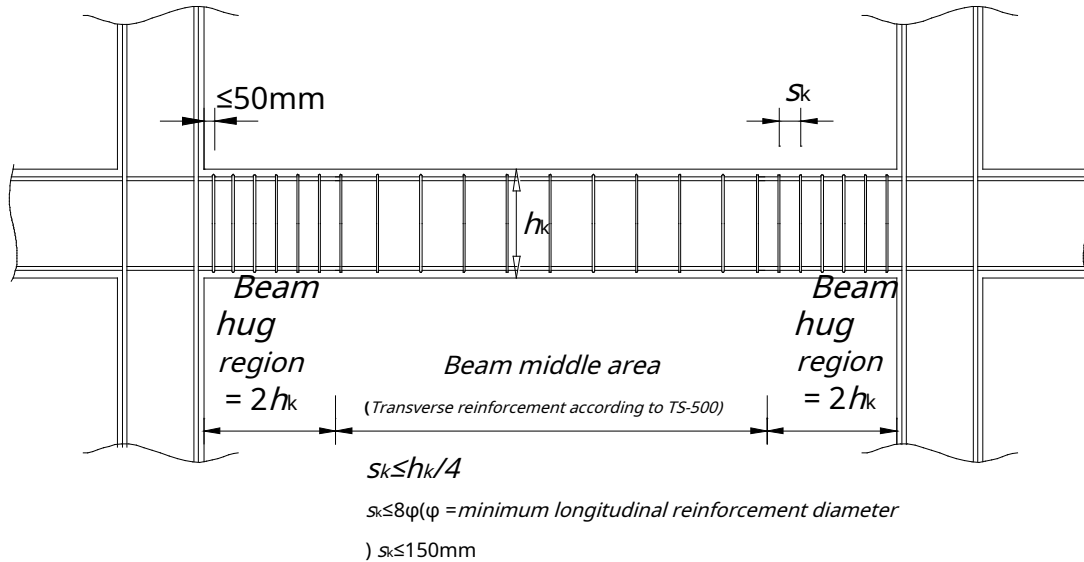


Figure 7.8

7.4.5.2 – 7.4.5.1 The cutting force calculated according to V_o , **Equation (7.10)** conditions given with will provide. **Equation (7.10)** If the second condition in is not met, the earthquake calculation will be repeated by enlarging the section dimensions as necessary.

$$\begin{aligned} V_{to} &\leq V_r \\ V_{to} &\leq 0.85 b_w d \sqrt{f_{ck}} \end{aligned} \quad (7.10)$$

7.4.5.3 – In the calculation of beam transverse reinforcement, the contribution of concrete to shear strength, V_c , to TS 500 will be determined according to. However, **7.4.4** In the calculation of transverse reinforcement in beam confinement zones defined in , if the shear force resulting from only earthquake loads is greater than half of the total shear force in the seismic condition, the contribution of concrete to the shear strength $V_c = 0$ will be taken. The contribution of the piles to the shear strength in the frame beams will not be taken into account.

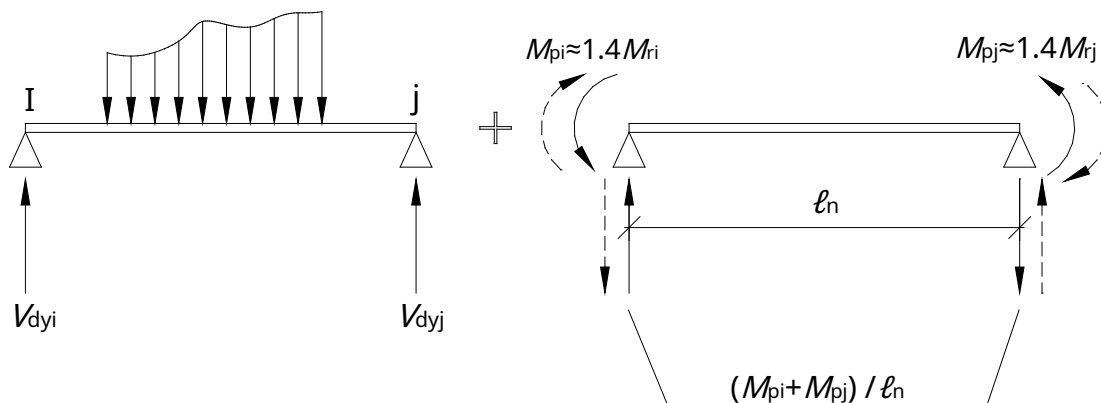


Figure 7.9

7.5. COLUMN JOINT AREAS IN FRAME SYSTEMS WITH HIGH DUCTILITY LEVEL

7.5.1. Encircled and Unencircled Combinations

In frame systems formed by columns and beams with high ductility level, column-beam connections are as follows: (a) And (b) As defined in , it will be divided into two classes.

(a) If the beams are connected to the column from four sides and the width of each beam is not less than 3/4 of the width of the column they are connected to, then the column-beam connection *surrounded union* will be defined as.

(b) Above (a) All combinations that do not satisfy the conditions in , *unenclosed union* will be defined as.

7.5.2. Shear Safety of Column-Beam Joint Areas

7.5.2.1 –Shear force in the column-beam joint areas in the earthquake direction considered, Equation (7.11) will be calculated with (Figure 7.10).

$$V_{to} = 1.25 f_{yk} (A_{s1} + A_{s2}) - V_{arm} \quad (7.11)$$

For cases where the beam penetrates the column only from one side and does not continue on the other side $A_{s2} = 0$ will be taken.

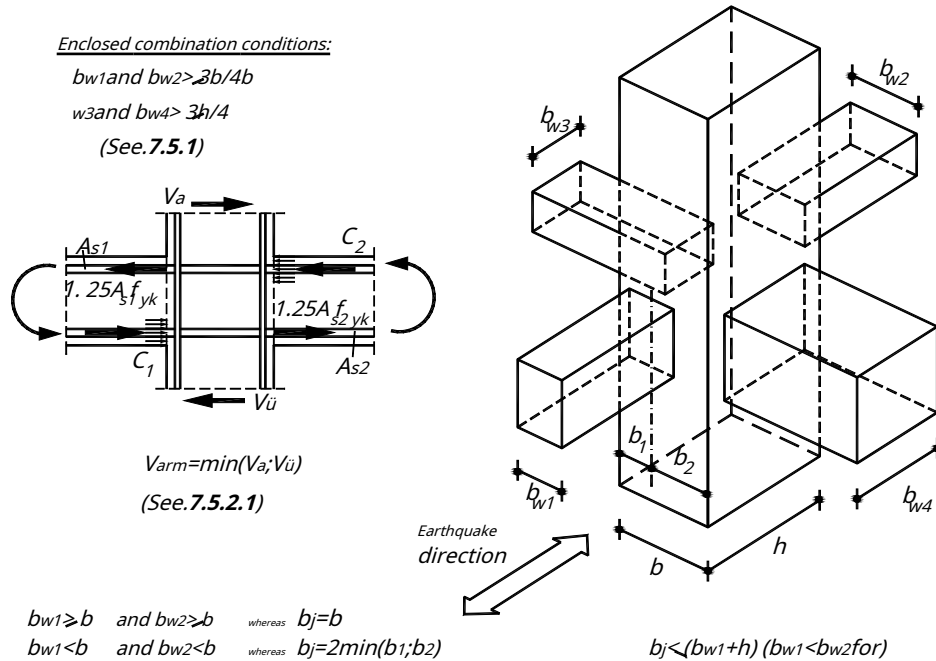


Figure 7.10

7.5.2.2 –In any junction area Equation (7.11) The shear force calculated with is never in the earthquake direction considered. Equation (7.12) And Equation (7.13) It will not exceed the limits given in ((Figure 7.10) If these limits are exceeded, the seismic calculation will be repeated by enlarging the column and/or beam section dimensions.

(a) In encircled combinations: $V_{to} \leq 1.7 b_j h f_{ck} \sqrt{\quad}$ (7.12)

(b) In unenclosed combinations: $V_{to} \leq 1.0 b_j h f_{ck} \sqrt{\quad}$ (7.13)

7.5.2.3 –Minimum transverse reinforcement conditions in the beam-column joint area are given below: (a) And (b) is given in (Figure 7.3):

(a) In confined connections, at least 40% of the transverse reinforcement amount calculated for the confinement zone of the lower column shall be used throughout the connection zone. However, the diameter of the transverse reinforcement shall not be less than 8 mm and its spacing shall not exceed 150 mm.

(b) In unconfined connections, at least 60% of the transverse reinforcement amount calculated for the confinement zone of the lower column shall be used throughout the connection zone. However, in this case, the diameter of the transverse reinforcement shall not be less than 8 mm and its spacing shall not exceed 100 mm.

7.6. WALLS WITH HIGH DUCTILITY LEVEL

7.6.1. Cross-Section Conditions

For walls other than basement walls, the following cross-sectional conditions shall be met.

7.6.1.1 –The net cross-sectional area of the curtain after removing the gaps, N_{dm} Taking into account the live load reduction coefficients defined for live loads in TS 498, G And Q vertical loads and T the combined effect of the earthquake effect $G+Q+T$ being the maximum of the axial compressive forces calculated under, $A_c \geq N_{dm} / (0.35 f_{ck})$ condition will be met. Connecting beam (gapped) curtains A_c And N_{dm} In calculating the values, the entire hollow wall section (total of the wall parts) will be taken into account.

7.6.1.2 –Curtains are vertical carrier system elements with a ratio of the longest side to the thickness in plan of at least six.

(a) **7.6.1.3** Except for the special cases specified in , the wall thickness in the body area of rectangular and U, L and T type walls shall not be less than 1/16 of the floor height and 250 mm.

(b) The thickness of a rectangular curtain or curtain arm shall not be less than 1/30 of the unsupported length of the curtain or curtain arm in the lateral direction in plan.

(c) If the curtain arm is held by a curtain in the lateral direction from both ends, the curtain arm thickness shall not be less than 1/20 of the floor height and 250 mm.

7.6.1.3 –In buildings whose carrier system consists of curtains, **Equation (7.14)** If both of the conditions given are met, the wall thickness shall not be less than 1/20 of the height of the highest floor in the building and 200 mm. In addition, **7.6.1.1** The condition in shall be complied with.

$$\frac{\sum A_g}{\sum A_p} \geq 0.002 \quad V_t$$

$$/ \sum A_g \leq 0.5 f_{ctd} \quad (7.14)$$

Equation (7.14), will be applied at the ground floor level in buildings with very rigid reinforced concrete shear walls around the basement floors, and at the upper foundation level in other buildings.

7.6.2. Wall End Zones and Critical Wall Height

7.6.2.1 – $H_w/l_w > 2.0$ at both ends of the plan of the curtains *curtain end zones*

will be created (**Figure 7.11**). The end zones can be formed within the thickness of the curtain end zone itself or can be arranged within another curtain that is connected to the curtain.

7.6.2.2 – From the top of the foundation or from the level where the plan length of the curtain is reduced by more than 20% *critical curtain height*, $2-l_w$ not to exceed its value, **Equation (7.15)** Given in will be determined in a way that provides for the most unfavorable of conditions.

$$2-l_w \geq H_{cr} \geq \max[-l_w; H_w/6] \quad (7.15)$$

Here H_w , above the foundation or the gross section bending rigidity of the wall is halved (in plan length is reduced by more than 20% or section width is reduced by more than half) is the height of the wall measured from the level. In the basement floors, reinforced concrete perimeter walls with a very high rigidity compared to the upper floors and in the basement floors

In buildings where the floors operate as rigid diaphragms in the horizontal plane, H_w And H_{cr} their size shall be taken into account upwards from the ground floor slab. In such buildings the critical shear wall area shall also be extended downwards along the height of at least the first basement floor below the ground floor.

7.6.2.3 – In rectangular section curtains, **7.6.2.2** defined in *critical curtain height* Along the wall section above the critical wall height, the plan length of each of the wall end zones shall not be less than 20% of the total plan length of the wall and twice the wall thickness. Along the wall section above the critical wall height, the plan length of each of the wall end zones shall not be less than 10% of the total plan length of the wall and twice the wall thickness (**Figure 7.11**).

7.6.2.4 – In cases where the curtain end zones are arranged inside another curtain that is connected to the curtain; each curtain end zone shall extend into the curtain body by at least the thickness of the curtain, but not less than 300 mm (**Figure 7.11**). Cross-sectional area of the curtain end area, for rectangular section curtains **7.6.2.3** It will be arranged in a way that is not less than the area defined in .

7.6.3. Body Reinforcement Conditions

7.6.3.1 – Total cross-sectional area of web reinforcement on both sides of the bulkhead, for each of the longitudinal and transverse reinforcement, between the bulkhead end zones *curtain body* gross cross-sectional area

It shall not be less than $0.0025 \cdot H_w/l_w \leq$ In case of 2.0, the curtain body area is the curtain will be taken into account as the entire section. The longitudinal and transverse reinforcement spacing in the curtain body will not be more than 250 mm (**Figure 7.11**).

7.6.3.2 – **7.6.1.3** also **Equation (7.14)** In buildings where both the conditions given by are met, the total longitudinal and transverse web reinforcement ratios can be reduced to 0.002 each. However, in this case, the reinforcement spacing will not exceed 300 mm.

7.6.3.3 – Except for the end zones, the reinforcement networks on both sides of the curtain walls shall be at least four for each square metre of curtain wall surface. *special earthquake kipper* will be mutually connected with. However, **7.6.2.2** defined in *critical curtain height* At least ten special earthquake cross ties will be used for each square meter of curtain surface, except for the end areas. The diameter of the cross ties shall be

will be at least as much as the diameter of the horizontal reinforcement. However, the number of crossties per unit area is ϕ_{body}/ϕ_{weedy}

The diameter can be reduced by increasing the ratio.

7.6.4. Arrangement of Body Reinforcements

The web reinforcements will be interlocked in the curtain end area. The web end area will consist of wrapping reinforcement, closed stirrups and crossties. In addition, horizontal web reinforcements with ends connected to the longitudinal reinforcement with 135 degree hooks can be used as web reinforcement in the curtain end area. Horizontal or vertical square (90 degree hook) can be made to ensure that horizontal web reinforcements are interlocked in the curtain end areas. The distance between the ends or squares of horizontal web reinforcements and the outer edge of the curtain will not be greater than 150 mm. In cases where overlapping joints are required to be made to horizontal web reinforcements in the curtain body, overlapping joints

The attachments will be made staggered along the length of the curtain body, overlap length 1.5- ϕ from will not be small, 90-degree hooks will be formed at the ends of horizontal reinforcement in the overlapped joint. If hooks are not used at the ends of horizontal web reinforcement, these reinforcements will be arranged in a way that they will remain on the inside of the longitudinal web reinforcements, there will be at least six longitudinal web reinforcements along the overlapped joint, the horizontal distance between the longitudinal web reinforcements in the overlapped joint area will not exceed 200 mm (Figure 7.11).

7.6.5. Reinforcement Conditions in Wall End Zones

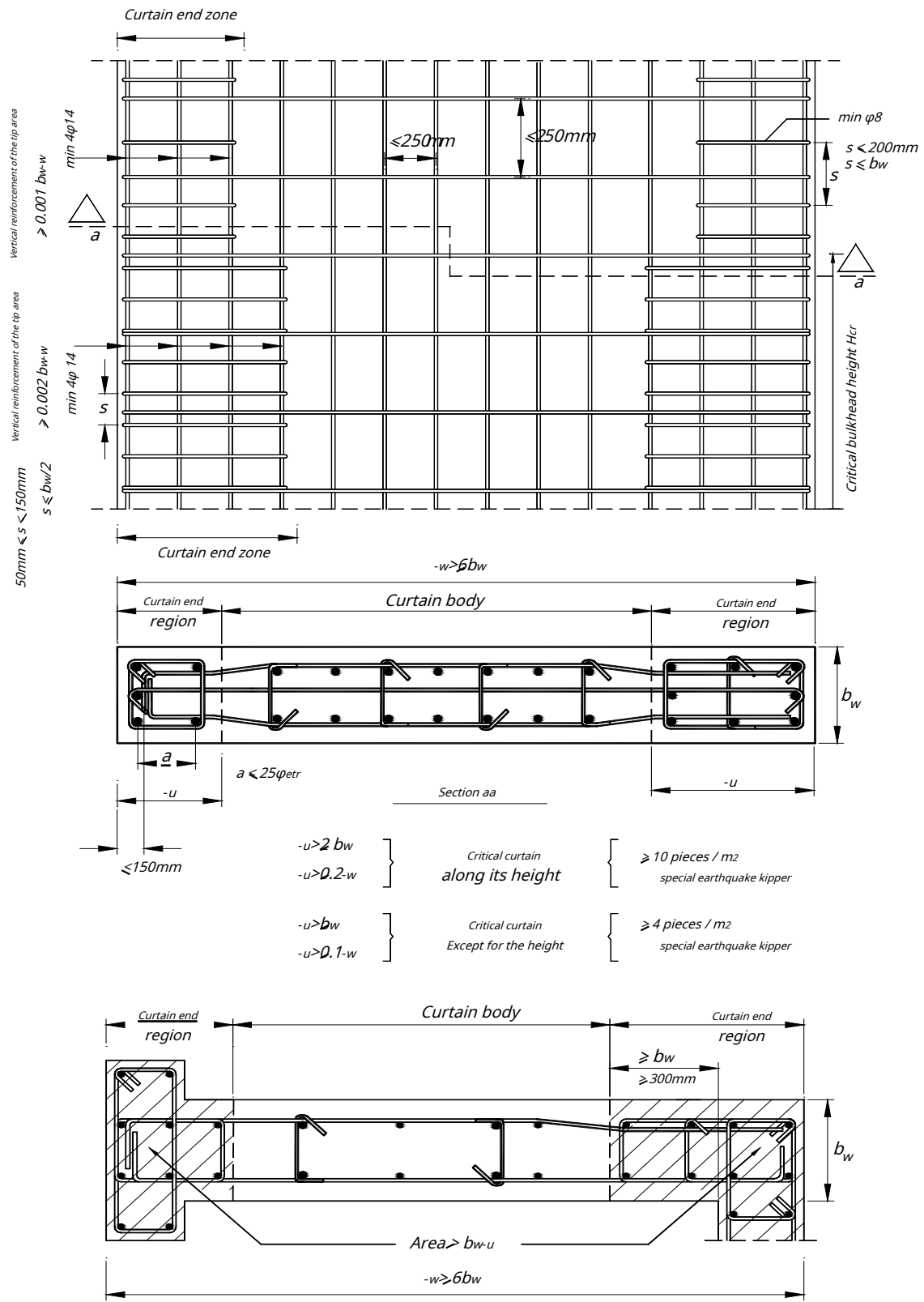
7.6.5.1 – 7.6.2.2 defined in *critical curtain height* The ratio of the total vertical reinforcement area to the gross cross-sectional area of the wall in each of the wall end zones along the wall shall be at least 0.002. This ratio shall not be less than 0.001 outside this height. The transition in the geometry and reinforcement of the wall end zone shall be made gradually throughout three floors. In addition, the amount of longitudinal reinforcement in each of the wall end zones shall not be less than $4\phi 14$. In the wall end zones The longitudinal reinforcement ratio shall not exceed 0.03 (0.06 in the overlap zone) (Figure 7.11).

7.6.5.2 – Vertical reinforcement at the curtain end areas, below (a), (b) and (c) In accordance with the rules in , it will be surrounded by transverse reinforcement consisting of stirrups and/or crossties as in columns.

(a) The diameter of the transverse reinforcement to be used in the end areas shall not be less than 8 mm. The horizontal distance between the stirrup arms and/or crossties shall be, a , It will not be more than 25 times the diameter of the stirrup and crossbar.

(b) **7.6.2.2** defined in *critical curtain height* along the shear wall end areas, for the wrapping areas of the columns **7.3.4.1** also **Equation (7.1)** At least $2/3$ of the transverse reinforcement determined by the second condition of shall be placed. The stirrup and/or crosstie spacing in the vertical direction shall not be taken greater than 150 mm and less than 50 mm (Figure 7.11). This distance shall not exceed 6 times the diameter of the longitudinal reinforcement and $1/3$ of the wall thickness. The transverse reinforcement in the wall end area shall be continued within the foundation for a height not less than 300 mm and the wall thickness.

(c) In the wall end areas outside the critical wall height, the vertical stirrup and/or crosstie spacing will not be taken as greater than the wall thickness and 200 mm (Figure 7.11).



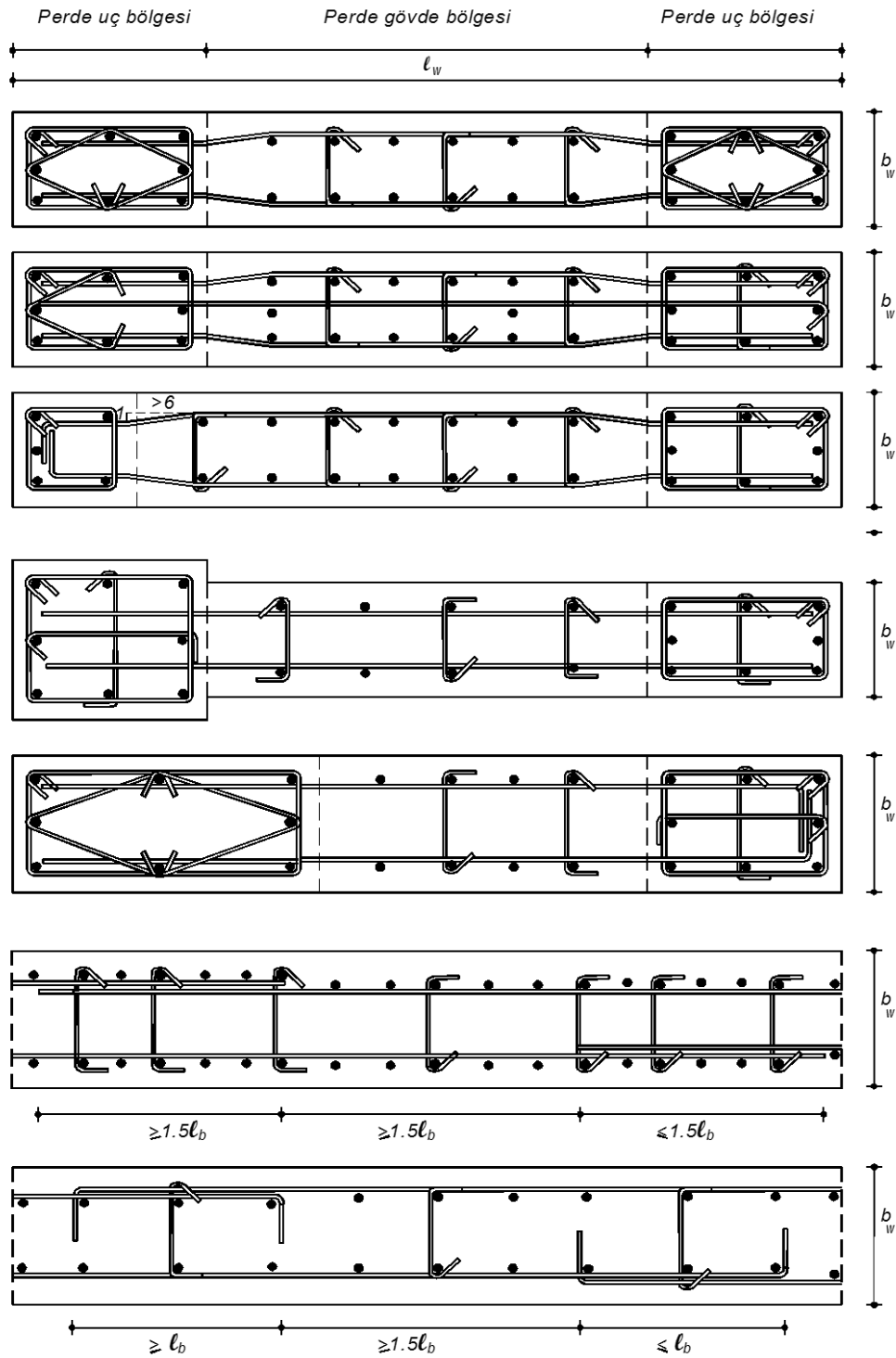


Figure 7.11

7.6.6. Design Bending Moments and Shear Forces

7.6.6.1 $-H_w/-w$ Design basis bending moments for walls that meet the 2.0 condition,

7.6.2.2 As a constant value along the critical wall height determined according to , at the wall base

Chapter 4 It will be taken as equal to the bending moment calculated according to . Above the section where the critical wall height ends, **Chapter 4** According to , a linear moment diagram parallel to the line connecting the moments calculated at the base and top of the wall will be applied (**Figure 7.12**).

3.3.1.1 In buildings with basements that meet the conditions given in the fixed

curtain moment, **7.6.2.2** It will be taken into account along the critical wall height defined in . $H_w / -w \leq$
Design bending moments in all sections of the shear walls with 2.0, **Chapter 4** According to
will be taken equal to the calculated bending moments.

7.6.6.2 - $H_w / -w$ In case of > 2.0 , the bearing capacity of the shear wall sections at each floor
moments for columns in the strong direction of the wall **Equation (7.3)** It is mandatory to meet the condition
given by . Otherwise, the earthquake calculation will be repeated by increasing the wall dimensions and/or
reinforcement.

7.6.6.3 - $H_w / -w >$ Design shear force to be taken as basis in the calculation of transverse
reinforcement in any section considered in walls meeting the 2.0 condition, V_{to} , **Equation (7.16)**
will be calculated with .

$$V_{to} = \beta \frac{(M_{pt})_d V_d}{v(M_d)_t} \quad (7.16)$$

The dynamic amplification factor of the cutting force in this equation is $\beta_v = 1.5$ will be taken. However, in
buildings where the entire earthquake load is carried by reinforced concrete walls, $\beta_v = 1.0$ can be taken. In
cases where a more precise calculation is not made, $(M_p)_t \leq 1.25 (M_r)_t$ acceptable. Vertical loads
with **Chapter 4** According to the shear force calculated from the earthquake is $1.2D$ (gapless curtains) or
 $1.4D$ (the value obtained by multiplying the value of the shear walls with tie beams, **Equation**
(7.16) calculated with V_{to} If it is less than , V_{to} This cutting force will be used instead.

$H_w / -w \leq$ Design shear forces in all sections of 2.0 shear walls, **Chapter 4** It will be taken equal to the
cutting forces calculated according to .

7.6.7. Cutting Safety of Curtains

7.6.7.1 - Shear strength of curtain sections, V_r , **Equation (7.17)** will be calculated with .

$$V_r = A_{ch}(0.65 f_{ctd} + \rho_{s,h} f_{ywd}) \quad (7.17)$$

7.6.6.3 defined in V_{to} design shear force **Equation (7.18)** will meet the conditions given in:

$$\begin{aligned} V_{to} &\leq V_r \\ V_{to} &\leq 0.85 A_{ch} \sqrt{f_{ck}} \quad (\text{Gapless curtains}) \quad (7.18) \\ V_{to} &\leq 0.65 A_{ch} \sqrt{f_{ck}} \quad (\text{Tie beam curtains}) \end{aligned}$$

Otherwise, the wall transverse reinforcement and/or wall section dimensions will be increased to meet these
conditions.

7.6.7.2 - Vertical reinforcement in horizontal construction joints to be made at the connection level to the
foundation and on the upper floors shall be determined as defined in TS 500, taking into account the shear force
transferred in that section. *cutting friction method* In the calculation of cutting friction, the curtain body and the tip
All vertical reinforcement of the region A_s and contribution of concrete to roughened surface f_{ctd}
will be taken into consideration. V_{to} friction shear force **Equation (7.19)** will meet the conditions
given in:

$$V_{to} \leq f_{ctd} A_c + \mu A_s f_{yd} \quad (7.19)$$

$$V_{to} \leq \min[0.2 f_{ck} A_c; (3.3 + 0.08 f_{ck}) A_c]$$

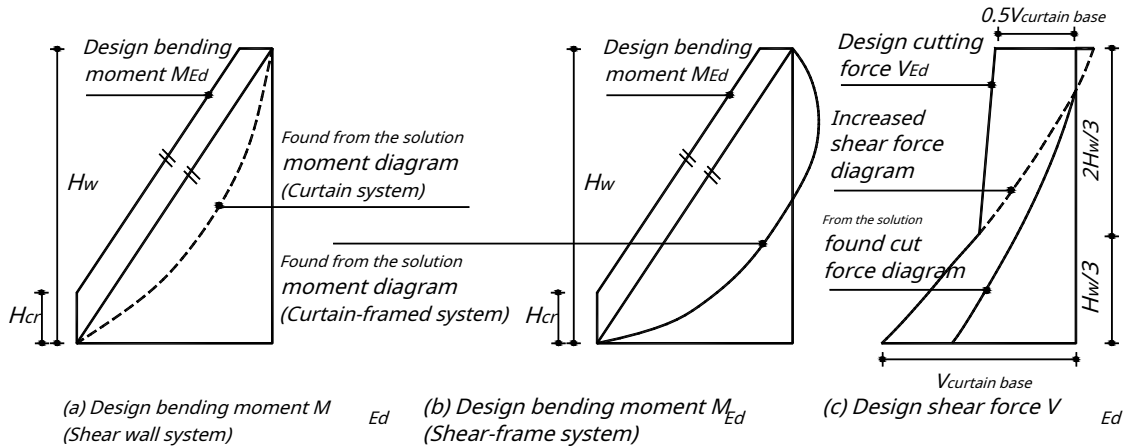


Figure 7.12

In the shear friction calculation, the yield stress of the reinforcement will not be exceeded.

$f_{yk} = 500$ MPa value

7.6.8. Rules and Conditions Regarding Connecting Beam (Slotted) Shear Walls

7.6.8.1 –All the rules and conditions given above for curtains also apply to each of the curtain parts that make up the tie beam curtains.

7.6.8.2 –Rules for shear reinforcement of tie beams are given below:

(a) Equation (7.20) If any of the conditions given above are met, the calculation of shear reinforcement of the tie beams **7.4.5** It will be done according to.

$$n > 2h_k \quad (7.20a)$$

$$V_d \leq 1.5 b_w d f_{ctd} \quad (7.20b)$$

(b) Equation (7.20) If both of the conditions given above cannot be met, cross reinforcement will be used to resist the shear force in the connecting beam and the bending moment it creates (**Figure 7.13**). Total reinforcement area in each cross reinforcement bundle **Equation (7.21)** will be determined by.

$$A_{sd} = V_d / (2 f_{yd} (\sin \gamma)) \quad (7.21)$$

There will be at least four pieces of reinforcement in the cross reinforcement bundles and these reinforcements will be at least $1.5b_w$ will be extended to . Reinforcement bundles will be wrapped with special earthquake stirrups and the diameter of the stirrups to be used shall not be greater than 8 mm and their spacing shall not be greater than 8 times the diameter of the cross reinforcement and 100 mm. In addition to the cross reinforcements, the minimum amount of stirrups and horizontal reinforcement specified in TS 500 shall be placed on the tie beam. In cases where the reinforcement bundle is not wrapped with special earthquake stirrups, the spacing of the beam stirrups shall not exceed 6 times the diameter of the cross reinforcement and 150 mm. In addition, horizontal and vertical crossties shall be used at intervals not exceeding 200 mm along the beam height and 200 mm along the beam width (**Figure 7.13**).

Stirrups and crossties placed in this way, both vertically and horizontally, **Equation (7.1)** will meet the conditions given in .

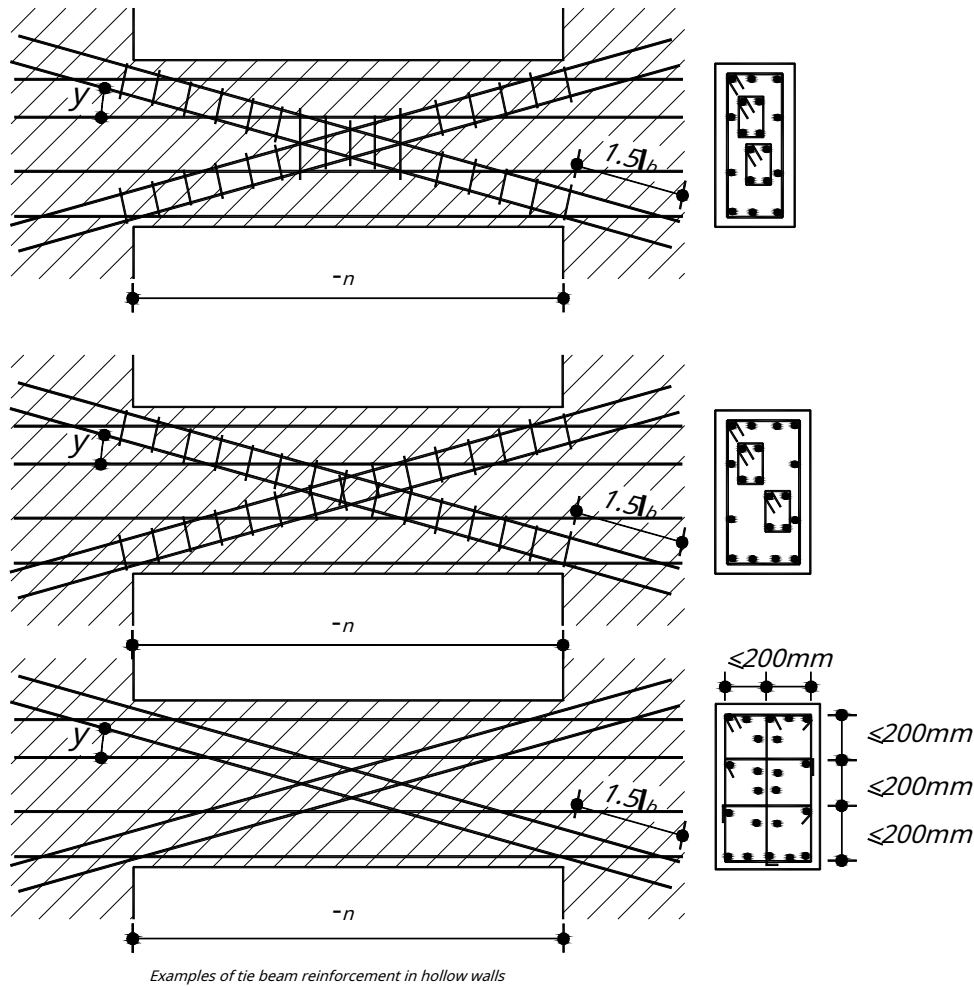


Figure 7.13

7.6.8.3 –Shear force acting on the tie beams in tie beam walls V_d **Equation (7.22)** will not exceed the upper limit given by .

$$V_d \leq 0.85 b_w d f_{ck} \sqrt{\quad} \quad (7.22)$$

7.6.9. Gaps in Curtains

Gaps such as windows and installations within the curtain will be created in the middle third of the curtain in the plan, the horizontal dimension of the gap will not be greater than 20% of the curtain width and the vertical dimension will not be greater than 20% of the floor height. Additional vertical and horizontal reinforcement wrapped with stirrups will be placed on the edges, top and bottom of the gap; the total cross-sectional area of the additional reinforcement placed in these areas in each direction will not be less than the total cross-sectional area of the reinforcement not placed in the gap area and the stirrup spacing will not be greater than 150 mm. (**Figure 7.14**).

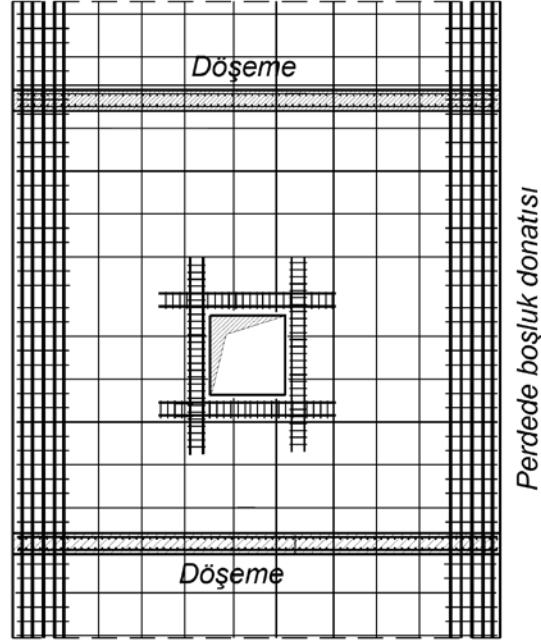


Figure 7.14

7.7. COLUMNS WITH LIMITED DUCTILITY LEVEL

7.7.1. Cross-Section Conditions

Gross cross-sectional area of the column, N_{dm} Taking into account the live load reduction coefficients defined for live loads in TS 498, G and Q vertical loads and T the combined effect of the earthquake effect $G+Q+T$ being the maximum of the axial compressive forces calculated under, **Equation (7.23)** will satisfy the condition given in .

$$A_c \geq N_{dm} / (0.40 f_{ck}) \quad (7.23)$$

Apart from this, for columns with high ductility level regarding cross-section dimensions **7.3.1.1** The conditions specified in are also valid for columns with limited ductility levels.

7.7.2. Longitudinal Reinforcement Conditions

For columns with high ductility level regarding longitudinal reinforcement **7.3.2** The conditions specified in are also valid for columns with limited ductility levels.

7.7.3. Arrangement of Longitudinal Reinforcement

For columns with high ductility level regarding the arrangement of longitudinal reinforcement **7.3.3** The conditions specified in are also valid for columns with limited ductility levels.

7.7.4. Transverse Reinforcement Conditions

Conditions regarding the minimum transverse reinforcement to be used in columns, *column wrapping regions* for **7.7.4.1** and in *column mid region* for **7.7.4.2** In all column regions, **7.2.8** defined in *special earthquake stirrups* and *special earthquake ties* will be used.

7.7.4.1 –For the length of each of the column wrapping regions **7.3.4.1** The definition given in is also valid for columns with limited ductility level. In columns with limited ductility level, the transverse reinforcement spacing in the confined zone shall not be greater than 1/3 of the smallest cross-section dimension, 8 times the smallest longitudinal reinforcement diameter and 150 mm. Transverse reinforcement **Equation (7.1)** And **Equation (7.2)** It will not be less than half of the values given in .

7.7.4.2 –For columns with high ductility level regarding the column middle region **7.3.4.2** The definition and minimum transverse reinforcement conditions given in are also valid for columns with limited ductility level. Transverse reinforcement in the column middle region, **7.7.5.3** It will be determined according to.

7.7.5. Shear Safety of Columns

7.7.5.1 –In columns with limited ductility level, vertical loads and **Chapter 4** and Excess Strength Coefficient determined in D The joint effect of the earthquake effects calculated by increasing with The shear force obtained under V_d , will be taken as basis in the calculation of transverse reinforcement.

7.7.5.2 –For columns with high ductility level regarding the upper limit of shear force **Equation (7.7)** The condition given in , V_{d0} instead of Excess Strength Coefficient D increased with V_d It is also valid for columns with limited ductility level.

7.7.5.3 –Column transverse reinforcement **7.7.5.1** Contribution of concrete to shear strength in its calculation according to the shear force defined in , V_c , according to vertical loads and earthquake loads smallest calculated N_d It will be determined according to TS 500, taking into account the axial force.

7.7.6. Conditions Regarding Short Columns

For columns with high ductility level in relation to short columns **7.3.8** The conditions specified in are also valid for columns with limited ductility levels.

7.8. BEAMS WITH LIMITED DUCTILITY LEVEL

7.8.1. Cross-Section Conditions

For beams with high ductility level in relation to cross-section dimensions **7.4.1.1** The conditions specified in are also valid for beams with limited ductility.

7.8.2. Longitudinal Reinforcement Conditions

For beams with high ductility level regarding longitudinal reinforcement **7.4.2** The conditions specified in are also valid for beams with limited ductility.

7.8.3. Arrangement of Longitudinal Reinforcement

For beams with high ductility level regarding the arrangement of longitudinal reinforcement **7.4.3** The conditions specified in are also valid for beams with limited ductility.

7.8.4. Transverse Reinforcement Conditions

The area twice the beam height from the column face in beam supports, *spurification zone* and throughout this region **7.2.8** defined in *special earthquake stirrups* Transverse reinforcement with diameters smaller than $\phi 8$ shall be used in confinement zones.

will not be used and the distance of the first stirrup to the column face will be maximum 50 mm. **7.8.5** Unless a more unfavorable value is obtained according to , the stirrup spacing shall not exceed 1/4 of the beam height, 8 times the diameter of the smallest longitudinal reinforcement and 200 mm. Except in the confinement zone, the transverse reinforcement conditions given in TS 500 shall be complied with. The distance between stirrup arms in beams shall not be greater than 350 mm.

7.8.5. Shear Safety of Beams

7.8.5.1 –In beams with limited ductility level, vertical loads and **Chapter 4** and Excess Strength Coefficient determined in D shear obtained under the combined effect of earthquake effects increased by force V_d , will be taken as basis in the calculation of transverse reinforcement.

7.8.5.2 –For beams with high ductility level regarding the upper limit of shear force **Equivalent.** **(7.10)** The condition given in , V_{d0} instead of Excess Strength Coefficient D increased with V_d be taken. It is also valid for beams with limited ductility level.

7.8.5.3 –Beam transverse reinforcement **7.8.5.1** Contribution of concrete to shear strength in its calculation according to the shear force defined in , V_c , will be determined according to TS 500. In frame beams The contribution of the piles to the shear strength will not be taken into account.

7.9. COLUMN-BEAM JOINT AREAS IN FRAME SYSTEMS WITH LIMITED DUCTILITY LEVEL

Regarding the column-beam connections of frame systems formed by columns and beams with high ductility level **7.5** The terms and conditions given in , **7.5.2.1** and **7.5.2.2** It is also valid for beam-column connections of systems with limited ductility level, with the exception of

7.10. WALLS WITH LIMITED DUCTILITY LEVEL

The net cross-sectional area of the curtain after removing the gaps, N_{dm} Taking into account the live load reduction coefficients defined for live loads in TS 498, G and Q vertical loads and T_0 the combined effect of the earthquake effect $G+Q+T_0$ being the maximum of the axial compressive forces calculated under, **Equation (7.24)** will satisfy the condition given in .

$$A_c \geq N_{dm} / (0.35 f_{ck}) \quad (7.24)$$

Shear walls with limited ductility level will be dimensioned and equipped according to the internal forces resulting from the combined effects of vertical loads and earthquakes. For shear walls with high ductility level **7.6.6**, **7.6.8.2** and **7.6.8.3** Except for the terms and conditions given in , **7.6** All other rules and regulations given in conditions, V_{d0} in its place D Being the Excess Strength Coefficient $V_{d0} = D V_d$ It is also valid for shear walls with limited ductility level.

7.11. FLOORINGS

7.11.1 –Floors will have the rigidity and strength to ensure that the earthquake loads acting on the masses on the floors are safely distributed to the vertical load-bearing system elements.

7.11.2 –In in-situ cast-in-place floor systems with or without infill under the effect of an earthquake, the slab thickness shall not be less than 70 mm. However, it shall ensure that the shear forces resulting from vertical loads and the earthquake forces within the slab plane are safely transferred.

between the teeth and the plaque to ensure *shear force connections* will be made and it will be shown by calculation that these connections are sufficient. For the thickness of other floors, the conditions given in TS 500 are valid.

7.11.3 –In-plane axial and shear stresses in the floors of buildings with and without beams shall be calculated by assuming an elastic diaphragm. The average in-plane tensile, compressive and shear stresses in the floors of these buildings under the effect of an earthquake shall be calculated by the Excess Strength Factor. *D* Tensile stress within the floor plane will be applied. the value of f_{ctd} In the case where is greater than , the in-plane tensile stress value is given by ρ , where ρ is the excess reinforcement ratio required for the flexural strength of the slab. f_{yd} the limit

In cases where this condition is not met, the lack of reinforcement will be completed with additional in-plane reinforcement. The compressive stress value within the floor plane shall be $0.85 f_{cd}$ the limit

Horizontal shear stresses within the floor plane in both directions **Equation (7.25)** will not exceed the limit given by .

$$\tau_r = 0.65 f_{ctd} + \rho f_{yd} \quad (7.25)$$

In this calculation, ρ is the ratio of the floor reinforcement to be placed in the direction parallel to the shear stress and remaining from what is required for the bending strength. The shear stress in the plane is $0.65 f_{ctd}$ will not exceed the limit.

The in-plane shear stresses between the slab and the wall shall not exceed the in-plane shear friction strength at these sections. The shear friction strength is defined in terms of stress, where ρ is the ratio of the floor reinforcement that is stuck into the wall and has sufficient interlocking length and remains above that required for flexural strength. **Equation (7.26)** will be calculated with .

$$\tau_r = \mu \rho f_{yd} \quad (7.26)$$

For the coefficient of shear friction in cast joints μ = The value 1.0 will be used.

7.11.4 –It must be shown through calculations that the earthquake loads are transferred safely from the floors to the vertical load-bearing elements. **A2** And **A3** in buildings with beamed floors where irregularities are present, for floors **7.11.3** In the section calculation and detailing of the beams that transfer the earthquake loads from the floor to the curtain wall, the axial pressure and tensile forces formed in these beams will be calculated as the Excess Strength Factor. *D* Taking into account The axial compressive stress in these beams shall be taken into account. f_{ck} is greater than beam section in cases **7.3.4** for column wrapping regions in **Equation 7.1** The transverse reinforcement conditions given in , shall be met. The axial pressure force formed in these beams shall not exceed the axial pressure carrying capacity of the section calculated as a column in any case.

7.11.5 –In buildings with floors without beams or in buildings with floors with beams where it is necessary to show with calculations that the earthquake loads are transferred safely from the floors to the vertical load-bearing elements, the earthquake force to be transferred from the floor to the curtain or curtain arm in the strong direction is the earthquake force that occurs in the upper and lower sections of the floor level and the earthquake effects are calculated as Excess Strength.

Coefficient *D* calculated taking into account *DV* as the difference of the shear forces

This force difference is the sum of the axial tensile strengths of the beams or floor reinforcements that penetrate the wall from both sides in the strong direction and remain beyond what is required for the bending strength. $A_h f_{yd}$ bending at the junction of the floor and the curtain

the shear friction resistance of the floor reinforcement remaining beyond that required for its strength $\mu A_{sb} f_{yd}$ will not exceed the total.

There will be no reduction in the amount of axial reinforcement (transfer reinforcement) inserted into the wall in the strong direction along the calculated wall length, and the shear friction (connection) reinforcement will provide interlocking conditions both inside the wall and inside the slab.

Shear friction in floor and wall joints will be calculated in accordance with TS 500, for the friction coefficient $\mu \leq 1.0$ will be used. The amount of reinforcement in the transfer reinforcement

It should be taken into account that an appropriate amount of reduction can be made as it moves away from the curtain surface, starting from the curtain surface where it is embedded.

Shear friction control will also be carried out along the length of the transfer element between the transfer elements formed in this way and the slab (**Figure 7.15**).

7.11.6 –In cases where there are large gaps in the slab, additional horizontal reinforcement wrapped with stirrups will be placed on both sides of the gap in each direction; the total cross-sectional area of the additional horizontal reinforcement in these areas will not be less than the total cross-sectional area of the horizontal reinforcement not placed in the gap area and the stirrup spacing will not be greater than 150 mm. These additional horizontal reinforcements will be at least three times the length of the gap in each direction.

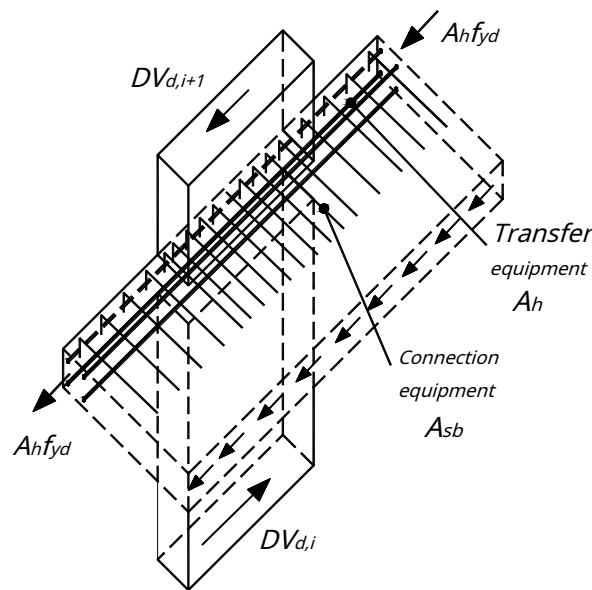


Figure 7.15

7.11.7 –Punching control in beamless slabs and beamless slab foundations, Excess Strength Factor obtained from finite element model solution and vertical loads D It can be done by taking the vertical shear stress formed under the earthquake effects magnified by γ . The effective height of the floor or foundation for the shear stress value obtained from the finite element solution d including, **Equation (7.27)** The condition given in will be met.

$$\tau_{pdf} = v_{pdf} / d \leq f_{ctd} \quad (7.27)$$

Here v_{pdf} is the shear force calculated per unit width. The basis for the design is τ_{pdf} value, due to column or curtain d It will be calculated at 2 distances.

7.11.8 –In floor-column connections in beamless plate systems, the Excess Strength Factor with vertical loads D Under the earthquake effects magnified by the column

The total bending moment transferred in the direction considered at the level y_f The value obtained by multiplying the coefficient with the bending reinforcement, $1 - \gamma_f$ obtained by multiplying by the coefficient Assuming that the obtained value is transferred through punching (shear) stresses and assuming that the shear stresses in the slab change linearly according to the geometric center of the punching perimeter in the direction of loading, the punching perimeter (u_p) acting along The idealized distribution of shear stresses can be calculated. The highest shear stress value obtained from this calculation is the design basis τ_{pdf} as the value of the flooring, the stapling can be used in control.

7.11.9 –In the calculation of the total bending moment transferred to the column at the floor level, the bending moments balancing the bending moments in the column sections above and below the floor-column joint moment will be taken into account. γ_f In calculating the coefficient, the dimensions of the rectangular punching perimeter are taken into account in the direction of the loading taken into account. b_1 and perpendicular to the load b_2 including, **Equation (7.28)** In circular section columns, $\gamma_f = 0.60$ will be used.

$$\gamma_f = \frac{1}{1 + (2/3) \sqrt{b_1/b_2}} \quad (7.28)$$

7.11.10 – $\tau_{pdf} \leq f_{ctd}$ condition is not met and the slab thickness is not less than 250 mm

In this case, the punching strength of the slab can be increased by using reinforcement. However, in this case, the contribution of the concrete to the punching strength in terms of shear stress is $0.5f_{ctd}$ Will be reduced to, contribution of reinforcement to punching strength f_{ctd} Not less than and the punching strength of the reinforced plate shall not be more than $1.5f_{ctd}$ When shear wedge is used as punching reinforcement, its contribution to the punching strength of concrete in terms of shear stress will be $0.75f_{ctd}$ will be, the contribution of the reinforcement to the punching strength f_{ctd} Not less than and the punching strength of the reinforced plate shall not be more than $1.75f_{ctd}$ Will be.

7.11.11 –In case the punching reinforcement is used as evenly distributed crossties or trestle reinforcement, the punching strength of the reinforced plate is expressed in terms of shear stress. **Equation (7.29a)** with, when using shear keys **Equation (7.29b)** will be calculated with .

$$\tau_p = 0.5f_{ctd} + \rho f_{yd} \leq 1.5f_{ctd} \quad (7.29a)$$

$$\tau_p = 0.75f_{ctd} + \rho f_{yd} \leq 1.75f_{ctd} \quad (7.29b)$$

Here ρ is the punching crossbar (and/or the stand to grasp the horizontal reinforcement) per unit area. is the cross-sectional area of the reinforcement and the punching reinforcement is at least four piece/m^2 will be, maximum due to column or wall d The distance between the punching reinforcements to be placed in a uniform spread starting at $4d/2$ will not exceed (**Figure 7.16**).

7.11.12 –In cases where closed stirrup strips or shear wedge rails are used in at least two perpendicular directions instead of placing the punching reinforcement evenly, the punching strength of the reinforced slab is expressed in terms of shear stress. **Equation (7.30a)** with; when using shear keys **Equation (7.30b)** will be calculated with .

$$\tau_p = 0.75 f_{ctd} + A_{wp} f_{yd} / (u_p s) \leq 1.75 f_{ctd} \quad (7.30b)$$

(b) Punching reinforcement in the form of shear wedges in slab and slab foundations

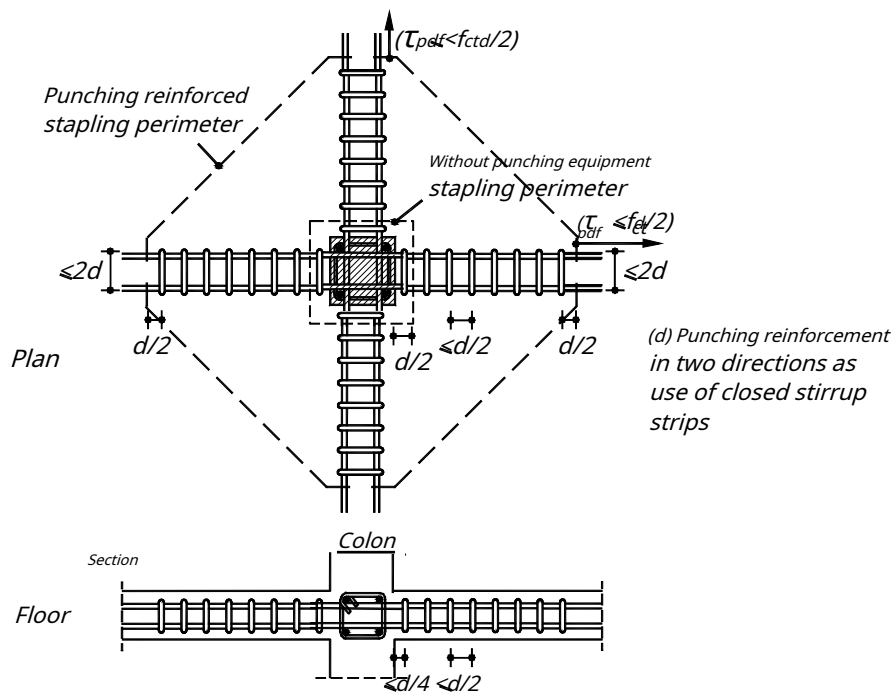
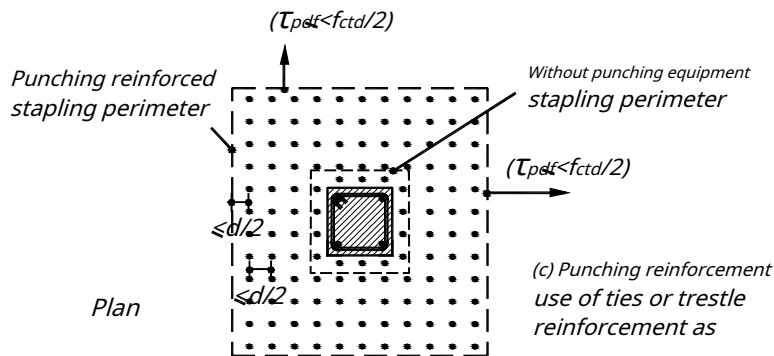


Figure 7.16

7.11.13 –Punching reinforcement shall be placed at a distance of at least four times the slab or raft thickness from the column or wall surface to the end of the punching reinforcement. from the section it reaches d /Calculated shear stress at 2 distances f_{ctd} /It will be continued until the region where it drops to the value of 2.

7.11.14 –In column strips and middle strips of beamless slabs **Figure 7.17** The reinforcement rules given in will be followed. No reduction will be made in the slab bottom reinforcement along the column strips, and no overlapping joints will be made in the slab top and bottom reinforcement in areas where punching is critical.

7.12. WALL BANDS

For horizontal and vertical reinforced concrete beams arranged within elements such as partition walls, parapets and shield walls in reinforced concrete buildings, **Chapter 11** The rules given in will be followed.

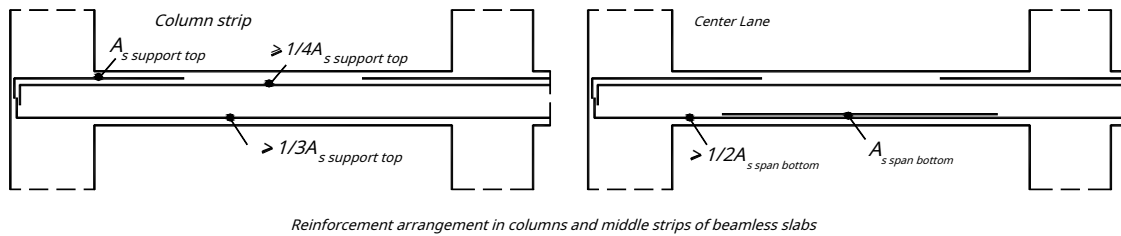


Figure 7.17

7.13. RULES FOR REINFORCED CONCRETE APPLICATION PROJECT DRAWINGS

7.13.1. General Rules

7.13.1.1 –Concrete strength class, reinforcement class and environmental impact class in accordance with TS EN 206 will be specified on all drawing sheets.

7.13.1.2 –Considered in the design *Map Spectral Acceleration Coefficients S_s And S_1 , Local Ground Class (Table 16.1), Building Usage Class (BKS, Table 3.1), Building Importance Coefficient (I , Table 3.1) And Building Height Class (BYS, Table 3.3), Earthquake Design Classes (DTS, Table 3.2), Carrier System Behavior Coefficient (R , Table 4.1) And Excess Strength Coefficient (D , Table 4.1) will be specified on all formwork plan sheets.*

7.13.1.3 – 7.2.8 The hook bend details of the special earthquake stirrups and special earthquake crossties defined in will be shown on each of the column, wall and beam detail sheets (**Figure 7.1**).

7.13.2. Column and Wall Details

7.13.2.1 –In column layout plans, the location, diameter and number of vertical reinforcements within the cross-section shall be shown in detail. Column and shear reinforcements protruding from the foundation, the number, diameter and spacing of the transverse reinforcement associated with them and their openings shall be indicated on the drawing.

7.13.2.2 –For each column type with all the same longitudinal and transverse reinforcement, longitudinal sections will be taken and vertical openings of the reinforcement will be made. Longitudinal section in columns will include reinforcement joint areas and overlap lengths.

7.13.2.3 –For each column type, the lengths of the confinement zones, the diameters, numbers and spacing of the transverse reinforcement placed in these zones, the column middle zone and the upper column-beam joint zone, and their openings in the cross section will be shown on the drawing.

7.13.2.4 –In addition to showing the location, diameter and number of vertical reinforcements in the curtain body and curtain end areas in the curtain layout plans, vertical openings of reinforcements will be made by taking longitudinal sections for each curtain type. The critical curtain height will be clearly indicated in the curtain longitudinal section. The diameter, number and spacing of the transverse reinforcements used along this height and in other curtain sections and their openings will be shown on the drawing.

7.13.3. Beam Details

In beam detail drawings, the lengths of the confinement zones in the beam supports, the diameters, numbers, spacings and openings of the transverse reinforcement placed in these zones and in the middle area of the beam shall be shown on the drawing, separately for each beam.

CHAPTER 8 – PREFABRICATED REINFORCED CONCRETE BUILDING UNDER EARTHQUAKE IMPACT

SPECIAL RULES FOR THE DESIGN OF CARRIER SYSTEMS

8.0. ICONS

D	= Excess Strength Coefficient
$\bar{F}_H$	= Equivalent seismic force acting on the hinged connection =
f_{CD}	Design strength of concrete
f_{yd}	= Design strength of reinforcement steel (pin) =
I	Building Importance Factor
$-n$	= Clear span of the beam
M_{pi}	= Moment capacity of the left end of the beam
M_{pj}	= Moment capacity of the right end of the beam
m_k	= Effective mass for joint
n	= Number of pins
R	= Carrier System Behavior Coefficient
$S_{ae}(T)$	= Natural vibration period T Horizontal elastic design spectral acceleration
V_r	corresponding to = Shear force capacity of pin in frame plane
μ	= Coefficient of shear friction =
φ	Pin diameter

8.1. SCOPE

8.1.1 – Specific rules for the design of precast reinforced concrete building load-bearing systems under earthquake effects are given in this Section.

8.1.2 – The types of column-beam connections and panel-floor systems used in prefabricated reinforced concrete buildings and the rules for their design are also within the scope of this Section.

8.1.3 – The rules for creating roof and floor diaphragms in prefabricated reinforced concrete buildings and transferring earthquake effects through diaphragms are explained in this Section.

8.2. GENERAL CHARACTERISTICS OF PRE-ENGINEERED BUILDINGS

8.2.1. Prefabricated Reinforced Concrete Carrier System Elements and Connections

8.2.1.1 – Reinforced concrete prefabricated buildings are buildings consisting of columns, beams, curtains, panels, double-walled panels, floors, walls or cell elements that are produced in production facilities and transported to the construction site and assembled.

8.2.1.2 – Reinforced concrete prefabricated building elements can be connected to each other in a way that will transfer moment, or they can be connected to each other in a way that will not transfer moment, such as by pinning, welding or in other ways, and they are designed in accordance with these features.

8.2.1.3 –Different connection types than those defined in this section can also be designed and implemented. However, these different connection types; **1.4**It will be documented by showing that it has sufficient rigidity, strength and ductility properties, can form stable cyclic cycles and can consume energy equivalent to a monolithic connection with similar strength, through experimental studies to be carried out according to .

8.2.2. General Rules for Precast Reinforced Concrete Building Design

8.2.2.1 –In the design of reinforced concrete prefabricated buildings, it is essential to make calculations and details by taking into account the assembly and usage stages.

8.2.2.2 –Reinforced concrete prefabricated buildings *Design by Strength* in the scope of,

(a) For linear earthquake calculation, depending on the connection types and the type of carrier system *Carrier System Behavior Coefficients (R)*, *Excess Strength Coefficients (D)* with permitted *Building Height Classes (BYS)* **Table 4.1** is given in .

(b) For columns, beams, walls and their casting connection areas with high ductility level **7.3, 7.4, 7.6** And **7.5** The conditions given in will be complied with.

(c) For columns, beams, walls and their casting connection areas with limited ductility level **7.7, 7.8, 7.10** And **7.9** The conditions given in will be complied with.

(d) Connections of reinforced concrete prefabricated elements **8.3** And **8.4** is given in.

8.2.2.3 – Table 3.4(a) According to *Earthquake Design Class* $DTS=1a$, $DTS=2a$ and *Building Height Class* In prefabricated reinforced concrete buildings with $BYS=2$, $BYS=3$;

(a) Under the influence of DD-2 earthquake ground motion $I=$ Taking **1.5** **Chapter 4A** design made with DGT accounting principles in *preliminary design* will be taken into consideration.

(b) The pre-designed building carrier system is now under the influence of the DD-1 earthquake ground motion. **Table 3.4(a)** in *Advanced Performance Target* defined as *Controlled Damage (KH)* performance target and also under the DD-3 earthquake *Limited Damage (SH)* suitable for various connection types to meet the performance target *modeling and evaluation rule* taking into account **Chapter 5** According to *Evaluation and Design Based on Shape shifting* (The design will be evaluated with the (ŞGDT) approach and, if necessary, the design will be repeated to ensure the targeted performance.

8.2.2.4 – 8.4.3 Earthquake calculation for preliminary design of fully post-tensioned prefabricated buildings as defined in **Chapter 4** After it is done with DGT approach according to **Chapter 5** According to , performance evaluation will be made with the ŞGDT approach.

8.2.3. Material Properties in Prefabricated Buildings

8.2.3.1 – 7.2.5 The material specifications given in will be complied with.

8.2.3.2 –The lowest concrete quality to be used in prefabricated reinforced concrete buildings will be C30.

8.2.3.3 –The cement mortar to be used during the assembly phase and post-tensioning applications will be of a type that gains strength quickly and does not shrink. The mortar compressive strength will not be less than the design compressive strength of the combined load-bearing elements.

8.2.3.4 –Prefabricated elements and connections subjected to earthquake loads, in relation to the selected ductility level **4.3.3** and **4.3.4** it is necessary to meet the conditions given in .

8.3. ARTICULATED JOINTS (MFB)

8.3.1. General Features

8.3.1.1 –Articulated connections are connections that cannot transfer moment in the main frame plane but can transfer shear force and axial force.

8.3.1.2 –Hinged connections can only be used in multi-storey buildings with cast-in-place or prefabricated void-free and/or tie beam (void) reinforced concrete shear walls.

8.3.1.3 –It is not allowed for horizontal carrier elements to be supported independently of other horizontal carriers, columns and walls in their jointed connections.

8.3.1.4 –Ignoring the friction force that occurs on the supports where the horizontal carriers sit, *ideal joint connection* assumption will be made.

8.3.1.5 –A steel reinforced elastomer cushion with a thickness of at least 10 mm will be used in connections formed with pins, sockets or different forms of these, where sheet metal is not used in their supports.

8.3.1.6 –It is mandatory that all pin connections, except purlin-truss connections, are filled with cement mortar and their free ends are finished with washers and nuts.

8.3.1.7 –Purlin-truss connections will be arranged using details with shear capacity that will transfer the forces that will occur within the roof plane and using wet or mechanical connection methods. The use of elastomer cushions in purlin-truss connections is not mandatory.

8.3.1.8 –Calculation of connection forces in all jointed connections **Table 4.1** defined in *Excess Strength Coefficient* (D) will be taken into consideration.

8.3.1.9 –In systems consisting of columns that are hinged at the roof plane and fixed at the bottom, hinged connection is required. *capacity design* will be applied to ensure that it remains in the linear elastic region.

8.3.1.10 –Lateral stability calculations will be made under the influence of vertical and horizontal loads on all carrier roof beams that have hinged connections to columns or work as hinged during the assembly phase.

8.3.1.11 –Beam top flange width at least ≥ 50 . Here n shows the beam clear span.

8.3.2. Overturning Stability in Articulated Connections

8.3.2.1 – 8.6 In the roof planes of buildings where the horizontal diaphragm condition given in is met, out-of-plane tipping control of the beams with hinged connections will be carried out. Equivalent earthquake load to be used in control **Equation (8.1)** will be calculated with .

$$F_H = \frac{m_{\text{dead}} S_e D}{(R/I)} \quad (8.1)$$

Here m_k shows the calculated mass for the connection point. F_H equivalent earthquake load, Overturning control will be carried out by acting on the beam end from the mass center level in a way that will give the most unfavorable internal forces in horizontal and vertical directions (**Figure 8.1** Pin diameters, weld areas and socket thicknesses determined by this investigation will be checked with the calculation methods defined in the relevant sections.

8.3.2.2 –For different assembly stages, lateral stability checks of prefabricated elements will be carried out.

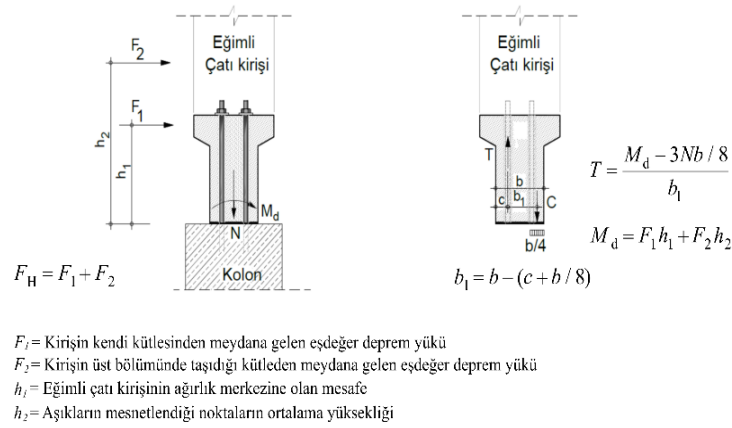


Figure 8.1

8.3.3. Pin Connections (MFB1)

Detail of a typical pin connection **Figure 8.2** is given in .

8.3.3.1 –Pin holes shall be at least 6 times the pin diameter from both the beam end and the free edge of the column support.

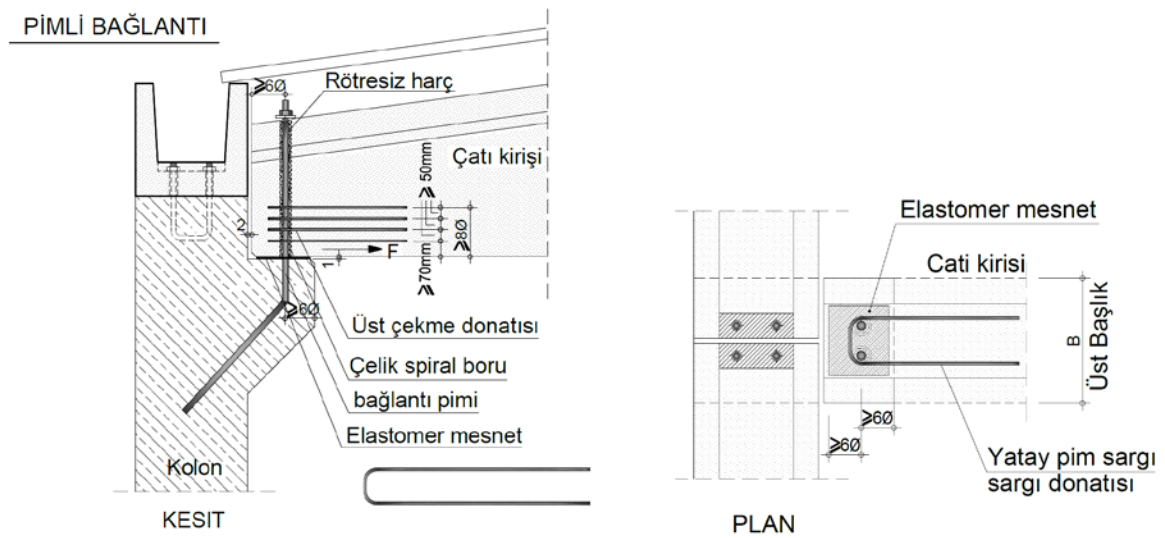


Figure 8.2

8.3.3.2 –The diameter of the pin-bolt holes to be left in the beams will be at least twice the pin diameter.

8.3.3.3 –No material or detail that will create a smooth surface can be used in pin slots.

8.3.3.4 –The cement mortar used to fill the pin holes will have the strength to provide the necessary interlocking.

8.3.3.5 –Sufficient wrapping reinforcement shall be used against tensile stress concentration in the column and beam end areas connected with pins. Wrapping shall be done with closed stirrups in the column end area and with horizontal U-stirrups in the beam end areas. Wrapping reinforcement shall be placed at least as long as the effective pin length, and the clear distance between them shall not exceed 50 mm. Pin sockets shall be placed so that they coincide with the corners of closed and U-stirrups, and if necessary, pin sockets shall be kept in the horizontal plane with closed crossties.

8.3.3.6 –The tensile strength of the concrete will not be taken into account in capacity calculations perpendicular to the pin axis.

8.3.3.7 –Shear force capacity of pin in main frame plane V_r , **Equation (8.2)** will be calculated with d/ϕ ratio shall be greater than 6.0. ϕ is the pin diameter, d beam longitudinal The smallest distance from the pin center to the beam or column edge in the direction, n indicates the number of pins.

$$V_r = 0.7 n \phi^2 f_{cd} f_{yd} \quad (8.2)$$

8.3.4. Welded Joints (MFB2)

Welded joint connection detail **Figure 8.3** is given in.

8.3.4.1 –Transverse wrapping reinforcement will be placed around the anchor reinforcement of the connection plate placed on the column.

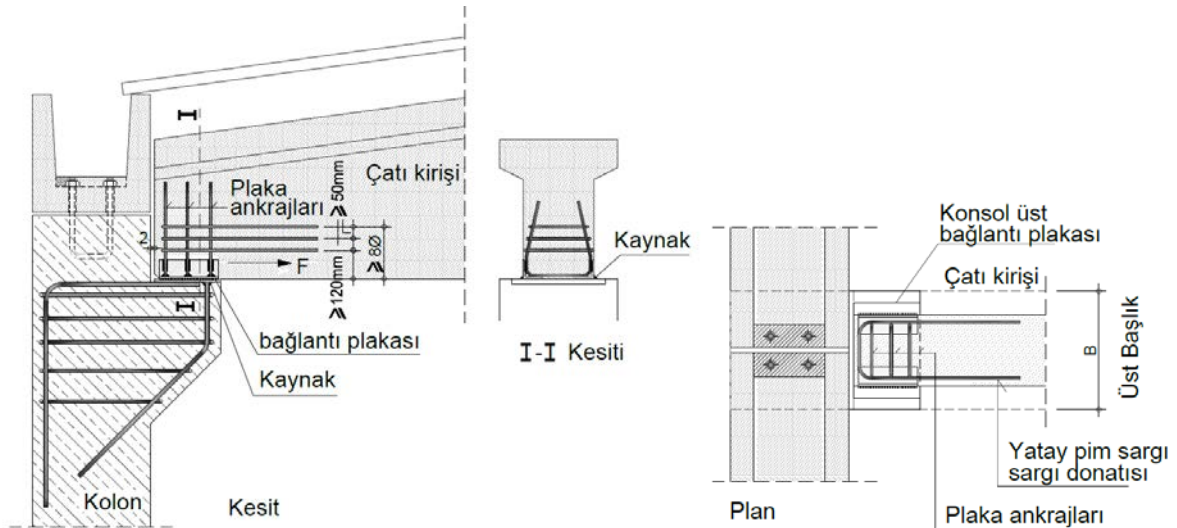


Figure 8.3

8.3.4.2 –At the joint, a welding capacity of up to 45% of the joint metal (electrode) yield strength is permissible, provided that it is equal to or greater than the yield strength of the materials being joined.

8.3.5. Socket Connections (MFB3)

Slotted joint connection details Figure 8.4 And Figure 8.5 is given in.

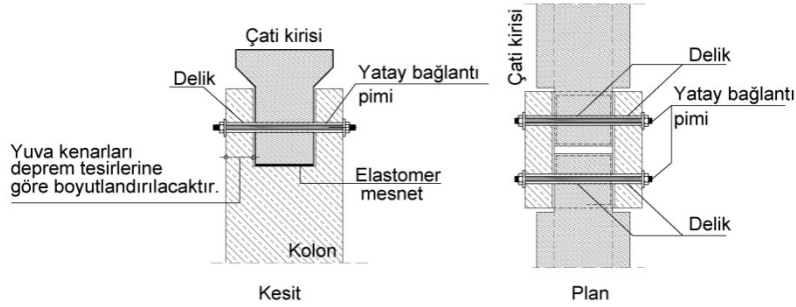


Figure 8.4

8.3.5.1 –The minimum thickness of the socket cheeks will be 15 cm and pins will be used.

8.3.5.2 –The bending reinforcement on the socket cheeks shall be calculated to resist 1.2 times the internal forces resulting from the overturning moment of the supported beam.

8.3.5.3 –For the direction perpendicular to the main frame plane on the housing cheeks Equation (8.1) The shear force obtained from the equivalent earthquake load calculated with will be resisted only by the reinforcements located on the socket sides.

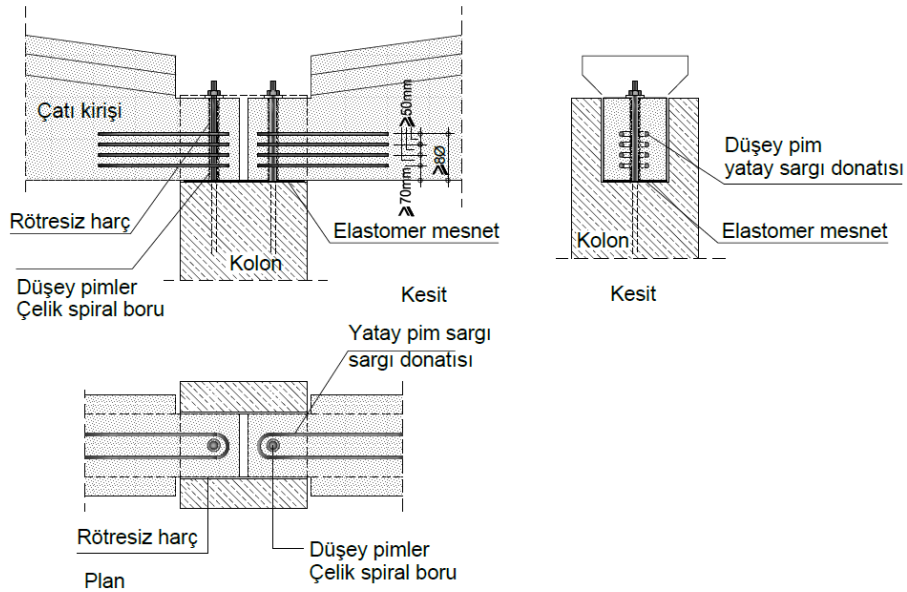


Figure 8.5

8.3.6. Other Articulated Connections

Articulated connections to be formed with different forms of the connection types defined above, **8.3.1** And **8.3.2** It will meet all the conditions defined in .

8.4. MOMENT TRANSFERRING CONNECTIONS (MAB)

8.4.1. General Features

8.4.1.1 –Moment-resisting connections of precast reinforced concrete buildings will be designed to exhibit ductile behavior under the influence of reversible cyclic earthquake loads.

8.4.1.2 –It will be shown that moment-resisting connections can transfer internal forces at the carrying capacity of the connected elements without loss of strength and ductility.

8.4.1.3 –Wet column-beam connection(**MAB1**), fully post-tensioned column-beam connection(**MAB2**), top wet-bottom welded column-beam connection(**MAB3**)and sleeve-pin column-beam connection(**MAB4**)for prefabricated reinforced concrete buildings R And D coefficients **Table 4.1** is given in .

8.4.1.4 –Fully post-tensioned column-beam connection(**MAB2**)Earthquake calculations for preliminary design of prefabricated reinforced concrete buildings **Chapter 4** According to , the earthquake calculation for the final design is for the column-beam connection mechanism (opening-closing) **8.4.3** Considering the nonlinear properties given in **Chapter 5** It will be done according to.

8.4.1.5 –Internal forces from earthquake effects in the calculation of reinforcement-plate and plate-plate welds and sleeve-pin connections. **Table 4.1** Given in *excess strength coefficient* (D) will be enlarged by multiplying by .

8.4.2. Wet Column-Beam Connection (MAB1)

Wet connections, which are made to exhibit equivalent behavior to cast-in-place column-beam connections, are connections where positive and negative moment capacities are provided by cast-in-place concrete and continuity reinforcement (**Figure 8.6** Beam-column geometric seating can be created with or without a gusset or with a temporary gusset.

8.4.2.1 –The interlocking lengths of continuity reinforcement will be determined in accordance with the relevant conditions of this Regulation and TS 500.

8.4.2.2 –In the combined beam and column sections; all necessary checks such as the interlocking length of longitudinal reinforcement, shear capacity, etc. will be made for both directions of the bending moment.

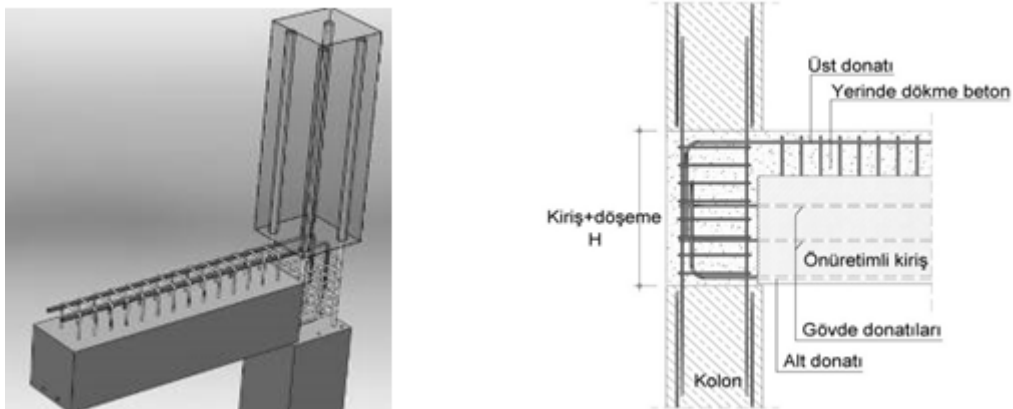


Figure 8.6

8.4.3. Fully Post-Tensioned Connection (MAB2)

Prefabricated beams can only be connected to columns by post-tensioning to transfer moments. (Figure 8.7) The sheaths through which the post-tensioning ropes are passed will be filled with cement mortar after the post-tensioning process.

8.4.3.1 –The space between columns and beams will be filled with non-shrinking cement mortar.

8.4.3.2 –In post-tensioned connections, the shear force transfer between the beam and the column face is realized by the clamping force and friction created by the post-tensioning ropes.

8.4.3.3 –The initial tensile force to be given to the post-tensioning ropes, taking into account the tension losses;

$$\frac{M_{pi} + M_{pj}}{-n} \quad (8.3)$$

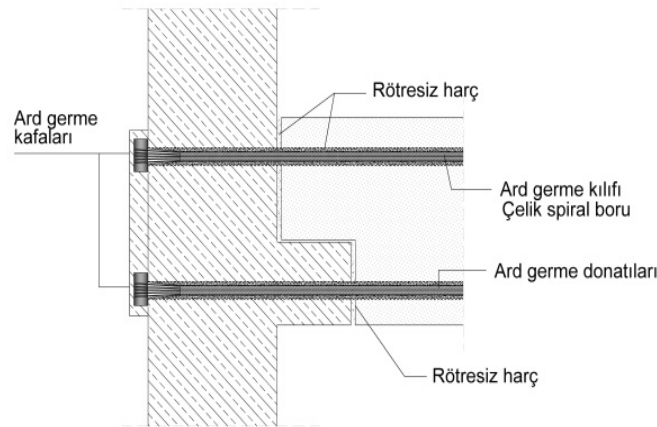


Figure 8.7

The cutting force determined by the expression ($\mu = 0.5$) will be calculated taking into account the transportation. Equation (8.3) in M_{pi} and M_{pj} bending moment capacities of the left and right ends of the beam, respectively, $-n$ indicates the beam free span.

8.4.3.4 –If the beam is supported on the column gusset, the contribution of the gusset to the shear force resistance can be taken into account. However, it will be shown that the shear force that will occur in the opposite direction to the direction of gravity due to the earthquake can only be carried by friction.

8.4.3.5 –Initial post-tensioning shall not exceed 50% of the rope yield stress.

8.4.3.6 –The internal cross-sectional area of the spiral sheath cannot be less than twice the total cross-sectional area of the post-tensioning ropes passing through it.

8.4.3.7 –For post-tensioned connection *cyclical behavior model envelope* and definitions of critical moment and rotation Figure 8.8 is given in. Here M_1 and θ_1 the moment at which the post-tensioning force is overcome and to return, M_2 and θ_2 shows the moment and rotation at the stage where 90% of the yield stress is reached in post-tensioning ropes.

8.4.3.8 –In the case of nonlinear time history analysis for post-tensioned connections *center-oriented cyclical behavior model* will be used (**Figure 8.8**).

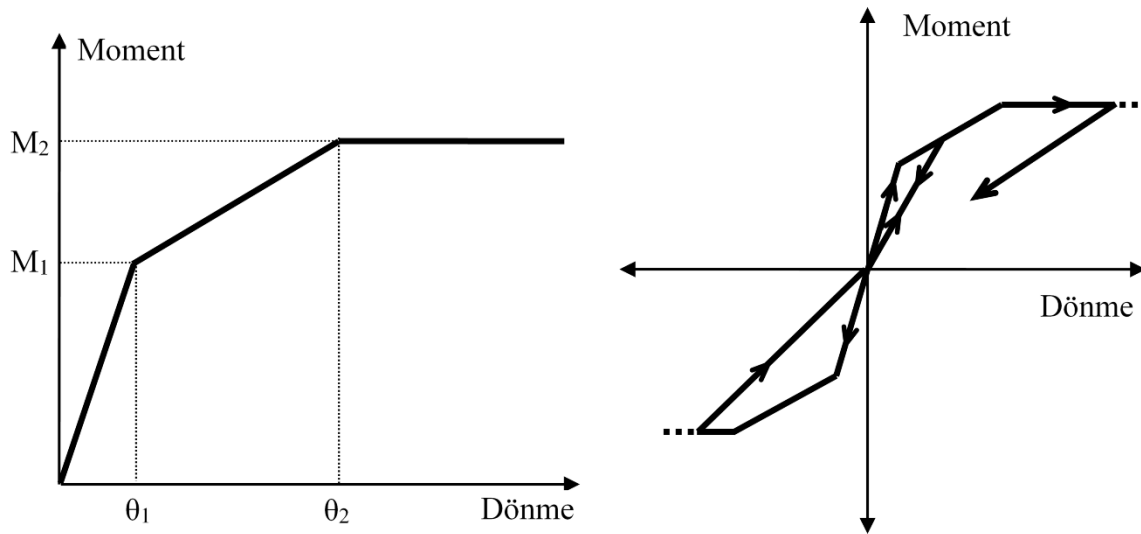


Figure 8.8

8.4.3.9 – Chapter 5 In case of non-linear earthquake calculation according to , the allowed opening-closing rotation in the connection is, *Limited Damage Performance Target* for θ_2 from returning, *Controlled Damage Performance Target* for 0.02 radians, *Migration Prevention Performance Target* shall not exceed 0.03 radians.

8.4.4. Wet Top – Welded Bottom Connections (MAB3)

These are the connections where the negative moment capacity continuity is provided by cast-in-place reinforced concrete and the positive moment capacity continuity is provided by welding. (**Figure 8.9**).

8.4.4.1 –The space left in the column for negative moment continuity reinforcement will be carefully filled during the pouring of the structural coating concrete.

8.4.4.2 –If a separate hole is left for each reinforcement, the hole diameter will be at least twice the diameter of the reinforcement that will pass through it. The hole will be filled with non-shrinking cement mortar, leaving no gaps. The distance between the holes will be at least as much as the diameter of the reinforcement.

8.4.4.3 –The space between columns and beams will be filled with cement mortar.

8.4.4.4 –Reinforcement arrangement will be made to ensure that plasticization occurs outside the connection area.

8.4.4.5 –To increase the deformation capacity of the beam longitudinal reinforcement welded to the steel plates, at least 12ϕ from the plate to the beam span (ϕ (related longitudinal reinforcement diameter) Plastic sheathing will be used in the long area to prevent adhesion with the concrete.

8.4.4.6 –The stirrup spacing at the distance where the beam sits on the console created in the column is at most $4\phi_L$ and will be 75 mm.

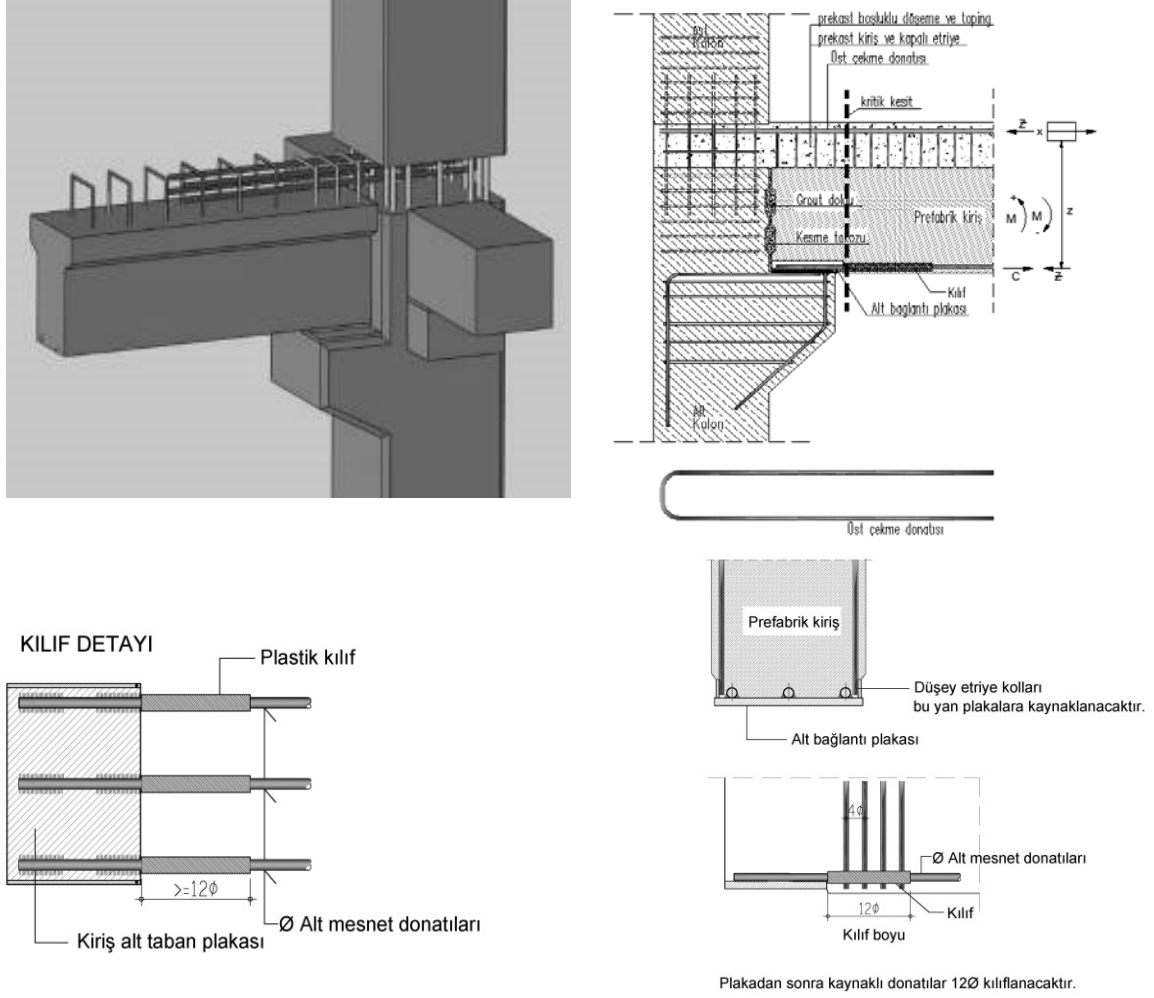


Figure 8.9

8.4.4.7 -In the area where the beam sits on the console formed in the column, the inverted U-shaped beam stirrups will be connected to the steel plates of at least 50 mm height welded to the bottom plate with a fillet weld seam. Welding of stirrups directly to the bottom plate is not allowed.

8.4.4.8 -In the combined beam and column sections; all necessary checks such as longitudinal reinforcement clamping length, horizontal shear capacity, welded bottom connection, etc. will be made for both directions of bending moment.

8.4.5. Sleeve-Pin Connections (MAB4)

In this type of connection, the negative moment is carried by the cast-in-place reinforced concrete, and the positive moment is carried by the shear capacity of at least 4 pins left in the column bracket. (Figure 8.10). Reinforcements carrying negative moment are concreted by screwing them into the space left in the column or into special sleeves left in the column.

8.4.5.1 -Adherence plates are used at the ends of the reinforcement when necessary to ensure adherence of the pins and sleeve anchor reinforcement.

8.4.5.2 -For the anchoring of the pins used in positive moment continuity to the beam, special grooved sheet metal pipes will be used in the beams. The inner diameter of the grooved sheet metal pipes will be at least twice the pin diameter. The pin holes will be filled with non-shrinking cement mortar.

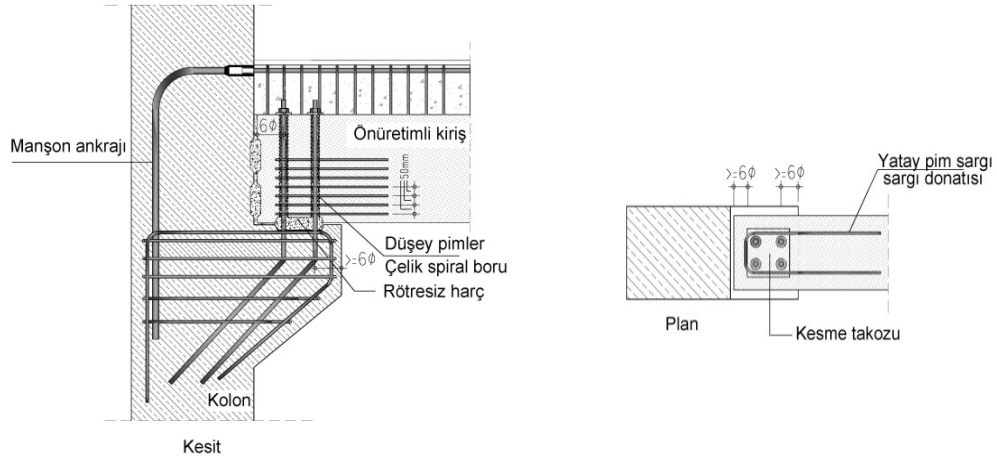


Figure 8.10

8.4.5.3 –The ends of the pins will be threaded and will be tightened using washers and nuts of adequate sizes immediately after the grooved pipe is filled with mortar.

8.4.5.4 –The total shear capacity of the pins will be greater than the total yield capacity of the beam support lower reinforcement.

8.4.5.5 –To ensure contact between the column console and the beam bottom surface, special cutting teeth will be left on the console and under the beam, and all gaps between the column and the beam will be filled with cement mortar.

8.4.5.6 –It will be shown that in coupled connections, failure will occur ductilely outside the coupled region.

8.4.6. Reinforcement Insertion Tools to be Used in the Joints of Moment Resisting Elements

To be used in moment transferring column-column, beam-beam and column-foundation connections *cutted, tube (with cement mortar or special adhesive), mechanical (with special bolts) or welding* rules regarding reinforcement attachments **ANNEX 8A** It is given in.

8.5. PRE-PRODUCTION CARRIER PANEL-FLOORING SYSTEMS

8.5.1. Double Walled Panels

8.5.1.1 –They are wall elements produced by combining two thin-walled panels with cage reinforcements. After the double-walled panels and floor elements are assembled, panel-panel, panel-floor bond reinforcements are placed and the panel gaps are filled with concrete together with the floor.

8.5.1.2 –The minimum wall thickness will be 50 mm.

8.5.1.3 –Typical details for different panel combinations **Figure 8.11** is given in .

8.5.2. Single Wall Panels

8.5.2.1 –After the single-skin prefabricated wall panel and floor elements are installed, panel-panel and panel-floor bonding reinforcements are placed to form the floor concrete.

8.5.2.2 –Minimum panel thickness will be 120 mm.

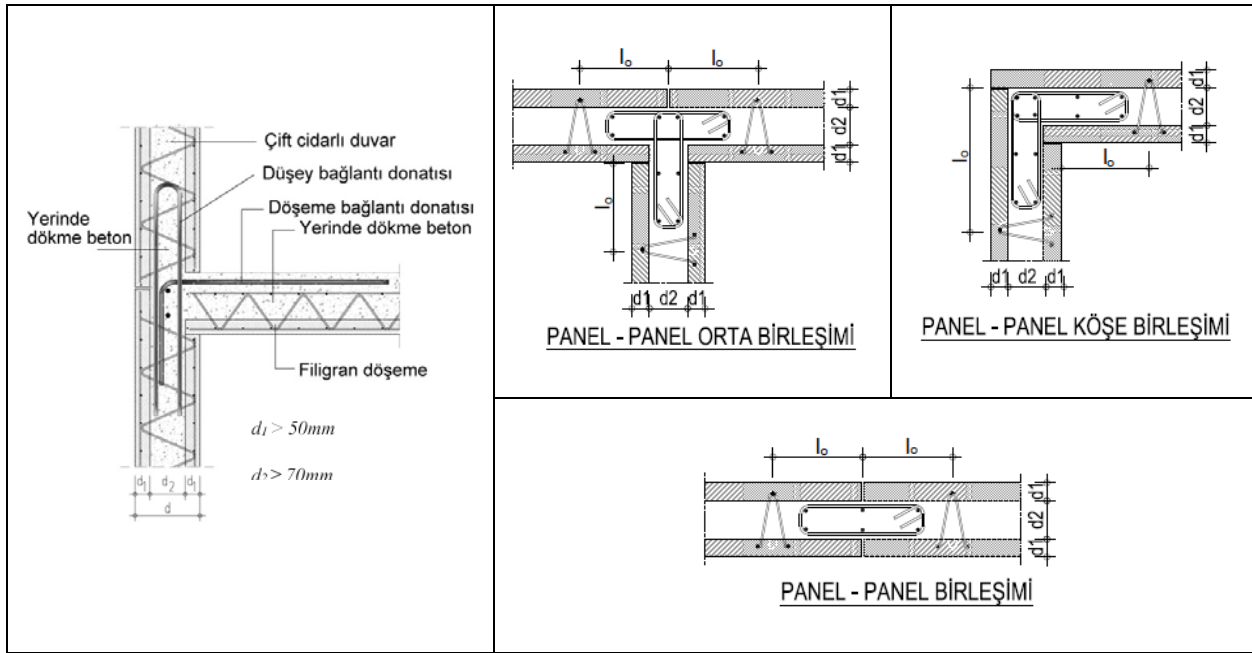


Figure 8

8.5.2.3 –Typical detail for different panel connections

8.6. DIAPHRAGM IN PRE-ENGINEERED BUILDINGS

In prefabricated single or multi-storey buildings, the diaphragm will be created as defined and the floor will be

And 8.6.2' also

8.6.1. Roof Diaphragm

In the roof planes of single-storey building systems where all earthquake loads are carried by the roof level, the metal roof cover will be taken in-plane and taken into account. Diaphragm formation

by

I connected
rak attention
has been given.

8.6.2. Creation of Floor Diaphragm

8.6.2.1 –Internal diaphragm forces as a basis for the design of prefabricated reinforced concrete buildings, Chapter 4 And Chapter 7 It will be determined by the diaphragm calculation and design made in accordance with the additional conditions given in and listed below.

8.6.2.2 –While creating the diaphragm model; the effective stiffness of the floor elements such as prefabricated reinforced concrete TT slab, hollow slab, filigree slab will be taken as the stiffness of the composite section if the horizontal shear friction conditions are met, otherwise it will be taken as the stiffness of only the structural coating concrete.

8.6.2.3 –Floors will not be assumed to be infinitely rigid in their own planes, finite element modeling will be performed. It will be shown that vertical and horizontal loads are transferred safely. Internal forces obtained with diaphragm calculations will be counteracted by tie reinforcements calculated in accordance with the shear friction calculation method. Additional reinforcement will be used in areas where there are tensile forces on the diaphragm.

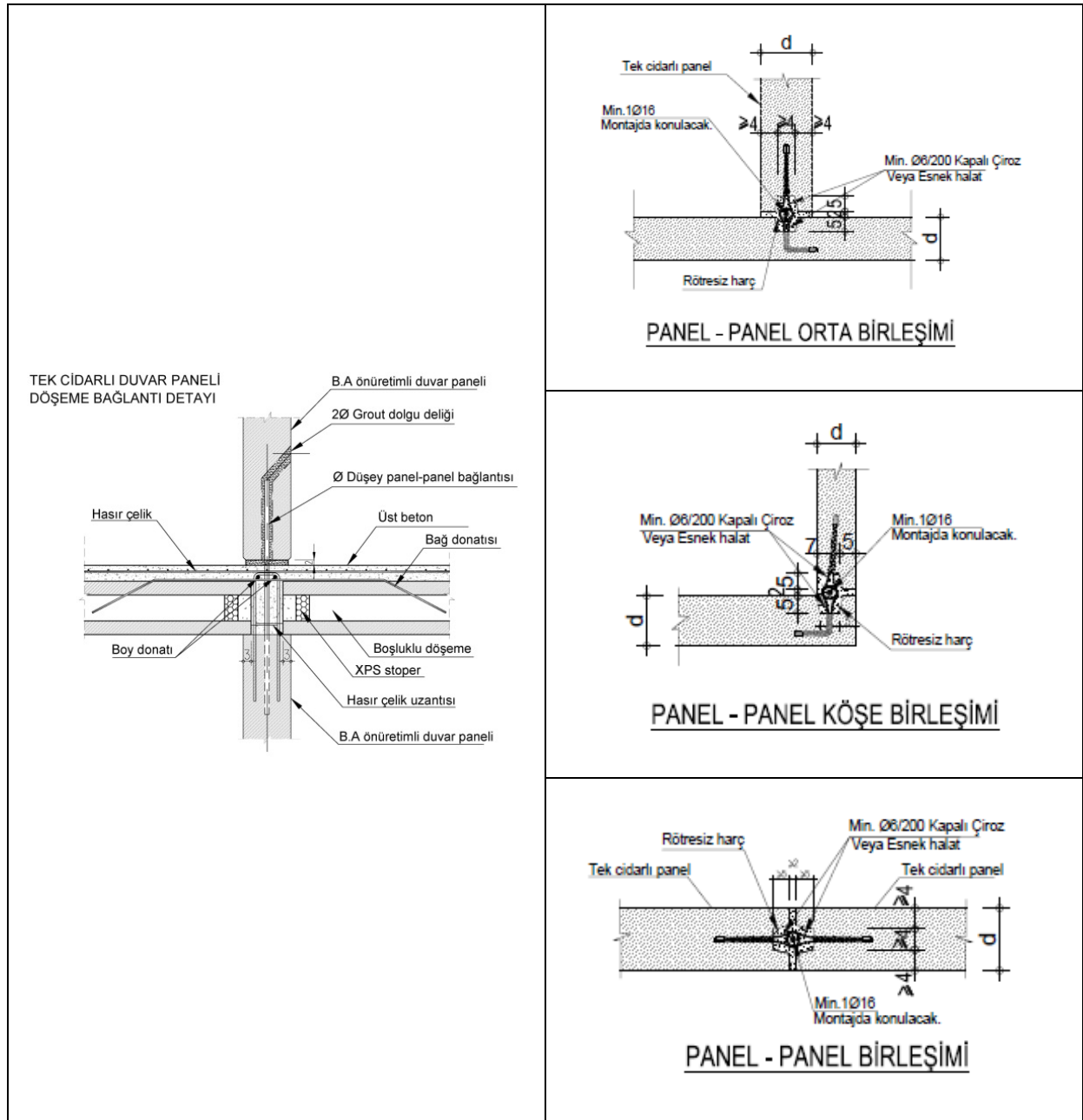


Figure 8.12

8.6.2.4 -It is mandatory to limit prefabricated reinforced concrete floor systems with shear walls and/or beams. For this purpose, frame beams will be used in both directions and on all axes.

8.6.2.5 -In areas where prefabricated reinforced concrete floor systems are directly connected to the walls, calculations will show that the internal forces of the diaphragm are transferred through the reinforcement and appropriate details within the structural coating concrete.

8.6.2.6 -Structural coating concrete application is mandatory together with precast concrete floor elements. Structural coating concrete thickness will be at least 70 mm.

8.6.2.7 -Structural coating concrete quality **8.2.3.2** It will be suitable for.

8.6.2.8 –Longitudinal reinforcement of at least 0.0015 will be placed in the structural coating concrete separately in both main directions.

8.7. ADDITIONAL CONDITIONS FOR BUILDINGS WITH ARTICULATED COLUMNS AT THE TOP

8.7.1 –In prefabricated reinforced concrete buildings consisting of single-storey frames with columns hinged at the roof level; in case a partial mezzanine floor is created, provided that it does not exceed 25% of the building base area, the additional internal forces that will occur in the roof diaphragm due to the horizontal rigidity of this partial mezzanine floor and the torsional irregularity in the plan will be taken into account or the partial mezzanine section will be separated with a joint to eliminate this irregularity.

8.7.2 –The connection of the intermediate floor carrier system to the main carrier system will be made jointly or monolithically.

8.7.3 –In the three-dimensional structural model, the in-plane rigidity provided by the roof covering material will be taken into account.

8.7.4 –In case of using single-storey frames with columns hinged at the roof level as the top floor (attic) of multi-storey cast-in-place reinforced concrete or prefabricated reinforced concrete buildings.

4.3.6The method given in will be applied.

ANNEX 8A – REINFORCEMENT ADDING TOOLS

In moment transferring column-column, beam-beam and column-foundation connections *cuffed, tube (with cement mortar or special adhesive), mechanical (with special bolts) or welding* can be formed as reinforcement attachment.

8A.1 –It shall be shown that the strength and ductility capacities provided in the section by means of reinforcement splicing are compatible with the strength and ductility of the connected element(s).

8A.2 –Shear force control will be performed on the joints.

8A.3 –For welded reinforcement joints **7.2.7.1** is valid.

8A.4 –High strength used in mechanical connections ($f_{yk} > 800$ MPa) In case of post-tensioning of bolts, it will be shown that they remain elastic under earthquake effects.

8A.5 –The expected behavior in sleeved or tubed (high strength cement mortar or special adhesive) reinforcement splices is; *pull, pressure And cyclic loading* For these cases, plasticization occurs in the reinforcement connected away from the joint area.

8A.6 –A sample consisting of two pieces of reinforcement and a splicing tool placed between them. *pull, pressure, abrasion, fatigue And cyclic loading* tests will be applied. In these tests; if reinforcement bar breakage is observed in all examined samples, the splicing tool will be considered successful.

8A.7 –For each different type of reinforcement attachment tool, 2% of the total number of uses will be tested. This number cannot be less than 5 pieces.

8A.8 –If the yield strength of the reinforcement determined by a sufficient number of tensile tests is less than or equal to the yield strength declared by the splicing tool manufacturer. *cyclic loading experiment* does not need to be implemented.

8A.9 –*In the tensile test* The sample must reach 100% of the characteristic breaking strength of the reinforcement, 135% of the characteristic yield strength and 120% of the determined yield strength of the reinforcement.

8A.10 –*In the pressure test*; the sample must reach 125% of the characteristic yield strength of the reinforcement.

8A.11 –*Cyclic loading* in the experiment, **Table 8A.1** The cycles defined in will be applied.

8A.12 –Created with high strength mortar *in tube fittings attachments* At least four cube mortar samples shall be taken for each reinforcement joint to be tested. Mortar samples and tubular reinforcement joint shall be stored under the same conditions. At least two mortar samples shall be subjected to pressure tests at the beginning and end of the tests applied on the reinforcement joint. In order to determine the mortar strength, the average of the results of at least four samples shall be taken and this value shall be evaluated as the minimum mortar strength. This strength shall be higher than the minimum mortar strength declared by the manufacturer for the reinforcement joint.

8A.13 –If the rib shapes of the reinforcement bars added with the mortar tube are different, separate tests will be carried out for each rib shape.

Table 8A.1. Cyclic Loading Tests to be Applied to Reinforcement Splices

My name	Pull	Pressure	Number of Cycles
1	$0.95 f_{yk}$	$0.5 f_{yk}$	20
2	$2\epsilon_y$	$0.5 f_{yk}$	4
3	$5\epsilon_y$	$0.5 f_{yk}$	4
4	Tensile will be applied until breakage		

f_{yk} : characteristic yield strength of reinforcement bar

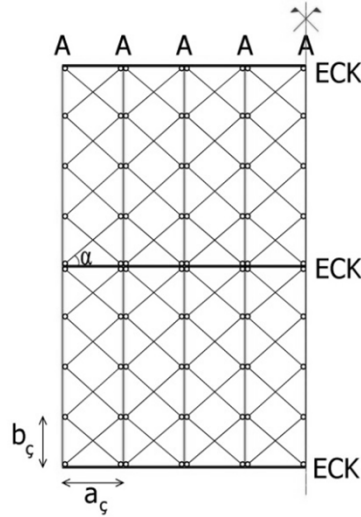
ϵ_y : reinforcement unit strain corresponding to the true yield stress

8A.14 –Freeze-thaw tests for mortar shall be carried out in accordance with TSE CEN/TR 15177. After at least 300 freeze-thaw cycles, the ratio of the elasticity modulus to its initial value shall be at least 90%.

8A.15 –The rules given in this section, *lifting* And *purpose of carrying* It does not apply to sleeves used with.

ANNEX 8B – DEFINITION OF ROOF PLANE DIAPHRAGM

8B.1 –The three-dimensional carrier system model to be created for the calculation, *trapezoidal sheet metal* or *sandwich panel* type roofing materials will also be included. The roofing material will be represented by elastic cross pairs with jointed ends placed between the purlins.



ECK = Inclined Roof Beam; A = Purlin; a_c = purlin spacing $\leq 2.0\text{m}$, $\alpha \approx 45^\circ$

Figure 8B.1

8B.2 –Equivalent axial rigidities of the braces representing the trapezoidal sheet metal or sandwich panel type roof covering made of steel or aluminum alloy (EA)_{to} [kN] **Equation (8B.1)** will be calculated.

$$(EA)_{to} = 3.5t \quad (8B.1)$$

Here, t [mm] shows the wall thickness of the sheet forming the cover material. In the case of trapezoidal sheet metal, it is equal to the sheet thickness, and in the case of sandwich panels, it is equal to the sum of the bottom and top sheet thicknesses. **Equation (8B.1)** For the application of , the purlin spacing should be maximum 2 m, cross angle α 'of 45° should be around .

8B.3 –Additional internal forces arising in purlins, gutter beams and inclined roof beams (OCB) due to the diaphragm formed in the roof plane will be taken into account in the dimensioning of the relevant elements and in the connection calculations.

8B.4 –The number of connecting elements connecting the roof to the purlins will be determined by using the largest axial force formed in the crosses representing the roof cover. The largest cross force N_{Max} Taking the projection of , the number of required connectors is **Equation (8B.2)** will be calculated with .

$$n = \frac{2N_{Max} \cos \alpha}{Q_{suck}} \quad (8B.2)$$

Here Q_{suck} the shear force that a fastener can safely carry, n whereas b_c indicates the number of connecting elements to be used in length (**Figure 8B.1**).

CHAPTER 9 – STEEL BUILDING SUPPORT SYSTEMS UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGN

9.0. ICONS

In the dimensional expressions in this section where the following symbols are used, the forces are: *Newton*[[N], lengths *millimeter*[[mm], angles *radian*[rad] and stresses *MegaPascal*[MPa] = [N/mm²] unit.

A	= Cross-sectional area
A_c	= Concrete cross-sectional area =
A_{to}	Effective net cross-sectional area =
A_{fb}	Header cross-sectional area =
A_g	Lossless cross-sectional area
A_g	= Total cross-sectional area of the composite element
A_{os}	= Cross-sectional area of spiral wound reinforcement =
A_s	Cross-sectional area of steel
$A_{s.c.}$	= Cross-sectional area of the steel core showing yield deformation
$A_{s.h.}$	= Minimum total stirrup area in the confinement zone in composite columns with stirrups = Longitudinal
A_{sr}	reinforcement area
A_w	= Body cross-sectional
A_w	area = Fillet weld area
a	= Horizontal distance between stirrup arms and/or
b	crossties = Width (half head width)
b_{bf}	= Head width of beam section
b_k	= Distance between the outermost transverse reinforcement axes in the column cross-section
C_a	= Ratio of required axial force strength to design axial force strength = Outer
D	diameter in circular ring sections
D	= Excess strength coefficient
D	= Circular column core diameter (distance between spiral wound reinforcement axes) = Beam
d_b	cross-section height
d_c	= Column cross-section height
TO	= Structural steel elasticity modulus, $TO=200000$ N/mm ²
TO_d	= Earthquake effect
to	= Tie beam length
F_{cr}	= Critical buckling stress

F_{cre}	= Critical buckling stress calculated with possible yield stress =
F_{TO}	Characteristic tensile strength of the weld material (electrode) =
F_u	Characteristic tensile strength of structural steel
F_y	= Characteristic yield stress of structural steel
F_{yb}	= Characteristic yield stress of the beam material
F_{yc}	= Characteristic yield stress of column material
F_{ysc}	= Characteristic yield stress of steel core or tensile test of steel core True yield stress determined by
F_{YSR}	= Characteristic yield stress of reinforcement steel
f_{ck}	= Characteristic compressive strength of concrete
f_{ywk}	= Characteristic yield strength of transverse reinforcement
G	= Constant load
H	= Floor height
H	= Ground horizontal
H_c	thrust = Column height
$H_{average}$	= Average of column heights above and below the node = Cross-
h	section height
h_i	= Floor height of the i'th floor of the building
h_{he}	= Distance between the centers of the section headings = Building
I	importance coefficient
I	= Radius of inertia
I_y	= Radius of inertia of the beam cross-section relative to the weak axis
K	= buckling coefficient
L	= Bar length
L_b	= Where the pressure head is supported in the lateral direction or the cross-section is protected against distortion distance between supporting points
-h	= Distance of the possible plastic hinge point at the beam end to the column face (See Annex 9B)
-n	= Distance between possible plastic hinge points at the beam ends (See Fig. Annex 9B) =
M_a	(GKT) required bending strength determined based on load combinations
$M_{bending}$	= (GKT) Possible plastic hinge at the beam ends based on load combinations Additional bending moment occurring on the column axis due to the shear force at the point
M_p	= Characteristic plastic flexural strength

M_{pb}^*	= Characteristic for each of the connected beams in the beam-column joint area Plastic moment strengths are $1.1 R_p$ possible plasticity at the beam end with solid bending moment determined by the sum of the additional bending moment occurring in the column axis due to the shear force at the joint
M_{pc}	= Characteristic plastic bending strength of the column
M_{pc}^*	= Column bending in the beam-column joint area in accordance with the direction of the earthquake. The bending moment strength of the upper or lower floor columns is calculated by taking into account the design axial forces that minimize the moment strengths.
M_p	= Possible bending moment capacity
M_r	= Required flexural strength
M_u	= (YDKT) required bending strength determined based on load combinations
M_{fly}	= Required bending moment of the beam on the column face at the beam-column connection strength
M_{uv}	= (YDKT) possible plastic hinge at the beam ends based on load combinations Additional bending moment occurring on the column axis due to the shear force at the point
N_{dm}	= Column axial force calculated under the combined effect of vertical loads and earthquake
P_a	= (GKT) required axial force strength calculated with load combinations
P_{hungry}	= Taking into account the increased earthquake effects, for (GKT) load combinations Calculated required axial compressive force
P_{br}	= Required axial force resistance of lateral support members = Axial
P_{no}	compressive force resistance of composite column cross-section =
P_r	Required axial force resistance
P_u	= (YDKT) required axial force strength calculated with load combinations
P_{fly}	= Taking into account the increased earthquake effects, for (YDKT) load combinations Calculated required axial compressive force
P_y	= Axial force strength at yield limit state
$P_{y_{sc}}$	= Steel core axial force yield strength
Q	= Live load
R	= Carrier system behavior coefficient
R_t	= Ratio of possible tensile strength to characteristic tensile strength
R_y	= Ratio of possible yield stress to characteristic yield stress
R_{y_b}	= Ratio of possible yield stress in the beam to characteristic yield stress
R_{y_c}	= Ratio of possible yield stress in the column to characteristic yield stress
S	= Snow load

S	=Transverse reinforcement spacing in composite
S	column =Spiral wound reinforcement pitch = Thickness
t	
t_{bf}	= Beam cross-section flange thickness =
t_{cf}	Column cross-section flange thickness =
t_t	Reinforcement plate thickness
t_w	= Body thickness
U	= Length of the shear zone perimeter
V_d	=Possible plastic joint at beam ends at points, $(1.2G+0.5Q+0.2S)$ load Calculated shear force under the effect of vertical load with the combination
V_{TO}	= Design shear force resulting from earthquake effects
V_n	= Characteristic shear force strength
V_p	= Plastic shear strength
V_{fly}	= Required shear strength of the beam-column joint on the column face =
V_{up}	Required shear strength of the beam-column joint shear (panel) zone
W_p	= Plastic strength moment
W_{pb}	= Beam plastic strength moment
W_{pc}	= Column plastic strength moment
W_{RBS}	= Plastic strength moment of weakened beam cross-section =
ΔI	Relative story drift at the i'th floor of the building
β	= Compressive strength correction coefficient
γ_p	= Tie beam rotation angle
θ_p	= Relative story drift angle
λ_{hd}	= Cross-section condition limiting value for elements with high ductility level =
λ_{md}	Cross-section condition limiting value for elements with limited ductility level
μ	= Coefficient of friction
ρ_s	= Volumetric ratio of spiral reinforcement in circular column, $\rho_s = \frac{4A_{os}}{D_s}$
φ	Strength coefficient
ω	= Hardening effect correction
Ω	factor = Safety factor

9.1. SCOPE

9.1.1 –The dimensioning and arrangement of the joints of the structural system elements of the steel building under the effect of earthquakes shall be carried out in accordance with the relevant standards and regulations in force on this subject, as well as the special rules specified in this section.

9.1.2 –Horizontal load-bearing systems of steel buildings within the scope of this section may consist of only moment-resisting steel frames, only centrally or eccentrically braced steel frames, only buckling-restrained braced steel frames, or combinations of moment-resisting steel frames with braced steel frames or reinforced concrete shear walls. In addition, steel-reinforced concrete composite columns of moment-resisting frame-type load-bearing systems are within the scope of this section.

9.1.3 –Rules for the design of systems created within normal floor and roof planes and transferring horizontal forces to horizontal load-bearing systems.**9.10** is given in .

9.1.4 –Rules for foundation design of buildings with steel and steel-reinforced concrete composite columns **Chapter 16** is given in .

9.2. GENERAL RULES

9.2.1. Classification of Steel Carrier Systems

Horizontal load-bearing systems of steel buildings, in terms of their earthquake resistance,**9.2.1.1**, **9.2.1.2** and **9.2.1.3** It is divided into three classes defined in .

9.2.1.1 –The following horizontal load-bearing systems for steel buildings: *Systems with High Ductility Level* has been defined as.

(a) **9.3** and **9.11.1.1** meeting the conditions specified in *Moment Resisting Steel Frame* type carrier systems.

(b) **9.6** meeting the conditions specified in *Central Braced Steel Frame*, **9.8** meeting the conditions specified in *Eccentric Braced Steel Frame* and **9.9** meeting the conditions specified in *Buckling Resistant Braced Steel Frame* type carrier systems.

(c) **9.3** With the moment resisting steel frames defined in (b) Systems consisting of a combination of braced steel frames as defined in .

9.2.1.2 –The following horizontal load-bearing systems for steel buildings: *Systems with Limited Ductility Level* has been defined as.

(a) **9.4** and **9.11.1.2** meeting the conditions specified in *Moment Resisting Steel Frame* type carrier systems.

(b) **9.7** meeting the conditions specified in *Central Braced Steel Frame* type carrier systems.

(c) **9.4** With the moment resisting steel frames defined in (b) Systems consisting of a combination of braced steel frames as defined in .

9.2.1.3 – *Limited ductility level* moment-resisting steel frame systems *high ductility level* Systems created by using steel braced frames or reinforced concrete shear walls *Ductility Level Mixed Systems* defined as. However, **9.11.1.2(a)** meeting the conditions specified in *limited ductility level* Composite column moment resisting frame systems are only allowed to form a mixed system with reinforced concrete shear walls.

9.2.1.4 –To be applied in the design of systems falling into these three classes under earthquake effects. *R* carrier system behavior coefficients And *D* excess strength coefficients with permitted building height classes (BYS) **Table 4.1** Special conditions and circumstances regarding the mixed use of these systems are given in **4.3.4** It is located in .

9.2.1.5 –Steel buildings containing at most two different horizontal load-bearing systems in the vertical direction or mixed buildings consisting of reinforced concrete and steel load-bearing systems and those to be applied to them. *R, D* Conditions for coefficients **4.3.6** is given in .

9.2.1.6 –In case there are reinforced concrete walls with high ductility level in the carrier system, for the design of reinforced concrete walls **7.6** The rules given in will be applied.

9.2.2. Relevant Standards and Design Principles

9.2.2.1 –The design of steel carrier systems within the scope of this section; **Part 2, 3, 4** And **5** It will be made according to the relevant standards and especially the rules in this section, taking into account the calculation rules given in and the loads foreseen in TS 498.

9.2.2.2 –Steel structure elements and connections must perform all expected functions throughout the operational life of the structure. *security* It shall have sufficient strength, stability and rigidity to perform under the required conditions.

9.2.2.3 –In the design of a steel building, provided that only one of them is applied, the ones defined in the "Regulation on the Design, Calculation and Construction Principles of Steel Structures" published in the Official Gazette dated 04/02/2016 and numbered 29614 *Design with Load and Resistance Coefficients* (YDKT) method or *Design with Safety Factors* (The GKT method can be used. However, **Annex 9B** The moment resisting beam-column connections located in , **9.11** Only the YDKT method will be used in the design of composite column bearing systems and foundations given in . **Annex 9B** In the design of moment resisting beam-column connections to be dimensioned in accordance with the YDKT method **Annex 9B** The strength coefficients defined in will be used.

9.2.2.4 –Within the scope of this Regulation, the design rules to be applied for section calculations, joint and joint calculations to be made under the combined effect of vertical loads and earthquake are given separately for the relevant YDKT and GKT methods.

9.2.3. Material Conditions

9.2.3.1 –Within the scope of this regulation, all structural steels defined in the "Regulation on Design, Calculation and Construction Principles of Steel Structures" can be used. However, the steel material to be used in the elements of the horizontal load-bearing system must also meet the following conditions.

(a) The characteristic yield stress of structural steel to be used in elements expected to exhibit nonlinear behavior is 355 N/mm² should not exceed its value. In addition, **9.4** The characteristic yield stress of structural steel to be used in columns of horizontal load-bearing systems defined in this section, except for columns of steel frames with limited moment resisting ductility level specified in , is 460 N/mm² Cannot exceed .

(b) Minimum Charpy-V-Notch (CVN) strength (in tests carried out in accordance with ASTM A673 or equivalent standards) in rolled profiles with a minimum thickness of 40 mm, sheets with a minimum thickness of 50 mm and artificial profiles manufactured from these sheets. *Notch Toughness* value is 21°C will be 27 Nm (27 J).

9.2.3.2 –High strength bolts of (8.8) or (10.9) quality are used in the connections and joints of elements under the effect of earthquake loads. *full preload* In the connections where the bolts are under the effect of shear force, the unthreaded body cross-section of the bolts shall be in the shear plane. The contact surface between the joined parts shall provide a friction coefficient of at least $\mu=0.20$ in the connections where crushing is effected.

Normal bolts defined in the "Regulation on Design, Calculation and Construction Principles of Steel Structures" can be used in the connections and joints of elements that are not affected by earthquake loads.

9.2.3.3 –In welded joints, weld metal suitable for the steel material and welding method will be used and the yield stress of the weld metal will not be less than the yield stress of the base metal to be joined. In all welds of elements and joints under the effect of earthquake loads. **Table 9.1** Weld metal that meets the conditions specified in will be used.

9.2.3.4 –Bolts and welds cannot be used together in such a way that they share the force transferred to the joint or a component of this force (See Fig. **Annex 9A**).

**Table 9.1 – Connections and Joints of Elements Under the Effect of Earthquake Loads
Properties of Weld Metal to be Used**

Weld Metal Class	E480	E 550
Characteristic Yield stress, N/mm ²	400	470
Min. Tensile strength, (F_{T0}) N/mm ²	480	550
Min. Elongation, %	22	19
Notch Toughness (CVN)	- 18°C de min. 27J	

9.2.4. Possible Material Strength

In the design of steel structural elements and connections where deemed necessary in the relevant articles of this section. *possible (expected) material strengths* For possible collapse limit states in the element with a determined strength, *characteristic strength* in their statements characteristic yield stress, F_y and characteristic tensile strength, F_u instead of the possible yield stress, $R_y F_y$ and possible tensile strength, $R_t F_u$ will be used. Possible yield stress and

The coefficients to be applied in the calculation of possible tensile strength depend on the class of structural steel and the type of element, **Table 9.2** is given in .

Table 9.2 – R_y And R_t Coefficients

Structural Steel Class and Element Type	R_y	R_t
Rolled profiles and sheets made of S 235 steel	1.4	1.1
Rolled profiles and sheets made of S 275 steel	1.3	1.1
Rolled profiles and sheets made of S 355 steel	1.25	1.1
Rolled profiles made of S 460 steel	1.1	1.1
Pipe and Box profiles	1.4	1.3
Reinforcement steel	1.2	1.2

9.2.5. Load Combinations Including Earthquake Effects

Steel structure elements will be dimensioned based on Section 5.3 of the “Regulation on Design, Calculation and Construction Principles of Steel Structures”. In this dimensioning, **9.2.6** As load combinations that include earthquake effects, except for the special cases specified in **9.2.5.1** or **9.2.5.2** The load combinations given in will be used.

9.2.5.1 –YDKT (*Design with Load and Resistance Coefficients*) when applied *Load Combinations Involving Earthquake Effects* for **Section 4.4.4.2(a)** Live load value is 5.0 kN/m² If is less than or equal to **Equation (4.11)** in *Q*, %50 percent
However, this reduction will not apply to garages and buildings with high density of people.

9.2.5.2 –GKT (*Design with Safety Factors*) when applied *Load Combinations Involving Earthquake Effects* below **Equation (9.1)** It will be taken as in .

$$G+0.75Q+0.75S\pm0.75(0.7TO_d) \quad (9.1a)$$

$$G\pm0.7TO_d \quad (9.1b)$$

$$0.6G+0.75H\pm0.7TO_d \quad (9.1c)$$

9.2.6. Earthquake Effects Magnified by Excess Strength Factor

Where explained in the relevant articles of this section, the required strength of steel structural elements, joints and additional details shall be taken into account in order to withstand earthquake effects. *excess strength coefficient D* Internal forces determined by multiplying and magnifying by **9.2.5** will be obtained by using the load combinations defined in . However, these internal forces, *capacity design principle* As required by , the effects of hardening and material strength increase are also taken into account. *flowing (mechanism) compatible with the situation* shall not be taken greater than the internal forces. *D excess strength coefficients*, depending on the types and ductility levels of the carrier systems **Table 4.1** is given in .

9.2.7. Cross-Section Conditions

S *High level of ductility* or *limited ductility level* cross-section of the system elements to be designed as *head width/thickness, body height/thickness* And *diameter/wall thickness* rates **Table 9.3** It will not exceed the relevant limit values given in .

9.2.8. Stability Connections in Beams

9.2.8.1 –*High ductility level* or *limited ductility level* Bottom and/or top heads of steel beams of horizontal load-bearing systems designed as **Equation (9.2)** or **Equation (9.3)** It will be supported against lateral buckling in a way that meets the conditions given by . In steel carrier systems where reinforced concrete slabs work as composite with steel beams, it is not mandatory to comply with this condition in the heads of the beams connected to the reinforced concrete slab.

Maximum distance between lateral support members

(a) Earthquake load carrying systems *high ductility level* in beams designed as elements

$$L_b \leq 0.086 I_y F_y \frac{TO}{\dots} \quad (9.2)$$

(b) Earthquake load carrying systems *limited ductility level* In beams designed as elements

$$L_b \leq 0.17 I_y F_y \frac{TO}{\dots} \quad (9.3)$$

will provide the condition.

Also, during nonlinear deformation of the system *plastic joint* possible points, At points where concentrated loads act and at points where the beam cross-section changes suddenly, beam flanges will be supported against lateral displacement and torsion.

9.2.8.2 – High level of ductility or limited ductility level In the elements to be designed as lateral stability support elements will be dimensioned to have sufficient strength and rigidity. The conditions to be taken into account for this are given below.

(a) The required strength and rigidity of the elements to be used against lateral displacement and torsion in the beam heads of horizontal load-bearing systems are determined in the “Design, Calculation and Analysis of Steel Structures”. “Regulation on Construction Principles” Section 16.3 $C_d = 1.0$ for the beam **Equation (9.4)** It will be determined based on the bending strength given by .

$$M_r = M_a = R_y F_y W_p 1.5 / \quad (GKT) \quad (9.4a)$$

or

$$M_r = M_u = R_y F_y W_p \quad (YDKT) \quad (9.4b)$$

(b) In possible plastic hinge areas, the elements to be used against lateral displacement and torsion in beam flanges will be sized to have sufficient strength and rigidity under the special conditions given below.

(1) Required strength of lateral support elements for both heads **Equation (9.5)** will be calculated with .

$$P_{br} = (0.06 \ 1.5) R_y F_y W_p h_{fe} \quad (GKT) \quad (9.5a)$$

or

$$P_{br} = 0.06 R_y F_y W_p h_{fe} \quad (YDKT) \quad (9.5b)$$

(2) The required strength of the elements used in the beam against torsion is specified in Section 16.3 of the “Regulation on the Design, Calculation and Construction Principles of Steel Structures”. $C_d = 1.0$ is taken **Equation (9.6)** will be calculated with .

$$M_a = (0.06 \ 1.5) R_y F_y W_p \quad (GKT) \quad (9.6a)$$

or

$$M_u = 0.06 R_y F_y W_p \quad (YDKT) \quad (9.6b)$$

(3) The required rigidity of these elements is specified in Section 16.3 of the “Regulation on Design, Calculation and Construction Principles of Steel Structures”. C_d For = 1.0, **Equation (9.4)** The bending strength of the beam is given by

will be determined by taking into account the

Table 9.3 – Cross-Section Conditions

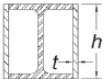
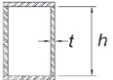
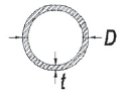
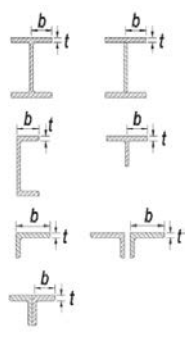
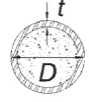
	Element Definition	Slenderness The rate	Limit values		Explanation
			Ductility Level Higher Elements, λ_{hd}	Ductility Level Bounded Elements, λ_{md}	
Rigidized Elements	Rectangle box sections Don't do that rectangle box sections And I from the cross section box shaped made constitute sections Titles I from the cross section box shaped made constitute sections side plates And cross member aspect to be used fake box sections	b/t b/t h/t	$0.55 \sqrt{\frac{TO}{F_y}}$	$0.64 \sqrt{\frac{TO^{[a]}}{F_y}}$	  
	Cross as an element to be used I or don't I sections bodies	h/t_w	$1.49 \sqrt{\frac{TO}{F_y}}$	$1.49 \sqrt{\frac{TO}{F_y}}$	 
	Beam or colon aspect to be used I or don't I sections bodies [b] Beam or colon aspect to be used I from your profile box shaped said constitute cross sections side plates	h/t_w h/t	$C_a \leq \text{If } 0.125$ $2.45 \sqrt{\frac{TO}{F_y}} (1 - 0.93 C_a)$ $C_a \text{ If } > 0.125$ $0.77 \sqrt{\frac{TO}{F_y}} (2.93 - C_a) \geq 1.49 \sqrt{\frac{TO}{F_y}}$	$C_a \leq \text{If } 0.125$ $3.76 \sqrt{\frac{TO}{F_y}} (1 - 2.75 C_a)$ $C_a \text{ If } > 0.125$ $1.12 \sqrt{\frac{TO}{F_y}} (2.93 - C) \geq 1.49 \sqrt{\frac{TO}{F_y}}$	  
	Beam or colon aspect to be used fake box cross sections bodies	h/t	$C_a = \frac{\Omega P_{Ca}}{F_y}, \Omega_c = 1.67 \text{ (GKT)}$	$C_a = \frac{P_u}{\phi_c (F_y)}, \phi_c = 0.90 \text{ (YDKT)}$	
	Pipe cross-section elements	D/t	$0.038 \frac{TO}{F_y}$	$0.044 \frac{TO^{[k]}}{F_y}$	

Table 9.3 (continued)

	Element Definition	Slenderness The rate	Limit values		Explanation
			Ductility Level Higher Elements, λ_{hd}	Ductility Level Bounded Elements, λ_{md}	
Non-Rigidified Elements	I or don't I sections titles, U or T sections, Corner (L) or split pair Corners arms, continuous combined pair Corners arms	b/t	$0.30 \sqrt{\frac{TO}{F_y}}$	$0.38 \sqrt{\frac{TO}{F_y}}$	
Composite Elements	Box cross-section composite elements walls	b/t	$1.4 \sqrt{\frac{TO}{F_y}}$	$2.26 \sqrt{\frac{TO}{F_y}}$	
	Pipe cross-section composite elements wall	D/t	$0.076 \frac{TO}{F_y}$	$0.15 \frac{TO}{F_y}$	

(a) In rectangular box cross-sections used as beams or columns, artificial box cross-sections and the limit value of the cross-section condition in the headings of the sections formed in the form of a box from the cross-section is $1.12 TO/F_y$ will be taken as.

(b) $C_s \leq 0.125$, high ductility level moment resisting frames In the sections, the cross-section condition (h/t_w) limit value $2.45 TO/F_y \sqrt{C_s}$ cannot exceed. $C_s \leq 0.125$ is limited to moment resisting frames In the sections, the cross-section condition (h/t_w) limit value $3.76 TO/F_y \sqrt{C_s}$ cannot exceed.

(c) The limit value of the cross-section condition in pipe profiles used as beams or columns is $0.07 TO/F_y$ can be taken as.

9.2.9. Capacity Preserved Zones for Horizontal Load Bearing Systems with High Ductility Level

For horizontal load-bearing systems with high ductility level *regions with preserved capacity*, It covers the possible plastic hinge regions in moment resisting frames and the end and middle regions of the tie beams of eccentrically braced steel frames and the cross members of centrally braced steel frames (**Figure 9.1** Sudden cross-section changes, application of additional details, gaps (including gaps for temporary assembly purposes), element connections and the use of headed sliding elements that adversely affect the non-linear behavior of the relevant elements are not allowed within these zones.

9.2.10. Column and Beam Splices

9.2.10.1 –For all columns, including those outside the scope of the load-bearing systems under the influence of earthquake loads. *column suffix* shall be made at least 1.2 m away from the connection planes of the beam flanges to the column. However, if the column net height is less than 2.4 m, the column splice shall be formed in the middle of the net height. However, the splices where the column web and flanges are connected with full penetration butt welds may be allowed to be formed in areas closer to the connection planes of the beam flanges to the column, provided that they are not less than the column cross-section height.

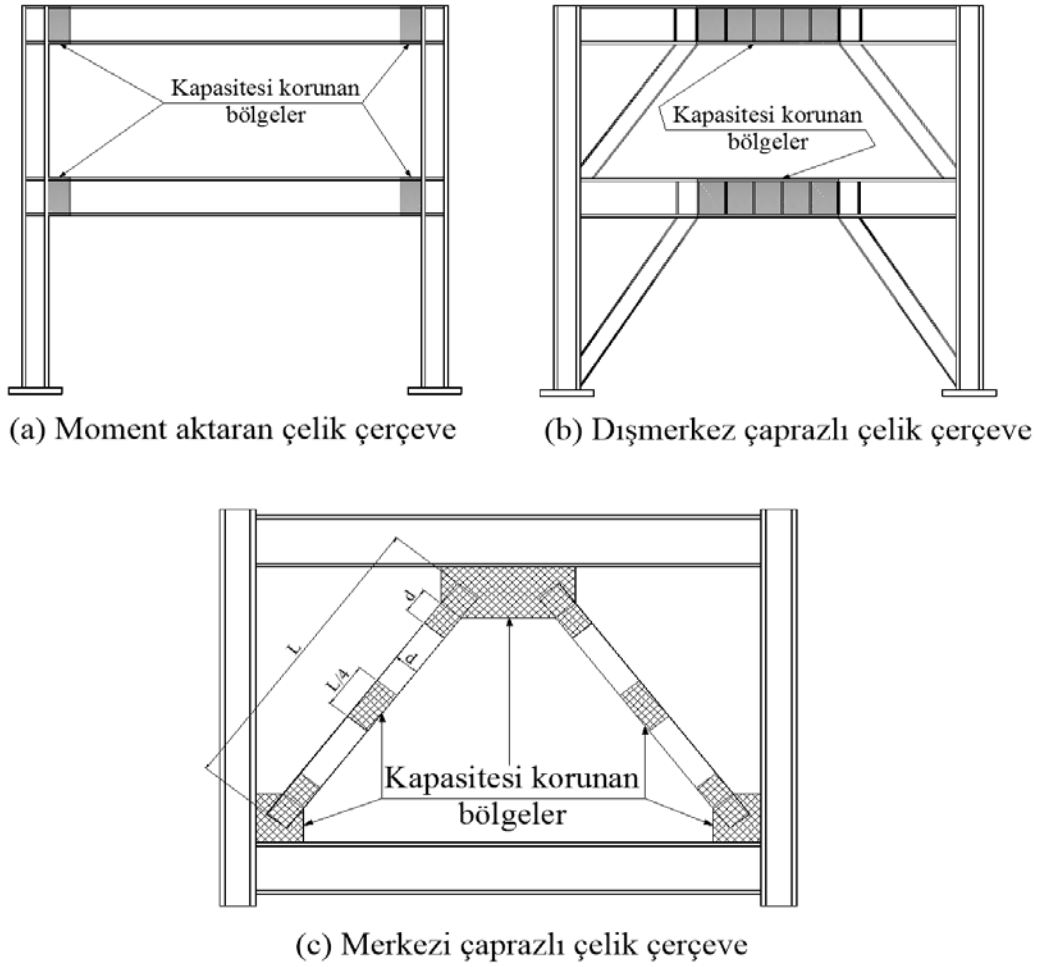


Figure 9.1

9.2.10.2 –In dimensioning column joints, the required strength of the joint detail will be determined based on the internal force status taken as a basis for dimensioning the column and the rules specified in the relevant articles.

9.2.10.3 –Column joints can be made with bolts, corners and butt welding. In case the joint is butt welded, full penetration butt welding will be used (See Fig. **Annex 9A**).

9.2.10.4 –Beam attachments within the scope of load-bearing systems under the effect of earthquake load, **9.2.9**It will be established outside the areas with protected capacity defined in .

9.2.10.5 –The required strength of the joint detail in dimensioning the beam joints, **9.2.6**It will be determined based on the internal force status defined in .

9.2.10.6 –In the joint details, the plates can only be connected to one element by bolting and to the other by welding.

9.3. STEEL FRAMES WITH HIGH DUCTILITY LEVEL THAT TRANSFER MOMENTS

High ductility moment resisting steel frames are carrier systems in which the beams have the ability to undergo significant non-linear deformation under earthquake effects, and the shear zone of the beam-column connections has the ability to undergo limited non-linear deformation. Ductility

The rules to be followed in dimensioning high moment resisting steel frames are given below.

9.3.1. General Conditions

9.3.1.1 –In beams and columns of steel frames that transfer moments with high ductility level, *title width/thickness* And *body height/thickness* rates **Table 9.3** Given λ_{hd} will not exceed its value.

9.3.1.2 –Beam flanges of high ductility level moment resisting steel frames are reinforced against lateral drift and torsion, for high ductility level elements. **9.2.8** It will be supported based on the relevant conditions given in .

9.3.1.3 –In dimensioning steel frame columns that transfer moments with high ductility level, the most unfavourable internal forces to be determined by taking into account the conditions given below will be taken as basis.

(a) 9.2.5 Shear force, axial force and bending moments to be obtained by taking into account the load combinations including the earthquake effect given in .

(b) Without considering the bending moments, **9.2.6** Axial tensile and compressive forces to be obtained in accordance with. However, if there are horizontal forces acting between the column ends, the bending moments that will be created by these will also be taken into account.

9.3.1.4 –Beams of steel frames that transfer moments with high ductility level, **9.2.5** shall be dimensioned under the most unfavourable internal forces to be obtained on the basis of

9.3.2. Condition that Columns are Stronger than Beams

9.3.2.1 –At each beam-column node point of the frames of high ductility level moment transferring steel frames or reinforced concrete shear wall steel frame systems, in the earthquake direction taken into consideration, **Equation (9.7)** It will be shown that the condition that columns are stronger than beams is met in accordance with **(Figure 9.2)**.

$$\frac{\sum M_{pc}^*}{\sum M_{pb}^*} > 1.0 \quad (9.7)$$

Equation (9.7) in the M_{pc} column bending moment capacities **Equation (9.8a)** or **Equation (9.8b)** with will be calculated. M_{pb} beam bending moment capacities, **Equation (9.9a)** or **Equation (9.9b)** And In case of using weakened beam cross-section connection **Equation (9.9c)** or **Equation (9.9d)** will be calculated with .

In the calculation of column bending moment capacities, minimizing these bending moment capacities in accordance with the direction of the earthquake, **9.3.1.3** obtained with P_{hungry} (GKT) or P_{fly} (YDKT) the required axial compressive force shall be taken into account.

$$\sum M_{pc} = \sum W_{pc} (F_{yc} - 1.5 P_{hungry} / A_g) \quad (GKT) \quad (9.8a)$$

or

$$\sum M_{pc} = \sum W_{pc} (F_{yc} - P_{fly} / A_g) \quad (YDKT) \quad (9.8b)$$

$$\sum M_{pb}^* = \sum (1.1 R_y F_y b W_{pb} + 1.5 M_{\text{hunting}}) \quad (\text{GKT}) \quad (9.9a)$$

or

$$\sum M_{pb}^* = \sum (1.1 R_y F_y b W_{pb} + M_{uv}) \quad (\text{YDKT}) \quad (9.9b)$$

$$\sum M_{pb}^* = \sum (1.1 R_y F_y b W_{RBS} + 1.5 M_{\text{hunting}}) \quad (\text{GKT}) \quad (9.9c)$$

or

$$\sum M_{pb}^* = \sum (1.1 R_y F_y b W_{RBS} + M_{uv}) \quad (\text{YDKT}) \quad (9.9d)$$

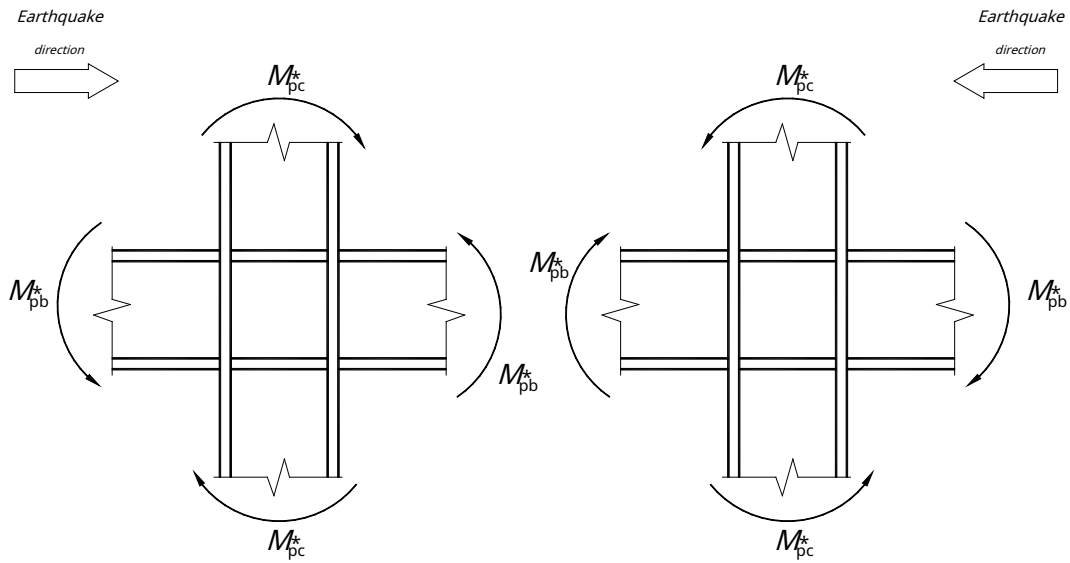


Figure 9.2

9.3.3. Case where the condition that the columns are stronger than the beams is not met

9.3.3.1 – 9.2.5 Taking into account the load combinations including the earthquake effect given in **Equation (9.10)** In columns where, the condition that the columns are stronger than the beams will not be required for the following situations.

$$P_a < 0.3 (F_y A_g / 5) \quad (\text{GKT}) \quad (9.10a)$$

or

$$P_u < 0.3 (F_y A_g) \quad (\text{YDKT}) \quad (9.10b)$$

(a) Single-storey buildings and top floor columns of multi-storey buildings.

(b) Not satisfying the condition that columns are stronger than beams in the considered direction
columns on a floor The situation where the total of the shear force resistances is less than 20% of the total of the shear force resistances of the moment resisting frame columns in the same direction at that floor and the total of the shear force resistances of the columns on each moment resisting frame axis, *on the column axis* The situation where the total shear force resistance of the moment resisting frame columns is less than 33%.

For this condition *column axis*, is defined as the axis where parallel row columns are located within 10% of the plan dimension in the perpendicular direction to a column axis. The design shear strength of the column is the sum of the bending moment strengths of the beams connected to both ends of the columns that meet the condition of the columns being stronger than the beams or the sum of the bending moment strengths of the column ends in columns where this condition is not met. H will be obtained by dividing by the storey height.

9.3.3.2 –If the ratio of the existing shear strength of the columns on any i 'th floor of the building to the required shear strength in the earthquake direction considered is 50% greater than the ratio of the existing shear strength of the columns above the relevant floor to the required shear strength, the condition that the columns are stronger than the beams will not be required.

9.3.4. Beam - Column Joint Areas

9.3.4.1 –The following three conditions shall be met together in beam-column connections of steel frames that resist moments of high ductility level.

(a) The combination is at least 0.04 radians *relative story drift angle* (relative floor drift/story height). For this purpose, details whose validity has been proven by experimental and/or analytical methods will be used. Various examples of bolted and welded connection details whose validity has been proven and their application limits **Annex 9B** is given in .

(b) Joint on the column face M_{fly} required bending moment resistance, **Annex 9B** In the details defined in , the possible plastic bending moment of the beam connected to the node point is, M_p with beam

Taking into account the location of the possible plastic joint at the end, **Equation (9.11)** It will not be less than the bending moment strength calculated by .

$$M_{fly} = M_p + V_{fly-h} \quad (9.11)$$

Required bending moment resistance In determining , the contribution of the additional bending moment from the vertical loads for the section of the beam between the plastic hinge and the column face may also be taken into account.

(c) To be used in sizing the joint V_{fly} shear force, yield (mechanism)

at the plastic hinge points at the beam end with the shear force determined based on the situation $(1.2G + 0.5Q + 0.2S)$ by adding the shear force to be calculated under the load combination

Equation (9.12) will be obtained with G required shear strength In determining , the contribution of additional shear forces from vertical loads can also be taken into account for the section of the beam between the plastic hinge and the column face.

$$V_{fly} = V_{d \pm} (M_{pri} + M_{prj}) - n \quad (9.12)$$

9.3.4.2 –In the beam-column connection detail, the shear zone limited by the column and beam flanges (**Figure 9.3**) V_{up} required shear strength, possible beams connecting to the column

Based on the column shear force resulting from the plastic moments effect **Equation (9.13)** will be calculated with .

$$V_{up} = \sum M_{fly} - \frac{1}{d_b - t_{bf}} - \frac{1}{H_{average}} \quad (9.13)$$

(a) The shear zone will be dimensioned to have sufficient shear strength. For this, the design of the shear zone will be made by taking into account the yield limit state under shear effect.

shear strength $\phi_v = 1.0$ (YDKT) will be determined. Failure to meet this condition

In this case, the shear zone will be strengthened by adding web reinforcement plates or, for example, plates in diagonal direction.

(b) Minimum thickness of the column web plate and each of the reinforcement plates, if used, $t_{\min} \geq u/V$ will provide the 180 condition (Figure 9.4). This condition is not met

In such cases, the reinforcement plates and the column web plate shall be welded to each other in such a way that they work together and the total of the plate thicknesses shall be $\sum t \geq u/V$. It will be checked that it meets the 180 condition.

(c) If reinforcement plates are used in the shear zone, full penetration is required to connect these plates to the column header plates. directional kU ka weld or fillet weld will be used (Figure 9.4). These welds shall be dimensioned taking into account the shear force resisted by the reinforcement plate.

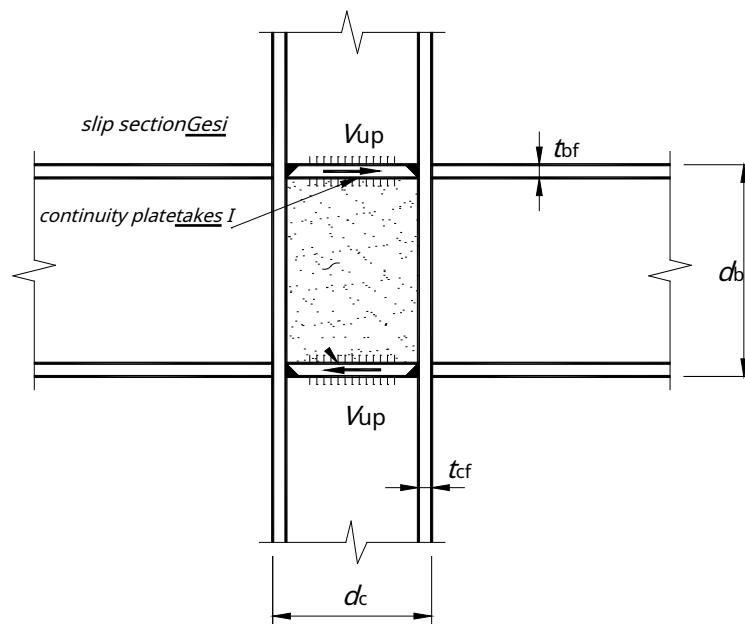


Figure 9.3

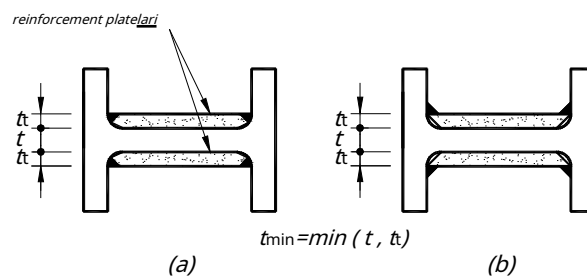


Figure 9.4

9.3.4.3 -In moment transferring beam-column connection details, on both sides of the column body, at the beam flange level continuity sheets By placing the tensile and compressive forces on the beam flanges, the safe transfer of these forces to the column (and to the adjacent beam in two-sided beam-column joints) will be ensured. (Figure 9.3).

(a) The thickness of the continuity plates shall not be less than the flange thickness of the combined beam in single-sided beam connections, and shall not be less than the larger of the flange thicknesses of the combined beams in the case of beam connections to the column from two sides.

(b) Full penetration butt welding shall be used for the connection of continuity plates to column webs and flanges. Fillet welding may also be used for the connection of continuity plates to column webs. However, this welding shall be of a length and thickness that will transfer a force equal to the shear capacity of the continuity plate in its own plane to the column web.

(c) Column heading thickness

$$t_{cf} \geq 0.4 \sqrt{.8 B T_{bf} \frac{R_y b F_y b}{R_{yc} F_{yc}}} \quad (9.14a)$$

And

$$t_{cf} \geq \frac{b_{bf}}{6} \quad (9.14b)$$

If both conditions are met, a continuity plate may not be required.

9.3.4.4 –In dimensioning the beam-column connection detail, **Annex 9B** The calculation principles given in will be used.

9.3.4.5 –Regions with preserved capacity also, **9.2.9** The conditions given in shall be observed. The use of capped sliding elements is not permitted within these zones.

9.3.5. Column Attachments

Column appendices **9.2.10** It will be sized to meet the relevant conditions given in . In addition, if a bolted joint detail is applied, the required bending strength of the column joints will be determined by taking into account the smallest cross-section of the added elements. $R_y F_y W_p / 1.5$ (GKT) or $R_y F_y W_p$ (YDKT) will be calculated as the required shear strength of the joints, $\sum M_{pc} / 1.5 H_c$ (GKT) or $\sum M_{pc} / H_c$ (YDKT), **9.3.1.3(a)** And **9.3.1.3(b)** will be taken as the largest of the shear force values obtained from . Here, $\sum M_{pc}$ is the sum of the column bending moment resistances at the lower and upper ends of the added column.

9.3.6. Stability of Beam-Column Connections

In order to prevent rotations that may occur outside the frame plane, columns will be supported in beam-column joint areas, taking into account the following conditions.

(a) In cases where beam-column joint areas are supported laterally by the floor system or roof stability connections, if the beam and column bodies are in the same plane and it is shown that the column remains elastic outside the shear zone, the column flanges will be supported laterally only at the level of the beam top flanges. **Equation (9.7)** In cases where the ratio given in is greater than 2.0, the columns can be assumed to remain elastic. In other cases, the column flanges shall be supported laterally from the alignment of the beam flanges. Each lateral support element shall be $F_y A_{fb} / 1.5$ (GKT) or $F_y A_{fb}$ (YDKT) to 2% of the calculated value

shall be dimensioned under the influence of a uniform axial force.

(b) In cases where the beam-column joint areas are not laterally supported in a direction perpendicular to the frame plane where they are located, the columns will be dimensioned using the length between the points where they are laterally supported and in accordance with the following conditions, in accordance with the principles of Section 11 of the "Regulation on Design, Calculation and Construction Principles of Steel Structures".

(1) Required strength of the column, $9.2.6$ It will be obtained by using the load combinations defined in .

(2) The slenderness ratio of the column, L/λ It will meet the 60 condition.

(3) The required bending moment resistance of the column in the direction perpendicular to the frame plane is for the lateral support element. (a) It will be calculated as the second order bending moment based on the lateral displacement that will occur under the effect of the required axial force resistance defined in .

9.4. STEEL FRAMES WITH LIMITED DUCTILITY LEVEL, THAT TRANSFER MOMENTS

Limited ductility moment resisting steel frames are load-bearing systems in which the shear zone of beams, columns and beam-column connections has the ability to undergo limited non-linear deformation under earthquake effects. The rules to be followed in dimensioning limited ductility moment resisting steel frames are given below.

9.4.1. General Conditions

9.4.1.1 – In beams and columns of steel frames with limited ductility level that transfer moments, *title width/thickness* And *body height/thickness* rates **Table 9.3** Given λ_{md} will not exceed the values.

9.4.1.2 – Beam flanges of steel frames with limited ductility level resisting moment, against lateral drift and torsion, for elements with limited ductility level **9.2.8** It will be supported based on the relevant conditions given in .

9.4.1.3 – Dimensioning of steel frame columns with limited ductility level resisting moments **9.3.1.3** The rules given in apply.

9.4.1.4 – Dimensioning of steel frame beams with limited ductility level resisting moment **9.3.1.4** The rules given in apply.

9.4.1.5 – For moment resisting steel frames with limited ductility level For moment resisting steel frames with high ductility level **9.3.2** Given in, *columns are stronger than beams* Compliance with the condition is not mandatory.

9.4.2. Beam - Column Joint Areas

9.4.2.1 – In moment resisting beam-column connections of moment resisting steel frames of limited ductility level, the following three conditions shall be met simultaneously:

(a) The combination is at least 0.02 radians *relative story drift angle* (relative floor drift/story height). For this purpose, details whose validity has been proven by experimental and/or analytical methods will be used. Various examples of bolted and welded connection details whose validity has been proven and their application limits **Annex 9B** is given in .

(b) The combination M_{fy} *required bending moment resistance* in your account, **9.3.4.1(b)** The rules given in apply.

(c) The basis for sizing the joint V_{fy} in determining the cutting force

9.3.4.1(c) The rules given in apply.

9.4.2.2 –In the beam-column connection detail, the shear zone limited by the column and beam flanges(**Figure 9.3**)The required shear strength will be determined by the beam end moments that will give the most unfavourable result under (YDKT) or (GKT) load combinations.

9.4.2.3 –In moment transferring beam-column connection details,**9.3.4.3**The rules given in apply.

9.4.2.4 –In dimensioning the joint detail,**Annex 9B**The calculation principles given in will be used.

9.4.2.5 –In regions with preserved capacity**9.3.4.5**The conditions given in will be complied with.

9.4.3. Column Attachments

For column splices of high ductility moment resisting steel frames**9.3.5**The conditions given in are equally valid for column splices of steel frames resisting moments of limited ductility level.

9.5. CENTRAL AND DISCENTRAL BRACED STEEL FRAMES

Braced steel frames are horizontal load-bearing systems consisting of articulated jointed or moment-transmitting frames and centrally and eccentrically connected crosses. The horizontal load-bearing capacities of such systems are provided by the axial force resistance of the elements, in addition to their bending strength. Braced steel frames are divided into two depending on the arrangement of the crosses.

(a)Centrally braced steel frames(**Figure 9.5**).

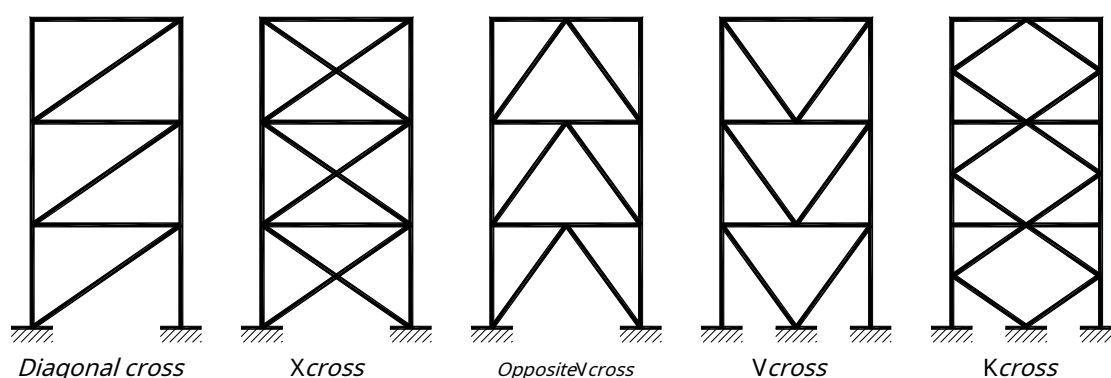


Figure 9.5

(b)Eccentrically braced steel frames(**Figure 9.6**).

Rules for the design of systems that are formed within normal floor and roof planes and that transfer horizontal forces to horizontal load-bearing systems**9.10**will be evaluated within the scope of .

The diagonals are centrally connected to the frame nodes.*centrally braced steel frames*They can be dimensioned as systems with high or limited ductility levels. On the other hand, the diagonals are eccentrically placed at the frame nodes.

connected to *eccentrically braced steel frames* They will be dimensioned as a system with high ductility level.

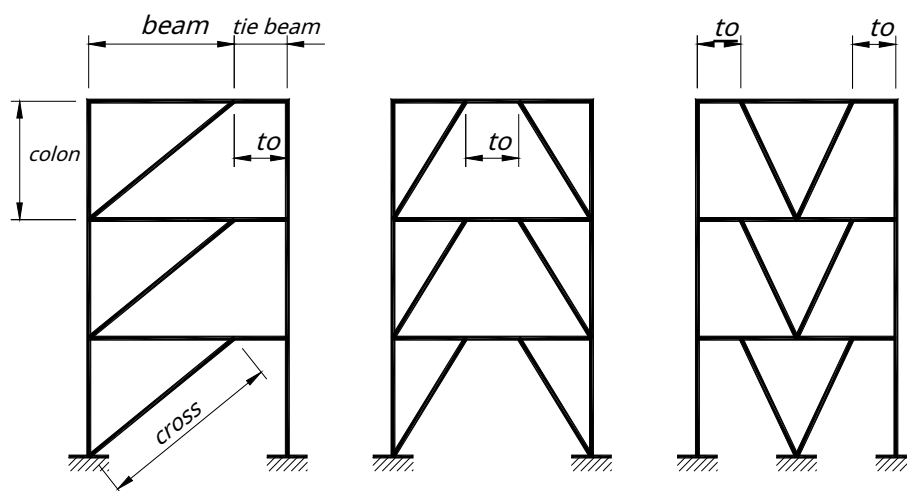


Figure 9.6

9.6. STEEL FRAMES WITH CENTRAL BRACES WITH HIGH DUCTILITY LEVEL

Centrally braced steel frames with high ductility level are dimensioned in such a way that even if some of the diagonal members buckle under the effect of compression, no significant loss of strength will occur in the system. The rules to be applied in the dimensioning of these systems are given below.

9.6.1. General Conditions

9.6.1.1 –In beams, columns and braces of central braced steel frames with high ductility level, *head width/thickness, body height/thickness* And *diameter/thickness* rates

Table 9.3 Given λ_{hd} will not exceed the limit values.

9.6.1.2 –The central diagonal system elements on one axis of the building will be arranged in such a way that at least 30% and at most 70% of the horizontal forces acting in both earthquake directions in that axis will be resisted by the diagonals working in pressure.

9.6.2. System Analysis

9.6.2.1 –Required strengths in dimensioning columns, beams and connections of central braced steel frame systems with high ductility level, **9.2.6** will be calculated based on .

9.6.2.2 –Columns, beams and connections **9.2.6** The internal forces of the cross members are given below for the calculation of the internal forces compatible with the yield (mechanism) condition defined in **(a)** And **(b)** It is defined in .

(a) In the case of a typical mechanism that opposes the buckling moment of diagonal members under the influence of axial compression force (**Figure 9.7a**), which causes plasticity of the cross members and **Equation (9.15)** And **Equation (9.16)** Possible axial tensile and compressive strengths determined by .

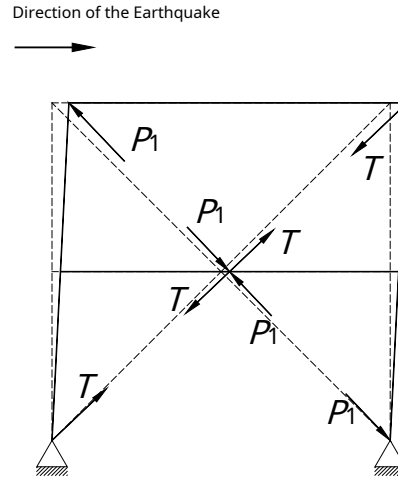


Figure 9.7a

(b) Typical mechanism case against post-buckling of diagonal members under axial compressive force (Figure 9.7b), which causes plasticity of the cross members and Equation (9.15) and Equation (9.17) Possible axial tensile force and possible axial compressive force strengths after buckling determined by .

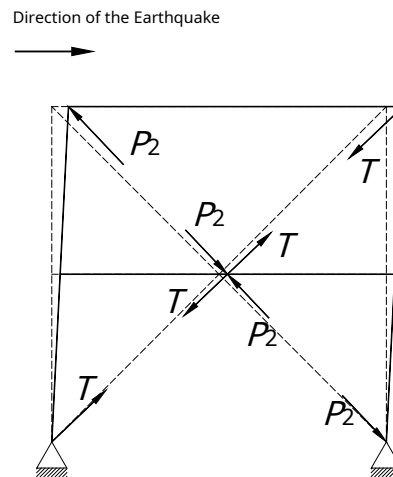


Figure 9.7b

$$T = R_y F_y A_g \quad (9.15)$$

$$P_1 = 1.14 F_{cre} A_g \quad (9.16)$$

$$P_2 = 0.30 (1.14 F_{cre} A_g) \quad (9.17)$$

Equation (9.15), Equation (9.16) and Equation (9.17) The possible axial tensile and compressive strengths determined by (YDKT) will be used as is, and by dividing by 1.5 for (GKT). F_{cre} , possible critical The length of the diagonal element itself can be taken as basis in the calculation of buckling stress.

9.6.2.3 –Axial forces of columns, beams and connections, 9.6.2.2(a) and (b) It will be calculated with the help of equilibrium equations for each of the mechanism cases on which is based.

9.6.2.4 –In determining the required strength of columns, beams and connections,**9.2.5** Instead of earthquake effects in the load combinations given in **9.6.2.3** The most unfavourable internal forces defined in will be used. These earthquake effects,**9.2.6** According to the amplified earthquake effects will not be greater than

9.6.3. Crosses

The diagonal members of centrally braced steel frames with high ductility level shall be dimensioned to meet the following conditions.

9.6.3.1 –Slenderness ratio in diagonal members (bar buckling length / radius of inertia) KL/I
 $\leq$ It will meet the 200 condition.

9.6.3.2 –Braces with multi-section cross-sections can only be used if their parts are continuously joined.

9.6.3.3 –Crosses,**9.2.5**'given in *Load combinations involving earthquake effects* will be sized taking into account.

9.6.3.4 –The breaking strength calculated using the effective net area of the cross, $R_t F_u A_{tn}$
 Yield strength calculated using lossless cross-sectional area, $R_y F_y A_g$ less than its value
 If this condition is not met, the net area cross-section will be reinforced by taking into account the following conditions.

(a) The characteristic yield stress of the reinforcement member shall be at least equal to the characteristic yield stress of the diagonal member.

(b) Connection detail of the reinforcement element to the cross element, $R_y F_y A_{fb} / 1.5$ (GKT) or $R_y F_y A_{fb}$
 It will be sized under the effect of axial force calculated with (YDKT).

9.6.4 Additional Conditions for Special Cross Patterns

9.6.4.1 – Vor reverse V In the form of cross systems, additional conditions that the beams to which the crosses are connected must meet are given below.

(a) Beams will be continuous between the columns they connect.

(b) Top and bottom headings of beams **9.2.8** It will be supported in the lateral direction based on the relevant conditions given in. In addition, the upper and lower flanges of the beams will be supported in the lateral direction at the points where the crosses are connected.

(c) The required strength of beams and their connections are determined according to the load combinations including the earthquake effects. **9.6.2.2** It will be calculated by taking into account the unbalanced forces and vertical loads caused by the internal forces specified in .

9.6.4.2 –In frames with central steel braces with high ductility level K Diagonal arrangement (where diagonals are connected to the column midpoint) is not allowed.

9.6.4.3 –In steel frames with central braces of high ductility level, the use of tension-only braces is not permitted.

9.6.5. Cross – Beam – Column Connections

The following conditions shall be complied with in the connection of the cross member or the joint plate to the beam-column joint area. In addition, the following internal forces shall be applied:**9.10.3** It will be evaluated together with the internal forces of the struts (transmission elements) calculated as explained in .

(a) When the beam-column connection is designed as hinged, the connection detail will be created to provide a rotation angle of at least 0.025 radians. For this connection detail **Annex 9C** One of the beam-column connection details given in and proven to meet this condition can be used.

(b) When the beam-to-column connection is designed as rigid, the connection detail will be sized to transfer the smaller of the bending moment values given below.

(1) The beam M_p plasticity moment $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ Bending moment resulting from (YDKT) storey.

(2) Columns that join to the node $\sum M_p$ total plasticity moment $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ Bending moment resulting from (YDKT) storey.

9.6.6. Required Strength of Cross End Joints

Cross end joints, excess strength coefficient, D It will be sized based on the internal earthquake forces multiplied by . These internal forces will not exceed the axial tensile and compressive forces and bending strengths given below.

(a) Tensile strength

In case the cross is in tension, the tensile strength $R_y F_y A_g$ (YDKT) or

$0.7 R_y F_y A_g /$ It will be calculated with 1.5 (GKT).

(b) Compressive strength

In case the cross is under pressure, the compressive strength is, $R_y F_y A_g$ and $1.14 F_{cre} A_g$

The smallest of the possible axial compressive strength calculated with 1.1 for (YDKT) or 0.7 for (GKT) ($1.1 / 1.5$) will be obtained by multiplying by .

(c) Compatibility with cross buckling of joint plate

The diagonal joint shall have the strength to resist the bending effects that will occur due to the buckling of the diagonal. This shall be achieved by complying with one of the following conditions.

(1) Required bending moment resistance

When the end plastic hinges are intended to be formed in the cross member itself, the bending strength of the cross connection must be greater than the possible bending strength of the cross member. The possible bending moment strength of the cross is the minimum buckling strength corresponding to the According to the axis $(1.1 / 1.5) R_y M_p$ (GKT) or $1.1 R_y M_p$ It will be calculated with (YDKT).

(2) Rotation capacity

When the end plastic joints are intended to be formed at the joint point or the connection plate, the cross connection shall have sufficient rotation capacity. Sufficient rotation capacity shall be provided by creating sufficient areas to allow plastic rotation in the joint point plate or connection plate to be used at the end connection of the cross member. Details that can be applied to ensure this condition is met **Annex 9C** is given in .

9.6.7. Column Attachments

Column appendices 9.2.10 It will be sized to meet the conditions given in . The bending moment resistance of the column splices shall not be less than 50% of the bending capacity of the smaller section of the added elements. The required shear force resistance shall be, $\sum M_{pc} / 1.5 H_c$ (GKT) or $\sum M_{pc} / H_c$ (YDKT) will be calculated as follows. Here $\sum M_{pc}$, crop top and shows the sum of the characteristic plastic bending moment strengths of the columns below.

9.7. CENTRAL BRACED STEEL FRAMES WITH LIMITED DUCTILITY LEVEL

The rules to be applied in the dimensioning of centrally braced steel frames of limited ductility level are given below.

9.7.1. General Conditions

9.7.1.1 –In beams, columns and braces of centrally braced steel frames with limited ductility level, *title width/thickness, body height/thickness And diameter/thickness rates*
Table 9.3 Given λ_{md} Will not exceed the limit values.

9.7.2. System Analysis

Required strengths in dimensioning columns, beams and connections of centrally braced steel frame systems with limited ductility level, 9.2.6 The excess strength coefficient defined in , D It will be determined by taking into account the load combinations that include the earthquake effect magnified by .

9.7.3. Crosses

The diagonal members of centrally braced steel frames of limited ductility level shall be dimensioned to meet the following conditions.

9.7.3.1 –Slenderness ratio of diagonal members (bar buckling length/radius of inertia) $KL/I \leq 200$ will not exceed the limit value. However, V or reverse V In case of using cross arrangements in the form of 4.0 for the slenderness ratio EF_y condition will be complied with.

9.7.3.2 –In centrally braced steel frames with limited ductility, the use of multi-piece elements may be permitted provided that the cross-sectional parts of the brace elements behave as one piece. This condition should be provided by equidistantly placed brace plates in multi-piece bars with a distance between elements equal to the thickness of a brace plate, and in other cases by lattice connections (braid elements). In the design of the interconnection elements of multi-piece braces, Section 8.4 of the “Regulation on the Design, Calculation and Construction Principles of Steel Structures” shall be taken as basis.

9.7.3.3 –Crosses, 9.2.5' given in *Load combinations that include earthquake effects* will be sized taking into account.

9.7.4 Additional Conditions for Special Cross Patterns

9.7.4.1 – Vor reverseVIn the shaped brace systems, the additional conditions that the beams to which the braces are connected outside the beam-column joint area must meet are given below.

(a) Beams will be continuous between columns.

(b) Top and bottom headings of beams **9.2.8** It will be supported in the lateral direction based on the relevant conditions given in. In addition, the upper and lower flanges of the beams will be supported in the lateral direction at the points where the crosses are connected.

(c) Beams shall be dimensioned under vertical loads and diagonal axial forces defined below, if diagonals are ignored.

In dimensioning the crosses under tension effect, the smallest of the following axial forces will be taken.

(1) $R_y F_y A_g / 1.5$ (GKT) or $R_y F_y A_g$ Possible axial tensile force calculated as (YDKT).

(2) Excess strength coefficient, D tensile force magnified by .

30% of the characteristic compressive strength in diagonals under compression ($0.3 F_{cr} A_g$) will be taken as the axial compressive force.

(d) In addition, it will be shown that the beams to which the braces are connected will safely carry the vertical loads acting on them even if the braces are ignored.

9.7.4.2 – In centrally braced steel frames with limited ductility level **K** Diagonal arrangement (where diagonals are connected to the column midpoint) is not allowed.

9.7.4.3 – In steel frames with central braces of limited ductility, the use of tension-only braces is permissible. However, in braces sized to carry only tension forces, the slenderness ratio (bar buckling length/radius of inertia) shall not exceed 300. This limit shall not apply to steel cables and shafts.

9.7.5. Cross Joints

Required strengths in dimensioning cross connections, **9.7.2** However, the axial tensile and compressive forces to be taken as basis in dimensioning the cross connection need not be greater than the axial tensile and compressive strengths defined below.

(a) In case the cross is in tension, $R_y F_y A_g / 1.5$ (GKT) or $R_y F_y A_g$ Possible tensile strength calculated with (YDKT).

(b) In case the cross is under pressure effect, $R_y F_y A_g$ and $1.14 F_{cr} A_g$ determined by the possible The smaller of the cross axial compressive strengths is 1.1 for (YDKT) or 0.7 for (GKT) ($1.1 / 1.5$) Possible cross strength calculated by multiplying by .

9.7.6. Column Attachments

For column splices of centrally braced steel frames with high ductility level **9.6.7** The conditions given in are also valid for column splices of centrally braced steel frames of limited ductility level.

9.8. STEEL FRAMES WITH HIGH DUCTILITY LEVEL ECCENTRIC BRACED

High ductility eccentrically braced steel frames are horizontal load-bearing systems in which the tie beams have the ability to undergo significant nonlinear deformation under earthquake effects. These systems are dimensioned in such a way that during the plastic deformation of the tie beams, the columns, braces and other beams other than the tie beam remain in the elastic region. The rules to be applied in the dimensioning of high ductility eccentrically braced steel frames are given below.

9.8.1. General Conditions

9.8.1.1 –In beams, columns and braces of eccentrically braced steel frames with high ductility level, *title width/thickness, body height/thickness And diameter/thickness rates* **Table 9.3** Given λ_{hd} Will not exceed the limit value.

9.8.2. Tie Beams

9.8.2.1 –In eccentrically braced steel frames with high ductility level, there shall be a tie beam at at least one end of each diagonal element.

9.8.2.2 –Tie beams, **9.2.5** Given in *Load combinations that include earthquake effects* It will be dimensioned under the internal forces (bending moment, axial force and shear force) calculated based on the dimensions.

9.8.2.3 –Design shear strength of the tie beam, V_n/Ω_v (GKT) or $\phi_v V_n$ (YDKT),

Equation (9.18) As stated in , the minimum shear strength found from the yield limit states of the body under the effect of shear force and the yield limit states of the section under the effect of bending moment For both limit cases, $\Omega_v = 1.67$ (GKT) or $\phi_v = 0.90$ (YDKT) will be taken as.

$$V_n = \min(V_p; 2M_p / t_o) \quad (9.18)$$

V_p And M_p The following expressions will be used for .

$$\frac{P_r \leq}{P_y} \text{ For } 0.15 \quad V_p = 0.6 F_y A_w \quad \text{And } M_p = F_y W_p \quad (9.19)$$

$$\frac{P_r >}{P_y} \text{ For } 0.15 \quad V_p = 0.6 F_y A_w \quad \sqrt{1 - (P_r / P_y)^2} \quad \text{And } M_p = F_y W_p - \frac{1 - P_r / P_y}{0.85} \quad (9.20)$$

At the yield limit, P_y axial force resistance, $F_y A_g / 1.5$ (GKT) or $F_y A_g$ (YDKT) will be taken as.

9.8.2.4 –The web plate of the tie beam will be a single piece, there will be no reinforcement plates in the web plane. No gaps are allowed in the web plate.

9.8.2.5 –Tie beam wide head mill I profile or make I In case of using artificial cross-sectional tie beam, header and body plate connections will be provided with full penetration butt welding.

9.8.3. Supporting the Tie Beam in Lateral Direction

9.8.3.1 –The upper and lower heads of the tie beam are at both ends of the beam, and in tie beams arranged at the edge of the column, they are at one end of the beam.**9.2.8.2(b)**It will be supported laterally based on the special conditions given in .

9.8.4. Rotation Angle of the Tie Beam

The i'th floor where the tie beam is located,**Chapter 4**defined in Δ_i depending on the relative storey drift

$$\theta_p = \frac{R\Delta_i}{I h_i} \quad (9.21)$$

Due to the relative story drift angle found with the equation, the y occurring between the connecting beam and the floor beam in the extension of this beam, The tie beam rotation angle is given below as the limit values will not exceed,(**Figure 9.8**).

(a)The length of the tie beam is $1.6 M_p / V_p 0.08$ radians if less than or equal to .

(b)The length of the tie beam is $2.6 M_p / V_p 0.02$ radians if greater than or equal to .

If the tie beam length is between these two limit values, linear interpolation will be performed.

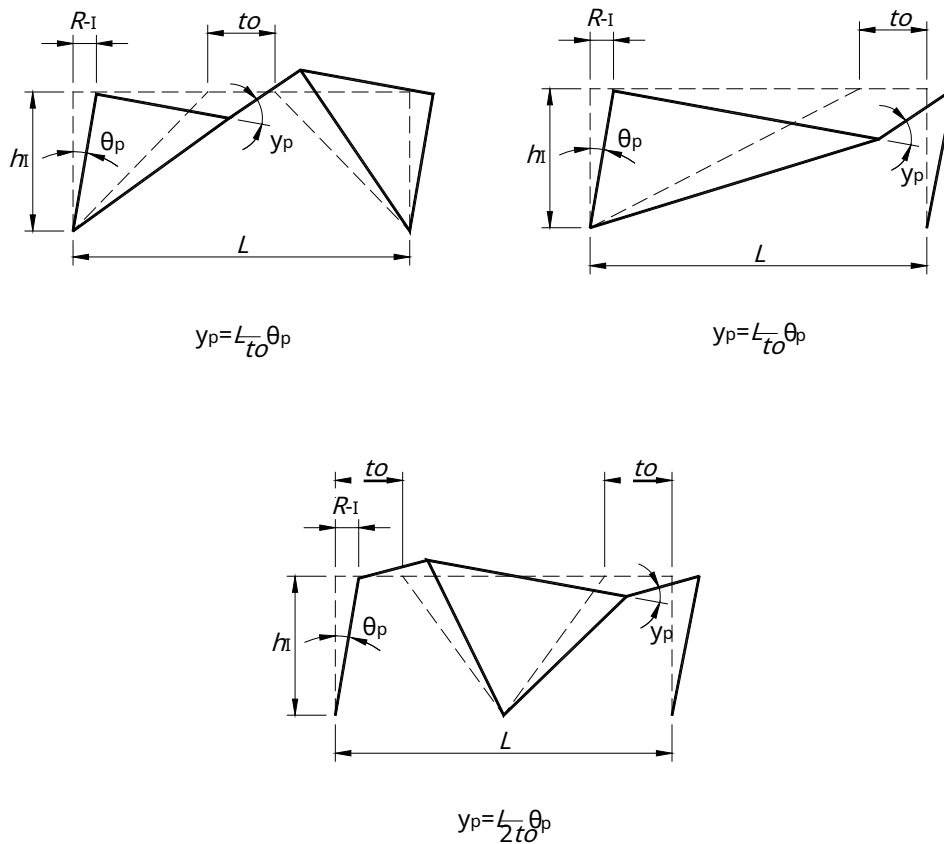


Figure 9.8

9.8.5. Rigidity (Reinforcement) Plates

9.8.5.1 –At the ends of the cross members where they transfer load directly to the connecting beam and its extensions *rigidity plates* Stiffness plates shall be arranged. Unless otherwise specified, the connecting beam web to be placed on both sides of the plate, at the height of the body plate and $(b_{bf} - t_w) / 2$ will be wide (Figure 9.9). The thickness of the stiffness plates shall not be less than 0.75 of the web plate thickness and 10 mm. The continuous fillet welds connecting the stiffness plates to the web of the connecting beam shall be capable of transferring the forces resulting from the product of the cross-sectional area of the stiffness plate and the yield stress of the material.

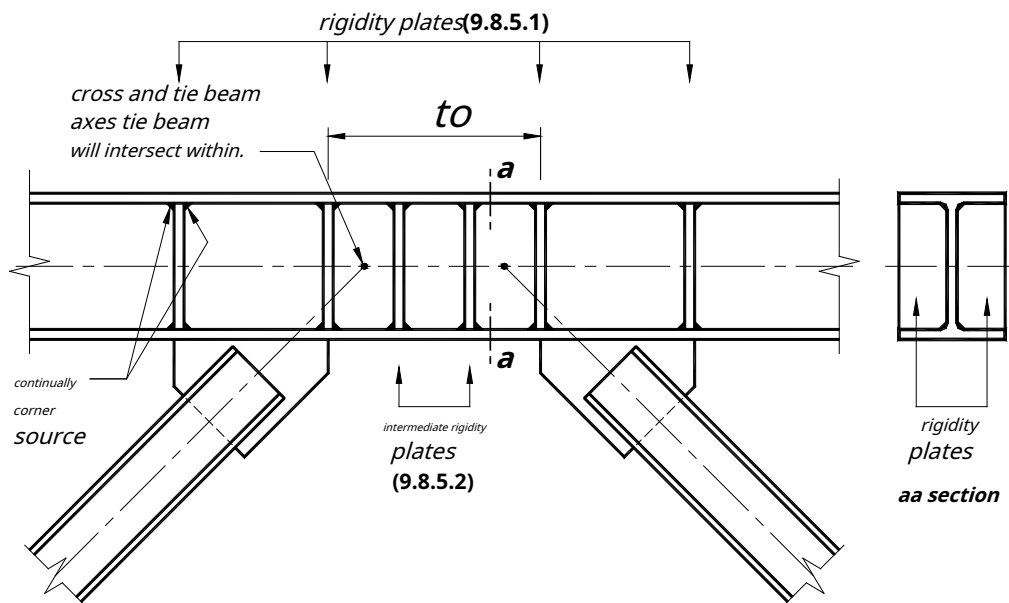


Figure 9.9

9.8.5.2 –In addition to the stiffness plates at the tie beam ends, the following are described: *intermediate rigidity plates* will be placed.

(a) Height $1.6 M_p / V_p$ or shorter spacing of intermediate rigidity plates in case the spacing of the tie beam rotation angle is 0.08 radians $(30 t_w - d_b / 5)$ tendon, ligament rotation. If the angle is less than 0.02 radians $(52 t_w - d_b / 5)$ It will not be more than . Linear interpolation will be done for intermediate values of the tie beam rotation angle.

(b) Height $2.6 M_p / V_p$ Equal to or longer than and $5 M_p / V_p$ In tie beams shorter than , 1.5 from the tie beam ends b_{bf} Stiffness plates will be placed at a distance of 1.5 m.

(c) Height $1.6 M_p / V_p$ and $2.6 M_p / V_p$ in the connecting beams between, **(a)** And **(b)** The intermediate rigidity plates specified in will be used together.

(d) Height $5 M_p / V_p$ or longer, it is not necessary to use intermediate stiffness plates in tie beams.

9.8.6. Diagonals, Floor Beams and Columns

9.8.6.1 –Loading that causes plasticity of the tie beam,**9.2.5**The earthquake effects in the load combinations given in V_{design} shear force and

$V_n = \min(V_p; 2M_p/t_o)$ including, V_n/V_o defined as *design magnification factor* It will be determined by increasing it in a way that is compatible with the .

9.8.6.2 –In determining the required strength of the crosses,**9.2.5**The earthquake effects in the load combinations given in , the loading causing the plasticity of the connecting beam is $1.25R_y$ will be enlarged with a solid. Cross-sections with multiple parts can only be used if their parts are continuously joined.

9.8.6.3 –In determining the required strength of the section of the floor beam other than the tie beam,**9.2.5**The earthquake effects in the load combinations given in , in the case where the beams work as composites with reinforced concrete slabs, the loading causing the plasticity of the connecting beam is $1.1R_y$ with a multiple, otherwise $1.25R_y$ will be enlarged by multiples.

9.8.6.4 –In determining the required strength of columns,**9.2.5**The earthquake effects in the load combinations given in , the loading causing the plasticity of the connecting beam is $1.1R_y$ will be enlarged by multiples.

9.8.7. Column Attachments

For column splices of centrally braced steel frames with high ductility level**9.6.7**The conditions given in are equally valid for column splices of eccentrically braced steel frames with high ductility level.

9.8.8. Cross - Tie Beam Joint

Detail of the connection of the crosses with the connecting beam**9.8.6.2**It shall be dimensioned according to the internal forces calculated as specified in .

9.8.9. Connection Beam - Column Connection

9.8.9.1 –Length of the connecting beam connecting to the column

$$t_o \leq 1.6 M_p / V_p \quad (9.22)$$

will provide the condition, ((**Figure 9.10**).

9.8.9.2 –Detail of welded moment transferring connection in the connection of the tie beam to the column (See Fig.**Annex 9C**) will be used. Full penetration butt welding will be applied for the connection of the connecting beam heads to the column. The bending moment resistance of the connecting beam to the column M_p value, shear strength is $1.1 V_p$ The bending moment strength here shall not be less than M_p and shear strength V_p , **9.8.2.3** in **Equation (9.19)** or **Equation (9.20)** will be calculated with .

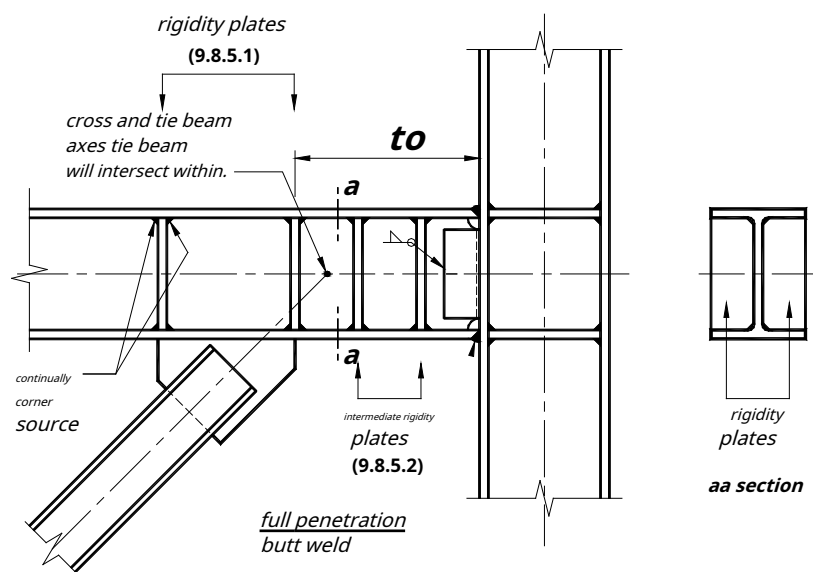


Figure 9.10

9.8.10. Cross – Beam – Column Connection

The following conditions shall be complied with in the connection of the cross or joint plate to the beam-column joint area. In addition, the following internal forces shall be applied: **9.10.3** It will be evaluated together with the internal forces of the struts (transmission elements) calculated as explained in .

(a) When the beam-column connection is taken as a hinged connection, the connection detail will be created to provide a rotation angle of at least 0.025 radians. For this connection detail **Annex 9C** One of the beam-column connection details given in or analytically proven to satisfy this condition can be used.

(b) When the beam-to-column connection is taken as rigid, the connection detail will be sized to transfer the smallest of the bending moment values given below.

(1) Plasticity moment of the beam M_p of $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ (YDKT) Bending moment magnified by multiple.

(2) Total plasticity moment of columns connected to the node $\sum M_p$ of $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ (YDKT) Bending moment magnified by multiple.

9.9. BRACED STEEL FRAMES WITH ANTI-BURST

Created by connecting specially produced buckling prevented cross elements centrally to beams or beam-column joint areas. *buckling-resistant braced steel frames*, are dimensioned in a way that ensures that plastic deformations occur in the cross members under the effects of tension and compression. The rules to be applied in the dimensioning of these systems are given below.

9.9.1. General Conditions

9.9.1.1 – In beams and columns of braced steel frames with buckling restraints, *head width/thickness, body height/thickness* And *diameter/thickness* rates **Table 9.3** Given λ_{hd} will not exceed the limit values.

9.9.1.2 –Eccentricity not exceeding the cross-section height of the beam may be permitted at the end joints of buckling-restrained braces, provided that it can be demonstrated that the internal forces obtained by taking this effect into account in the design of the members and joints do not alter the plastic deformation of the brace members.

9.9.1.3 –Buckling restrained braces shall be dimensioned based on the largest of the deformation conditions corresponding to at least 2% relative story drift and twice the design relative story drift. In determining the amount of deformation of the element, the displacement condition due to vertical load effects in braced frames shall also be taken into account.

9.9.1.4 –Buckling restrained braced steel frames shall be dimensioned based on the condition that the braced members reach yielding under tension and compression forces under the effect of design earthquake ground motion.

9.9.2. System Analysis

9.9.2.1 –Required strengths in dimensioning columns, beams and connections of buckling-resistant braced steel frame systems, **9.2.6** will be calculated based on the following:

(a) Columns, beams and connections **9.2.6** For the calculation of internal forces compatible with the yield (mechanism) condition defined in , the internal forces of the diagonal members are given below. **Equivalent. (9.23)** And **Equivalent. (9.24)** is defined with.

$$T = \omega R_y P_{y_{sc}} \quad (9.23)$$

$$P = \beta \omega R_y P_{y_{sc}} \quad (9.24)$$

(b) The hardening effect correction factor, ω , is the maximum tensile strength for the possible displacement condition, divided by the yield strength ($R_y P_{y_{sc}}$) rate will be determined by the manufacturer.

(c) Axial yield strength of steel core, $P_{y_{sc}}$ If the yield stress obtained by tensile testing is used in determining , R_y coefficient does not need to be applied.

(d) The compression strength correction factor, β , is the maximum pressure for the possible displacement condition. It will be determined as the ratio of the maximum tensile force to the maximum tensile force. This ratio determined by the manufacturer cannot be less than 1.0 in any case.

9.9.2.2 –Buckling restrained crosses will not be taken into account in transferring vertical load effects.

9.9.2.3 –In columns of braced steel frames with buckling restraints, bending moments occurring under earthquake effects may be allowed to be neglected. However, if there are horizontal forces acting between the column ends, the bending moments that will be generated by these forces shall be taken into account.

9.9.3. Crosses

The diagonal members of buckling-restrained braced steel frames shall be dimensioned to meet the following conditions.

9.9.3.1 –Diagonal members will consist of a steel core that meets the conditions given below and a wrapping system that prevents buckling by wrapping this core.

(a) If 50 mm or thicker plates are used as steel core, these plates

9.2.3.1(b) It will satisfy the minimum notch toughness condition given in .

(b) Joints in the steel core are not allowed.

(c) The buckling restraint system should limit local and general buckling of the steel core in the event of possible displacement.

9.9.3.2 – The steel core shall be dimensioned to resist the entire axial force acting on the cross member. The axial force resistance of the cross member shall be, P_{ysc} , Equation (9.25) with will be calculated.

$$P_{ysc} = F_{ysc} A_{s.c.} \quad (9.25)$$

Design axial force resistance, ϕP_{ysc} , (YDKT) or safe axial force resistance, P_{ysc}/Ω , (GKT), $\phi = 0.90$ (YDKT) or $\Omega = 1.67$ (GKT). It will be determined by taking 1.67 (GKT).

9.9.4. Additional Conditions for Special Cross Patterns

9.9.4.1 – Vor reverse V In the form of cross systems, additional conditions that the beams must meet are given below.

(a) The required strengths of beams and their connections will be determined according to the relevant load combinations, assuming that there are no crosses under the effect of dead and live loads. In load combinations that include earthquake effects, the earthquake effects will be calculated using the possible tensile and compressive strengths of the cross elements.

(b) Beams will be continuous between columns. Top and bottom heads of beams 9.2.8.1(b) It will be supported in the lateral direction based on the conditions given in .

9.9.4.2 – In buckling prevented braced steel frames K The cross layout (where the crosses are connected to the column midpoint) will not be used.

9.9.5. Beam – Column Connections

In the beam-column joint area where the cross or joint plate is connected to the beam and column, the joint shall satisfy one of the following conditions. In addition, the following internal forces, 9.10.3 It will be evaluated together with the internal forces of the struts (transmission elements) calculated as explained in .

(a) When the beam-column connection is designed as hinged, the connection detail will be created to provide a rotation angle of at least 0.025 radians. For this connection detail Annex 9C One of the beam-column connection details given in or analytically proven to satisfy this condition can be used.

(b) When the beam-to-column connection is designed as rigid, the connection detail will be sized to transfer the smaller of the bending moment values given below.

(1) The beam M_p plasticity moment $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ Bending moment resulting from (YDKT) storey.

(2) Columns that join to the node $\sum M_p$ total plasticity moment $1.1 R_y / 1.5$ (GKT) or $1.1 R_y$ Bending moment resulting from (YDKT) storey.

9.9.6. End Joints of Crosses

9.9.6.1 –Cross end joints,**9.9.2.1**It will be dimensioned under the internal forces obtained by taking into account the situations given in . These internal forces are 1.1 for (YDKT) or (GKT) for 0.7(1.1/ 1.5)will be taken into account by multiplying by .

9.9.6.2 –In the design of the joint, the buckling limit states of the joint plate will be taken into account. When necessary, the joint plate will be laterally supported.

9.9.7. Column Attachments

Column appendixes**9.2.10**It shall be sized to meet the conditions given in . In case of using butt weld in the joint design, full penetration welding shall be used. The bending strength of the column joints shall not be less than 50% of the bending capacity of the smaller section of the added element. The required shear strength shall be, $\sum M_{pc} / 1.5 H_c$ (GKT) or $\sum M_{pc} / H_c$ (YDKT)

It will be calculated as follows.

9.10. DIAPHRAGM INNER HEAD AND PILLAR ELEMENTS

Diaphragms are carrier systems that are formed within normal floor and roof planes and ensure that horizontal forces are safely transferred to horizontal load-bearing systems. A typical diaphragm system containing a reinforced concrete floor system, for example, has header zones and struts (transfer elements) created for earthquake effects in the (y) direction. **Figure 9.11** is shown schematically in . **Table 4.1** The elements and connections of the horizontal cross systems and space frame roof systems supported by one of the horizontal load-bearing systems defined in will also be dimensioned under the diaphragm forces defined in this section.

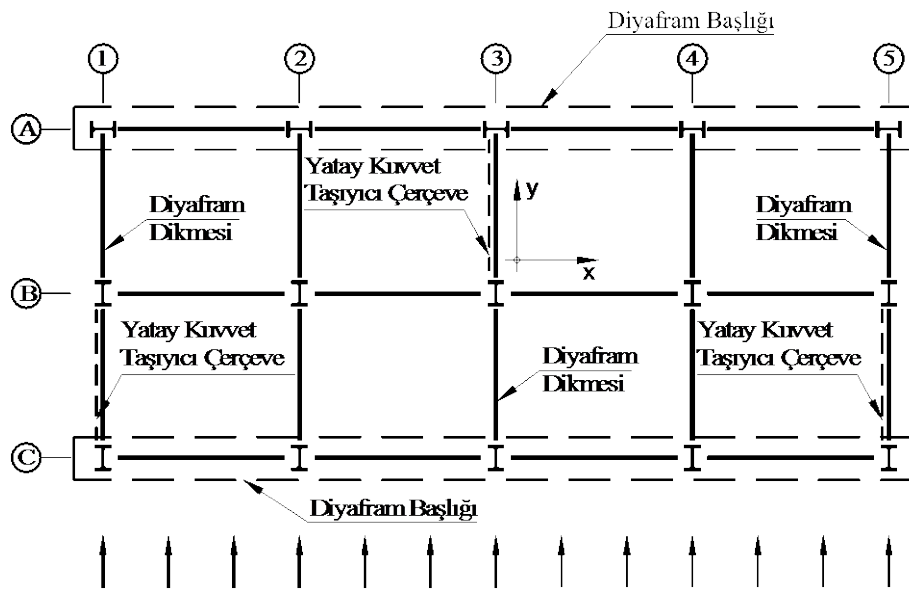


Figure 9.11

9.10.1. General Conditions

9.10.1.1 –Diaphragms shall be dimensioned to meet the shear and bending effects of horizontal design forces on the floor plane. If there are discontinuities in the diaphragms such as gaps and recessed corners, additional effects due to these discontinuities

By taking into consideration the shear and bending effects in these areas, it will be ensured that they are transferred safely.

9.10.1.2 –In dimensioning the reinforced concrete structural elements forming the diaphragm, in addition to the rules given in this section, **7.11** The relevant design rules given in shall also be taken into account.

9.10.2. Diaphragm Cap

Diaphragm heads will be dimensioned by taking into account the force pair that creates tension and compression effects due to the bending effect arising from horizontal forces on the floor plane. In reinforced concrete slabs, the diaphragm head tension force will be carried safely by additional tension reinforcement to be placed inside the reinforced concrete slab or only by the steel element. In reinforced concrete slabs, the pressure on the outermost concrete fiber in the pressure zone will be stress value is $0.20f_{ck}$. If it does not exceed , additional pressure and winding equipment may be used.

Internal forces to be used in the design of the header areas **9.2.5** It will be calculated by taking into account the load combinations including the earthquake effect given in .

9.10.3. Diaphragm Posts

Diaphragm struts and their connections shall be dimensioned in a way that ensures safe load transfer from the structural sections outside the horizontal load-bearing systems to the horizontal load-bearing systems. The internal forces to be taken as basis in the dimensioning of these elements and their connections shall be, **9.2.6** The excess strength coefficient given in , D It will be calculated by taking into account the load combinations that include the earthquake effect magnified by .

9.10.4. Horizontal Bracing Systems and Space Roof Truss Systems

The elements and connections of horizontal cross systems and space roof truss systems shall be dimensioned in a way that ensures safe load transfer from the building sections outside the horizontal load-bearing systems to the horizontal load-bearing systems. The internal forces to be taken as basis in the dimensioning of these elements and their connections shall be: **9.2.6** The excess strength coefficient given in , D It will be calculated based on load combinations that include the earthquake effect magnified by .

9.11. STEEL – REINFORCED CONCRETE COMPOSITE COLUMNS OF MOMENT RESISTING FRAMES

9.11.1. General

Steel-reinforced concrete composite columns of moment-resisting frame type carrier systems are discussed in this section. **Chapter 7** and “Regulation on Design, Calculation and Construction Principles of Steel Structures” Section 12. The transfer of the external load effect applied to steel embedded and concrete filled composite columns through longitudinal shear forces between the common surfaces of concrete and steel components will be evaluated within the scope of Section 12.7 of “Regulation on Design, Calculation and Construction Principles of Steel Structures”. Horizontal load carrying systems in buildings with steel reinforced concrete composite columns, in terms of their behavior against earthquakes, **9.11.1.1** And **9.11.1.2** It is divided into two classes defined in .

9.11.1.1 –In buildings with steel-reinforced concrete composite columns, the following horizontal load-bearing systems *Systems with High Ductility Level* has been defined as.

(a) 7.4 With reinforced concrete beams that meet the conditions given in 9.11.2.1 And 9.11.3 Moment transferring frame type carrier systems consisting of steel embedded composite columns that meet the conditions specified in .

(b) 9.3 With steel beams that meet the conditions given in 9.11.2.2 And 9.11.4 Moment transferring frame type carrier systems consisting of concrete filled composite columns that meet the conditions specified in .

9.11.1.2 –In buildings with steel-reinforced concrete composite columns, the following horizontal load-bearing systems *systems with limited ductility* has been defined as.

(a) 7.8 With reinforced concrete beams that meet the conditions given in 9.11.5 Moment transferring frame type carrier systems consisting of steel embedded composite columns that meet the conditions specified in .

(b) 9.4 With steel beams that meet the conditions given in 9.11.6 Moment transferring frame type carrier systems consisting of concrete filled composite columns that meet the conditions specified in .

9.11.2. Condition of Columns Being Stronger Than Beams

9.11.2.1 –In high ductility level moment transferring frames consisting of reinforced concrete beams and steel embedded composite columns, each beam is- The total bending moment capacities of the columns connected to the column node point shall be at least 20% greater than the total bending moment capacities of the beams connected to the column face sections. In columns where this condition is not met, 7.3.6 The rules given in will be taken as basis.

9.11.2.2 –In high ductility level moment transferring frames consisting of steel beams and concrete filled composite columns, in the earthquake direction taken into consideration, Equation (9.7) It will be shown that the condition of columns being stronger than beams is met in accordance with . In columns where this condition is not met, 9.3.3 The conditions given in , Equation (9.10) It will be applied based on the coefficient of 0.10 instead of the coefficient of 0.30.

9.11.2.3 –In calculating the bending moment capacities of composite columns, the design axial forces that minimize these moment capacities will be taken into account in accordance with the direction of the earthquake.

9.11.2.4 –In steel-reinforced concrete composite column moment resisting frames of limited ductility level, it is not mandatory to meet the requirement that the columns be stronger than the beams.

9.11.3. Steel Embedded Composite Columns of High Ductility Moment Resisting Frames

9.11.3.1 –Cross-section of columns of high ductility moment resisting frames consisting of reinforced concrete beams and steel embedded composite columns under vertical loads, TS 498' Taking into account the load reduction coefficients defined for live loads, vertical loads and 1.2 under the combined effect of the earthquake $G+Q+0.2S \pm TQ$ Maximum axial compressive forces calculated with load combination N_{dm} including

$$N_{dm} \leq 0.40 P_{no} \quad (9.26)$$

Here, the axial compressive strength of the composite column cross-section is, P_{no} , Equation (9.27) will be calculated with .

$$P_{no} = 0.85A_c f_{ck} + A_s F_y + A_{sr} F_{YSR} \quad (9.27)$$

9.11.3.2 –Special wrapping zones shall be created at the lower and upper ends of each column. The length of each wrapping zone shall not be less than 1/6 of the column clear height, 1.5 times the largest dimension of the column cross-section and 500 mm, measured upwards from the top floor level and downwards from the bottom face of the deepest beam connected to the column.

9.11.3.3 –Transverse reinforcement with a diameter smaller than 10 mm shall not be used in confinement zones. In this zone, the stirrup and crosstie spacing in the longitudinal direction shall not be greater than 1/3 of the smaller dimension of the column cross-section, six times the diameter of the longitudinal reinforcement and 150 mm, and shall not be smaller than 50 mm. The horizontal distance between the stirrup arms and/or crossties surrounding the longitudinal reinforcement shall not be less than 1/3 of the smaller dimension of the column cross-section, six times the diameter of the longitudinal reinforcement and 150 mm, and 50 mm. a_s shall not be taken greater than twenty-five times the stirrup diameter. The pitch of continuous circular spirals shall not be greater than 1/5 of the hub diameter and 80 mm. In circular columns, all wrapping reinforcement shall be provided by transverse reinforcement placed around the perimeter.

9.11.3.4 –The column middle zone is the area between the confinement zones defined at the lower and upper ends of the column. The confinement reinforcement spacing in this region **9.11.3.3** It cannot exceed twice the range specified in .

9.11.3.5 –Column appendices **9.2.10.2**, **9.2.10.3** And **9.3.5** It will be sized to meet the conditions specified in .

9.11.3.6 –Reinforcement steel splices **Chapter 7** It will be created in a way that meets the conditions specified in .

9.11.3.7 –Total longitudinal reinforcement cross-sectional area A_{sr} , $0.01A_g < A_{sr} < 0.04A_g$ will provide the condition. Here A_g is the total cross-sectional area of the composite column. In overlapped sections, the total longitudinal reinforcement cross-sectional area is $0.06A_g$ will not exceed its value.

9.11.3.8 –Axial force in stirrup columns $N_{dm} > 0.20P_{no}$ (pressure) in case of minimum total transverse reinforcement area in confinement zones, **Equation (9.28)** will satisfy the condition given by .

$$A_{s.h.} = 0.075 s b_k \left[1 - \frac{F A_{ys} - f_{ck}}{P_{no} - F_{YSR}} \right] \quad (9.28)$$

9.11.3.9 –In circular columns with spiral reinforcement $N_{dm} > 0.20P_{no}$ (pressure) in case of Minimum volumetric ratio of transverse reinforcement in confinement zones, **Equation (9.29)** will satisfy the condition given by .

$$\rho_s \geq 0.12 \left[1 - \frac{F A_{ys} - f_{ck}}{P_{no} - f_{ywk}} \right] \quad (9.29)$$

9.11.3.10 – $N_{dm} \leq 0.20P_{no}$ in the column wrapping regions in case of **Equation (9.28)** And **Equation (9.29)** At least 2/3 of the transverse reinforcement given shall be used as minimum transverse reinforcement.

9.11.3.11 –Transverse reinforcement shall be continued within the foundation to a height not less than the minimum column dimension.

9.11.4. Concrete Filled Composite Columns of High Ductility Moment Resisting Frames

9.11.4.1 –In the elements of high ductility moment transferring frames consisting of steel beams and concrete filled composite columns, *header width/thickness, body height/thickness And diameter/wall thickness* rates **Table 9.3** Given λ_{hd} will not exceed the limit values.

9.11.4.2 –Column cross-section under vertical loads TS 498 Taking into account the load reduction coefficients defined for live loads, under the combined effect of vertical loads and earthquake $1.2G + Q + 0.2S \pm 70d$ Maximum axial compressive forces calculated with load combination N_{dm} including, **Equation (9.26)** will satisfy the condition given by .

In this equation, the axial compressive strength of the composite column cross-section is, P_{no} , **Equation (9.27)** Here, the coefficient 0.95 will be used instead of the coefficient 0.85.

9.11.4.3 –The characteristic shear strength of the composite column shall be calculated considering only the effective shear area of the structural steel cross-section.

9.11.4.4 –Column appendixes **9.2.10.2** And **9.2.10.3** It will be sized to meet the conditions specified in .

9.11.4.5 –Beam-column connections will be formed using diaphragm plates at the beam flange alignment, outside the column or inside the column. The thickness of these plates will be taken as at least equal to the beam flange thickness. Diaphragm plates will be connected to the column along its entire perimeter using full penetration butt welding or double-sided corner welding. Circular gaps will be opened in the diaphragm plates formed inside the column to provide concrete passage.

9.11.4.6 –Steel beam-composite column connections at least 0.04 radians *relative story drift angle* (relative story drift/story height). For this purpose, details whose validity has been proven by experimental and/or analytical methods will be used.

9.11.5. Steel Embedded Composite Columns of Moment Resisting Frames with Limited Ductility Level

9.11.5.1 –For columns of high ductility moment-resisting frames consisting of reinforced concrete beams and steel embedded composite columns, **9.11.3** All the conditions given in are equally valid for the columns of moment resisting frames of limited ductility level consisting of reinforced concrete beams and steel embedded composite columns.

9.11.5.2 –Column cross-section under vertical loads TS 498 Taking into account the load reduction coefficients defined for live loads, under the combined effect of vertical loads and earthquake $1.2G + Q + 0.2S \pm 70d$ Maximum axial compressive forces calculated with load combination N_{dm} including, **Equation (9.26)** In this equation, the axial compressive strength of the composite column cross-section will be, P_{no} , **Equation (9.27)** will be calculated with .

9.11.6. Concrete Filled Composite Columns of Moment Resisting Frames with Limited Ductility Level

9.11.6.1 –Columns of moment resisting frames of limited ductility level consisting of steel beams and concrete filled composite columns shall meet the following cross-sectional conditions.

(a)In frame columns, *head width/thickness, body height/thickness* And *diameter/wall thickness* rates **Table 9.3** Given λ_{md} will not exceed the limit values.

(b)Column cross-section under vertical loads TS 498 Taking into account the load reduction coefficients defined for live loads, under the combined effect of vertical load and earthquake $1.2G + Q + 0.2S \pm 70d$ Maximum axial compressive forces calculated with load combinations N_{dm} including, **Equation (9.26)** In this equation, the axial compressive strength of the composite column cross-section will be, P_{no} , **Equation (9.27)** Here, the coefficient 0.95 will be used instead of the coefficient 0.85.

9.11.6.2 –For columns of high ductility moment-resisting frames made of steel beams and concrete-filled composite columns, **9.11.4.3, 9.11.4.4** And **9.11.4.5** The conditions given in are equally valid for the columns of moment resisting frames of limited ductility level consisting of steel beams and concrete filled composite columns.

9.11.6.3 –Steel beam-composite column connections at least 0.02 radians *relative story drift angle* (relative story drift/story height). For this purpose, details whose validity has been proven by experimental and/or analytical methods will be used.

9.12. BASIC CONNECTION DETAILS

Unless otherwise stated below, all foundation connection details, including foundation connection details that are outside the scope of load-bearing systems under the influence of earthquake loads, shall be dimensioned according to the rules given in this section.

9.12.1 –Required strengths in dimensioning the basic connection details, **9.2.6** will be determined on this basis.

9.12.2 –The required axial tensile and compressive strength of the connection details of the horizontal load-bearing system columns to the foundation shall be calculated by taking into account the vertical components of the required strengths of the other element connections connected to the foundation in question. However, these forces cannot be less than the greater of the axial pressure and tensile forces defined for the column joints in the relevant articles.

9.12.3 –The required shear strength of the foundation connection details will be calculated by taking into account the matters defined below.

(a) The sum of the horizontal components of the required strength of the connections of the diagonals of braced frame systems connected to the foundation.

(b) In columns of moment resisting frames, the shear force resistance specified for column splices.

9.12.4 –The required bending strength of the connection details of horizontal load-bearing systems to the foundation will be calculated by taking into account the issues defined below.

(a) The resultant of the required bending strengths of the joints of the columns and braces connected to the foundation of the braced frame systems.

(b) In columns of moment resisting frames, the smaller of the bending moment values defined below.

(1) Columns, $1.1 R_y F_y W_p / 1.5$ (GKT) or $1.1 R_y F_y W_p$ Flexural strength calculated with (YDKT).

(2) 9.2.6 The excess strength coefficient given in DB Bending moment calculated by taking into account load combinations that include the earthquake effect magnified by .

9.12.5 – In case of using full penetration butt welding in the connection of the column to the base plate, the starting and finishing plates and the base plates shall be removed after the welding process. However, the inner surface of the heads and If the base plates in the body of the cross-section profiles are fixed to the base plate with fillet welds of at least 6 mm thickness, the base plates do not need to be removed. Welding of base plates to column heads is not allowed.

9.12.6 – The elongation rate at break of the anchor rod material shall not be less than 14%.

9.13. PROJECT ACCOUNTING REPORT AND RULES FOR IMPLEMENTATION PROJECTS

9.13.1. Project Account Report

9.13.1.1 – In the project calculation report, regarding the earthquake calculation report, **Chapter 2, 3, 4** and **5** The main information given in will be included.

9.13.1.2 – The project accounting report will also include the information listed below.

(a) Material qualities and characteristic strength values of the profiles and sheet metal panels that constitute the building carrier system and the bolts used in joints and connections.

(b) Characteristic tensile strength of the weld material (weld metal).

(c) Load combinations taken into account in design, earthquake design load combinations and augmented earthquake effects.

9.13.1.3 – In addition to the dimensioning calculations and stability verifications of the structural elements, the calculations of the joints and additional details will also be given in detail within the scope of the project calculation report.

9.13.2. Rules Regarding Steel Application Project Drawings

9.13.2.1 – The following sheets will be included in the steel application project.

(a) General construction plans for roof slabs and floor slabs.

(b) Column application (layout) plan.

(c) Anchor plan and details.

(d) Sufficient number of elevations and sections.

(to) Detailed drawings of the columns and beams forming the structural system and the roof, horizontal plane and vertical plane crosses.

(f) All combinations and additional details.

9.13.2.2 –The type and material quality of the profiles and sheets used in the steel construction of the building, as well as the bolts used in the connections, as well as the type of electrode to be used and the characteristic tensile strength will be specified on all sheets.

9.13.2.3 –Considered in the design *Building Usage Class* (BKS), *Earthquake Design Class* (DTS) and local ground features **Table 4.1** Determined according to *carrier system behavior coefficient, R* And *excess strength coefficient, D* will be specified on all general construction sheets.

9.13.2.4 –Information will be given on the type of bolt used in bolted joints and additional details, bolt and hole diameters, washer and nut features, and the pre-tension force and friction surface to be applied to the bolts.

9.13.2.5 –In welded joint and joint details, the type of weld to be applied, the weld thickness and length, and in butt welds where a weld bevel must be opened, the geometric dimensions of the weld bevel, the backing plate and weld access hole details will be given.

9.13.2.6 –In the anchor details, information about the properties of the anchor rod to be applied, rod and hole diameters, and anchor length will be given.

ANNEX 9A – EARTHQUAKE FORCE AT THE JOINTS OF CARRIER SYSTEMS

GENERAL CONDITIONS

9A.1 –Principles to be followed in typical joint and joint details to prevent the sharing of the same force between bolts and welds **Figure 9A.1** is given in .

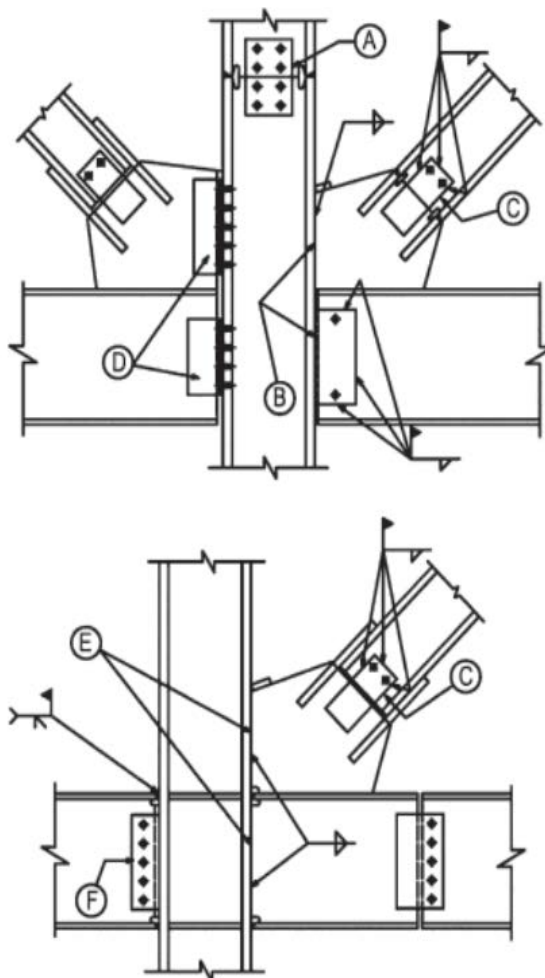


Figure 9A.1

A.In the bolted connection of the body and the welded connections of the heads, the body connection should be sized by taking into account only the shear force, and the welded connections of the heads should be sized by taking into account the column axial force (tension or compression) and bending moments.

B.In in-situ welded cross-beam-column connections, the vertical component of the force in the cross member should be appropriately distributed to the welded connections between the joint plate-column flange and the beam web-column flange. Mounting bolts may be used in the beam web.

C.In cross-node plate connections, the welds on the heads and body should be sized according to the axial force of the cross. Bolts can be used for assembly purposes.

D.In bolted cross-beam-column connections, the horizontal and vertical components of the force in the cross member should be distributed appropriately to the bolted connections between the joint plate-column flange and the beam web-column flange.

T.O.In shop-welded cross-beam-column connections, the vertical component of the force in the cross member should be appropriately distributed to the welded connections between the joint plate-column flange and the beam web-column flange.

F.In moment-resisting beam-column connections, the bolted web connection should be dimensioned taking into account the shear force, and the welded heads should be dimensioned under bending moments and axial forces.

9A.2 –Bolts and welds,9.2.3.4As stated in , they cannot be used together in a way that they share a force or a component of a force in the connection. Situations encountered in bolted and welded connections that are not suitable in terms of force sharing**Figure 9A.2**It is explained in .

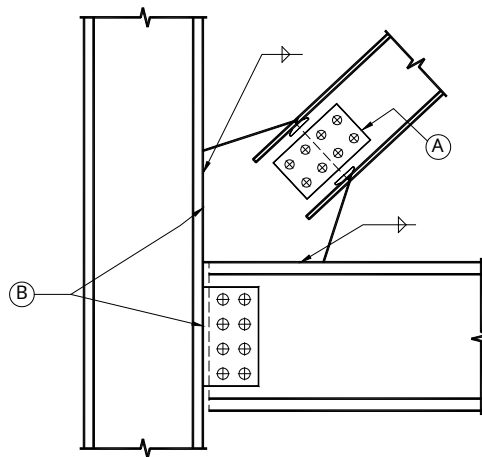


Figure 9A.2

A.In the bolted connection of the body and the welded connection of the heads, bolts and welds cannot be used together to transfer the axial force in the cross.

B.In case the joint plate is welded to the column face and the beam web is bolted, the load transfer will be different compared to the case where the connection to the column face is completely welded or bolted. In such a case, the welded joint plate connection on the column face will tend to transfer the total of the vertical component of the force in the cross member and the vertical support reaction in the beam. At the same time, the transfer of the horizontal force to the column by the bolted connection in the beam web will be prevented due to the more rigid welded joint plate connections and thus the joint plate-beam flange welded connection will tend to transfer the entire horizontal component of the force in the cross. Thus, the forces that should be transferred to the column in the beam-column-brace connection will be transferred to a large extent by the welded joint plate to the column face and the shear plate in the beam web will not be able to transfer the force. In this case, when the balance of the connection is taken into account, additional moments will occur in the beams and columns and in order to transfer these moments, greater forces will act on the welds in the joint plate-column face and the joint plate-beam flange.

ANNEX 9B – BEAM-COLUMN IN MOMENT-RESISTING STEEL FRAMES JOINING DETAILS

9B.0. ICONS

b	= Width of cross-section part
b_{bf}	= Beam flange width
b_{cf}	= Column head width
b_p	= End plate width
b_p	= Header plate width
C_p	= Considering the conditions such as hardening in the joint, etc. and the maximum possible bending A coefficient used to calculate moment strength
d	= Characteristic height of the section =
d_b	Beam cross-section height
d_c	= Column cross-section height
TO	= Structural steel modulus of elasticity ($TO=200000\text{MPa}$)
F_u	= Structural steel characteristic tensile strength =
F_y	Structural steel characteristic yield stress
F_{yp}	= Characteristic yield stress of the plate
g	= Horizontal distance between bolts
h	= Element body height
h_p	= Body skid plate height
h_{st}	= Height of the stiffness plate
$-h$	= Distance between possible plastic hinge points at beam ends and column
$-n$	face = Beam span between possible plastic hinge points at beam ends =
L_{st}	Minimum length of stiffness plates
M_p	= Possible bending moment capacity
M_{pri}	= Left end of beam/Possible bending moment capacity at
M_{prj}	= Right end of beam/Possible bending moment capacity at
M_{fly}	= Required bending moment strength of the connection on the column
n	face = Total number of bolts used in one flange of the beam
R_t	= Ratio of possible tensile strength to characteristic tensile strength =
R_y	Ratio of possible yield stress to characteristic yield stress = Distance
S	between the centers of two consecutive holes
S_1	= Distance of the closest hole to the column head to the column head surface

t	= Element thickness
t_{bf}	= Beam head thickness =
t_{bww}	Beam body thickness
t_{cf}	= Column section header thickness
t_p	= End plate thickness
t_p	= Body shear plate thickness
t_s	= Body rigidity plate thickness
$t_{s.c.}$	= Column web stiffness plate thickness =
t_w	Element web thickness
V_d	= Simple beam formed by vertical loads at the plastic hinge point of the beam shear force
V_{fly}	= Required shear force resistance of the connection on the column face
W_p	= Plastic strength moment according to the bending axis
φ_d	= Strength coefficient for ductile collapse case
φ_n	= Strength coefficient for non-ductile collapse case

9B.1. SCOPE AND GENERAL MATTERS

9B.1.1 –In this section, **9.3.4.1(a)** in and **9.4.2.1(a)** At least 0.04 radians or 0.02 radians as prescribed in *relative story drift angle* Various examples of bolted and welded connection details that have been proven by experimental and/or analytical methods to be capable of providing (relative storey drift / storey height) are given.

9B.1.2 –These details can be used in moment resisting beam-column connections of frames with high and limited ductility levels, within the application limits given for them in this section.

9B.1.3 –The cross-sections of the combined beams and columns shall satisfy the conditions defined below.

(a) Rolled product wide head profiles shall be used in accordance with the boundary conditions given for them in this section.

(b) Making double axis symmetry **I** Cross-sectional elements can be used if the following conditions are met.

(1) The width, height and thickness of the heads and bodies must be compatible with the rolled profiles that meet the relevant conditions given in this section.

(2) Continuous welding should be used in the connection of the body to the heads. The application principles of these welds for artificial beams and columns are given in the following articles.

(i) In artificial beams, the body and heads are replaced by plastic joints starting from the beam end. **(-h)** One

It shall be connected with full penetration butt welding together with fillet weld reinforcement applied to both sides of the web in a way that is not less than the beam height. The thickness of the fillet weld reinforcement shall not be less than 6 mm and 0.70 times the beam web thickness.

(ii) Don't do that In cross-section columns, the column web and flanges shall be connected with full penetration butt welding together with the applied fillet weld reinforcement in the area covering the section up to 300 mm from the top and bottom flanges of the connected beam. The thickness of the fillet weld reinforcement shall not be less than 6 mm and 0.70 times the thickness of the column web.

(iii) Box cross-sectioned wide head If the sheets are used as headings in profiles, the width / thickness of these sheets, b/t rate, b_{net} width between headings including $0.6EF\sqrt{h}$ In case the boards are used as a body,

height / thickness, h/t rate, **Table 9.3** It cannot be greater than the relevant limit value given in Box The flange and web plates of the columns formed into cross-sections shall be joined with full penetration butt welding in the area covering the section up to 300 mm from the top and bottom flanges of the connected beam. Outside this area, the joint of the plates shall be provided with continuous corner or butt welding.

(iv) Columns with headed () cross-sections can be constructed using rolled or artificial cross-section elements. The body of the cross-sectional element, continuous The cross-sectional element body will be connected with full penetration butt weld with fillet weld reinforcement applied on both sides. The minimum fillet weld thickness will be taken as equal to the smaller of 6 mm and 0.70 times the column body thickness. Continuous plates will meet the conditions given for wide-head columns.

9B.1.4 –The design of the connections within the scope of this section will be carried out within the framework of the principles explained below.

(a) Only the (YDKT) method will be used as a basis for dimensioning the connections. In the connection design, depending on the collapse limit state taken into consideration, the strength coefficients given below will be used for the ductile collapse limit state (yield limit state) and the non-ductile collapse limit state (fracture limit state).

(1) For the ductile failure limit state $\varphi_d = 1.0$

(2) For the non-ductile failure limit state $\varphi_n = 0.9$

(b) Distance of plastic joint from column face, $-h$, are defined within the relevant application limits for each type of joint in this section.

(c) The maximum possible moment in the plastic hinge is, M_p

$$M_p = C_p R_y F_y W_p \quad C_p = \frac{F_y + F_u \leq 1.2}{2 F_y} \quad (9B.1)$$

It will be calculated as follows. The joint on the column face required bending moment strength M_{fy} and the required shear strength V_{fy} (Figure 9B.1), will be determined by taking into account the possible plastic hinge points at the beam end.

(d) **9.3.4.3** In accordance with the , continuity plates will be used at beam flange alignments when necessary.

9B.1.5 –The shear zone limited by the beam-column joint, high ductility level or limited ductility level for moment resisting steel frames, respectively, **9.3.4.2** or **9.4.2.2** will provide the conditions in .

9B.1.6 –The capacity protected areas shall be defined for each connection type. Unless otherwise specified, the capacity protected areas shall start from the column face and extend to the plastic hinge.

It is defined as the area covering a distance of 1.5 times the beam height from the point. In the beam web, bolt holes arranged in accordance with the conditions in the relevant sections for each connection type may be allowed.

9B.1.7 –Welding details of welded joint applications shall be arranged to meet the conditions given below.

(a)When the continuity plates are connected to the column web with a full penetration butt weld, there is no need to remove the backing plate. However, when connected to the column head with a full penetration butt weld, the backing plate will be connected to the column head with a continuous fillet weld with a weld thickness of 6 mm. This fillet weld will be applied to the side where the butt weld is located, so that it remains under the full penetration butt weld. If the backing plate is removed, a groove will be opened by grinding at the weld root and this area will be reinforced with a continuous fillet weld with a weld thickness of 6 mm.

(b)If a backing plate is used in the full penetration butt welded connection of the beam bottom flange to the column flange, the backing plate shall be removed after welding. After the backing plate is removed, a groove shall be opened by grinding at the weld root and this area shall be reinforced with a continuous fillet weld with a minimum weld thickness of 6 mm. The thickness of this fillet weld reinforcement shall be determined to include the beam flange surface.

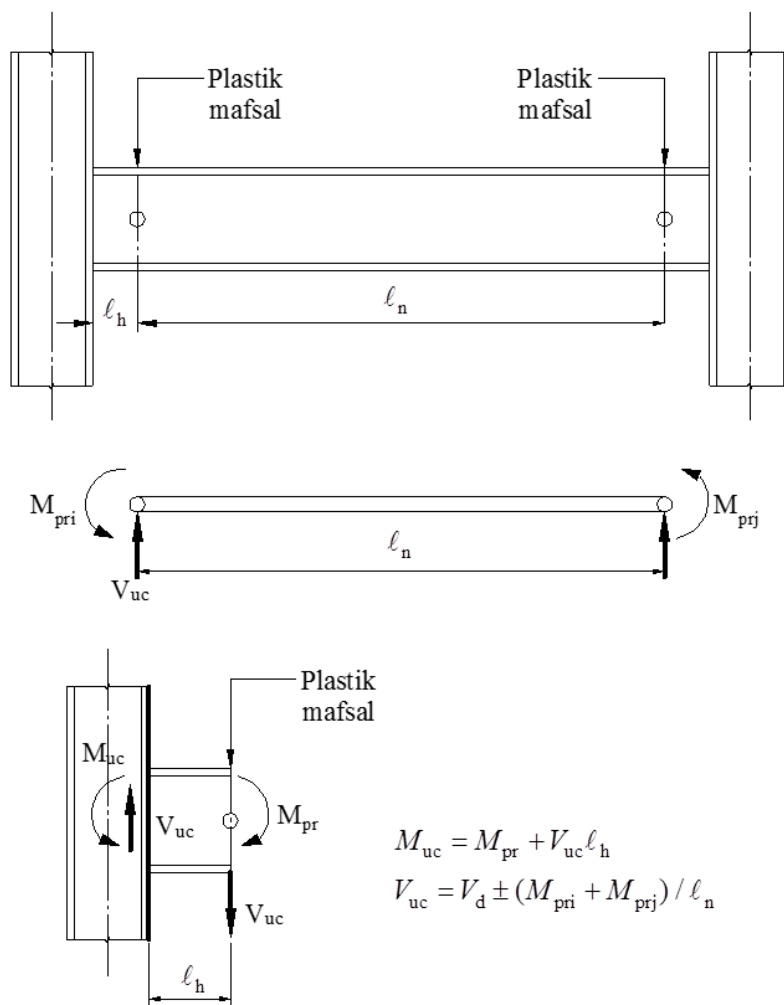


Figure 9B.1

(c) If a backing plate is used in the full penetration butt welded connection of the beam top flange to the column flange, the backing plate will be left in place and welded to the column flange with a continuous fillet weld with a weld thickness of 6 mm. This fillet weld will be applied to the side where the butt weld is located, so that it remains under the full penetration butt weld.

(d) Especially in full penetration butt welding applications, starting and finishing plates will be used at the ends. After the welding process, these plates will be cut and removed with a suitable cutting method. The surfaces will be processed in a way that does not leave any notches or sharp corners.

(to) In areas where capacity is maintained, temporary welds (tack welds) connecting the base plate to the start and end plates shall remain in place if they remain within the welded joints.

(f) At the junction of continuity plates to the body and head 9.3.4.3 The conditions in will be complied with. However, the welding process to be applied to the column body will be terminated at least 40 mm away from the curvature end points in the column cross-section neck area. The welding process to be applied to the column head will be terminated at least 15 mm away from the curvature end points in the column cross-section neck area.

9B.2. FULL STRENGTH BOLT-FOAM PLATE JOINTS

9B.2.1 – In this section, three different types of face plate joint details are given, the adequacy of which has been proven by experimental and/or analytical methods. (Figure 9B.2).

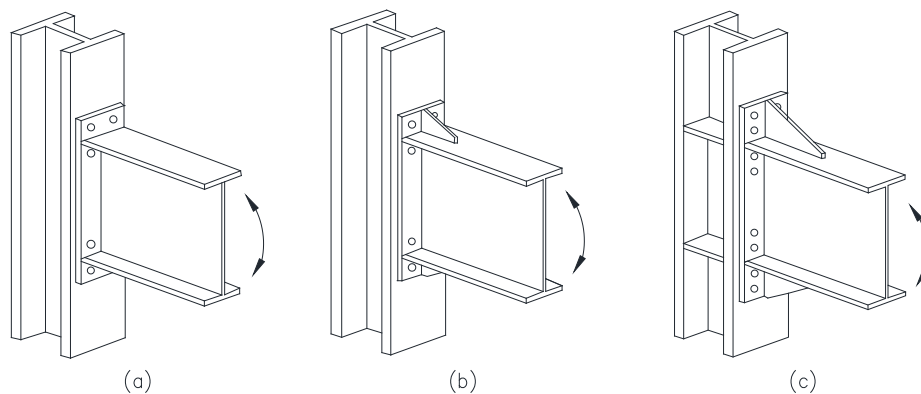


Figure 9B.2

The limiting conditions that determine the behavior of this type of beam-column connections can be listed as; beam cross-section yielding under bending effect, end plate yielding under bending effect, yielding of column panel area, bolts breaking under tension effect, bolts breaking under shear effect or fractures occurring in welded joint areas. The design principles defined here are based on the principle that the connection elements have sufficient strength to ensure that inelastic deformations occur due to yielding in the beam cross-section.

9B.2.2 – These connection detail types are used in the connections of high ductility level and limited ductility level moment resisting frames. 9B.1.1 In order to be used as combinations defined in , Table 9B.1 Within the limits of application given in , subject to the additional conditions given below

Geometric variables to be taken as basis in the evaluation of the application limits of these types of joints **Figure 9B.3** is given in.

9B.2.3 –In contact with reinforced concrete slab **Figure 9B.2** If the connections in are used in frames with high ductility level and moment resisting frames with limited ductility level, these connections can only be used if the following special conditions are met.

(a) Shear connections shall not be used in the section of the beam flange between the column flange face and 1.5 times the beam height.

(b) The reinforced concrete slab shall be finished at least 25 mm away from both flange faces of the column. The space between the column face and the edge of the reinforced concrete slab shall be filled with a compressible material.

9B.2.4 –In the application of these connections, the end plate will be connected to the column head.

9B.2.5 –The width of the end plate must be equal to or greater than the flange width of the connected beam. However, in end plate calculations, the effective width must be taken as maximum 25 mm more than the flange width of the beam.

9B.2.6 –In 4 and 8 bolted butt plate connections with stiffness plates, minimum length of stiffness plates L_{st} **Equation (9B.2)** It will not be less than the value calculated with .

$$L_{st} = \frac{h_{st}}{\tan 30^\circ} \quad (9B.2)$$

9B.2.7 –The following conditions shall be complied with in the welded connection of the beam and the end plate.

(a) Weld access holes will not be used.

(b) In rigid plate connections, full penetration butt welds shall be used in the connection of the rigid plate to the butt plate. If the rigid plate thickness is less than 10 mm, the use of fillet welds with a strength equal to the rigid plate strength may be permitted.

9B.3. BOLT-HEADING-PLATE JOINT

9B.3.1 –Bolted flange plate moment transferring connections are formed by flange plates welded to the column flange and bolted to the beam flange. **(Figure 9B.4)** The conditions defined here are based on the principle that the joint elements must have sufficient strength to ensure the formation of a plastic hinge in the region near the end of the header plate.

9B.3.2 –This connection detail is used in the connections of frames with high ductility level and limited ductility level moment transferring frames, **9B.1.1** In order to be used as combinations defined in , **Table 9B.2** Within the scope of application given in , the additional conditions defined below must be adhered to.

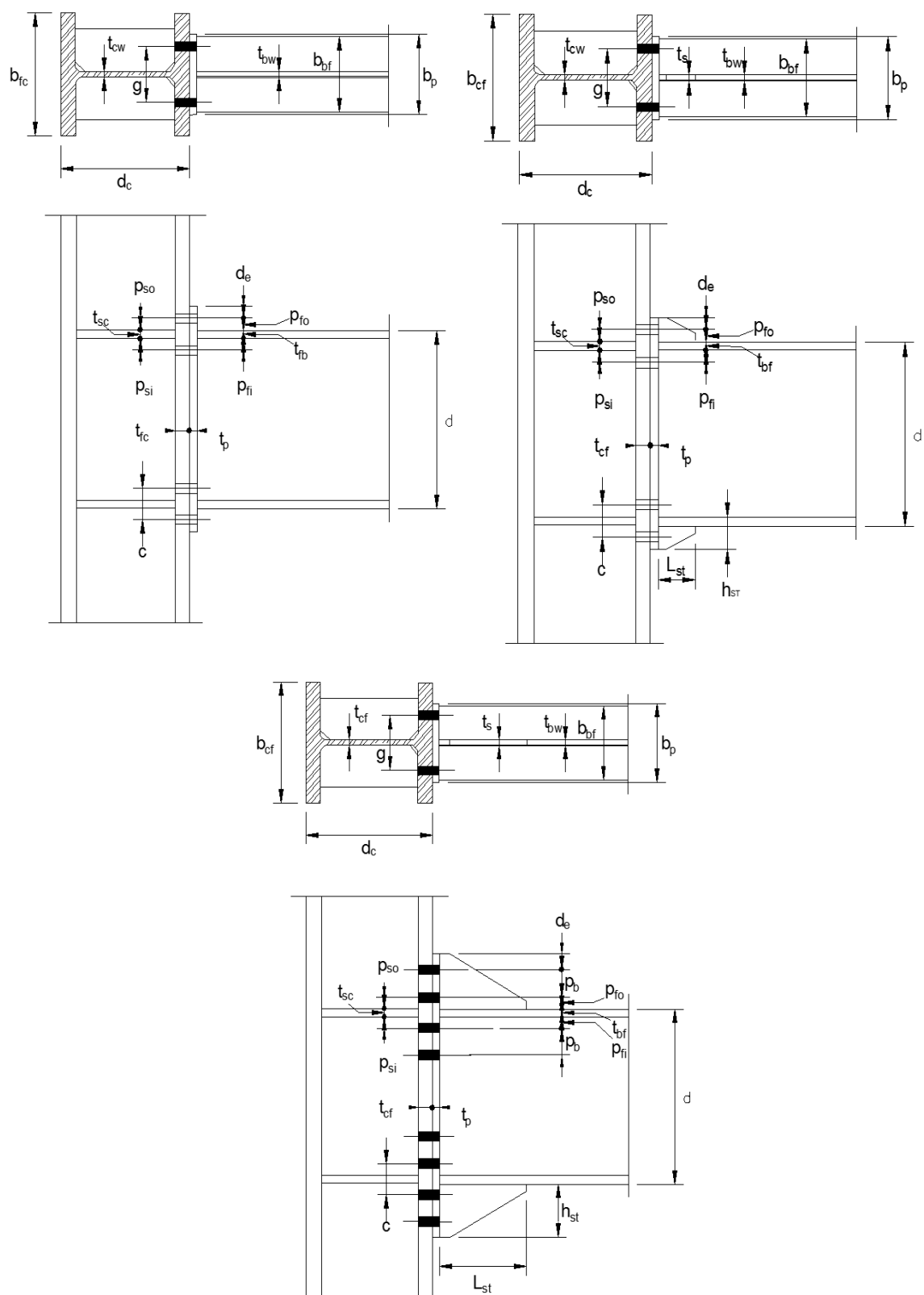


Figure 9B.3

Table 9B.1 – Application Limits for Full Strength Bolted Face Plate Connections

Join Detail Variables	Application Limits
Distance of plastic joint from column face, a_h	$\min(d_b / 2, 3d_{bf})$
Beam span / cross-section height ratio	≥ 7 (frames with high ductility level) ≥ 5 (frames with limited ductility level)
Face plate thickness, t_p	$12 \leq t_p \leq 60$ mm (without 4-bolt rigidity plate)
	$12 \leq t_p \leq 40$ mm (with 4 bolted rigidity plate)
	$18 \leq t_p \leq 65$ mm (with 8 bolted rigidity plate)
Face plate width, b_p	$160 \leq b_p \leq 300$ mm (without 4-bolt rigidity plate)
	$160 \leq b_p \leq 300$ mm (with 4 bolted rigidity plate)
	$215 \leq b_p \leq 400$ mm (8 bolts (with rigidity plate)
Horizontal distance between bolts, g	$100 \leq g \leq 155$ mm (without 4-bolt rigidity plate)
	$70 \leq g \leq 155$ mm (with 4 bolted rigidity plate)
	$125 \leq g \leq 155$ mm (with 8 bolted rigidity plate)
$p_r(p_{phi} \text{ And } p_{ro})$	$40 \leq p_r \leq 115$ mm (without 4-bolt rigidity plate)
	$45 \leq p_r \leq 140$ mm (with 4 bolted rigidity plate)
	$40 \leq p_r \leq 50$ mm (with 8 bolted rigidity plate)
p_b	$90 \leq p_b \leq 100$ mm (with 8 bolted rigidity plate)
Beam cross-section height, d_b	$270 \leq d_b \leq 1400$ mm (without 4-bolt rigidity plate)
	$270 \leq d_b \leq 610$ mm (with 4 bolted rigidity plate)
	$450 \leq d_b \leq 950$ mm (with 8 bolted rigidity plate)
Beam head thickness, t_{br}	$10 \leq t_{br} \leq 25$ mm (without 4 bolted rigidity plate)
	$10 \leq t_{br} \leq 25$ mm (with 4 bolted rigidity plate)
	$14 \leq t_{br} \leq 35$ mm (with 8 bolted rigidity plate)
Column cross-section height	≤ 920 mm (for I profiles)
Bolt class	8.8 or 10.9
Bolt pre-tension conditions	Full pre-pull
Fascia material class	S235, S275 or S355
Title plate source	Full penetration butt weld
Capacity protected area	For connections without rigid plates, the area between the column face and the beam height or the area between the column face and the beam flange width three times the smaller of the distant region
	For rigidity plate connections, column face and column The area between the end point of the stiffener and half the beam depth from the face or three times the beam flange width from the column face the smaller of the area as far away as

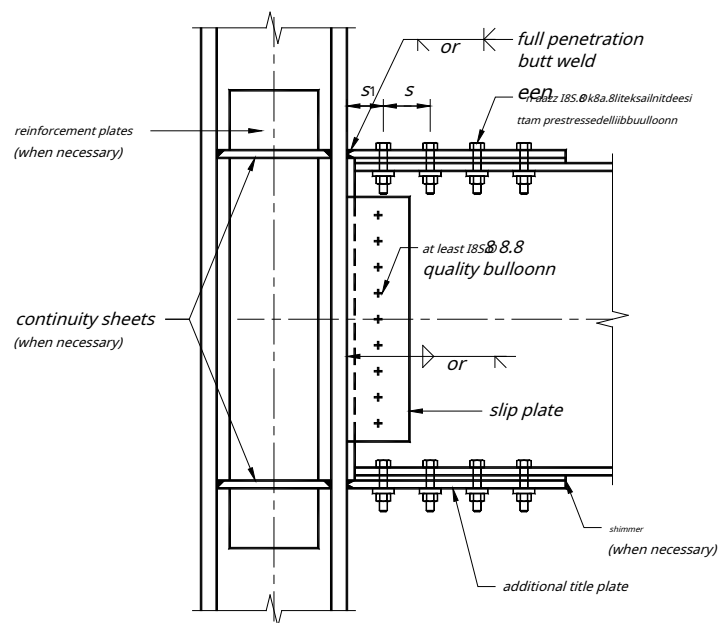


Figure 9B.4

Table 9B.2 – Application Detail of Moment Resisting Connection with Bolted Head Plate

The limits of

Join Detail Variables	Application Limits
Distance of plastic joint from column face, s_h	$s_h = s_1 + s - \frac{n}{2} - 1$
Beam span / cross-section height ratio	≥ 9 (frames with high ductility level) ≥ 7 (frames with limited ductility level)
Beam cross-section height, d_b	$\leq 920\text{mm}$
Beam head thickness	$\leq 25\text{mm}$
Column cross-section height	$\leq 920\text{mm}$ (for I profiles) $\leq 610\text{mm}$ (for artificial box profiles)
Bolt class	8.8 or 10.9
Maximum bolt size	M27
Bolt pre-tension conditions	Full pre-pull
Title plate material class	S235, S275 or S355
Title plate source	Full penetration butt weld
Capacity protected area	From the column face to the bolt farthest from the column face area as far as the beam depth

9B.3.3 –If the header plate bolted connections in contact with the reinforced concrete slab are used in moment resisting frames with high ductility level and limited ductility level, these connections can only be used if the following special conditions are met.

(a) Shear connections shall not be used in the section between the beam flange face and the distance of 1.5 times the beam height from the column face.

(b) The floor concrete shall be finished at least 25 mm away from the face of the column heads. The gap between the column face and the edge of the floor concrete shall be filled by placing a compressible material.

9B.3.4 –Both heads will have the same geometric properties.

9B.3.5 –The welding of the header plates to the column head shall be a full penetration butt weld. When the base plate is used, this plate shall not be left in place.

9B.3.6 –The body slip plate should be welded to the column head. The column head-body plate connection will be provided with full penetration butt welding or double-sided fillet welding.

9B.3.7 –The length of the bolt group to be used in the header plate cannot exceed the beam height. Standard hole diameter will be used in the beam header. In the header plate, standard or large circular hole diameter is allowed. It is not allowed to open the bolt holes by punching method.

9B.3.8 –For the bolt diameter that will prevent the beam flange from reaching the tensile failure limit state in the joint **Equation (9B.3)** The condition in will be met.

$$d_{bf} \leq \frac{b}{2} - 1 - \frac{R_{fy}}{R_t F_u} - 3 \text{ mm} \quad (9B.3)$$

9B.3.9 –A feed plate can be used between the beam header and the header plate if necessary; however, the thickness of the feed plate cannot exceed 6mm.

9B.3.10 –In steel frame systems with high ductility level and limited ductility level moment transferring systems, if the beam flange is connected to the reinforced concrete slab with steel anchors with a center distance not exceeding 300 mm along the span between the areas with protected capacity, there is no need to use special stability connections on the upper and lower flanges of the beam in the plastic joint areas.

9B.4. FULL PENETRATION BUTT WELDED JOINT

9B.4.1 –The conditions given below are based on the principle that the joint must have sufficient strength to ensure that the plastic joint is formed in the area close to the column head, with the application of special details given for column head - beam head welds, column head - beam web welds and weld access holes. Joint detail **Figure 9B.5** It is seen in .

9B.4.2 –This connection detail is used in the connections of frames with high ductility level and limited ductility level moment transferring frames, **9B.1.1** In order to be used as combinations defined in , **Table 9B.3** Within the scope of application given in , the additional conditions defined below must also be complied with.

9B.4.3 –In dimensioning the joint, **Equation (9B.1)** In the calculation of the maximum possible moment in the plastic hinge, $C_p = 1.4$ will be taken.

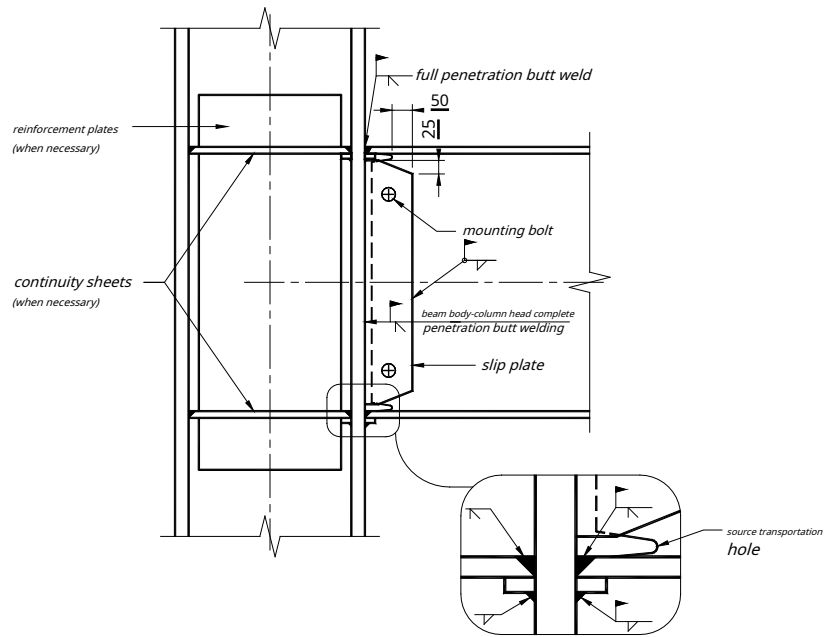


Figure 9B.5

Table 9B.3 – Application Limits of Full Penetration Butt Weld Joint Detail

Join Detail Parameters	Application Limits
Distance of plastic joint from column face, $-h$	$-h = 0$
Beam cross-section height	$\leq 920\text{mm}$
Beam span / cross-section height ratio	≥ 7 (frames with high ductility level) ≥ 5 (frames with limited ductility level)
Beam head thickness	$\leq 25\text{mm}$
Column cross-section height	$\leq 920\text{ mm}$ (for I profiles) ≤ 610 mm (for artificial box profiles)
Source access hole	Necessary
Title plate source	Full penetration butt weld
Capacity protected area	Beam cross-section height from column face the area encompassing the distance

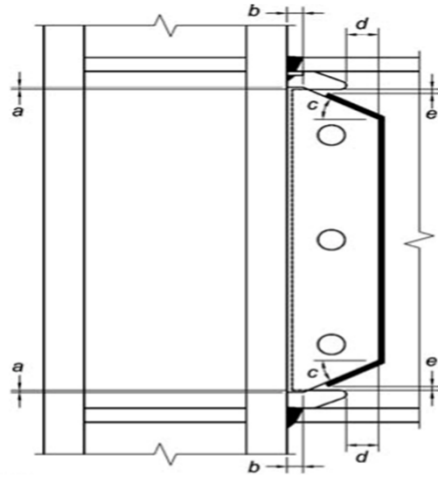
9B.4.4 –In plastic joint areas, lateral supports should be formed on both heads in the area covering a distance of 1 to 1.5 times the beam height from the column face. Lateral support is not allowed in the area covering a distance of the beam height from the column face.

9B.4.5 –In steel frame systems with high ductility level and limited ductility level moment transferring systems, if the beam flange is connected to the reinforced concrete slab with steel anchors with a center distance not exceeding 300 mm along the span between the areas with protected capacity, there is no need to use special stability connections on the upper and lower flanges of the beam in the plastic joint areas.

9B.4.6 –The following conditions shall be complied with at the column head - beam head connection.

(a) The beam flange will be connected to the column flange with a full penetration butt weld.

(b) Weld access holes, **Figure 9B.6** It will have the geometry and application conditions given in .



$$6\text{mm} \leq a \leq 12\text{mm}$$

$$b \geq 25\text{mm}$$

$$c = 30^\circ (\pm 10^\circ)$$

$$d \geq 50\text{mm}$$

$$12\text{mm} \leq e \leq 25\text{mm}$$

e: Kaynak ulaşım deliği kenarının köşe kaynak bitiş noktasına uzaklığı

Figure 9B.6

9B.4.7 –The following conditions shall be complied with at the column head - beam body connection.

(a) The thickness of the web shear plate shall be at least equal to the beam web thickness. The height of the web shear plate shall be arranged so that there is a minimum distance of 6 mm and a maximum distance of 12 mm between the edges of the plate and the edges of the source access hole for the top and bottom heads. Its width shall be determined so that it ends at least 50 mm away from the source access hole. (**Figure 9B.6**).

(b) The body shear plate will be welded to the column head. The design strength of the welds, h_p plate height and t_p being the plate thickness, $h_p t_p (0.6 R_y F_{yp})$ will not be less than the value calculated as follows.

(c) Body shear plate to beam body **Figure 9B.6** Sleeve length as per the details given in t_p – Will be joined with 2mm corner welds.

(d) If necessary, the use of mounting bolts with standard holes in the beam body or short oval holes in the direction of the beam may be permitted.

(to) Column flange and beam web welds shall be full penetration butt welds applied over the entire length of the area between the weld access holes. Start and end plates are not required for welding.

9B.5. WELDED WEAKENED BEAM CROSS SECTION BEAM – COLUMN COMBINATION

9B.5.1 –In welded, weakened beam cross-section moment transferring connection details, the width of the beam flanges in the area close to the beam-column connection is reduced with an appropriate cut. Connection details **Figure 9B.7** The conditions defined here are based on the principle that the connection must have sufficient strength to ensure the formation of a plastic hinge in the reduced beam cross-section.

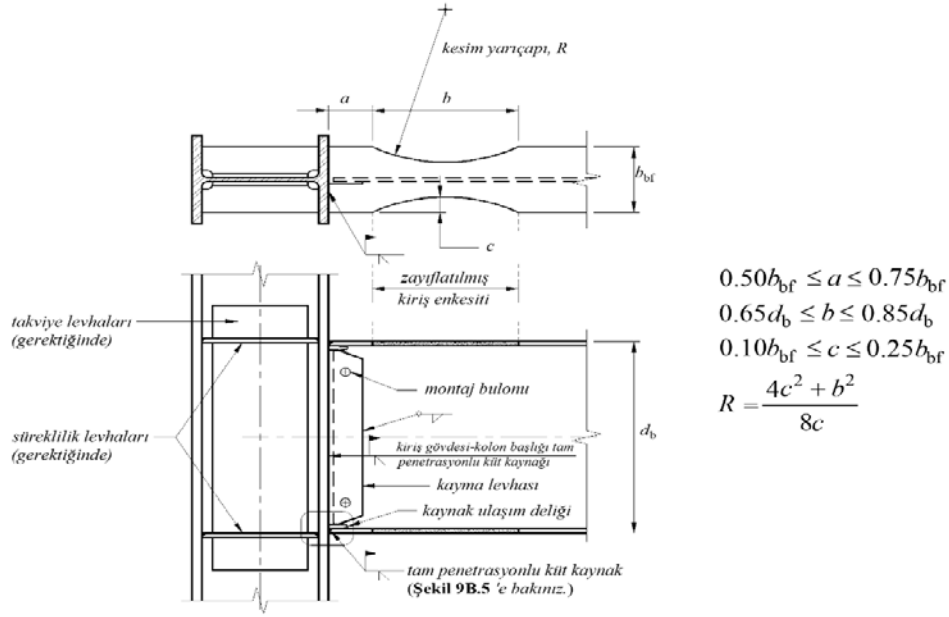


Figure 9B.7

9B.5.2 –This connection is used in frames with high ductility level and limited ductility level that transfer moments,**9B.1.1**In order to be used as combinations defined in ,**Table 9B.4**Within the scope of application given in , the following additional conditions must be observed.

9B.5.3 –Since it is anticipated that a plastic hinge will form due to yielding in the reduced beam cross-section of the joint,**9.2.8.1**According to , additional lateral support should be provided against lateral torsional buckling in both flanges. However, if the upper flange of the beam is connected to the reinforced concrete slab with steel anchors with a center distance not exceeding 300 mm, these additional lateral supports are not required in the lower and upper flanges of the beam. In case of lateral support, this support should be provided at a distance of at most half the height of the beam from the most distant point of the area where the beam cross-section is reduced from the column. The connection of the lateral support to the beam is not allowed within the area between the column face and the farthest end of the reduced beam cross-section area from the column face.

Table 9B.4 – Beam – Column Connection Detail of Welded Weakened Beam Cross Section Application Limits

Join Detail Parameters	Application Limits
Beam cross-section height	$\leq 920\text{mm}$
Beam unit length weight	$\leq 450 \text{ kg/m}$
Beam span / cross-section height ratio	≥ 7 (frames with high ductility level) ≥ 5 (frames with limited ductility level)
Beam head thickness	$\leq 44\text{mm}$
Column cross-section height	$\leq 920 \text{ mm}$ (for I profiles) $\leq 610 \text{ mm}$ (for artificial box profiles)
Source access hole	necessary
Additional header plate source	Full penetration butt weld
Capacity protected area	The most weakened beam cross-section with column face the area between the far point

9B.5.4 –Full penetration butt welding will be used in the connection of beam flanges to column flanges.

9B.5.5 –The following conditions shall be complied with in the connection of the beam web to the column head.

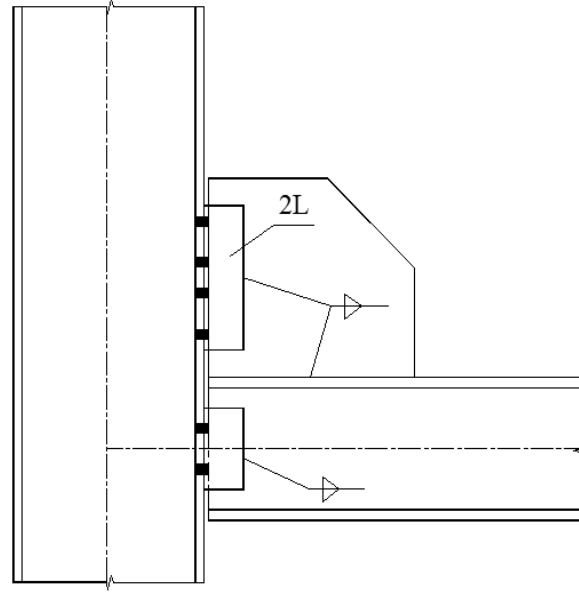
(a)In frames with high ductility level, the connection of the beam web to the column head should be provided with full penetration butt welding between the weld access holes. For full penetration butt welding, it is allowed to use a shear plate with a minimum thickness of 10 mm between the weld access holes to act as a backing plate. Bolt holes can be drilled in the beam web for assembly purposes.

(b)In frames with limited ductility level**(a)**The conditions given in can be applied. However, in these systems, bolted shear plate web connection is also allowed. In this case, the connection will be sized as a friction effective connection and its application will be ensured. Full penetration butt weld or double-sided fillet weld will be used in the connection of the shear plate to the column head. In case of fillet weld, the minimum thickness of each fillet weld should be 0.70 of the plate thickness. Bolt holes will be standard holes. However, it is allowed to use a short oval hole opened parallel to the beam head provided that it is opened in either the beam body or the shear plate.

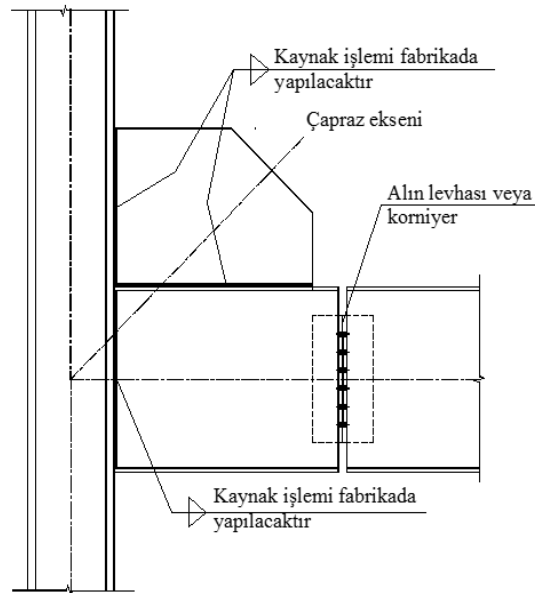
(c)The reduced beam cross-section must be created by thermal cutting.

ANNEX 9C – CROSS - BEAM - COLUMN JOINT DETAILS

(a) In the connection of the cross or nodal plate to the beam - column joint area, 9.6.5 in and 9.8.9 When the beam to column connection is designed as hinged or rigid, as defined in , the following connection details shall be used that have been proven to provide the required rotation capacity. (Figure 9C.1).



(a) Mafsallı Birleşim



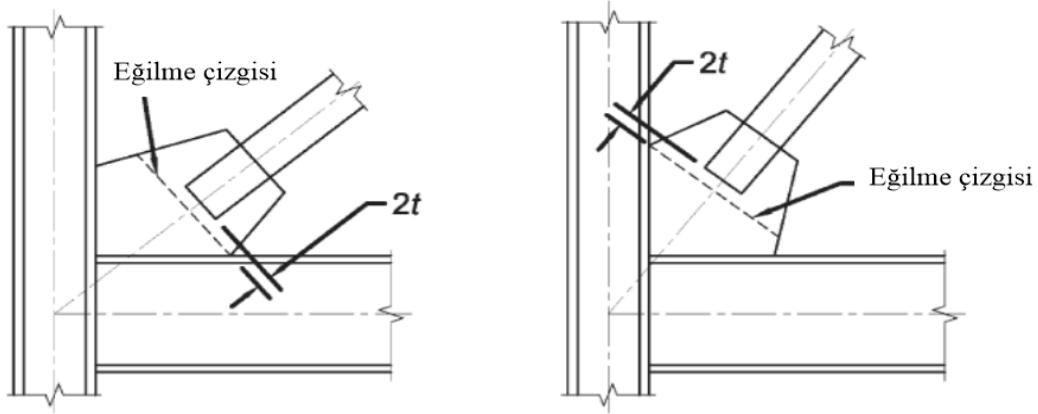
(b) Rijit Birleşim

Figure 9C.1

(b) Plastic joints 9.6.6(c)(2) When intended to occur at a joint or connecting plate, the cross joint shall have sufficient rotation capacity as defined in

Sufficient rotation capacity will be provided by creating sufficient areas to allow plastic rotation in the joint plate or connection plate to be used in the cross member end joint. Details that can be applied to ensure this condition is met **Figure 9C.2** and in **Figure 9C.3** is given in.

(c) Welded reinforced full strength moment transferring connection detail that can be used in the connection of the tie beam to the column **Figure 9C.4** is given in.



$t = \text{Düğüm noktası levhası kalınlığı}$

Figure 9C.2

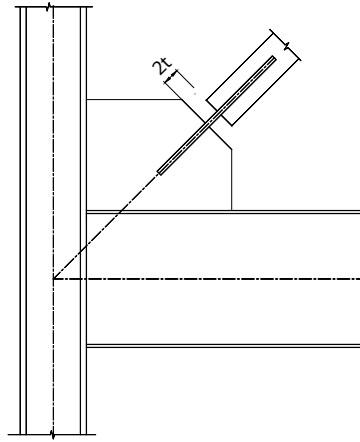


Figure 9C.3

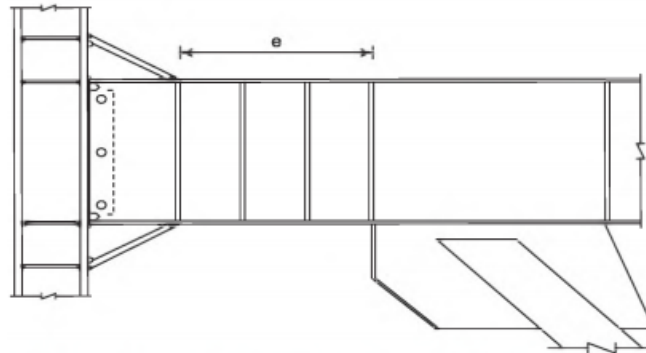


Figure 9C.4

CHAPTER 10 – LIGHT STEEL BUILDING CARRIER UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGNING SYSTEMS

10.0. ICONS

In the dimensional expressions in this section where the following symbols are used, the forces are: *Newton*[[N], lengths *millimeter*[[mm], angles *radian*[rad] and stresses *MegaPascal*[[MPa]=[N/mm²] unit.

A	= Cross-sectional area
A_{effect}	= F_n effective area in tension
A_g	= Gross cross-sectional area
A_{net}	= Net cross-sectional area
b	= Profile header width
b_p	= Covering width
C, C_i	= Compressive force
C	= Length of the first side reinforcement perpendicular to the
C_d	header = Maximum distance between two uprights
D	= Excess strength coefficient
d	= Horizontal length of the second side
d_s	reinforcement = Profile body height
d_v	= Characteristic screw diameter
d_w	= Larger diameter of washer or screw head
TO	= Steel elasticity modulus =
TO_d	Earthquake effect
F_{to}	= Characteristic compressive strength for bending buckling limit state
F_n	= Characteristic compressive strength
F_u	= Steel tensile strength
F_{u1}	= Tensile strength of the sheet metal in contact with the screw head
F_{u2}	= Tensile strength of the sheet metal that is not in contact with the screw head
F_y	= Steel yield stress
G	= Shear modulus of the coating material =
G	Dead load
H	= Ground horizontal
h	thrust = Panel height

h_l	= Profile body height
I	= Moment of inertia
K	= Effective buckling length
L	coefficient = Unconstrained length of
-	the strut = Panel width
-I	= Width of panel partitions at full floor height
M_t	= Panel tipping moment
M_n	= Strut axial compressive strength
P_n	= Strut axial compressive strength
$P_{n,cross}$	= Axial force calculation strength of cross member
$P_{clear,cross}$	= Net cross-sectional calculation strength of the diagonal member
P_{Nov}	= Characteristic pull-out strength of the screw head from the sheet metal
P_{notes}	= Characteristic pull-out strength of the screw tip from the sheet metal
P_{ns}	= Characteristic shear strength of the screw
P_{ns1}	= Characteristic shear strength of the screw for the tilting limit state =
$P_{ns2,3}$	Characteristic shear strength of the screw for the crushing limit state
P_{nt}	= Characteristic pull-out strength of the
$P_{n,design}$	screw = Design basis axial force
P_s	= Screw design shear strength
P_t	= Screw design pull-out resistance
p	= Spacing of connection elements at intermediate posts
Q	= Live load
R	= Carrier system behavior coefficient =
r	Radius of inertia of the strut cross section =
r_p	Profile inner bending radius
S	= Snow load
s	= Spacing of connection elements at side posts =
$T, \bar{T}$	Tensile force
t	= Sheet metal thickness
t_c	= Thickness of the sheet metal that is not in contact with the screw head or washer
$t_{planting}$	= Strut wall thickness

t_{covering}	= Coating material thickness
t_1	= Thickness of the sheet metal in contact with the screw head
t_2	= Thickness of the sheet metal that is not in contact with the screw head
V_d	= Total horizontal load acting on the wall
V_c	= Panel shear force capacity
v_c	= Characteristic shear strength per unit length
v_d	= Shear demand per unit length
λ	= Slenderness ratio
φ	= Strength coefficient
Ω	= Safety factor

10.1. SCOPE

10.1.1. The dimensioning and arrangement of the joints of the load-bearing system elements of all light steel buildings made of cold-formed profiles under the effect of earthquakes will be carried out primarily according to the rules specified in this section, together with the relevant standards and regulations in force on this subject.

10.1.2. Horizontal load-bearing systems of light steel buildings within the scope of this section shall be constructed in two ways: coated panel systems and braced panel systems. In case the horizontal load-bearing system includes rolled steel elements, **Chapter 9** The rules given in will be followed.

10.1.3. Rules for light steel building foundations **Chapter 16** is given in .

10.2. GENERAL RULES

10.2.1. Classification of Light Steel Building Carrier Systems

Horizontal load-bearing systems of light steel buildings, in terms of their earthquake resistance,

10.2.1.1 And **10.2.1.2** It is divided into two classes defined in .

10.2.1.1 –Light steel buildings where all earthquake effects are resisted by screwed, bolted OSB or plywood wall panels. (**Figure 10.1**). Such buildings *Systems with High Ductility Level* will be taken as.

10.2.1.2 –Braced panels with all earthquake effects ((**Figure 10.2**) or light steel buildings covered with cladding panels containing gypsum boards. Such buildings *Systems with Limited Ductility Level* will be taken as.

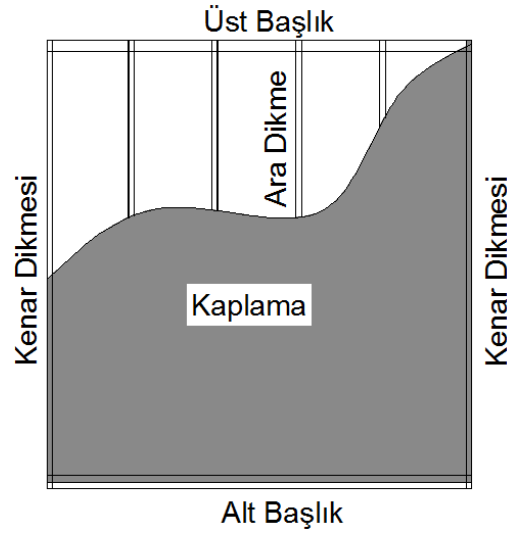


Figure 10.1

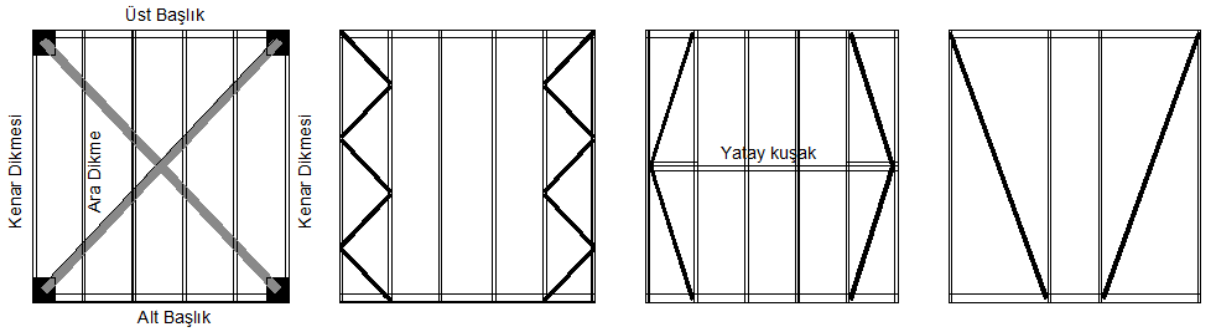


Figure 10.2

10.2.1.3 -The following will be applied in the design of systems falling into these two classes: *Carrier System Behavior Coefficients* (R) And *Excess Strength Coefficients* (D) with permitted *Building Height Classes* (BYS) **Table 4.1** is given in .

10.2.1.4 -The carrier panels of light steel buildings will be placed as regularly as possible in the plan and symmetrically or nearly symmetrically with respect to the main axes. The carrier panels will be arranged on top of each other on all floors.

10.2.2. Relevant Standards

10.2.2.1 -The design of light steel building load-bearing systems constructed with cold-formed steel elements within the scope of this section is defined in this Regulation. **Part 2, Chapter 3** And **Chapter 4** It will be made according to the calculation rules given in and the loads foreseen in the TS 498 Standard, and according to other relevant standards and primarily the rules in this section.

10.2.2.2 -For matters not included in this section and other than earthquake design, internationally accepted standards and regulations can be used.

10.2.3. Conditions Regarding Materials and Assembly Tools

10.2.3.1 –The material of cold-formed steel elements is the construction material obtained by bending steel flat products with wall thicknesses ranging from 0.45 mm to 16 mm in forming machines, as defined by TS EN 10025, TS EN 10346 and TS EN ISO 1461.

10.2.3.2 –The material properties of cold formed steel members shall meet the following minimum conditions.

(a)The minimum yield stress will be 235 MPa.

(b)The ratio of breaking strength to yield stress shall be at least 1.08.

(c)Materials with a minimum elongation rate of 10% *normally ductile material*, materials with a minimum elongation ratio of 16% *high ductile material* They are called.

(d)Normal ductile materials can only be used in purlins, facade girts and non-load-bearing struts.

(to)In materials with a tensile strength of 550 MPa and above, **(b)**And **(c)** It will be checked by tests that the conditions specified in paragraphs are met.

(f)Self-drilling screws and/or bolts shall be used in the connections. These screws shall be tested in accordance with ASTM C1513 or equivalent standards. **Table 10.1** It will provide the minimum torque strength, Rockwell surface hardness, and Rockwell core hardness values given in .

Table 10.1 – Conditions for Self-Drilling Screws

International No.	Screw Diameter (mm)	Minimum Torque Strength (Nm)	Minimum Rockwell Surface Hardness	Minimum Rockwell Core Hardness
6	3.5	2.7	C 50	C 32
8	4.2	4.7		
10	4.8	6.9		
12	5.5	10.4		
¼"	6.3	16.9		

10.2.3.3 –Wood-based panels to be used in horizontal load-bearing systems of light steel buildings will comply with TS EN 12369-1, 2, 3 standards.

10.2.4. Cross-Section Conditions

10.2.1.1And**10.2.1.2**The cross-sectional conditions that the profiles to be used in the creation of the panel systems specified in are given in the following items.

10.2.4.1 –In elements consisting of cold formed profiles *header width/header thickness, body height/body thickness* And *edge reinforcement length/edge reinforcement wall thickness* Limit values for rates **Table 10.2** is given in .

10.2.4.2 – Table 10.2In addition to the conditions given in , the following conditions shall also be complied with to ensure the effective contribution of the edge reinforcements.

$$0.20 \leq c/b \leq 0.60 \quad (10.1a)$$

$$0.10 \leq d/b \leq 0.30 \quad (10.1b)$$

Table 10.2 – Cross-Section Conditions

<i>ENKESİT ÖZELLİKLERİ</i>		<i>Sınır Değerler</i>
		$b/t < 50$
		$b/t < 60$ $c/t < 50$
		$b/t < 90$ $c/t < 60$ $d/t < 50$
		$b/t < 500$
		$h_i/t < 500 \cdot \sin(\phi)$ $45 < \phi < 90$

10.2.4.3 – $c/b < 0.20$ or $d/b < 0.10$, the contribution of the edge reinforcements will be ignored ($c=0$ or $d=0$ will be taken as 0).

10.2.4.4 – $c/b > 0.60$ or $d/b > 0.30$, the effective cross-section properties will be calculated with the equations given in the TS EN 1993-1-3 standard and verified by experiment.

10.2.4.5 – Internal bending radii of profiles $r_p < 5t$ and $r_p < 0.1b$ their effect on the cross-sectional properties, $r_p = 0$ and will be ignored.

10.2.4.6 – Conditions to be followed when opening holes in profiles **Annex 10A** is given in .

10.2.4.7 – The minimum header width of the profiles shall meet the following limit values.

(hungry and/or Σ To ensure that the coatings to be connected to the profiles can be screwed, the head width will be a minimum of 30 mm (**Figure 10.3**).

(This The header width on the profiles will be minimum 20 mm. (**Figure 10.3**).

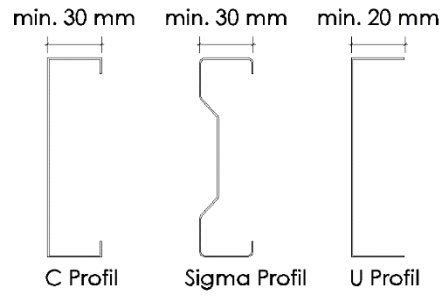


Figure 10.3

10.2.5. Design Method and Load Combinations

10.2.5.1 –Cold-formed steel structural elements and connections must perform all expected functions throughout the operational life of the structure. *security* It shall have sufficient strength, stability and rigidity to perform under the required conditions.

10.2.5.2 –In the design of a light steel building carrier system, only one application provided that, the principles and load combinations are published in the Official Gazette dated 04/02/2016 and numbered 29614. defined in the "Regulation on Design, Calculation and Construction Principles of Steel Structures" published *Design with Safety Factors*(GKT) method or *Design with Load and Resistance Coefficients*(YDKT) method can be used. However, only YDKT method will be used in the design of foundations.

10.2.5.3 –Within the scope of this Regulation, the design rules required for section calculations, joint and joint calculations to be made under the combined effect of vertical loads and earthquake are given separately for both GKT and YDKT methods.

10.2.5.4 –*Design with Safety Factors*(In the GKT method, load combinations to be applied under vertical loads and earthquake effects **Equation (10.2a)** And **Equation (10.2b)** has also been given.

$$G+0.75Q+0.75S+0.75(0.7TO_d) \quad (10.2a)$$

$$0.6G+0.7TO_d+H \quad (10.2b)$$

In these equations, G fixed load, Q moving load, S snow load, H ground horizontal thrust, TO_d whereas

4.4It shows the earthquake effect calculated according to .

10.2.5.5 –*Design with Load and Resistance Coefficients*(Load combinations to be applied under vertical loads and earthquake effects in the YDKT method **4.4.4.2** is given in .

10.2.6. Earthquake Calculation

10.2.6.1 –In earthquake calculation **4.7** Given in *Equivalent Earthquake Load Method* and for the calculation of elastic design spectral acceleration, $T=0.2s$ is taken. **4.7.1** The equivalent earthquake load found according to will be taken as basis. The equivalent earthquake load will be distributed to the carrier system elements depending on the horizontal load carrying capacity of the coating materials or braced panels per unit length and their location in the plan.

10.2.6.2 –To be applied in earthquake analysis and dimensioning *Carrier System Behavior Coefficient*(R) And *Excess Strength Coefficient*(D) **Table 4.1** will be taken from.

10.2.6.3 –For panels made of OSB, plywood, plasterboard and sheet metal cladding types

10.3.1 The conditions specified in will be taken into account.

10.2.6.4 –Shear strength of panels formed using other coating materials, **Annex 10C** According to the principles set out in and **1.4**It will be determined and documented by testing according to .

10.2.7. Height Limits of Light Steel Building Systems

10.2.7.1 –Height limits of light steel building systems **Table 4.1** is given in .

10.2.7.2 –Created with Plasterboard *Coated Panel Systems*'s are not allowed to meet the earthquake demand alone. These systems have the highest earthquake demand meeting rates depending on the number of floors. **Table 10.3** The remaining earthquake demand is given in other horizontal load-bearing systems. **10.2.6.1** It will be met by distributing it with the approach explained in .

**Table 10.3 – Earthquake for Coated Panel Systems Made of Plasterboard
Maximum Demand Meeting Rates**

Floor	Meeting Maximum Earthquake Demand The rates(%)		
	Number of Floors		
	3	2	1
3rd Floor	80	-	-
2nd Floor	60	80	-
1st Floor	40	60	80

10.3. DESIGN PRINCIPLES OF PANELS

A typical slip panel with coating **Figure 10.4** The shear stiffness and strength of the panel are determined by the spacing of the connection elements. The design approach for panels made of cold-formed steel profiles is based on the distribution of the total shear demand at a floor according to the relative bearing capacities of the panels at that floor.

(a) The shear capacity of a panel will be greater than the shear force demand on that panel.

(b) The axial force carrying capacity of the uprights will be greater than the axial forces acting on the uprights.

(c) The design of anchors transferring shear force and foundation and floor connection anchors shall be made to meet the base shear force, floor shear force and overturning moment.

10.3.1. Creation of Coated Panel Systems

Coated Panel Systems The rules to be followed in the creation of are given below.

10.3.1.1 –In order for the coating to be accepted as a carrier, the largest strut spacing is “ c_d ” It will be 625 mm (**Figure 10.4**).

10.3.1.2 –The carrier pillars are at least 0.8 mm thick and have a body height of at least 70 mm. **C-** or **Σ-** –It will consist of cross-sectional elements and the edge reinforcement length of these elements will be at least 9 mm. If a different cross-section is used, it must be shown that the cross-section used can safely carry the design load.

10.3.1.3 –Panel bottom and top header, minimum wall thickness 0.8 mm **U** or **C**-It will consist of cross-sectional elements **(Figure 10.4)** If a different cross-section is used, it must be demonstrated that the selected cross-section can safely carry the design load.

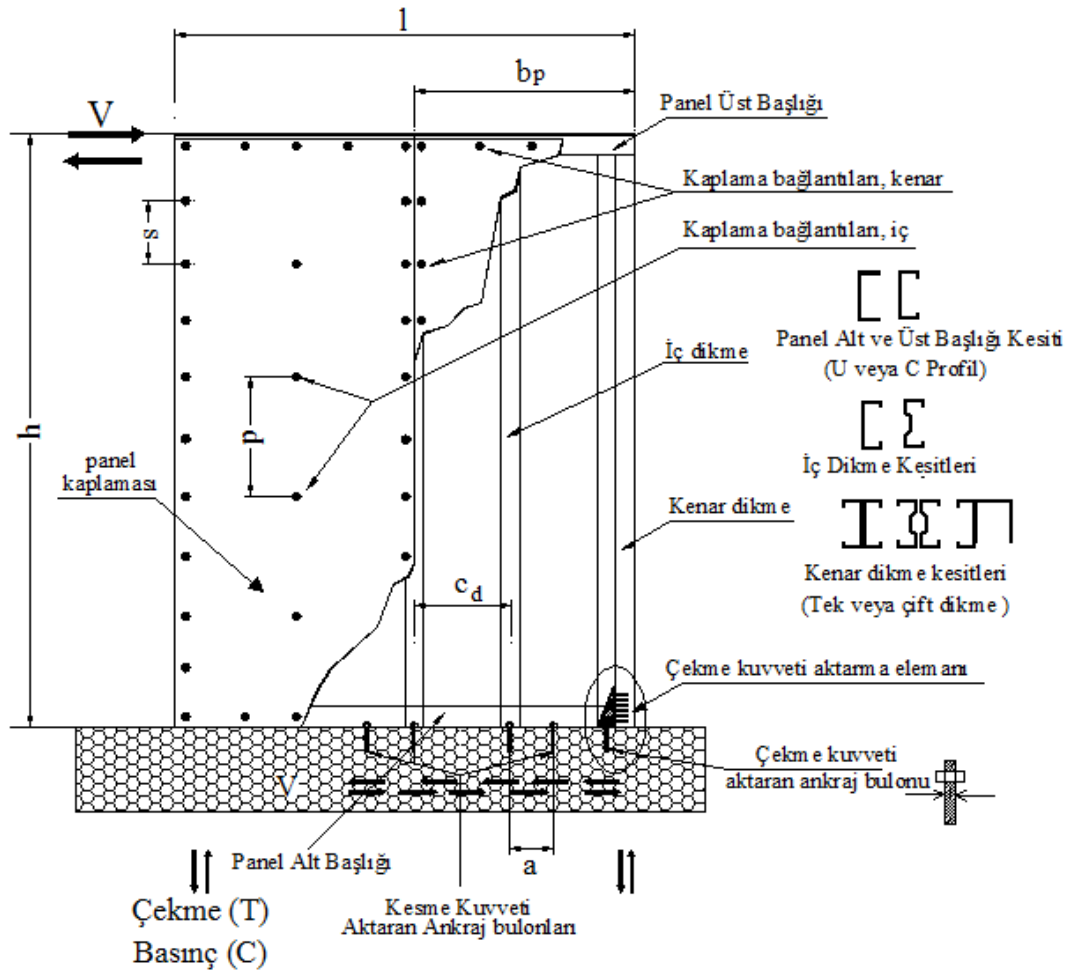


Figure 10.4

10.3.1.4 –The coating will generally be used with the long side in the vertical direction. **(Figure 10.4)** If the coatings are used in horizontal direction, a horizontal belt will be placed inside the panel and the coating will be screwed to this element in a way that it will transfer the shear force. **(Figure 10.5)**.

10.3.1.5 –It is assumed that the carrier panel sections at full floor height have the capacity to withstand shear force. **(Figure 10.6)**. As seen in the figure, for each carrier panel section, tensile force transfer anchors will be used in the side posts. In addition, in case of using coated panels and braced panels side by side, there will be at least one tensile force transfer anchor in the common side posts of the panels. **(Figure 10.6)**.

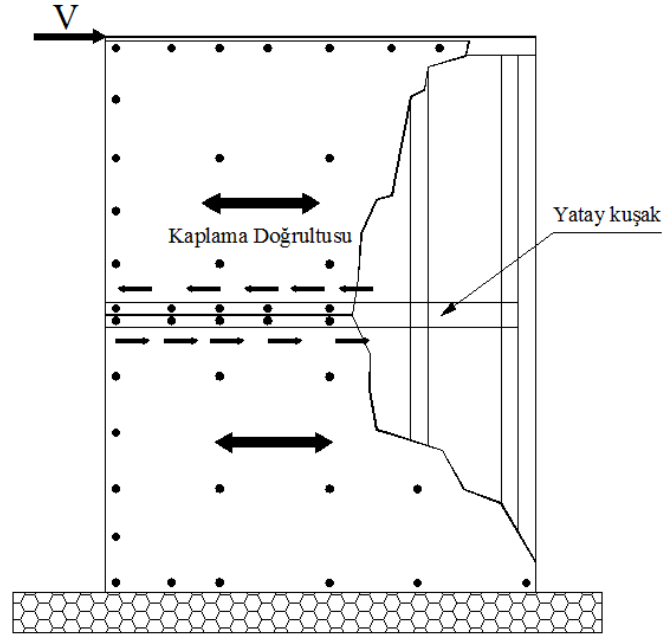


Figure 10.5

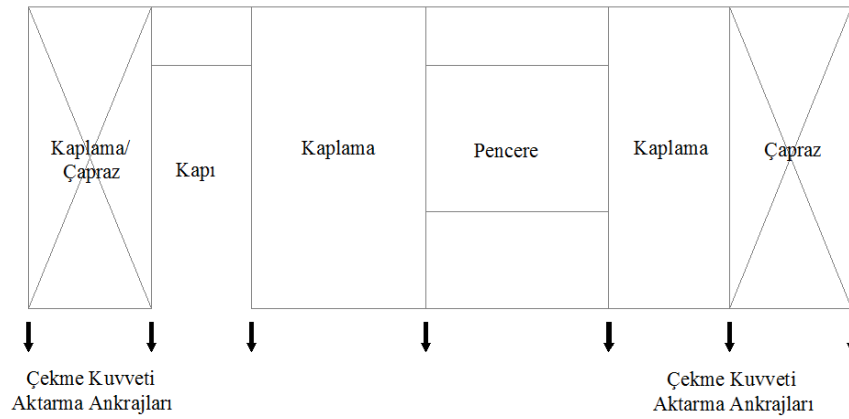


Figure 10.6

10.3.2. Panel Design

10.3.2.1 –The shear force demand on the panel depends on the design method to be selected, **Equation (10.3a)** And **Equation (10.3b)** It will be smaller than the shear force capacity calculated by selecting the appropriate one from:

(a) *Design with Safety Factors (GKT)* for method:

$$V_c = v_c \sum_{I=1}^n \quad (10.3a)$$

(b) *Design with Load and Resistance Factors (YDKT)* for method:

$$V_c = \phi v_c \sum_{I=1}^n \quad (10.3b)$$

Characteristic shear strength per unit length v_c , **Table 10.5** and the relevant coefficients for the earthquake situation will be obtained from **Table 10.4** Taken from **Equation (10.3a)** or **Equation (10.3b)** also

In case wind loads are more effective in the design, for GKT and YDKT methods, respectively, $\Omega = 2.0$ and $\varphi = 0.65$ will be taken.

Table 10.4 – Safety and Strength Factors

Design Method		Earthquake
Design Method with Safety Factors (DFT)	Ω	2.5
Design Method with Load and Resistance Factors (YDKT)	φ	0.6

10.3.2.2 –In design, the width ≥ 30 cm and *height/width rate* $h/l \leq 2$ The contribution of panels meeting condition 4 will be taken into account. **(Figure 10.7).** *Height/Width rate* $2 \leq h/l \leq 4$ Unit length of panels in the range of 4 κ shear strength, **Table 10.5** Values obtained from $2 - 1/h$ will be obtained by multiplying the ratio by .

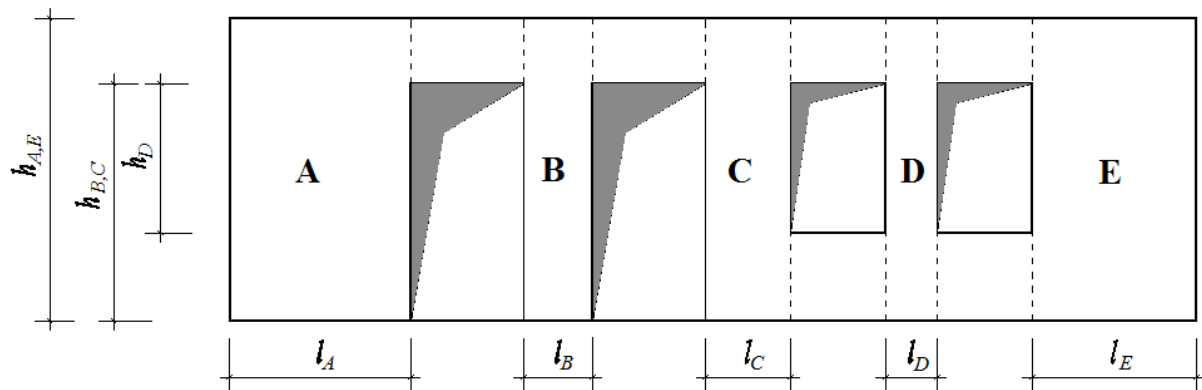


Figure 10.7

10.3.2.3 –Shear strengths of panels per unit length and minimum conditions used to obtain these values **Table 10.5** is given in. **Table 10.5** The values given in are the values corresponding to coating on one side of the panel and the rules stated below will be followed.

(a) If more than one coating is applied to the same face of the panel, only the material in contact with the panel face **Table 10.5** The strength values given in will be used.

(b) If the same coating is applied to both sides of the panel with the same screw placement, **Table 10.5** given for one face in *shear strength values per unit length* are used by adding them together for both faces. In this case, the calculation will show that the panel edge struts and the lower and upper header profiles safely withstand the shear force that occurs.

(c) In case of using different coating materials on both sides of the panel, for the calculation of shear strength per unit length, **Table 10.5** Using the strength values given in , the shear strength of the coating material with a low shear strength will be compared with the shear strength of the coating material with a high shear strength, and the larger value will be taken.

(d) Minimum screw diameters to be used in panel construction **Table 10.5** is given in.

Table 10.5 – Shear Strengths of Panels Per Unit Length(*,)(kN/m)**

Panel The material	h/l	For panel edge mullions/intermediate mullions Covering screw spacing (mm)				Planting minimum meat thickness (mm)	Minimum screw diameter (mm)
		Shear strength per unit length (kN/m)					
12.5 mm Plaster Plate	2:1	200/300	150/300	100/300	100/100		
		2.7	3.1	3.4	-	0.9	3.5
12mm plywood	2:1	150/300	100/300	75/300	50/300		
		11.4	14.4	-	-	0.9	4.2
		13.0	19.4	25.9	32.0	1.1	4.2
11mm OSB	2:1	150/300	100/300	75/300	50/100		
		10.2	13.4	-	-	0.9	4.2
		12.0	18.0	22.6	30.0	1.1	4.2
		18.0	27.0	33.7	45.0	1.4	4.8
		150/300	100/300	75/300	50/300		
0.46 mm sheet metal	2:1	5.7	-	-	-	0.9	4.2
0.68 mm sheet metal	4:1		14.6	15.8	17.1	0.9	4.2

* For panel materials that are included in this table but do not have values for coating screw spacing and for other materials that are not included in this table. **Annex 10.C** The test will be carried out according to and the characteristic strength values will be determined in accordance with the results obtained from the test, **10.3.2.3(a), (b) and (c)** will be taken into consideration in the cases specified in paragraphs.

* * Values in this table **Annex 10.C** It will be used if it is verified by experiments performed according to . If the test results differ from the values in the table, the values obtained from the experiment will be used.

(to) When covering the panels, the connection screws will not be allowed to be buried in the coating and the screws will be advanced into the profile/sheet metal by a minimum of 3 screw steps. In practice, the screw will first enter the coating and then the sheet metal. **(Figure 10.8).**

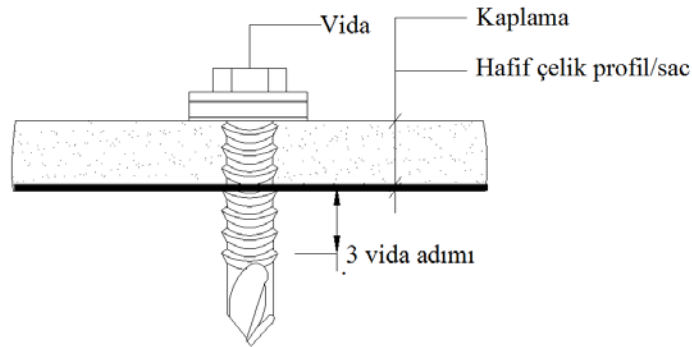


Figure 10.8

10.3.3. Strut Design

In coated panel systems, the axial forces occurring on the side posts due to horizontal loads are as follows: **Equation (10.4)** and **Equation (10.5)** will be calculated with **(Figure 10.9)**.

$$M_t = v_d \cdot I \cdot h \quad (10.4)$$

$$T_t = C_t = \frac{M_t}{-I} \quad (10.5)$$

Under the sum of the axial forces calculated in this way and the axial forces consisting of vertical loads, the strut design will be made as follows.

Axial compressive strength of the strut P_n , will be determined as follows.

Design with Load and Resistance Coefficients (For the YDKT method:

$$P_n = 0.85 A_{\text{effect}} F_n \quad (10.6a)$$

Design with Safety Factors (For the GKT method:

$$P_n = 0.56 A_{\text{effect}} F_n \quad (10.6b)$$

In these expressions A_{effect} The effective area will be determined by using the relevant equations given in TS EN 1993-1-3 Section 5.5.

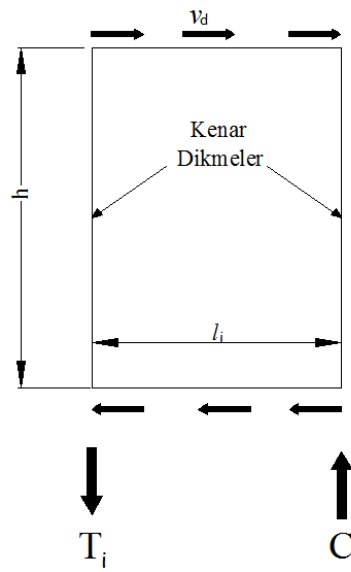


Figure 10.9

10.3.3.1 -Sewing F_n The calculation of characteristic compressive strength is explained below.

(a) Sewing F_n characteristic compressive strength for the bending buckling limit state of the strut

Equation (10.7) or **Equation (10.8)** will be calculated with:

$$F_n = \left(0.658 \lambda_c^2 \right) F_y \quad (\lambda_c \leq 1.5 \text{ for }) \quad (10.7)$$

$$F_n = \left(\frac{-0.877 -}{\lambda_c^2} \right) F_y \quad (\lambda_c > 1.5 \text{ for }) \quad (10.8)$$

Here,

$$\lambda_c = \sqrt{\frac{F_y}{F_{to}}} \quad (10.9)$$

$$F_{to} = \frac{\pi^2 TO}{\left(\frac{KL}{r} \right)^2} \quad (10.10)$$

$$r = I \sqrt{\frac{1}{A}} \quad (10.11)$$

will be determined by the equations.

(b) According to the torsional and torsional-bending buckling limit states in the uprights, F_n Characteristic compressive strength according to TS EN 1993-1-3 Section 6.2 or AISI S100 Section C4 according to will be determined.

10.3.4. Design of Joints

Cold-formed steel elements shall be connected with screws and/or bolts. In the connections, screws or bolts shall be designed under the forces acting on them. The forces to be carried by screws or bolts shall be calculated using different equations for different failure modes. Alternative strength calculations for self-drilling screws and bolts **Annex 10B** is given in .

For each screw *design shear strength* P_s And *design pull-out strength* P_t It will be calculated with the following equations assuming that the screw is in contact with the coating.

(a) *Design with Load and Resistance Coefficients* (For YDKT:

$$P_{ns} = \min(P_{ns1}, P_{ns2}, P_{ns3}) \quad P_s = \varphi(P_{ns}) \quad (10.12a)$$

$$P_{nt} = \min(P_{Nov}, P_{notes}) \quad P_t = \varphi(P_{nt}) \quad (10.12b)$$

$\varphi = 0.5$ (*Strength coefficient*)

(b) *Design with Safety Factors* (For GKT:

$$P_{ns} = \min(P_{ns1}, P_{ns2}, P_{ns3}) \quad P_s = \left(\frac{P_{ns}}{\Omega} \right) \quad (10.13a)$$

$$P_{nt} = \min(P_{Nov}, P_{notes}) \quad P_t = \left(\frac{P_{nt}}{\Omega} \right) \quad (10.13b)$$

$\Omega = 3.0$ (*Safety factor*)

10.3.4.1 - *Characteristic shear strength* P_{ns} will be calculated with the following equations and **(a)**, **(b)**, **(c)** will be determined according to the conditions stated in paragraphs.

$$P_{ns1} = 4.2 t_3 \sqrt{2 d_v F_{u2}} \quad (\text{for side-lying limit state}) \quad (10.14)$$

$$P_{ns2} = 2.7 t_1 d_v F_{u1} \quad (10.15)$$

$$P_{ns3} = 2.7 t_2 d_v F_{u2} \quad (10.16)$$

(a) $t_2/t_1 \leq 1$; In case of 1;

$$P_{ns} = \min(P_{ns1}, P_{ns2}, P_{ns3}) \quad (10.17)$$

(b) $t_2/t_1 \geq 2.5$; In case of 2.5;

$$P_{ns} = \min(P_{ns2}, P_{ns3}) \quad (10.18)$$

(c) $1.0 < t/t_1$ In case < 2.5 , (a) And (b) calculated with paragraphs P_{ns} Linear interpolation will be done between the values.

10.3.4.2 –Characteristic pull-out strength of sheet metal screw head **Equation (10.19)** will be calculated with:

$$P_{Nov} = 1.5 t_1 d_w F_{u1} \quad (10.19)$$

Here d_w , shall not be greater than 13 mm and the thickness of the flake used shall not be less than 1.3 mm.

10.3.4.3 –Characteristic pull-out strength of screw tip from sheet metal **Equation (10.20)** will be calculated with:

$$P_{notes} = 0.85 t_c d_v F_{u2} \quad (10.20)$$

10.3.4.4 –*Coated Panel Systems* The following restrictions shall apply to combinations of :

(a) The screw spacing in the side posts is the maximum s It will be 200 mm.

(b) The screw spacing in the inner uprights is the maximum p It will be 300 mm.

(c) The distance between the screw edges along the edges of the sliding panel shall be at least 10 mm.

(d) The length of the screw to be used in the connection of the sheet metal elements forming the panel should be at least $2 t_{planting} + 10$ mm. In addition, the screws will enter the sheet metal that the screw head does not touch, at least 3 screw steps.

(to) The length of the screw to be used in the connection of the cladding material to the panel should be at least $t_{covering} + 2 t_{planting}$ It will be +10mm. Also, the screws should be driven into the sheet metal where the screw head does not come into contact. at least 3 screw steps will enter **(Figure 10.8)**.

10.3.4.5 –Slip and connection anchors will comply with TS EN ISO 898-1 standard.

10.3.4.6 –Shear anchors will be placed in sufficient numbers to transfer the shear force that occurs.

10.4. CONDITIONS TO BE COMPLIED WITH FOR HORIZONTAL LOAD CARRYING PANELS

Coated Panel Systems And *Braced Panel Systems* The conditions that must be followed in the design of are given below.

10.4.1. Conditions for Coated Panel Systems

The following conditions shall be taken into account in the design of panels covered with sheet metal, OSB, plywood and plasterboard.

10.4.1.1 –Joints will be sized according to the conditions given below.

(a) Panel edge posts, bottom header, top header elements and, if any, belt profiles **Table 4.1** The relevant strength excess coefficient given in D shall be sized to withstand the forces resulting from increased loading conditions.

(b) The pull-out strength of screws cannot be used to resist tensile forces resulting from earthquake loads.

(c) The panels will be connected in areas where there are edge posts, in a way that will safely transfer the resulting tensile forces to the system below with anchors.

(d) In sizing the anchors, **Table 4.1** Given in *D* Internal forces magnified by applying the excess strength coefficient will be used.

10.4.1.2 –The Foundations **Table 4.1** Given in *D* will be dimensioned under internal forces magnified by applying the excess strength coefficient.

10.4.1.3 –Wall thickness of panel struts **Table 10.5** It will not be smaller than the strut wall thicknesses specified in .

10.4.2. Conditions for Braced Panel Systems

10.4.2.1 –The elements and connections of braced panel systems will be sized in accordance with the rules given below.

(a) Panel edge posts, header elements and connection elements to which cross elements are connected, are subject to horizontal loads. $P_{n,design}$ axial forces **Table 4.1** Given in *D* They will be sized to meet the calculated forces by enlarging them by applying the excess strength coefficient.

(b) Axial force calculation strength of cross member $P_{n,cross}$, in case the axial force is tensile,

Design with Safety Factors (For the GKT method:

$$P_{n,cross} = \frac{1}{\Omega_t} A_g F_y \quad (10.21a)$$

$$\Omega_t = 1.67$$

Design with Load and Resistance Coefficients (For the YDKT method:

$$P_{n,cross} = \phi_t A_g F_y \quad (10.21b)$$

$$\phi_t = 0.90$$

It will be calculated as follows.

In case the axial force is pressure $P_{n,cross}$ calculation strength, design to be selected

According to the method **Equation (10.6a)** or **Equation (10.6b)** will be calculated using .

(c) Under the effect of tensile force, the net cross-section calculation strength control will be carried out at the point where the crosses are connected to the side posts and header elements. **Equation (10.23)** condition will be shown to be met.

$$P_{clear,cross} = A_{net} F_u \quad (10.22)$$

$$P_{clear,cross} \geq 1.20 P_{n,cross} \quad (10.23)$$

Equation (10.23) The coefficient of 1.20 in is obtained from the ratio of the net cross-sectional design strength to the safety coefficients (2/1.67) and strength reduction coefficients (0.9/0.75) taken as basis for the cross-sectional design strength.

(d) The pull-out strength of screws is not used in earthquake load calculations.

10.4.3. Additional Special Conditions for Braced Panel Systems with Braced Members in Tension Only

(a) If the strip is not applied crosswise on both sides of the panel, the moment effect that will occur from eccentricity " $T \times t_o$ " will be taken into account in the planning calculations. (Figure 10.10).

(b) The connections of the strip crosses will be made in a tense manner.

(c) The Foundations Table 4.1 Given in D They will be dimensioned under internal forces magnified by applying the strength excess coefficient.

(d) In horizontal load-bearing panel posts, h including panel height h / Horizontal belt will be placed at 2 intervals ((Figure 10.2).

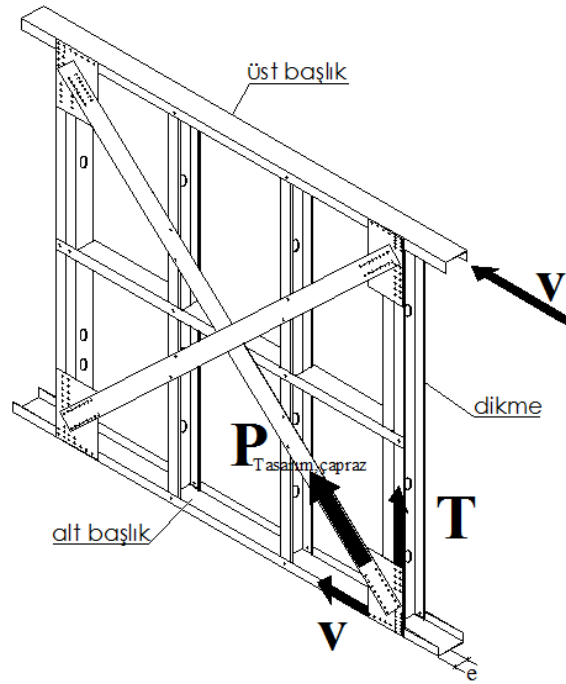


Figure 10.10

10.5. FLOORINGS

Floors will be calculated to safely transfer the forces generated in their own planes due to earthquakes to the horizontal load-bearing (HL) panels. (Figure 10.11). In light steel structures, the floor system can be formed with beams and shaped steel sheet (trapezoidal), wood or cement-based elements. Necessary precautions will be taken to ensure that the floors are rigid enough to safely transfer earthquake forces to the horizontal load-bearing system.

The necessary conditions for the floor covering to fulfill its diaphragm function are given below.

(a) In dimensioning the slabs under earthquake effects, Table 4.1 Given in D excess strength coefficient will be applied.

(b) It will be shown through calculations that diaphragm forces are safely transferred to horizontal load-bearing vertical panels, floor beams and header elements with appropriate connections (screws, bolts).

(c) Longitudinal transfer elements will be used where necessary to carry the head forces resulting from the diaphragm behavior.

(d)There is no need to make a separate calculation for the gaps to be opened in the floors, as long as the gap area does not exceed 3% of the floor area.

(to)In supporting floor beams, the minimum seating distance will be 40 mm.

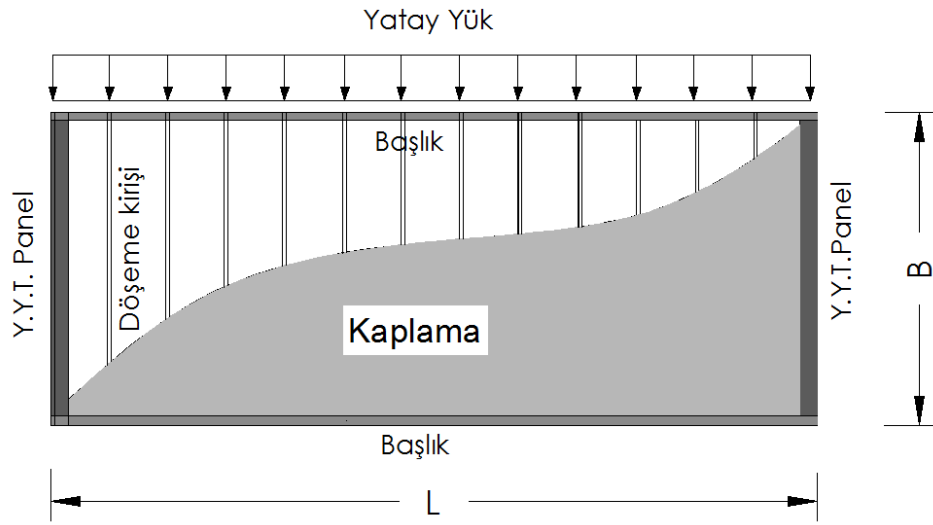


Figure 10.11

ANNEX 10A – RULES FOR HOLES TO BE OPENED IN ELEMENTS

10A.0. ICONS

h	= Profile flat body height =
R_d	Circular hole diameter
r_d	= Inner bend radius of corners in non-circular holes =
t	Profile wall thickness

10A.1. SCOPE AND GENERAL MATTERS

In this section, the conditions to be observed regarding the holes to be opened in the carrier elements for mechanical and electrical installations or other reasons are given.

10A.2. RULES FOR HOLES

When it is necessary to drill holes in the profile bodies, the following rules will be followed and hole losses will be taken into account in the cross-section calculations.

- (a) The holes to be opened in the bodies of the profiles must be centered on the vertical axis of the body.
- (b) Circular hole diameter in the profile, R_d and the height of the flat body profile, h including $R_d/h < 0.70$.
- (c) In case of a hole being opened $h/t < 200$ cross-section condition will be met.
- (d) The distance between the holes will be maximum 600 mm from axis to axis.
- (e) The clear distance between holes will be at least 450 mm.
- (f) The diameter of the circular holes shall be maximum 150 mm.
- (g) Holes will be opened at a clear distance of at least 300 mm from the end of the profiles.
- (h) $R_d/h < 0.38$, the effect of the hole can be neglected in the strength calculations.
- (i) In non-circular holes, the hole height shall be maximum 65 mm and the hole length shall be maximum 115 mm.
- (j) Inner bend radius of corners in non-circular holes $r_d > 2t$ will provide the condition.

ANNEX 10B – STRENGTH FOR DRILL SCREWS AND BOLTS ACCOUNTS

10B.0. ICONS

A_{net}	= Net cross-sectional area
A_s	= Bolt tooth base area
d_b	= Characteristic bolt diameter
d_{he}	= Hole diameter
d_v	= Characteristic screw diameter
d_w	= Larger diameter of washer or screw head
t_{01}	= The connection element screw or bolt hole is at the end of the joined sheet metal or profile. distance (Parallel to the load transfer direction)
t_{02}	= The connection element screw or bolt hole is aligned with the edge of the joined sheet metal or profile. distance (perpendicular to the load transfer direction)
$F_{b,Rd}$	= Crushing strength
$F_{n,Rd}$	= Net cross-sectional strength
$F_{o,Rd}$	= Resistance to pulling out the screw tip from the sheet metal =
$F_{p,Rd}$	Resistance to pulling out the screw head from the sheet metal
$F_{t,Rd}$	= Screw tensile strength determined by tests
$F_{U,sup}$	= Tensile strength of the profile or sheet metal to which the screw is attached
F_u	= Minimum breaking strength of sheet metal
F_{Ub}	= Bolt shear strength =
$F_{v,Rd}$	Screw shear strength
$F_{v,Rk}$	= Characteristic screw shear strength determined by the test result
s_v	= Screw pitch
ρ_1	= The distance between the screws or bolt holes of the connection element and the axis to axis distance (Parallel to the load transfer direction)
ρ_2	= Distance of the connection element screws or bolt holes from axis to axis (Perpendicular to the load transfer direction)
t_i	= Thickness of the thin sheet metal at the joint
t_k	= Thickness of the thickest sheet metal in the joint
t_{sup}	= The wall thickness of the profile or sheet metal to which the screw is attached
α	= Variable coefficient used for crushing strength

10B.1. SCOPE

10.B.1.1 –In this section, the equations to be used for the strength calculations of self-drilling screws and the application conditions of these equations are given.

A strength reduction coefficient of 1/1.25 was used in these equations.

(a) In cases where the screws are under the influence of shear force,

Crushing strength Equation (10B.1) will be calculated with .

$$F_{b,Rd} = 0.80 \alpha F_u d_v t \quad (10B.1)$$

In this equation, the coefficient α will be determined as follows.

$t = t_k$ be or $t_k \geq 2.5 t$ And t In case of $< 1.0 \text{ mm}$

$$\alpha = 3.2 \sqrt{\frac{t \leq 2.1}{d_v}} \quad (10B.2a)$$

$t \geq 2.5 t$ And $t \geq$ In case of 1.0 mm

$$\alpha = 2.1 \quad (10B.2b)$$

$t < t_k < 2.5 t$ In this case, α will be obtained by linear interpolation.

Net cross-section strength Equation (10B.3) will be calculated with .

$$F_{n,Rd} = 0.80 A_{net} F_u \quad (10B.3)$$

Screw shear strength:

The shear strength of the screws shall be determined by testing and shall also meet the following conditions.

$$F_{v,Rd} = 0.80 F_{v,Rk} \quad (10B.4)$$

$$F_{v,Rd} \geq 1.20 F_{b,Rd} \text{ or } \sum F_{v,Rd} \geq 1.20 F_{n,Rd} \quad (10B.5)$$

(b) In cases where the screws are under the influence of tensile force,

Resistance to pulling out the screw tip from the sheet metal:

$$t_{sup}/s_v < 1.0 \quad \text{for} \quad F_{o,Rd} = 0.36 d_v t_{sup} F_{u,sup} \quad (10B.6)$$

$$t_{sup}/s_v \geq 1.0 \quad \text{for} \quad F_{o,Rd} = 0.52 d_v t_{sup} F_{u,sup} \quad (10B.7)$$

Resistance to pull out screw head from sheet metal:

Under static loads,

$$F_{p,Rd} = 0.80 d_w t F_u \quad (10B.8)$$

Under the influence of the earthquake,

$$F_{p,Rd} = 0.40 d_w t F_u \quad (10B.9)$$

The tensile strength of the screws shall be determined by testing and shall also meet the following conditions.

$$F_{t,Rd} \geq F_{o,Rd} \quad \text{or} \quad \sum F_{p,Rd} \leq F_{t,Rd} \quad (10B.10)$$

(c) (a)And(b)All equations given in paragraphs will also satisfy the following conditions.

$$t_{o1} \geq 3d_v$$

$$p_1 \geq 3d_v$$

$$3\text{mm} \leq d_v \leq 8\text{mm}$$

$$t_{o2} \geq 1.5d_v$$

$$p_2 \geq 3d_v$$

Under tensile effect: $0.5\text{mm} \leq t \leq 1.5\text{mm}$ and $t_k \geq 0.9\text{mm}$

$$F_u \leq 550\text{MPa}$$

(a)And(b)In the equations given in paragraphs, it is assumed that the screw heads are in contact with the thin sheet metal. In addition, when calculating the pull-out strength from the screw head, it is assumed that if a washer head screw is used, the screw head will provide the necessary strength.

10.B.1.2 -In this section, the equations that can be used for the strength calculations of bolts and the application conditions of these equations are given.

In these equations, a strength reduction coefficient of 1/1.25 was used.

(a)In cases where bolts are under the influence of shear force: *Crushing strengthEquivalent(10B.11)* will be calculated with .

$$F_{b,Rd} = 2\alpha_b k_t F_u d t \quad (10B.11)$$

α_b for the value of, $t_{o1}/3d_b$ or 1.0, whichever is smaller, will be used.

$0.75\text{mm} < t$ In case of $<1.25\text{ mm}$

$$k_t = (0.8t + 1.5)/2.5 \quad (10B.12)$$

t In case of $>1.25\text{ mm}$

$$k_t = 1 \quad (10B.13)$$

Net Cross Section Strength:

$$F_{n,Rd} = 0.80 \left[1 + 3r(d_{he}/u - 0.3) \right] A_{net} F_u \quad (10B.14)$$

$$F_{n,Rd} \leq 0.80 A_{net} F_u \quad (10B.15)$$

$r = [\text{Number of bolts in the cross section}] / [\text{Total number of bolts in the joint}]$ u

$= 2t_{o2}$ And $u \leq p_2$ **(Figure 10B.1)**

Bolt shear strength **Equation (10B.16)** or **Equation (10B.17)** will be calculated with.

For bolt quality 4.6, 5.6 and 8.8 **Equation (10B.16)** is valid.

$$F_{v,Rd} = 0.48 F_{ub} A_s \quad (10B.16)$$

For bolt quality 4.8, 5.8 and 10.9 **Equation (10B.17)** And **Equation (10B.18)** is valid.

$$F_{V,Rd} = 0.40 F_{Ub} A_s \quad (10B.17)$$

$$F_{V,Rd} \geq 1.20 F_{b,Rd} \text{ or } \sum F_{V,Rd} \geq 1.20 F_{n,Rd} \quad (10B.18)$$

(b) In bolts under tensile force *pull-out strength* not calculated. *Bolt tensile strength*:

$$F_{p,Rd} = 0.72 F_{Ub} A_s \quad (10B.19)$$

$$\sum F_{p,Rd} \leq F_{t,Rd} \quad (10B.20)$$

(c) (a) And (b) The following conditions shall also be met for all equations and bolts in the paragraphs: **(Figure 10B.1)**.

$$t_{o1} \geq 1.0 d_0$$

$$p_1 \geq 3 d_0$$

$$0.75 \text{ mm} \leq t < 3 \text{ mm}$$

$$t_{o2} \geq 1.5 d_0$$

$$p_2 \geq 3 d_0$$

$$F_u \leq 550 \text{ MPa}$$

Minimum bolt size will be M6 and bolt quality will be in the range of 4.6 - 10.9.

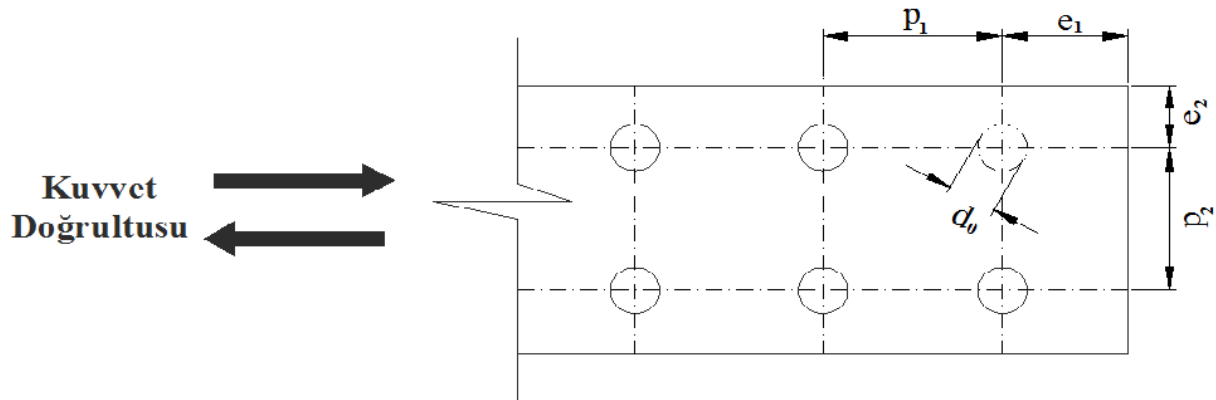


Figure 10B.1

ANNEX 10C – PANEL TESTING PRINCIPLES

10C.0. ICONS

D_{Max} = Maximum displacement

10C.1. SCOPE AND GENERAL MATTERS

10.C.1.1 –In this section, the determination of the strength of panel systems including shear and tensile anchors is discussed.**1.4**The principles of the experiments to be conducted according to are included.

10.C.1.2 –The strength of the panels shall be determined by displacement controlled tests and by averaging the results of at least three panel tests.*height / width*the ratio will be 2/1.

10.C.1.3 –The Equal Energy Elastic-Plastic (EEEP) rule will be applied. According to this rule, the areas under the envelope curve and the idealized curve are equal to each other.

10.C.1.4 –How to conduct the experiment is explained below, and the experimental steps and cycle numbers to be applied during each experiment are given.**Table 10.C.1**is given in .

Table 10.C.1- Experimental Steps and Cycle Numbers

My name	Number of Cycles	Initial Cycle Size(% D_{Max})
1	6	5
2	7	7.5
3	7	10
4	4	20
5	4	30
6	3	40
7	3	70
8	3	100
9	3	100+50
10	3	Additional 50 increments until model collapse

(a)In each step, the size of the cycles after the first cycle will be taken as 75% of the size of the first cycle. D_{Max} is the maximum expected displacement and this value is in the first cycle

It is obtained at the point where the applied force does not fall below 80% of the maximum force obtained during the test.

(b) D_{Max} Its value cannot also be greater than 2.5% of the panel height.

(c)Loading will be at constant frequency (in the range of 0.2-0.5) or speed (1.0-63.5 mm/s).

(d)The collapse of the model is considered to be the point where the ratio of the applied force to the maximum force is 0.8.

10.C.1.5 –The test result document includes the unit shear strength of the panel, as well as the material and geometric properties of the panel (profile type, dimensions, wall thickness, sheet material properties and coating).

thickness and properties) and coating application properties (screw type, screw strengths, screw spacing) shall be specified.

CHAPTER 11 – MASONRY BUILDING CARRIER UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGNING SYSTEMS

11.0. ICONS

A	= Horizontal cross-sectional area of solid wall piece
A_{si}	= Vertical reinforcement area in reinforced masonry and reinforced panel walls
A_{sw}	= Total shear reinforcement area in the wall
b	= Shape coefficient related to the distribution of shear stresses in the horizontal section of the wall =
EO_{wall}	Wall modulus of elasticity
to_{Ed}	= In-plane direction of the axial force acting on the calculated wall section eccentricity (M_{Ed}/N_{Ed})
f_b	= Standardized (size-adjusted) masonry unit 100 mm × 100 Average compressive strength (equivalent to a sample of mm size)
$f_{b,min}$	= Minimum compressive strength of the masonry unit in the direction perpendicular to the horizontal joints
$f_{bh,min}$	= Minimum compressive strength of the masonry unit in the direction parallel to the horizontal joints
f_d	= Masonry wall design compressive strength = Masonry
f_k	wall characteristic compressive strength = Mortar
f_m	average compressive strength (28 days) = Mortar
$f_{m,min}$	minimum compressive strength (28 days)
f_{Vko}	= Characteristic shear strength in the absence of axial stress
f_{Vdo}	= Characteristic shear strength in the absence of axial stress; γ_m value divided by the bulk material coefficient
$f_{etc.}$	= Wall design obtained using average vertical stresses on the wall shear strength
f_{Vk}	= Wall stresses obtained using average vertical stresses on the wall characteristic shear strength
f_{yd}	= Reinforcement design yield stress
G_{wall}	= Wall shear modulus
H	= Wall free height, from the upper floor level to the lower floor level (if any) length up to
h_{ef}	= Effective height of the wall
h_k	= Floor height
I	= Moment of inertia of solid wall part =
k_{wall}	Wall stiffness
-	= Wall section length

$-c$	= Length of the pressure-bearing part of the wall
M_{Ed}	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Bending moment calculated under
M_{Rd}	= γ_m and γ_s Bearing capacity moment calculated using material coefficients
N_{Ed}	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Calculated axial force under
N_{Rd}	= γ_m Axial force capacity calculated using material coefficient = Carrier
R	system behavior coefficient
$R_a(T_1)$	= Earthquake load reduction coefficient
$S(T_1)$	= Spectrum coefficient
t	= Wall thickness
t_{ef}	= Effective thickness of the wall
V_{Ed}	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Calculated wall shear force under
V_{Rd}	= Wall design shear strength
V_{Rd1}	= Contains vertical reinforcement but neglects the contribution of shear reinforcement Shear strength of reinforced masonry walls =
α	Masonry unit hole ratio
ϵ_{sit}	= Stacked boundary unit shortening
γ_m	= Masonry material strength reduction coefficient =
γ_s	Reinforcement steel yield stress reduction coefficient =
λ	Slenderness related capacity reduction coefficient
σ_d	= Joint effect of vertical loads and earthquake loads multiplied by load coefficients Calculated vertical compressive stress under

11.1. SCOPE AND DEFINITIONS

11.1.1 –This Section covers the rules for earthquake resistant design and construction of unreinforced, braced and reinforced masonry buildings and reinforced panel system buildings to be constructed in earthquake zones.

11.1.2 –Types of masonry buildings are given below:

(a) Unreinforced masonry building is a type of building with limited ductility level, constructed using only masonry units and mortar without using any reinforcement for load-bearing walls.

(b) Reinforced masonry buildings are buildings with high ductility levels obtained by placing reinforcement in load-bearing walls constructed using masonry units and mortar in accordance with the rules given in this Section.

(c) Enclosed masonry building, reinforced concrete-bonded load-bearing walls to each other and to the floor

Buildings with limited ductility level constructed by constructing horizontal and vertical bond beams prepared after they are laid and using them as formwork, in accordance with the rules given in this Section,

(d) The reinforced panel system building includes buildings with high ductility level, where the reinforcements in the grooves between the prefabricated reinforced aerated concrete panels are connected to the foundation and reinforced concrete beams at the floor levels, and the vertical aerated concrete panels are brought side by side to form the load-bearing walls, and the reinforced aerated concrete panels are connected to the reinforced concrete beams to form the floors.

11.1.3–Maximum number of floors allowed for masonry building types **Table 4.1** In addition to these floors, the attic area cannot be larger than 25% of the gross building area on the foundation. In addition, a single basement floor can be built.

11.1.4–Effective thickness of single-layer, double-layer and sandwich walls (t_{ef}), TS EN 1996-1-1 standard, **5.5.1.3** Will be determined according to the wall thickness (t) plaster thickness will not be taken into account when determining.

11.1.5–Effective height of the wall (h_{ef}), TS EN 1996-1-1 standard, **5.5.1.2** It will be determined according to.

11.2. MATERIALS AND DURABILITY

11.2.1 –In load-bearing masonry walls, all joints, both horizontal and vertical, will be filled with binding mortar. In buildings with reinforced panel systems, reinforcement extending in the longitudinal direction of the panel will be placed in the grooves between the panels and the grooves will be filled with binding mortar.

11.2.2 –Masonry load-bearing walls will be constructed with brick masonry units (TS EN 771-1), dense or lightweight aggregate concrete masonry units (TS EN 771-3), aerated concrete masonry units (TS EN 771-4), natural stone units (TS EN 771-6) or artificial stone units (TS EN 771-5). Masonry units are divided into two groups according to the type of material used and the hole ratio (**Table 11.1** The voids in the units where the reinforcement will be placed and completely filled with mortar will not be taken into account in the hole ratio calculation. However, this hole ratio that is not taken into account will not exceed 15%.

Table 11.1 - Grouping of Masonry Units According to Hole Ratios

Masonry Unit Type(*)	Group I	Group II
Brick	$\alpha \leq 35$	$35\% < \alpha \leq 50$
Concrete	$\alpha \leq 35$	$35\% < \alpha \leq 50$

11.2.3 –The minimum standardized compressive strength of the masonry units to be determined according to TS EN 772-1 is perpendicular to the horizontal joints. $f_{b,min} = 5.0$ MPa to horizontal joints in parallel direction $f_{bh,min}$ shall not be less than 2.0 MPa.

11.2.4 –Hollow concrete blocks, other bricks produced for infill walls, adobe, rubble stone, pumice stone and similar shaped blocks will never be used as load-bearing wall materials.

11.2.5 –The design and production of panels to be used in buildings with reinforced panel systems shall be in accordance with TS EN 12602. Aerated concrete quality lower than Aerated Concrete 5 class shall not be used in wall and floor panels. Reinforcements to be placed in the grooves between adjacent panels shall be of S420, B420C or B500C class and the reinforcement diameter shall not be less than 12 mm. The groove diameter shall not be less than 5 times the reinforcement diameter.

11.2.6 –The minimum cube compressive strength values of the mortar to be used shall be determined according to TS EN 1015-11 for unreinforced and enclosed masonry. $f_{m,min}$ = 5.0 MPa for reinforced masonry

$f_{m,min}$ = Not less than 10.0 MPa. For reinforced panel system, the distance between panels

The compressive strength of the cement grout to be filled into the grooves shall not be lower than 10.0 MPa.

11.2.7 –Characteristic compressive strength of masonry load-bearing walls, f_k , can be determined in two ways:

i) By means of tests to be carried out on wall samples according to TS EN 1052-1, ii) Compressive strengths obtained from separate tests to be carried out on the masonry unit and mortar according to TS EN 772-1 and TS EN 1015-11 standards, respectively. **(Table 11.2).**

Table 11.2 – Characteristic Compressive Strength of Masonry Bearing Walls, f_k (MPa)

Unit- Unit Class	Mortar class	Mortar pressure strength, f_m (MPa)*	Masonry unit compressive strength, f_b (MPa)					
			5	10	15	20	25	30
Group I	General purposeful mortar	M10-M20	3.4-4.2	5.5-6.8	7.3-9.0	8.9-11.0	10.4-12.9	11.9-14.6
		M2.5-M9	2.2-3.3	3.6-5.3	4.8-7.1	5.9-8.7	6.9-10.1	7.8-11.5
		M1-M2	1.7-2.1	2.8-3.4	3.7-4.5	4.5-5.5	5.2-6.4	5.9-7.3
Group II and Cut stone		M10-M20	2.8-3.4	4.5-5.5	6.0-7.4	7.3-9.0	8.5-10.5	9.7-12.0
		M2.5-M9	1.8-2.7	3.0-4.4	3.9-5.8	4.8-7.1	5.6-8.3	6.4-9.4
		M1-M2	1.4-1.7	2.3-2.8	3.0-3.7	3.7-4.5	4.3-5.3	4.9-6.0
Brick (Group I)	Thin stratified mortar**		2.9	5.3	7.5	9.6	11.6	13.5
Brick (Group II)			2.2	3.5	4.7	5.7	6.7	7.6
Concrete (Group I), Aerated concrete			3.1	5.7	8.0	10.2	12.3	14.4
Concrete (Group II)			2.6	4.6	6.5	8.3	10.0	11.7

(*) Mortars are named in such a way that the number following the letter M indicates their characteristic compressive strength in MPa.

(**) Thin-layered mortar is a mortar with an inter-unit mortar layer of 0.5 mm to 3.0 mm.

11.2.8 – Table 11.2' In order to use the values given in the coefficient of variation of the tests to be carried out on the masonry unit and mortar will not exceed 25%. Double layer

on the walls **Table 11.2** The values given in will be multiplied by the coefficient of 0.8. **Table 11.2** Intermediate values not given in will be found by linear proportioning.

11.2.9 –Characteristic shear strength of the wall f_{vk} , from tests to be performed on wall samples or **Equation (11.1)** will be obtained with

$$f_{vk} = f_{vko} + 0.4 \sigma_d \leq 0.10 f_b \quad (11.1)$$

11.2.10 –Characteristic initial shear strength, f_{vko} , with tests to be carried out in accordance with TS EN 1052-3 or TS EN 1052-4 standards or **Table 11.3** It will be determined according to.

Table 11.3 – Initial Shear Strengths of Walls, f_{vko} (MPa)

Masonry unit	General purpose mortar ^(*)		Thin layer mortar
Brick	M10-M20	0.30	0.30
	M2.5-M9	0.20	
	M1-M2	0.10	
Concrete	M10-M20	0.20	0.30
Aerated concrete	M2.5-M9	0.15	0.30
Natural or Artificial Stone	M1-M2	0.10	Unavailable

^(*)Mortars are named in such a way that the number following the letter M indicates their characteristic compressive strength in MPa.

11.2.11 –While determining the design strength of the walls, γ in the masonry material m and γ in the reinforcements Material coefficients (strength reduction coefficients) will be used. γ_s coefficient will be accepted as 1.15, the bulk material coefficient is γ_m It will be accepted as 1.75 for aerated concrete material and 2.0 for other materials.

11.2.12 –The concrete class in reinforced concrete components of unreinforced masonry buildings, reinforced masonry buildings, enclosed masonry buildings and buildings with reinforced panel systems shall be at least C25.

11.2.13 –Elasticity modulus of load-bearing walls, EO_{wall} , can be determined by tests to be carried out in accordance with TS EN 1052-1. In cases where these tests are not carried out EO_{wall} value 750 for structural analysis f_k The elasticity modulus of the walls constructed with reinforced panels is 450 f_k will be taken as. Wall shear modulus, G_{wall} will be taken as 40% of the modulus of elasticity.

11.3. EARTHQUAKE CALCULATION

11.3.1 –Calculation of earthquake loads **Chapter 4** It will be done according to.

11.3.2 –The building load-bearing system will be modeled and structural analysis will be performed under the combined effects of vertical and horizontal loads. Structural analysis can be done using either finite element or equivalent bar methods.

11.3.3 –In the analysis to be made using the finite element method, the load-bearing wall can be modeled using detailed micro modeling, simplified micro modeling or macro modeling techniques. In the detailed micro modeling technique, the masonry wall components (masonry unit, horizontal and vertical mortar joints) are considered separately. In the simplified micro modeling technique, the horizontal and vertical mortar joints are neglected and the expanded masonry units are separated from each other by average interface lines. In the macro modeling technique, the masonry wall is considered as a composite material.

11.3.4 –In the analysis to be made using the equivalent bar method, the rigidity of the masonry wall will be calculated by taking into account the shear and bending deformations. The free height of the wall, H , is taken as the length from the upper floor level to the lower floor level (if any) of the beam. The wall length is taken as the length of the wall part remaining between the gaps. For a rectangular section wall part, the elastic rigidity is calculated by assuming that both ends are fixed. **Equation (11.2)** will be calculated with .

$$k_{wall} = \frac{1}{\frac{H^3}{12 T_{Owall} I} + \frac{H}{1.2 G_{wall} A}} \quad (11.2)$$

T_{Owall} and G_{wall} , **11.2.13** It will be calculated according to . A is the horizontal cross-sectional area of the solid wall segment and I represents the moment of inertia of the solid wall segment.

11.3.5 –In unreinforced masonry buildings, reinforced masonry buildings, enclosed masonry buildings and buildings consisting of reinforced panels, the effective stiffness of the load-bearing walls shall be determined by reducing the gross stiffness by 50%. The relative story drift ratio resulting from the displacements calculated using these effective stiffnesses shall be R/I . It will be ensured that the multiplication of the coefficient is below 0.007.

11.3.6 –Redistribution on each load-bearing wall will be allowed provided that the shear force on the wall does not decrease by more than 25% and does not increase by more than 33%.

11.3.7 –In unreinforced masonry buildings, reinforced masonry buildings and enclosed masonry buildings, a reinforced concrete floor of at least 100 mm thickness shall be constructed to provide the rigid diaphragm effect. This floor shall be supported by horizontal bond beams with a section height of at least 300 mm and longitudinal reinforcement of 6 ϕ 12 and transverse reinforcement of ϕ 8/150 mm. The width of the horizontal bond beams shall be at least as thick as the wall. The vertical spacing of the horizontal bond beams shall not exceed 4 m.

11.3.8 –In buildings consisting of reinforced panels, if the floors are again formed with reinforced panels, arrangements will be made to ensure that the panels work as rigid diaphragms in both directions.

11.3.9 –In enclosed masonry buildings, reinforced concrete vertical and horizontal bond beams will not be taken into account in structural modelling.

11.3.10 –Walls whose wall section length is less than twice the wall thickness will not be considered as load-bearing in the calculations.

11.4. CALCULATION ACCORDING TO THE BEARING CAPACITY METHOD

11.4.1 –Design force acting vertically on the masonry wall, N_{Ed} , the wall vertical load design resistance N_{Rd} It shall not be greater than. Single layer masonry load bearing wall vertical load design strength of unit length **Equation (11.3)** Masonry wall design compressive strength is given in f_d , **Equation (11.4)** will be determined by.

$$N_{Rd} = \lambda A m n e s t y_d \quad (11.3)$$

$$f_d = f_k / \gamma_m \quad (11.4)$$

11.4.2 –Depending on the delicate λ The capacity reduction coefficient is the ratio of the slenderness (h_{ef}/t_{tot}) from the value of 6 If it is smaller, it will be taken as 1.0, if it is between 6 and 10, it will be taken as 0.8 and if it is between 10 and 15, it will be taken as 0.7. The slenderness ratio is allowed to be greater than 15. will not be given. Strength reduction factor for masonry wall γ_m , **11.2.11** It will be determined according to.

11.4.3 –Load-bearing masonry wall design shear force resistance in unreinforced masonry walls V_{Rd} , the design shear force acting on the wall, V_{Ed} It will be greater than the wall shear strength V_{Rd} , **Equation (11.5)** will be taken as the smaller of the values calculated with . In this equation f_b The value shall not be taken greater than 25 MPa. b The coefficient is the ratio of the wall height to the wall length and will not be taken less than 1 or greater than 1.5.

$$V_{Rd} = f_{etc} \cdot t \cdot c \quad (11.5a)$$

$$V_{Rd} = -t \frac{1.5 f_{do} 1}{b} \sqrt{\frac{N_{Ed}}{1.5 - t f_{do}}} \quad (11.5b)$$

11.4.4 –During the design of reinforced masonry walls and panels subjected to in-plane bending and vertical loads, the tensile strength of the masonry and aerated concrete material will be neglected. Deformation and stress distribution in the section under the compressive effects of reinforced masonry walls **Figure 11.1'** also taking into account as given M_{Rd} Design strength moment will be calculated. Reinforced

The moment capacity of the wall formed with gas concrete panels will be taken as the sum of the bending moment capacities of the panels forming the wall, which will be calculated separately. While calculating the moment capacity of each panel, the axial load acting on the panel and the contributions of the vertical groove reinforcement remaining in the tension region of the panel will be taken into account, and the groove not remaining between the two panels will be taken into account.

The contributions of the reinforcement will be neglected. Masonry wall design moment of resistance M_{Rd} ,

the design moment acting on the wall, M_{Ed} It will be greater than . Under the influence of pressure

Limiting pressure deformation for non-concrete sections, in case the masonry unit is Group I $\epsilon_{is} it = 0.0035$ will not be taken greater than . For other cases $\epsilon_{is} it =$ more than 0.002

It should not be large. It should be shown through calculations that there will be no pressure failure in the wall before the reinforcement in the load-bearing walls yields.

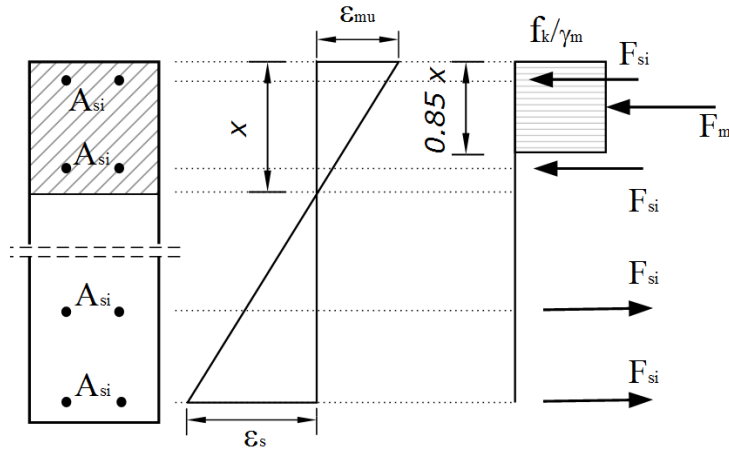


Figure 11.1

11.4.5 –Load-bearing masonry wall design shear force resistance in the calculation of reinforced masonry walls subjected to in-plane shear effect V_{Rd} , design shear force acting on the wall V_{Ed} from Shear strength of reinforced masonry walls containing vertical reinforcement but neglecting the contribution of shear reinforcement V_{Rd1} 's account, **11.4.3**'also

In reinforced masonry walls containing vertical reinforcement and taking into account the contribution of shear reinforcement, **Equation (11.6)** In order to take into account the contribution of the shear reinforcement, the continuity of the horizontal reinforcement along the wall has been ensured.

The contribution of the shear reinforcement should be, V_{Rd2} , **Equation (11.7)** In cases where the contribution of the shear reinforcement is taken into account, **Equation (11.8)** will be provided.

$$V_{Rd1} + V_{Rd2} \geq V_{Ed} \quad (11.6)$$

$$V_{Rd2} = 0.9 A_{sw} f_{yd} \quad (11.7)$$

$$(V_{Ed}) / (t) \leq 2.0 \text{ MPa} \quad (11.8)$$

11.4.6 –Wall shear strength in buildings constructed with reinforced panels **Equation (11.9)**, **Equation (11.10)** And **Equation (11.11)** It will be taken as the smallest of the values obtained from .

$$V_{Rd} = 0.15 f_d)^{0.5-t} \quad (11.9)$$

$$V_{Rd} = N_{Ed} + 0.5 A_{si} f_{yd} \quad (11.10)$$

$$V_{Rd} = 0.2 f_d - t \quad (11.11)$$

11.4.7–For unreinforced masonry buildings **11.4.3** For reinforced masonry buildings calculated according to, **Equation (11.6)** And **Equation (11.7)** for buildings constructed with reinforced panels, **Equation (11.9)**, **Equation (11.10)** And **Equation (11.11)** The adequacy of the shear strength obtained with **Chapter 4** Given in D will be controlled by taking into account the excess strength coefficient.

11.4.8–In confined masonry buildings, the calculation of elements subject to vertical loads and/or bending shall be made according to the calculation principles given for reinforced masonry walls. In the part of the wall section subject to pressure, only the masonry strength shall be taken into account, and the contribution of the reinforcement and concrete in the pressure zone shall be neglected. The vertical reinforcements in the vertical bond beams remaining in tension shall be taken into account as reinforcement.

11.4.9–The shear strength of the encircled masonry walls shall be obtained by adding the shear strengths of the wall and the vertical encircling beams. The calculation of the masonry shear strength shall be made by adopting the approach given for unreinforced masonry. When calculating the shear strength of the beams, only the concrete contribution shall be taken into account and the axial force on the beams shall be neglected.

11.5. OTHER DESIGN CRITERIA

11.5.1 –Walls to be considered as load-bearing elements **Table 11.4** They will meet the conditions given in. Walls that do not meet these conditions will not be considered as load-bearing elements.

Table 11.4 – Geometrical Design to be Applied to Masonry Walls Under Shear Force Effect

Conditions		
Stacking Type	$(t_{ef})_{min}$ (mm)	$(h_{ef}/t_{ef})_{Max}$
Unreinforced masonry, natural or artificial cut stone	350	9
Unreinforced masonry, with other masonry units	240	12
Besieged stockade	240	15
Reinforced masonry	240	15
Reinforced panel systems	200	15

11.5.2 –Maximum unsupported lengths of load-bearing walls and distances between vertical beams in unreinforced and girded masonry buildings, **Figure 11.2** It will comply with the conditions given in. In reinforced masonry and reinforced panel system buildings **Figure 11.2** The dimension limits given in can be increased by 20%. Verifications of wall elements that do not comply with these conditions in the direction perpendicular to the plane will be made according to the conditions given in TS EN 1996-1-1.

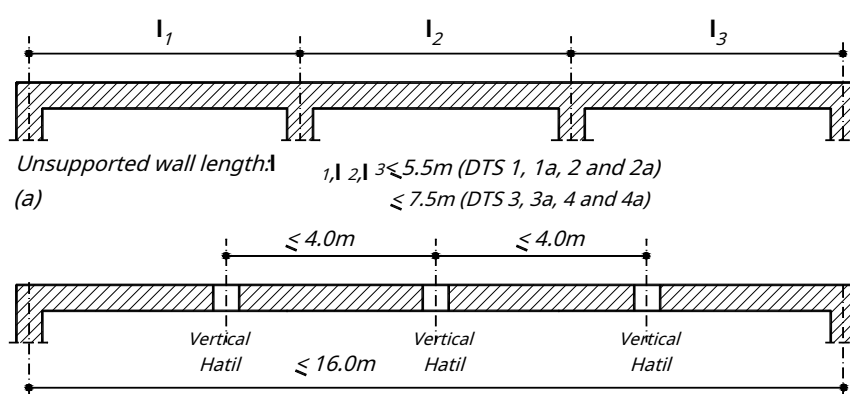


Figure 11.2

11.5.3 –Reinforced concrete lintels will be made over the door and window openings. The length of the sections of the lintels that sit on the wall will not be less than 200 mm. The lintel height will not be less than 150 mm. In the door and window openings to be left on the load-bearing walls **Figure 11.3** The rules given in will be followed.

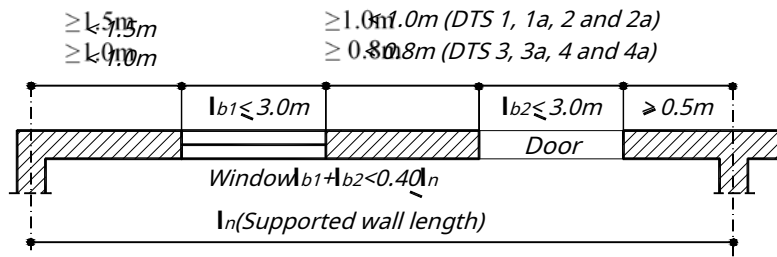


Figure 11.3

11.5.4 –In enclosed masonry buildings, horizontal and vertical beams will be cast after the masonry walls are built and the walls will be used as formwork (**Figure 11.4** and **Figure 11.5**).

11.5.5 –In enclosed masonry buildings, vertical beams shall be constructed in the following cases:

- (a) on the free edges of load-bearing walls,
- (b) area 1.5 m^2 On both sides of wall openings larger than ,
- (c) every 4 m wall length, the distance between the beams not exceeding 4 m,
- (d) Where two load-bearing walls meet each other (if the nearest vertical beam is more than 1.5 m away).

11.5.6 –In enclosed masonry buildings, the cross-sectional heights of vertical bond beams shall not be less than 300 mm and the longitudinal reinforcement in these beams shall not be less than $6\phi 12$ and the transverse reinforcement shall not be less than $\phi 8/150 \text{ mm}$. The cross-sectional width of vertical bond beams shall be at least as thick as the wall. Lapped joints in vertical bond beam reinforcement shall not be shorter than 60ϕ .

11.5.7 –In reinforced masonry buildings, horizontal reinforcements shall be placed inside horizontal joints or in suitable notches. The vertical spacing of these reinforcements shall not exceed 600 mm. The horizontal reinforcement ratio on the wall shall not be less than 0.05% of the gross cross-sectional area of the wall. (**Figure 11.6**).

11.5.8 –In reinforced masonry buildings, vertical reinforcements shall be placed in pockets or holes in the masonry units. The ratio of these reinforcements shall not be less than 0.08% of the gross cross-sectional area of the wall. The vertical reinforcement spacing shall be at most 600 mm. At least $2\phi 12$ additional reinforcement shall be placed along each side of door and window openings. (**Figure 11.6**). Horizontal reinforcements in the form of open stirrups with a diameter of not less than 4 mm will be passed around the reinforcements used at the extreme ends of the wall sections. (**Figure 11.7**).

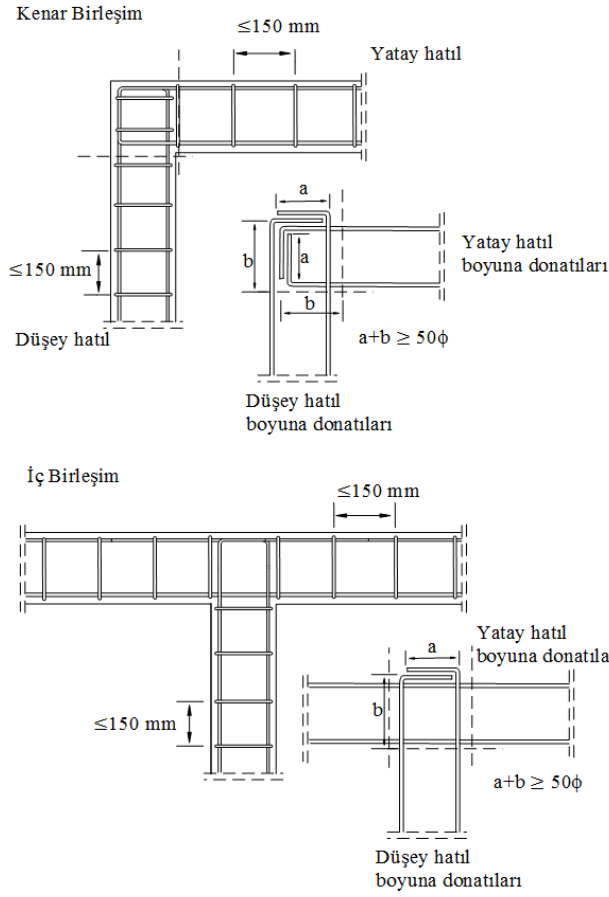
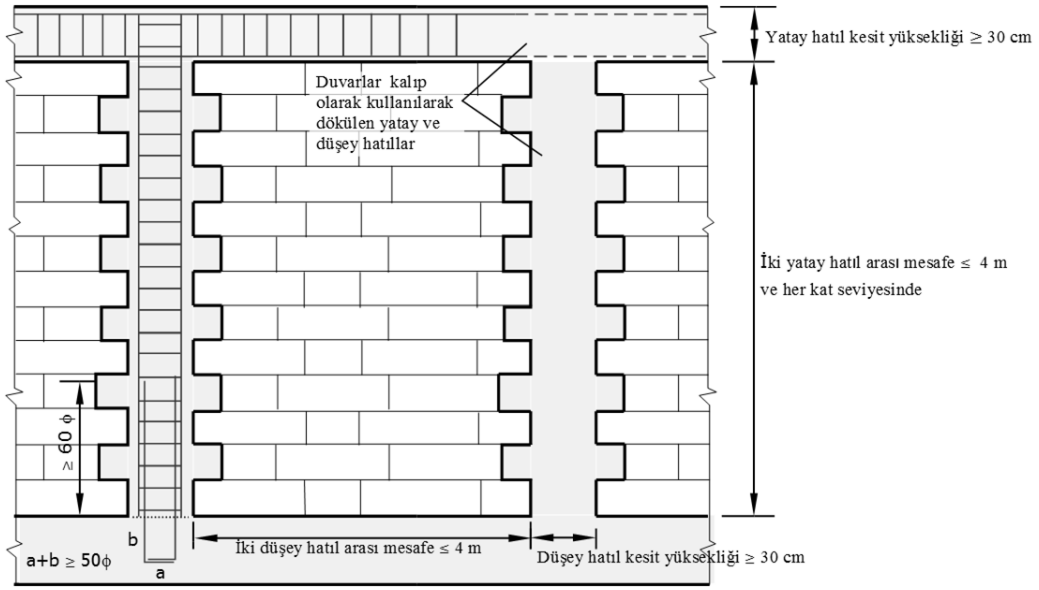


Figure 11.4

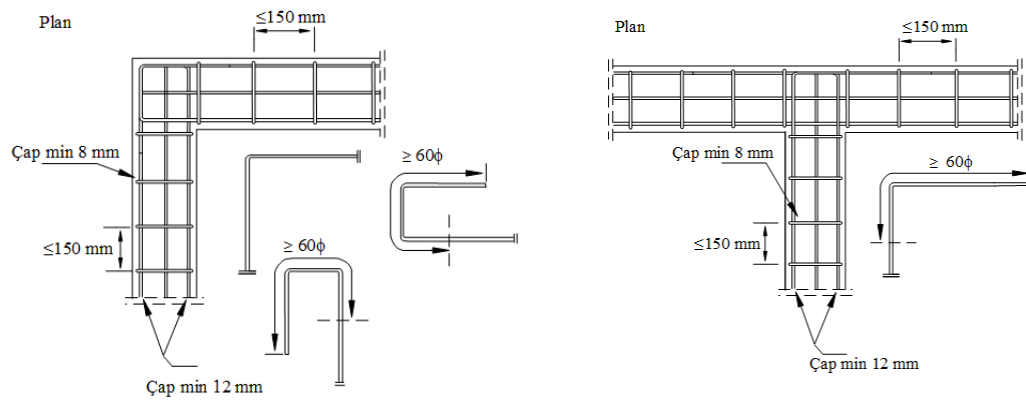


Figure 11.5

11.5.9 –In reinforced masonry buildings, in addition to the vertical reinforcement mentioned above, vertical reinforcement shall be placed for the following cases, not less than 4φ12, and the pockets or holes where these reinforcements are placed shall be filled with mortar.

- (a) free edges of load-bearing walls,
- (b) wall-to-wall joints,
- (c) every 5 m

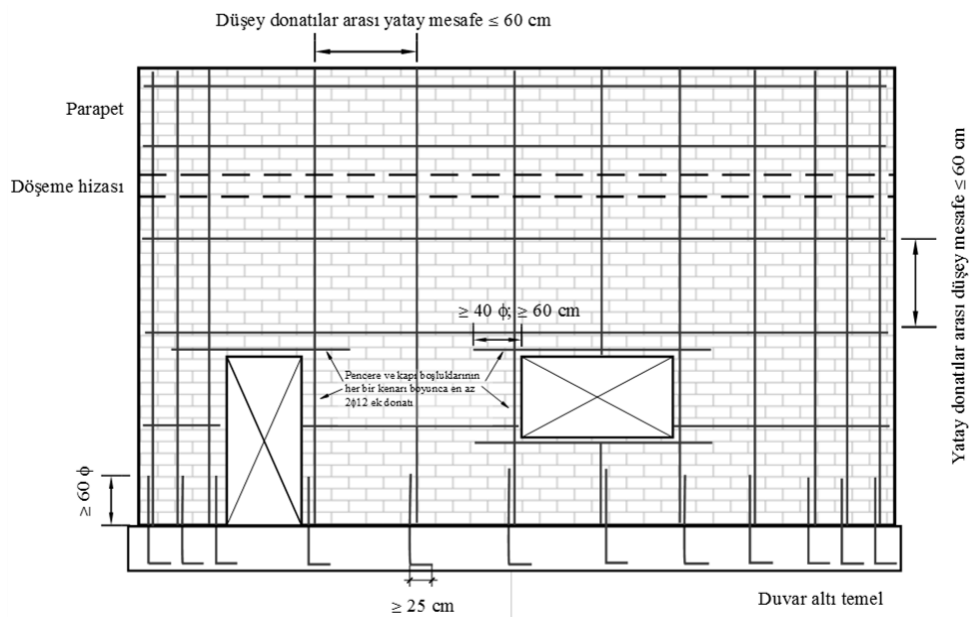


Figure 11.6

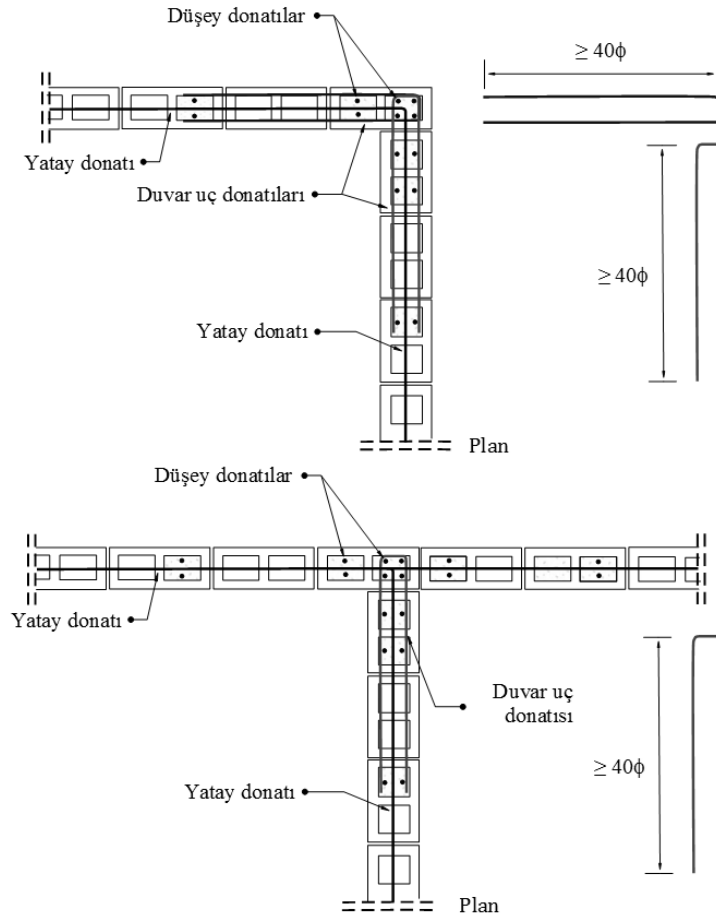


Figure 11.7

11.5.10 –Connection details of floor panels, reinforced concrete beams and foundations on which they are supported in buildings constructed with reinforced panels **Figure 11.8** will meet the conditions given in .

11.5.11 –If the height of the roof gable wall resting on the horizontal beam on the top floor is greater than 0.80 m, vertical and inclined beams will be made (**Figure 11.9**).

11.5.12 –The maximum unsupported lengths of the top floor parapet walls in unreinforced and girded masonry buildings and the distances between vertical beams, **Figure 11.2** It will comply with the conditions given in. In reinforced masonry and reinforced panel system buildings **Figure 11.2** The size limits given in can be increased by 20%. If the parapet height is greater than 1.00 m, a horizontal joist will be created every 1.00 m along the parapet height. If these conditions are not met, the height of the parapet wall will not exceed four times the wall thickness.

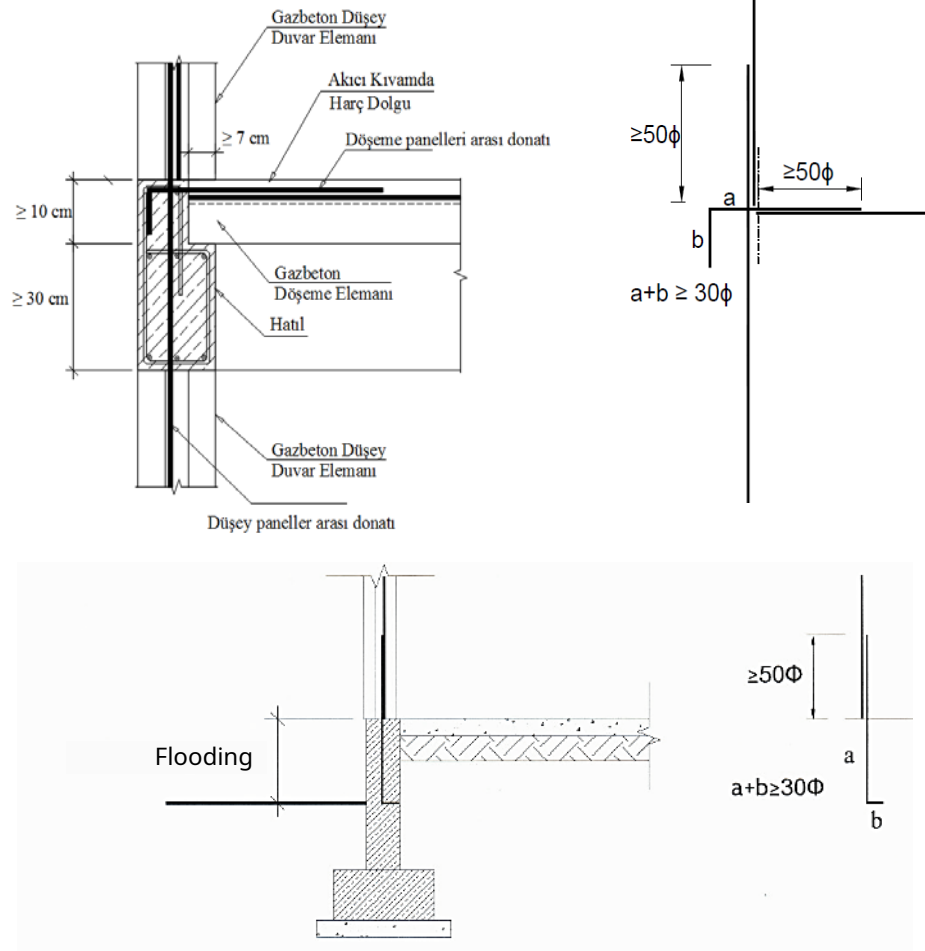


Figure 11.8

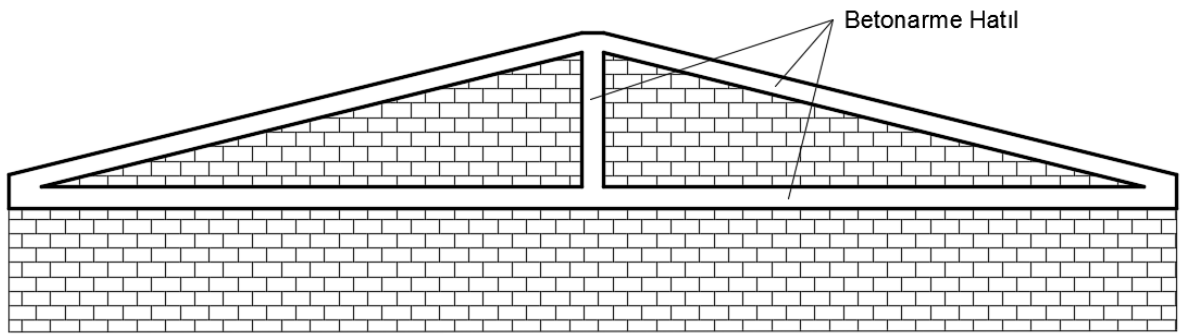


Figure 11.9

CHAPTER 12 – WOODEN BUILDING CARRIER UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGNING SYSTEMS

12.0. ICONS

a_{he}	= Intermediate beam spacing in wooden floors [m]
b_{he}	= Intermediate post spacing [m]
C	= Compressive force [kN] = Size effect
α	coefficient = Connection element
d	diameter [mm] = Shear capacity of a
f	nail [kN] = Panel height [m]
h	
h_b	= Beam height [m]
$-I$	= Width of panel partitions at full floor height [m]
M	= Bending moment [kNm]
ρ	= Nail spacing in intermediate uprights
S	[m] = Nail spacing in main uprights [m]
t_b	= Beam thickness [mm]
T	= Tensile force [kN]
v_c	= Safe shear strength per unit length [kN/m] =
v_p	Shear force demand per unit length [kN/m]
V_p	= Total horizontal load acting on the panel [kN]
y_M	= Partial safety factor

12.1. SCOPE

12.1.1 –The dimensioning and arrangement of the joints of the load-bearing system elements of all wooden buildings to be constructed under the effect of earthquakes will be made in accordance with the rules specified in this section, together with the relevant standards and regulations in force on this subject.

12.1.2 –Horizontal load-bearing systems of timber buildings within the scope of this section may consist of only clad panels or only timber braced panels.

12.1.3 –Rules for wooden building foundations **Chapter 16** is given in .

12.2. GENERAL RULES

12.2.1. Classification of Wooden Carrier Systems

Horizontal load-bearing systems in wooden buildings in terms of their behavior against earthquakes **12.2.1.1** And **12.2.1.2** It is divided into two classes defined in .

12.2.1.1 – Coated Panel Systems: Coated panel systems are listed below **(a)** And **(b)** It is divided into two as shown in .

(a) The entire earthquake effect is resisted by nailed or screwed OSB and plywood panels, **12.2.2.4** And **12.2.3.3** Wooden buildings that meet the conditions given in. Such buildings *Systems with High Ductility Level* will be taken as.

(b) Wooden buildings where the entire earthquake effect is resisted by glued, curtain and floor panel elements connected with nails, screws and bolts. Such buildings *Systems with Limited Ductility Level* will be taken as.

12.2.1.2 – Braced Panel Systems: Wooden buildings where the entire earthquake effect is resisted by cross members. Such buildings *Systems with Limited Ductility Level* will be taken as.

12.2.1.3 – To be applied in the design of systems falling into these two classes under earthquake effects. *Carrier System Behavior Coefficients* And *Excess Strength Coefficients* (*D*) with permitted *Building Height Classes* (BYS) **Table 4.1** is given in .

12.2.1.4 – The load-bearing curtain panels of wooden buildings will be placed as regularly as possible in the plan and symmetrically or nearly symmetrically with respect to the main axes. The load-bearing curtain panels will be arranged on top of each other on all floors.

12.2.2. Relevant Standards and Design Principles

12.2.2.1 – The design of the timber carrier systems within the scope of this section is carried out in accordance with the specified rules. **Part 2, Part 3** And **Chapter 4** It will be done using the rules given in and the rules given in TS EN 1995.

12.2.2.2 – Wooden building elements and joints must fulfill all the expected functions throughout the life of the building. *security* It shall have sufficient strength, stability and rigidity to perform under the required conditions.

12.2.2.3 – In timber buildings within the scope of this section, nonlinear behaviour shall be permitted only in materials and connections known to exhibit ductile behaviour. Design of such elements **Table 4.1** also given. It will be constructed according to the carrier system coefficients and reduced earthquake loads.

12.2.2.4 – Below are the joining elements **(a)** And **(b)** Elements that meet the conditions given in will be considered ductile.

(a) The thickness of the wooden elements combined with steel pins, bolts and nails, wood-wood and wood-steel joints must be at least $10d$ Elements with a joint element diameter not exceeding 12 mm.

(b) Wood-based coating thickness in flooring and curtain panel elements is at least $4d$ Elements with a nail diameter not exceeding 3.1 mm.

12.2.2.5 – 12.2.2.4 Combinations that do not meet the conditions specified in and **12.2.3.3** The properties of wooden elements that do not meet the conditions specified in , will be determined according to the tests defined in TS EN 12512.

12.2.2.6 – 12.2.2.5 The ductility level of the elements with defined cyclic properties is given below depending on the displacement ductility of these elements. **(a)** And **(b)** It will be determined as defined in .

(a) Elements with displacement ductility greater than 4 and less than 6 *limited ductility level* will be accepted.

(b) Elements with displacement ductility greater than or equal to 6 *high ductility level* will be accepted.

12.2.3. Material Conditions

12.2.3.1 –Within the scope of this regulation, the strength classes and material properties of all structural timber elements will be determined according to TS EN 1995.

12.2.3.2 –Within the scope of this regulation, the material properties of the steel elements to be used in the connections **9.2.3** will comply with the conditions set forth herein.

12.2.3.3 –Below are the wooden wall and floor coverings (a), (b), (c) Ductile behavior is permissible if the conditions given in are met:

(a) The density of particle board panels is 650 kg/m³ to be,

(b) The thickness of plywood and OSB wooden coatings should be at least 9 mm,

(c) Particleboard and fibreboard thickness must be at least 13 mm.

12.2.3.4 –Within the scope of this regulation *carrying capacity approach* For the material properties and strength of the wooden elements to be used in the design to be made with **Table 12.1** The material safety factors given in will be used.

Table 12.1. Material Safety Factors for Wooden Elements

Material type	Material security coefficient
Solid Wood	1.3
Glued Laminated Wood	1.25
Plywood, OSB	1.2
Particleboard	1.3
Fibreboard	1.3
Combinations	1.3
Toothed Plate	1.25

12.3. STRUCTURAL MODELING AND CALCULATION UNDER EARTHQUAKE EFFECT

12.3.1 –Modeling and calculation of wooden structures within the scope of this section are given below. (a) to (d) It will be done in a way that meets the conditions in .

(a) Upright elements, base and head beams, intermediate girder beams and cross elements that form the main skeleton of wooden panel *two-dimensional finite elements* with, wood cladding *two-dimensional finite elements* In simple systems, timber cladding can be modelled with equivalent cross elements.

(b) Wooden beams and wooden intermediate belt beams that form the main skeleton of wooden floors *rod finite elements* with, wood cladding *two-dimensional finite elements* In simple systems, wooden floors can be modeled as rigid diaphragms in their own planes.

(c) The loss of rigidity that will occur due to stripping in the connection elements at the joint points of all timber elements will be determined according to TS EN 1995 – 2.3.2 and will be reflected in the calculation models.

(d) Wooden elements in modeling *modulus of elasticity* The value corresponding to the sudden loading condition will be taken as .

12.3.2 – 12.3.1' Within the framework of the modeling rules defined below: (a) to (d) Design of elements in , **Table 4.1** also given *R carrier system behavior coefficients* will be made according to the internal forces obtained from the reduced earthquake loads:

(a) **12.2.2.4'** The joint elements defined in

(b) **12.2.3.3'** The wood cladding elements defined in

(c) **12.2.2.6'** or high ductility level connections and wooden elements,

(d) Wooden diagonal members carrying horizontal loads.

12.3.3 – 12.3.1' Within the framework of the modeling rules defined below: (a) to (f) Design of the elements specified in , **Table 4.1** Given in *R* Internal forces obtained from earthquake loads reduced by the load-bearing system behavior coefficients *D excess strength coefficients* will be made according to the enlarged values

(a) **12.2.2.4** And **12.2.3.3'** and does not meet the conditions defined in **12.2.2.6** All connection elements and wooden elements defined as having limited ductility level according to ,

(b) All glued joints,

(c) Panel edge posts, upper and lower header elements and, if any, belt beams,

(d) Anchors that transfer the tensile forces formed in the areas where the side posts are located to the system below and anchors that transfer the base shear force to the foundation,

(to) Diagonal steel tension elements used to increase in-plane rigidity,

(f) Timber diaphragm heads and girder beams.

12.3.4 – Load combinations to be applied under vertical loads and earthquake effects **4.4** is given in.

12.4. DESIGN PRINCIPLES OF PANELS

12.4.1. Elements that Make Up the Panels

The wooden skeleton of the panels is below (a) to (f) It will be constructed from the elements specified in:

(a) *Main pillars,*

(b) *Intermediate pillars,*

(c) To be placed under the pillars *floor beams,*

(d) To be placed on the pillars *header beams,*

(to) Horizontal beams that create rectangular eyes in the wall between the base and head beams and connect the studs along the wall. *girdle beams,*

(f) Crosses that divide the rectangular eyes into triangular eyes or horizontal load-bearing ones *cladding panels.*

A typical panel frame with cladding **Figure 12.1** is a typical braced panel skeleton **Figure 12.2** is given in .

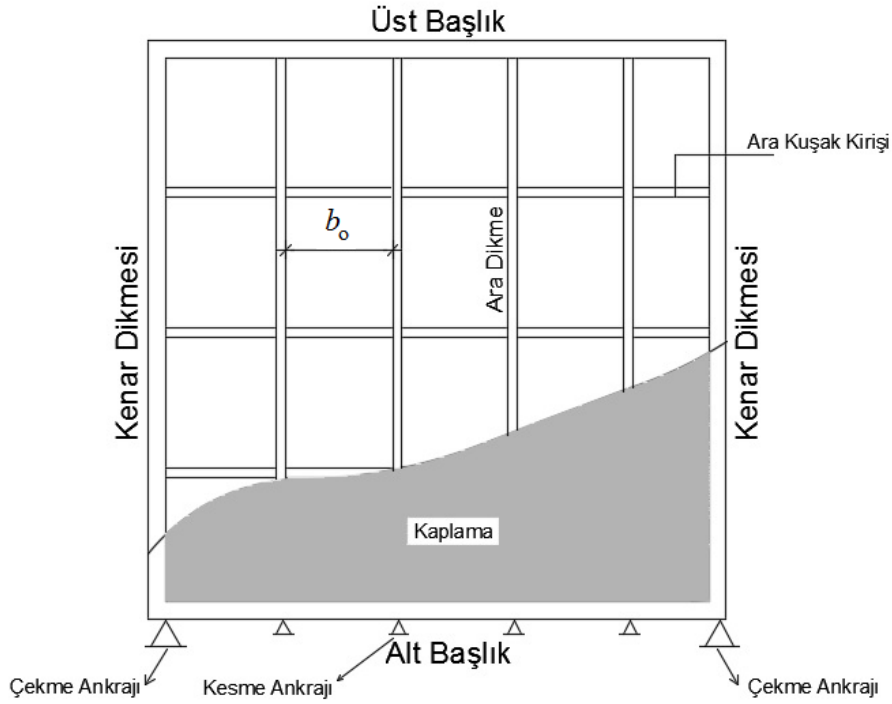


Figure 12.1

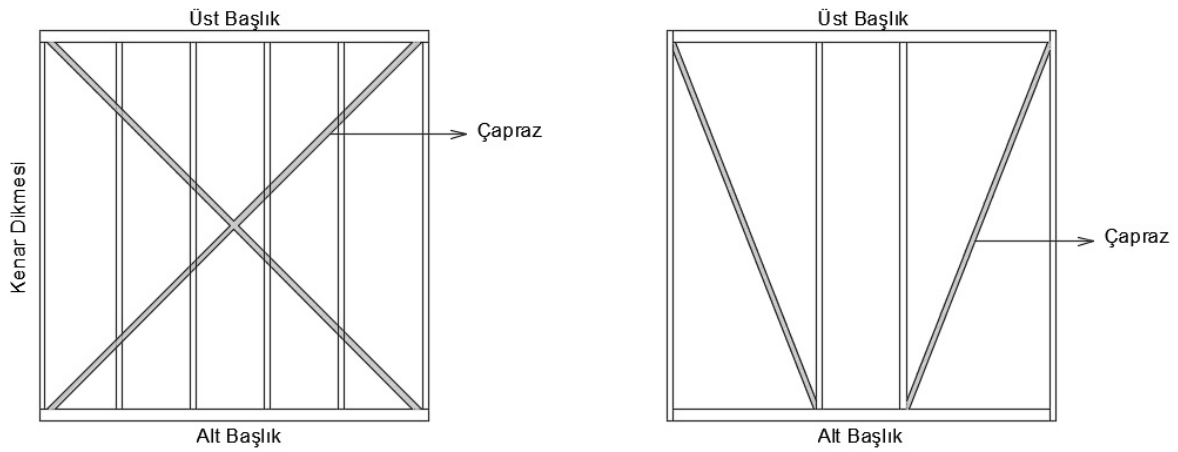


Figure 12.2

12.4.2. Rules Regarding the Creation of Panels

The rules to be followed in the creation of panel systems are given below.

12.4.2.1 -hpanel height and -iincluding panel width((**Figure 12.1**), $h/i \leq$ Condition 4

Panel elements providing *horizontal load-bearing panel elements* Panel elements that do not meet this condition will not be accepted as horizontal load carrying elements.

12.4.2.2 – t wood veneer thickness and b_{he} including the vertical spacing ((**Figure 12.1**), all wooden cladding elements $b_{he}/t \leq 100$ It will meet the 100 condition.

12.4.2.3 – Panels in areas where there are door and window openings will not be considered as horizontal load-bearing panels.

12.4.2.4 – In order for the wood cladding to be considered as load-bearing, b_{he} The maximum spacing between struts will be 0.625 m.

12.4.2.5 – Gypsum panel coverings will not be accepted as horizontal load bearing.

12.4.2.6 – In single-storey wooden buildings, the cross-sectional dimensions of the uprights, base and head beams and diagonals must be at least 100 mm. $\times$ 100 mm, the cross-sectional dimensions of the intermediate belt beams are at least 50 mm $\times$ It will be 100 mm.

12.4.2.7 – Main uprights and crosses will be continuous (jointless) throughout the floor. If the crosses are in X shape, one of the diagonal elements will be continuous and the other interrupted diagonal element will be connected to each other with the help of steel plates.

12.4.2.8 – Additional beams can be added to the base beams and head beams. Calculations will show that the additional beams have sufficient strength.

12.4.2.9 – Even if the main posts are connected to the base beam and the head beam by interlocking, they will be reinforced with nails. If interlocking connections are not made, nailed corner wedges, steel connection elements or specially produced elements will be used. It is mandatory to show that the created system transfers loads with calculations.

12.4.2.10 – Other element connections can be made in the form of straight or oblique tenons, and auxiliary steel connecting elements can also be used.

12.4.2.11 – Unless special precautions are taken to operate in tension, cross members shall be considered as members operating only in compression.

12.4.3. Rules Regarding the Design of Panels

12.4.3.1 – In the design of wooden curtain panels:

(a) The shear capacity of the panel will be greater than the shear force demand on this panel.

(b) The axial force carrying capacity of the uprights will be greater than the axial forces acting on the uprights.

(c) The design of anchors transferring shear force and foundation and floor connection anchors shall be made to meet the base shear force, floor shear force and overturning moment.

12.4.3.2 – Only the full-story height carrier panel sections will be considered to have shear force carrying capacity. For each carrier panel section, tensile force transfer anchors will be used in the side posts. ((**Figure 12.3**).

12.4.3.3 – The shear force demand on the coated panel, **Equation (12.1)** The calculated shear will be less than the design strength according to:

$$V_p \leq \frac{v_p \sum h}{y_M} \quad (12.1)$$

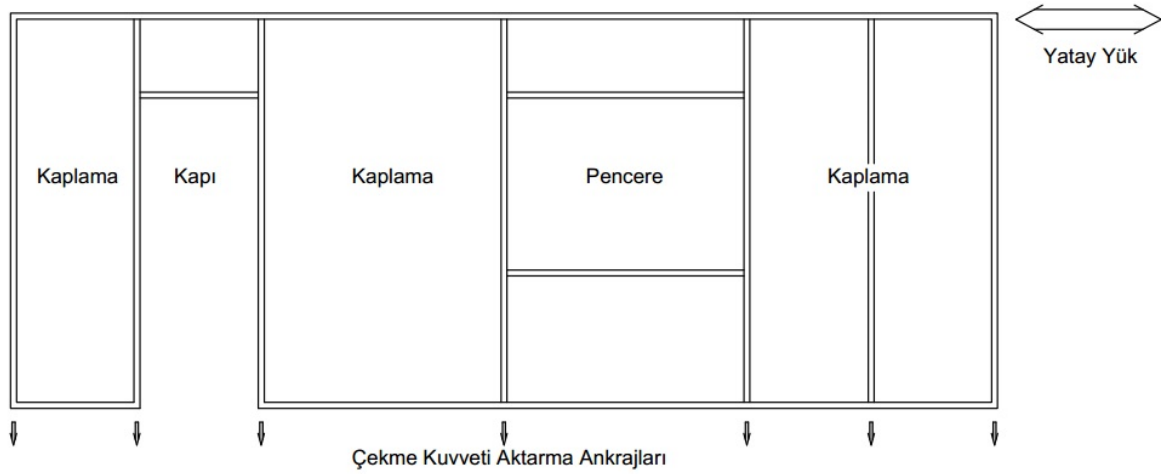


Figure 12.3

Characteristic shear strength per unit length v_p , Equation (12.2) It will be calculated according to .

$$v_p = \frac{f_{if}}{s} \quad (12.2)$$

Cutting capacity of the nail f_{if} 12.2.3.1 It will be determined according to the size effect coefficient α

Equation (12.3) will be calculated according to .

$$\alpha = \frac{2h \leq 1.0}{h} \quad (12.3)$$

12.4.3.4 -The shear strengths of the panels per unit length are the values corresponding to the coating applied to one side of the panel. Otherwise, below (a),(b),(c) The rules specified in will be followed.

(a) If more than one coating is applied to the same face of the panel, only the strength value of the material in contact with the panel face will be used.

(b) If the same coating is applied to both sides of the panel with the same nail or screw placement, 12.4.3.3 The shear strength values of the unit length calculated according to are added up for both faces and used. In this case, the calculation will show that the panel edge struts and the upper and lower header beams safely withstand the resulting shear force.

(c) In case of using different coating materials on both sides of the panel, the shear strength per unit length is; 12.4.3.3 According to , the coating material with a small shear strength will be taken as twice its strength, and the coating material with a large shear strength will be taken as the greater of its strength.

12.4.3.5 -In horizontal load-bearing panels, an anchor or a connection that transfers tensile force will be used at most every 5 m.

12.4.4. Design of Pillars

In coated panel systems, axial forces occurring on the side posts due to horizontal loads **Equation (12.4)** will be calculated with **(Figure 12.4)**.

$$C_i = T_i = v_d h \quad (12.4)$$

Under the sum of the axial forces found in this way and the axial forces resulting from the vertical loads, the axial compressive strength of the strut will be calculated according to TS EN 1995 Part 6.

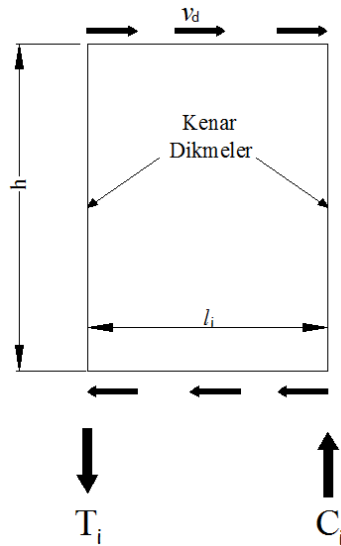


Figure 12.4

12.4.5. Design of Joints

12.4.5.1 –The wooden structural elements forming the panel will be connected with nails/screws and/or bolts. The forces to be carried by nails, screws or bolts will be calculated according to TS EN 1995 Section 8 under different collapse types.

12.4.5.2 –*Coated Panel Systems* Below are the combinations of **(a)** to **(d)** The limits given in will be observed:

- (a)** The nail/screw spacing in the side posts is the maximum s It will be 150 mm.
- (b)** The nail/screw spacing in interior uprights is the most p It will be 300 mm.
- (c)** The distance between the screws along the edges of the panel will be at least 10 mm.
- (d)** Shear anchors will be placed in sufficient numbers to transfer the shear force that occurs.

12.4.5.3 –Other rules to be followed in the joints of coated and braced panel systems are given below. **(a)** And **(b)** is given in .

(a) The pull-out strength of nails and screws cannot be used to resist tensile forces resulting from earthquake loads.

(b) The panels will be connected in a way that will safely transfer the tensile forces occurring in the areas where the edge posts are located to the system below with anchors.

12.4.5.4 –The foundations, **Table 4.1** also given D will be sized under forces magnified by the coefficient.

12.5. DESIGN PRINCIPLES OF FLOORING

12.5.1. Elements that Make Up Wooden Floors

12.5.1.1 –The skeleton of the wooden floors is below(a),(b),(c)It will be constructed from the elements specified in:

(a)Wooden floor beams,

(b)Horizontal intermediate girder beams connecting wooden floor joists to each other,

(c)Wooden floor covering.

Appearance of a typical wooden floor frame**Figure 12.5**is given in.

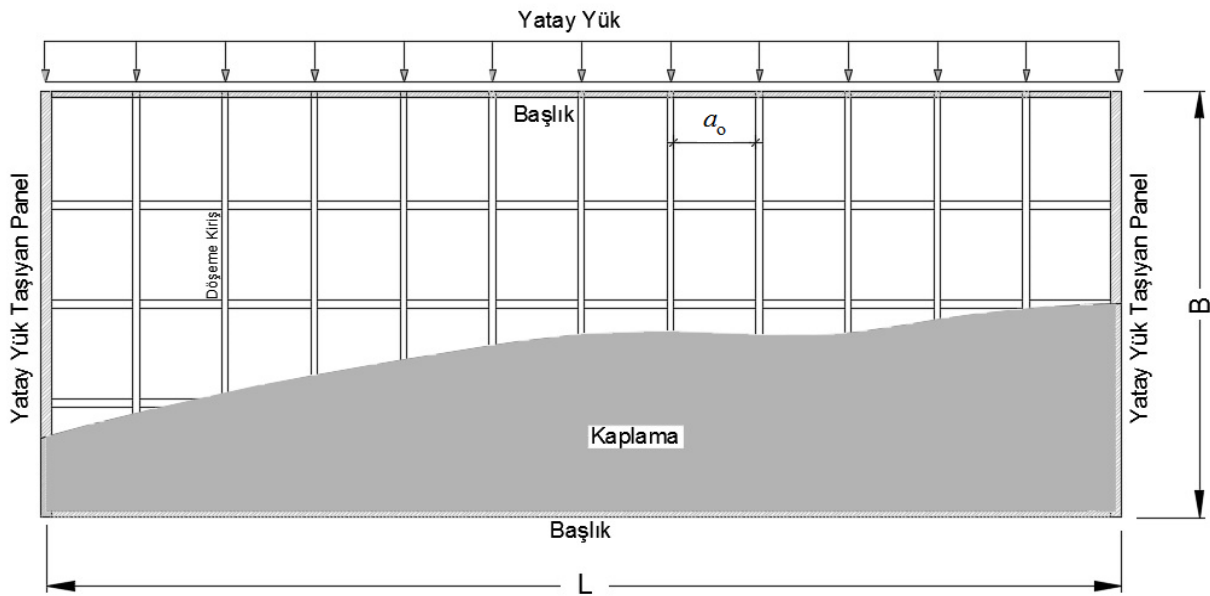


Figure 12.5

12.5.1.2 –The floor beams on the ground floor will be placed on the base beams, and the floor beams and roof trusses on the other floors will be placed on the head beams; they will be connected with nails, screws, bolts or specially designed steel connection elements. A typical connection detail**Figure 12.6**is given in .

12.5.2. Rules for Design of Floors

12.5.2.1 –All floor panel elements that are not connected to the frame elements will be connected to the transverse connection (wedge) elements to be created between the wooden floor beams. These elements will also be arranged at the upper level of the horizontal load-bearing walls.

12.5.2.2 –In case of a gap in the floor, continuity should be ensured in the beams surrounding the gap.

12.5.2.3 –In case transverse connection elements are not used in beams, the ratio of beam height to width (h_b/t_b) will be less than 4.

12.5.2.4 - t panel coating thickness and a_{he} All wooden cladding elements, including the beam spacing $a_{he}/t \leq 100$ It will meet the 100 condition.

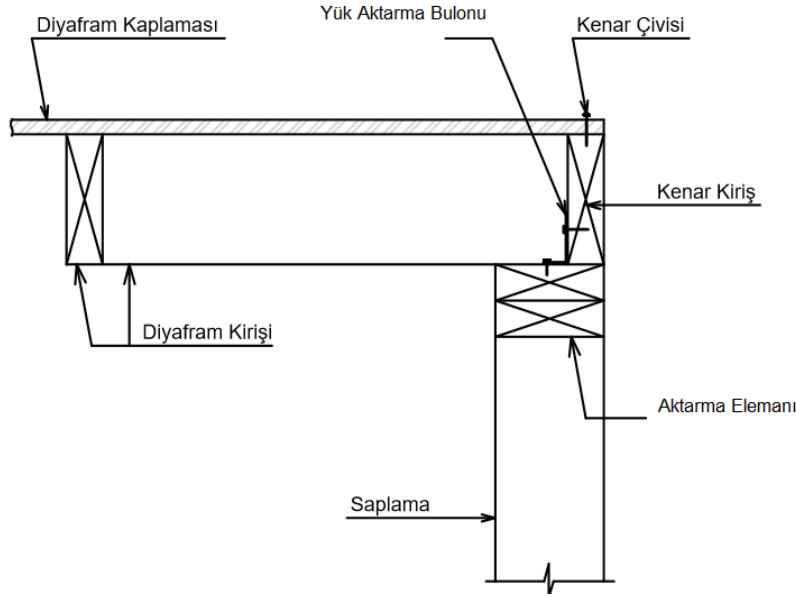


Figure 12.6

12.6. DESIGN OF JOINTS

12.6.1 -The elements will be designed in such a way that they will not separate from each other and will maintain their positions due to the loss of load carrying capacity caused by deformations that may occur in the pressure elements and their joints (gear joints, etc.) due to repeated loads.

12.6.2 -In wood-wood and steel-wood joints where no wood joining elements are used, large bolts and wedges ($d > 16\text{mm}$) will not be used.

12.6.3 -In case of wood-wood or wood-steel panel joint with nails or screws, it is mandatory to show through calculations that the nail or screw has sufficient strength.

CHAPTER 13 – HIGH-RISE BUILDING CARRIER UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGNING SYSTEMS

13.0. ICONS

BYS	= <i>Building Height Class</i>
D	= Excess Strength Coefficient =
DTS	= <i>Earthquake Design Class</i>
DD-1	= Earthquake location with a 2% probability of being exceeded in 50 years (recurrence period 2475 years) movement level
DD-2	= Earthquake location with a 10% probability of being exceeded in 50 years (recurrence period 475 years) movement level
DD-3	= Earthquake location with a 50% probability of being exceeded in 50 years (recurrence period 72 years) movement level
DD-4	= Earthquake location with a 68% probability of being exceeded in 50 years (recurrence period 43 years) movement level
$TQ_{d(H)}$	= Directional combination applied <i>design basis</i> horizontal earthquake effect =
G	dead load effect
g	= Gravitational acceleration [m/s ²]
H_N	= Above the basement floors of the building <i>upper section</i> Total height of [m] =
I	Building Importance Factor
m_t	= Above the basement floors of the building <i>upper section</i> Total mass of [t] =
n	Live load participation coefficient
R	= Carrier System Behavior Coefficient
Q	= Live load effect
Q_{to}	= Effective live load effect
S_{DS}	= Short period <i>design spectral acceleration coefficient</i> [[dimensionless]]
$V_{tE}^{(X)}$	= (X) earthquake direction affecting the entire building <i>total equivalent earthquake load (base shear force)</i> [kN]
α	= Empirical coefficient used for earthquake joint spacing =
η_{bi}	Torsional irregularity coefficient at the i'th floor
y_{TO}	= Used to determine the equivalent base shear force magnification coefficient empirical coefficient

13.1. SCOPE AND DEFINITION

13.1.1 – This part of the regulation covers the special rules to be applied for the design of reinforced concrete and steel high-rise building load-bearing systems under earthquake effects.

13.1.2 – 3.3.1 Based on the height definition given in **Table 3.3** According to below (a), (b), (c) The buildings mentioned in *high rise building* and is classified as $BYS = 1$:

(a) DTS = height for 1, 1a, 2, 2a H_N Buildings > 70 m;

(b) DTS = 3, height for 3a H_N Buildings > 91 m;

(c) DTS = 4, height for 4a H_N Buildings > 105 m.

13.1.3 – High-rise building carrier systems, **13.1.5** Except for the carrier systems defined in , **Chapter 4, Chapter 7, Chapter 8** and/or **Chapter 9** According to the definitions given in *high ductility level* will be organized as a system. *Limited ductility level* And *ductility level mixed* other systems are not allowed.

13.1.4 – High-rise building carrier systems, **Table 4.1** also **A12, A13, A14, A15, B12, B13** identified by the symbols *high ductility level* reinforced concrete or **C12, C14, C15** identified by the symbols *high ductility level* will be constructed from steel carrier systems.

13.1.5 – Also in high rise buildings with DTS=4 only **Table 4.1** also **A21, A22** And **C21, C22** identified by the symbols *ductility level mixed* Reinforced concrete and steel carrier systems can also be used.

13.1.6 – The thickness of reinforced concrete walls in high-rise building load-bearing systems shall not be less than 300 mm. **7.6.1.3** And **7.6.3.2** will not be applied.

13.1.7 – In high-rise building reinforced concrete carrier systems only **B420C** or **B500C** quality ribbed reinforcement steels will be used.

13.2. PERFORMANCE TARGETS AND DESIGN PHASES

Design of high-rise buildings, **13.2.1, 13.2.2, 13.2.3** It will be carried out in three stages specified in . The order of Stage II and Stage III may be changed.

13.2.1. Design Phase I: Preliminary Design Dimensioning Under DD-2 Earthquake Ground Motion

13.2.1.1 – At this stage, under the influence of DD-2 earthquake ground motion *Controlled Damage* (KH) high-rise building to meet the performance target *Design by Strength* (Preliminary design with DGT approach – sizing will be done.

13.2.1.2 – Details of the compulsory calculation and design principles for this stage **13.4** is given in.

13.2.1.3 – At this stage **Chapter 4** together with **Chapter 7, Chapter 8** and/or **Chapter 9** The rules given in and also in this Section shall be taken as basis.

13.2.2. Design Phase II: Evaluation – Improvement for Continuous Use or Limited Damage Performance Target under DD-4 or DD-3 Earthquake Ground Motion

13.2.2.1 – At this stage, the preliminary design of the high-rise building has been completed;

(a) Table 3.4(b) According to *normal performance target* for DD-4 earthquake ground motion *Uninterrupted Use*(KK) to achieve the performance target, *Design by Strength* Performance evaluation will be done with the (DGT) approach.

(b) Table 3.4(b) According to *advanced performance target* for DD-3 earthquake ground motion *Limited Damage*(To achieve the performance target of SH, *Evaluation and Design Based on Shapeshifting*) Performance evaluation will be made with the ŞGDT approach.

(c) Details of the accounting principles for this stage **13.5** is given in.

13.2.2.2 –At this stage **Chapter 4** or **Chapter 5** together with **Chapter 7**, **Chapter 8** and/or **Chapter 9** The rules given in and also in this Section will be taken as basis. If deemed necessary as a result of the evaluation, the preliminary design will be improved and the evaluation will be repeated.

13.2.3. Design Phase III: Evaluation – Improvement – Final Design for Prevention of Collapse or Controlled Damage Performance Target under DD-1 Earthquake Ground Motion

13.2.3.1 –At this stage, the first two design stages of the high-rise building have been completed **Table 3.4(b)** According to DD-1 earthquake ground motion under *normal performance target* aspect *Prevention of Immigration*(GO), *advanced performance target* as if *Controlled Damage*(KH) to achieve the performance target *Evaluation and Design Based on Shapeshifting* (Performance evaluation will be made with the ŞGDT approach.

13.2.3.2 –Details of the accounting principles for this stage **13.6** is given in .

13.2.3.3 –If deemed necessary as a result of the evaluation, the design will be improved and the evaluation will be repeated to reach the final design. At this stage **Chapter 5** together with **Chapter 7**, **Chapter 8** and/or **Chapter 9** The rules given in and also in this Section shall be taken as basis.

13.3. BEHAVIORAL CHARACTERISTICS OF HIGH-RISE BUILDING CARRIER SYSTEM ELEMENTS

13.3.1. Deformations Related to Nonlinear Ductile Behavior

General description **4.2.2** Given in *capacity design principles* Within the scope of application, the element types and behavior modes for which nonlinear ductile behavior can be defined are listed below:

(a) *Reinforced concrete walls with high ductility level*: Yielding under bending and axial force (PMM) in two-way bending in shear wall base areas (in areas immediately above the foundation, basement or plinth) and in areas at a reasonable height upwards from these areas. Yielding under bending and/or shear in reinforced concrete or steel shear wall tie beams in the case of tie beam shear walls.

(b) *Reinforced concrete or steel frames with high ductility level*: Yielding under the effect of bending in beam end regions, yielding under the effect of bending in two directions and axial force (PMM) in column base sections (in the sections just above the foundation, basement or plinth).

(c) *High ductility level eccentrically braced steel frames*: Shear-bending yielding in tie beams (preferably only shear yielding), yielding in column base sections (in sections just above the foundation, basement or plinth) under the effect of bending in two directions and axial force (PMM).

(d) High ductility central braced steel frames: Yielding in tension braces, buckling in compression braces, bending in two directions in column base sections (in the sections just above the foundation, basement or plinth) and yielding under the effect of axial force (PMM).

(to) Buckling-resistant braced steel frames with high ductility level: Yielding in tension and compression diagonals, yielding in column base sections (in sections just above the foundation, basement or plinth) under the effect of bending in two directions and axial force (PMM).

13.3.2. Internal Forces Related to Linear Behavior

13.3.2.1 –General description **4.2.2** Given in *capacity design principles* In the application of , it is essential to provide sufficient strength for the internal forces related to linear behavior.

13.3.2.2 –The following internal forces related to linear behavior are, *critical internal forces* will be defined as:

(a) Shear forces in walls, basement walls, columns and beams (excluding cross-reinforced tie beams),

(b) Axial forces in columns subjected to axial forces only,

(c) Internal forces transferred to vertical load-bearing system elements (walls and columns) through transfer elements in floor slabs and transfer floor slabs

(d) Punching forces in floor and raft foundation slabs without special punching reinforcement

(to) Shear forces in foundations

(f) Internal forces in steel connections having less strength than the strength of the connected members.

13.3.2.3 –The following internal forces related to linear behavior are, *non-critical internal forces* will be defined as:

(a) Bending moments in basement walls

(b) Bending moments in foundations

(c) Bending moments in slabs.

13.4. DESIGN STAGE I: PRELIMINARY DESIGN – CALCULATION PRINCIPLES FOR DIMENSIONING

13.4.1. Scope

13.4.1.1 –In Phase I, the selected high-rise building structural system *preliminary design – sizing* purpose earthquake calculation, standard *design earthquake ground motion* Under the influence of the DD-2 earthquake ground motion, which is defined as, taking into account the additional rules defined herein, **Chapter 4** Given in *Design by Strength* (DGT) will be made according to the accounting principles.

13.4.1.2 –In this Section and with the calculation results **Chapter 7**, **Chapter 8** and/or **Chapter 9** Based on the rules given in , the selected high-rise building carrier system *preliminary design* (pre-sizing) will be done.

13.4.2. Stage I Structural System Modelling

13.4.2.1 –Stage I carrier system modeling **4.5** It will be done according to the rules given in .

13.4.2.2 –Load combinations including earthquake effects to be considered in Phase I **4.4.4** It is defined in .

13.4.3. Stage I Earthquake Calculation

13.4.3.1 –Before Stage I earthquake calculation *taking into account the construction stages* Vertical load calculations, wind calculations and creep calculations in reinforced concrete buildings will be made.

13.4.3.2 –In the Stage I earthquake calculation, **13.1.4** or **13.1.5** For the high-rise building carrier system selected according to **Table 4.1** Given in *Carrier System Behavior Coefficient R* And *Excess Strength Coefficient D* to be used and **4.3.2.4** The rules given in will be applied.

13.4.3.3 –Within the scope of Stage I earthquake calculation, under DD-2 earthquake ground motion **4.8.2** According to *Mod Merge Method* or **4.8.3** According to the time domain *Mod Collection Method* Three-dimensional linear calculation will be done with .

13.4.3.4 – **4.8.4** Made according to *Magnification of reduced internal forces relative to the Equivalent Base Shear Force* process, in the Stage I earthquake calculation of high-rise buildings **Equation (13.1)** defined by *Minimum Base Shear Force* will be made on the basis of.

$$V_{t,min} = 0.04 \alpha_H m_t S_{DS} g \quad (13.1)$$

Here m_t including the base and tower *upper section* The total mass of , S_{DS} DD-2 earthquake location for the level of movement **2.3.2** defined in *short period design spectral acceleration coefficient's*, g the acceleration due to gravity, α_H whereas **3.3.1** Based on the height definition given in **Equation (13.2)** building with height H_N Shows the coefficient calculated depending on .

$$\begin{aligned} \alpha_H &= 1.0 & H_N &\leq 105\text{m} \\ \alpha_H &= 2.05 - 0.01 H_N & 105\text{m} < H_N \leq 155\text{m} \\ \alpha_H &= 0.5 & 155\text{m} < H_N \end{aligned} \quad (13.2)$$

Equation (13.1) calculated with $V_{t,min}$, **Equation (4.25)** in $V_{t,E}$ will be used instead and in the same equation $y_{TO}=1$ will be taken.

13.4.4. Additional Conditions Regarding Pre-Dimensioning of the Carrier System

13.4.4.1 –Dimensions and reinforcement of structural elements, **Chapter 7**, **Chapter 8** and/or **Chapter 9** Taking into account the principles given in **13.4.3** It will be determined according to the preliminary design internal forces calculated according to .

13.4.4.2 –Preliminary design of high-rise building foundations **4.10.3** Based on the forces defined in **Chapter 16** It will be done according to.

13.5. DESIGN STAGE II: CALCULATION BASIS FOR UNINTERRUPTED USE OR LIMITED DAMAGE PERFORMANCE EVALUATION

13.5.1. Scope

13.5.1.1 –In Phase II, the pre-designed load-bearing system elements, **Table 3.4(b)** According to *normal performance target* for DD-4 earthquake ground motion *Uninterrupted Use* (KK) performance target, *advanced performance target* For the DD-3 earthquake ground motion *Limited Damage* (SH) to achieve the performance target *linear or very close to linear behavior* will be proven to be the case.

13.5.1.2 –*Uninterrupted Use* (KK) evaluation for performance target, **Chapter 4** According to will be done with DGT approach based on linear calculation. *Limited Damage* (SH) is the evaluation for the performance target, **Chapter 5** According to will be done with the GDT approach based on non-linear calculation.

13.5.2. Stage II Structural System Modeling

13.5.2.1 –In Phase II **Table 3.4(b)** According to *normal performance target* In case of applying the DGT approach based on linear calculation for **4.5** Given in, *advanced performance target* In case of application of the nonlinear calculation based GDT approach for **5.4** The modeling rules given in will be followed. However, below **13.5.2.2** to **13.5.2.5** The different rules given in will be followed in both cases.

13.5.2.2 –In Phase II, for the definition of load combinations **4.4.4** or **5.2.2** will be taken as basis.

13.5.2.3 –In Phase II *normal performance target* In case of applying the DGT approach based on linear calculation for reinforced concrete elements, **Table 13.1** Given in *effective section stiffness factors* will be used. **Table 13.1** The factors given in, *advanced performance target* it can be used for slabs and basement walls in case the nonlinear calculation based GST approach is applied.

13.5.2.4 –The effect of additional eccentricity will not be taken into account.

13.5.2.5 –The damping ratio will be taken as 2.5%.

13.5.3. Stage II Earthquake Calculation

13.5.3.1 – **13.4.3.1** According to , before the Stage I earthquake calculation was made and *taking into account the construction stages* The vertical load calculation results will also be used in Phase II.

13.5.3.2 –Within the scope of Stage II earthquake calculation, *normal performance target* for **13.5.1.2** According to DD-4 earthquake ground motion under **4.8** According to *linear calculation with modal calculation methods* will be done. *Advanced performance target* For the DD-3 earthquake ground motion **5.7** According to *nonlinear time history computation* will be done.

13.5.3.3 –Earthquake calculation *normal performance target* for **4.8** According to *modal calculation methods* In case it is done with;

(a) In the calculation of internal forces $R/I=1$ and $D=1$ will be taken.

(b) In Phase I *minimum base shear force* condition will not apply.

(c) Earthquake calculation **4.8.2** According to *Mod Merge Method* In case of doing with, **13.5.2.5** Horizontal elastic design spectral acceleration corresponding to 2.5% damping ratio defined in values $S_{ae}(T)$ 'In order to obtain the **Equation (2.2)** The spectral values given for 5% extinction ratio are acceleration values will be multiplied by a factor of 1.25. Also, to combine modal contributions, **4B.2.4** According to *Perfect Square Combination Rule* In case of use of, **Equation (4B.5)** In the calculation of the cross-correlation coefficients given by, the damping ratio for all modes will be taken as 2.5%.

(d) Earthquake calculation **4.8.3** According to the time domain *Mod Collection Method* In case of doing with, modal single degree of freedom system for each vibration mode **Equation (4B.10)** In the equation of motion given by, the damping ratio will be taken as 2.5%.

Table 13.1. Effective Methods to be Applied in Stage II for Reinforced Concrete Bearing System Elements
Section Stiffness Factors

Reinforced Concrete Carrier System Element	Effective Section Stiffness The multiplier	
<i>Curtain – Floor (In-Plane)</i>	<i>Axial</i>	<i>Slip</i>
Curtain	0.75	1.00
Basement curtain	1.00	1.00
Floor	0.50	0.80
<i>Curtain – Floor (Out of Plane)</i>	<i>Bending</i>	<i>Cutting</i>
Curtain	1.00	1.00
Basement curtain	1.00	1.00
Floor	0.50	1.00
<i>Rod element</i>	<i>Bending</i>	<i>Cutting</i>
Tie beam	0.30	1.00
Frame Beam	0.70	1.00
Frame Column	0.90	1.00
Curtain (equivalent rod)	0.80	1.00

13.5.3.4 –Earthquake calculation *advanced performance target* for **5.7** According to *nonlinear time history computation* In case it is done as, **5.7.3.2** According to *Rayleigh proportional damping matrix's or modal damping matrix* In the modes to be considered for the creation of, the damping ratio will be taken as 2.5%.

13.5.4. Demands for Transformation and Internal Forces Based on Evaluation in Phase II

13.5.4.1 –Stage II earthquake calculation *normal performance target* for **13.5.3.3** in the scope of **4.8** linear according to *modal calculation methods* In case of doing this, the internal forces obtained in this calculation *internal forces based on evaluation* will be taken into consideration.

13.5.4.2 –Stage II earthquake calculation *advanced performance target* for **5.7** According to *nonlinear time history computation* In case it is done as follows;

(a) In elements with ductile behavior, the deformation requests that form the basis for evaluation are the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated as the average of the largest absolute values of the results obtained from each.

(b) *Critical internal forces* aspect **13.3.2.2** The basic demands for the evaluation of internal forces defined in , the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated as 1.4 times the average of the largest absolute values of the results obtained from each.

(c) *Non-critical internal forces* aspect **13.3.2.3** The basic demands for the evaluation of internal forces defined in , the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated as the average of the largest absolute values of the results obtained from each.

13.5.5. Stage II Performance Evaluation

13.5.5.1 – 13.5.1.1 According to *normal performance target* for DD-4 earthquake ground motion *Uninterrupted Use* performance target or *advanced performance target* for DD-3 earthquake ground motion *Limited Damage* To achieve the performance target, the following **13.5.5.2** The internal force limits given in or **13.5.5.3** It is necessary not to exceed the deformation and internal force limits given in . In both cases, instead of the characteristic material strengths, the internal force capacities are calculated. **Table 5.1** Given in *average (expected) material strength* values will be used.

13.5.5.2 –Stage II Earthquake calculation *normal performance target* for **4.8** linear according to *modal calculation methods* In case it is done with;

(a) **13.3.1** The demand/capacity ratio for internal forces in elements with ductile behavior defined in $TO/K=$ It will not exceed the value of 1.5.

(b) **13.3.2.2** And **13.3.2.3** The demand/capacity ratio for internal forces defined in and not having ductile behavior $TO/K=$ It will not exceed the value of 0.7.

13.5.5.3 –Stage II Earthquake calculation *advanced performance target* for **5.7** According to *nonlinear time history computation* In case it is done as follows;

(a) In reinforced concrete members with ductile behavior **13.5.4.2(a)** Calculated according to *shapeshifting demands* **5.8.1** It will not exceed the limit values given for the relevant performance level.

(b) In steel members with ductile behavior **13.5.4.2(a)** Deformation requests calculated according to , **ANNEX 5C** It will not exceed the limits given for the relevant performance level.

(c) In elements that do not have ductile behavior **13.5.4.2(b)** And **13.5.4.2(c)** Internal force demands calculated according to for the relevant reinforced concrete elements **Chapter 7** for steel elements **Chapter 9** It will be shown that the internal force capacities are smaller than the internal force capacities defined in . In the calculation of internal force capacities, instead of the characteristic material strengths, **Table 5.1** Given in *average (expected) material strength* will be taken as basis.

13.5.5.4 –Above **13.5.5.2** And **13.5.5.3** If the conditions given in are not met, the preliminary design of the carrier system will be changed and Stage I earthquake calculations will be repeated.

13.6. DESIGN STAGE III: CALCULATION BASIS FOR MIGRATION PREVENTION OR CONTROLLED DAMAGE PERFORMANCE EVALUATION

13.6.1. Scope

13.6.1.1 –In Phase III, the preliminary design and dimensioning of which was completed in Phase I and in Phase II *Uninterrupted Use* (KK) or *Limited Damage* (SH) The high-rise building load-bearing system has been shown to meet the performance target, *the largest earthquake considered* referred to as

and under DD-1 earthquake ground motion with a 2% probability of exceedance in 50 years (recurrence period 2475 years)**Table 3.4(b)** According to *normal performance target* aspect *Prevention of Immigration* (GO) or *advanced performance target* aspect *Controlled Damage* (It will be demonstrated that the performance target (KH) has been achieved.

13.6.1.2 –Above 13.6.1.1 In order to achieve the performance target defined in, as a result of the three-dimensional nonlinear calculation of the high-rise building load-bearing system in the time history domain in Phase III, **13.6.4** According to the obtained *basis for evaluation* deformation, internal force and relative storey drift demands **13.6.5** It will be shown to be smaller than the performance limits defined in . If these conditions are not met, the necessary improvements will be made in the carrier system and the analysis will be repeated and the design will be developed and finalized in this way.

13.6.2. Stage III Structural System Modeling

13.6.2.1 –In Phase III, for nonlinear computation in time history 5.4 The modeling rules given in will be applied. Rules that differ from these rules **13.6.2.2** to **13.6.2.4** It is stated in .

13.6.2.2 –For load combination in Stage III 5.2.2 will be taken as basis.

13.6.2.3 –Additional eccentricity effect 13.6.3.3 will be taken into consideration according to.

13.6.2.4 –The damping ratio will be taken as 2.5%.

13.6.3. Stage III Earthquake Calculation

13.6.3.1 –Stage III Earthquake Calculation of the high-rise building structural system under the influence of DD-1 earthquake ground motion **5.7** Explained in *nonlinear calculation method in time history* will be done with.

13.6.3.2 – 5.7.3.2 According to *Rayleigh proportional damping matrix's* or *modal damping matrix* In the modes to be considered for the creation of , the damping ratio will be taken as 2.5%.

13.6.3.3 –In Phase I calculations, on any i-th floor of the building Table 3.6 defined in **A1** type irregularity and torsional irregularity coefficient $\eta_{bi} > 1.5$

in case, **4.5.10.2** The additional eccentricity effect defined in will be taken into account in the Stage III earthquake calculation. At each floor *additional floor mass moment of inertia 4.5.10.2(b)* and will be calculated according to **4.5.10.3** will be taken into consideration.

13.6.4. Formation and Internal Force Demands Based on Evaluation in Phase III

13.6.4.1 –In elements with ductile behavior, the deformation requests that form the basis for evaluation are the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated as the average of the largest absolute values of the results obtained from each.

13.6.4.2 –Critical internal forces aspect **13.3.2.2** The basic demands for the evaluation of internal forces defined in , the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated by adding one standard deviation to the mean of the largest absolute values of the results obtained from each, but will not be taken more than 1.5 times and less than 1.2 times the mean.

13.6.4.3 – Non-critical internal forces aspect **13.3.2.3** The basic demands for the evaluation of internal forces defined in , the calculations made (at least $2 \times 11 = 22$ calculations) will be calculated as the average of the largest absolute values of the results obtained from each.

13.6.5. Stage III Performance Evaluation

13.6.5.1 –Under the influence of DD-1 earthquake ground motion **13.6.1.1** According to *normal performance target* aspect *Prevention of Immigration (GO)*, *advanced performance target* as if *Controlled Damage* (To achieve the performance target (KH) below **(a)** And **(b)** and the deformation limits given in **(c)** It is necessary not to exceed the internal force limits given in .

(a) In reinforced concrete members with ductile behavior **13.6.4.1** Calculated according to *shape shifting* demands **5.8.1** It will not exceed the limit values given for the relevant performance level.

(b) In steel members with ductile behavior **13.6.4.1** Deformation requests calculated according to , **ANNEX 5C** It will not exceed the limits given for the relevant performance level.

(c) In elements that do not have ductile behavior **13.6.4.2** And **13.6.4.3** Internal force demands calculated according to for the relevant reinforced concrete elements **Chapter 7** for steel elements **Chapter 9** It will be shown that the internal force capacities are smaller than the internal force capacities defined in . In the calculation of internal force capacities, instead of the characteristic material strengths, **Table 5.1** Given in *average (expected) material strength* will be taken as basis.

13.6.5.2 –In the third stage, as a result of the nonlinear calculations made for the high-rise building structural system under the DD-1 earthquake ground motion, $2 \times 11 = 22$ The average relative storey drift ratio obtained from an earthquake will not exceed 0.03, and the largest relative storey drift ratio obtained from a single earthquake will not exceed 0.045.

13.7. FINALIZATION OF HIGH-RISE BUILDING DESIGN AND DESIGN OF FOUNDATIONS

13.7.1. Finalization of High-Rise Building Design

The dimensions and reinforcement of the vertical load-bearing elements (wall and column) and foundations determined in Stage I will not be reduced in Stage III. If a reduction is desired in these elements, the sections will be changed and all three calculation stages will be repeated. For other load-bearing elements (beam, tie beam, etc.), if deemed necessary, reductions can be made in Stage III. In this case, only the Stage III calculation will be repeated.

13.7.2. Performance Evaluation / Design of Foundations

13.7.2.1 –The force demands based on the performance evaluation/design of the high-rise building foundations, which were pre-designed in Phase I, under DD-1 earthquake ground motion in Phase III. **13.6.4** It will be calculated according to .

13.7.2.2 –The foundations **Chapter 16** Reinforced concrete material strengths in performance evaluation/design according to *average (expected) strength* will be taken into consideration.

13.7.2.3 –Regarding nonlinear soil behavior **16.8.3.4(b)** The calculations defined in will be made.

13.7.2.4 –In tall buildings with pile foundations;

(a) In Phase III *structure-pile-soil dynamic interaction* calculations, both under the earthquake ground motion defined in the bedrock and between the soil environment and the piles *kinematic interaction*ⁱ, and also under the effects transferred from the superstructure to the soil-pile system *inertial interaction*^j, including, **16.10** It will be done with the methods defined for high-rise buildings.

(b) If the effects occurring in the superstructure carrier system as a result of interaction calculations are more favorable than those obtained in the superstructure analyses performed without taking interaction into account, the interaction effects on the superstructure will not be taken into account.

13.8. BUILDING HEALTH MONITORING SYSTEM TO BE INSTALLED IN HIGH-RISE BUILDINGS

13.8.1 –DTS = 1, 1a, 2, 2a class buildings $H_N > 105$ m, the records are real

in a manner that allows it to be kept simultaneously by the Disaster and Emergency Management Presidency and the building owner, in accordance with the directive published by the Presidency *building health monitoring system* will be installed. It will be shown in the system application projects.

13.8.2 –Building owners will be responsible for the maintenance and preservation of the monitoring system.

INFORMATION ANNEX 13A – HIGH-RISE BUILDING SUPPORT SYSTEMS

13A.1. TYPICAL HIGH-RISE BUILDING SUPPORT SYSTEMS

A general description of typical high-rise building load-bearing systems is given below. In practice, load-bearing systems that fit this description are generally used.

13A.1.1 –A typical high-rise building, usually sunken into the ground *basement* a low-rise building located above the floors *base* (podium) and a tower rising above it *tower* The vast majority of tall buildings fit this definition, but some tall buildings may not have a plinth.

13A.1.2 – *Tower* The carrier system is a central structure that absorbs a significant portion of the earthquake and wind effects, or even almost all of them, and is generally constructed of reinforced concrete. *core curtain* It consists of frames formed by reinforced concrete, steel or composite columns and beams together with the system and/or beamless slabs supported directly on columns or walls.

13A.1.3 –Care should be taken to ensure the necessary torsional rigidity and strength for the tower carrier system, with frames of sufficient rigidity and strength arranged all around.

13A.1.4 –Low-rise buildings located above the basement floors buried in the ground and occupying a larger area in plan compared to the tower section above. *base* The load-bearing system of the section is generally a system consisting of walls and frames and/or beamless slabs in addition to the tower load-bearing system continuing from top to bottom. The load-bearing system of the basement floors buried in the ground consists of the surrounding rigid walls in addition to the load-bearing system continuing from top to bottom.

13A.1.5 –The fact that the tower is located in a common carrier system without being separated from the base and rigid surrounding walls basement floors causes the effects coming from the tower to be transferred to the base and to a large extent to the rigid basement walls. Thus, both the height of the tower is shortened and the overturning moment of the tower at the foundation level is significantly reduced. However, in return, the transition from the tower to the base and especially to the surrounding basement wall *transition floors (transfer floors)* s floors are subjected to great strain within their own planes.

13A.1.6 –The main element of the tower carrier system *core curtain* The system is formed by independently operating U, E or similar sectioned shears and usually connecting them to each other with reinforced concrete or steel tie beams with high rigidity and strength. *tie beam curtain(s)* It is a carrier system consisting of .

13A.1.7 –In special cases, it is supported by the core curtain group in the center and strong columns on the outside in certain floors or floors. *outriggers* systems can also be included in the tower carrier system.

13A.1.8 –Multi-piece steel *mega-columns* And *mega-beams* Created from, when necessary *mega crosses* s also included *mega-frame systems* can also be used as high-rise building carrier systems.

13A.2. CONSIDERATION IN THE ARRANGEMENT OF THE HIGH-RISE BUILDING CARRIER SYSTEM MATTERS TO BE TAKEN

13A.2.1 –High-rise building load-bearing systems should be as regular as possible both in plan and vertical direction. Care should be taken to place the core shear wall in the center of the building and to select load-bearing systems that are symmetrical or nearly symmetrical with respect to both main axes in the plan.

13A.2.2 –In case of using reinforced concrete beamless floor systems, it should be a principle to create strong external frames by connecting the surrounding columns or walls to each other with as rigid beams as possible, to work together with the core wall in the center, and thus to provide sufficient torsional rigidity and strength in the carrier system (See Figure 2). **13A.1.3** In this context, it should be ensured that the period of the dominant torsional mode of the carrier system is not longer than the period of the dominant translational mode.

13A.2.3 –In the carrier system *outrigger* In case of use (See **13A.1.7**) measures should be taken to limit the excess strength of the support beams so that the axial forces they will transfer to the core walls and support columns on which they are supported do not exceed the anticipated levels. In this context, the support beams should be designed with controllable yield levels. *buckling-resistant steel braces* It may be preferred to arrange them as cage systems using .

CHAPTER 14 – INSULATED BUILDING CARRIER UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGNING SYSTEMS

14.0. SYMBOLS AND DEFINITIONS

14.0.1. Icons

A	= Gross cross-sectional area of the elastomer insulation unit
A_L	= Elastomer layer edge surface area
A_P	= Area of lead core in lead core elastomer insulation units
A_r	= Surface area of a single layer of elastomer bonded to a steel plate, subjected to load
b	= The longest plan length of the structure d and the shortest plan length perpendicular to it =
B	The diameter of the elastomer bonded to the steel plate
B'	= Dimension defined in Figure 14A.1
B_L	= Lead core diameter
d	= Longest plan length of the building
D	= Displacement value for insulation unit
D_D	= Insulation in the direction considered at the design earthquake ground motion level displacement at the effective rigidity center of the system
D_{TD}	= Insulation in the direction considered at the design earthquake ground motion level Total displacement at the effective rigidity center of the system
D_M	= Insulation at the maximum earthquake ground motion level in the direction considered displacement at the effective rigidity center of the system
D_{TM}	= Insulation at the maximum earthquake ground motion level in the direction considered Total displacement at the effective rigidity center of the system
D_y	= Effective yield displacement of the insulation unit
to	= External distance measured between the superstructure center of gravity and the insulation system rigidity center The sum of the eccentricity and the additional eccentricity calculated as 5% of the plan dimension in the direction perpendicular to the earthquake loading
TO_d	= Total earthquake effect based on design
TO_c	= Compressive modulus of the elastomer insulation unit
TO'_c	= Compression modulus based on constant volume assumption of the elastomer insulation unit
TO_{he}	= Modulus of elasticity
TO_v	= Vertical stiffness modulus
F	= In the insulation unit resistance against displacement =
F_S	earthquake force acting on the superstructure

F_y	= Effective yield strength
F_Q	= Characteristic strength of the insulation unit (force-displacement cycle) force corresponding to zero displacement)
g	= Gravitational acceleration
G	= Constant load
G_v	= Shear modulus of elastomer material
G_{to}	= Effective shear modulus
H	= Total height of insulation unit
h_i	= distance between i'th floor and (i-1)th floor (floor height)
h_x	= height of x floor
I	= Building Importance Coefficient
K	= Bulk modulus of rubber material =
KK	Uninterrupted Use Performance Level
K_D	= Isolation system at design earthquake ground motion displacement level nominal effective stiffness
K_M	= The insulation system at the level of the largest earthquake ground motion displacement nominal effective stiffness
k_1	= Initial stiffness of the elastomer insulation unit (elastic stiffness)
k_2	= Secondary (extra-elastic) stiffness of the elastomer insulation unit
k_{to}	= Deffective stiffness corresponding to displacement
k_v	= Vertical rigidity of the elastomer insulation unit
P	= The largest load occurring under the combined effect of vertical load and earthquake load. vertical load
P_{cr}	= Elastomer insulation units in case of no horizontal displacement buckling load
$P_{c'r}$	= Buckling of elastomer insulation units in case of horizontal displacement burden
P_{str}	= Axial load due to unit displacement of elastomer insulation units capacity
R_c	= Radius of curvature of the insulation unit sliding surface
R	= Earthquake load reduction coefficient =
S	Shape coefficient
SH	= Limited Damage Performance Level

$S_{ae}^{(DD-2)}$	= Horizontal spectral for 5% damping at design earthquake ground motion level acceleration [g]
$S_{ae}^{(DD-2)} (T_d)$	= At design earthquake ground motion level T_d Spectral acceleration in the period [g]
$S_{ae}^{(DD-1)}$	= Horizontal spectral for 5% damping at the maximum earthquake ground motion level acceleration [g]
$S_{ae}^{(DD-1)} (T_M)$	= At the level of the largest earthquake ground motion T_M Spectral acceleration in the period [g]
t	= Elastomer layer thickness
T_D	= Effective vibration at the design displacement level of the earthquake isolated building period
T_p	= Calculated using the gross section stiffness of the uninsulated building dominant mode vibration period
T_M	= Effective vibration of the earthquake isolated building under the largest displacement period
T_r	= Total elastomer thickness
T_v	= Vertical vibration of the structure insulated with elastomer insulation units period
V_D	= Design earthquake ground motion level DD-2 calculated for the superstructure force
V_M	= Maximum earthquake ground motion level DD-1 calculated for the superstructure force
y	= Seismic loading between the insulation system rigidity center and the relevant element distance in perpendicular direction
Q	= Reduced live load according to TS 498
W	= Weight of the building against the total mass during the earthquake
W_d	= Consumed energy (Area inside the force-displacement curve)
w_i	= Dead load of the i 'th floor of the building
w_x	= total weight of x floor
β_{to}	= Effective damping ratio
τ_{yp}	= Shear yield stress of lead material
ε_b	= Elastomer material strain value at break = Angular
y_c	strain caused by vertical loads
$y_{c,E}$	= DD-1 and DD-2 earthquakes are caused by pressure under the influence of ground motion. incoming unit shapeshifting

$y_{c,st}$	= Angular deformation caused by pressure
$y_{r,st}$	= Relative rotation between the top and bottom plates of the insulation unit unit deformation
y_s	= Total earthquake design from displacement originating from angular shapeshifting
$y_{s,E}$	= Horizontal earthquake ground motions under the influence of DD-1 and DD-2 unit deformation resulting from displacement
$y_{s,st}$	= Horizontal displacement caused by effects other than earthquakes (expansion, wind, etc.) angular deformation resulting from displacement
μ	=Effective coefficient of friction
η_{bi}	= Torsional Irregularity Coefficient defined at the i'th floor
η_D	= Damping scaling factor for design earthquake ground motion level
η_M	= Damping scaling factor for the maximum earthquake ground motion level
Δ_s	=The amount of horizontal displacement caused by horizontal loading
θ_s	=Design rotation angle
ξ	= For the isolation unit at the DD-1 or DD-2 earthquake ground motion level calculated effective damping percentage

14.0.2. Definitions

Insulation System:It is the entirety of the components that provide vertical rigidity, horizontal flexibility and energy consumption to the structure at the insulation interface.

Insulation Interface:It is the interface where the insulation system is placed.

Infrastructure:The part of the building's insulation interface that lies beneath it, including the foundation system.

Superstructure:It is the structural part of the building that remains above the insulation interface.

Insulation Unit:Each of the elements that make up the insulation system is called an insulation unit. The insulation unit acts as a support that allows the insulation system to act horizontally flexible and vertically rigid under earthquake loads and to make large horizontal displacements. Insulation units must have energy dissipation properties.

Design Earthquake Ground Motion: 2.2.2DD-2 earthquake ground motion level defined in .

Design Displacement:Design Earthquake It is the maximum relative displacement (relative displacement between the bottom of the superstructure and the top of the substructure) occurring at the effective stiffness center at the interface of the isolation system in the earthquake direction considered under the effect of Ground Motion.

Total Design Displacement:Design Earthquake Ground Motion is the maximum relative displacement (relative displacement between the bottom of the superstructure and the top of the substructure) obtained in each of the insulation units at the insulation interface, in the earthquake direction considered, taking into account the torsional behavior of the structure.

Largest Earthquake Ground Motion: 2.2.1DD-1 earthquake ground motion level defined in .

Maximum Displacement:It is the maximum relative displacement (relative displacement between the bottom of the superstructure and the top of the substructure) occurring at the effective stiffness center at the interface of the isolation system under the effect of the maximum earthquake ground motion, in the direction of the earthquake considered.

Total Maximum Displacement:It is the maximum relative displacement (relative displacement between the bottom of the superstructure and the top of the substructure) obtained in each of the insulation units at the insulation interface under the effect of the maximum earthquake ground motion, in the earthquake direction considered, taking into account the torsional behavior of the structure.

Effective Damping Ratio:It is the effective damping ratio that the insulation system will exhibit under displacements of certain amplitudes (determined separately for design displacement and maximum displacement).

Effective Stiffness:It is the stiffness value obtained by dividing the total shear force occurring in the insulation system by the relative displacement value at the insulation system rigidity center (determined separately for design displacement and maximum displacement).

Effective Center of Rigidity:It is the stiffness center calculated based on the effective horizontal stiffness of the units in the isolation system against the displacement under the design earthquake ground motion.

Effective Insulation Period:It is the natural vibration period of a single degree of freedom system whose mass is equal to the superstructure mass and whose stiffness is equal to the effective horizontal stiffness of the isolation system (determined separately for design displacement and maximum displacement).

14.1. SCOPE

14.1.1 –This Section covers the principles of earthquake design for new buildings to be seismically isolated and existing buildings to be strengthened by seismically isolated.

14.1.2 –This Chapter covers the use of elastomer and curved surface friction isolation units in earthquake isolation.

14.2. PURPOSE

14.2.1 –The main purpose of earthquake insulation is to reduce the earthquake forces acting on the building's load-bearing system. For this purpose, it is necessary to extend the natural vibration period of the isolated building and/or increase its energy consumption capacity.

14.3. GENERAL DESIGN PRINCIPLES

14.3.1 –In designs falling within the scope of this Section, the insulation system shall be located at an insulation interface located beneath the superstructure.

14.3.2 –In buildings with earthquake insulation, Building Importance Factor $I=1$ will be taken.

14.3.3 –Below in design (a) And (b) and stated in 2.2 Earthquake ground motions at two different levels defined in will be considered.

(a) DD-2 Earthquake Ground Motion Level (Design Earthquake Ground Motion)

(b) DD-1 Earthquake Ground Motion Level (Maximum Earthquake Ground Motion)

14.3.4 –Infrastructure and superstructure of buildings with earthquake isolation can be designed according to limited ductility level.

14.3.5 –In order to reduce torsional effects, the effective rigidity center of the isolation system and the vertical projection of the superstructure mass center at the isolation interface will be as close as possible to each other.

14.3.6 –The maximum displacement calculation of the insulation system is the parameters of the insulation units at the DD-1 earthquake ground motion level. *lower limit values* and the calculation of the maximum horizontal force acting on the superstructure is based on the parameters of the isolation units at the DD-2 earthquake ground motion level. *upper limit values* will be done with.

14.3.7 –The difference between the internal strength of the insulation system at the maximum displacement under the DD-1 earthquake ground motion and the internal strength at 50% of the maximum displacement is at least 0.025 M . In addition, the insulation system will be designed to be able to create horizontal centering force in displacement and the vibration period calculated using secondary (extra-elastic) rigidity shall not be greater than 6 seconds. Forces independent of displacement shall not be taken into account in providing the centering force and tangential rigidity conditions.

14.3.8 –In the entire building model, it is possible to model only the insulation units for nonlinear behavior, while the superstructure and substructures are modeled as linear elastic.

14.3.9 –Effective section stiffness factors to be used in modeling the structure **Table 13.1** will be determined using .

14.3.10 –Nominal values of relevant parameters of the insulation system can be used in determining floor accelerations in the building and in equipment verifications.

14.3.11 –The isolation system must have an unobstructed movement area equal to the largest calculated displacement.

14.3.12 –For matters not specified in this section, TS EN 1337-1 and TS EN 15129 standards will be taken as basis.

14.4. BASIC PROPERTIES OF INSULATION UNITS

14.4.1 –The insulation units that make up the insulation system to be used will show increasing resistance to increasing horizontal displacements under the combined effects of vertical load and earthquake. On the other hand, any changes that may occur in their physical properties will remain within the limits specified in this Section.

14.4.2 –Upper and lower limit values for insulation units are for elastomer insulation units.

14.12.1 According to , for curved surface friction isolation units **14.13.1** It will be determined according to.

14.4.3 –In the design of insulation unit elements, precautions will be taken for effects such as wind, temperature, aging, creep, fatigue and humidity, in addition to earthquakes, and these effects will be taken into account in the calculations.

14.4.4 –In the design of the insulation system, it is essential that no tensile force occurs in the insulation units at any earthquake ground motion level. However, in special cases, special quality insulation units can be used to withstand the tensile forces. The rigidity of such insulation units under tension and their ability to withstand the tensile forces will be verified by tests.

14.5. STABILITY OF THE INSULATION SYSTEM

14.5.1 –The stability of the insulation system and the elements under the insulation system will be demonstrated by the calculation using the lower limit values of the parameters of the insulation units at the DD-1 earthquake ground motion level.

14.5.2 –All elements of the isolation system shall have the capacity to carry the axial forces at the maximum displacement in a stable manner.

14.5.3 –The building insulation system shall have a safety factor of at least 1.0 against overturning at the DD-1 earthquake ground motion level. All seismic and non-seismic load combinations shall be taken into account in the safety factor calculation. In the event of any lifting or shrinkage in any unit in the insulation system, the calculation shall show that the stability of the building is not compromised.

14.6. WIND, FIRE AND ENVIRONMENTAL EFFECTS

14.6.1 –The maximum relative drift that will occur at the insulation interface under wind load will not exceed the relative floor drift value allowed for the building at the DD-2 earthquake ground motion level. In cases where the insulation system cannot provide this limit on its own, locking mechanisms that will only work under wind load can be used.

14.6.2 –The fire resistance of the insulation system shall be equal to or greater than the fire resistance required for other vertical load-bearing elements of the building.

14.6.3 –Changes that may occur in the mechanical properties of the insulation units over time due to production and placement or due to physical and environmental effects will be taken into account for the most unfavorable situations during the design phase.

14.7. INSPECTION, MONITORING AND MAINTENANCE OF THE INSULATION SYSTEM

14.7.1 –After the insulation units are placed on the insulation interface, a conformity inspection will be carried out on site and it will be checked that the insulation units are placed and connected in a way that will provide the displacements specified in the project, and that the installation elements that cut the insulation interface have the features specified in the project.

14.7.2 –Insulation units must be accessible throughout the life of the building, and sufficient work space and transition space must be left for the insulation unit to be replaced if necessary.

14.7.3 –Procedures for periodic maintenance and replacement of insulation units when necessary will be defined for a period covering the life of the building in a report prepared by the engineer responsible for the design.

14.8. APPROVALS AND DOCUMENTS

14.8.1 –Insulation units will have the CE mark.

14.9. PERFORMANCE TARGETS

14.9.1 –Insulated buildings *Earthquake Design Class* Performance targets and design approaches to be applied according to **Table 3.5** is given in .

14.9.2 –In the design of buildings with earthquake insulation **Table 3.5** In order to achieve the targeted performance levels under DD-2 earthquake ground motion, earthquake load reduction coefficient R and excess strength coefficient D **Table 14.1** It will be determined according to.

Table 14.1– Earthquake Load Reduction and Strength Excess Factors

Performance Target	R	D
KK	1.2	1.2
SH	1.5	1.5

14.10. LOAD COMBINATIONS TO BE USED IN DESIGN

14.10.1. Load Combinations to be Used in the Design of the Insulation Interface and the Substructure Elements Below It

In the calculation of the loads acting on the elements located at the insulation interface and below it. **Equation (14.1), Equation (14.2), Equation (14.3)** The load combinations given in will be taken into account.

$$1.4G+1.6Q \quad (14.1)$$

$$1.2G+Q\pm TO_d \quad (14.2)$$

$$0.9G\pm TO_d \quad (14.3)$$

14.10.2. Load Combinations to be Used in the Design of Superstructure Elements on the Isolation Interface

Load combinations to be used in superstructure design and including earthquake effects **4.4.4** is given in.

14.10.3. Load Combinations to be Used in Insulation Unit Tests

In the calculation of loads to be used in insulation unit tests **Equation (14.4) to Equation (14.8)** The load combinations given in will be taken into account.

$$1.4G+1.6Q \quad (14.4)$$

$$G+0.5Q \quad (14.5)$$

$$1.2G+0.5Q\pm TO_d \quad (14.6)$$

$$1.2G+Q\pm TO_d \quad (14.7)$$

$$0.9G\pm TO_d \quad (14.8)$$

14.11. INSULATION SYSTEM CALCULATION PRINCIPLES

14.11.1 –The insulation units will have the capacity to withstand the displacements and axial loads obtained by taking into account the nonlinear behavior of the insulation system in the time history under the effect of DD-1 earthquake ground motion.

14.11.2 –Unless otherwise determined by test data, changes in insulation unit properties due to effects such as axial force, temperature, speed, etc. will be taken into account in the design.

14.12. ELASTOMER INSULATION UNITS

14.12.1. Lower and Upper Limit Values to be Used in Design

The upper and lower limit values of the insulation unit parameters to be used in the calculations are the nominal values of the insulation unit parameters, respectively λ_{top} and λ_{lower} . Multiplying by the coefficients will be obtained with λ_{top} and λ_{lower} values. **Equation (14.9)** and **Equation (14.10)** It will be calculated with the relations given in .

$$\lambda_{top} = [1 + 0.75(\lambda_{ae, top} - 1)] \lambda_{test, top} \lambda_{Spec, Top} \quad (14.9)$$

$$\lambda_{lower} = [1 - 0.75(1 - \lambda_{ae, bottom})] \lambda_{experiment, sub} \lambda_{Spec, Bottom} \quad (14.10)$$

Any λ_{top} and λ_{lower} If the value cannot be determined by prototype or production control tests, the calculations **Table 14.2** The lower and upper limit values given in will be used.

Table 14.2 – Elastomer Insulation Units Strength and Stiffness Parameters Lower and Upper Limit Recommended Values

	Medicine	F_Q		k_2	
		<i>lower</i>	<i>top</i>	<i>lower</i>	<i>top</i>
λ_{ae}	KCE	1.00	1.10	1.00	1.30
	YSE	1.00	1.30	1.00	1.40
$\lambda_{experiment}$	KCE	0.70	1.30	0.90	1.30
	YSE	0.70	1.30	0.90	1.30
λ_{Spec}	KCE	0.85	1.15	0.85	1.15
	YSE	0.85	1.15	0.85	1.15

Table 14.2 In , KÇE indicates lead core elastomer type insulation unit and YSE indicates high damping elastomer type insulation unit. Here F_Q lead core rubber type insulation unit characteristic strength, k_2 indicates the secondary (extra-elastic) stiffness of the lead-core rubber type insulation unit. λ_{ae} aging and environmental effects, $\lambda_{experiment}$ effects such as loading speed and heating, λ_{Spec} represents the multipliers to be used due to variations in production.

14.12.2. Unit Strain Limits of Elastomer Insulation Units

14.12.2.1 –Unit deformations and stability conditions that will occur under earthquake, fixed loads and moving loads in elastomer insulation units will be checked.

14.12.2.2 –Unit deformation components occurring in the non-earthquake situation **Equation (14.11)**, **Equation (14.12)** And **Equation (14.13)** is given with.

(a) Angular deformation caused by pressure (**Figure 14.1a**):

$$y_{c,st} = \frac{6SP_{K1}}{A_r TO_c} \quad (14.11)$$

P_{K1} **Equation (14.1)** The vertical force obtained under the load combination given in , A_r a single elastomer The surface area of the layer bonded to the steel plate and exposed to the load. Calculation **Equation (14A.5)** And **Equation (14A.6)** is given in . **SANNEX 14A.6** The shape coefficient defined in , TO_c **Equation (14A.11)** The elastomer insulation unit calculated according to represents the pressure modulus.

(b) Angular deformation resulting from horizontal displacement due to effects other than earthquakes (expansion, wind, etc.) **Figure 14.1b**:

$$y_{s,\bar{s}} = \frac{\Delta}{T_r} \quad (14.12)$$

Δ_s is the amount of horizontal displacement resulting from horizontal loading, T_r is the total thickness of the elastomer layers.

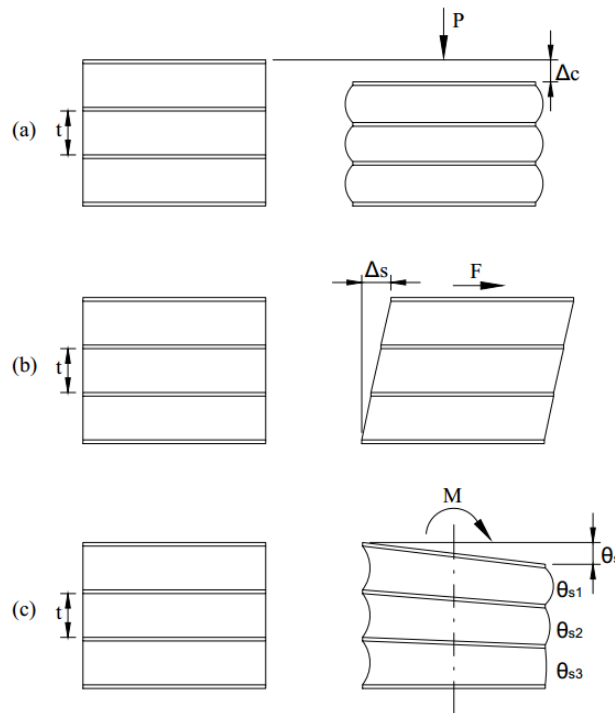


Figure 14.1

(c) Unit deformation resulting from relative rotation between the top and bottom plates of the insulation unit (**Figure 14.1c**):

$$y_{r,st} = \frac{B_2 \theta_s}{2tT_r} \quad (14.13)$$

Design rotation angle θ_s fixed load, moving load and rotation effects during manufacturing
This value will be taken as at least 0.005 radians. B the diameter of the elastomer adhered to the steel plate, t elastomer layer thickness, T_r represents the total elastomer thickness
is continuing.

14.12.2.3 –The unit deformation values that will occur in the elastomer material under the effects of earthquake and vertical load at the DD-1 and DD-2 earthquake ground motion levels are given below. **(a)** And **(b)** It will be calculated as defined in .

(a) DD-1 and DD-2 Unit deformation caused by pressure under the influence of earthquake ground motion **Equation (14.14)** will be calculated with .

$$Y_{c,E} = \frac{6SP_{k2}}{A_{re} T_{oc}} \quad (14.14)$$

P_{k2} , **Equation (14.2)** is the vertical force obtained with the load combination given in .

(b) Unit deformation resulting from horizontal displacement under the influence of DD-1 and DD-2 earthquake ground motion. **Equation (14.15)** will be calculated with .

$$y_{s,E} = \frac{D}{T_r} \quad (14.15)$$

D , It is the horizontal displacement occurring at the relevant earthquake ground motion level.

14.12.2.4 –Unit deformations that will occur in elastomer insulation units will satisfy the following conditions.

(a) Unit deformations in elastomer materials caused by vertical loads other than earthquake loads. **Equation (14.16)** And **Equation (14.17)** the smallest of the boundaries in
In these equations, ϵ_b is the rupture unit strain value of the elastomer material.

$$y_{c,st} \leq 3.5 \quad \text{or} \quad y_{c,st} < \epsilon_b / 3 \quad (14.16)$$

$$y_{c,st} + y_{s,st} + y_{r,st} \leq 5.0 \quad \text{or} \quad y_{c,st} + y_{s,st} + y_{r,st} < 0.75 \epsilon_b \quad (14.17)$$

(b) Total unit deformation resulting from horizontal displacement and rotation at the interface of the elastomer material **Equation (14.18)** And **Equation (14.19)** will provide the boundaries in .

$$y_{c,E} + y_{s,E} + 0.5 y_{r,st} \leq 6.0 \quad (14.18)$$

$$y_{s,E} \leq 2.0 \quad (14.19)$$

14.12.3. Vertical Stability of Elastomer Insulation Units

14.12.3.1 –In the absence of horizontal displacement, the buckling load of the elastomer insulation units (P_{cr}) for circular cross-section, lead-free insulation units **Equation (14.20)**, square for sectional, lead-free insulation units **Equation (14.21)**, for lead core insulation units **Equation (14.22)** will be calculated with .

$$P_{cr} = 0.218 G_v B^4 / (t T_r) \quad (14.20)$$

$$P_{cr} = \frac{0.340 G_v B^4}{t T_r} \quad (14.21)$$

$$P_{cr} = 0.218 \frac{G_v B^4 (1 - B_L / B) (1 - B_L^2 / B^2)^2}{t T_r (1 + B_L^2 / B^2)} \quad (14.22)$$

G_v elastomer material shear modulus, B_L indicates the lead core diameter.

14.12.3.2 -In cases where there is horizontal displacement, under the influence of DD-1 earthquake ground motion buckling load of elastomer insulation units P_{cr} Equation (14.23) will be calculated with .

$$P_{cr} = P_{cr} (A_{re} / A) \quad (14.23)$$

A_{re} In determining the **Figure 14.2** defined in D for the value of D_{TM} will be used.

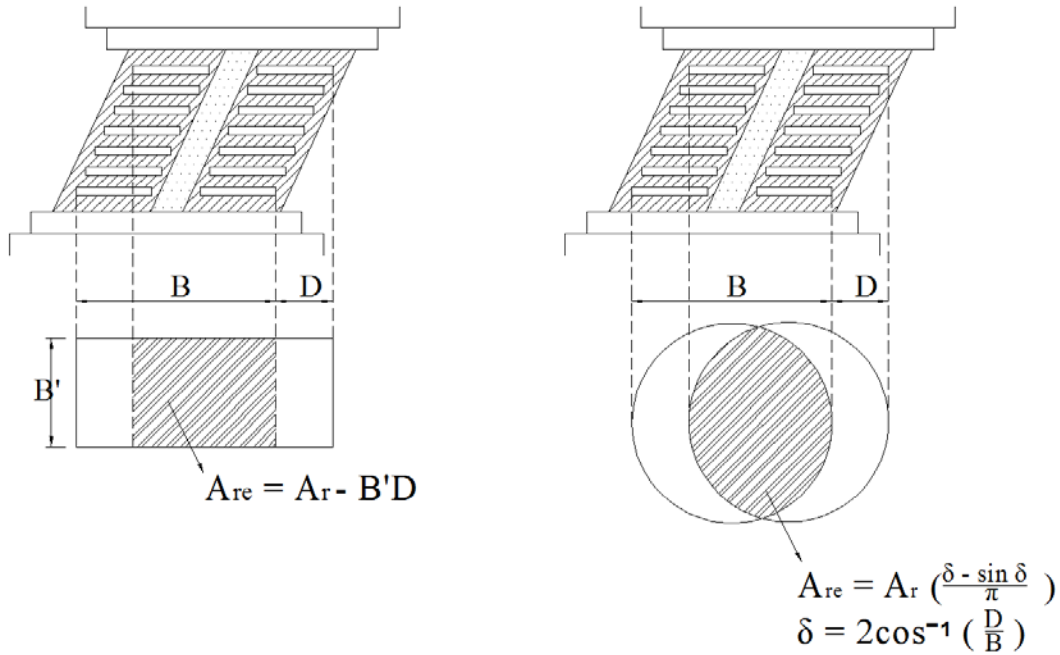


Figure 14.2

14.12.3.3 -Axial load capacity of elastomer insulation units due to unit displacement Equation (14.24) will be calculated with .

$$P_{str} = \frac{3.5 A_r E_c}{6 S} \quad (14.24)$$

14.12.3.4 -Buckling load limit of elastomer insulation units in the absence of horizontal displacement Equation (14.25) is given with.

$$\frac{\min(P_{cr}, P_{str})}{P_{K1}} \geq 2.0 \quad (14.25)$$

14.12.3.5 –Buckling load limit of elastomer insulation units under horizontal displacement occurring at the maximum earthquake ground motion level **Equation (14.26)** is given with.

$$\frac{\min(P_{cr}, P_{str})}{P_{K2}} \geq 1.1 \quad (14.26)$$

14.12.4. Connections of Elastomer Insulation Units

14.12.4.1 –Design of top and substructure connections of elastomer insulation units under maximum total displacement **14.10.1** It will be constructed to carry the horizontal and vertical forces calculated under the loads obtained by multiplying all load combinations given in by 1.1.

14.10.1' also TO_d , shows the earthquake load for the DD-1 earthquake ground motion level. Earthquake effect TO_d In determining the vertical earthquake component in the design of the isolation unit,

The friction effect between the connection surface and the insulation unit will not be taken into account in the connection calculation.

14.12.4.2 –For top and substructure connections of elastomer insulation units **Figure 14.3a** If the standard bolted connection is used, there is no need to ensure rolling stability.

The insulation units and the upper and substructure connections are not in a tension-capable arrangement (**Figure 14.3b**) and ensuring rolling stability in elastomer insulation units where horizontal loads are supported by shear wedges **Equation (14.27)** It will be checked according to.

$$D_M - \frac{(2/3)P_{K3} - B}{-k_{to}H + P_{K3}} \quad (14.27)$$

The earthquake load required to calculate this force TO_d Vertical earthquake in determining

The horizontal displacement of the insulation unit under the effect of the largest earthquake ground motion will also be taken into account. D_M and the corresponding effective linear horizontal stiffness is k_{to} with

shown. H total height of the insulation unit, P_{K3} **Equation (14.3)** represents the vertical force calculated with .

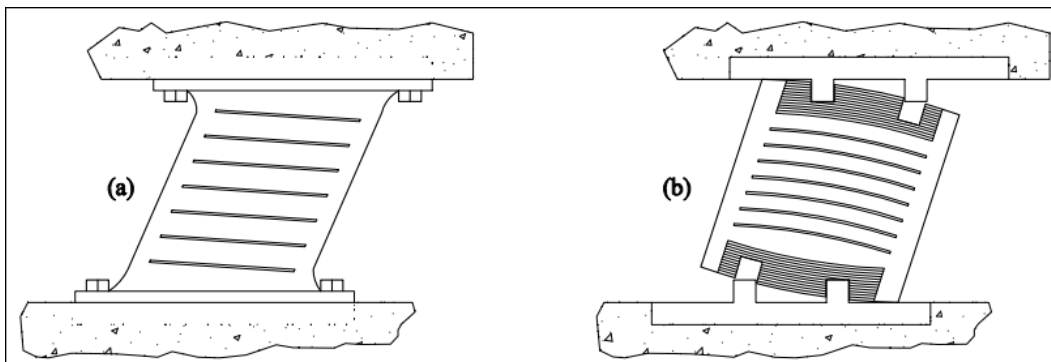


Figure 14.3

14.12.4.3 – **Figure 14.3** The adequacy of applications other than the connection types shown in will be proven separately.

14.13. CURVED SURFACE FRICTION INSULATION UNITS

14.13.1. Lower and Upper Limit Values to be Used in Design

14.13.1.1 –The upper and lower limit values of the insulation unit parameters to be used in the calculations are the nominal values of the insulation unit parameters, respectively λ_{top} and λ_{lower} with coefficients will be obtained by multiplying λ_{top} and λ_{lower} values **Equation (14.9)** And **Equation (14.10)** It will be calculated with the relations given in .

14.13.1.2 –Any λ_{top} and λ_{lower} the value of prototype or production control experiments In case it cannot be determined, the accounts **Table 14.3** The lower and upper limit values given in will be used.

Table 14.3 – Lower and Upper Limit Recommendations for Curved Surface Friction Isolation Units Values

	μ	
	<i>lower</i>	<i>top</i>
λ_{ae}	1.00	1.20
$\lambda_{experiment}$	0.70	1.30
λ_{Spec}	0.85	1.15

Lower and upper limit values are given for curved surface friction isolating units that are protected (isolated) from environmental influences and contain an unlubricated PTFE friction layer.

Here μ The curved surface friction isolation unit indicates the effective friction coefficient. λ_{ae} aging and environmental effects, $\lambda_{experiment}$ effects such as loading speed and heating and λ_{Spec} refers to the multipliers to be used due to variations in production.

14.13.2. Connections of Curved Surface Friction Isolation Units

14.13.2.1 –Design of top and substructure connections of curved surface friction isolation units under maximum total displacement, **14.10.1** To carry the horizontal and vertical forces calculated under the loads obtained by multiplying all load combinations given in by 1.1 will be done. Here TO_d , shows the total earthquake load for the DD-1 earthquake ground motion level.

Earthquake load TO_d The vertical earthquake component is also taken into account in determining

The friction effect between the connection surface and the insulation unit will not be taken into account in the connection calculation.

14.14. CALCULATION METHODS

14.14.1. Selection of Calculation Method

14.14.1.1 –If all the conditions defined below are met, seismic calculation of the superstructure and infrastructure **14.14.2** Given in *Effective Earthquake Load Method* can be done with.

(a) The building is located on ground type ZA, ZB, ZC or ZD.

(b) The effective period of the insulated building under the influence of DD-1 earthquake ground motion is less than 4.0 seconds.

(c) The maximum number of floors of the building on the insulation system is 4, and the total height above the insulation interface is 20 meters.

(d) There is no lifting or shrinkage in the insulation units.

(to) The effective damping ratio is below 30%.

(f) Torsional irregularity coefficient η on each floor of the building on the insulation system $\eta \leq 2.0$ condition is met and in the building **B2** There is no type of irregularity.

(g) Vibration period of the building in the vertical direction $T_v \leq 0.1$ s is.

14.14.1.2 – 14.14.1.1 Given in (a), (b) And (d) superstructure and infrastructure in cases where the conditions are met **14.14.3** It can be calculated by the Mod Combination Method given in .

14.14.1.3 – Superstructure and infrastructure, in all cases 5.7 It can be calculated by the Nonlinear Analysis Method in Time History given in .

14.14.2. Effective Earthquake Load Method

14.14.2.1 – Isolation unit displacement for design earthquake ground motion level DD-2 D_D

Equation (14.28) will be calculated with .

$$D_D = 1.3 \frac{g}{4\pi^2 \eta} S_{ae}^{(DD-2)} (T_D) \quad (14.28)$$

14.14.2.2 – Building effective period for design earthquake ground motion level DD-2 T_D

Equation (14.29) will be calculated with .

$$T_D = 2\pi \sqrt{\frac{W}{gK_D}} \quad (14.29)$$

14.14.2.3 – Horizontal displacement of the isolation unit for the maximum earthquake ground motion level DD-1 D_M **Equation (14.30)** will be calculated with .

$$D_M = 1.3 \frac{g}{4\pi^2 \eta} T_M^2 S_{ae}^{(DD-1)} (T_M) \quad (14.30)$$

14.14.2.4 – Building effective period for the maximum earthquake ground motion level DD-1 T_M

Equation (14.31) will be calculated with .

$$T_M = 2\pi \sqrt{\frac{W}{gK_M}} \quad (14.31)$$

14.14.2.5 – Damping scaling coefficients η_D and η_M **Equation (14.32)** will be calculated with .

$$\eta = \sqrt{\frac{10}{5 + \xi}} \quad (14.32)$$

Here ξ , It is the effective damping percentage calculated for the isolation unit displacements at the DD-1 or DD-2 earthquake ground motion level.

14.14.2.6 –Total displacement of the isolation interface at DD-2 and DD-1 earthquake ground motion levels, **Equation (14.28)** And **Equation (14.30)** It will be calculated by adding the additional displacements due to the torsion of the superstructure and insulation system to the displacement values calculated from . In case of a uniform distribution, the total displacements will be **Equation (14.33)** And **Equation (14.34)** cannot be less than the values calculated with .

$$D_{TD} = D_{D-1} + y \frac{12 t o}{b_2 + d_2} \quad (14.33)$$

$$D_{TM} = D_{M-1} + y \frac{12 t o}{b_2 + d_2} \quad (14.34)$$

Calculated displacement values taking into account torsions **Equation (14.33)** And **Equation (14.34)** If the values are smaller than the values calculated with **Equation (14.33)** And **Equation (14.34)** Instead of the values calculated with 1.1, the values obtained from the calculation D_{Mand} and 1.1 D_o values greater than these will be taken into account.

14.14.2.7 –The force acting on the superstructure is for the design earthquake ground motion level DD-2. **Equation (14.35)** will be calculated with .

$$V_D = \frac{S_{ae}^{(DD-2)} (T_D) W \eta_D}{R} \quad (14.35)$$

14.14.2.8 –The force acting on the superstructure is for the maximum earthquake ground motion level DD-1. **Equation (14.36)** will be calculated with .

$$V_M = \frac{S_{ae}^{(DD-1)} (T_M) W \eta_M}{R} \quad (14.36)$$

14.14.2.9 –In no case shall the force acting on the superstructure be less than the design wind force value or the force value required for the insulation system to pass into secondary rigidity.

14.14.2.10 –The force acting on the superstructure is **Equation (14.37)** It will be distributed according to.

$$F_{Sx} = \frac{V_{xx} w h}{\sum_{i=1}^n w h_i} \quad (14.37)$$

14.14.2.11 –Force acting on the superstructure for design earthquake ground motion level DD-2 **Equation (14.35)** will be calculated with, R value **Table 3.5** According to performance targets in **Table 14.1** will be taken from.

14.14.2.12 –The force acting on the superstructure is for the maximum earthquake ground motion level DD-1 **Equation (14.36)** will be calculated with, R value **Table 3.5** According to performance targets in **Table 14.1** will be taken from.

14.14.2.13 –Force acting on infrastructure R = Taking 1 **Equation (14.36)** will be calculated with .

14.14.3. Mode Combination Method

14.14.3.1 –In case of using Mod Merge Method **4.8.2** In addition to the conditions given in , the rules in this Section will also be taken into account.

14.14.3.2 –The insulation system units shall be modeled in a way that shows the vertical and horizontal displacements of the system in both directions and the torsional effects with sufficient accuracy.

14.14.3.3 –Calculated displacements, if there is no irregularity in the building

14.14.2.6According to 80% of the calculated values, in the building **A1, B2** or **B3** If at least one of the following irregularities is present: **14.14.2.6** It will not be less than 90% of the values calculated according to .

14.14.3.4 –Calculated superstructure and infrastructure forces in case there is no irregularity in the building

14.14.2.7According to 80% of the calculated values, in the building **A1, B2** or **B3** If at least one of the following irregularities is present: **14.14.2.7** It will not be less than 90% of the values calculated according to .

14.14.3.5–The effective damping ratio will be limited to 30%.

14.14.4. Nonlinear Calculation Method in Time History

14.14.4.1 –Nonlinear analysis method in time history **5.7** will be applied according to **14.14.1.1(g)** In cases where is provided, the two horizontal components of the earthquake will be used, and in cases where is not provided, the vertical component will be used in addition to the two horizontal components of the earthquake.

14.14.4.2 –Selection of earthquake records to be used in earthquake calculations in the time history of earthquake-isolated buildings **2.5** All the rules given in apply. However, **2.5.2** also defined $0.2 T_p$ instead of $0.5 T_M$ (T_M earthquake isolated building under the largest displacement *upper limit values* effective vibration period calculated with), $1.5 T_p$ instead of $1.25 T_M$ (T_M -most earthquake isolated building under large displacement/*lower limit values* (the effective vibration period calculated with) will be used.

14.14.4.3 –In cases where the soil is weaker than ZD, three-dimensional dynamic structure-soil interaction calculations will be made and the results will be taken into account in the building calculations.

14.14.4.4 –Nonlinear calculation for each earthquake ground motion level and in each direction **5.7.2.1** According to , the averages of the largest values obtained in the calculations made at the relevant earthquake level and direction, which will be repeated for at least eleven record pairs, will be taken into account as the basic value for the design.

14.14.4.5 –The insulation unit displacements obtained as a result of the nonlinear calculation in the time history domain will be determined as the vectorial resultant of the displacements in perpendicular directions occurring at each time step.

14.14.4.6 –Calculated displacements at the insulation interface as a result of nonlinear calculation in the time history **14.14.2.6** It will not be less than 80% of the values calculated according to .

14.14.4.7 –The superstructure and substructure forces obtained as a result of nonlinear calculations in the time history domain are in the absence of irregularities in the building. **14.14.2.7** It will not be less than 80% of the values calculated according to .

14.14.5. Relative Floor Drift Limits and Earthquake Joints in Insulated Buildings

14.14.5.1 –Relative floor drifts in the superstructure, *Uninterrupted Use*(KK) performance level for $0.005h$, *Limited Damage*(SH) performance level for $0.01h$, *Controlled Damage*((KH) and 0.015 for performance level h will not exceed its value.

14.14.5.2 –In determining the dimensions of earthquake joints to be left in insulated buildings, in addition to the total maximum displacement of the insulation system, relative floor drifts will also be taken into account. The joint width will not be less than the sum of the absolute maximum displacements calculated for the structures on both sides of the joint.

14.15. INSULATION UNIT TESTS

14.15.1. Test Conditions

14.15.1.1 –Prototype Tests will be conducted to determine the force-displacement properties, effective damping ratios, effective horizontal and vertical rigidities and other parameters of the insulation units used in the insulation system or to control the parameters taken in the calculations, and Production Control Tests will be conducted to confirm that the change in production remains within the predetermined limits.

14.15.1.2 –Tests will be conducted under the supervision and control of the design engineer and the test results will be documented.

14.15.1.3 –Insulation units with the same features and dimensions**14.15.2**In case of application of the Prototype Tests specified in , it is not mandatory to perform these tests again. These tests of the insulation units shall be documented by the manufacturer.

14.15.1.4 –The design engineer determines the vertical load and displacement values to be used in prototype and production tests.**Table 14.4**And**Table 14.5**will be prepared in accordance with.

14.15.2. Insulation Unit Prototype Experiments

Prototype tests will be applied dynamically to at least 2 insulation units for each type of insulation unit. The displacement and period values to be used in the tests will be determined using the nominal parameters of the insulation unit. Prototype tests will be carried out in laboratories accredited according to TS ISO/IEC 17025, independent of the manufacturer. Content of prototype experiments**Table 14.4**is given in .

14.15.3. Insulation Unit Production Control Experiments

14.15.3.1 –Production control tests will be applied to 30% of each type of insulation unit produced, which is randomly selected. Any of these selected insulation units**Table 14.5**In case the acceptance conditions given in are not met, production control tests will be applied to all of these types of insulation units produced.

14.15.3.2 –The displacement and period values to be used in the tests will be determined using the nominal parameters of the insulation unit. Content of production control tests**Table 14.5**In Experiment No.2, T_0 In case the experiment is not performed with the period**Table**

14.4The Prototype Experiment No. 3 given in will be repeated at the production control Experiment No. 2 speed to determine the speed correction coefficients.

Table 14.4 – Insulation Unit Prototype Experiments

Experiment No	Vertical Load or Displacement	Horizontal Displacement / Horizontal Force	Number of Cycles	Experimental Period/ Cycle Period (s)	Acceptance Condition
1	Equation (14.4) Maximum value of vertical load calculated with	0	-	Static (180 seconds)	a
2	Equation (14.5) Average value of vertical load calculated by	Per insulation unit falling horizontal wind force or 25 mm	20	T_D	b
3	Equation (14.5) Average value of vertical load calculated by	$0.25D_D$ $0.50D_D$ $1.00D_D$ $1.00D_M$	3 3 3 3	T_D T_D T_D T_M	c,d
4	Equation (14.6) Average value of vertical load calculated by	$0.25D_D$ $0.50D_D$ $1.00D_D$ $1.00D_M$	3 3 3 3	T_D T_D T_D T_M	c,d
5	Equation (14.8) Average value of vertical load calculated by	$0.25D_D$ $0.50D_D$ $1.00D_D$ $1.00D_M$	3 3 3 3	T_D T_D T_D T_M	c,d
6	Equation (14.5) Average value of vertical load calculated by	$1.00D_{TM}$	3	T_M	c, d, e
7	Equation (14.5) Average value of vertical load calculated by	$1.00D_{TD}$	10	T_D	f
8	Equation (14.7) Maximum value of vertical load calculated with	$1.00D_{TM}$	Unidirectional Push	-	g
9	Equation (14.8) Minimum value of vertical load calculated with	$1.00D_{TM}$	Unidirectional Push	-	g
10*	Between the top and bottom plates largest vertical opening	$1.00D_{TM}$	Unidirectional Push	-	g

*Applicable to insulation units subjected to tensile stress at any loading combination or losing compressive contact between the top and bottom plate. D_{TM} after reaching the displacement Eq. 14.7 By loading the average axial load obtained with , the insulation unit will be brought back to its center position.

Table 14.5 – Insulation Unit Production Control Experiments

Experiment No	Vertical Load or Displacement	Horizontal Displacement / Horizontal Force	Cycle Number of	Experimental duration (s)/ Test speed (m/s)/ Cycle Period (s)	Acceptance Condition
1	Equation (14.4) Maximum value of vertical load calculated with	-	-	Static (180s)	a
2	Equation (14.5) Average value of vertical load calculated by	D_b	3	$T_{\phi r}$ $\geq 0.05 \text{ m/s}$	h, i

14.15.4. Insulation Unit Acceptance Conditions

The acceptance conditions of the insulation units are specified in the following items. During all tests, the insulation units shall remain stable and the force-displacement curve shall always have a positive slope.

(a) The vertical rigidity of the elastomer insulation units shall be determined on the test element. The ratio of the difference between the determined vertical rigidity and the vertical rigidity taken into account in the calculation to the value obtained from the test shall not exceed 15%. There shall be no permanent deformation in the elastomer insulation unit or on the sliding surfaces of curved surface frictional insulation units.

(b) The values obtained as a result of the experiment are, 14.6.1 will meet the conditions given in .

(c) The secondary stiffness value calculated for each cycle in all loading and displacement steps is λ used in the design, $\lambda_{\text{experiment,sub}}$ and $\lambda_{\text{test,top}}$ values of nominal secondary stiffness will remain between the lower and upper limits obtained by multiplying.

(d) For two insulation units of the same type tested, the difference between the calculated effective stiffness values in each cycle and the average effective stiffness value shall remain below 15%.

(to) Average secondary stiffness and average energy consumed; λ used in the design $\lambda_{\text{Spec, Bottom}}$ and $\lambda_{\text{Spec, Top}}$ It will remain between the lower and upper limits obtained by multiplying the nominal values with the specified values, with a tolerance of +/- 5%.

(f) The effective stiffness and effective damping ratio calculated in the second cycle for both insulation units tested shall not vary by more than 30% in all further cycles.

(g) There will be no permanent loss of pressure, tension or horizontal rigidity in the insulation unit and the insulation unit will maintain its stability.

(h) The secondary stiffness calculated for each cycle was used in the design $\lambda_{\text{experiment,min}}$ and $\lambda_{\text{test,max}}$ will remain between the lower and upper limits obtained by multiplying the nominal value by the values.

(i) The largest and smallest average secondary stiffness values obtained from each production test were used in the design $\lambda_{\text{Spec, min}}$ and $\lambda_{\text{Spec, max}}$ obtained by multiplying the nominal value by will remain between the specified lower and upper limits.

14.15.5. Design Verification

14.15.5.1 –In cases where the insulation unit acceptance conditions cannot be met, the lower and upper limit values will be reviewed and changed, all calculations will be repeated or the insulation unit design will be improved and it will be shown that the building meets the conditions specified in this Section. If necessary, prototype tests will be repeated according to the new axial load and displacement conditions.

ANNEX 14A – BASIC PROPERTIES OF ELASTOMER INSULATION UNITS

14A.1 –Elastomer insulation units are made using a combination of steel and natural rubber (or neoprene) plates.(**Figure 14A.1**).

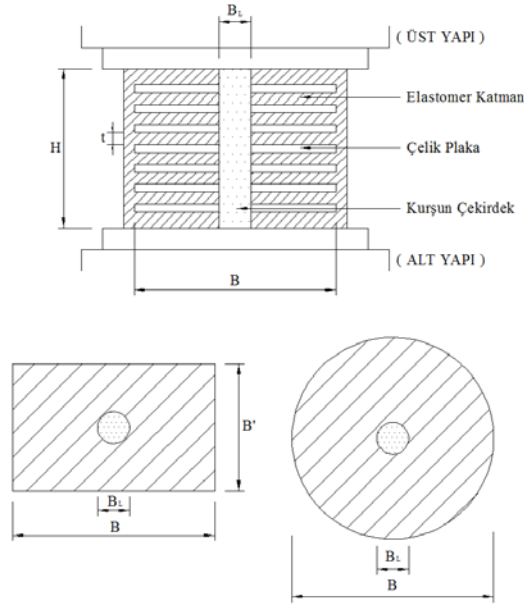


Figure 14A.1

Simplified force-displacement (loading cycle) curve of such insulation units **Figure 14A.2** is given in .

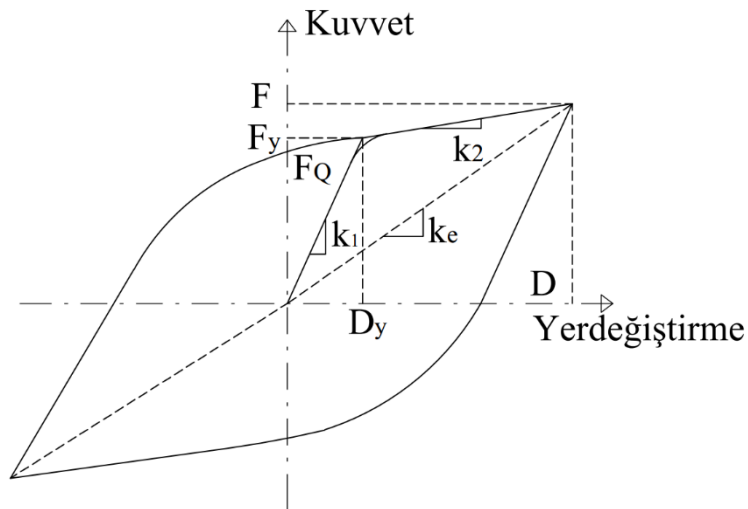


Figure 14A.2

Figure 14A.2 also:

F_Q = Characteristic strength,

k_1 = Initial (Elastic) stiffness,

k_2 = Secondary (inelastic) rigidity,

k_{to} = The effective stiffness corresponding to the displacement is,
 F = Horizontal force corresponding to displacement, =
 F_y Effective yield strength,
 D_y = is the effective yield displacement.

14A.2 –Effective stiffness of the insulation unit (k_{to}) is the horizontal force applied in the loading cycle (F) is the horizontal displacement corresponding to this force (D) is obtained by dividing by the value.

$$k_{to} = \frac{F}{D} \quad (14A.1)$$

14A.3 –Effective damping ratio β_{to} Eq.(14A.2) will be determined by. W_d is the amount of energy consumed in one charging cycle.

$$\beta_{to} = \frac{1 - W_d}{2\pi F D} \quad (14A.2)$$

14A.4 –Characteristic strength of lead core elastomer insulation unit F_Q , Equation (14A.3) will be obtained with .

$$F_Q \cong F_y = A_p \tau_{yp} \quad (14A.3)$$

14A.5 –The initial stiffness of a lead core insulation unit will be determined by the elastic stiffness of the lead core. Secondary (extra-elastic) stiffness of an elastomer insulation unit Equation (14A.4) will be obtained with .

$$k_2 = G_v (A_r / T_r) \quad (14A.4)$$

For elastomer insulation units with circular cross-section and lead core A_r Equation (14A.5) with, for elastomer insulation units with rectangular cross-section and lead core A_r Equation (14A.6) will be determined by.

$$A_r = (\pi / 4) (B_2 - B_{2L})^2 \quad (14A.5)$$

$$A_r = B B - (\pi B L / 4) \quad (14A.6)$$

14A.6 –Elastomer insulation units Shape Coefficient (S) the surface area of each elastomer layer bonded to the steel plate (subject to load) (A_r), elastomer layer edge surface area (A_L) will be found by dividing.

$$S = A_r / A_L \quad (14A.7)$$

For elastomer insulation units with circular cross-section and lead core S value Equation (14A.8) will be calculated with .

$$S = (B_2 - B_{2L}) / (4 B T) \quad (14A.8)$$

Here,

B = Diameter of elastomer sheet bonded with steel plate,

B_L = Lead core diameter,

t = Elastomer layer thickness.

14A.7 –Vertical rigidity of elastomer insulation units k_v **Equation (14A.9)** will be calculated with .

$$k_v = \frac{EA}{T_r} \quad (14A.9)$$

Equation (14A.9) Vertical stiffness modulus in TO_v **Equation (14A.10)** will be calculated with .

$$TO_v = \frac{1}{\frac{1}{TO_c} + \frac{1}{K}} \quad (14A.10)$$

Here K is the bulk modulus of the rubber material and will be taken as 2000 MPa. TO_c is the compressive modulus of the elastomer insulation unit and **Equation (14A.11)** will be calculated with .

$$TO_c = TO_0(1 + 2kS^2) \quad (14A.11)$$

In this equation TO_0 modulus of elasticity of rubber material is, $TO_0 = 4G_v$ with the relation will be calculated. k The coefficient will be taken as 0.75, 0.60 and 0.55 in cases where the hardness value of the rubber material is 50, 60 and 70, respectively.

ANNEX 14B – BASIC FEATURES OF CURVED SURFACE FRICTION INSULATION UNITS FEATURES

14B.1 –Curved surface friction isolation units consist of one or more concave steel elements with a sliding element inside.(**Figure 14B.1**). Simplified force-displacement curve (loading cycle) of this type of insulation units**Figure 14B.2**is shown in .

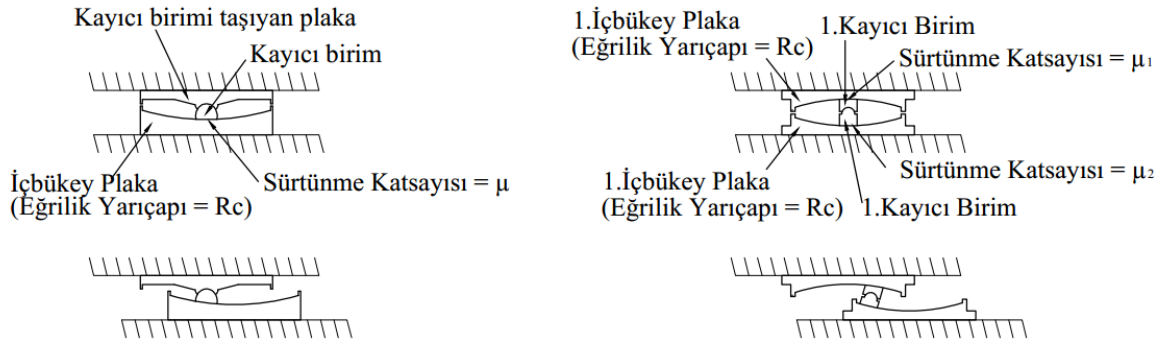


Figure 14B.1

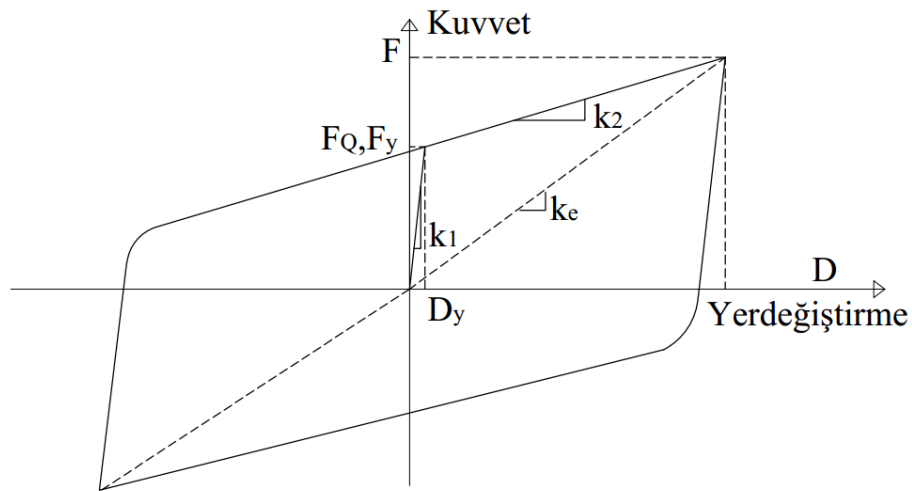


Figure 14B.2

Figure 14B.2'also:

$F_Q = F_y$ = Characteristic strength or effective yield strength,

k_1 = Initial stiffness,

k_2 = Secondary stiffness,

k_{to} = Effective stiffness opposing the displacement,

$F = D$ the force opposing the displacement, D_y = is the effective yield displacement.

14B.2 –Characteristic strength or effective yield strength **Equation (14B.1)** The effective friction coefficient (at zero displacement) is μ as stated in t_0 with vertical force acting on the insulation unit force (P) is equal to the product of .

$$F_Q = F_y = \mu_{t_0} P \quad (14B.1)$$

14B.3 –Initial stiffness is a virtual stiffness which is a very high value in calculations. Secondary stiffness **Equation (14B.2)** As stated in , the vertical force (P) curved surface friction insulation The effective radius of curvature of the sliding surfaces (R_c) will be determined by dividing by .

$$k_2 = \frac{P}{R_c} \quad (14B.2)$$

14B.4 –The effective stiffness at a given loading cycle (k_{t_0}), **Equation (14B.3)** As stated in , the maximum horizontal force applied in the cycle in question (F) the maximum horizontal displacement reached (D) will be obtained by dividing.

$$k_{t_0} = \frac{F}{D} = \frac{P}{R_c} + \frac{\mu_{t_0} P}{D} \quad (14B.3)$$

14B.5 –Effective damping ratio β_{t_0} **Equation (14B.4)** As stated in , the energy consumed in a displacement cycle (W_d) , $2\pi F D$ will be obtained by dividing by the value.

$$\beta_{t_0} = \frac{1 - W_d}{2\pi F D} = \frac{2 - \frac{\mu_{t_0}}{\pi - \mu_{t_0} + D/R_c}}{2} \quad (14B.4)$$

CHAPTER 15 – EFFECT OF EXISTING BUILDING SYSTEMS UNDER EARTHQUAKE SPECIAL RULES FOR EVALUATION AND REINFORCEMENT DESIGN

15.0. ICONS

In the dimensional expressions in this section where the following symbols are used, the forces are: *Newton*[[N], masses [tons], lengths *millimeter*[mm] and stresses *MegaPascal*[MPa] = [N/mm²] unit.

b	= Width of horizontal plates in steel coil =
b_w	Width of web of beam
BKS	= Building Usage Class =
BYS	Building Height Class
d	= Useful height of beam and column
ECHO	= Impact/capacity ratio
$(ECHO)_I$	= in column i 15.5.2.3 Effect/capacity ratio calculated according to =
f_{ctm}	= 15.2.4.3 Tensile strength of existing concrete defined according to =
f_{wow}	Yield strength of steel in steel wrapping
GÖ	= Pre-Collapse Damage =
I	Building Importance Factor =
KH	Controlled Damage
L_p	= Plastic joint length
LP	= Fibrous Polymer
L_s	= Cutting clearance
M_y	= Effective yield moment
R_a	= Earthquake Load Reduction Factor =
S	Spacing of horizontal plates in steel coil =
SH	Limited Damage
t_j	= Thickness of horizontal plates in steel coil
V_{to}	= Design shear force based on column, beam and wall
V_i	= Calculated shear force in column i = Additional
V_j	shear strength provided by steel coil
V_{arm}	= Above and below the node Chapter 4 Column shear calculated according to the smaller of the forces
V_r	= Shear strength of column, beam or wall section =
β_v	Dynamic magnification factor of shear force in wall =
φ_t	Total curvature

φ_y	= Equivalent yield curvature
θ_p	= Plastic return demand
θ_k	= Displaced axis rotation =
θ_y	Yield rotation
$\rho_{s,h}$	= Volumetric ratio of transverse reinforcement in the direction considered =
W_f	Width of the fibrous polymer strip

15.1. SCOPE

15.1.1 –The calculation rules to be applied in the evaluation of the performance of all existing and to be strengthened buildings and building-type structures under earthquake effects, the principles to be taken as basis in strengthening decisions and the strengthening design principles of the buildings decided to be strengthened are defined in this Section.

15.1.2 –Capacity calculation of existing and retrofitted masonry buildings, **Chapter 11** It will be done within the framework of the principles in. However, the information on existing masonry buildings will be compiled according to this Section.

15.1.3 –Capacity calculation of existing and reinforced steel buildings, **Chapter 4, Chapter 5** And **Chapter 9** It will be done within the framework of the principles defined for new buildings to be built in. However, information on existing steel buildings will be compiled according to this Section.

15.1.4 –Capacity calculation of existing prefabricated reinforced concrete buildings, **Chapter 4, Chapter 5** And **Chapter 8** It will be done within the framework of the principles defined for new buildings in . However, the information for existing prefabricated reinforced concrete buildings will be compiled according to this Section.

15.1.5 –The rules given in this Section do not apply to structures that are not buildings. Furthermore, the evaluation and reinforcement of registered structures and monuments with historical and cultural value are outside the scope of this Section.

15.1.6 –Following an earthquake that causes damage to the building, the seismic safety of the damaged building cannot be determined by the methods given in this Section.

15.1.7 –Following an earthquake that causes damage to the building, the principles given in this Section shall be applied for the strengthening of the damaged building and for determining the seismic performance of the strengthened building afterwards. The civil engineer responsible for the project shall decide to what extent the strength and rigidity of the existing elements shall be taken into consideration in the strengthening of the damaged building.

15.2. COLLECTING INFORMATION FROM BUILDINGS

15.2.1. Scope of Information to be Collected from Buildings

15.2.1.1 –Information on the details and dimensions of the elements to be used in determining the capacities of the load-bearing system elements of existing buildings and in evaluating their earthquake resistance, as well as the geometry and material properties of the load-bearing system, will be obtained from the projects and reports of the buildings, from observations and measurements to be made in the building, and from tests to be applied to material samples to be taken from the building.

15.2.1.2 –The procedures to be performed within the scope of collecting information from buildings are defining the structural system, determining the building geometry, foundation system and ground properties, determining the existing damage (if any) and previous changes and/or repairs, measuring the element dimensions, determining the material properties, and checking the conformity of all this information collected on site with the building's project, if any.

15.2.1.3 –The examination, data collection, compilation, evaluation, material sampling and experimentation processes defined within the scope of collecting information from buildings will be carried out under the responsibility of civil engineers.

15.2.2. Knowledge Levels

Depending on the scope of the current status information to be obtained from the examination of the buildings, the level of information for each building type and, accordingly, **15.2.12** The information level coefficients specified in will be defined. The information levels are as follows, *annoyed* And *comprehensive* The information levels obtained will be used in the calculation of the load-bearing element capacities.

15.2.2.1 –*Limited knowledge level* The carrier system features are determined by measurements to be made in the building. Limited level of knowledge only **Table 3.1** defined in *Other Buildings* (It will be applied for BKS=3).

15.2.2.2 –*Comprehensive knowledge level* More measurements are made based on limited information level.

15.2.3. Current Material Strength

The material strengths to be used in calculating the capacity of the load-bearing elements are specified in this Section of the Regulation. *existing material strength* is defined as.

15.2.4. Limited Level of Knowledge in Reinforced Concrete Buildings

15.2.4.1 –*Building Geometry*: The building's carrier system plan survey will be obtained through field work. If architectural projects are available, they will be used to assist in survey work. The information obtained should include the location and material of all reinforced concrete elements and partition walls on each floor, their axis openings, heights and dimensions, and should be sufficient to create a calculation model of the building. The foundation system will be determined by opening a sufficient number of inspection pits inside or outside the building. Short columns and similar negativities in the building will be processed on the floor plan and sections. The relationship of the building with neighboring buildings (separate, adjacent, joint present/absent) will be determined.

15.2.4.2 –*Element Details*: It is assumed that the amount and details of reinforcement in reinforced concrete elements provide the minimum reinforcement conditions on the date of construction of the building. In order to verify this assumption or to determine the extent to which it is realized, the concrete cover of at least 5% of the curtain walls and columns on each floor will be stripped and reinforcement will be determined. The concrete cover of one beam on each floor will be stripped for the purpose of reinforcement detection. The stripping process should be carried out in the middle third of the span length of the columns and beams. The stripped surfaces will then be covered with high-strength repair mortar. In addition, the number and placement of transverse and longitudinal reinforcement in 20% of the curtain walls and columns whose concrete cover is not stripped will be determined with reinforcement detection devices. The ratio of the existing reinforcement in the reinforced concrete curtain walls and columns where reinforcement detection is performed to the minimum reinforcement is expressed as *coefficient of reinforcement realization* This coefficient used in determining the wall and column capacities cannot be greater than 1. This coefficient is applied to all other walls and columns where no reinforcement is determined.

Possible reinforcement quantities will be determined. Only the reinforcement required under vertical design loads will be used for beams.

15.2.4.3 – Material Properties: At least three concrete samples will be taken from columns or walls on each floor in accordance with the conditions specified in TS EN 12504-1 and will be tested. The strength values found by testing cores with equal length and nominal diameter of 100 mm will be calculated without applying any coefficient. *existing concrete strength* can be used in the determination of. In converting the test results obtained from cores with different length/diameter ratios, appropriate conversion factors should be taken as basis. If the total number of samples is three, the lowest compressive strength obtained from the samples is taken without statistical evaluation. *existing concrete strength* If the number of samples is more than three, the larger of the values obtained from the samples (mean minus standard deviation) and (0.85 times the mean) will be taken as *existing concrete strength* It will be taken as. By evaluating the difference between the lowest value and the average of the remaining results among the test results of a group of concrete samples, it will be checked whether the lowest value is a statistically deviant result. For this purpose, in the evaluation of the sample results in the group, if the lowest single value is lower than 75% of the average of the remaining results, this sample will not be evaluated. Reinforcement class **15.2.4.2** The characteristic yield stress of this class of steel, to be determined by visual examination of the stripped surfaces as described in *available steel strength* In this examination, elements with corrosion observed in the reinforcement will be marked on the plan and this will be taken into account in the element capacity calculations.

15.2.5. Comprehensive Knowledge Level in Reinforced Concrete Buildings

15.2.5.1 – Building Geometry: If the reinforced concrete projects of the building are available, the conformity of the existing geometry to the projects is checked with measurements. If the projects show significant differences with the measurements, the project is ignored. If there is no project, the building's load-bearing system survey will be obtained with field work. The information obtained should include the location, openings, heights, dimensions and materials of all reinforced concrete elements and partition walls on each floor. Short columns and similar negativities in the building will be processed on the floor plan and sections. The relationship with neighboring buildings (separate, adjacent, joint present/absent) will be determined. The building geometry information should include the necessary details for the precise definition of the building mass. The foundation system will be determined with a sufficient number of inspection pits to be opened inside or outside the building.

15.2.5.2 – Element Details: If the building has reinforced concrete detail projects, to check the suitability of the reinforcement to the project. **15.2.4.2** The operations specified in will be applied to the same amount of reinforced concrete elements. In addition, the number and placement of transverse and longitudinal reinforcement in 20% of the walls and columns whose concrete cover is not stripped and in 10% of the frame beams will be determined with reinforcement detection devices. In case of inconsistency between the project and the application, the ratio of the existing reinforcement in the reinforced concrete elements to the reinforcement envisaged in the project will be determined. *coefficient of reinforcement realization* will be determined separately for walls, columns and beams. This coefficient used in determining element capacities cannot be greater than 1. This coefficient will be applied to all other elements where reinforcement is not determined and possible reinforcement quantities will be determined. If reinforced concrete projects or construction (application) drawings are not available, reinforcement will be determined by stripping the concrete cover of 10% of the columns and walls, at least two on each floor. The stripped surfaces will then be covered with high-strength repair mortar. In addition, the number and placement of transverse and longitudinal reinforcement will be determined with reinforcement detection devices in 30% of the columns and walls whose concrete cover is not stripped and in 15% of the beams.

15.2.5.3 – Material Properties: Not less than three columns or curtains on the ground floor and two on other floors in accordance with the conditions specified in TS EN 12504-1 and in the building

not less than nine in total, every 400 m² A concrete sample will be taken from and tested. The strength values found by testing cores with equal length and nominal diameter of 100 mm, without applying any coefficients. *existing concrete strength* can be used in the determination of. In converting the test results obtained from cores with different length/diameter ratios, appropriate conversion factors should be taken as basis. In calculating the capacities of the elements, the larger of the (mean minus standard deviation) value obtained from the samples and the (0.85 times the mean) value should be used. *existing concrete strength* It will be taken as. By evaluating the difference between the smallest value and the average of the remaining results among the test results of a group of concrete samples, it will be checked whether the smallest value is a statistically deviant result. For this purpose, in the evaluation of the sample results in the group, if the lowest single value is lower than 75% of the average of the remaining results, this sample will not be taken into consideration. The distribution of concrete strength in the building can be checked with concrete sample test results and adapted concrete hammer readings or similar non-destructive inspection tools. Reinforcement class,

15.2.5.2 As explained in the examination of the scraped surfaces, one sample will be taken for each class of steel (S220, S420, etc.) and the test will be carried out, the yield stress, rupture strength and deformation properties of the steel will be determined and its suitability for the project will be determined. If it is suitable for the project, the characteristic yield stress of the steel used in the project will be used in the element capacity calculations. *available steel yield stress* If it is not suitable, at least three more samples will be taken and the test will be carried out, the most unfavourable yield stress obtained will be used in the element capacity calculations. *available steel yield stress* In this examination, elements with corrosion observed in the reinforcement will be marked on the plan and this will be taken into account in the element capacity calculations.

15.2.6. Limited Level of Knowledge in Steel Buildings

15.2.6.1 – Building Geometry: **15.2.4.1** The conditions given in are valid as is. Only *reinforced concrete's* place *steel* will take.

15.2.6.2 – Element Details: If steel projects or manufacturing drawings are not available, all steel or other types of elements (columns, beams, joints, crosses, floors) on each floor will be dimensional checked, welding features and joint details will be determined in detail. If application projects or manufacturing drawings are available, 20% of the above-mentioned elements will be dimensional checked.

15.2.6.3 – Material Properties: If steel projects are not available, a sample of each type of steel structure element will be cut and tested, and the strength and deformation properties will be determined. Similarly, a weld sample will be cut from the building and tested. The removed samples will be filled and repaired. For bolted connections, a bolt sample will be taken and tested. In calculating the capacities of the elements, the average strengths obtained from the tests will be used. *available steel yield stress* will be taken as.

15.2.7. Comprehensive Knowledge Level in Steel Buildings

15.2.7.1 – Building Geometry: **15.2.5.1** The conditions given in are valid as is. Only *reinforced concrete's* place *steel* will take.

15.2.7.2 – Element Details: Steel detail projects of the building are available. The element dimensions and connection details specified in the projects will be verified by checking at least 20% of the total number of each element and connection type in the building.

15.2.7.3 –Material Properties: The steel class specified in the project will be checked by cutting a sample from at least one steel element and testing it. Similarly, a sample will be cut from a welded joint in the project and its strength will be checked by testing it to see if it complies with the project. The removed samples will be filled and repaired. For bolted joints, a bolt sample will be taken and tested. If compliance with the project is verified, the characteristic strengths foreseen in the project will be used in calculating the element capacities.*available steel yield stress*If compliance with the project cannot be achieved, at least three samples and source samples will be taken and the most unfavorable values obtained will be used in the element capacity calculations.*available steel yield stress*will be taken as.

15.2.8. Limited Level of Knowledge on Prefabricated Reinforced Concrete Buildings

For building geometry**15.2.4.1**The conditions in apply. Only*reinforced concrete's placeprefabricated reinforced concrete*For element details**15.2.6.2**The conditions in apply, but not in this clause*steel*In its place*prefabricated reinforced concrete*will be used in determining material properties.

15.2.4.3The conditions in apply.

15.2.9. Comprehensive Knowledge Level in Prefabricated Reinforced Concrete Buildings

For building geometry**15.2.5.1**The conditions given in apply. For element details**15.2.7.2**The conditions given in are valid. Only the relevant articles are valid.*steel*And*reinforced concrete*in its place*prefabricated reinforced concrete*will be used. For concrete compressive strength, every 600 m²At least one sample will be taken from the area and the test will be carried out. The total number of concrete samples taken from the building will be at least five. In the capacity calculations of the elements, the average concrete compressive strength obtained from the tests or the concrete compressive strength specified in the project (the lower one)*existing concrete strength*The yield stress of the reinforcement to be used in the capacity calculation of the elements will be the characteristic yield stress of the steel class specified in the project.

15.2.10. Limited Level of Knowledge in Masonry Buildings

15.2.10.1 –Building Geometry: If architectural projects are available, a visual inspection of the building will determine whether the existing geometry is compatible with the project. If there is no architectural project, a system survey of the building will be obtained. The information obtained should include the location of the masonry walls on each floor, their lengths, thicknesses, gaps and floor heights.

15.2.10.2 –Detail: The type of roof and floor, the connection types with the walls, the condition of the joists and lintels will be determined visually.

15.2.10.3 –Material Properties: The type of wall materials will be determined visually by removing the plaster of a section of the wall surface. In building strength calculations**Chapter 11**The wall shear strengths given for each wall type will be taken as basis.

15.2.11. Comprehensive Knowledge Level in Masonry Buildings

15.2.11.1 –Building Geometry: The system survey of the building will be prepared. The information obtained should include the location of the masonry walls on each floor, their lengths, thicknesses, gaps and floor heights. The foundation system will be determined by an inspection pit opened from outside the building.

15.2.11.2 –Detail: The type of roof and floor, the connection types with the walls, the condition of the joists and lintels will be visually examined. As a result of this examination, it will be determined whether the rigid diaphragm feature is provided on each floor. If this feature is not provided, it will be decided that the building is inadequate in terms of earthquake safety.

15.2.11.3 –Material Properties: The type of wall materials will be determined visually by removing the plaster of a section of the wall surface. At least two wall piece samples will be taken from the building to determine the wall material properties and **Chapter 11** In the calculations to be made according to , the average properties obtained from the compressive strength tests of these samples will be used. In the building strength calculations, **Chapter 11** The wall shear strengths given for each wall type will be taken as basis.

15.2.12. Knowledge Level Coefficients

(a) It will be applied to the staff capacities according to the information levels obtained from the examined buildings. *Knowledge Level Coefficients* **Table 15.1** is given in .

(b) Unless specifically stated, material strengths shall not be divided by the material coefficients given in the relevant design regulations. In the calculation of element capacities *available material strength* will be used.

Table 15.1 – Information Level Coefficients for Buildings

Level of Knowledge	Knowledge Level Coefficient
Annoyed	0.75
Comprehensive	1.00

15.3. DAMAGE LIMITS AND DAMAGE ZONES IN BUILDING ELEMENTS

15.3.1. Section Damage Situations

Three damage states and damage limits are defined at the section level for ductile elements. These are: *Limited Damage* (SH), *Controlled Damage* (KH) and *Pre-Collapse Damage* (GÖ) states and their limiting values. Limited damage defines a limited amount of inelastic behavior in the relevant section, controlled damage defines an inelastic behavior where the section strength can be safely provided, and pre-collapse damage defines an advanced inelastic behavior in the section. This classification is not valid for brittle damaged elements.

15.3.2. Section Damage Zones

Elements whose critical section damage does not reach SH *Limited Damage Zone* , the elements between SH and KH *Significant Damage Area* , the elements between KH and GÖ *Advanced Damage Zone* , the elements exceeding GÖ are *Migration Zone* They are located in (**Figure 15.1**).

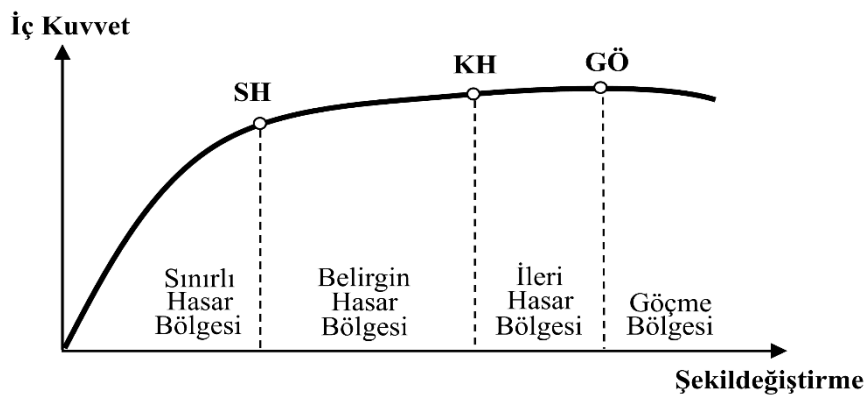


Figure 15.1

15.3.3. Identification of Section and Element Damages

15.5 or **15.6** Internal forces and/or deformations calculated by the methods defined in **15.3.1** It will be decided which damage zones the sections are in as a result of comparing them with the numerical values defined to correspond to the section damage limits in . Element damage will be determined according to the most damaged section of the element.

15.4. GENERAL PRINCIPLES AND RULES REGARDING EARTHQUAKE CALCULATION

15.4.1 –According to this Section of the Regulation, the purpose of the earthquake calculation is to determine the seismic performance of existing or retrofitted buildings. For this purpose, **15.5** defined in *linear* or **15.6** defined in *nonlinear* calculation methods can be used. However, it should not be expected that performance evaluations made with these methods, which are theoretically based on different approaches, will yield exactly the same results. The general principles and rules defined below apply to both types of methods.

15.4.2 –In the definition of earthquake effect, **2.2** For earthquake ground motion levels determined according to **2.3.4** or **2.4.1** The horizontal elastic design spectrum given in will be used in earthquake calculations. **3.1.2** The Building Importance Factor defined in will not be applied ($I=1.0$).

15.4.3 –The seismic performance of buildings will be evaluated under the combined effects of vertical loads and earthquake effects acting on the building. Masses in earthquake calculations **4.5.9** will be defined according to.

15.4.4 –Earthquake forces will be applied to the building in both directions and separately in both directions.

15.4.5 –The building's structural system model will be prepared with sufficient accuracy to calculate the internal forces, displacements and deformations that will occur in the structural elements under the combined effects of earthquake effects and vertical loads.

15.4.6 –In buildings where the floors operate as rigid diaphragms in the horizontal plane, two horizontal displacements and rotational degrees of freedom around the vertical axis will be taken into account at each floor. Floor degrees of freedom will be defined at the mass center of each floor, and additional eccentricity will not be applied.

15.4.7 –Uncertainties in the structural systems of existing buildings are determined according to the scope of the data collected from the building. **15.2** defined in *knowledge level coefficients* will be reflected in the accounting methods.

15.4.8 – 7.3.8 Columns defined as short columns according to will be defined with their actual free lengths in the carrier system model.

15.4.9 –The conditions for defining interaction diagrams of reinforced concrete sections subjected to uniaxial or biaxial bending and axial force are given below:

(a) Concrete and reinforcement steel in earthquake calculations **15.2** Determined according to the level of knowledge defined in *Current strength* will be taken as basis.

(b) The maximum compressive strain of concrete can be taken as 0.0035, and the maximum strain of the reinforcement steel can be taken as 0.01.

(c) Interaction diagrams can be modeled as multilinear or multiplanar diagrams by appropriately linearizing them.

15.4.10 –In defining the element dimensions of reinforced concrete systems, joint areas can be considered as rigid end areas.

15.4.11 –Cracked section in reinforced concrete members under bending effect *effective section stiffnesses* will be used. Effective section stiffnesses **4.5.8** It will be calculated according to .

15.4.12 –In the calculation of positive and negative plastic moments of reinforced concrete slab beams, the slab concrete and the reinforcement inside it can be taken into account.

15.4.13 –In case of insufficient interlocking or overlapping length in reinforced concrete elements, the yield stress of the relevant reinforcement in the calculation of the section capacity moment will be reduced in proportion to the deficiency in interlocking or overlapping length.

15.4.14 –In cases where ground deformations may affect the building behavior, ground properties will be reflected in the analysis model.

15.4.15 – Part 3, Chapter 4 And Chapter 5 Other principles regarding modelling given in are valid.

15.5. EARTHQUAKE CALCULATION WITH LINEAR CALCULATION METHODS

15.5.1. Calculation Methods

Linear calculation methods to be used to determine the seismic performance of buildings, **4.7** Equivalent Earthquake Load Method defined in and **4.8.2** The following additional rules apply to these methods.

15.5.1.1 –Buildings where the Equivalent Seismic Load Method can be applied **Table 4.4**. Additional eccentricity will not be taken into account in the earthquake calculation of buildings. Total equivalent earthquake load (base shear force) **Equation (4.19)** And **Equation (4.8)** According to his account $R_a = 1$ will be taken.

15.5.1.2 –With the Mod Merge Method **4.8.2** According to the earthquake calculation $R_a = 1$ will be taken.

In the calculation of the internal forces and capacities of the elements compatible with the applied earthquake direction and direction, the internal force directions obtained in the dominant mode in this direction will be taken as basis.

15.5.2. Determination of Damage Types in Structural Elements of Reinforced Concrete Buildings

15.5.2.1 –In determining the application limits of the following calculation methods for reinforced concrete systems, the bending of the sections of beams, columns and shear walls is taken into account. *impact/capacity ratios* Numerical values expressed as (ECO) will be used.

15.5.2.2 –Reinforced concrete elements, if the failure type is bending ““ductile”, if the cut “crispy” are classified as.

(a) In order for columns, beams and shear walls to be considered as ductile elements, the shear force calculated in accordance with the bending capacity at the critical sections of these elements. V_o 's, **15.2** also compatible with the defined knowledge level *existing material strength* using the values

Equation (7.10) It is necessary to meet the limits given in . V_o 's calculation for columns **7.3.7** to, for beams **7.4.5** and for curtains **7.6.6** will be made according to, but **Equation (7.16)** in $\beta_v = 1$ will be taken. In columns, beams and walls V_o In the calculation of , instead of hardening bearing capacity moments, bearing capacity moments will be used. With vertical loads $R_a = 1$, the total shear force calculated from the earthquake V_o If it is less than V_o This cutting force will be used instead.

(b) Above **(a)** Reinforced concrete elements that do not meet the ductile element conditions given in , *brittle damaged elements* will be defined as.

15.5.2.3 –Effect/capacity ratio (ECR) of ductile beam, column and shear wall sections under vertical loads and earthquake effects $R_a = 1$, the total section moment is calculated as the section moment is obtained by dividing it by its capacity. The direction of the applied earthquake force will be taken into account in the ECO calculation. The bending moment capacities of the column and shear sections correspond to axial forces, vertical loads and $R_a = 4$ will be calculated under the joint effect of the calculated earthquake loads.

15.5.2.4 –The effect/capacity ratio of reinforced partition walls is the ratio of the calculated shear force under earthquake effect to the shear force resistance. The shear forces formed in reinforced partition walls modeled with diagonal bars will be considered as the horizontal component of the axial force of the bar.

15.5.2.5 –In reinforced concrete column-beam connections, the forces affecting the connection for all limit states and **Equation (7.11)** Cutting forces to be calculated from **7.5.2.2** The shear strengths given in should not exceed. However, **Equation (7.11)** also V_{ar} in its place **7.3.7** Calculated without taking into account hardening according to V_o to be used, **Equation (7.12)** or **Equation (7.13)** In the strength calculation of f_{ck} in its place **15.2** Determined according to the level of knowledge defined in *existing concrete strength* In case the connection shear force demand exceeds the shear force resistance, the column-beam connection area *brittle damaged element* will be defined as.

15.5.3. Application Limits of Linear Calculation Methods

15.5.3.1 –Linear calculation methods cannot be applied if any of the following situations occur.

(a) Building height class is less than 5 ($BYS < 5$).

(b) In the building **3.6.2.4** The presence of the B3 disorder mentioned in .

(c) In reinforced concrete buildings, the average of the ECO values scaled with the shear force of the vertical ductile elements (columns, shear walls and reinforced partition walls) for each earthquake direction at any floor of the building other than the top floor is greater than the average ECO value of the beams in the earthquake direction.

(d) The average of the shear force and scaled ECO values of ductile shear walls, ductile columns and reinforced partition walls for each earthquake direction at any floor other than the top floor of the building is greater than 3.

(to) The average ECO value of ductile beams in each earthquake direction on any floor of the building other than the top floor is greater than 5.

15.5.3.2 –Above(c)And(d)ECO values scaled with the shear force specified in the items **Equation (15.1)**will be calculated with .

$$ECO = \frac{\sum V_i (ECHO)_i}{\sum V} \quad (15.1)$$

Equation (15.1) i is the column number on the relevant floor, V_i The calculated shear force in column i , $(ECO)_i$ whereas**15.5.2.3**According to , this column shows the calculated ECO value.

15.5.3.3 –In the building15.5.3.1 (a)-(e)In case of occurrence of any of the situations specified in the building**15.6**It will be evaluated by one of the methods specified in .

15.5.4. Determination of Unit Strain and Plastic Rotation Demands

15.5.4.1 –Unit deformation and plastic rotation demands of element sections,4.7or 4.8.2As a result of the calculation made according to , the total obtained at any element end displaced axis rotation θ_k Definition of displaced axis rotations at element ends **ANNEX 15A**is given in .

15.5.4.2 –Total curvature demand of the element end section φ_t ,Equation (15.2)It will be calculated from the relation in .

$$\varphi_t = \frac{(\theta_k - \theta_y)}{L_p} + \varphi_y \quad (15.2)$$

Equation (15.2)in θ_y displaced axis yield rotation at the element end section, φ_y is element is the yield curvature at the end section. Definition of displaced axis yield rotations at the ends of the element**ANNEX 15A**is given in . L_p is the plastic hinge length and the section in the effective direction will be taken equal to half the size.

15.5.4.3 –Effective yield curvature φ in reinforced concrete systems, and effective yield moment M_y , will be calculated by moment-curvature analysis.

15.5.4.4 –For wrapped or unwrapped concrete and reinforcement steel models, in cases where no other selection is made,ANNEX 5Acan be used.

15.5.4.5 –Plastic rotation demand of element sections θ_p ,Equation (15A.2)will be obtained from.

15.6. EARTHQUAKE CALCULATION WITH NON-LINEAR CALCULATION METHODS

15.6.1. Definition

The purpose of nonlinear calculation methods to be used for determining the structural performance of existing or strengthened buildings under earthquake effect and for reinforcement calculations is to calculate the plastic deformation and plastic rotation demands related to ductile behavior and the internal force demands related to brittle behavior for a given earthquake. Then, these demand magnitudes will be compared with the deformation and internal force capacities and structural performance evaluation will be made at the section and building level.

15.6.2. Calculation Methods

Non-linear calculation methods to be used in the evaluation of existing or retrofitted buildings, details 5.6 Explained in *Single Mode Propulsion Methods*, 5.6.6 Explained in *Multimodal Push Methods* with 5.7 Explained in *Nonlinear Calculation Method in Time History* is.

15.7. FORMATION LIMITS

15.7.1. Sectional Strain and Plastic Rotation Limits of Reinforced Concrete and Precast Reinforced Concrete Members

15.7.1.1 –Concrete and reinforcement steel 15.5.4 to or 15.6.2 The unit deformation demands obtained according to will be compared with the unit deformation capacities defined below to determine the structural system performance at the section level.

15.7.1.2 –If the longitudinal reinforcements of the reinforced concrete elements for which deformation calculations are made are arranged with non-ribbed (flat) reinforcement steel, the reinforcement steel unit deformation demand and plastic rotation demand will be increased by multiplying by 1.5.

15.7.1.3 –Upper limits (capacities) of unit deformation and plastic rotation allowed according to various section damage limits in reinforced concrete ductile beams, shear walls and columns where plastic deformations occur. 5.8.1.1, 5.8.1.2, 5.8.1.3 And 5.8.1.4 It is defined in .

Equation (5.4d) $p_{ins,h}$ 30% of the 90 degree closed stirrups can be included in the calculation.

5.8.1.2 Given in Equation (5.6) in L_s, L_p It will not be taken smaller than .

15.7.1.4 –Shear force ratio of the reinforced concrete section for which deformation calculation is made $V_{ol}/(b_w d f_{ctm})$ If < 0.65 15.7.1.3 The upper deformation limits calculated according to are valid.

If the shear force ratio is greater than 1.30 15.7.1.3 The upper limits of deformation calculated according to will be reduced by multiplying by 0.50. Linear interpolation will be applied for intermediate values.

15.7.2. Deformation and Plastic Rotation Limits of Steel Elements

Permissible limits for calculated deformations and member end rotations in existing or retrofitted steel buildings are determined for the relevant performance levels. ANNEX 5C As given in. Material strengths available material strength will be used.

15.7.3. Deformation Limits of Reinforced Partition Walls

Determination of seismic performance of reinforced partition walls in reinforced concrete buildings Table 15.2 The slip angle limits given in will be taken into account. The slip angle will be obtained by dividing the largest relative floor drift calculated at the relevant floor by the floor height.

Table 15.2 – Defining Performance Limits for Strengthened Partition Walls Slip Angles

Performance Limit		
SH	KH	GÖ
0.003	0.005	0.010

15.8. DETERMINATION OF EARTHQUAKE PERFORMANCE OF EXISTING BUILDINGS

15.8.1. Targeted Earthquake Performance in Existing or Reinforced Buildings

Earthquake ground motion levels to be taken as basis in determining the earthquake performance of existing or to be retrofitted buildings and the minimum performance targets for buildings at these earthquake ground motion levels. **Table 3.4** is given in .

15.8.2. Earthquake Performance of Existing or Retrofitted Buildings

The seismic performance of existing buildings is related to the expected damage to the building under the applied earthquake effect and is defined based on four different damage situations. **15.5** And **15.6** Application of the calculation methods defined in and **15.7** The building earthquake performance level is determined by deciding on the element performances according to . The rules to be applied to determine the earthquake performance of buildings are given below. The rules given here are valid for reinforced concrete, prefabricated reinforced concrete and steel buildings. The rules to be applied in masonry buildings **15.8.8** is given in .

15.8.3. Limited Damage Performance Level in Existing Buildings

In any floor of reinforced concrete buildings, maximum 20% of the beams should be supported as a result of the calculations made for each earthquake direction applied. *Significant Damage Area*, but all other carrier elements can pass to *Limited Damage Zone*. If there are brittle damaged elements, provided that they are strengthened, buildings in this situation *Limited Damage Performance Level* It is assumed that . These exceptions do not apply to steel and prefabricated reinforced concrete buildings.

15.8.4. Controlled Damage Performance Level in Existing Buildings

Buildings that meet the following conditions, provided that brittle damaged elements, if any, are strengthened: *Controlled Damage Performance Level* It is assumed that:

(a) As a result of the calculations made for each earthquake direction applied in any floor of reinforced concrete buildings, excluding secondary beams (not included in the horizontal load-bearing system), maximum 35% of the beams and vertical elements (columns, shear walls and reinforced partition walls) should be as follows: (b) as defined in paragraph *Advanced Damage Zone* can pass to. These exceptions do not apply to steel and prefabricated reinforced concrete buildings.

(b) *Advanced Damage Zone* The total contribution of the vertical elements to the shear force carried by the vertical elements at each floor should be below 20%. At the top floor *Advanced Damage Zone* The ratio of the total shear forces of the vertical elements in to the total shear forces of all vertical elements in that floor can be at most 40%.

(c) All other load-bearing elements *Limited Damage Zone* or *Significant Damage Area* However, in any floor, both the upper and lower sections are *Clear Damage Limit* The ratio of the shear forces carried by the exceeded vertical elements to the shear forces carried by all the vertical elements on that floor should not exceed 30% (In the calculation with the linear method, both the lower and upper nodes should not exceed 30%). **Equation (7.3)** Columns for which are not included in this calculation).

15.8.5. Collapse Prevention Performance Level in Existing Buildings

All elements damaged brittlely *Migration Zone* Taking into account that the buildings are located in the following conditions: *Migration Prevention Performance Level* It is assumed that:

(a) As a result of the calculation made for each earthquake direction applied in any floor of reinforced concrete buildings, maximum 20% of the beams, excluding secondary beams (not included in the horizontal load-bearing system) *Migration Zone* can pass to. These exceptions do not apply to steel and prefabricated reinforced concrete buildings.

(b) All other load-bearing elements *Limited Damage Zone, Significant Damage Area or Advanced Damage Zone* However, in any floor, both the upper and lower sections are *Clear Damage Limit* The ratio of the shear forces carried by the exceeded vertical elements to the shear forces carried by all the vertical elements on that floor should not exceed 30% (In the calculation with the linear method, both the lower and upper nodes should not exceed 30%). **Equation (7.3)** Columns for which are not included in this calculation).

(c) The use of the building in its current state is dangerous in terms of life safety.

15.8.6. Immigration Status

Building *Migration Prevention Performance Level* If he cannot provide *Immigration Status*'s. The use of the building is dangerous in terms of life safety.

15.8.7. Performance Levels of Reinforced Partition Walls in Reinforced Concrete Buildings

In the calculation made with linear and non-linear methods, the shear angle capacities of the reinforced partition walls at any floor of the building in each earthquake direction are calculated for each damage limit. **Table 15.2** It is necessary to meet the slip angle limits given in . Otherwise **15.8.3-15.8.6** Damage assessments made in will not be taken into account.

15.8.8. Determination of Earthquake Performance of Masonry Buildings

Performance level of masonry buildings, **15.2** According to the review and **Chapter 11** The decision will be made as a result of the calculation made according to If the shear strength of all walls of the masonry building in both directions is sufficient to withstand the shear forces formed under the applied earthquake effects, the building *Limited Damage Performance Level* It is concluded that the building is in compliance with the earthquake applied to any floor. If the contribution of the walls that do not meet this condition to the floor shear force is below 40%, the building is *Controlled Damage Performance Level* If this rate exceeds 40%, the building *Immigration Status* It is assumed that .

15.9. STRENGTHENING OF BUILDINGS

Strengthening of buildings includes operations such as eliminating defects that will cause earthquake damage, adding new elements to increase earthquake safety, reducing mass, improving the earthquake behavior of existing elements, and ensuring continuity in force transfer.

15.9.1. Determination of Earthquake Safety of Reinforced Buildings

In calculating the seismic safety of strengthened buildings and their elements, the calculation methods and evaluation principles given in this Section for existing buildings will be used.

15.9.2. Design of Elements to be Added to Buildings

In the design of new elements to be added to buildings for strengthening purposes, the special rules given in this Section shall be taken into account. **Chapter 7** and/or **Chapter 9** and other applicable standards and regulations will be complied with.

15.9.3. Types of Reinforcement

Reinforcement applications will be evaluated in two different scopes for each type of load-bearing system: at the element and building system level.

15.9.3.1 –Processes applied to increase the strength and deformation capacities of the building's elements that resist earthquake loads, such as columns, beams, curtains and joints, *element reinforcement* is defined as.

15.9.3.2 –Increasing the strength and deformation capacity of the building's carrier system and ensuring continuity in the distribution of internal forces, adding new elements to the building, strengthening joint areas, and reducing the mass of the building in order to reduce earthquake effects. *system strengthening* is defined as.

15.10. STRENGTHENING OF REINFORCED CONCRETE BUILDINGS

The element and system strengthening methods given in this Section cover the techniques frequently used in practice. However, types of strengthening not covered here may be applied provided they comply with the general approach and principles of this Section.

15.10.1. Wrapping of Columns

The following methods can be used to increase the shear and compressive strengths of columns and to eliminate the weaknesses of lap joints in order to increase their ductility. The bending capacity of columns cannot be increased with these methods.

15.10.1.1 –*Reinforced Concrete Wrap*: The concrete cover of the existing column will be scraped or its surfaces will be roughened. The reinforced concrete coating should be of sufficient thickness for the placement of both horizontal and vertical reinforcement, pouring of concrete and ensuring minimum concrete cover. The minimum coating thickness is 100 mm. The reinforced concrete coating starts above the floor slab and ends below the upper floor slab. In the coating made for the purpose of increasing the axial pressure resistance, the entire height of the column for the transverse reinforcement in the coating concrete **7.3.4.2** The rules given in will be applied. In the calculation of the shear and compressive strengths of the wrapped column, the wrapped gross section dimensions and the design strength of the jacket concrete will be used, but the obtained strengths will be reduced by multiplying by 0.9.

15.10.1.2 –*Steel Wrap*: Steel wrapping is formed by placing four longitudinal angles at the corners of rectangular reinforced concrete columns and welding the angles with horizontal plates arranged at certain intervals. There should be no gap between the angles and the reinforced concrete surfaces. Horizontal plates should be continuous on four surfaces. In order for the steel wrapping to increase the column bending axial load capacity, the angles should be continuous between the lower and upper slabs.

(gaps should be removed) and the pressure transfer to the slabs should be shown by calculation. If necessary, the axial pressure load of the existing reinforced concrete column section caused by vertical loads can be reduced by preloading the angles. Additional shear strength to be provided by steel wrapping **Equation (15.3)** will be calculated with .

$$V_{fj} = \frac{t \cdot b \cdot d}{s} \cdot f_{wov} \quad (15.3)$$

Equation (15.3) also t, b And s thickness, width and spacing of horizontal plates, d if the section is useful height. In order to eliminate the weaknesses of the overlapped joints with steel coils, the coil length should be at least 50% longer than the overlapped area length and the steel coil should be compressed with at least two rows of bolted anchors with a diameter of 16 mm arranged on opposite faces of the column in the reinforcement overlapped area. If the overlapped joint is made at the lower end of the column, at least two rows of bolted anchors should be made 250 and 500 mm above the sub-slab, respectively.

15.10.1.3 –Fibrous Polymer (LP) Wrap:Wrapping is achieved by wrapping and bonding the LP layer around the columns, with the fibers parallel to the transverse reinforcements. With LP wrapping, the ductility capacity, shear and pressure strengths of reinforced concrete columns and the reinforcement interlocking strength in cases where the longitudinal reinforcement overlap length is insufficient are increased. In reinforcements made with LP wrapping, the full wrapping (wrapping the entire cross-section) method should be used and at least 200 mm overlap should be made at the end of the wrapping. LP wrapping can be applied to rectangular columns by rounding the column corners with a radius of at least 30 mm. LP application should be carried out in accordance with the method recommended by the manufacturer. Calculation methods of the increase in ductility with the increase in shear, axial pressure and interlocking strengths obtained in columns wrapped with LP **ANNEX 15B** is given in .

15.10.2. Increasing the Bending Capacity of Columns

Column sections can be enlarged to increase the bending capacity of columns. This process also increases the shear and compression force capacities of the column. The continuity of the longitudinal reinforcement added to the enlarged column will be ensured between floors. Longitudinal reinforcement will be passed through holes opened in the floor slabs. Necessary transverse reinforcement will be placed by drilling the beams or anchoring the beams in the column-beam joint areas. Enlarged section of the column **7.3.4** It will be wrapped with transverse reinforcement according to . The concrete cover of the enlarged column section will be of sufficient thickness to cover the added vertical and horizontal reinforcement. In order to ensure the adhesion of the new and old concrete, the plaster layer on the surface of the existing column will be stripped and the concrete surfaces will be roughened. In the calculation of the bending, shear, compressive strength and bending rigidity of the enlarged column section, the gross section dimensions and the design properties of the added section concrete will be taken as basis, but the obtained rigidity and strengths will be reduced by multiplying by 0.9.

15.10.3. Wrapping of Beams

The purpose of wrapping reinforced concrete beams is to increase the shear strength and in some cases the ductility capacity of the beams. The bending capacity of the beams cannot be increased by the methods given below.

15.10.3.1 –External Stirrup Addition:In beam support areas with insufficient shear strength, the required number of stirrup bars are placed on both sides of the beam. **Figure 15.2** It will be added from outside as shown in . The bars connected to a steel profile placed under the beam with bolts will be placed by passing through the holes opened in the upper floor and bent into the slot opened in the upper surface of the floor. Then the gaps opened in the concrete will be filled with concrete. This method

can also be applied using different details with the same principles. The shear strength of the beams increased by the stirrups added from outside will be calculated according to TS 500. The stirrups added from outside have no wrapping effect and do not increase the ductility of the beam section. In this application, the profiles and bolts must be protected against external effects.

15.10.3.2 – Wrapping with Fibrous Polymer (LP): In order to increase the ductility and shear strength of the beam with LP wrapping, the full wrapping (wrapping around the entire cross-section) method should be used. Shear strength of beam strengthened with LP **ANNEX 15B** Given in **Equivalent. (15B.1)** In case of using discontinuous (in strips) LP, the intervals of the LP strips are

$(w+d/4)$ should not exceed the value. LP winding should be applied to the corners of the beams with a minimum radius of 30 mm.

will be applied by rounding. In windings made with LP, at least 200 mm overlap should be made at the end of the winding. LP application should be carried out in accordance with the method recommended by the manufacturer.

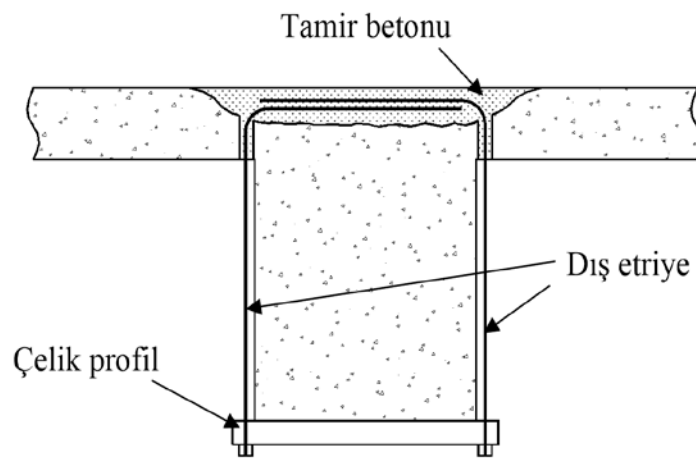


Figure 15.2

15.10.4. Reinforcement of Partition Walls

The rigidity and shear strength of the partition walls in the reinforced concrete frame, which are continuous from the top of the foundation to the top, **ANNEX 15C** If the strengthening of the building is mainly done by strengthening the partition walls, this method can only be applied to buildings with a maximum of three floors excluding the basement.

15.10.5. Strengthening of Reinforced Concrete Bearing Systems with Cast-in-Situ Reinforced Concrete Shear Walls

Reinforced concrete load-bearing systems with insufficient lateral rigidity and strength can be strengthened with cast-in-place reinforced concrete walls. Reinforced concrete walls can be arranged within the existing frame plane or adjacent to the frame plane.

15.10.5.1 – Adding Reinforced Concrete Shear Wall in Frame Plane: The walls to be added to the reinforced concrete system will be arranged within the frame axis and will be continuous from the foundation to the upper level of the wall. For this purpose, the longitudinal reinforcement in the wall end area and, if necessary, the longitudinal reinforcement in the wall body will be continuous along the height of the wall. The walls will be connected to the frame they are in with anchor rods to ensure that they work together. The anchor rods will reduce the shear stresses that occur under earthquake forces at the interfaces between the existing frame elements and the reinforced concrete wall element added.

They shall have sufficient strength to meet the requirements of the shear stresses at the interfaces. The distribution of shear stresses along the frame elements shall be calculated in accordance with known mechanical principles. The friction shear principles in TS 500 shall be used in the design of anchor rods. The smallest anchor rod diameter shall be 16 mm, the minimum anchor depth shall be ten times the rod diameter and the widest rod spacing shall be 400 mm. In the event that there is no existing column at the end of the curtain, **7.6.5** According to , the curtain end area will be created. If there is an existing column at the curtain end, the existing column can be used as the end area. If necessary, the existing column **15.10.2** The curtain end area will be created by enlarging it according to or by arranging a column inside the curtain adjacent to the existing column. In both cases, the continuity of the vertical reinforcements to be added to the curtain end area will be ensured between the floors. Under the curtain **16.7** And **16.8** The foundation will be made in accordance with the principles given in . The curtain foundation will be sized in a way that it will safely transfer the internal forces formed at the curtain base to the foundation soil. In order to reduce the eccentricity that may occur in the curtain foundation, the curtain foundation can be expanded to include neighboring columns and the axial pressure forces of the existing columns can be utilized. Necessary precautions will be taken for the curtain foundation to work together with the existing foundation system.

15.10.5.2 –Adding Reinforced Concrete Shear Adjacent to the Frame Plane: The curtains to be added to the reinforced concrete system will be arranged outside the outer frame axis, adjacent to the frame, and will be continuous from the foundation to the top level of the curtain. The curtains will be connected to the frame they are adjacent to with anchor rods to ensure that they work together. The anchor rods will have sufficient strength to resist the shear stresses that occur under earthquake forces at the interfaces between the existing frame elements and the eccentric curtain element added to the system. In the design of the anchor rods **15.10.5.1** The principles given in will be followed.

15.10.5.3 –If there is no existing column at the curtain end **7.6.5** According to , the curtain end area will be created. If there is an existing column at the curtain end, the existing column can be used as the end area. If necessary, the existing column **15.10.2** The curtain end area will be created by enlarging it according to . Under the curtain **16.7** And **16.8** The foundation will be made in accordance with the principles given in . The curtain foundation will be sized in a way that it will safely transfer the internal forces formed at the curtain base to the foundation soil. In order to reduce the eccentricity that may occur in the curtain foundation, the curtain foundation can be expanded to include neighboring columns and the axial pressure forces of the existing columns can be utilized. Necessary precautions will be taken for the curtain foundation to work together with the existing foundation system.

15.10.6. Adding New Frames to the Reinforced Concrete System

The foundations of the frames to be added outside the reinforced concrete system will be arranged together with the foundations of the existing building. In order for the new frames to work together with the carrier system of the existing building, these frames will be connected to the floors of the existing building in a way that will provide the necessary load transfer.

15.10.7. Reducing the Mass of the Reinforced Concrete System

The most effective methods that can be applied to reduce earthquake loads are to cancel and remove the upper floor or floors of the building, to replace the existing roof with a lightweight roof, to lower the weight of installations such as water tanks on the roof to the ground, and to replace heavy balconies, parapets, partition walls and facade cladding with lighter elements.

ANNEX 15A – COLUMNS, WALLS AND BEAM IN CALCULATION WITH LINEAR METHODS DEFINITION OF DISPLACED AXIS ROTATIONS IN ELEMENTS

15A.0. ICONS

-	= Element total length =
-c	Element clear span
-s	= Shear clearance (moment/shear force ratio at the section) =
M_y	Effective yield moment
$M_{good\ i}$	= effective yield moment at end i
$M_{y\ j}$	= effective yield moment at end j
Δ	= Interstory drift = i node
θ_i	rotation
θ_j	= j node rotation
$\theta_{that\ i}$	= displaced axis rotation at end i =
θ_{kj}	displaced axis rotation at end j
θ_{pi}	= plastic rotation at the end i
θ_y	= Flow rotation
$\theta_{good\ i}$	= yield rotation at end i
$\theta_{y\ j}$	= yield rotation at end j

15A.1. DEFINITIONS

Deformation characteristics of a typical flexural member under double curvature bending
Figure 15A.1 Here - is the total length of the element, -c net clearance, Δ inter-story translation, θ_i and θ_j rotations of nodes i and j, respectively, $\theta_{that\ i}$ and θ_{kj} denotes the displaced axis rotations at the i and j ends, respectively.

15A.2. DISPLACED AXIS ROTATION

Relationship between the displaced axis and nodal point rotations at the i end and the interstory drift when the flexural element is in linear elastic deformation state **Equation (15A.1)** It is defined in .

$$\theta_{that\ i} = \frac{\Delta}{-c} - \theta_i \quad (15A.1)$$

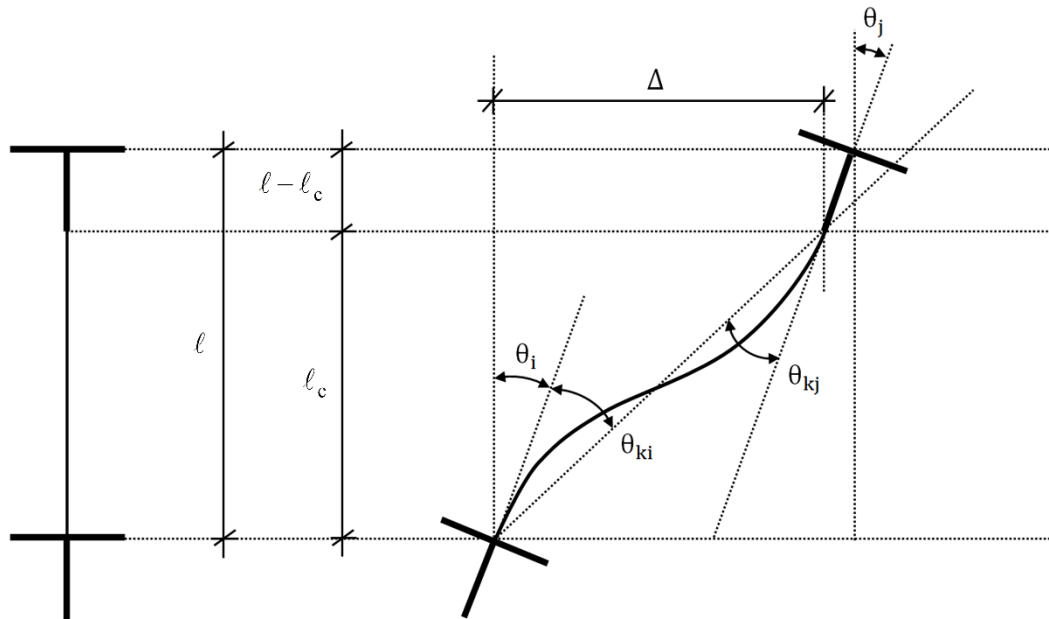


Figure 15A.1

In beam elements, the interstory drift value can generally be taken as zero ($\Delta = 0$). When yielding occurs at end i of the element, the total displaced axis rotation at end i is equal to the sum of yield rotation and plastic rotation at that end.

$$\theta_{\text{total}} = \theta_{\text{good}} + \theta_{\text{pi}} \quad (15A.2)$$

15A.3. YIELD ROTATION IN FRAME ELEMENTS

Relationships between the end yield rotations and end moments at the i and j ends of a flexural member undergoing yielding at both ends **Equation (15A.3)** Definition of yield rotations for elements with both ends in yield state, **15.5.4** According to , it corresponds to the most unfavorable situation in the calculation of unit deformation demands.

$$\theta_{\text{good}} = \frac{M_{\text{good}}}{3EI} - 1 - \frac{M_{\text{Yj}}}{2M_{\text{good}}} \quad (15A.3a)$$

$$\theta_{\text{Yj}} = \frac{M_{\text{Yj}}}{3EI} - 1 - \frac{M_{\text{good}}}{2M_{\text{Yj}}} \quad (15A.3b)$$

Equation (15A.3) also E flexural rigidity of the uncracked section, M_{good} and M_{Yj} and j respectively at the end *effective yield moments* The directions of yield moments are positive in the counterclockwise direction and negative in the clockwise direction. Therefore, **Equation (15A.3)** It covers both double and single curvature bending cases. *Effective yield moment* M_{Y} , **ANNEX 5A.1** obtained in accordance with the definition given in will be done.

15A.4. YIELD ROTATION IN CURTAIN ELEMENTS

4.5.3.2 The relationship between the yield rotation and yield moment of a bending element defined as a shear wall at any floor of a building **Equation (15A.4)** is given in .

$$\theta_y = \frac{M_{y-s}}{3EI} \quad (15A.4)$$

Here s is the shear span (ratio of moment/shear force at the section). It can be taken as approximately half the distance from the base of each storey to the top of the shear wall.

ANNEX 15B – STRENGTH AND RESISTANCE OF COLUMNS WRAPPED WITH FIBER POLYMER CALCULATION OF DUCTILITY INCREASE

15B.0. ICONS

A_s	= Column reinforcement area (for single bar) =
$A_{s.h.}$	Transverse reinforcement area
b	= Section width
b_w	= Beam web or column width =
d	Useful height of the section
d'	= Concrete cover thickness
TO_f	= Modulus of elasticity of fibrous polymer =
TO_s	Modulus of elasticity of steel reinforcement
f_{cc}	= Compressive strength of concrete wrapped with fibrous polymer =
f_{CD}	Compressive strength of concrete
f_{cm}	= 15.2.4.3 Existing concrete compressive strength defined according to
f_{hs}	= Lateral pressure corresponding to 0.001 unit elongation in transverse reinforcement
f_{hs}	= In case of 0.001 unit elongation in the transverse reinforcement, the transverse reinforcement and the fibrous polymer total lateral pressure required to provide
f_{ym}	= 15.2.4.3 Current steel yield strength defined according to
f_i	= Lateral pressure provided by the fibrous
h	polymer = Section size in the working direction
LP	= Fibrous polymer
L_s	= Current overlap length
n	= Number of overlapped reinforcement =
n_f	Number of LP winding layers on one side =
p	Core cross-section perimeter
r_c	= Rounding radius on corners
s_f	= Axis-to-axis spacing of fibrous polymer tapes
t_f	= Effective thickness for a sheet of fibrous polymer (making allowance for insufficient overlap length) (effective thickness of fibrous polymer required for
V_c	wrapping) = Contribution of concrete to shear strength
V_f	= Contribution of fibrous polymer to shear strength
V_{Max}	= Shear force defined to limit principal compressive stresses =
V_r	Shear strength of column or beam
V_w	= Contribution of transverse reinforcement to shear strength

w	= Width of the fibrous polymer strip
ϵ_{cc}	= Unit shortening corresponding to the compressive strength of the confined concrete =
ϵ_f	Effective unit elongation limit of the fibrous polymer
ϵ_{fu}	= Elongation at break of fibrous polymer =
K_a	Section shape efficiency coefficient
φ	= Longitudinal reinforcement diameter
ρ_f	= volume fraction of fibrous polymer
$\rho_{s,h}$	= Transverse reinforcement ratio, $\rho_{s,h} = \frac{A_{s,h}}{sb_w}$

15B.1. INCREASING THE SHEAR STRENGTH OF COLUMNS

Shear strength of columns and beams confined with LP **Equation (15B.1)** is calculated with.

$$V_f = V_c + V_w + V_f \leq V_{Max} \quad (15B.1)$$

Contribution of concrete to shear strength V_c , contribution of transverse reinforcement V_w and defined to limit the principal compressive stresses V_{Max} values recommended by TS 500 with equations, but **15.2** Determined according to *available material strengths* Contribution of LP winding to shear force resistance will be calculated using V_w wrapping in strips in case of **Equation (15B.2)** will be calculated with .

$$V_f = \frac{2ntwE_f d_{eff}}{s_f} \quad (15B.2)$$

Equation (15B.2) also n Number of LP winding layers on one side, t The effective thickness for a sheet of LP is, w Width of the LP strip, T LP modulus of elasticity, ϵ_f LP is the effective strain limit, d element useful height, s_f is the spacing of the LP strips from axis to axis (**Figure 15B.1**). In case of continuous wrapping, $w = s_f$ will be taken. Effective unit elongation value **Equation (15B.3)** and **Equation (15B.4)** It will be taken as the smaller of the values determined according to .

$$\epsilon_f \leq 0.004 \quad (15B.3)$$

$$\epsilon_f \leq 0.50\epsilon_{fu} \quad (15B.4)$$

Equation (15B.4) in ϵ_{fu} LP is the unit elongation at break. Using discontinuous (in strips) LP In case of LP strip spacing s_f , $(w + d/4)$ will not exceed its value.

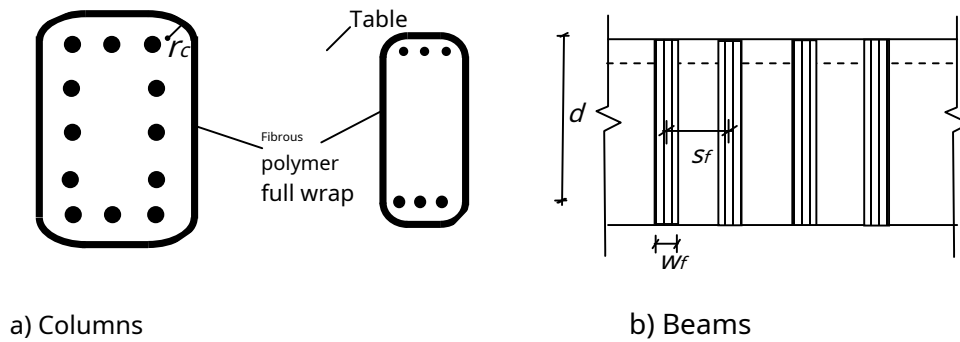


Figure 15B.1

15B.2. INCREASING THE AXIAL COMPRESSIVE STRENGTH OF COLUMNS

In order to increase the axial compressive strength of columns with LP confinement, the ratio of the long dimension to the short dimension of the column section should not exceed two and a half. The effectiveness of LP can be increased by changing the cross-sections of the columns from rectangular to elliptical. In elliptical sections, the ratio of the long dimension to the short dimension can be at most three. The axial load of a column confined with LP

When calculating the strength of concrete, the compressive strength f_{cc} in its place **Equation (15B.5)** determined by f_{cc} value will be used.

$$f_{cc} = f_{cm} [1 + 2.4 (f_l / f_{cm})] \geq 1.2 f_{cm} \quad (15B.5)$$

Equation (15B.5) also f_{cm} of unwrapped concrete *available compressive strength*, is the f_l LP winding provides amount of lateral pressure. **Equation (15B.6)** It will be calculated according to .

$$f_l = \frac{1}{2} \kappa \rho_f \epsilon_f \sigma_{of} \quad (15B.6)$$

Equation (15B.6) in ϵ_f **Equation (15B.3)** And **Equation (15B.4)** In this equation, κ is the smaller of κ_a section shape efficiency coefficient, ρ_f LP is the volumetric ratio. κ_a for various cross sections **Equation (15B.7)** is given in .

$$\kappa_a = \frac{1 - b/h}{1 - \frac{(b-2r_c)^2 + (h-2r_c)^2}{3b.h.}} \quad \begin{array}{ll} \text{Circular section} & - \\ \text{Ellipse section} & - \\ \text{Rectangular section} & -- \end{array} \quad (15B.7)$$

Equation (15B.7) in b And h for rectangular sections the short and long side dimensions, for elliptical sections the short and long dimensions the relevant dimensions of the ellipse, r_c in rectangular sections made at the corners is the radius of the rounding (**Figure 15B.2**).

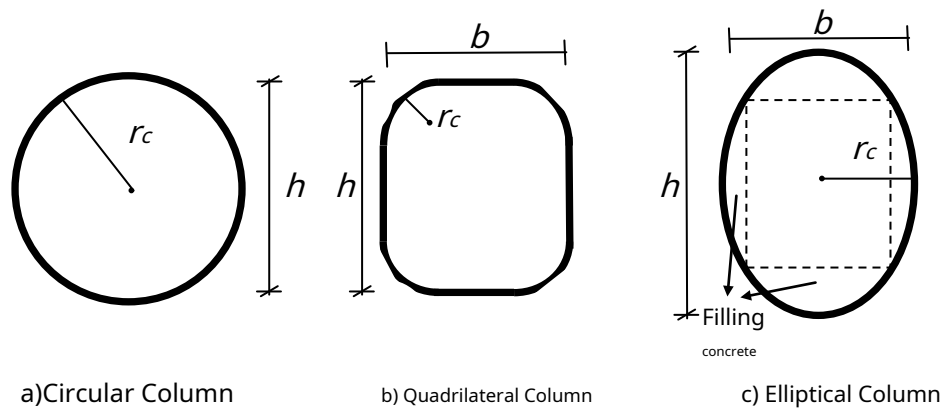


Figure 15B.2

15B.3. INCREASING THE DUCTILITY OF COLUMNS

In order to increase the ductility of columns with LP confinement, the ratio of the long dimension to the short dimension of the column section should not be more than two and a half. In ellipse sections, the ratio of the long dimension to the short dimension can be at most three. In a column confined with LP, the confined concrete compressive strength the corresponding unit shortening (ϵ_{cc}) Equation (15B.8) can be determined by .

$$\epsilon_{cc} = 0.002 - 1 + 15(f_t/f_{cm})^{0.75} \quad (15B.8)$$

In case of LP confinement in columns to increase ductility Equation (15B.5) And Equation (15B.8) 'also f_t Equation (15B.6) will be calculated with . Equation (15B.6) in ϵ Equation (15B.4) with The calculated value will be taken as the smaller of 0.01.

(a) When using linear calculation methods, any column Equation (15B.8) calculated with ϵ_{cc} If the value is greater than 0.018, the column in question is wrapped. otherwise it is assumed to be unwrapped.

(b) When obtaining the moment-curvature relationship of LP confined sections for nonlinear calculation methods, an idealized stress-strain relationship consisting of two lines can be used for LP confined concrete. In this relationship, at the bending point stress and strain values f_{cm} (capacity) and 0.002 can be taken. Stress-strain

values at the end of the relationship Equation (15B.5) And Equation (15B.8) In reinforced concrete carrier system elements wrapped with LP where plastic deformations occur, the maximum concrete unit shortening values allowed according to performance levels are calculated for the damage limit before section collapse. Equation (15B.8) for the controlled damage limit, equal to the value calculated by Equation (15B.8) 75% of the calculated value and 0.004 for the limited damage limit. These values and the unit elongation values of the reinforcement steel in the section 5.8 It cannot exceed the upper limits specified in .

15B.4. WRAPPING FOR INSUFFICIENT OVERLAPPING LENGTH IN COLUMNS

For columns with a section aspect ratio greater than two or with flat surface longitudinal reinforcement, the reinforcement of overlapping zones cannot be done with LP wrapping since the wrapping effect will be insufficient. In columns with ribbed longitudinal reinforcement, the required LP thickness to eliminate the overlap length insufficiency Equation (15B.9) It is calculated according to.

$$\epsilon_f = \frac{b}{w} \frac{f - f_{hs}}{2 TO_s 0.001} \quad (15B.9)$$

Equation (15B.9) in b_w the larger the section width, f_{hs} is the lateral pressure opposing 0.001 unit elongation in transverse reinforcement. **Equation (15B.9)** in f_{hs} **Equation (15B.10)** a, will be **Equation (15B.11)** According to

$$f_{hs} = 0.001 p_{s,h} TO_s \quad (15B.10)$$

$$f_{hs} = \frac{2r_c}{b_w} \frac{A_s f_{ym}}{\frac{p}{2n} + 2(\varphi + d') L_s} \quad (15B.11)$$

Equation (15B.10) in $p_{s,h}$ transverse reinforcement ratio (the smaller of the values obtained for the two directions), TO_s is the modulus of elasticity of the reinforcement steel. **Equation (15B.11)** also A_s column longitudinal reinforcement area (for the largest diameter single bar with lap splices), f_{ym} available steel for longitudinal reinforcement yield strength, p core cross-section perimeter, n number of overlapped reinforcement, φ longitudinal reinforcement diameter (if there are different diameter reinforcements, diameter of the largest diameter overlapped reinforcement), d' concrete cover thickness and L_s is the current overlap length. r_c the radius of the rounding made at the corners, b_w is the longitudinal dimension of the section.

15B.5. STRENGTHENING COLUMN-BEAM JOINT AREAS WITH LP

Reinforced concrete column-beam joints can be strengthened by bonding fibers parallel to the principal tensile stresses. The effectiveness of the reinforcement details to be applied must be proven experimentally. The LP effective unit elongation value to be taken into account in the calculations **Equation (15B.3)** must meet the condition in .

ANNEX 15C – STRENGTHENING OF FILLING WALLS

15C.0. ICONS

A_d	= Horizontal cross-sectional area of the infill wall
a_d	= Width of equivalent pressure bar [mm] =
TO_c	Elasticity modulus of frame concrete =
TO_d	Elasticity modulus of infill wall
TO_f	= Modulus of elasticity of fibrous polymer
f_d	strip = Compressive strength of infill wall
f_{yd}	= Design yield strength of mesh reinforcement steel
h_d	= Height of infill wall [mm] =
h_k	Column length [mm]
I_k	= Moment of inertia of the column [mm ⁴]
k_d	= Axial rigidity of equivalent compression
k_t	rod = Axial rigidity of tension rod =
$-min$	Minimum depth of anchor rod
LP	= Fibrous polymer
r_d	= Diagonal length of infill wall [mm]
s_{Max}	= Anchor rod maximum spacing = Thickness
t_d	of the infill wall [mm] = Thickness of the
t_f	fibrous polymer strip = Tensile strength of
T_f	the tensile rod = Shear strength of the infill
V_d	wall = Width of the fibrous polymer strip
w_f	
φ_{min}	= Anchor rod minimum diameter =
λ_d	Equivalent pressure rod coefficient
$\rho_{s.h.}$	= Horizontal body reinforcement in the curtain and wall to the gross cross-sectional area of the curtain body rate
θ	= Angle of equivalent pressure bar with
τ_d	horizontal = Shear strength of infill wall

15C.1. STRENGTHENING OF FILLING WALLS

15.10.4 According to , the rigidity and shear strength of infill walls within reinforced concrete frames that are continuous from the top of the foundation to the top can be increased by the strengthening methods described below.

15C.2. FILLING WALLS WITH SPECIAL PLASTER WITH WIRE MESH REINFORCED STEEL STRENGTHENING

The rigidity and shear strength of infill walls can be increased by applying a specially mixed plaster layer with steel mesh reinforcement to the wall surface.

(a) The thickness of the plaster layer should be at least 30 mm, and the mesh reinforcement concrete cover should be at least 20 mm. The compressive strength of the plaster will be at least 5 MPa.

(b) The ratio of the diagonal length of the walls to be strengthened to the thickness before strengthening should be less than 30. In such applications, the formation of a pressure bar should be provided within the existing frame and the necessary anchors should be arranged for the load transfer to the frame. For this, there should be a gap of at least 30 mm depth between the wall face where the application will be made and the outer face of the frame elements (**Figure 15C.1**). Otherwise, this type of wall reinforcement cannot be applied.

(c) The minimum diameter of the frame anchor rod to be used between the reinforced plaster layer and the frame elements should be 12 mm, the minimum anchor depth should be ten times the rod diameter and the widest rod spacing should be 300 mm. In addition, in order to ensure that the reinforced plaster layer and the existing infill wall work together, four body anchors will be made per square meter of wall area, perpendicular to the wall plane. The body anchor rods to be made perpendicular to the wall will be embedded into the mortar joints of the infill wall and the rod diameter will be at least 8 mm, and the anchor depth will be at least ten times the rod diameter. All anchor rods to be made parallel and perpendicular to the wall plane will be planted in the holes to be opened with an epoxy-based material and their ends will be bent 90 degrees in an L shape and inserted into the mesh reinforcement. Application details **Figure 15C.1** is shown in .

(d) The necessary foundation arrangement should be made to ensure that the forces generated in the reinforced infill walls are safely transferred to the ground. Walls reinforced with mesh steel reinforcement will be included in the structure model according to the principles given below.

15C.2.1 – Modeling Fundamentals: The stiffness and strength properties to be used for representing infill walls strengthened with wire mesh reinforcement in the structural model are defined below. Infill walls arranged within a reinforced concrete frame and having a ratio of diagonal length to thickness less than 30 will be considered in the structural model. Walls with voids whose ratio to wall surface area does not exceed 10% may be allowed to be included in the structural model, provided that the location of the voids does not prevent the formation of diagonal pressure bars. Infill walls strengthened with wire mesh reinforcement will be represented by equivalent diagonal bar elements receiving pressure force in the applied earthquake direction.

(a) **Rigidity:** The thickness of the equivalent pressure bar is equal to the thickness of the reinforced infill wall. Pressure bar width a_d **Equation (15C.1)** It will be calculated from .

$$a_d = 0.175 (\lambda h_c)^{0.4} r_d \quad (15C.1)$$

Here a_d bar width (mm), h_c column length (mm), r_d is the diagonal length of the infill wall (mm). λ_d

Equation (15C.2) It will be calculated from .

$$\lambda_d = \frac{-E_f f_{ds} \sin 2\theta - \frac{1}{4}}{-4 T O_c I_k h_d} \quad (15C.2)$$

Equation (15C.2) also TO_d And TO_c modulus of elasticity of infill wall and frame concrete, t_d And h_d Thickness and height of reinforced wall (mm), I_c moment of inertia of the column (mm⁴) and θ is the angle of the diagonal with the horizontal. The axial rigidity of the diagonal compression member **Equation (15C.3)** will be calculated with .

$$k_{\text{dd}} = \frac{\text{horse } TO_d}{r_d} \quad (15C.3)$$

(b) Shear Strength: The shear strength of the infill wall strengthened with mesh steel reinforcement will be considered as the horizontal component of the equivalent compressive strength of the diagonal bar. Horizontal cross-sectional area A_d , compressive strength f_d and shear strength τ_d Shear strength of reinforced infill wall V_d , **Equation (15C.4)** will be calculated with .

$$V_d = A_d (\tau_d + f_{yd} \rho_{s,h}) \leq 0.22 A_d f_d \quad (15C.4)$$

Here f_{yd} design yield strength of mesh reinforcement, $\rho_{s,h}$ the horizontal body reinforcements on the wall is the ratio of the wall gross cross-sectional area. Mesh reinforcement must have the same reinforcement area in horizontal and vertical directions.

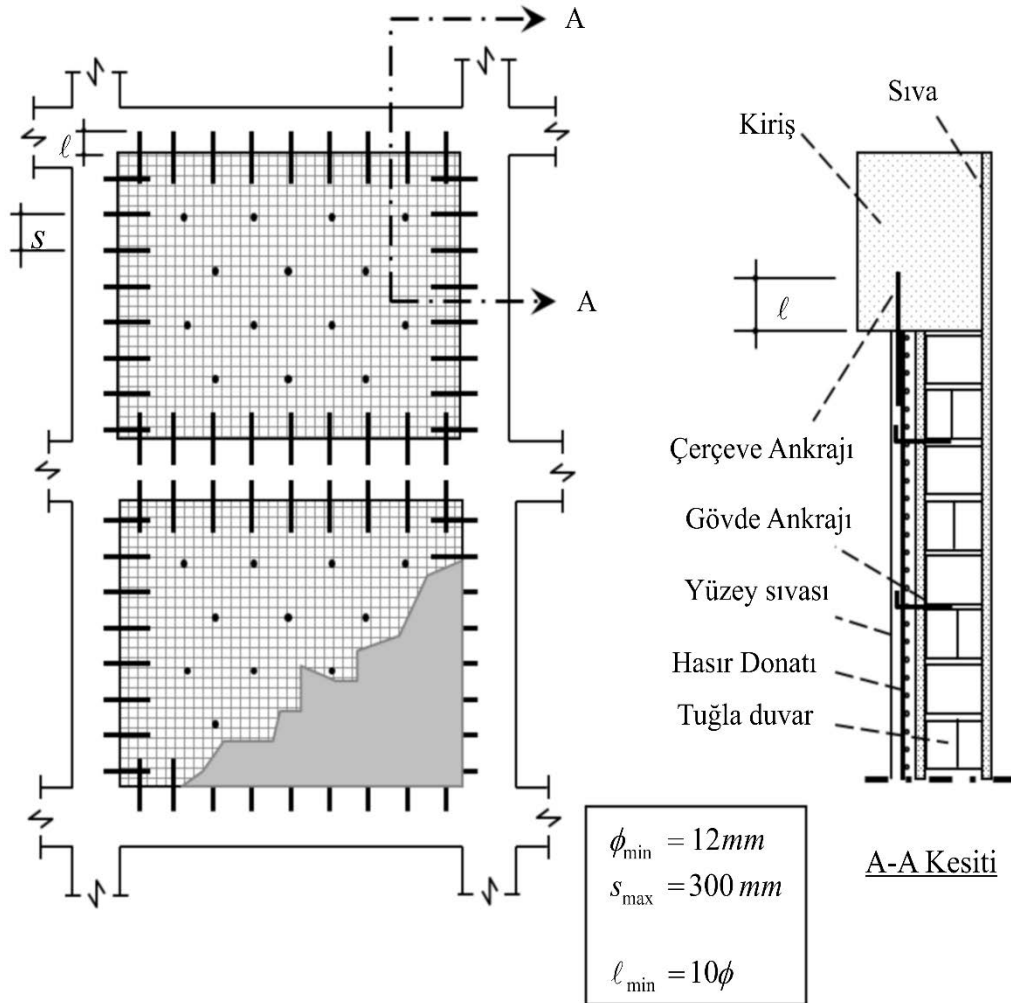


Figure 15C.1

15C.2.2 –Material Properties: In the equations given above TO_d , f_d and τ_d various bricks for Recommended values for infill walls made of types **Equation (15C.5)** The composite section structure of the reinforced wall can be taken into account in the calculation of the elasticity modulus, compressive and shear strengths.

Hollow factory brick: $TO_d = 2000 \text{ MPa}$; $f_d = 3.0 \text{ MPa}$; $\tau_d = 0.20 \text{ MPa}$ **(15C.5a)**

Solid blend brick: $TO_d = 2000 \text{ MPa}$; $f_d = 5.0 \text{ MPa}$; $\tau_d = 0.25 \text{ MPa}$ **(15C.5b)**

Aerated concrete block: $TO_d = 1000 \text{ MPa}$; $f_d = 2.5 \text{ MPa}$; $\tau_d = 0.20 \text{ MPa}$ **(15C.5c)**

15C.3. STRENGTHENING OF FILLING WALLS WITH FIBER POLYMERS

15C.3.1 –The rigidity and shear strength of infill walls with a length-to-height ratio of 0.5 to 2 can be increased by applying fibrous polymers (LP) to the wall surface.

(a) In such applications, the formation of pressure bars within the existing frame must be ensured and the necessary anchors must be arranged for load transfer to the frame. For this purpose, there must be a gap of at least 30 mm depth between the wall face where the application will be made and the outer face of the frame elements. Otherwise, this type of wall reinforcement cannot be applied.

(b) Detailing of diagonal fiber polymer strips **Figure 15C.2** In order to provide load distribution in corner areas and to place sufficient number of anchors between reinforced concrete frame and LP strips, square LP sheets with a width not less than 1.5 times the width of the strip shall be used. Fibrous polymer application shall be made on both sides of the wall and LP strips shall be fixed to the wall with LP bolts passing through the wall thickness. The distance between LP bolts shall not be more than 600 mm and the distance of the bolt to the edge of diagonal strip shall not be more than 150 mm. LP anchors shall be used to provide load transfer between diagonal LP strip and frame. LP anchors shall be formed by impregnating LP strips with epoxy and wrapping them around a silicone rod. The ends of LP anchors shall be shaped like a fan and at least four anchors shall be placed by injecting epoxy into the dust-free hole opened in the concrete in the direction of diagonal LP strip. The width of LP wrapped around the rod in anchorage construction shall not be less than 100 mm. The diameter of the anchor hole shall not be less than 10 mm and its depth shall not be less than 150 mm. Accordingly, the tensile strength of an anchor prepared shall be taken as 20 kN or 30% of the tensile capacity of the LP wrapped around the silicone rod, whichever is less.

(c) The necessary foundation arrangement should be made to ensure that the forces generated in the reinforced infill walls are safely transferred to the ground. Walls reinforced with fibrous polymers will be included in the building model according to the principles given below.

15C.3.2 –Modeling Fundamentals: Infill walls reinforced with fibrous polymers will be represented by diagonal compression and tension bar pairs in the structural model.

(a) Pressure Bars: Stiffness and shear strength of compression bars **15C.2.1(a)** And

(b) It will be calculated according to .

(b) Towbars: Tensile strength of the tension rod T_i **Equation (15C.6)** with will be calculated.

$$T_i = 0.003 TO_d w t \quad \textbf{(15C.6)}$$

The shear strength of the tension bar shall be considered as the horizontal component of the tensile strength. The axial rigidity of the tension bar shall be **Equation (15C.7)** will be calculated with .

$$k_t = \frac{w_f t_f TO_f}{r_d} \quad (15C.7)$$

In these equations TO_f , w_f And t_f are the elasticity modulus, width and length of the fibrous polymer tape, respectively. thickness, r_d is the diagonal length of the infill wall. w_f value **Equation (15C.1)** calculated with cannot be larger than the width.

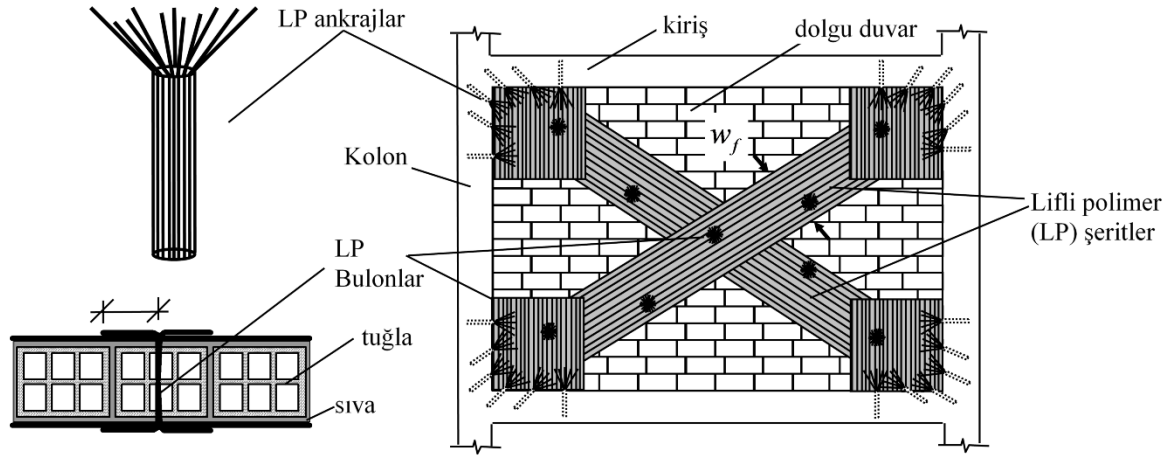


Figure 15C.2

CHAPTER 16 – FOUNDATION AND FOUNDATIONS UNDER EFFECT OF EARTHQUAKE SPECIAL RULES FOR DESIGN

16.0. ICONS

A_c	= Total area where compressive stresses occur under the foundation [m ²]
BYS	= Building Height Class
B	= Effective foundation
b	width = Slice width [m]
b_c, b_q, b_y	= Foundation base slope correction coefficients
C_B	= Drill hole diameter correction factor =
C_{TO}	Energy ratio correction factor
C_M	= Design earthquake moment magnitude correction factor
C_N	= Geological stress (depth) correction applied to cohesionless soils coefficient
CPT	= Cone Penetration Test = Rod
C_R	length correction coefficient
$CRR_{M7.5}$	= Cyclic resistance ratio against an earthquake with moment magnitude 7.5
C_S	= Sampler type correction factor
c	= Cohesion strength of soil [kPa]
c'	= Effective (drained) cohesive strength of the soil
CU	[kPa] = Consolidated-Drained
c_u	= Undrained shear strength [kPa]
$(c_u)_{30}$	= Average undrained shear strength of the upper 30 meters [kPa] =
$c_{u,i}$	Undrained shear strength of the i th sub-layer [kPa]
D	= Pile diameter [mm] =
DTS	Earthquake Design Class
d_c, d_q, d_y	= Foundation depth correction coefficients
d_{This}	= Underwater wall height [m]
TO_{giant}	= Total of the effects that force the retaining structure to collapse
TO	= Design effects related to loading cases including static and earthquake =
F_H	Horizontal inertia force acting on the slope [kN]
F_V	= Vertical inertia force acting on the slope [kN] =
G_{Max}	Maximum shear modulus [MPa]
g_c, g_q, g_y	= Ground slope correction coefficients H = Total height of retaining structure [m]

H_b	= Total height of basement wall [m]
H_c	= Vertical distance from the midpoint of the slice base to the slice center [m] =
h_i	i'th substrate thickness [m]
I_c, I_q, I_y	= Loading skew correction coefficients
K	= Typical total (static+dynamic) earth pressure coefficient =
K_a	Total (static+dynamic) active earth pressure coefficient = Total
K_p	(static+dynamic) passive earth pressure coefficient
k_h	= Static-equivalent horizontal earthquake coefficient =
k_v	Static-equivalent vertical earthquake coefficient =
M_w	Moment magnitude of the design earthquake
N	= Standard penetration test (SPT) blow number =
N_b	Foundation tie beam design axial force [kN]
N_c, N_q, N_y	= Bearing capacity coefficients
N_k	= The maximum axial force in the column or wall to which the foundation tie beam is connected [kN]
$(N_{60})_{30}$	= Average number of standard penetration blows in the upper 30 meters
$N_{60,i}$	= standard penetration stroke number of the ith substrate
$N_{1.60}$	= Corrected SPT hit count
$N_{1.60f}$	= SPT hit count corrected for fines content
PI	= Plasticity index
P_t	= Resultant of total (static and dynamic) earth pressure acting on the retaining structure
P_{this}	= Resultant static water pressure [kN/m ²]
P_{TV}	= Design vertical pressure force acting on the foundation base, vertical pressure force acting on the pile design force [kN]
P_{ty}	= Lateral design force acting on the pile [kN]
p_{this}	= Static water pressure [kN/m ²]
Q_{ks}	= Pile characteristic perimeter friction resistance [kN] =
Q_{ku}	Pile characteristic tip resistance [kN]
Q_{KTV}	= Pile characteristic total bearing capacity [kN] =
Q_s	Pile perimeter friction resistance [kN]
Q_{TV}	= Vertical design strength of pile [kN] =
Q_{ty}	Lateral design strength of pile [kN]
Q_u	= Pile tip resistance [kN] =
q	Additional load (surcharge) [kN/m ²]

q_k	= Foundation bearing capacity characteristic strength [kN/m ²]
q_{he}	= Vertical load acting at the foundation level, created by shear and moment effects foundation base pressure [kN/m ²]
q_t	= Foundation bearing capacity design strength [kN/m ²]
R_d	= Radius of circular failure surface [m]
R_{giant}	= Total of the effects and resistances resisting overturning in the retaining structure
R_k	= Characteristic strength [kN/m ²] = Characteristic passive resistance [kN/m ²]
R_{pk}	
R_{pt}	= Design passive resistance [kN/m ²]
R_t	= Design strength [kN/m ²]
R_{th}	= Design friction resistance [kN/m ²] = Static-
r	equivalent earthquake reduction factor
S_{DS}	= Short period design spectral acceleration coefficient [dimensionless]
SPT	= Standard Penetration Test =
S_T	Topographic amplification coefficient
S	= Ratio of pile row spacing to pile diameter =
S_c, S_q, S_y	Foundation shape correction coefficients
t	= Steel pipe pile wall thickness [mm] =
UU	Unconsolidated – Undrained
U	= Water pressure acting at the base of the slice [kN/m ²]
V_s	= Shear wave velocity [m/s]
$(V_s)_{30}$	= Average shear wave velocity in the upper 30 meters [m/s]
V_{si}	= Shear wave velocity of the i th sub-layer [m/s]
V_{th}	= Design horizontal force acting at the foundation base [kN]
W	= Natural water content
W	= Weight of sliding slope mass [kN] =
W_s	Weight of slice [kN]
Z	= Height measured from the water surface downwards [m]
$\bar{Z}$	= The resultant of static-equivalent dynamic water pressure measured from the water surface depth
α	= Angle of the slice base to the horizontal [degrees]
β	= Angle of inclination of the ground surface behind the wall relative to the horizontal [degrees]
β_G	= Empirical to approximate group effect in piles reduction coefficient
ΔP_{this}	= Resultant dynamic water pressure [kN/m ²]

Δp	= Additional ground pressure caused by the earthquake [kN/m ²]
$\Delta p_{\text{This}}(z)$	= Variation of static-equivalent dynamic water pressure along water depth [kN/m ²] = Friction
δ	angle between the foundation base and the ground [degrees]
δ_d	= Angle of friction between the ground and the wall [degrees] =
φ'	Effective shear resistance angle of the ground [degrees]
φ'_d	= Design shear resistance angle of the soil
γ	[degrees] = Natural unit weight of the soil [kN/m ³]
γ_d	= Unit volume weight of the soil saturated with water [kN/m ³]
γ_{This}	= Unit volume weight of water [kN/m ³]
γ^*	= Typical unit weight of soil [kN/m ³] =
γ_R	Strength coefficient
γ_{Rc}	= Coefficient of resistance for cohesion resistance
γ_{Rdev}	= Design safety factor against overturning in the retaining structure
γ_{Rh}	= Friction resistance strength factor
γ_{Rk}	= Design safety factor against slipping
γ_{Rp}	= Passive resistance coefficient
γ_{Rsb}	= Pile perimeter friction resistance (compression) strength coefficient
γ_{Rsc}	= Pile perimeter friction resistance (tensile) strength coefficient
γ_{Rt}	= Pile total bearing capacity (compression) resistance coefficient =
γ_{Ru}	Pile tip resistance resistance coefficient
γ_{Rv}	= Basic bearing capacity strength coefficient =
$\gamma_{R\varphi}$	Strength coefficient for friction resistance
ψ	= Angle between the back surface of the wall and the base of the wall [degrees]
ρ	= Unit volume mass of soil [t/m ³] =
$\tau_{cy,u}$	Undrained shear strength [kPa]
$\tau_{\text{Earthquake}}$	= Average cyclic shear stress in the ground caused by the earthquake [kPa]
τ_R	= Liquefaction resistance of soil [kPa]
θ	= Angle based on static-equivalent earthquake coefficient [rad]

16.1. SCOPE

16.1.1 –The scope of the necessary ground investigations for new buildings to be designed under the effect of earthquakes and existing buildings to be evaluated or strengthened for their seismic performance, determination of ground conditions, classes and parameters, design of building foundations and surrounding basement walls under the effect of earthquakes, structure-ground interaction analyses and ground analysis.

The assessment of liquefaction potential will be carried out in accordance with the rules and conditions given in this Section, together with the regulations and standards on these matters.

16.1.2 –The design principles of soil retaining structures and slopes under earthquake effects are also given in this section.

16.1.3 –General principles for in situ improvement of land soil properties to be applied if necessary **ANNEX 16D** is given in .

16.2. DESCRIPTION OF SOIL CONDITIONS

The studies to be carried out to define the construction site ground conditions, determine and report the geotechnical parameters required for structural design are specified below.

16.2.1. Soil Research

16.2.1.1 –In order to determine the ground conditions, ground investigations including field and laboratory studies will be carried out. The scope of ground investigations will be planned by taking into consideration the characteristics of the structure and its components, the characteristics of the geological structure and ground units, the status of the surrounding structures, the status of the groundwater, regional earthquake characteristics and environmental conditions, drilling wells and/or inspection pits of sufficient number and depth will be opened, necessary field tests will be carried out, disturbed and undisturbed samples will be taken and laboratory tests will be applied.

16.2.1.2 –Soil investigations **ANNEX 16A** It will be done according to the rules defined in .

16.2.2. Soil and Foundation Survey Reports

Soil and Foundation Survey Reports, where the results of ground investigations will be presented *Data Report* and will be prepared for design *Geotechnical Report* It will consist of.

16.2.2.1 –*Data Report*, is a report presenting the data obtained from the ground investigations carried out in the field and laboratory. Within the scope of this report, the geological structure of the region and the geological characteristics of the project area, research drillings and inspection pit logs, ground sections and groundwater level, field and laboratory test results, geophysical research findings, etc. ground investigation results will be presented.

16.2.2.2 –*Geotechnical Report*, is a report in which the ground model of the land is created by taking into account static, dynamic and earthquake effects, geotechnical design parameters for the ground layers are given, options for the selection of foundation types are examined, engineering analyses and evaluations and suggestions for foundation design are presented. Explanations regarding the content of the geotechnical report are given below:

(a) By evaluating the field soil research findings presented in the data report, a field soil model will be created, the behavior of the soil layers under building loads and earthquake effects will be examined, and geotechnical design parameters related to the design of the structure and its foundations will be given.

(b) The foundation system compatible with the structural features and expected performance level will be selected, bearing capacity and short and long term soil displacement-settlement values will be given, and within this scope, the swelling behavior of the soils, net foundation pressures and possible uplift forces will be taken into account.

(c) If ground improvement and/or reinforcement is required, possible methods will be examined, application principles for the proposed method will be defined, targeted ground properties for the improved ground, bearing capacity of the foundations and displacement values will be given.

(d) The soil parameters required for the design of the retaining systems to be applied in temporary or permanent foundation excavations will be given. For structures to be constructed on sloping lands with the risk of slope instability, slope stability analyses will be conducted by taking into account the excavation and construction steps and appropriate precautions to be taken against sliding will be determined.

16.3. DETERMINATION OF SOIL PARAMETERS

The following principles will be followed in determining the soil parameters to be used in the design of building foundations under earthquake effects, the definition of local soil classes, and the analysis of retaining structures and slopes.

16.3.1 – In the evaluations, strength parameters compatible with drained or undrained conditions will be used, which will be determined according to the loading rate and permeability conditions of the field soils.

16.3.2 – In cohesive soils, undrained shear is taken into account in total stress analyses, taking into account the strength losses/softening that may occur under the effect of an earthquake. c_u value will be used.

16.3.3 – In cohesionless soils, the undrained shear strength (τ) is calculated in the total stress analyses, taking into account the pore water pressure increases and the internal friction angle decreases that will occur under the effect of an earthquake. c_v, u value will be used.

16.3.4 – If pore water pressures can be determined under earthquake effects, valid for cohesive and non-cohesive soils, analyses can be performed using effective stress parameters.

16.3.5 – Minimum uniaxial compressive strength for rocks q_u , geological strength index (GSI) Appropriate strength parameters will be determined using rock mass classification parameters such as values etc.

16.3.6 – The maximum shear modulus to be used in the analysis is, G_{Max} Equation (16.1) will be obtained from.

$$G_{Max} = \rho V_s^2 \quad (16.1)$$

Shear wave velocity (V_s) is essential to be determined by geophysical methods. However, standard It can also be calculated indirectly using field test results such as the penetration test (SPT), cone penetration test (CPT) and generally accepted relations.

16.3.7 – 16.4 Shear wave velocity in determining local soil classes according to V_s , SPT pulse number N and for cohesive soils c_u value can be used.

16.3.8 – In site-specific soil response analyses and dynamic soil-structure interaction analyses, the earthquake waves formed during propagation within the soil profile are taken into account. *shear unit*

shear modulus compatible with deformation and equivalent hysteretic damping coefficient will be used (See 16.5.2).

16.4. DETERMINATION OF LOCAL SOIL CLASSES

16.4.1 – Part 2 According to , the definition of earthquake design spectra will be based on local soil classes, will be determined by ground investigations to be carried out in accordance with this Section.

(a) Soils requiring site-specific research and evaluation, are classified as ZF class soils. Definitions related to soils in this class Table 16.1 is given in the last line of .

(b) To determine other local soil classes defined as ZA, ZB, ZC, ZD and ZE, the properties related to these classes Table 16.1 are given in other lines of .

Table 16.1 – Local Soil Classes

Local Ground Class	Soil Type	Average in the upper 30 meters		
		$(V_s)_{30}$ [m/s]	$(N_{60})_{30}$ [impact /30 cm]	$(c_u)_{30}$ [kPa]
Z A	Solid, hard rocks	> 1500	–	–
ZB	Slightly weathered, moderately strong rocks	760 – 1500	–	–
ZC	Very dense layers of sand, gravel and hard clay or weak, weathered, highly fractured rocks	360 – 760	> 50	> 250
ZD	Medium dense – dense sand, gravel or very solid clay layers	180 – 360	15-50	70-250
ZE	Loose sand, gravel or soft to solid clay layers or $P > 20$ and $w > 50\%$ Soft clay with a total thickness of more than 3 meters meeting the conditions of 40 layer ($c_u < 25$ kPa) Profiles containing	< 180	< 15	< 70
ZF	Soils requiring site-specific research and evaluation: 1) Soils that have the risk of collapse and potential collapse under the influence of an earthquake (liquefiable soils, highly sensitive clays, collapsible weakly cemented soils, etc.), 2) Peat and/or clays with high organic content with a total thickness of more than 3 meters, 3) High plasticity with a total thickness of more than 8 meters ($P > 50$) clays, 4) Very thick (>35 m) soft or medium hard clays.			

16.4.2 – Table 16.1 The soil parameters given in shall be determined for the top 30 m thick section of the soil profile starting from the foundation or pile cap bottom level. In soil profiles containing distinctly different soil and rock layers, the layers in the top 30 meters shall be sufficiently separated into sub-layers so that $i = 1$ at the top and $i = N$ at the bottom. will be listed as follows. The top 30 meters average shear wave velocity $(V_s)_{30}$, average standard penetration blow count $(N_{60})_{30}$ And average undrained shear strength $(c_u)_{30}$ Equation (16.2) will be calculated with:

$$(V_s)_{30} = \frac{30}{\sum_{i=1}^N \frac{h_i}{V_{s,i}}} ; (N_{60})_{30} = \frac{30}{\sum_{i=1}^N \frac{h_i}{N_{60,i}}} ; (c_u)_{30} = \frac{30}{\sum_{i=1}^N \frac{h_i}{c_{u,i}}} \quad (16.2)$$

Here h value (i) is the thickness of the sublayer [m], V_{si} , $N_{60,i}$ And $C_{u,i}$ The values are in order of and , show the shear wave velocity [m/s], standard penetration test blow number [blow/30 cm] and undrained shear strength [kPa] of the same substrate.

16.4.3 –In shallow foundations, if there is a soil thickness of more than 3 m between the foundation base level and the top of the rock level, ZA and ZB class definitions will not be made.

16.5. SITE-SPECIFIC SOIL BEHAVIOR ANALYSES UNDER EARTHQUAKE IMPACT

16.5.1. Scope

16.5.1.1 –*Site specific ground behavior analyses*, is performed to determine the variation of the earthquake ground motion defined in the bedrock along the soil layers and the earthquake ground motion on the ground surface.

16.5.1.2 –For site-specific soil behavior analyses in cases where the ground environment in the building foundation and its immediate surroundings consists of approximately horizontal layers.

16.5.2's based on one-dimensional *horizontally layered free ground model* Otherwise, two or three dimensional ground models will be used.

16.5.1.3 – **Table 16.1** According to , it is mandatory to perform a site-specific ground response analysis to determine the earthquake ground motion on the surface of soils defined as ZF class.

16.5.1.4 –Field specific *free ground analysis* The results of for pile foundations **16C.3.2** And **16C.4.2** Explained in *kinematic interaction analyses* It will be used as earthquake data.

16.5.2. Nonlinear Free Ground Model and Earthquake Analysis

16.5.2.1 –In order to create a site-specific soil behavior model, horizontal soil layers *shear unit deformation* depending on *shear modules*'s and *equivalent hysteretic damping coefficients* It is necessary to define the nonlinear variations of . These parameters will be defined according to field and laboratory tests and generally accepted knowledge in the literature for similar soil conditions.

16.5.2.2 –Soil behavior analysis, horizontally layered *free ground model* Under the earthquake ground motion defined in the base rock within the framework of **16.5.2.1** defined in *nonlinear* It will be performed with one of the following analysis methods based on dynamic soil parameters:

(a) In the time domain *nonlinear analysis*,

(b) Provided that the shear strain does not exceed 1% in the frequency domain *equivalent linear calculation model* by using the sequential approach *nonlinear analysis*.

16.5.2.3 –The free ground model will be created as follows:

(a) Table 16.1 According to , the soil layer defined as local soil class ZA or ZB is the soil layer where the earthquake will affect the soil profile. *engineering bedrock* The depth of the engineering base rock defined in this way from the building foundation will not be less than three times the largest building width and the longest pile length in pile systems. In this case **2.3.4** to or **2.4.1** Spectral magnitudes describing the design earthquake ground motion according to , **Table 2.1** And **Table 2.2** It will be reduced by taking into account the local ground effect coefficients given for the ZA or ZB local class.

(b) However, if the bedrock is much deeper than the lower limit given above, the layered soil model, **Table 16.1** According to , it can be terminated with the soil layer with local soil class ZC or ZD. In this case, the earthquake effect will be defined on this layer instead of the base rock and **2.3.4** to **2.4.1** Spectral quantities describing the design earthquake ground motion according to **Table 2.1** and **Table 2.2** The local ground effect coefficients given for the ZC or ZD local class will be enlarged by taking into account this layer and the ground environment below it. This layer and the ground environment below it will be enlarged by taking into account the local ground effect coefficients given for the ZC or ZD local class in the one-dimensional ground profile model. *permeable boundary* will be idealized using the condition.

(c) For the sake of accuracy of the analysis, the soil layers will be subdivided into sufficiently fine sublayers. In addition to the uncertainties in the total depth of the soil model, the uncertainties in the dynamic soil parameters will also be taken into account by appropriate sensitivity analyses.

16.5.2.4 – 16.5.2.3 According to , the ground model base is to be affected by the one-dimensional layered ground model, **2.3.4** or **2.4.1** At least eleven earthquake records are consistent with the elastic acceleration spectrum defined according to **2.5.3** will be defined according to.

16.5.2.5 – In the context of determining the site-specific earthquake spectrum at the ground surface, the ratio of the ground surface spectral acceleration to the basement rock spectral acceleration for each spectral period will be calculated for each record, then these ratios will be averaged for at least eleven records to determine the earthquake spectral acceleration for the relevant period. *local ground effect coefficient* These coefficients will be defined as **2.3.4** or **2.4.1** As a result of multiplying with the base rock spectrum defined in , the ground surface *site specific earthquake spectrum* In soils other than local soil class ZF, the ordinates of the site-specific earthquake spectrum determined on the ground surface, **Table 2.1** and **Table 2.2** According to , it cannot be taken smaller than the spectral accelerations determined by taking into account the relevant local ground class.

16.5.2.6 – 16.6 According to , in soils with liquefaction potential **16.5.2.2(b)** According to , no calculation will be made in the frequency domain with the equivalent linear analysis model. In these cases **16.5.2.2(a)** According to , nonlinear analysis will be performed in the time history.

16.6. ASSESSMENT OF THE RISK OF SOIL LIQUEFICATION UNDER EARTHQUAKE IMPACT

16.6.1 – Chapter 3 According to *Earthquake Design Class* For buildings with $DTS=1$, $DTS=1a$, $DTS=2$ and $DTS=2a$ **Table 16.1** In , they are in the form of a continuous layer or thick lenses, falling into the ZD, ZE or ZF groups and **16.6.6** In sandy soils other than those described in *liquefaction potential* It is mandatory to examine the presence or absence of using appropriate analysis methods based on field and laboratory tests and to report the analysis results in detail.

16.6.2 – Soil liquefaction will be defined as a significant decrease in shear strength and stiffness of cohesionless or low cohesion ($PI < 12\%$) soils located below the groundwater level and up to 20 m below the surface, parallel to the increase in pore water pressure under earthquake shaking.

16.6.3 – The soil investigation studies to be carried out for the assessment of soil liquefaction will include, at a minimum, the determination of grain diameter distribution, water content and Atterberg limit values in the relevant soil layers in addition to the performance of standard penetration test (SPT) and/or cone penetration test (CPT).

16.6.4 –Potentially liquefiable soils will be defined as sand, gravelly sand, silty clayey sand, non-plastic silt, and silt-sand mixtures located below the groundwater table.

16.6.5 –The sub-base soils consist of potentially liquefiable soils and the corrected SPT hit count in these soil layers, $N_{1.60}$, Less than 30 strokes / 30 cm

In cases where soil liquefaction triggering assessment will be made.

16.6.6 –*Earthquake Design Class* Liquefaction trigger analysis may not be performed if $DTS = 4$ and at least one of the following is also met:

(a) In sandy soils where the clay content is more than 20% and the plasticity index is higher than 10%;

(b) fines percentage more than 35% and corrected SPT hit count, $N_{1.60}$, On sandy soils where it is higher than 20 strokes / 30 cm;

16.6.7 –In the assessment of soil liquefaction, in addition to the risk of triggering liquefaction, the loss of soil strength and stiffness after liquefaction and the displacements that may occur in the foundation soil should be taken into account.

16.6.8 –Method based on soil liquefaction assessment using SPT test results **ANNEX 16B** In case the evaluation is made according to CPT or shear wave velocity, generally accepted methods can be used in practice.

16.6.9 –Safety condition against liquefaction **Equation (16.3)** It is defined in .

$$\frac{\tau_R}{\tau_{\text{earthquake}}} \geq 1.10 \quad (16.3)$$

Here τ_R And $\tau_{\text{earthquake}}$, respectively, the liquefaction resistance and the average earthquake-induced stress on the ground

represents cyclic shear stress. **Equation (16.3)** If the condition given in is not met, the decrease in the strength and rigidity properties of the layers expected to liquefy, possible bearing capacity losses, stability disorders and ground movements such as settlement and lateral spreading will be evaluated.

16.6.10 –The effects of the determined post-liquefaction displacements on the superstructure/ infrastructure behavior will be evaluated and, if necessary, superstructure and/or ground improvements will be implemented.

16.7. GENERAL RULES FOR DESIGN OF FOUNDATIONS

16.7.1. Purpose

16.7.1.1 –The purpose of building foundation design under earthquake effects is to ensure that the foundation bearing capacity is not exceeded and that ground displacements remain within permissible limits. The following general principles will be followed in foundation design.

16.7.1.2 –If the foundations are located near slopes, landslide areas, excavations, tunnels, underground and mining excavations, foundation general stability and/or interaction analyses should be performed. General stability analyses **16.13** It will be done in accordance with the principles given in. The scope of interaction analyses and design conditions are outside the scope of this Regulation.

16.7.2. Bearing Capacity Principle in Foundation Design

In this regulation, for the geotechnical design of shallow and deep foundations, *carrying capacity principle* The foundation soil is based on possible collapse mechanisms *design carrying capacity* sufficiency of **Equation (16.4)** will be provided by the general expression given in:

$$TO \leq R_t \quad (16.4)$$

Here TO for static and seismic loading conditions *design effects* R_t which is against the relevant migration mechanism *design strength* It represents the .

16.7.3. Design Effects

16.7.3.1 –Static load combinations will be taken from the relevant regulations. Load combinations that include earthquake effects will be taken from the relevant regulations. **4.4.4** The effects occurring on the foundation ground are given in, TO , vertical load effects together with **4.10.3** According to , the calculation will be based on the forces transferred from the building carrier system to the foundation during an earthquake.

16.7.3.2 –The axial force and bending moment of the design are the vertical forces at the base of the foundation. *basic bearing capacity* will be met with.

16.7.3.3 –The horizontal shear force, which is the basis of the design, is the force between the ground and the foundation base. *friction resistance* along with the base side surface *passive earth pressure* It will be met by taking into account a maximum of 30% of the .

16.7.4. Design Strengths

Regarding loading cases including static and earthquake *design strength* R_t , *characteristic strength* R_k *coefficient of resistance* y_R It will be found by dividing by .

$$R_t = \frac{R_k}{y_R} \quad (16.5)$$

Strength coefficient The values of depend on the foundation type and the calculated strength component.

16.8.2 (Table 16.2) And **16.9.3 (Table 16.4)** is given in).

16.8. SURFACE FOUNDATIONS

16.8.1. Scope

The following rules shall be followed in dimensioning shallow foundations. The design principles given here cover single, continuous (strip) and raft (slab) foundations.

16.8.1.1 –The bearing capacity of the shallow foundations and their design resistance against horizontal shear will be calculated and it will be shown that they meet the design effects in static and earthquake loading cases.

16.8.1.2 –In soils where excessive pore water pressure increases may occur during an earthquake, the possible effects on undrained shear strength should be taken into account in total stress analyses, and pore water pressure should be taken into account in effective stress analyses.

16.8.2. Strength Coefficients for Shallow Foundations

For all loads multiplied by the load coefficients *design strength* R , **Table 16.2** Given in *coefficient of resistance* y_R will be calculated using the values.

Table 16.2. Strength Coefficients for Shallow Foundations

My endurance Type	Strength Coefficient Symbol	Strength Coefficient The value of
Basic Carrying Power	y_{Rv}	1.4
Friction Resistance	y_{Rh}	1.1
Passive Resistance	y_{Rp}	1.4

16.8.3. Bearing Capacity of Shallow Foundations

16.8.3.1 –In each of the loading cases including static and seismic effects **Equation (16.6)** The inequality in will be satisfied:

$$q_{he} \leq q_t \quad (16.6)$$

Here q_{he} It is the foundation base pressure created by the vertical load, shear and moment effects acting at the foundation level. q_t is the design strength R is the equivalent of the basic bearing capacity and **Equation (16.7)** is defined by:

$$q_t = k \frac{q}{y_{Rv}} \quad (16.7)$$

16.8.3.2 –Basic bearing capacity *characteristic strength* q_k **Equation (16.8)** will be calculated with .

$$q_k = c N_c s_c d_c I_c g_c b_c + q N_q s_q d_q I_q g_q b_q + 0.5 \gamma B' N_y s_y d_y I_y g_y b_y \quad (16.8a)$$

Equation (16.8a) Bearing capacity coefficients included in **Equation (16.8b)** It is defined in:

$$N_q = \frac{1}{\tan \varphi'} \tan^2(45 + \varphi'/2) \quad ; \quad N_c = (N_q - 1) \cot \varphi' \quad ; \quad N_y = 2 (N_q - 1) \tan \varphi' \quad (16.8b)$$

Equation (16.8a) dimensionless in *correction coefficients* located as *basic shape coefficients* s_c, s_q, s_y ; *depth coefficients* d_c, d_q, d_y ; *loading inclination coefficients* i_c, I_q, I_y ; *foundation ground slope coefficients* g_c, g_q, g_y And *foundation base slope coefficients* b_c, b_q, b_y will be calculated using generally accepted correlations based on literature.

16.8.3.3 –The presence of variable layers and/or discontinuities in the foundation soil within the effective depth of the foundation will be taken into account in the bearing capacity calculation.

16.8.3.4 –Displacements under the foundation will remain within permissible limits. In this context;

(a) Under the effect of an earthquake, sub-foundation displacements will be calculated using generally accepted geotechnical engineering approaches, taking into account the possible loss of strength and stiffness and the increases in pore water pressure under cyclic loading in soft clays and water-saturated loose-medium dense cohesionless soils.

(b) Section 13 In high-rise buildings defined in , floors other than ZA and ZB classes and *Earthquake Design Class* In other buildings with DTS = 1, 1a, 2, 2a, permanent deformations will be calculated by taking into account the non-linear soil behavior that may occur under shallow foundations on soils other than classes ZA, ZB and ZC.

16.8.4. Horizontal Slippage of Shallow Foundations

16.8.4.1 – Horizontal slip in each of the static and seismic loading cases with respect to **Equation (16.9)** The inequality in will be satisfied:

$$V_{th} \leq R_{th} + 0.3 R_{pt} \quad (16.9)$$

Here V_{th} acting on the foundation base *design horizontal force*, R_{th} *design friction resistance*, R_{pt} whereas *design passive resistance* It shows the .

16.8.4.2 – Design friction resistance R_{th} in drained condition **Equation (16.10)** can be calculated with:

$$R_{th} = \frac{P_{TV} \tan \delta}{y_{Rh}} \quad (16.10)$$

Here, P_{TV} represents the design vertical pressure force acting on the foundation base, and δ represents the friction angle between the foundation base and the soil.

16.8.4.3 – The coefficient of friction $\tan \delta$, unless otherwise determined by field tests, **Table 16.3** Values greater than those given in will not be taken.

Table 16.3. Coefficient of Friction Between Shallow Foundations and Soil

Friction Interface	$\tan \delta$
Cast-in-Situ Concrete – Compacted Foundation Base Soil	0.6
Precast Concrete – Compacted Foundation Base Soil	0.4
Cast-in-Situ Concrete – Concrete	0.5
Concrete – Base Rock	0.5

16.8.4.4 – Design friction resistance R_{th} in cohesive soils (in undrained condition)

Equation (16.11) can be calculated with .

$$R_{th} = \frac{c_u A_c}{y_{Rh}} \quad (16.11)$$

Here, A_c It refers to the total area where compressive stresses occur under the foundation.

16.8.4.5 – Design passive resistance R_{pt} , *characteristic passive resistance R_{pk} 's coefficient of resistance* It will be calculated by dividing by:

$$R_{pt} = \frac{R_{pk}}{y_{Rp}} \quad (16.12)$$

16.8.4.6 –In foundations below the groundwater level, the design friction resistance during an earthquake will be calculated based on the undrained shear strength of the soil.

16.8.5. Foundation Tie Beams

16.8.5.1 –In reinforced concrete and steel buildings, tie beams will be arranged to connect individual foundations or pile caps in both directions, and continuous foundations in column or wall alignments. **Table 16.1** According to , tie beams may not be made in foundation soils with local soil class ZA.

16.8.5.2 –Tie beams can be constructed at any level from the bottom of the foundation to the column base, in accordance with the foundation excavation.

16.8.5.3 –Axial force to be considered in the section design of the connecting beam N_b **Equation (16.13)** is given with.

$$N_b = 0.10 S_{DS} N_k \quad (16.13)$$

Here N_k The maximum axial force in the column or wall to which the tie beam is connected, S_{DS} whereas *short period design spectral acceleration coefficient* it shows the .

16.8.5.4 –In the section calculation, it will be taken into account that the tie beams will work under both compression and tension forces. In the case of tie beams covered by soil or base concrete working under compression, the buckling effect may not be taken into account. In the case of tension, it will be assumed that the tension force is carried only by the reinforcement. The minimum size of the tie beams will be 300 mm × 300 mm, the reinforcement ratio will be at least 0.5%, the stirrup diameter will be 8 mm and the stirrup spacing will be 200 mm.

16.8.5.5 –Reinforced concrete slabs can also be used instead of tie beams. In this case, the slab thickness will not be less than 150 mm. The slab and the reinforcement placed in it, **Equation (16.13)** It will be shown by calculation that it can safely transfer loads equal to the axial loads given by .

16.9. PILE FOUNDATIONS

16.9.1. Scope

The following rules shall be followed in dimensioning pile foundations. The design principles given here cover cast-in-situ and driven piles.

16.9.1.1 –The vertical and lateral bearing forces of the pile foundations will be calculated and it will be shown that they meet the design effects in static and earthquake loading cases.

16.9.1.2 –One of the following approaches can be applied in the calculation of the bearing capacity of pile foundations. Bearing capacity;

(a) Static loading can be calculated using data obtained from test results.

(b) It can be calculated using soil properties obtained from ground investigations.

(c) It can be calculated using the results obtained from static loading tests and validated dynamic loading tests.

16.9.2. Strength Coefficients for Pile Foundations

16.9.2.1 –In the calculation of vertical bearing capacity of pile foundations, the soil properties obtained from the results of soil research and loading tests will be used. *environmental friction* Q_s And *terminal resistance* Q_u In the account of **Table 16.4** The strength coefficients given in γ_R will be applied.

16.9.2.2 – Table 16.4 The values of the strength coefficient given for the pile loading test in the case of pile loading can be used if at least one loading test is performed under each building and at least 1% of the piles used in the project site. In case more tests are performed than these minimum numbers, the strength coefficients can be reduced by up to 0.10 in the light of regional experiences.

16.9.3. Vertical Bearing Capacity of Piles

16.9.3.1 –In each of the loading cases including static and seismic effects **Equation (16.14)** The inequality in will be satisfied:

$$P_{TV} \leq Q_{TV} \quad (16.14)$$

Here P_{TV} is the vertical design force acting on the pile. Q_{TV} is the vertical design strength of the pile and **Equation (16.15a)** or **Equation (16.15b)** is defined by:

$$Q_{TV} = \frac{Q_{ks}}{\gamma_{Rs}} + \frac{Q_{ku}}{\gamma_{Ru}} \quad (16.15a)$$

or

$$Q_{TV} = \frac{Q_{KT}}{\gamma_{Rt}} \quad (16.15b)$$

Here Q_{ks} And Q_{ku} the characteristic perimeter friction resistance and the characteristic tip resistance of the pile, Q_{KT} shows the characteristic total pile bearing capacity.

Table 16.4. Strength Factors for Pile Foundations

My endurance Type	Strength Coefficient Symbol	Strength Coefficient Value	
		Pile loading if the experiment has not been done	Pile loading if the experiment has been done
Environmental friction (pressure)	γ_{Rsb}	1.5	1.3
Environmental friction (tension)	γ_{Rsc}	1.6	1.4
Tip resistance	γ_{Ru}	2.0	1.5
Total bearing capacity (pressure)	γ_{Rt}	–	1.4

16.9.3.2 –Unit perimeter and unit tip resistances in cohesive and non-cohesive soils can be calculated using literature-based and generally accepted relations. The effect of the pile manufacturing method should be taken into account in the calculation of bearing capacity.

16.9.3.3 –In pile foundations that will exhibit group behavior, the bearing capacity will be selected as the smaller of the total bearing capacity of each pile in the group and the bearing capacity of the block consisting of the piles and the soil between them.

16.9.3.4 –At least two static loading tests will be performed on pile foundations of buildings with Earthquake Design Class DTS = 1, 1a, 2, 2a to prove that the design assumptions are verified on site.

16.9.4. Lateral Bearing Capacity of Piles

16.9.4.1 –In each of the loading cases including static and seismic effects **Equation (16.16)** The inequality in will be satisfied:

$$P_{ty} \leq Q_{ty} \quad (16.16)$$

Here P_{ty} is the lateral design force acting on the pile. Q_{ty} shows the lateral design strength of the pile.

16.9.4.2 –In the design of laterally loaded piles that are expected to exhibit non-linear load-displacement behavior, it shall be shown by calculation that the allowable displacements and the carrying capacity of the pile section are not exceeded.

16.10. EARTHQUAKE CALCULATION OF PILE FOUNDATIONS

16.10.1. Structure – Pile – Soil Interaction

16.10.1.1 –The calculation methods given in this section will be applied for the design of pile foundations, which work as carrier system elements together with the superstructure (building) under the effect of earthquakes, taking into account the effect of the ground.

16.10.1.2 –For earthquake calculation of pile foundations, dynamic *structure – pile – soil interaction* Within this framework, the following accounting approaches can be used:

(a) Common systems approach In this approach, where the superstructure (building), building foundation, piles and soil are modeled and analyzed together as parts of the load-bearing system, nonlinear behaviors of all parts can be taken into account.

(b) Subsystem approach: This approach, in which the superstructure-foundation subsystem and the foundation-pile-soil subsystem are modeled separately and analyzed by taking into account their interactions with each other, is based on linear behavior in principle. However, in engineering practice, this approach may be allowed to be applied to the nonlinear behavior of the foundation-pile-soil subsystem and the superstructure-foundation subsystem within the framework of certain methods.

16.10.1.3 –Within the framework of the subsystem approach, the structure-pile-soil interaction is considered in two stages. *Kinematic interaction* And *inertial interaction* Details of these stages, called as "calculation methods and design rules that can be used for these stages" **ANNEX 16C** The application areas of these methods, called Method I, Method II and Method III, are given in *Local Soil Class*'na, *Earthquake Design Class*' and *Building Height Class* Depending on **Table 16.5** Depending on the preference of the building owner / design engineer, Method I can be used instead of Method II, Method II can be used instead of Method III, and the application areas of the methods can be expanded.

16.10.2. Special Conditions Regarding the Design of Piles

16.10.2.1 –Due to the damage risks they carry within the scope of kinematic interaction, building foundations under the influence of earthquakes on *slanted stakes* not allowed.

16.10.2.2 –The minimum cross-sectional dimensions of reinforced concrete and prestressed reinforced concrete piles are 30 cm / 30 cm or ØIt will be 35 cm.

Table 16.5 – Application Areas of Interaction Analysis Methods

Analysis Method	Earthquake Design Class	Building Height Class	Local Soil Class
Method I	DTS = 1, 1a, 2, 2a	BYS = 1	ZD, ZE, ZF
Method II	DTS = 1a, 2a	BYS = 2, 3	ZD, ZE, ZF
	DTS = 3, 3a, 4, 4a	BYS = 1	
Method III	DTS = 1a, 2a	BYS ≥ 4	ZD, ZE, ZF
	DTS = 1, 2, 3, 3a	BYS ≥ 2	

16.10.2.3 –In reinforced concrete piles, the longitudinal reinforcement ratio in the top 1/3 of the pile length under the pile cap, not less than 3 meters, shall not be less than 0.01 in buildings with DTS = 1, 1a, 2, 2a, and 0.008 in buildings with DTS = 3, 3a, 4, 4a. The diameter of the spiral reinforcement (or stirrup for driven piles with rectangular section) to be placed in this area shall not be less than 10 mm in buildings with DTS = 1, 1a, 2, 2a, and 8 mm in buildings with DTS = 3, 3a, 4, 4a, and the spiral pitch (or stirrup spacing) shall not exceed 200 mm, but shall be reduced to 100 mm at a height of at least two pile diameters from the top.

16.10.2.4 –Wall thickness of steel pipe piles **Equation (16.17)** will satisfy the condition:

$$\begin{aligned}
 &\text{In buildings with DTS=1, 1a, 2, 2a: } \frac{D}{t} \leq 60 \\
 &\text{In buildings with DTS=3, 3a, 4, 4a: } \frac{D}{t} \leq 80
 \end{aligned}
 \tag{16.17}$$

16.10.2.5 –Monolithic connections of steel piles to the reinforced concrete foundation (pile cap) will be made with reinforced concrete plugs to be arranged at least two pile diameters deep at the top of the piles. The longitudinal reinforcement in the plugs will be arranged in a way that transfers the bending moment and axial force calculated at the pile top as a reinforced concrete section. Along the plug height

16.10.2.3 Spiral transverse reinforcement defined in will be used. It will be shown by calculation that the pile axial force is safely transferred from the plug to the steel pile section.

16.11. STATIC AND DYNAMIC GROUND PRESSURE AFFECTING THE BASEMENT CURTAINS OF BUILDINGS

In cases where the ground is idealized nonlinearly, excessive pore water pressures are taken into account and calculations are not made with wall-soil interaction models that include the structural elements forming the perimeter basement walls, the ground pressures acting on the rigid basement perimeter walls of the buildings can be calculated as given below.

16.11.1 –Ground pressures to be applied uniformly in static conditions (p) **Table**

16.6 It is defined in. Here H is the total height of the basement wall, y is the natural height of the ground

unit volume weight, γ_d saturated unit volume weight of water, q additional load (surcharge), z shows the height measured from the water surface downwards.

Table 16.6. Static Earth Pressures Acting on Basement Shear Walls

Type of Ground Outside the Basement Bulkhead	Pressure Affects Height	Ground Pressure (p)
Cohesionless soil	Throughout the entire height	$0.2(\gamma^* H_b + q)$
Soft to medium solid cohesive soil	Along the top 20%	$0.2(\gamma^* H_b + q)$
	Along the lower 80%	$0.3(\gamma^* H_b + q)$
Solid – hard cohesive soil	Throughout the entire height	$0.3(\gamma^* H_b + q)$
Note: In case there is no water behind the basement bulkhead, $\gamma^* = \gamma$ shall be taken. In case the basement bulkhead is partially submerged, above the water level $\gamma^* = \gamma$ and below the water level $\gamma^* = (\gamma_d - \gamma_{\text{This}})$ will be taken, and also the static water pressure will be applied to the ground pressure from the top of the water level downwards ($p_{\text{This}} = \gamma_{\text{This}} z$) will be added. All ground pressures other than static water pressure will be applied in a uniformly distributed manner.		

16.11.2 –Additional ground pressures under earthquake influence (Δp) **Equation (16.18)** will be calculated with:

$$\Delta p = 0.4 S_{DS} \gamma H_b \quad (16.18)$$

This pressure will be applied uniformly along the wall height.

16.11.3 –In cohesionless soils, if the basement is partially dry, between the water level and the basement floor, **Table 16.6** In addition to the static water pressure in the second line of , the variation of the static-equivalent dynamic water pressure along the water depth will be considered,

$\Delta p_{\text{This}}(z)$, **Equation (16.19)** will be determined by:

$$\Delta p_{\text{This}}(z) = \frac{7}{8} (0.4 S_{DS}) \gamma_{\text{This}} \sqrt{z} \quad (16.19)$$

Here S_{DS} is the short period design spectral acceleration coefficient defined at the ground surface, d_{This} whereas shows the height of the wall under water. **Equation (16.19)** By integrating with the water depth, the resultant static-equivalent additional dynamic water force and the resultant depth from the water surface are **Equation (16.20)** is obtained with:

$$\Delta P_{\text{This}} = \frac{7}{12} (0.4 S_{DS}) \gamma_{\text{This}} d_{\text{This}}^2 \quad ; \quad \bar{z} = 0.6 d_{\text{This}} \quad (16.20)$$

16.12. RULES FOR DESIGN OF RESISTANCE STRUCTURES UNDER EARTHQUAKE EFFECT

In the design of retaining structures under the effect of earthquakes, both limit equilibrium (collapse) and service conditions will be taken into account. Retaining structures can be designed to allow displacements at levels that will not harm their functions after the earthquake.

16.12.1. Stability of Retaining Structures

The effects (moments/forces) that force overturning and horizontal sliding will be compared with the balance of the effects and resistances that oppose them:

16.12.1.1 –To ensure the safety of the supporting structure against overturning **Equation (16.21)** The condition in must be met:

$$\frac{T_{O_{\text{giant}}}}{\gamma_{R_{\text{dev}}}} \leq \frac{R_{\text{giant}}}{\gamma_{R_{\text{dev}}}} \quad (16.21)$$

Here, $T_{O_{\text{giant}}}$ the sum of the forces that force the overturning, R_{giant} anti-rollover effects and sum of resistances, $\gamma_{R_{\text{dev}}}$ is the design safety factor against rollover. This factor is $\gamma_{R_{\text{dev}}} \geq 1.3$ will be selected by the engineer responsible for the project, taking into account the risks that may arise from collapse.

16.12.1.2 –Verifications against exceeding the ground bearing capacity and horizontal sliding on the foundation of the retaining structure, **16.8.3** And **16.8.4** According to , investigations regarding wholesale migration are **16.13** It will be done according to.

16.12.2. Earth Pressures

16.12.2.1 –Horizontal and vertical to be used in calculating soil pressures *static-equivalent earthquake coefficients* **Equation (16.22)** It is defined in:

$$k_h = \frac{0.4 S_{ds}}{r} ; \quad k_v = 0.5 k_h \quad (16.22)$$

Equation (16.22) Located in r coefficient for different types of retaining structures **Table 16.7** is given in .

16.12.2.2 –In the presence of saturated soils where high pore water pressure increases may occur, r The value of the coefficient should not be greater than 1.

Table 16.7. For Retaining Structures r coefficients

Type of Retaining Structure	r
Up to $120 S_{ds}$ (mm) allowable displacement of gravity type walls	2.0
Up to $80 S_{ds}$ (mm) allowable displacement of gravity type walls	1.5
Anchored walls, gravity type walls that are not allowed to move	1.0

16.12.2.3 –The resultant of total (static and dynamic) earth pressure acting on the retaining structure **Equation (16.23)** will be calculated with:

$$P_t = K(1-k) \left(\gamma^* H \frac{1}{2} + qH \right) + P_{\text{This} + \Delta P_{\text{This}}} \quad (16.23)$$

Here H is the wall height, γ^* is the typical unit weight of the soil, q additional load (surcharge), K total (static+dynamic) active (K_a) or passive (K_p) is the earth pressure coefficient, k vertical static-equivalent earthquake coefficient, P_{This} And ΔP_{This} shows the resultant static and dynamic water pressures.

16.12.2.4 –Total active pressure coefficient **Equation (16.24)** will be calculated with:

(a) $\beta \leq \phi'_d$ In case – θ :

$$K_a = \frac{\cos \theta \sin^2 \psi \sin(\psi - \theta - \delta_d)}{\cos^2(\psi + \varphi'_d - \theta) \sqrt{\frac{\sin(\varphi'_d + \delta_d) \sin(\varphi'_d - \beta - \theta)}{\sin(\psi - \theta - \delta_d) \sin(\psi + \beta)}}} \quad (16.24a)$$

(b) $\beta > \varphi'_d$ In case $\beta > \varphi'_d$:

$$K_a = \frac{\cos \theta \sin^2 \psi \sin(\psi - \theta - \delta_d)}{\cos^2(\psi + \varphi'_d - \theta)} \quad (16.24b)$$

16.12.2.5 – Total passive pressure coefficient, assuming no friction between the floor and the wall, **Equation (16.25)** will be calculated with:

$$K_p = \frac{\cos \theta \sin^2 \psi \sin(\psi + \theta)}{\cos^2(\psi + \varphi'_d + \theta) \sqrt{\frac{\sin \varphi'_d \sin(\varphi'_d + \beta - \theta)}{\sin(\psi + \theta) \sin(\psi + \beta)}}} \quad (16.25)$$

16.12.2.6 – **Equation (16.24)** and **Equation (16.25)** φ'_d is design shear resistance angle of the soil, δ_d

is the friction angle between the ground and the wall, β is the inclination angle of the ground surface behind the wall with respect to the horizontal, and ψ is the angle of inclination of the wall with respect to the horizontal (from the horizontal in front of the wall to the back of the wall).

Angle θ depending on the static-equivalent earthquake coefficient and **Equation (16.23)** The values of the soil unit weight γ^* in , indicate whether there is water behind the retaining structure.

depending on the absence of **16.12.3** is given in.

16.12.2.7 – In static condition **Equation (16.24)** and **Equation (16.25)** In the relations given by , $\theta = 0$ will be taken. The dynamic soil pressure coefficient will be obtained by subtracting the static pressure coefficient from the total pressure coefficient.

16.12.2.8 – The point of action of the resultant force related to dynamic earth pressures will be taken as the midpoint of the wall height. In walls that can rotate freely around their toes, it can be assumed that the dynamic force will act at the same point as the static force.

16.12.2.9 – Static and dynamic soil pressures are maximum in case of active pressure, according to the normal behind the wall. (2/3) φ' It will be assumed that it acts with an angle of inclination and zero inclination in the passive state.

16.12.3. Effect of Water Level Behind Retaining Structure

16.12.3.1 – It is essential to ensure that the soil mass held by the retaining structure does not remain below the water table with drainage measures. In this context, the drainage system should be arranged in a way that will adapt to the temporary and permanent displacements caused by the earthquake without damaging its function.

16.12.3.2 – Depending on whether there is water behind the retaining structure, the angle θ and **Equation (16.23)** The values of the soil unit weight γ^* in are defined below.

(a) If the water level is below the foundation base level ($P_{\text{This}} = \Delta P_{\text{This}} = 0$);

$$\theta = \arctan \frac{k_h}{1 - k_v} \quad ; \quad \gamma^* = \gamma \quad (16.26a)$$

Here γ represents the natural unit volume weight of the soil.

(b) If the water level is above the foundation base level and the ground is dynamically impermeable (permeability coefficient is 5×10^{-4} (On grounds where it is less than m/s):

$$\theta = \gamma_{\text{dawn}} - \frac{\gamma_d}{\gamma_d - \gamma_{\text{This}}} \frac{k_h}{1 - k_v} ; \quad \gamma^* = \gamma_d - \gamma_{\text{This}} \quad (16.26b)$$

Here γ_d indicates the unit volume weight of saturated soil. $\Delta P_{\text{This}} = 0$ will be taken.

(c) If the water level is above the foundation base level and the ground is dynamically permeable:

$$\theta = \gamma_{\text{dawn}} - \frac{\gamma}{\gamma_d - \gamma_{\text{This}}} \frac{k_h}{1 - k_v} ; \quad \gamma^* = \gamma_d - \gamma_{\text{This}} \quad (16.26c)$$

In this case, the resultant static-equivalent additional dynamic water force ΔP_{This} and the depth of the resultant from the water surface **Equation (16.20)** will be calculated with .

16.13. STABILITY OF SLOPE UNDER EARTHQUAKE IMPACT

16.13.1 –In order to maintain the safety and serviceability of structures to be constructed on or near natural or artificial slopes under the effect of a design earthquake, it is necessary to verify that the slope will remain stable and functional (serviceable) under the effect of earthquake loads. For buildings with Earthquake Design Class DTS=4, slope stability analysis under earthquake effect may not be performed.

16.13.2 –In slopes supported by structural elements (for example: slopes supported by flexible walls or piles, slopes with anchors or nails, etc.), soil-structure interaction analyses should be performed, taking into account the relative stiffness differences of the soil and structural elements.

16.13.3 –In slope stability analysis under earthquake effect, soil properties *strength coefficients* (for cohesion and friction resistance, respectively, c' / γ_{Rc} and $\tan \phi' / \gamma_{R\phi}$)

In the design of structural elements such as piles or anchors located within the slope, soil resistances should also be taken into account by applying the same strength coefficients.

16.13.4 –In the stability analysis of slopes, a collapse model suitable for the type of soil is accepted and the stability of the soil or rock mass is investigated. In order for a slope to be safe against total collapse, **Equation (16.27)** The condition given in will be met:

$$TQ \leq \frac{R_t}{\gamma_{Rk}} \quad (16.27)$$

Here, TQ the sum of the effects that force migration, R_t effects and resistances opposing migration sum, $\gamma_{Rk} (\geq 1.0)$ against slipping *design safety number* is defined as.

16.13.5 –Stability of slopes in case of earthquake, undrained shear strength in clayey soils (c_u) Total stress analysis using effective stress in sandy (cohesionless) soils will be calculated by stress analysis.

16.13.6 –In old landslide areas that are at risk of being re-triggered by earthquakes, values corresponding to large changes in ground strength parameters

In loose to medium dense cohesionless soils that may be subject to cyclic pore water pressure increase, the maximum pore water pressure that can be expected to occur should be taken into account.

16.13.7 –Slope stability control under earthquake effects can be done by equivalent static limit equilibrium analyses, finite element method or dynamic response analyses to be performed in the time domain. In cases where the topography and soil stratification show sudden irregularities and for liquefiable or sensitive soils that may experience significant rigidity loss under the effect of cyclic loading, the equivalent static analysis method should not be used.

16.13.8 –Where allowable displacement criteria apply, permanent displacements can be calculated using soil strength parameters compatible with dynamic effects, using the Newmark sliding rigid block method or equivalent methods. The calculations should take into account the possible effects of material hardening or softening behavior and pore water pressure increase that will occur with increasing deformations.

16.13.9 –In equivalent static analyses, the soil mass and the forces present on the slope, horizontal (F_H) and vertical (F_V) effect of inertial forces **Equation (16.28)** will be taken into consideration.

$$F_H = 0.5 W (0.4 S_{DS} S_T) \quad ; \quad F_V = \pm 0.5 F_H \quad (16.28)$$

Here W the weight of the sliding mass, S_{DS} is the short period design spectral acceleration coefficient, S_T is the topographic amplification coefficient.

16.13.10 –To be constructed on and near the slope, *Earthquake Design Class* $DTS = 1, 1a$, In the earthquake stability analyses to be carried out for structures with $2, 2a$, topographic magnification effect will be taken into account. Topographic magnification factor (S_T) two-dimensional topographic It will be used in slopes with irregularities (e.g. long valleys, hills with an altitude of more than 30 m) and in cases where the potential failure surface passes close to the crest of the slope. Topographic amplification effect, slope angle $\beta \leq 15^\circ$ can be neglected on slopes; crown For ridges with a (top) width smaller than the base width, the slope angle $\beta \geq 30^\circ$ on the slopes $S_T \geq 1.4$, at smaller slope angles $S_T \geq$ It will be used as 1.2. It can be neglected in potential landslide areas where the possible collapse surface passes close to the base.

16.13.11 –In equivalent static limit equilibrium analyses based on the method of slices, in determining the critical surface and calculating the safety factor against sliding with iterative methods, the mobilized values of the soil resistances are reached by dividing the soil strength parameters by the safety factor ($\gamma_{Rc} = \gamma_{R\phi} = \gamma_{Rk}$) is accepted:

$$\tau_{mob} = \frac{c'}{\gamma_{Rc}} + \frac{\tan \phi'}{\gamma_{R\phi}} = \frac{c'}{\gamma_{Rk}} + \frac{\tan \phi'}{\gamma_{Rk}} \quad (16.29)$$

(a) In the static case:

$$\gamma_{Rk} = \sum \frac{1}{W_s \sin \alpha} \sum \left[c' b + \frac{1}{m_\alpha} W_s - \frac{c' b \sin \alpha}{\gamma_{Rk}} - U b - \tan \phi' \right] \quad (16.30)$$

(b) In case of an earthquake, the effect of horizontal and vertical inertia forces will be taken into account in the calculation of the factor of safety against sliding as follows.

$$y_{Rk} = \frac{1}{\sum W_s (1 - k_h) \sin \alpha + k_h W_s \cos \alpha - \frac{H_c}{R_d} \sum c' b + \frac{1 - W_s (1 - k_v)}{m_\alpha} - \frac{c' b \sin \alpha}{y_{Rk}} - U b \tan \phi'} \quad (16.31)$$

Here, b slice width, W is the slice weight, α is the angle that the slice base makes with the horizontal, u water pressure acting on the base of the slice, $k_h = 0.25 b_s$ and $k_v = \pm 0.5 k_h$ are the horizontal and vertical equivalent earthquake coefficients, respectively, H_c the vertical distance from the midpoint of the slice base to the slice center, R_d represents the radius of the circular failure surface. **Equation (16.30)** and **Equation (16.31)** located in **Equation (16.32)** is given in:

$$m_\alpha = \cos \alpha + \frac{\tan \alpha \tan \phi'}{y_{Rk}} \quad (16.32)$$

(c) Safety number against slipping (y_{Rk}) obtained by sequential approach. Special risks Except for cases involving $y_{Rk} \geq 1.10$ can generally be considered as a sufficient level of safety against slipping.

ANNEX 16A – GENERAL RULES FOR SOIL INVESTIGATIONS

Soil investigations will be carried out according to the general rules defined below.

16A.1. SOIL INVESTIGATION DRILLINGS

16A.1.1 – Drilling wells, ground drilling will be done with percussion or rotary systems using auger, sand or clay bucket, and rock drilling will be done with rotary systems that proceed by continuously taking cores from the rock surface.

16A.1.2 – Drilling Numbers: Every 300 m, taking into account the variability of ground conditions, the base area of the structure and the loads it transfers and the surrounding conditions. It will be determined as at least one for each floor area of . However, *Building Height Class* In areas where buildings with $BYS \leq 3$ will be constructed, the number of drillings will not be less than three. The number of drillings will be 1000 m² of floor area. In buildings exceeding , it will be planned to have at least five units, one at each corner of the building and one in the middle.

16A.1.3 – Drilling Locations: If topographic and geomorphological conditions do not indicate special places, drilling locations will be selected at least in the corners and middle of the structure plan. If structure types and locations are certain in large areas, drilling points will be selected in accordance with the structure layout. In project areas with uncertain layout, drilling locations will be planned on a grid. In sloping terrain with slope instability potential, drilling locations will be planned along lines that will provide data for slope stability analysis, and drilling locations will be selected outside the structure area.

16A.1.4 – Drilling Depths: Drilling depth is at least 1.5 times the width of the building starting from the foundation base for building foundations or the depth of the soil caused by the net foundation base pressure.

Stress increase ($\Delta\sigma$) is the effective stress due to the soil's own weight. (σ'_{vo})

From the depth where it is equal to 10% ($\Delta\sigma = 0.10\sigma'_{vo}$) will be selected as the less favorable one.

(a) If a layer with high strength and rigidity is encountered at shallower depths, drilling can be terminated by entering at least 3 meters into this layer and determining the layer continuity.

(b) In cases where pile foundation application may be required, the drilling depth will be selected to allow pile bearing capacity and settlement calculations.

(c) In slope stability studies, the drillings will go deep enough below the possible slip surface depths, and in deep excavations, they will go from the excavation base to at least half the excavation depth.

(d) In regions with earthquake design classes $DTS=1$, $DTS=1a$, $DTS=2$ and $DTS=1a$, in areas where the groundwater level is higher than 10 meters from the foundation base and where loosely settled coarse-grained soils are located, the drilling depth will be at least 20 m from the ground surface.

16A.2. FIELD TESTS

16A.2.1 – Field tests (SPT, winged cutter, pressuremeter, etc.) will be carried out at appropriate intervals (e.g. every 1.50 m) in boreholes, compatible with the ground conditions and meeting the project requirements. For the continuous and accurate determination of the ground profile and properties, it will be useful to carry out Cone Penetration Tests (CPT and pore water pressure measurement - CPTu) planned in parallel with the drillings.

16A.2.2 – Sample Collection: Disturbed and undisturbed soil/rock samples will be taken at appropriate intervals in boreholes, compatible with the ground conditions and meeting the project requirements.

16A.2.3 – Disturbed Examples: In boreholes, disturbed samples will be taken from all types of soil during the Standard Penetration Test (SPT). If sufficient amount of disturbed sample cannot be obtained from the SPT spoon in coarse-grained or non-plastic silt soils below the groundwater level, disturbed samples, the finest of which is not washed, will be taken with special tubes with a soil holder attached to their mouths (basket-type).

16A.2.4 – Undisturbed Samples: In drilling wells, from cohesive soils, **16A.1.4A** sufficient number of undisturbed samples shall be taken to determine the engineering properties of each soil layer within the depth of influence defined in . Undisturbed soil samples shall be taken from the in accordance with its consistency and thin walled (area ratio = wall thickness/inner tube area < 10%) and the inner diameter is will be taken using a tube or piston tube of at least 76 mm. In addition, undisturbed samples can be prepared from blocks that are cut and taken appropriately from research pits and trenches.

16A.2.5 – In rock drilling, the core barrel and bit type will be selected according to the characteristics of the lithological units and the diameter of the cores will be at least NX (54 mm) or NQ (47.6 mm).

16A.2.6 – Measuring Groundwater Level: The groundwater level will be determined by measurements to be made in the borehole. In case there is more than one cohesive and cohesionless layer and/or water-bearing layer (aquifer) level within the depth of the study, it will be appropriate to use standard piezometers to measure the groundwater level or pressure. In all cases, observations and measurements in the borehole will be made for a period long enough for the groundwater level to reach equilibrium in the well.

16A.3. LABORATORY EXPERIMENTS

16A.3.1 – Laboratory tests will be planned by the civil engineer responsible for the project according to the characteristics of the structure and ground, in sufficient numbers and depths to define the ground properties in the research area with maximum 2 m intervals, and will be carried out in authorized laboratories.

16A.3.2 – Classification Experiments: On disturbed and undisturbed samples, consistency limits, natural water content (on preserved samples), sieve, pipette/hydrometer and specific gravity tests will be performed in numbers that will allow the soil classification of the layers. On undisturbed samples, natural water content and unit volume weight determination tests will also be performed. In problematic soils (liquefaction, swelling, collapse, softening, sensitive clays etc.), classification tests (including pipette/hydrometer) will be performed on all samples taken from the first 15 m.

16A.3.3 – Engineering Properties Experiments: Strength (uniaxial compression, shear box, triaxial compression-UU/CU) tests will be carried out on undisturbed samples to determine the engineering properties of the layers, and drained and undrained shear strength parameters will be determined in accordance with the project requirements. In addition, oedometer-consolidation tests will be carried out on undisturbed samples taken from clayey soils to determine the parameters for settlement calculations.

ANNEX 16B – SIMPLIFIED SOIL LIQUEFICATION ASSESSMENT

16B.1. SCOPE

Within the scope of soil liquefaction triggering analyses, the shear resistance of field soils against liquefaction, τ ; will be found using empirical soil liquefaction resistance relations and curves (with necessary corrections applied) based on ground penetration tests or shear wave velocity values developed for flat sites. The proposed method for calculating liquefaction resistance using SPT test results **16B.3** The shear stresses developed during earthquake shaking are given in $\tau_{\text{earthquake}}$, using the simplified method will be calculated.

16B.2. CORRECTION OF SPT DATA

16B.2.1. Correction of Raw SPT Data

16B.2.1.1 –Raw SPT data obtained from the field, N , will be **Equation (16B.1)** by using $N_{1.60}$ corrected to the value.

$$N_{1.60} = N C_N C_R C_S C_B C_T \quad (16B.1)$$

Here C_N is the geological stress (depth) correction factor applied in cohesionless soils, C_R rod length correction coefficient, C_S sampler type correction factor, C_B drilling hole diameter correction factor, C_T represents the energy ratio correction coefficient.

16B.2.1.2 –Depth correction coefficient C_N **Equation (16B.2)** It will be calculated with the relation given in .

$$C_N = 9.78 \sqrt{\frac{1}{\sigma'_{\text{he}}}} \leq 1.70 \quad (16B.2)$$

Equation (16B.2) In the relation given in , the effective vertical stress at the test depth is σ'_{he} (kN/m²), It is calculated according to the field conditions when the Standard Penetration Test (SPT) is performed. Effective stress changes caused by additional filling, foundation stress or ground excavation and similar reasons after the test will not be taken into account.

16B.2.1.3 – Eq.(16B.1) Other correction coefficients included in **Table 16B.1** is given in .

16B.2.2. Correction of SPT Data According to Fine Grain Content

Fine grain content (ID) Corrected impact numbers $N_{1.60f}$ **Equation (16B.3a)** will be calculated with:

$$N_{1.60f} = \alpha + \beta N_{1.60} \quad (16B.3a)$$

Equation (16B.3a) The coefficients α and β in **Equation (16B.3b)** is given in:

$$\begin{aligned} \alpha &= 0 ; \beta = 1.0 & (ID \leq 5\%) \\ \alpha &= \exp_{-1.76 - (190 / ID)} - ; \beta = 0.99 + ID / 1000 & (5\% < ID \leq 35\%) \\ \alpha &= 5.0 ; \beta = 1.2 & (ID \geq 35\%) \end{aligned} \quad (16B.3b)$$

Table 16B.1. SPT Correction Coefficients

Correction Coefficient	Variable	Value
C_R	Between 3m and 4m	0.75
	Between 4m and 6m	0.85
	Between 6m and 10m	0.95
	Deeper than 10m	1.00
C_S	Standard sampler (with inner tube)	1.00
	Sampler without inner tube	1.10-1.30
C_B	Diameter between 65mm-115mm	1.00
	Diameter 150mm	1.05
	Diameter 200mm	1.15
C_{TO}	Safe mallet	0.60-1.17
	Ring mallet	0.45-1.00
	Automatic impact rammer	0.90-1.60

16B.3. CALCULATION OF LIQUEFACTION RESISTANCE

16B.3.1 –Liquefaction resistance τ_R , cyclic strength ratio against an earthquake with moment magnitude 7.5 ($CRR_{M7.5}$), design earthquake moment magnitude correction factor (C_M) and effective vertical stress (σ'_{he}) It will be calculated by multiplying by .

$$\tau_R = CRR_{M7.5} C_M \sigma'_{he} \quad (16B.4a)$$

16B.3.2 –Cyclic strength ratio **Equation (16B.4b)** It will be calculated with the relation given in:

$$CRR_{M7.5} = \frac{1}{34 - N_{1.60f}} + \frac{N_{1.60f}}{135} + \frac{50}{[10N_{1.60f} + 45]^2} - \frac{1}{200} \quad (16B.4b)$$

Earthquake magnitude correction factor (C_M), design earthquake magnitude (M_w) depending on **Equation (16B.4c)** It will be calculated with the relation in .

$$C_M = \frac{10^{2.24}}{M_w^{2.56}} \quad (16B.4c)$$

16B.4. CALCULATION OF SHEAR STRESS IN AN EARTHQUAKE

16B.4.1 –Shear stress in the ground **Equation (16B.5)** It will be calculated with the relationship in .

$$\tau_{\text{earthquake}} = 0.65 \sigma_{vo} (0.4 S_{ds}) r_d \quad (16B.5)$$

Here σ_v the total vertical stress at the depth where liquefaction is assessed, r_d is the stress reduction coefficient at the relevant depth, S_D is the short period design spectral acceleration coefficient.

16B.4.2 –Stress reduction coefficient, r_d the depth examined (z) depending on
Equation (16B.6) will be obtained by the relation in .

$$\begin{aligned}
 r_d &= 1.0 - 0.00765z & z \leq 9.15\text{m} \\
 &= 1.174 - 0.0267z & 9.15\text{m} < z \leq 23\text{m} \\
 &= 0.744 - 0.008z & 23\text{m} < z \leq 30\text{m} \\
 &0.50 & z > 30\text{m}
 \end{aligned}
 \tag{16B.6}$$

ANNEX 16C – STRUCTURE – PILE – GROUND INTERACTION UNDER EARTHQUAKE EFFECT

16C.1. STRUCTURE – GROUND INTERACTION

16C.1.1 –Generally *structure-soil interaction*, is defined as the mutual interaction of the structure and the ground in the structure-ground joint system where the ground environment (and piles, if any) and the superstructure are considered together under the effect of an earthquake. The earthquake waves, which are formed by the earthquake ground motion defined in the bedrock and spread in the ground environment and reach the foundation of the structure, are partially reflected from the foundation of the structure and return to the ground environment, while some of them pass to the superstructure, causing it to vibrate and return to the ground environment again.

16C.1.2 –In the case of shallow foundations, dynamic soil-structure interaction for building-type structures on relatively soft soils generally affects the building in a favorable way and reduces the equivalent seismic loads generated by earthquake waves passing from the ground to the building. For the sake of staying on the safe side, the soil-structure interaction can therefore be neglected in buildings with shallow foundations (including buildings with basements).

16C.1.3 –On the other hand, in buildings constructed with piles on relatively soft ground, *structure – pile – soil interaction*, especially affects the seismic behavior of piles significantly. Piles are exposed to significant deformations and strains due to the effects of earthquake waves generated by the earthquake ground motion defined in the bedrock and propagating in the ground environment. In return, piles are exposed to significant changes in shape and strains due to the earthquake waves generated in the building foundation. *effective base movement* They can also significantly change the amplitude and frequency content of the superstructure. The inertia of the superstructure is not taken into account. *foundation – pile – ground system* This formation in *kinematic interaction* It is called Piles. Also called *inertial interaction* They are exposed to additional stresses due to the earthquake waves returning from the vibrating superstructure to the pile-soil environment. For these reasons, it is necessary to take into account the structure-pile-soil interaction in buildings with pile foundations on weak soils.

16C.1.4 –For buildings with pile foundations *kinematic interaction* And *inertial interaction* in their analysis **16.10** There are three methods to be used according to **16C.2**, **16C.3** And **16C.4** It is explained in .

16C.2. CALCULATION OF STRUCTURE – PILE – GROUND INTERACTION WITH METHOD I

16C.2.1. Method I – Scope

Method I, **Table 16.5** According to the classification in , when the local soil class is ZD, ZE or ZF, DTS = 1, 1a, 2, 2a is used for piled soils. *High-Rise Buildings* (For BYS = 1) **Chapter 13** In Phase III of , DD-1 will be applied in the nonlinear time history calculation under the influence of earthquake ground motion.

16C.2.2. Method I – Kinematic Interaction Calculation

16C.2.2.1 –The inertia of the superstructure is not taken into account. *foundation – pile – soil kinematic interaction calculation model* will be created as follows:

(a) *Soil environment*, **16.5.2.3** Defined according to *engineering bedrock* On the , *Discrete analysis methods* (finite elements or finite differences) will be modeled in three dimensions. The depth of the engineering base rock from the building foundation will not be less than three times the largest building width and the longest pile length. Along the outer perimeter of the finite soil model and not classified as ZA or ZB *engineering*

bedrock In order to prevent the reflection of the waves propagating outwards from the ground environment, it is appropriately selected. *permeable borders* The horizontal distance of the permeable boundaries to the building shall not be less than three times the width of the building foundation, separately in both directions and on both sides. Three-dimensional ground environment *nonlinear* behavior will be taken into account using models that are compatible with experimental data. **16.6** According to , in soils with liquefaction potential, in the time history *effective stress* enabling analysis to be carried out *nonlinear* ground structure models will be used.

(b) Stakes, 5.3.1 It will be modeled as rod elements with plastic joints defined in , placed at appropriate intervals along the depth. Appropriately selected between piles and soil environment *Interface elements* Before the earthquake calculation, the axial forces on the piles due to vertical loads will be determined by non-linear static calculation.

(c) Building foundation, together with the basement floor perimeter walls (up to ground floor level), will be modeled as a massless, infinitely rigid element.

(d) Earthquake records, at least eleven (two-component) pairs **2.5.1** Selected according to **2.5.3** It will be transformed according to and the engineering will be influenced by the system from the bedrock.

16C.2.2.2 – In Method I, as a result of the foundation-pile-soil kinematic interaction analysis to be performed in the time history domain, the following outputs will be obtained:

(a) Internal force and deformation demands in piles are determined by the analysis performed (at least $2 \times 11 = 22$ analysis) will be calculated as the average of the largest absolute values of the results obtained from each.

(b) $2 \times 11 = 22$ for analysis **16C.2.2.1(c)** For degrees of freedom defined as rotation about two horizontal axes and in the base of the rigid foundation modeled according to *effective basic movements* Changes in displacements with respect to time (*foundation level earthquake records*) and the acceleration spectra generated from them (*Base level earthquake spectra*) will be obtained. These outputs will be **16C.2.3** Explained in *inertial interaction calculation* They will be used as inputs of .

(c) Since the inertia of the superstructure is not taken into account and the building foundation is taken as rigid, no internal force occurs in the superstructure from kinematic interaction.

16C.2.3. Method I – Inertial Interaction Calculation

In Method I *inertial interaction* The calculation will be made as follows:

16C.2.3.1 – In the foundation – pile – ground system **16C.2.2.2(b)** In , separate static singular forces are applied incrementally to the degrees of freedom where effective foundation movements are defined (for rotation around and along two horizontal axes at the base of the rigid foundation), and nonlinear forces are obtained for each degree of freedom. *incremental static calculation* From these analyses, the foundation - pile - soil system representative *interaction arcs* Skeleton curves of the nonlinear force-displacement relationships will be obtained.

16C.2.3.2 – Building (superstructure) carrier system, **16C.2.2.1(c)** defined in *together with the building foundation and basement floors*, and taking into account their masses, **13.6.2** It will be modeled according to. **16C.2.3.1** Defined in *interaction arcs* will be placed at the base of the building foundation and a hysteretic behavior model suitable for these springs will be adapted. In this way, the superstructure forming the foundation system **16C.2.2.2(b)** Defined in *effective basic movements* By applying the acceleration records of the system in the time history, nonlinear *inertial interaction calculation* will be done.

16C.2.3.3 –In Method *Inertial interaction* calculation, superstructure (building) carrier system **5.7** The nonlinear time history calculation according to , *structure – pile – soil interaction* It goes against the way it is taken into account.

16C.2.3.4 –In Method *Inertial interaction* The following outputs will be obtained as a result of the calculation:

(a) The time-dependent changes of internal force, displacement and deformation demands in the superstructure and superstructure foundation (including basements) will be obtained.

(b) Formed in the springs defined at the base of the foundation *interaction displacements* and *interaction forces* The time dependent changes of will be obtained. From these and **16C.2.3.1** Found in *incremental static calculation* Using the results, the internal forces and deformations occurring in the piles during inertial interaction will be calculated.

(c) The internal forces and deformations obtained in the piles as a result of the inertial interaction calculation are compared with those obtained from the kinematic interaction calculation. **16C.5** Will be combined according to. Allowable unit deformation and internal force limits for reinforced concrete and steel piles **5.8.3** It is defined in .

(d) If the effects occurring on the superstructure (building) carrier system as a result of inertial interaction are more favorable than those obtained from building calculations made without taking interaction into account, the interaction effects on the superstructure (building) carrier system will not be taken into account.

16C.2.4. Method I – Interaction Calculation with Common Systems Approach

In Method I, interaction calculations are performed for superstructure, foundation, piles and soil depending on computer hardware and software capabilities. **16C.2.2** And **16C.2.3** obtained by combining the kinematic interaction and inertial interaction models defined in *Common system* It can also be performed as a time history calculation under the effect of an earthquake defined in the bedrock.

16C.3. CALCULATION OF STRUCTURE – PILE – GROUND INTERACTION WITH METHOD II

16C.3.1. Method II – Scope

Method II, **Table 16.5** According to the classification in , where the local soil class is ZD, ZE or ZF,

(a) DTS = 3, 3a, 4, 4a with pile *High-Rise Buildings* (For BYS = 1) **Chapter 13** In Phase III of , DD-1 will be applied in the nonlinear time history calculation under the influence of earthquake ground motion.

(b) For pile buildings with DTS = 1a, 2a and BYS = 2, **Table 3.4(a)** And **5.1.3.2** According to , DD-1 will be applied in nonlinear calculations made in the time history under the influence of earthquake ground motion.

16C.3.2. Method II – Kinematic Interaction Calculation

16C.3.2.1 –In Method II, the inertia properties of the superstructure are not taken into account. *foundation – pile – soil kinematic interaction calculation model* will be created as follows:

(a) *Soil environment* No modelling will be done for the soil environment. The effect of the soil environment is indirectly determined by the following (d) will be taken into account in paragraph .

(b) Stakes, 5.3.1 The relative force-displacement relationship between the pile and the soil medium at each pile node point will be modeled with nonlinear p_y , t_z springs and at the pile tip with QZ spring (**See 16C.6**). p_y arcs, one pair at each node and one in each direction *pressure-only spring* can be defined as. Before the earthquake calculation, the axial forces on the piles due to vertical loads will be calculated using the t_z and QZ springs. *group effect's* approximately considered

to be taken **Equation (16C.1)** The reduction coefficients β calculated with the empirical relation given in G , The force-displacement curves defined for the springs p_y will be applied to the force (p) axis:

$$\beta_G = 0.2 [(1 - \beta_{G1})s - (1 - 6\beta_{G1})] \quad (16C.1)$$

Here s , pile row spacing in the direction considered in diameter rate
($s \leq 6$), β_{G1} indicates a coefficient that will be applied differently for each pile row.

Pressure-only springs For the front row of piles subjected to lateral pressure to be applied to $\beta_{G1} = 0.7$, for the second row of stakes behind it $\beta_{G1} = 0.45$ for the third row of stakes $\beta_{G1} = 0.3$, and for the fourth and further back rows of piles $\beta_{G1} = 0.2$ will be taken. For displacements in the opposite direction, these coefficients will be applied to the same piles in reverse order. β for $G = 1$ will be taken.

(c) Building foundation, 16C.2.2.1(c) It will be modeled as a massless, infinitely rigid element, including the basement floors, according to the definition in . In order to represent the horizontal force-displacement relationship between the basement walls and the ground, a nonlinear force such as p_y selected from the literature is used in the direction perpendicular to the wall at the wall finite element nodes. *pressure-only springs* will be used.

(d) Earthquake records aspect, 16.5.2 In the free soil response analyses performed according to , the earthquake record pairs applied separately in both horizontal directions in the base rock were obtained at the pile node points along the soil profile. *Changes in total displacements with time* will be used (*total displacement*, is the sum of the displacement of the bedrock and the relative displacement in the ground). The total displacements at the instant will be applied simultaneously in both horizontal directions to the other ends of the p_y springs connected to the pile nodes at one end in the direction of pressure (*dynamic displacement loading*). These displacements will be applied to the other ends of the p_y -like springs, one end of which is connected to the shear wall finite element nodes in buildings with basements, in the direction of pressure, and thus the earthquake analysis will be performed in the time history of the kinematic interaction model.

16C.3.2.2 -As a result of the foundation – pile – soil kinematic interaction analysis in Method II, the following outputs will be obtained:

(a) Internal force and deformation demands in piles are determined by the analysis performed (at least $2 \times 11 = 22$ analysis) will be calculated as the average of the largest absolute values of the results obtained from each.

(b) $2 \times 11 = 22$ for analysis **16C.2.2.1(c)** For degrees of freedom defined as rotation about two horizontal axes and in the base of the rigid foundation modeled according to *effective basic movements* Changes in total displacements with respect to time (*foundation level earthquake records*) and the acceleration spectra generated from them (*Base level earthquake spectra*) will be obtained. These outputs will be **16C.3.3** Explained in *inertial interaction calculation* They will be used as inputs of .

(c) Since the inertia of the superstructure is not taken into account and the building foundation is taken as rigid, no internal force occurs in the superstructure during kinematic interaction.

16C.3.3. Method II – Inertial Interaction Calculation

16C.3.3.1 – The steps of inertial interaction calculation in Method II are the same as for Method I. **16C.2.3.1** And **16C.2.3.2** The calculation steps explained in are the same. The only difference is in the calculation model. Instead of the foundation-pile-soil environment model in Method I, the foundation-pile-soil springs (py, tz and QZ springs) model is taken into account in Method II.

16C.3.3.2 – In Method II *inertial interaction* calculation of the building carrier system **5.7** According to the nonlinear calculation made with the time history calculation method, *structure – pile – soil interaction* It goes against the way it is taken into account.

16C.3.3.3 – In Method II *inertial interaction* as a result of the calculation, exactly for Method I **16C.2.3.4** The outputs specified in will be obtained.

16C.3.3.4 – The internal forces and deformations obtained in the piles as a result of the inertial interaction calculation are compared with those obtained from the kinematic interaction calculation. **16C.5** Will be combined according to. Allowable unit deformation and internal force limits for reinforced concrete and steel piles **5.8.3** It is defined in .

16C.3.3.5 – If the effects occurring in the superstructure (building) carrier system as a result of inertial interaction are more favorable than those obtained in building calculations made without taking interaction into account, the interaction effects will not be taken into account in the superstructure (building) carrier system.

16C.3.4. Method II – Interaction Calculation with Common Systems Approach

In Method II, interaction calculations are performed for superstructure, foundation, piles and soil depending on computer hardware and software capabilities. **16C.3.2** And **16C.3.3** obtained by combining the kinematic interaction and inertial interaction models defined in *Common system* can also be implemented as a time domain calculation of . In this case, **16C.3.2.1(d)** As explained in , the results obtained from free soil response analyses *total displacements*, by applying pressure in the direction of the other ends of the py springs connected to the pile nodes at one end, simultaneously in both horizontal directions (*dynamic displacement loading*) calculation is done in the time domain of the common system. This type of interaction calculation can be preferred as a practical calculation in most cases.

16C.4. STRUCTURE – PILE – SOIL INTERACTION CALCULATION WITH METHOD III

16C.4.1. Method III – Scope

Method III, **Table 16.5** According to the classification in , where the local soil class is ZD, ZE or ZF,

(a) For piled buildings with DTS = 1a, 2a and BYS ≥ 44.7 or **4.8.2** According to , DD-2 will be applied in linear calculations under the influence of earthquake ground motion.

(b) For pile buildings with DTS = 1, 2, 3, 3a and BYS ≥ 24.7 or **4.8.2** According to , DD-2 will be applied in linear calculations under the influence of earthquake ground motion.

16C.4.2. Method III – Kinematic Interaction Calculation

Foundation – pile – soil kinematic interaction calculation to be made with Method III, **16C.3.2** According to , it corresponds to the simplified form of the calculation made with Method II.

16C.4.2.1 –In Method III, the foundation – pile – soil kinematic interaction calculation model, in which the inertia properties of the superstructure are not taken into account, will be created as follows.

(a) *Soil environment* No modelling will be done for the soil environment. The effect of the soil environment is indirectly determined by the following (d) will be taken into account in paragraph .

(b) *Stakes*, based on linear behavior in the earthquake direction considered **4.5.2** The relative force-displacement relationship between the pile and the soil medium at each pile node will be modeled with nonlinear p_y , t_z springs and at the pile tip with QZ spring (See **16C.6**). p_y arcs, one pair at each node and one in each direction *pressure-only spring* can be defined as. Before the earthquake calculation, the axial forces on the piles due to vertical loads will be calculated using the t_z and QZ springs. *group effect* To take into account approximately **Equation (16C.1)** The empirical relation given in will be used.

(c) *Building foundation*, **16C.2.2.1(c)** It will be modeled as a massless, infinitely rigid element, including the basement floors, according to the definition in . In order to represent the horizontal force-displacement relationship between the basement shear walls and the ground, nonlinear springs such as p_y selected from the literature will be used in the direction perpendicular to the shear wall at the shear wall finite element nodes.

(d) **16.5.2** For each of the earthquakes (eleven earthquake records) affected horizontally from the bedrock in the earthquake direction taken into consideration in the one-dimensional free soil response analyses carried out according to , the pile node alignments obtained along the soil profile *total displacements* The envelope of the maximum of with respect to time will be taken. The total displacement envelope is the envelope of the piles connected to the pile nodes at one end. *nonlinear* to the other ends of the p_y arcs *in the direction of pressure* starting from scratch *statically incrementally* will be applied (*static incremental displacement loading*). The same displacement envelope will be applied to the other ends of the p_y -like springs connected to the shear wall finite element nodes in buildings with basements in the pressure direction, and thus the kinematic interaction model will be approximated. *incremental static calculation* will have been done.

16C.4.2.2 –In Method III, as a result of the foundation-pile-soil kinematic interaction analysis, the internal force demands in the piles will be calculated as the average of the maximum absolute values of the results obtained from each of the analyses made separately for eleven earthquake records in each earthquake direction. The pile bending moments obtained in this way R It will be reduced by dividing by a factor of 2.5. No reduction will be made for shear forces and axial forces.

16C.4.2.3 –Obtained in Method I and Method II *effective basic movements* The displacements and spectra related to are not calculated in Method III, which is based on static kinematic interaction calculation. Instead, they are calculated approximately **16.5.2** The average of the spectra obtained from the free soil response analyses at the foundation base according to , **16C.4.3** In the inertial interaction calculation given in *design spectrum* can be used as . However, the ordinates of this spectrum are in the basic base **16.4** For the local soil class defined according to **2.3.4** It will not be taken smaller than the ordinates of the design spectrum given in .

16C.4.3. Method III – Inertial Interaction Calculation

16C.4.3.1 –In Method III, the inertial interaction calculation is based on linear behavior and consists of superstructure-foundation-pile-soil springs (p_y , t_z and Q_Z springs). *common system model* can be done with. However, in this model, the floor springs *linear* behavior (initial stiffness) will be taken into account. The masses of the foundation and basement floors will be taken into account in the model, but the masses of the piles will not be taken into account. As earthquake data, the earthquake direction taken into account

16C.4.2.3 The earthquake spectrum defined according to will be used.

16C.4.3.2 –In Method III *inertial interaction* calculation of the building carrier system **4.7** or **4.8.2** According to the linear earthquake calculation, *structure – pile – soil interaction* It goes against the way it is taken into account.

(a) 3.3.1 In buildings with basements surrounded by rigid external walls according to the definition given in **4.7.5** or **4.8.5** According to , applied for linear earthquake calculation *two-step accounting approach* Piles will be taken into consideration together with the basement elements and the basements will be applied R and D The coefficients will also be applied to the internal forces occurring in the piles due to inertial interaction.

(b) In buildings without a basement, the building (superstructure) carrier system is applied. R and D The coefficients shall be applied identically to the internal forces occurring in the piles due to inertial interaction.

16C.4.3.3 –As a result of the inertial interaction calculation, for each earthquake direction in the piles

16C.4.3.2 Internal forces obtained according to , from the kinematic interaction calculation for piles in each earthquake direction **16C.4.2.2** With the internal forces obtained according to **16C.5** The pile internal forces resulting from earthquakes in perpendicular directions will then be combined according to **4.4.2** will be merged according to.

16C.4.3.4 –If the effects occurring in the superstructure (building) carrier system as a result of inertial interaction are more favorable than those obtained in building calculations made without taking interaction into account, the interaction effects will not be taken into account in the superstructure (building) carrier system.

16C.5. BASIS FOR EVALUATION / DESIGN IN PILES SHAPES AND INTERNAL FORCES

16C.2, **16C.3** and **16C.4** As a result of the calculations made with the methods explained in , the deformations and internal forces that form the basis for evaluation / design in piles will be calculated as the sum of the absolute values of the kinematic and inertial interaction results.

16C.6. NON-LINEAR GROUND FOR PILE-GROUND INTERACTION SPRINGS

16C.6.1. Definition

For the purpose of modeling the pile–soil interaction, nonlinear p – y springs will be used for lateral response, t – z springs for axial response (friction) and Q – Z springs for axial pile toe response.

(a) For soft clay within the p – y arcs ($s_u < 50$ kPa) Matlock (1970), water level

Nonlinear relations developed for static and cyclic loadings by Reese et al. (1975) for solid clay below the water level, by Welch and Reese (1972) for solid clay above the water level, and by Reese et al. (1974) for sand can be used.

(b)For the t-z and Q-Z arcs, the nonlinear relations defined in the American Petroleum Institute API WSD RP-2A (2000) can be used.

16C.6.2. Sources for py Springs

(a)Matlock,H. (1970). "Correlation for design of laterally loaded piles in soft clay." Proc.,2nd Annu. Offshore Technology Conference, Paper No. OTC 1204, Houston, Texas, pp. 577-594.

(b)Reese, L. C. and Welch, R. C. (1975). "Lateral loading of deep foundation in stiff clay." Journal of Geotechnical Engineering Division, ASCE, 101(7), pp. 633-649.

(c)Reese, L.C., Cox, W.R., and Koop, F.D. (1975). "Field testing and analysis of laterally loaded piles in stiff clay." Proc. 7th Offshore Technology Conference, Paper No. OTC 2321, Houston, Texas, pp. 671-690.

(d)Welch, R. C. and Reese, L. C. (1972). "Laterally loaded behavior of drilled shafts." Research Report 3-5-65-89, Center for Highway Research, University of Texas, Austin.

(to)Reese, L.C., Cox, W.R., and Koop, F.D. (1974). "Analysis of laterally loaded piles in sand." Proc. 6th Offshore Technology Conference, Paper 2080, Houston, Texas, pp. 473-483.

16C.6.3. Source for TZ and QZ Springs

API WSD RP-2A (2000). Planning, Designing and Constructing Fixed Offshore Platforms, American Petroleum Institute.

ANNEX 16D – ON-SITE IMPROVEMENT OF LAND SOIL FEATURES

In cases where the effects cannot be resisted by the resistances, in grounds where liquefaction is triggered during an earthquake and/or where a large loss of strength/rigidity is expected, in situ improvement or reinforcement of the ground properties can be resorted to. Appropriate ground improvement or reinforcement method, *geotechnical* It is determined by the civil engineer responsible for the field, using the general principles listed below.

16D.1 –In loose and medium dense cohesionless (coarse grained) soils, increasing the degree of compaction is the most effective way of improvement. Methods that apply vibrations to the ground are more effective in increasing the degree of compaction. Vibration methods can be listed as deep compaction, vibroflotation, compaction piles, vibrated probes, blasting, compaction with vibrated rollers from the surface, etc.

16D.2 –In soft-solid cohesive (fine-grained) soils, effective improvement can be achieved by methods that reduce the water content of the soil, increase the consolidation pressure and strengthen the internal structure. In this context, drainage, groundwater level reduction, consolidation methods (preloading, electro-osmotic, vacuum effect) can be used. In addition, solidification (deep mixing) and injection methods can be used by mixing binding additives such as cement and lime. In cohesive soils that are not saturated with water and do not have high plasticity, the deep compaction (compression) method can also be used.

16D.3 –In cohesive and cohesionless soils, by creating rigid elements (stone columns, jet grout columns, concrete columns, etc.) within the soil, it can be ensured that static and seismic loads are mainly carried by these elements.

16D.4 –In the design of ground improvement, the soil properties to be improved (density, shear strength, compression modulus, etc.) and the targeted improvement degree should be clearly specified. The field application of the selected ground improvement method should be closely monitored with instrumental observations, and the degree of improvement achieved according to the current situation and the realization of the targeted improvement in the design should be checked.

16D.5 –The measures that can be applied for ground improvement under an existing structure are more limited. In addition to the difficulty of working under the foundations of the structure, possible ground displacements that may develop during the application of improvement methods must be taken into account. In some cases, measures taken around the structure can be used to limit lateral ground displacements that may occur during an earthquake.

16D.6 –Instead of (or together with) ground reinforcement methods, changing the foundation system (pile or barrette foundations, perimeter diaphragm wall, rigid basement foundation design, etc.) and making design changes in the superstructure can also be used.

SECTION 17 – FOR REGULAR CAST-IN-SITU REINFORCED CONCRETE BUILDINGS SIMPLIFIED DESIGN RULES

17.0. ICONS

In dimensional expressions in this section the following symbols are used, unless otherwise stated: *kilonewton*[[kN], length*metre*[m], area*square meters*[m²] and distributed loads *kilonewton/square meters*[kN/m²] unit.

A_i	= Cross-sectional area of the column considered =
A_g	Cross-sectional area of the column or wall
A_{pt}	= Building floor area
A_s	= Area of flexural reinforcement under tension in the beam cross-section
A_{sg}	= Internal row longitudinal reinforcement in the direction considered for the column cross-section Total of the areas or total of the longitudinal reinforcement areas in the wall body for the wall cross-section
$A_{s,h}$	= In the middle region of the column along the height corresponding to the transverse reinforcement spacing, all The cross-sectional area values of the stirrup arms and crossties are taken into account according to the column cross-section width (b) is the sum of the projections in the perpendicular direction
A_{this}	= Outermost row longitudinal reinforcement in the direction considered for the column cross-section total of the areas or total of the longitudinal reinforcement areas at the end zones for the wall cross-section
A_{st}	= Total longitudinal reinforcement area in column cross section
A_{sw}	= In the middle area of the beams along the length corresponding to the transverse reinforcement spacing, all sum of the cross-sectional area values of the stirrup arms and crossties
a	= Horizontal distance between stirrup arms and/or crossties in the column middle area =
BKS	Building Usage Class
b	= Cross-section width of column or wall for the direction
b_w	considered = Beam web width or wall thickness
DTS	= Earthquake Design Class
d	= From the concrete surface under the effect of pressure in the beam, column or wall cross-section distance from the centre of the flexural reinforcement under tension
d'	= From the concrete surface under the effect of pressure in the beam, column or wall cross-section distance from the centre of the flexural reinforcement under compression
TO_d	= Earthquake effect in the horizontal direction taken into consideration for design purposes
$TO_d^{(z)}$	= Earthquake effect in vertical direction
to_x, to_y	= x and y distance between the floor mass center and rigidity center in the directions
F_i	= Equivalent earthquake load acting on the i 'th floor of the building

$\bar{F}_j$	= Equivalent earthquake load acting on the jth floor of the building
$F_{per,j}$	= Horizontal earthquake force transmitted to the shear wall at the jth floor
f_{ck}	= Characteristic compressive strength of concrete
f_{CD}	= Design compressive strength of concrete
f_{yk}	= Characteristic yield strength of reinforcement steel
f_{yd}	= Design yield strength of reinforcement steel
G	= Constant load effect
g	= The area share of the vertical load-bearing element taken into consideration is affected by each floor or Average distributed dead load [kN/m ²] not magnified by the load coefficients acting on the whole ²
H	= Horizontal ground thrust effect
H_i	= Height of the i'th floor of the building measured from the top of the foundation or four floors in the basement The total number of floors measured from the top of the ground floor slab of the ith floor of the building surrounded by rigid basement walls on the sides.
H_w	= Total height of any wall measured from the top of the foundation or basement Total height of the building surrounded by rigid basement curtains on four floors, measured from the top of the ground floor slab
H	= Building ground floor height
h	= Cross-section height of column or wall for the direction considered =
h_i	Floor height of the ith floor (floor under the ith floor slab) of the building
I	= Building importance coefficient
I_i	= Cross-sectional inertia of a single column on the ground floor in the direction considered moment
I_{arm}	= Moment of inertia of column cross-section in the direction considered =
I_{per}	Moment of inertia of wall cross-section in the direction considered
k_T	= Torsional rigidity of the building
k_x	= CurtainxHorizontal rigidity in the direction =
k_y	Shear wallyHorizontal rigidity in direction
$-k_n$	= Clear span of the beam in the short direction of the slab part
$-n$	= Beam clear span
$-w$	= Length of the curtain on the floor plan
M_{br}	= For column or wall sections f_{CD} And f_{yd} calculated taking into account the balanced bending moment against the situation

M_d	= Design basis bending moment obtained from the analysis
M_{giant}	= Total overturning moment at the base of the building
M_i	= Total overturning moment occurring on the i'th floor of the building
M_{dirt}	= Bending moment due to vertical loads in any beam section
$M_{dirt,i}$	= Any beam section on the i'th floor of the building due to earthquake effects bending moment
M_{arm}	= Bending moment due to vertical loads in any column section
M_{parcel}	= Any column section on the i'th floor of the building due to earthquake effects bending moment
M_p	= For column or beam sections f_{ck} , f_{yk} and considering the increase in strength of steel calculated moment strength
M_{pa}	= At the lower end of the free height of the column f_{ck} , f_{yk} and the increase in strength of steel Moment strength calculated by taking into account
M_{pu}	= At the upper end of the free height of the column f_{ck} , f_{yk} and the increase in strength of steel Moment strength calculated by taking into account
M_{pi}	= At the column or wall face at the i end of the beam f_{ck} , f_{yk} and the increase in strength of steel Positive or negative moment strength calculated by taking into account
M_{pj}	= On the column or wall face at the j end of the beam f_{ck} , f_{yk} and the increase in strength of steel Positive or negative moment strength calculated by taking into account
M_{per}	= Bending moment due to vertical loads on any shear wall
$M_{per,giant}$	= Bending caused by earthquake effects at the base of any curtain moment
M_{fairy}	= Any shear wall on the i'th floor of the building due to earthquake effects. bending moment
M_r	= For column, beam or wall sections f_{cd} And f_{yd} calculated taking into account load-bearing capacity moment
M_{ri}	= At the column or wall face at the i end of the beam f_{cd} And f_{yd} taking into account calculated positive or negative bearing capacity moment
M_{rj}	= On the column or wall face at the j end of the beam f_{cd} And f_{yd} taking into account calculated positive or negative bearing capacity moment
$M_{r.a.}$	= Bearing force moment at the upper end of the column free height
M_{ru}	= Bearing capacity moment at the lower end of the column free height
m	= Long side / short side ratio for the slab piece

N	= Total number of floors of the building from the top of the foundation or from four sides in the basement The total number of floors above the ground floor slab in a building surrounded by rigid basement walls.
M_{br}	= For column or wall sections f_{cd} And f_{yd} calculated taking into account the balanced pressure force opposing the situation
N_d	= Design basis axial force obtained from the analysis
N_{en}	= For column or wall sections f_{yd} Calculated axial tension taking into account strength
N_{or}	= For column or wall sections f_{cd} And f_{yd} Calculated axially taking into account Compressive strength
n	= Live load participation coefficient
n_{drink}	= Number of interior columns in the building in the direction considered
$n_{external}$	= Number of external columns in the building in the direction considered
p	= Total of distributed fixed and moving loads acting on the floor ($g+q$)
Q	= Live load effect
q	= The area share of the vertical load-bearing element taken into consideration is affected by each floor or Average distributed live load [kN/m ²] not magnified by the load coefficients acting on the whole ²]
R_a	= Earthquake load reduction coefficient
S	= Snow load effect
S_{DS}	= Short period design spectral acceleration coefficient [dimensionless]
S	= Transverse reinforcement and crosstie spacing in the middle area of the column or
T_i	beam = Total torsional moment occurring at the i'th floor of the building
$T_{floor,j}$	= Torsional moment acting on the jth floor of the building
T_t	= Total torsional moment at the base of the building =
V_{dy}	Simple beam shear force
V_{to}	= Design basis shear force for columns, beams and walls
V_i	= Floor shear force occurring on the i'th floor of the building
V_{dirt}	= Shear force due to vertical loads in any beam section
V_{parcel}	= Any column on the i'th floor of the building due to earthquake effects. shear force
V_{fairy}	= Any shear wall on the i'th floor of the building due to earthquake effects. shear force
V_t	= Total equivalent earthquake load acting on the building (base shear force)

W	= Total weight of the building found using the live load participation coefficient =
w	Uniformly distributed (linear) vertical load transferred from the floor to any beam
w_i	= Calculated using the live load participation coefficient of the i th floor of the building weight
x	= From the reference point of the cross-section geometric center of any curtain distance in direction
y	= From the reference point of the cross-section geometric center of any curtain distance in direction
$\bar{x}$	= From the reference point of the floor rigidity center distance in direction
$\bar{y}$	= Floor rigidity center from the reference point distance in direction
ρ	= Beam top or bottom longitudinal reinforcement ratio =
$\rho_{s,h}$	Transverse reinforcement ratio in the column middle area
ρ_t	= Total longitudinal reinforcement ratio of column
ρ_w	= Transverse reinforcement ratio in the middle area of the beam
$\Delta V_{T,i}$	= Any shear wall on the i 'th floor of the building due to torsional effects on the floor additional shear force generated
$\sum A_i$	= Total of column cross-sectional areas on the ground floor
$\sum A_{pi}$	= Total building floor areas
$\sum A_w$	= Cross-sectional areas of curtain or curtain arm operating in the direction considered total
$\sum A_{0i}$	= Sum of area shares accumulated across all floors for the column under consideration
$\sum (I_i/H_i^2)$	= In the direction considered on the ground floor (column cross-section moment of inertia / floor height ²) values are summed
$\sum V_{\text{parcel}}$	= Shear caused by earthquake effects on the columns on the i 'th floor of the building the sum of the forces

17.1. SCOPE

17.1.1 –The basement is surrounded by rigid basement walls on three or four sides. *buildings with basements* in **3.3.1** According to *building base*'s definition of basement walls at the floor slab level on the upper level, in buildings without a basement or where the basement is surrounded by rigid basement walls on two sides *buildings with basements* In **3.3.1** According to *building base* By defining at the foundation top level, BYS for $DTS=1.2 \geq 6$ and $DTS=3.4$ for $BYS \geq 7$, cast-in-place reinforced concrete buildings without a basement or with a basement, **17.2** In the design of horizontal load-bearing system elements of buildings that meet the conditions in , the simplified calculation (structural analysis) and design rules defined in this section, together with the relevant standards and regulations in force on this subject, may be used, depending on the preference of the building owner and the engineer responsible for the project.

17.1.2 –The rules and conditions defined in this section may only be used for design purposes within the scope of this section.

17.1.3 –Scope of application of the rules defined in this section **Table 3.1** It is limited to residential or business type buildings where the Building Use Class given in is defined as BKS=3.

17.1.4 –The simplified calculation (structural analysis) and design rules defined in this section are not used, but **17.1.1** DTS=1,2 for DTS as defined in ≥ 5 and DTS=3.4 for BYS $\geq$ For residential or business type buildings with no basement or with a basement floor and where the Building Use Class is defined as BKS=3, **17.3** And **17.4** The dimensioning rules and lower boundary conditions of cross-section and reinforcement defined in can be used for preliminary design purposes at the discretion of the building owner and the engineer responsible for the project. However, in the final design of such buildings, it is mandatory to use the calculation and design rules defined in other sections of this regulation together with the relevant standards and regulations in force on this subject.

17.2. GENERAL RULES

The application limits of the design rules defined in this section are defined below:

17.2.1 –The horizontal load-bearing system of reinforced concrete buildings within the scope of this section shall consist only of high-ductility level frames or of the combined use of high-ductility level solid shear walls and high-ductility level frames.

17.2.2 –The scope of application of the design rules defined in this section is limited to the dimensioning of the cross-sections of the load-bearing elements and the determination of their reinforcement in order to provide sufficient moment carrying capacity and sufficient shear strength in the load-bearing system elements (wall, column and beam) under earthquake effects and vertical loads. Design rules according to strength, detailing rules, geometric rules, minimum conditions and upper limit conditions that are outside the scope of the design rules defined in this section, **Chapter 7** and is subject to all the rules given in the relevant standards and regulations in force. However, for buildings to which the design rules in this section are applied, it is not necessary to calculate the relative floor drifts under earthquake effects and to compare the relative floor drift ratios with the limit values.

17.2.3 –Where the design rules defined in this section are used for buildings whose load-bearing system consists solely of frames, infill walls or façade elements made of brittle materials and frame elements shall not be used. **4.9.1.3(b)** Flexible joints or connections should be created according to .

17.2.4 –The building should have a floor plan close to a rectangle. In the floor plan, the long side of the building should be at most 30 m long, and the ratio of the long side to the short side should be at most 4. The building carrier system (frame and curtains) should be created along orthogonal axes perpendicular to each other in the floor plan, and there should be no discontinuity or shift off-axis in the carrier system axes.

17.2.5 –The largest opening in the building floor plan should be 7.5 m, and the smallest opening should be 3 m. There should be at least two openings in each direction of the carrier system. Except for the opening between the free ends of U or C cross-section curtains that can be created around elevators or stairs, the difference between two adjacent openings in each direction should not exceed the length of the two openings.

If there are cantilever beams in the building, the console length should not be more than 2 m and 1/3 of the adjacent span.

17.2.6 –There should be no floor higher than 4 m in the building, and the height difference between floors should not be more than 1/5 of the height of the higher floor.

17.2.7 –In the building **Table 3.6** As defined in , there should be no irregularities such as A2 - Floor Discontinuities, A3 - Presence of Protrusions in the Plan, B1 - Strength Irregularity Between Adjacent Floors (Weak Floor) and B3 - Discontinuity of Vertical Elements of the Carrier System.

17.2.8 –For each earthquake direction on all floors except the partial basement

17.5.1 The eccentricity between the center of mass and the center of rigidity of the building calculated according to should not exceed 1/5 of the floor plan length of the building perpendicular to the relevant direction. In cases where reinforced concrete shear walls are used in the load-bearing system, there should be at least two rectangular shear walls or shear arm in each direction, at least one on each side of the floor center of mass in the building plan.

17.2.9 –Vertical load-bearing element (column, curtain) cross-sections will not be reduced along the building height, different beam cross-sections will not be used on different floors of the building, and no changes will be made to beam cross-sections along any horizontal load-bearing system axis.

17.2.10 –In case of using beamed slab floor system in the building, the slab thickness should be at least 150 mm. In case of using toothed or filled (asmolen) floor systems, the slab thickness should be at least 70 mm, the beams between the vertical load-bearing elements should be dimensioned and equipped as frame beams with high ductility level. The teeth in such floors should be formed perpendicular to the frame beams that are stuck to the rectangular curtains or curtain arms in a strong direction and transfer the earthquake loads from the floor to the curtains.

17.2.11 –The building foundation should be constructed as a raft or continuous foundation when the load-bearing system consists of frames, and as a raft foundation when the load-bearing system consists of shear walls and frames.

17.2.12 –The accounting rules in this section apply only to $G+Q+0.2S \pm TO_d$ and $0.9G \pm TO_d$ defined to be applied for load combinations. Snow load S , affecting the roof of the building can be added to the live load value. Vertical earthquake effect $TO_{(Z)}$ may not be taken into account in the calculation. Directional coupling will not be applied in horizontal earthquake effects.

17.2.13 –*Earthquake Design Class* In the horizontal load-bearing system of buildings with $DTS=1$ and $DTS=2$, the use of high-ductility, gapless shear walls is mandatory. In the design of such buildings, the condition that the total equivalent seismic load acting on the building is borne entirely by high-ductility shear walls and 33% by high-ductility frames is taken into account in the calculations. In buildings whose horizontal load-bearing system consists of high-ductility frames, the total equivalent seismic load acting on the building is borne entirely by the frames.

17.2.14 –In buildings with a basement surrounded by rigid basement walls on two sides, *building base* Since it is defined at the upper foundation level, the basement floor will be considered as the ground floor in the structural analysis, and the internal force values formed in the structural system elements in the basement due to earthquake effects will also be valid for the structural system elements on the upper floor.

17.2.15 –In buildings with a basement surrounded by rigid basement walls on three or four sides, *building base* Since the basement walls are defined at the floor slab level on the upper level, the floor above the basement floor will be considered as the ground floor in the structural analysis, and the internal force values formed in the load-bearing system elements on the ground floor due to earthquake effects will also be valid for the load-bearing system elements in the basement floor.

17.2.16 –Horizontal ground thrust effects (H) has not been taken into account in the calculation rules defined in this section. Therefore, in buildings exposed to unbalanced horizontal ground thrusts from two sides in the basement, it must be additionally demonstrated that the ground thrust effects are safely transferred to the foundation.

17.2.17 –In buildings to be designed in accordance with the rules defined in this section, concrete class with strength lower than C25 or higher than C50 and **7.2.5.3** Reinforcement steel other than B420C or S420 classes defined in cannot be used.

17.2.18 –In all load-bearing elements in buildings to be designed in accordance with the rules defined in this section, the interlocking length and overlapping length of longitudinal reinforcement cannot be less than 50 times the reinforcement diameter, and the interlocking length of 90-degree hooks cannot be less than 25 times the reinforcement diameter.

17.2.19 –In buildings to be designed in accordance with the rules defined in this section, the critical curtain height shall be continued for at least one floor above the foundation in buildings without a basement; and for at least two floors above the foundation in buildings with basements surrounded by basement curtains on two, three or four sides.

17.2.20 –Buildings to which the calculation and design rules defined in this section will be applied **17.3** And **17.4** It is necessary to meet the rules defined in .

17.3. DIMENSIONING OF VERTICAL CARRIER ELEMENTS

Cross-sectional areas of vertical bearing elements **17.3.1** And **17.3.2** The sum of the average distributed dead load and average distributed moving load values to be used in determining the equations given in $(g+q)$ value is 15 kN/m² from, $(g+0.3q)$ value is 13 kN/m² It shall not be taken less than . The distributed dead load values shall include the weight of horizontal and vertical load-bearing elements (beams, columns, curtains) and non-load-bearing elements (partition walls) along with the floor loads. The live load acting on the roof of the building shall also include the snow load reduced by the load coefficient of 0.2.

17.3.1. Buildings with Frame Structures

17.3.1.1 –To limit the axial compressive stresses in the columns, the cross-sectional area of each column A_i , **Equation (17.1)** will satisfy the condition in the equation. g And q

The values are the average fixed and live distributed loads acting on the area share of the column considered at each floor, $\sum A_{0i}$ The value is the sum of the area shares accumulated along all the floors carried by the arm of the column considered.

$$A_i \geq 0.00014(g+q) \sum A_{0i} \quad (17.1)$$

17.3.1.2 –In order for the columns to have sufficient shear strength, the cross-sectional area of each column A_i , **Equation (17.2)** The condition in the equation must be satisfied. S_{Ds} value, short period is the design spectral acceleration coefficient defined for the region.

$$A_i \geq 0.00022 S_{Ds}(g+0.3q) \sum A_{0i} \quad (17.2)$$

17.3.1.3 –In order for the carrier system to have sufficient lateral rigidity, the sum of the values of the ratio of the cross-sectional moment of inertia of the building ground floor columns to the square of the floor height.

$\sum (I_i / H^2)$ value in both directions of the building **Equation (17.3)** will satisfy the condition in .

In the equation $\sum A_{pi}$ value is the sum of the building floor areas.

$$\sum (I_i / H^2) \geq 4.44 \times 10^{-3} S_{DS} (g + 0.3q) \sum A_{pi} \quad (17.3)$$

17.3.2. Buildings with a structural system consisting of walls and frames

17.3.2.1 –To limit the axial compressive stresses in the columns, the cross-sectional area of each column A_i , **Equation (17.4)** will satisfy the condition in .

$$A_i \geq 0.00012 (g + q) \sum A_{oi} \quad (17.4)$$

17.3.2.2 –In order for the columns to have sufficient shear strength, the cross-sectional area of each column A_i , **Equation (17.5)** will satisfy the condition in .

$$A_i \geq 0.0001 S_{DS} (g + 0.3q) \sum A_{oi} \quad (17.5)$$

17.3.2.3 –In order for gapless curtains to have sufficient shear strength, the total of the rectangular curtain or curtain arm cross-sectional areas operating in each direction of the building $\sum A_w$

value in both directions of the building **Equation (17.6)** and **Equation (17.7)** will satisfy both conditions in the equation. $\sum A_{pi}$ value is the sum of the building floor areas, A_{pt} The value is the building base

is the field.

$$\sum A_w \geq 0.0002 S_{DS} (g + 0.3q) \sum A_{pi} \quad (17.6)$$

$$\sum A_w \geq 0.0007 S_{DS} (g + 0.3q) A_{pt} \quad (17.7)$$

17.3.2.4 –In order for the building load-bearing system to have sufficient shear force resistance at the base, the sum of the cross-sectional areas of the ground floor columns and the cross-sectional areas of the rectangular curtain or curtain arm working in each direction of the building are added together.

the result obtained $(\sum A_i + \sum A_w)$ value in both directions of the building **Equation (17.8)** will satisfy the condition in .

$$\sum A_i + \sum A_w \geq 0.0003 S_{DS} (g + 0.3q) \sum A_{pi} \quad (17.8)$$

17.4. CROSS SECTION AND REINFORCEMENT LOWER LIMITS

17.4.1. Lower Cross-Section Limits for Carrier System Elements

17.4.1.1 –The smallest square column cross-section will be 300 mm × 300 mm. In rectangular column cross-sections, the short side will be at least 300 mm long, and the ratio of the long side to the short side will not be more than 2.

17.4.1.2 –The smallest beam cross-section shall be 300 mm wide and 500 mm high. The beam height shall not be less than 1/11 of the span in buildings with a carrier system consisting of frames, and 1/12 of the span in buildings with a carrier system consisting of shear walls and frames. The beam height shall also not exceed 1/4 of the clear span.

17.4.1.3 –Rectangular wall thickness or wall arm thicknesses in L, T, U, C cross-section walls, excluding basement floor perimeter walls, shall not be less than 1/25 of the plan length of the wall and 250 mm. The thickness of basement floor perimeter walls shall not be less than 200 mm.

17.4.1.4 –The length of rectangular curtains or curtain arms in L, T, U, C cross-section curtains on the building plan shall not be less than 1/8 of the height of the building measured from the ground floor.

17.4.2. Lower Reinforcement Limits for Carrier System Elements

17.4.2.1 –Ratio of total vertical reinforcement area to column cross-section area in square cross-section columns ρ_t , shall not be less than 0.01. Rectangular cross-section with a ratio of long side to short side of 2

In columns, this ratio shall be at least 0.015. For intermediate values of the long side / short side ratio, it can be assumed that the lower limit of the vertical reinforcement ratio changes linearly.

17.4.2.2 –In square cross-section columns **7.3.4.2** Transverse reinforcement ratio in the column middle region defined in $\rho_{s,h} = A_{s,h} / (sb)$, shall not be less than 0.00165. The ratio of the long side to the short side is 2

In columns with rectangular cross-sections, this ratio shall be at least 0.0025. For intermediate values of the long side / short side ratio, it can be assumed that the lower limit of the transverse reinforcement ratio changes linearly. In any case, the stirrup and crosstie spacing in the column middle area shall not be more than half of the smallest cross-section dimension of the column and 200 mm, and the horizontal distance between the stirrup arms and/or crossties shall not be more than a , shall not exceed 25 times the stirrup diameter.

17.4.2.3 –In beam support sections where beams meet columns or shear walls, the upper and lower longitudinal reinforcement ratios of the beams $\rho = A_s / (b_w d)$, shall not be less than 0.006 and 0.004 respectively.

17.4.2.4 – **7.4.4** Transverse reinforcement ratio in the middle area of the beam defined in $\rho_w = A_{sw} / (sb_w)$, not less than 0.0025, the stirrup and crossbar spacing in this region is the usefulness of the beam shall not exceed half of its height.

17.5. SIMPLIFIED STRUCTURAL ANALYSIS

17.5.1. Calculation of Earthquake Effects

17.5.1.1 –For each direction considered, the total equivalent seismic load acting on the building (base shear force occurring on the ground floor), the equivalent seismic loads acting on the floors, the floor shear forces occurring on the floors, the total overturning moments occurring on the floors and the total overturning moment occurring at the building base (ground floor base), **Eq.(17.9–17.13)** will be calculated with **(Figure 17.1)**. In the calculation of total equivalent earthquake load *earthquake load reduction coefficient* for

$R_a = 4$ value is used in the calculation of the total weight of the building *live load participation coefficient* for $n = 0.30$

The dead load values to be used in the building weight calculation should include the weight of horizontal and vertical load-bearing elements (beams, columns, curtains) and non-load-bearing elements (partition walls) along with the floor loads. The live load acting on the roof of the building should also include the snow load reduced by the load coefficient of 0.2.

$$V_{fDS} = \frac{SW}{R_a} \quad (17.9)$$

$$F_i = V_i \frac{w_i H_i}{\sum_{j=1}^N w_j H_j} \quad (17.10)$$

$$V_i = \sum_{j=i}^N F_j \quad (17.11)$$

$$M_i = \sum_{j=i}^N F_j (H_j - H_{i-1}) \quad (17.12)$$

$$M_1 = M_{\text{giant}} = \sum_{j=1}^N F_j H_j \quad (17.13)$$

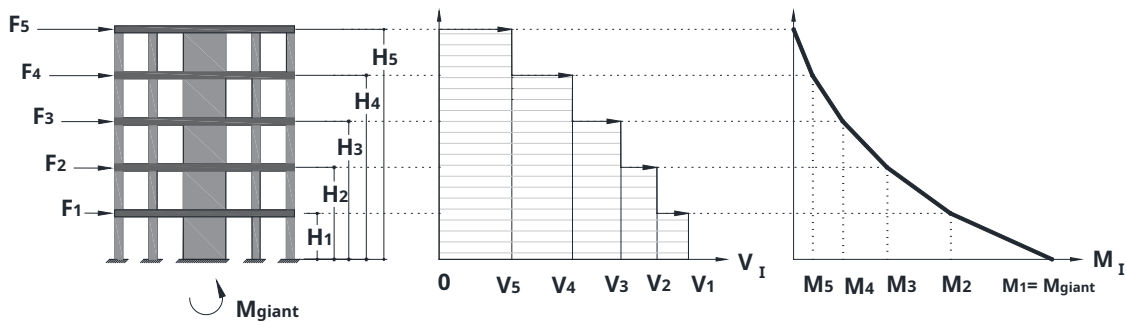


Figure 17.1

17.5.1.2 -In order to calculate the torsional moments acting on the building floors, the building floor rigidity center is selected from a reference point in the plan. x and y distance in directions (Figure 17.2), Equation (17.14) will be calculated with .Equation (17.14) in k_x and k_y stiffness values, x

For rectangular cross-section curtains in the direction Equation (17.15), y For rectangular cross-section curtains in the direction of Equation (17.16) In buildings without shear walls, it will be assumed that the floor rigidity center coincides with the floor mass center.

$$\bar{x} = \frac{\sum (k_{\text{good}} x_i)}{\sum k_{\text{good}}} \quad \bar{y} = \frac{\sum (k_x y_i)}{\sum k_x} \quad (17.14)$$

$$k_x = (b_w^3 / h_i) \quad k_y = (b_w^3 / h_i) \quad (17.15)$$

$$k_x = (b_w^3 / h_i) \quad k_y = (b_w^3 / h_i) \quad (17.16)$$

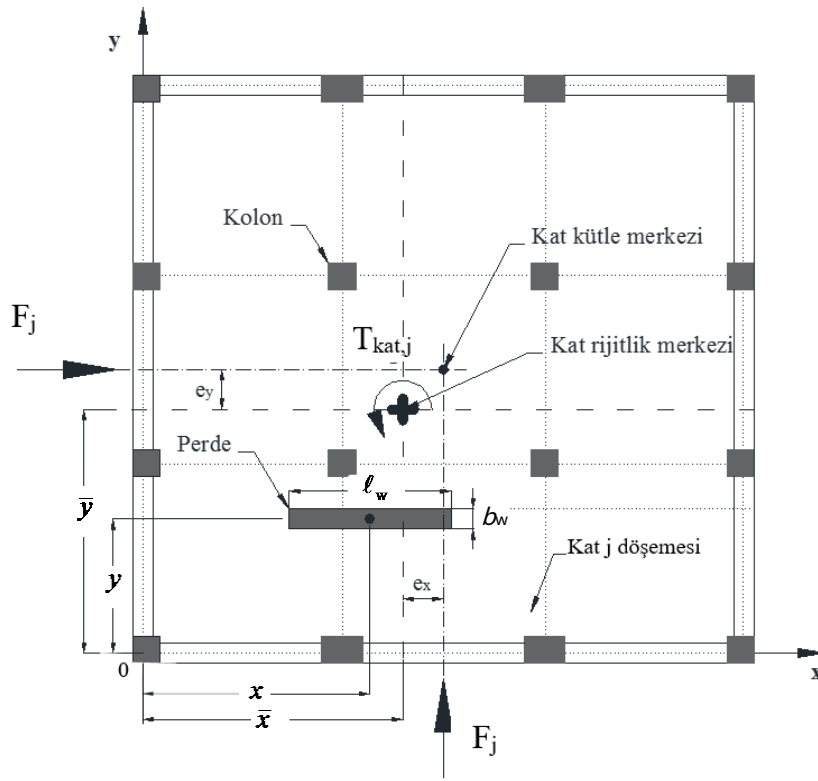


Figure 17.2

17.5.1.3 -The building/Torsional moment acting on the 'th floor slab, **Equation (17.17)** will be calculated with . **Equation (17.17)** in x and y. The eccentricity values in the directions are the distances between the floor mass center and the floor rigidity center (**Figure 17.2**). The floor mass center will be accepted as the geometric center of the floor slab and the gaps with sides larger than 2.0 m in the slab will be taken into account in the calculation of the floor mass center. Equivalent earthquake loads acting on the floors will be applied to the floor mass center. In buildings without shear walls, there is no need to calculate the torsional moments acting on the floors.

$$T_{\text{floor},j} = \max \{ (F_j t_{ox}), (F_j t_{oy}) \} \quad (17.17)$$

17.5.1.4 -The building/Total torsional moment occurring at the 'th floor and the base of the building, **Equation (17.18)** and **Equation (17.19)** will be calculated with .

$$T_I = \sum_{j=1}^N T_{\text{floor},j} \quad (17.18)$$

$$T_t = \sum_{j=1}^N T_{\text{floor},j} \quad (17.19)$$

17.5.1.5 -Torsional rigidity of the building, **Equation (17.20)** will be calculated with .

$$K_T = \frac{\sum (k_{\text{good}} x_i^2 + k_x y_i^2)}{\sum k_{\text{good}} - \sum y_i^2 k_x} \quad (17.20)$$

17.5.1.6 -The building/The shear force due to earthquake effects on a wall parallel to the loading direction considered on the 'th floor, **Equation (17.21)** will be calculated with . **Equation (17.21)** In , the floor shear forces are distributed to the walls in proportion to their moments of inertia.

assumed and additional shear forces that will occur on the walls due to torsional effects were taken into account. In the calculation of additional shear forces that occur on the walls due to torsional effects, x for curtains in line **Equation (17.22)**, y for curtains in line **Equation (17.23)** will be used (**Figure 17.2**).

$$V_{\text{airy}} = V_i \sum \frac{b_{w-w}}{b_{w-w}} + \Delta V_{T,i} \quad (17.21)$$

$$(\Delta V_{T,i})_x = T_i \frac{|y - \bar{y}| k_x}{k_T} \quad (17.22)$$

$$(\Delta V_{T,i})_y = T_i \frac{|x - \bar{x}| k_y}{k_T} \quad (17.23)$$

17.5.1.7 -The building/Bending moment due to earthquake effects on a shear wall parallel to the loading direction considered at the 'th floor **Equation (17.24)** and the overturning moment at the base of the curtain is **Equation (17.25)** will be calculated with **Equation (17.24)** and **Equation (17.25)** in F_{per} The force represents the horizontal load transferred from the floor to the wall at that floor.

The building is on the 'th floor or attic (N) The horizontal load transferred to the shear wall (on the 'th floor) is the difference between the shear forces occurring on the shear wall on the adjacent floors, **Equation (17.26)** or **Equation (17.27)** will be calculated with (**Figure 17.3**).

$$M_{\text{airy}} = \sum_{j=1}^N F_{\text{per},j} (H_j - H_{j-1}) \quad (17.24)$$

$$M_{\text{per},1} = M_{\text{per},\text{giant}} = \sum_{j=1}^N F_{\text{per},j} H_j \quad (17.25)$$

$$F_{\text{per},j} = V_{\text{per},j+1} - V_{\text{per},j} \quad (17.26)$$

$$F_{\text{per},N} = V_{\text{per},N} \quad (17.27)$$

17.5.1.8 -In buildings with a load-bearing system consisting of walls and frames, the building is placed in line with the load taken into account. The sum of the shear forces in the columns on the 'th floor, **Equation (17.28)** In buildings with a frame structure, this total will be calculated as: **Equation (17.29)** As stated in , it will be equal to the floor shear forces occurring in the building in the direction considered.

$$\sum V_{\text{parcel}} = (1 - \beta) V_i \quad (17.28)$$

$$\sum V_{\text{parcel}} = V_i \quad (17.29)$$

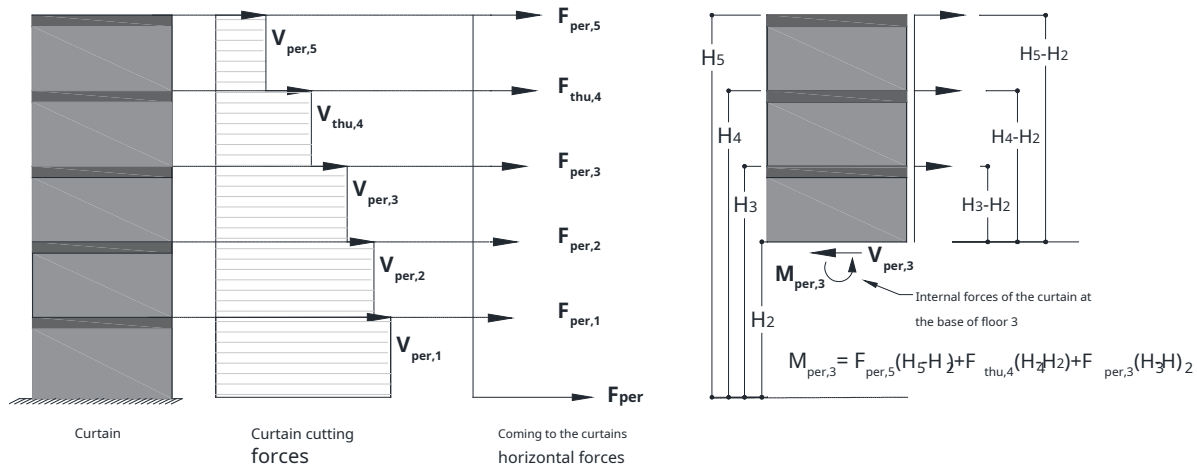


Figure 17.3

17.5.1.9 -Building for the loading direction considered/The shear force developed in any column on the 'th floor is the same as for the interior columns in the relevant direction.**Equation (17.30)**and for external columns**Equation (17.31)**will be calculated with (Figure 17.4).

$$V_{parcel} = \frac{2 \sum V_{parcel}}{2n_{drink} + n_{external}} \quad (17.30)$$

$$V_{parcel} = \frac{\sum V_{parcel}}{2n_{drink} + n_{external}} \quad (17.31)$$

17.5.1.10 -Building for the loading direction considered/The bending moment in any column on the 'th floor,**Equation (17.32)**will be calculated with (Figure 17.4).

$$M_{parcel} = V_{parcel} h / 2 \quad (17.32)$$

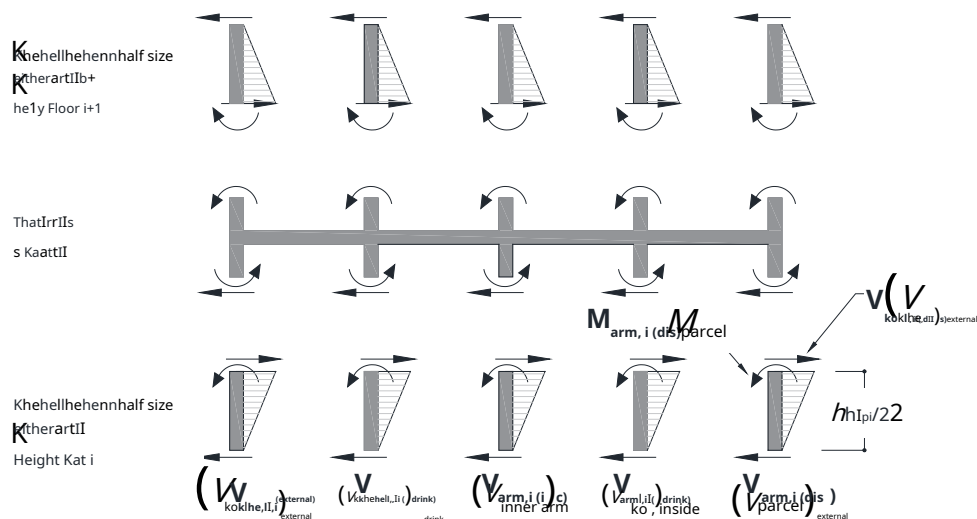


Figure 17.4

17.5.1.11 -Building for the loading direction considered/The bending moment occurring in any beam section on the 'th floor, at the beam - interior column joints**Equation (17.33)**and in beam - external column connections**Equation (17.34)**will be calculated with (Figure 17.5). On the top floor beams **Equation (17.33)**And**Equation (17.34)**The bending moment values calculated with will be halved.

$$M_{dirt,i} = \frac{4}{5} M_{parcel} \quad (17.33)$$

$$M_{dirt,i} = \frac{8}{5} M_{parcel} \quad (17.34)$$

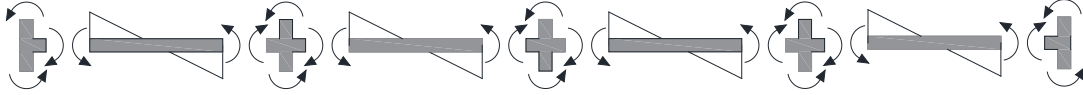


Figure 17.5

17.5.1.12 –The bending moment caused by earthquake effects in the beam sections at the beam-wall joints will not be calculated. The upper and lower longitudinal reinforcement to be placed on the beam in these sections **17.4.2.3** It will be in a quantity that will satisfy the lower boundary conditions in and will resist the bending moments caused by vertical loads.

17.5.2. Calculation of Vertical Load Effects

17.5.2.1 –Uniformly distributed (linear) vertical load transferred to any beam from the floor piece operating in each pair of directions (w), for the beam in the short direction of the slab piece, unless a more detailed calculation is made **Equation (17.35)** and for the beam in the long direction of the slab part, **Equation (17.36)** All of the load transferred from the plate (or toothed) slab pieces operating in one direction will be transferred to the beam in the long direction of the plate slab piece (or the beam into which the slab teeth are embedded). **Equation (17.35)** And **Equation (17.36)** in p , calculated by taking into account the load coefficients of the distributed fixed and live loads acting on the floor ($g+q$ or $1.4g+1.6q$) value. If there is a partition wall on the beam, the distributed load created by the partition wall is added to the beam load (w) load coefficient related to the load

is added by taking into consideration.

$$w = \frac{p-kn}{3} \quad (17.35)$$

$$w = \frac{p-kn-1.5}{3} - \frac{0.5}{m^2} \quad (17.36)$$

17.5.2.2 –Calculation values for shear force and bending moment due to vertical loads in any beam section are given below unless a more detailed calculation is made. **Table 17.1** It can be calculated using the relations given in .

17.5.2.3 –Unbalanced vertical load bending moments occurring at the beam ends in column-beam or column-wall connections (ΔM_{dirt}) (**Figure 17.6**), upper and lower floor columns or to your curtains **Equation (17.37)** And **Equation (17.38)** As defined in , the upper and lower floor column or wall rigidities will be distributed in proportion. In the calculation of the bending moment at the upper ends of the columns or walls on the top floor **Equation (17.39)** will be used.

$$M_{arm,upper} = M_{dirt} \frac{(I_{arm}/h)_{top}}{(I_{arm}/h)_{top} + (I_{arm}/h)_{lower}} \quad (17.37)$$

$$M_{arm,bottom} = \Delta M_{dirt} \frac{(I_{arm}/h)_{lower}}{(I_{arm}/h)_{top} + (I_{arm}/h)_{lower}}$$

$$M_{\text{pair,top}} = \Delta M_{\text{dirt}} \frac{\left(\frac{I_{\text{per}}}{h_1} \right)_{\text{top}}}{\left(\frac{I_{\text{per}}}{h_1} \right)_{\text{top}} + \left(\frac{I_{\text{per}}}{h_1} \right)_{\text{lower}}} \quad (17.38)$$

$$M_{\text{pair,sub}} = \Delta M_{\text{dirt}} \frac{\left(\frac{I_{\text{per}}}{h_1} \right)_{\text{lower}}}{\left(\frac{I_{\text{per}}}{h_1} \right)_{\text{top}} + \left(\frac{I_{\text{per}}}{h_1} \right)_{\text{lower}}}$$

$$\begin{aligned} M_{\text{arm}} &= \Delta M_{\text{dirt}} \\ M_{\text{per}} &= \Delta M_{\text{dirt}} \end{aligned} \quad (17.39)$$

Table 17.1 – Calculation Values for Vertical Loads in Beams

Inner Force	Account Value
Shear force on the first interior column or wall face of the outer span beam	$V_{\text{dirt}} = 1.15 \frac{W_n}{2}$
Shear force on all column or wall faces except cantilever beam	$V_{\text{dirt}} = \frac{W_n}{2}$
Shear force at cantilever beam support	$V_{\text{dirt}} = W_n$
Opening moment at outer opening	$M_{\text{dirt}} = + \frac{W_n^2}{14}$
Opening moment in internal opening	$M_{\text{dirt}} = + \frac{W_n^2}{16}$
Support moment on external column or external vertical wall face	$M_{\text{dirt}} = - \frac{W_n^2}{16}$
Support moment on the inner column or inner vertical wall face (in case of two openings)	$M_{\text{dirt}} = - \frac{W_n^2}{9}$
Support moment on the inner column or inner vertical wall face (in case of more than two openings)	$M_{\text{dirt}} = - \frac{W_n^2}{10}$
Support moment on the wall face of the beam stuck in the strong direction to the wall	$M_{\text{dirt}} = - \frac{W_n^2}{12}$
Cantilever beam support moment	$M_{\text{dirt}} = - \frac{3W_n^2}{4}$

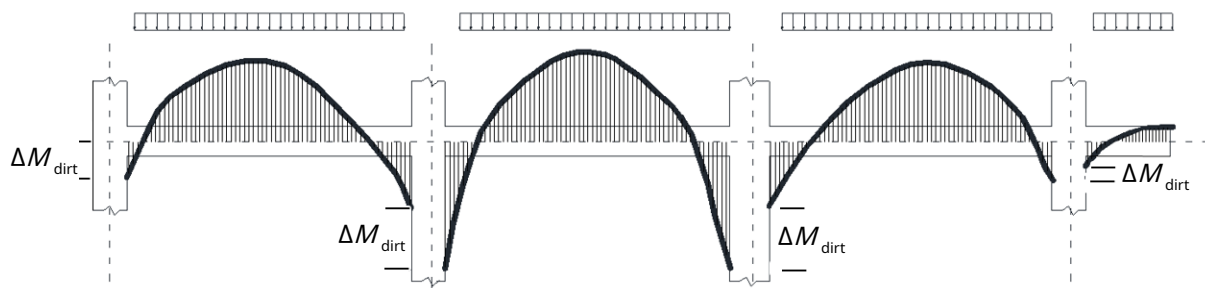


Figure 17.6

17.5.2.4 –Axial forces caused by vertical loads on columns or walls, fixed and mobile distributed loads acting on the floors ($g+q$) values and the total of the accumulated area shares along all floors carried by the column or curtain in question. In this case, the distributed dead load values to be considered in the calculation will include the weight of the horizontal and vertical load-bearing elements (beams, columns, curtains) and non-load-bearing elements (partition walls) along with the floor loads. The live load acting on the attic of the building should also include the snow load reduced by the load coefficient of 0.2.

17.6. INTERNAL FORCES BASED ON DESIGN

17.6.1 –The design of vertical reinforcement at the ends and body of the walls is taken into account for the combined effect of the axial force due to the vertical loads on the wall and the total bending moments caused by the vertical loads and earthquake effects in the rigid direction of the wall. will be done. $H_w / w \leq 2.0$ shear walls, bending moments due to earthquake effects, $-3 / (1 + H_w / w)$)–It will be enlarged by multiplying it by the coefficient. However, this coefficient will not be greater than 2.

17.6.2 –The design of longitudinal reinforcement in columns will be made for the combined effect of the axial force due to vertical loads on the column and the bending moments caused by the vertical loads and earthquake effects.

17.6.3 –The design of the flexural reinforcement in beams will be made for the sum of the bending moments occurring in the beam due to vertical loads and earthquake effects.

17.6.4 –The shear force that is the basis for the design of horizontal web reinforcement in walls, **Equation (17.40)** will be calculated with .

$$V_{to} = 2 V_{fairy} \quad (17.40)$$

17.6.5 –Shear force as a basis for the design of transverse reinforcement in the middle region of columns V_{to} , **Equation (17.41)** It will be calculated with the capacity design approach given in **(Figure 17.7)**. At the lower and upper ends of the column free height M_{pa} and M_{pu} values, bearing capacity of column section will be taken as 1.4 times the moment ($M_p = 1.4 M_r$). However V_{to} earthquake in column more than twice the calculated shear force due to the effects ($2 V_{parcel}$) It is not necessary to take a larger value.

$$V_{to} = (M_{pa} + M_{pu}) / n \quad (17.41)$$

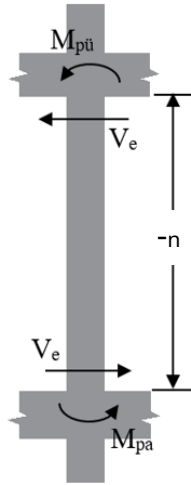


Figure 17.7

17.6.6 –Shear force as a basis for the design of transverse reinforcement in the middle region of beams V_o , **Equation (17.42)** and given in **Figure 7.9** It will be calculated with the capacity design approach shown in . At the left and right ends of the beam, the column or wall face M_{pi} and M_{pj} values, The bearing capacity of the beam section will be taken as 1.4 times the moment ($M_p = 1.4 M_r$).

$$V_{to} = V_{dy} \pm (M_{pi} + M_{pj}) / -n$$

$$V_{dy} = \frac{W \cdot n}{2}$$
(17.42)

17.6.7 –In foundation design and ground verification, internal forces formed at the lower ends of the vertical load-bearing elements connected to the foundation will be taken into account, bending moment at the lower ends of the curtain values ($M_{per, giant}$), in buildings without a basement 2, the basement is surrounded by curtains on four sides

It will be increased to 1.5 times in buildings where the basement is surrounded by curtains on two or three sides, and to 1.75 times in buildings where the basement is surrounded by curtains on two or three sides.

17.7. SIMPLIFIED SECTION BEARING CAPACITY RULES

17.7.1 –The bearing capacity moment of the beam sections, if there is no compression reinforcement in the section **Equation (17.43)** with, if there is compression reinforcement in the section **Equation (17.44)** can be calculated with .

$$M_r = A_s f_{yd} (0.85 d)$$
(17.43)

$$M_r = A_s f_{yd} (d - d')$$
(17.44)

17.7.2 –Vertical loads on column sections ($g + q$) axial compressive force under $N_{d, Max} = 0.35 f_{ck} b \cdot h$. value, and in shear walls, axial loads under vertical loads is the pressure force $N_{d, Max} = 0.30 f_{ck} b \cdot h$. will not be greater than the value of .

17.7.3 –Axial compressive and tensile strengths of column and wall sections, **Equation (17.45)** and **Equation (17.46)** will be calculated with .

$$N_{or} = 0.85 f_{cd} (A_g - A_{st}) + A_{st} f_{yd}$$
(17.45)

$$N_{en} = A_{st} f_{yd} \quad (17.46)$$

17.7.4 –Compressive force and bending moment values corresponding to the balanced condition in column and wall sections, **Equation (17.47)** And **Equation (17.48)** will be calculated with **Equation (17.48)** in A_{this}

value, the sum of the areas of the outermost row longitudinal reinforcement in the direction considered in the column section or the sum of the areas of the longitudinal reinforcement in the end regions in the wall section, A_{sg} value

is the sum of the areas of the longitudinal reinforcement in the inner row in the direction considered in the column section or the sum of the areas of the longitudinal reinforcement in the wall body in the wall section (**Figure 17.8**).

A_{this} And A_{sg} The sum of the values is equal to the total longitudinal reinforcement area in the column or wall section.

$$N_{br} = 0.42 f_{cd} b \cdot h \quad (17.47)$$

$$M_{br} = N_{br} 0.32 h + 0.6 A_{this} + 0.15 A_{sg} - f_{yd} [0.5 h - d'] \quad (17.48)$$

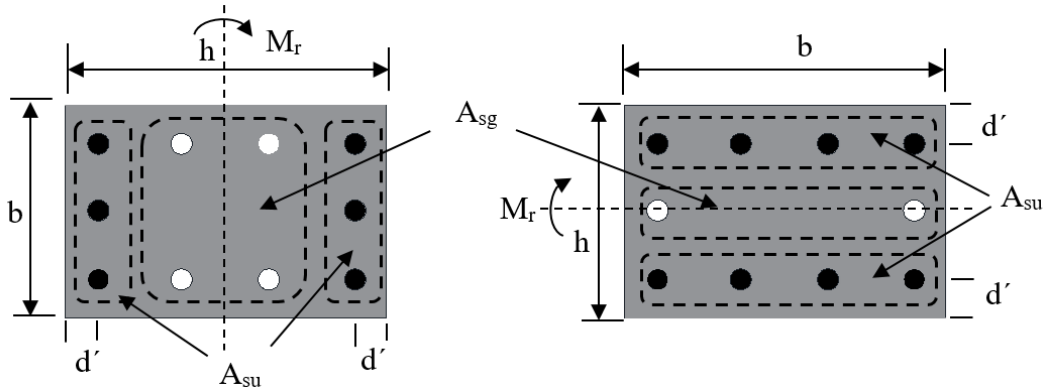


Figure 17.8

17.7.5 –Design based bending moment value obtained from the analysis in column and wall sections (M_d), the design basis axial pressure force value is less than the balanced pressure force value

in case of large ($N_d \geq N_{br}$) **Equation (17.49)** If the condition given by is small, ($N_d \leq N_{br}$)

Equation (17.50) will satisfy the condition given by (**Figure 17.9**).

$$M_d \leq M_r = \frac{N_{or} - N_d}{N_{or} - N_{br}} M_{br} \quad (17.49)$$

$$M_d \leq M_r = \frac{N_d + N_{en}}{N_{br} + N_{en}} M_{br} \quad (17.50)$$

17.7.6 –In corner columns **Equation (17.51)** The given condition will be met.

$$\frac{(M_d)_x}{(M_r)_x} + \frac{(M_d)_y}{(M_r)_y} \leq 1.0 \quad (17.51)$$

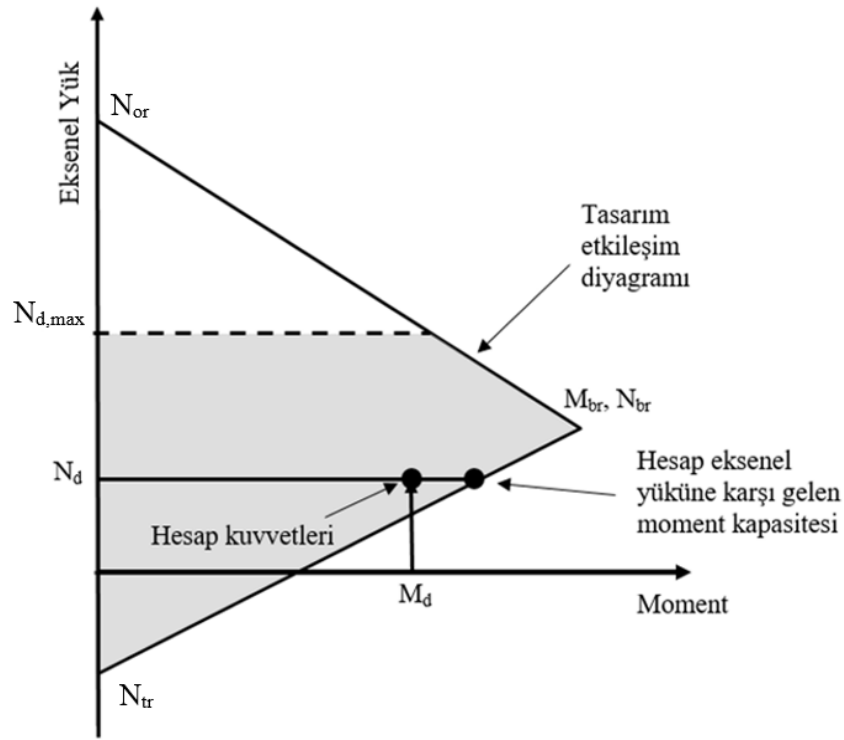


Figure 17.9

17.7.7 -Column sections, except the section above the foundation (or ground floor slab) and the interior column section under the top floor beam, in all other column-beam joint areas (**Figure 17.10**), **Equation (17.52)**It will satisfy the condition that the columns are stronger than the beams defined in .

$$(M_{r.a.} + M_{ru}) \geq 1.2 (M_{ri} + M_{rj}) \quad (17.52)$$

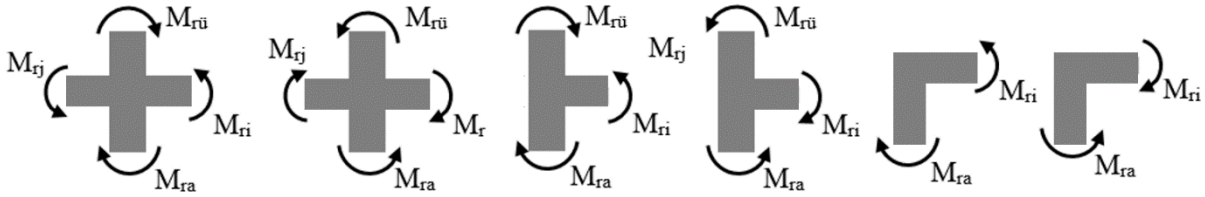


Figure 17.10

17.8. CURTAINS WITHOUT RECTANGULAR SECTION

17.8.1 -Special rules defined for structural analysis and shear wall design in buildings containing L, T, U, and C cross-section shear walls are given in this section.

17.8.2 -The arms of U and C cross-section curtains held only at one end in the lateral direction will be of the same length.

17.8.3 -In the analysis of the load-bearing system, these walls will be considered as rod elements located at the cross-section geometric centers and having horizontal stiffness in both directions. Wall horizontal stiffness, floor stiffness center, floor torsional moment, floor torsional stiffness

and in the calculation of shear forces, the moment of inertia values calculated in both directions with respect to the cross-sectional geometric centre of such shears will be used.

17.8.4 –In calculating the shear force resistance of such curtains, only the curtain arms that are in the same direction as the curtain shear force considered will be taken into account.

17.8.5 –In the calculation of the carrying capacity moment of such curtains, when the curtain head is under tension due to the bending effect (**Figure 17.11**), the load-bearing moment of the wall, for rectangular wall sections **17.7.4** and **17.7.5** With the method given in will be calculated. In this calculation b , as the total width of the curtain body, A_{th} is in the title will be taken as the total longitudinal reinforcement area. The curtain head is under pressure due to the bending effect (**Figure 17.11**) or in cases where one of the wall flanges is in tension and the other in compression, the wall's bearing capacity moment is for the beam sections.

17.7.1 It will be calculated with the method given in , the effect of axial load on the wall carrying capacity moment will not be taken into account.

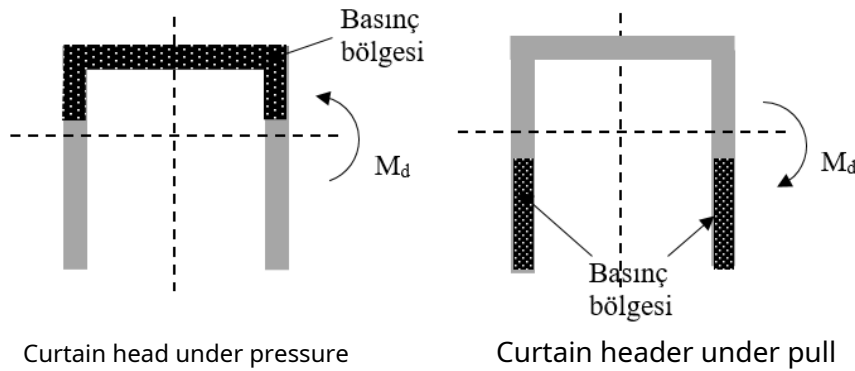


Figure 17.11

17.9. EARTHQUAKE LOAD TRANSFER FROM FLOORING TO FRAMES

17.9.1 –Equivalent earthquake loads acting on the building floors will be transferred from the floors to the curtain walls (or curtain arms) by beams embedded in the strong direction of the curtain.

17.9.2 –The axial tensile strength of the beam stuck in any shear wall in the strong direction on the shear wall face section, From the horizontal earthquake force transmitted to the 't' th floor ($F_{per,j}$) should not be less. This force, **17.5.1.7** As defined in , It is calculated as the difference between the shear force values occurring in the shear wall sections at the top and bottom of the floor.

17.9.3 –In the case where two beams are stuck into the curtain from both ends in the strong direction $F_{per,j}$ The load is distributed to the two beams embedded in the wall in proportion to the lengths of the beams on the floor plan.

17.9.4 –For this purpose, the longitudinal reinforcements (transfer reinforcements) to be placed on the beams will be ensured to be continuous along the length of the wall or overlapping additions will be made to these reinforcements within the length of the wall.

17.9.5 –The amount of transfer reinforcement calculated on the shear wall face can be reduced linearly, changing towards zero until the point where the beam ends on the floor plan. (**Figure 17.12**).

17.9.6 –Reinforcements that provide the interlocking conditions at any section of the beam and are left over from those required for the flexural strength of the beam can be taken into account as transfer reinforcement in the calculations.

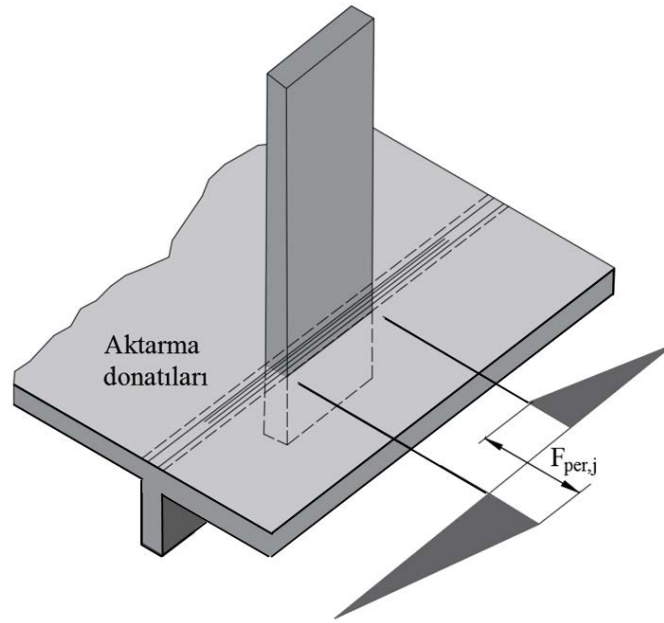


Figure 17.12